REAL TOEFL PASSAGES

Compiled by Mhd.Jawad

Table of Contents

Was There Water on Ancient Mars?

Despite Mars's bleak, cold climate today, there is abundant geologic evidence that liquid water once flowed on the Martian surface (and even recent evidence that some water still periodically flows today). The evidence of past flows includes river channels, now dry, that were once carved by powerful floods, valley networks of branching streams and tributaries indicating precipitation falling over wide areas, river deltas and layered sediments suggesting flow into standing bodes of liquid water, and salt and other minerals that dissolved in liquid water and were left behind when water evaporated. When that ancient wet period existed and when it ended is difficult to say.

The best method for dating a surface uses radioactivity, the process by which certain elements decay over time. One collects rocks from the surface and measures their radioactive elements to see how much of the decay products have accumulated in the rock since it formed. Transferring this age to the age of the surface can be tricky unless one knows where the rock came from. And one needs to collect the samples and bring them to a well-equipped laboratory.

The other way to date a planetary surface, which is used when rock samples are unavailable, is by counting craters. The longer a surface has been exposed to meteorite bombardment, the more craters it will have. This gives the relative age, and allows one to determine which surfaces are older and which surfaces are younger. To get the actual age one needs a standard. This was established for the Moon when lunar samples were returned to Earth as part of the United States Apollo program. The crater counts of the areas of the Moon from which the samples came were carefully noted. The samples were dated in terrestrial laboratories using radioactive decay, and the relation between absolute age in years and crater density--numbers per unit area-was determined. This relation is valid for the Moon, and it is possible that it is valid for Mars as well. Then counting craters on a Martian surface should yield its actual age. It is possible, however, that the number of impacting objects is different for Mars. The problem has been carefully studied, and adjustments have been made, but the general conclusion is that a wet period occurred early in Mars's 4.5-billon year history.

This causes a problem for Mars climate science, because the Sun was supposed to be fainter. perhaps by 20 or 30 percent, during the Mars wet period. Yet a watery Mars arguably would have required relatively warm temperatures. The power output of the Sun-its luminosity over time-is based on models of its interior and evolution. The models are well tested and agree with the luminosities of stars like the Sun whose ages are known. The same problem exists for Earth, which appears to have been warmer billions of years ago than it is today. Geologists have called this the "faint young Sun paradox." The solution to the paradox probably lies in climate science and not in the theory of stellar evolution, but the issue has not been entirely settled.

If Mars was wet billions of years ago, was it also warm? There could have been catastrophic floods released from underground aquifers (layers of rocks that can contain water) where the top surface of the water froze like the top of a Hawailan lava flow. The solid crust forms an insulating layer, trapping the heat of the liquid and allowing it to flow for great distances without freezing. Lava tubes on Hawalil are empty channels where lava once flowed in this way. Thus, the climate of Mars could have been wet but not warm, with the water flowing entirely under the ice. Ice sheets and glaciers on Earth today have extensive plumbing systems that allow water to flow and then drain away. Liquid water at the base of glaciers acts as a temporary lubricant and allows them to surge and slide downhill. Flood channels, river deltas, and standing bodies of liquid water all could have existed on Mars under thick layers of ice. The wet-but-not-warm hypothesis has the most trouble with areas that imply sources like precipitation over a wide area rather than the bursting of an underground aquifer. Further study of Mars's geology will clarify these issues.

Guam and the Brown Tree Snake

Guam is the largest island of Micronesia and, like many islands, once provided a home to a fantastic assemblage of native birds, reptiles, and mammals. The vast majority of these species are now extinct, and the remaining ones are threatened with extinction. Although many factors contributed to the loss of species, the principal cause is the invasive brown tree snake.

During the Second World War, the United States Navy established a large naval base on Guam. With the end of the war in 1945, the base proved useful for recovering abandoned war materials from the region, in particular, vehicles, aircraft, and other supplies from New Guinea where these items may have sat unused for some time. The brown tree snake is native to New Guinea and other regions of Australasia. It is typically nocturnal, and, during the day, rests within crevices and holes that provide good cover. It is commonly found in wheel wells on airplanes, under the hoods of cars, and in boxes of cargo, and most biologists think the brown tree snake was a hitchhiker within surplus Navy equipment brought to Guam from New Guinea in the postwar years of 1945-1950.

The brown tree snake was first noticed along Guam's southern shore near Apra Harbor. It then spread, somewhat slowly, until it occupied the entire island by 1970.Guam has no native snakes and no native predators that could have controlled tree snake numbers. Instead, the native vertebrates were all small species vulnerably to predation by the generalist tree snakes. The decline of most native forest animals was immediate and dramatic. Guam's Division of Aquatic and Wildlife Resources discontinued surveys of native birds and bats in the 1970s,as there were so few individuals of these species to count. The three native birds and mammals that persisted the longest were the Mariana fruit bat, Guam rail, and island swifitiet. The fruit bat and rail were relatively long-lived species that likely persisted simply because some individuals escaped predation and lived out the remainder of their lives; however, neither species had any reproductive success in the presence of the tree snake. The island swifitiet persists to this day but is confined to one cave on Guam. The swifitiet builds nests on the walls of caves using its own saliva and mud to adhere the nest to the cave wall. Despite this unusual habit, swifitiet are still vulnerable to tree snake predation, as tree snakes can easily capture prey in total darkness and climb most surfaces. The one cave where swifitiet persist is unique in that the cave walls are not textured enough for snakes to support themselves. This does not inhibit the swifitiet ability to build nests, but it does prohibit tree snakes from reaching those nests.

There have been many other losses of native forest animals on Guam. Consequently, the food web of Guam's forests (the dominant native habitat) has shifted dramatically. The most striking change is the reorganization of the web to one in which most components are nonnative. Beyond the brown tree snake, the Polynesian rat, Philippine turtledove, and the house mouse, among other nonnative species, have been successively established. Three native lizards survive in Guam's forests. All the lizards are small and active during daylight. One would assume that, with the destruction of the native food web, the brown tree snake would suffer from a lack of food resources, but the available pool of invasive prey and native lizards maintains tree snake densities. Consequently, native species went extinct with no corresponding negative effect on the tree snake. The tree snake even seems to be adapting to the diurnal habits of the remaining native lizards, since biologists have documented a shift in the tree snake's previously nocturnal activity pattern to one that Increasingly includes activity during daylight hours.

The reduction in the complexity of the Guam food web has had consequences beyond the loss of native animals. The loss of mammalian and avian insectivores presumably increased insect abundances at some cost to agricultural crops and to crop production. And newly introduced nonnative species may find it easy to invade Guam given the many "open niches “left by the loss of native species.

The Most Common Bird on Earth

The red-billed quelea is small, mostly brown bird found in the annual grasslands of Africa. Other than ornithologists and the people who see the bird in its native habitat, almost no one has ever heard of the little quelea, although, with a population estimated at 10 billion, it may be the most common bird on Earth.

Given the seemingly stressful conditions in the environments where the birds are found, it is surprising that the red-billed quelea survives at all, much less that it is so remarkably successful. Queleas feed on native annual grasses. Any animal that feeds primarily on the seeds of annual grasses has a potentially major problem-its food supply periodically appears and then disappears. A wide range of adaptations allows annual-seed eaters to survive. For example. rodents in annual grasslands often store their extra seeds to survive lean times, an adaptation that was taken up by human populations once they domesticated and became dependent on annual grasses, such as barley and wheat. If they are to survive, queleas must also cope with the challenging feast-or-famine situation attending annual-grass feeders. One might expect such species to be regularly wiped out (at least locally) when fluctuations in the growing season result in seed failures of their food. The red-billed quelea copes well enough to be the world's most common bird.

Part of the quelea’s success arises from its life-history adjustments to its environment. Over the African dry season, the species has to subsist on the ever-diminishing supply of seeds produced by the annual grasses at the end of the last wet season. As food becomes scarcer, the species feeds actively and gains sufficient weight reserves to be able to migrate to more auspicious areas. With the onset of rains in the African wet season, the seeds that queleas eat germinate (develop) and so are no longer available for food. The resulting severe food shortage lasts six to eight weeks. The birds are forced to migrate to better situations (if they can find them). For red-billed queleas, the savannas in which they live are an ever-shifting mosaic of patches of varying suitability depending on their recent rainfall history: some areas have dry seeds: others have immature grasses, mature grasses with abundant green seeds or old grasses with no seed. By moving over distances of 30 to 120 miles (50 to 200 km). the birds subsist in an inhospitable universe by moving from one patch of habitat with suitable food to another.

The so-called early-rains migration ends when queleas return to formerly abandoned locations. By this time the rains have come and grass seeds have germinated. The new grasses mature, and eventually fresh green seeds become available. ending the local food shortage. Then queleas establish communal breeding colonies in appropriate locations. They typically nest in thorny acacia trees. Their breeding colonies can contain several million pairs of breeding birds and can cover tens of hectares. The entire breeding cycle, from nesting colony to independent fledged young, requires six weeks-an exceptionally short interval for birds. The biological synchrony found among the birds in a breeding colony can be remarkable. Millions of eggs in millions of nests hatch on the same day. The fall of eggshells from subsequent dropping of the shells from the nests after these synchronous hatchings has been likened to a snowstorm. Sometimes the dry season comes early and the breeding areas dry out prematurely. In such times, the breeding colonies are abandoned. In other instances, the rains may sustain a prolonged green period and several episodes of breeding can occur in the same general area.

Since red-billed queleas feed on the seeds of annual grasses, they are preadapted to be effective feeders on the seeds of domesticated annual grasses as well. As human agriculture has planted more and more cereal grains across Africa, the numbers of quelea have exploded in response to the abundance of suitable food. The quelea's magnitude as pest animal has increased correspondingly. Although the numbers may be overestimated, the species is certainly capable of destroying 10 to 20 percent of the production of large farms and the entire crop of small, independent farmers. The red-billed quelea is not only the most abundant bird; it can also be described as the most destructive pest bird on Earth.

Iron in Ancient Africa

Metal-production technologies (metallurgy) have had a profound influence on the course of human history. Aside from gold, used for jewelry and ornamentation, the first metal to be widely used was copper, with smelting (the process of extracting metal from ore by heating) beginning by 5000 B.C.E. in modern-day Serbia. Production of bronze from copper and tin began by 2800 B.C.E. in Anatolia, in what is modern-day Turkey. Next was iron, with smelting starting around 1500 B.C.E. in Anatolia and spreading to Europe by 1100 B.C.E. Although iron production was difficult and complex, this metal's greater hardness made tools and weapons far superior to those made from copper and bronze.

For a long time, scholars believed that iron production came to sub-Saharan Africa from

Anatolia. One reason for this is that iron smelting requires considerable knowledge and experience, yet there was no evidence for previous metals technology in Africa. In particular, there seemed to have been no copper metallurgy, a likely precursor to iron production. Since Anatolia did possess the necessary knowledge and skill, it seemed to be a plausible source.

Some of the first evidence for African iron production was found in Meroe and dated to around 500 B.C.E.: iron tools and weapons, the remains of furnaces, and a large number of heaps of slag (waste left behind after processing metallic ores). This was significant, because the kingdom of Meroe had close contact with Egypt to the north, where iron production had been imported from Anatolia and was known by 670 B.C.E. Therefore, it seemed logical that iron technology had diffused southward through Egypt to Meroe, and from Meroe to sub-Saharan Africa. Some archaeologists have proposed a different source: Carthage, a north African city on the Mediterranean coast founded around 800 B.C.E. by the Phoenicians, whose homeland was close to Anatolia. However, no direct archaeological evidence for iron smelting has been found in Carthage, and it is not easy to explain how such knowledge and skill could have crossed the Sahara Desert to reach the sub-Saharan regions. Also, it appears that the earliest iron production south of the desert predates Carthage's founding.

Although it seems clear that iron technology reached Meroe from Anatolia via Egypt, recent research shows that the rest of Africa developed the technology independently. There is evidence of iron working as early as 1000 B.C.E. in Central Africa, and iron smelters were in use in the Great Lakes region of East Africa around 900 B.C., both earlier than would have been possible if the technologies had diffused from Anatolia through Egypt. Iron production was widespread in West and East Africa as early as 600 B.C.E., and it reached southern Africa a few hundred years later. Not only do such dates precede those of Meroe's iron production, but the design of the smelters differs significantly from that of the smelters at Mero. Also, there are recent indications that sub-Saharan Africa had copper technology in place as early as 2200. C.E., long before iron production began. Aside from the knowledge gained from copper technology, some experience may have come from pottery production, since around 4000 B.C.E. crushed iron ore was used to coat pottery in Egypt, and the temperatures necessary for this were almost as high as those needed for iron smelting.

Diffusion of iron technology through Africa was relatively slow, possibly as a result of the scarcity of iron ore. It also did not have the immediate impact of other technologies (such as steam power in the modem era), but iron tools made hunting easier and facilitated tasks such as shaping wood, building houses, and digging wells. Iron technology certainly helped farmers and so accelerated the spread of agricultural communities in West Africa after 400 B.C.E., such as those of the upper Niger River, where farming settlements began to form clusters of villages that specialized in rice, cotton, or dried fish, and eventually developed into large market towns. However, the rarity of major iron deposits meant that iron tools, weapons, and other goods had to be transported over long distances Along with the difficulty of iron production, this probably gave iron workers an honored place in the community.

Vasari, Art, and the Renaissance

The Italian artist, architect, and author Giorgio Vasari was an important influence on the modem Western conception of art. In his Lives of the Most Eminent Painters. Sculptors and Architects, a series of biographies of artists published in 1550, Vasari coins a word to describe the art made by the genius Michelangelo Buonarotti (1475-1564) and other painters, sculptors, and architects only slightly less talented than Buonarotti: Renaissance (rinascita, or "rebirth," in Italian). Although the artists Vasari discussed were not the first to take great interest in the art of ancient Greece and Rome-writers and thinkers of the fourteenth and early fifteenth centuries such as Petrareh (1304-1374) and orenzo Valla (1407-1457) had already seen themselves as reinvigorators of the Greek and Roman past-Vasari thought the rebirth of the late fifteenth century had gone beyond the original. In Vasari's eyes, Buonarotti and his contemporaries were not simply skilled and highly trained artisans, but "rare men of genius “who should sign their name to and take full credit for their work. (Traditionally, artists did not sign their name to art works.) This notion of the artist as creative genius did not apply to all branches of art, but to those in Vasari's title-painting, sculpture, and architecture-which were subsequently judged to be the major arts. Other types of art, such as needlework porcelain manufacture. goldsmithing and furniture making, were minor arts. decorative arts, or crafts and the names of those who made them were not considered important.

Along with inventing the term "Renaissance “Vasari influenced how Westerners understand other aspects of art and culture. He is often regarded as the first art historian, and his categories continue to shape the way that art history is taught and museums are arranged. Vasari's term "Renaissance" came to be used for a whole era and not simply its art. Because it derived from broad cultural changes and not specific events, the Renaissance happened at different times in different parts of Europe. "Renaissance" is used to describe fifteenth-century Italian paintings, sixteenth-century English literature, and seventeenth-century Scandinavian architecture. Some scholars see the Renaissance as the beginning of the modern era, while others see it as a sort of a transition between medieval and modem.

Vasari's distinction between art(works made by "rare men of genius") and craft(works made by everyone else) has been extended to other cultural realms in Europe: certain forms of writing, such as poetry, history, and epics, came to be defined as literature, while other types of writing, such as letters and diaries, were excluded from this category; certain forms of music became classical, while everything else was popular or folk: instruction that occurred in institutional settings was education, while that going on in the family or workshop was training or tradition. Scholars traced a growing split between professional and amateur, and, to a lesser degree, between learned and popular culture.

Research into all aspects of cultural life over the last several decades, however, has pointed out that the divisions between art and craft and learned and popular are more pronounced in hindsight than they were at the time. Vasari may have prized painting. sculpture, and architecture but patrons carefully ordered and paid enormous amounts for candlesticks, silver and gold tableware enamel or jewel- encrusted dishes, embroidery (cloth with decorative needlework), and enormous tapestries, along with paintings and statues. Embroiderers as well as painters experimented with perspective (the representation of three-dimensional space on a two-dimensional plane: paid great attention to proportion. shadowing, and naturalistic representation: and took their subjects from Greek and Roman antiquity. Writers paid as much, or even more, attention to literary conventions (standard practices) in their letters than in their poetry or drama. Folktales told orally for centuries became part of literary works in many countries, and people telling stories increasingly included those that someone had read in a book. Highly learned individuals participated in festivities involving all kinds of people that poked fun at their own intellectual pretensions and satirized various forms of power and status. Educated individuals did form a community among themselves with concerns different from the vast majority of the population, but they also shared many values and traditions with their neighbors who lacked extensive formal education.

Life in an Estuary

An estuary is the wide part of a river where it flows into the sea. It is affected by both marine influences (tides, waves, and the influx of saline water) and riverine influences (flows of fresh water and sediment) and is therefore not a perfectly stable environment. As a result, an estuary contains fewer resident species than the nearby marine or freshwater ecosystems, resulting in less competition for food and space. Because there is less competition, many estuarine species tend to be generalists that is, they are able to consume a variety of foods, depending on what is available. Species that can tolerate the salinity and temperature changes in estuaries can exploit the area's high productivity grow rapidly, and multiply into enormous populations.

Many marine animals have body fluids that contain about the same concentration of salts as seawater and that are essentially isosmotic to the surrounding water; that is, the pressure of their body fluids is equal to the pressure of the seawater, and they neither gain nor lose water Because the marine environment remains relatively constant, they do not have a problem maintaining water balance. Animals that live in estuaries, however, must have some physiological mechanism for dealing with the varying salinity; otherwise, their tissues and cells would absorb water and lose salts as they encountered an environment with lower salinity than the sea. Thus, estuarine animals are either osmoconformers, which survive by having tissues and cells that tolerate the loss of salts through dilution, or osmoregulators, which maintain an optimal salt concentration in their tissues regardless of the salt content of their environment.

Animals such as jellyfish are unable to actively adjust the amount of water in their tissues. When their environment becomes less saline, their body fluid gains water and loses ions until it is isosmotic to the surroundings. These organisms are examples of osmoconformers. The ability of osmoconformers to inhabit estuaries is limited by their tolerance for changes in their body fluid.in contrast to osmoconformers, osmoregulators employ a variety of strategies to maintain a constant salt concentration in their bodies. Osmoregulators that live in estuarine waters concentrate salts in their body fluids when the concentration of salts in the surrounding water decreases. For instance, some crabs and fish regulate their salt content in less-saline water by actively absorbing salt ions through the gills to compensate for salt ions lost from their body. This helps them to maintain a relatively constant body fluid. Some animals can either concentrate salts when their environment is less saline or excrete salts when the environment is extremely salty. The latter are generally animals that live partly on land or in areas such as salt marshes and mangrove swamps that occasionally receive large amounts of rain. Other animals, such as the blue crab, are osmoregulators at lower environmental salinity and osmoconformers at higher environmental salinity. Many fish species are osmoregulators that can adjust to both high-salt and low-salt environments.

Some estuarine organisms wall themselves off from their external environment to decrease water and salt exchange with their surroundings. Many estuarine animals have body surfaces that are less permeable than those of purely marine forms. This decreased permeability can be the result of increased amounts of calcium in the exoskeleton (outer skeleton) or increased numbers of mucous glands in the skin.

In addition to changes in salinity, the problem of remaining stationary in a changing environment affects the distribution of organisms in estuaries. The more or less constant movement of water in an estuary makes it difficult for some organisms to remain stationary long enough to feed and carry on other vital functions. Because of this, survival favors organisms that are benthic - those that live at the bottom. Marine plants and algae in estuaries have substantial root systems, or holdfasts, to prevent moving water from pulling them up and carrying them out to sea. Animals live attached to the bottom, either in the available spaces around other sedentary animals and plants or buried in the small crevices between sediment particles.

Evidence for Continental Drift

Continental drift refers to the idea that the present continents once formed a single. giant continent called Pangaea, and since that time have been slowly drifting apart. In 1915 the originator of this idea, the German meteorologist Alfred Wegener, was impressed by the close resemblance of coastlines of continents on opposite sides of the Atlantic Ocean, particularly South America and Africa. However, the configuration of the coastlines results from erosional and depositional processes and therefore is constantly being modified, so even if Wegener was right and the continents had separated in the distant past, it is not likely that the coastlines would fit exactly. But decades later it was shown that the continents fit together well along the continental slope (the broad underwater shelf on the edges of continents), where erosion is minimal, and recent studies have confirmed the close fit between continents when they are reassembled to form Pangaea.

If the continents were at one time joined, then the rocks and mountain ranges of the same age in adjoining locations on the opposite continents should closely match. Such is the case for the continents thought to have together formed the southern supercontinent Gondwana when Pangaea broke up into a northern and a southern supercontinent, mostly during the Jurassic period (200-146 million years ago) Antarctica, South America, Africa, Australia, New Guinea, and India comprised Gondwana. Marine, nonmarine, and glacial rock sequences of the Pennsylvanian epoch (325-299 million years ago) to the Jurassic period are almost identical for all five Gondwana continents, strongly indicating that they were joined at one time. The trends of several major mountain ranges also support the hypothesis of continental drift. The folded Appalachian Mountains of North America, for example, trend northeastward through the eastern United States and Canada and terminate abruptly at the Newfoundland coastline. Mountain ranges of the same age and deformational style occur in eastern Greenland, Ireland Great Britain, and Norway. Even though these mountain ranges are currently separated by the Atlantic Ocean, they form an essentially continuous mountain range when the continents are positioned next to each other.

During the later part of the Paleozoic era (544-255 million years ago massive glaciers covered large continental areas of the Southern Hemisphere, leaving behind layers of till (sediment deposited by glaciers) and striations (scratch marks) in the bedrock beneath the till. Fossils and sedimentary rocks of the same age from the Northern Hemisphere, however, give no indication of glaciation. Fossil plants found in coals indicate that the Northern Hemisphere had a tropical climate during the time that the Southern Hemisphere was glaciated.

All the Gondwana continents except Antarctica are currently located near the equator in subtropical to tropical climates. Mapping of glacial striations in bedrock in Australia India, and South America indicates that the glaciers moved from the areas of present-day oceans onto land. This would be highly unlikely because large continental glaciers (such as occurred on the Gondwana continents during the late Paleozoic era) flow outward from their central area of accumulation toward the sea. If the continents had not moved in the past, one would have to explain how glaciers moved from the oceans onto land and how large-scale continental glaciers formed near the equator. But if the continents are reassembled as a single landmass with South Africa located at the South Pole. the direction of movement of late Paleozoic continental glaciers makes sense. Furthermore, this geographic arrangement places the northern continents nearer the tropics, which is consistent with the fossil and climatologic evidence.

Finally, some of the most compelling evidence for continental drift comes from the fossil record. For example, fossils of Glossopteris, a group of woody shrubs, are found in equivalent Pennsylvanian and Permian coal deposits (299-251 million years ago) on all five Gondwana continents, Glossopteris shrubs produced seeds too large to have been carried by winds, and even if the seeds had floated across the ocean, they would not have remained viable for any length of time in salt water. The present-day- climates of the southern continents range from tropical to polar and are much too diverse to support the type of plants in the Glossopteris flora. These continents must once have been joined so that these widely separated localities were all in the same latitudinal climatic belt.

Han Dynasty Tomb Sculpture

Stone sculpture was something of a latecomer to Chinese art, starting a thousand years after figures were being made in jade and bronze Under the first period of the Han dynasty, known as the Western Han (206 B.C.E. 9 C.E.), it was used mainly for the tombs (burial chambers) of emperors or local rulers, but by the Eastern Han (25 C.E. -220 C.E.), the second period of Han rule, it had spread more widely. This change was largely the result of the increased importance of the tomb in the political philosophy of the time. The early Western Han emperors, faced with the problem of forging a unified empire threatened by uprisings on the part of ambitious rival kingdoms. had retained many of the first emperor's policies based on military force and harsh laws. But by the middle of the first century B.C.E., these were being replaced by an adaptation of the ideas of the fifth-century B.C.E. philosopher, Confucius, whose philosophy emphasized personal and governmental morality, to the conditions of a united empire. Believing that in the long run, stability depended on an acceptance of the legitimacy of the ruler (and dynasty) rather than on military force, Han intellectuals and officials adopted a moral philosophy based on the belief that humans are perfectible through education, and that a hierarchical society consists of a network of reciprocal duties and obligations: the subject's duty to obey the ruler was matched by the ruler's obligation to care for his subjects.

The increased emphasis on civic duty and order encouraged stability, loyalty, and obedience to the state, reinforcing central power. At the same time, the insistence on the value of education attracted intellectuals into state service, providing a well-qualified administration. In an age of rising standards of living with the growth of upwardly mobile merchant and artisan classes, the importance of filial piety (respect for parents and ancestors), one of the greatest of Confucian virtues, led to competitive tomb building. The imperial mausoleums (aboveground, freestanding tombs constructed as memorials) set the example and their extravagance was copied downwards. But why was there such urgent building of mausoleums?

A change in ritual had increased both the importance of the tomb and the scope for display. The ancestral rites previously held in city or palace temples were transferred to the tomb itself, making it necessary to build a hall south of the grave mound where the sacrifices could be performed. To emphasize the importance of the site, the approach was lined with an avenue of stone monuments known as the “spirit road" since it was along this road that the deceased would travel to the grave. This innovation became popular in Han society. It spread from the emperor to other parts of Han society, eventually crossing into neighboring lands such as Korea and Vietnam. The result was a manifold expansion in the use of stone statuary. No longer the prerogative of a few, it was now open to citizens anxious to display their piety and wealth by erecting freestanding stone statues on their fathers' graves. The use of statuary spread so rapidly that, in order to prevent a complete devaluation of its status, it was controlled by imperial decree, and henceforth the number and subject matter of spirit road statues were regulated according to the social rank of the deceased.

Stone elephant on Han Dynasty spirit road

More tomb statuary has survived than any other form of Han statuary, and the easiest way to see the development of sculpture during the first and second centuries is to use the tomb as a starting place. During the first century C.E. there appears to have been a remarkable increase in the use of stone in connection with the tomb. While free-standing statues and monuments were placed on the tomb above ground, the interior of stone and brick tomb chambers below were adorned with carvings of figures on walls and engravings on walls and doors. Features previously made in wood, such as coffins and steles (vertical markers placed in the ground to memorialize the dead), were now carved in stone. Above and below ground, tomb layout and ornamentation followed a coordinated plan. The same images and themes reappear in different places, and the tasks of the tomb are clearly allocated between different media. The result is an unparalleled picture of contemporary life and thought.

Aztec Artisans

Most of the material objects produced by the Aztec people of central Mexico during the Late Postclassic period (1350-1521) were manufactured in household settings by household members. Craft skills and knowledge, from weaving to metalwork, were passed from parent to child. Some objects were produced in small amounts strictly for household consumption, but many goods were produced in larger quantities and destined for market exchange. Specialization in the production of goods reached a high level of diversity and intensity during these times. In part, this was possible because of the food surpluses created by increasingly intense systems of agriculture, complemented by an abundance of nonagricultural resources. These surpluses released some persons, either partially or fully, from food-getting activities, allowing them to concentrate their time and energy in pursuing other production activities. High levels of craft production were also related somewhat to the population surge in the Late Postclassic period, providing large groups of producers and consumers of goods. And also in part, these specializations were stimulated and maintained by the increasing commercialization of the Late Postclassic period, facilitating relatively efficient and effective flows of raw materials and finished goods on local, regional, and extra regional levels.

Where artisans produced their crafts reflected, to some extent, their organization and sociopolitical context. Artisans were either "attached " or “independent." The former was situated in or near palaces, suggesting elite economic, social, and/or political relationships. This particularly applied to some luxury artisans who enjoyed the patronage(support) of the local ruler. These artisans were assured access to raw materials as well as guaranteed consumer. For instance, the Tenochtitlan-Tlate featherworkers attached to the Tenochtitlan ruler's palace enjoyed access to the ruler's aviary and stored tributes, and were specifically employed to produce ornate featherwork for the ruler's fine attire, for exquisite gifts for the ruler's diplomatic guests, and to adorn the god Huitzilopochtli. It appears that these highly esteemed artisans were resettled in or near the palace, yet it also seems that the household structure of production was maintained. That is, it is likely that entire households were relocated, allowing for the household's division of labor to be maintained other artisans attached to the royal palace in Tenochtitlan included goldworkers and silversmiths, coppersmiths, painters, cutters of stones, greenstone mosaic workers, and woodcarvers, all of whom worked at or near the totocalli, or bird house.

Artisans continued to be attached to native nobles even after the Spanish conquest of the Aztec empire. Don Juan de Guzman, a native governor of Coyoacan (an area near Tenochtitlan) who was appointed by the Spanish in the mid-sixteenth century, enjoyed the privilege of having “all the artisans and craftsmen... be attached to the royal house to do what is needed." Ten carpenters and ten stonemasons were also attached to Don Juan, but we know little about them beyond that he was having a house built, and “they are to do what is needed. "Prior to the Spanish conquest, when the king Motecuhzoma Xocoyotzin employed sculptors to carve a statue of him, he paid them handsomely for their work. Perhaps these were independent artisans, although the payment could as well represent compensation to attached artisans. While royal or elite sponsorship provided attached artisans with a predictable livelihood, it perhaps offered them less opportunity for creativity than the independent luxury artisans working beyond the reach of the palace and selling their finery in the markets.

Some of these independent artisans, producing both luxury and utilitarian goods, were concentrated in specific neighborhoods, or calpolli. Independent artisans could not necessarily depend on the generosity of the state in obtaining raw materials for their crafts. They relied on merchants and markets for access to raw materials and as outlets for their finished products. The Tlatelolco featherworkers maintained especially close relations with their equally exclusive neighbors, the entrepreneurial pochteca (professional merchants), who provided them with the shimmering tropical feathers essential in their craft. While undocumented, it is likely that similarly close relations existed between luxury artisans and merchants in other city-states.

Whether attached to elite palaces or living in exclusive neighborhoods, specialized artisans required enduring and predictable links to several other areas of Aztec life: they gained status through participation in society-wide ceremonies, served royal and other noble palaces, dealt with long-distance merchants, and were consistent fixtures in any major marketplace.

Why Water Bugs Practice Parental Care of Eggs

Although paternal care of young (fathers caring for their young) is common among fishes, the trait is rare among other animals, vertebrates and invertebrates alike. Among the few paternal insects are the giant water bugs. In Lethocerus water bugs, males guard and moisten clutches of eggs (eggs deposited during a single episode of laying) that females glue onto the stems of aquatic vegetation located above the waterline. Other kinds of male water bugs (e.g., Abedus and Belostoma) permit their mates to lay eggs directly on their backs, after which the male assumes responsibility for their welfare. A male Abedus caring for its eggs spends hours perched near the water's surface, pumping his body up and down to keep well-aerated water moving over the eggs. Clutches of Abedus eggs that are experimentally separated from a male attendant do not develop, demonstrating that male parental care is essential for offspring survival in this case.

Bob Smith has explored both the history and the adaptive value of these unusual paternal behaviors. Since the closest relatives of the family that contains the paternal water bugs are the Nepidae, a family of insects without male parental care, we can be confident that the brooding species (species caring for eggs) evolved from nonpaternal ancestors. Whether out-of-water brooding and back brooding evolved independently, or whether one preceded the other, is not known, although some evidence suggests that back brooding came later. In particular, female Lethocerus sometimes lay their eggs on the backs of other individuals, male or female, when they cannot find suitable vegetation for that purpose. This unusual behavior indicates how the transition from out-of-water brooding to back brooding might have occurred. Females with the tendency to lay their eggs on the backs of their mates could have reproduced in temporary ponds and pools where aquatic vegetation sticking out of the water was scarce or absent.

But why do the eggs of water bugs require brooding? Huge numbers of aquatic insects lay eggs that do perfectly well without a caretaker of either sex. However, Smith notes that the eggs of giant water bugs are much larger than the standard aquatic insect egg, with correspondingly large requirement for oxygen, which is needed to sustain the high metabolic rates underlying embryonic development. But the relatively low surface-to-volume ratio of large aquatic egg leads to an oxygen deficit inside the egg. Since oxygen diffuses through air much more easily than through water, laying eggs out of water can solve that problem. But this solution creates another problem, which is the risk of drying out that the eggs face when they are high and dry. The solution, brooding by males that moisten the eggs repeatedly, sets the stage for the evolutionary transition to back brooding at the air-water interface.

Wouldn't things be simpler if water bugs providing parental care simply laid small eggs with large surface-to-volume ratios? To explain why some water bugs, produce eggs so large that they need to be brooded, Smith points out that water bugs are among the world's largest insects almost certainly because they specialize in grasping and stabbing large vertebrates, including fish, frogs, and tadpoles. Water bugs, like all other insects, grow in size only during the immature stages. After the final molt (loss of outer covering layer) to adulthood, no additional growth occurs. As an immature insect loses its outer covering as it grows from one stage to the next, it acquires a new flexible outer covering that permits an expansion of size, but no immature insect grows more than 50 or 60 percent per molt. One way for an insect to grow large, therefore, would be to increase the number of molts before making the final transition to adulthood. However, no giant water bug molts more than six times. This observation suggests that these insects are locked into a five-or six-molt sequence, just as the bird species the spotted sandpiper evidently cannot lay more than four eggs per clutch. If a water bug is to grow large enough to kill a frog in just five or six molts, then the immature insect that hatches from the egg must be large, because it will get to undergo only five or so 50-percent expansions.

Imitation in Monkeys and Apes

Almost all animals learn novel tasks more easily if they can observe a knowledgeable demonstrator Biologist Tom Langen trained individual magpie jays to pry open a door on box that contained food. Subsequently, birds whose social groups included a demonstrator learned how to open doors much more rapidly than birds whose groups did not Indeed, birds in groups that lacked a demonstrator did not even realize that there was food in the boxes Similarly, in captivity, many monkeys can learn to use rudimentary tools to obtain food, and they do so more quickly and accurately in the presence of a demonstrator. If monkeys learned like humans do. it would be safe to assume that the monkeys learn to perform the tasks by watching the demonstrator and imitating his actions. This would imply that the monkeys understand the demonstrator's intentions and goals. But this does not seem to be the case.

Most animal species show very little evidence of purposeful copying by imitation. In the laboratory, monkeys are attracted to tools and often begin experimenting with them after observing another monkey do so, suggesting that social companions enhance and facilitate tool use. But learning about a tool's use through “social facilitation" typically requires extensive practice through trial and error. As a result, different individuals adopt different idiosyncratic styles, and the spread of the skill is very slow. Although many monkey species can learn to use tools in captivity, there are very few examples of tool use in the wild. Capuchin monkeys, which inhabit Central and South America, are the only monkeys that regularly use sticks or stones to pry into trees or break open nuts under natural conditions. Capuchin also have comparatively large brains compared to other monkeys. The relative lack of spontaneous tool use in monkey species suggests that monkeys have difficulty recognizing the relation between actions and objects.

In the wild, baboon monkeys seem to use tools only in aggressive contexts. When displaying, male baboons occasionally wave or throw sticks in the direction of their rivals. Whether they recognize the potential function of these weapons, though, seems doubtful. When one group of baboons in Namibia dislodged stones from a cliff when they were disturbed by humans, they did so not only when people were under the cliff but also when they were too far away to be struck. At Gombe, where chimpanzees compete with baboons for food, chimpanzees throw branches at baboons. Baboons however, never throw objects at chimpanzees.

Chimpanzees and orangutans are different. In captivity these animals, which are apes rather than monkeys, attend closely to a demonstrator when learning to use tools to open boxes, and they require very few trials to learn to copy his actions. They seem to recognize the intentions and goals of the demonstrator, and they rapidly learn a tool's function from attending to his behavior. Although they do not copy the demonstrator's exact motor patterns as closely as children do, they do tend to conform to his technique.

Under natural conditions, chimpanzees and orangutans also use a variety of tools for different purposes. In fact, different populations of chimpanzees and orangutans use different kinds of tools for different purposes, and the use of specific tool types appears to be socially transmitted. Two points about tool use in chimpanzees and orangutans seem relevant. First, in marked contrast to monkeys, no population of chimpanzees has been reported not to use tools. Second, unlike monkeys, chimpanzees and orangutans often show foresight and planning in selecting and modifying tools in advance of their use. Before fishing for termites, chimpanzees often search some distance from the termite mound to find an appropriate prodding stick and strip the bark from it. Similarly, when preparing to crack open nuts, chimpanzees must carry both stones and nuts to suitable hard surfaces. Often, this means that a chimpanzee will carry both nuts and stones over considerable distances before beginning a nut-cracking session.

In their ability to plan, understand a tool's function, and appreciate a demonstrator's goals, then, apes are strikingly different from most monkeys. This is not to say, however, that tool use and manufacture are unique to apes, or that monkeys are completely incapable of imitation.

Roman and Chinese Metalworking

Chinese steel was highly valued in the Roman Empire (which controlled much of the western world between 27 B.C.E. and 476 A..), since Roman workshops could not mass-produce metals with comparable strength and sharpness. Steel is a metal alloy (mixture)manufactured by heating iron ore (raw iron) to high temperatures, then combining the molten(melted) metal with carbon or other strengthening elements. The quality of steel produced by these techniques depends on the sustained temperature of the furnace and the quantity and condition of the added compounds. Good-quality steel is harder than iron and has much greater flexibility and strength. Blades made from steel therefore maintain a sharper edge for a longer time and have a greater resistance to rust than their equivalent in iron. Western civilizations did not have sufficient knowledge of the carbon compounds that need to be added to molten iron to produce reliable steel. Instead, the Romans produced wrought iron (iron with very low carbon content) by first heating iron ore in a furnace (oven) to separate metal from impurities in a process known as smelting. This technique was used throughout the Roman Empire to make tools and bladed weapons. The resultant iron also acquired a trace of strengthening carbon from the charcoal fueling the furnaces.

The situation was different in the Chinese Han Empire (206b.c.e.220a.d., where metalworkers using more advanced furnaces had identified the natural compounds that created better-quality iron during the smelting process. Through continual practice they had refined the measurements needed to ensure that the compounds added to iron ore introduced sufficient carbon to create a reliable quality of steel. The Chinese had developed large enclosed furnaces that included bamboo nozzles (round openings) to produce steady streams of air. This made it easier to keep the fire at a steady heat and control reactions within the furnace. Chinese furnaces also burned compressed coal, which further increased temperatures and reduced fuel costs. This was significant because mass-produced cast iron (iron with carbon content greater than 2 percent) could be transformed into steel by applying blasts of cool air that provided oxygen to the molten metal (the “Hundred Refinings Method"). The Chinese also knew how to turn wrought iron into steel. Blades were wrapped in fruit skins rich in carbon containing a small amount of impurities. These packages were then sealed in clay containers and heated at high temperatures over a sustained period (up to twenty-four hours) until the metal absorbed the necessary carbon and strengthening elements. In China, these techniques were used to mass-manufacture a variety of tools, including knives, hammers, and cooking pots.

This important development allowed the Han regime to mass-manufacture high-quality metal, including armor-piercing weapons. The Han understood that steel manufacturing techniques provided Chinese troops with superior armor and weaponry. Consequently, the regime tried to restrict the spread of this technology to foreign peoples and prevent the export of Chinese steel supplies to neighboring nations. However, there were strong incentives for Chinese merchants to break the embargoes(restrictions)an offer steel at high prices to foreign traders Smuggling became a problem, and the warlike Xiongnu people of Mongolia were able to capture Han metalworkers when they raided Chinese territory. From these prisoners the Xiongnu learned how to produce supplies of their own steel weaponry, which they offered for exchange with other peoples. In this way supplies of superior eastern steel reached the Parthian Empire in Iran.

The superiority of this steel weaponry was demonstrated on the battlefield of Carrhae in 53 B.C.E., when a Roman army encountered Parthians from eastern Iran. The steel-tipped arrows carried by the Parthians easily punched through Roman shields and armor. The early Roman Empire possessed no weapons or armor comparable to that of the Parthians. Consequently, Roman authorities did not fully understand the significance of the metal used by their Iranian rivals. It was assumed that Parthian steel was the product of unique iron ores that could only be found in the distant east, and no initiative was taken to determine how these metals could be produced. This explanation seemed reasonable when the Romans considered how geography and climate provided the distant east with better fabrics more precious stones, and more potent flavorings in the form of spices.

Early Chinese Silk Production

China is justly famous for its silk production which probably began more than 6,000 years ago. Excavations of Neolithic sites have revealed clay artifacts with impressions made apparently by silk cloth, as well as stone ornaments carved in the shape of silkworms. The earliest find of silk itself, dated to around 2800 B.C., is from Zhejiang province, where a fragment of cloth was preserved in damp conditions inside a bamboo box. Zhejiang province, in southeast China, has always been an important silk-producing area and was probably also the center of one of the major prehistoric cultures.

Later writers attributed a well-organized system of production to the Western Zhou dynasty (1050-771 B.C.). A text compiled or revised in the Han period (206 B.C.-A.D. 220) describes a silk supervisor, a hemp supervisor, dyers, and weavers working in the women's section of the palace. Although one silkworm, making its cocoon, spins a pair of filaments that may be up to one kilometer in length, because of the fineness of these filaments, thousands of silkworms are required to produce enough silk to weave a length of cloth. From an early date in China, it was found best to carry out the manufacture of silk cloth on a large scale, with division of labor according to the various tasks: picking the mulberry leaves to feed the silkworms, reeling the filaments from the cocoons, weaving, etc. In the eighteenth and nineteenth centuries, this highly organized and subdivided industry fascinated Westerners, who collected sets of illustrations of the different stages. Silk production has traditionally been associated with women in China, and in imperial days the empress would perform an annual mulberry-leaf-picking ceremony outside the Hall of Sericulture in Beijing.

By the Han dynasty the silk industry was already highly specialized, producing in addition to plain cloth, self-patterned monochrome cloth figured gauzes, and thicker, multicolored cloth. The last was highly valued, costing up to fifteen times as much as plain silk. Chain stitch embroidery also became widespread at this time and was used to decorate clothes, wall hangings, pillows, and horse trappings. Elaborate examples have been found in Han dynasty tombs in Hubei and Hunan provinces, where waterlogged land has preserved much organic material.

Although produced in large quantities, silk remained an expensive luxury. Both woven silk and raw silk were used as tribute to the emperor and as offerings to foreign tribes, such as the Xiongnu on the northwestern borders of China. From here Chinese silk must have been traded along the Silk Route through Central Asia and the Middle East as far as Europe. Most early examples of silk cloth in the West appear to have been woven locally, but from cultivated silk fiber of the Bombyx mori worm. The fact that China was the origin of this fiber was acknowledged by the ancient Greeks and Romans, who referred to silk as serica, literally, "Chinese. "By the end of the Han period, silk cultivation and knowledge of silk processing must have begun to spread westward, and by the fifth or sixth century had reached the Mediterranean area. But Western silk production throughout the Middle Ages (fifth to fourteenth centuries A.d.) was never sufficient to satisfy demand from Europe and Mediterranean areas, and imported Chinese silk remained very important.

Trade along the Silk Route was at its most vigorous during China's Tang dynasty (A.D. 618-906), and travelers record the bazaars of the Middle East as being full of Chinese patterned cloth and embroideries. Simultaneously, we are told that the Tang capital of Chang'an was populated by large numbers of Iranian craftsmen. A silk weave now known as weft-faced compound twill appears among Chinese textiles for a few centuries from about A.D. 700. This may well have been a technique introduced by foreign weavers, as it seems to have developed originally in Iran. In the West it was particularly associated with repeating designs of roundels(circles) enclosing paired or single animals, with flower heads or rosettes between the roundels. The paired animals, rosettes, and the ring of pearl-like dots that often made up the roundel frame all passed into Chinese design.

Indeed, trade in textiles or other items may have carried the motif of the pearl roundel to China before any transfer of technology, since it first appears in China as early as the fifth century a d. on stone carvings at Buddhist cave temples.

Murals, Frescoes, and Easel Paintings

Murals are pictures on walls or ceilings that become a part of the building's architectural decoration. Archaeologists have excavated and saved mural paintings from Pompeii and Herculaneum, cities that were buried by an eruption of Mount Vesuvius in 79 C.E. Unfortunately, outdoor painting of any kind has little chance of surviving the effects of the environment for very long Richard Haas's famous mural Homage to Cincinnatus, which was painted in 1983 and is displayed on the Brotherhood Building in Cincinnati in the United States, has faded considerably. Indoors, where walls can be kept dry, murals have traditionally done much better when they were painted in fresco. The Italian word fresco means "fresh." Michelangelo's Sistine Chapel ceiling, painted in the sixteenth century, is a perfect example of fresco painting. In true fresco, water-based paints are applied to fresh, wet plaster so that the pigment (color) soaks into the plaster. The paint actually becomes part of the wall. The lime of the plaster, changing to calcium carbonate when it dries, binds the pigments permanently in the wall. The colors are matte, not glossy. They stay bright for hundreds of years, and as long as the wall lasts, the fresco lasts.

Paint applied to the wall when it is dry is not very permanent. Since some pigments will not combine properly in fresco, they have to be applied by the artist. The blue pigment that Giotto used in The Nativity, fresco portraying the birth of Jesus Christ, would not mix with the lime in plaster, so he had to paint it on the dry plaster. As a consequence, the blue of the sky and the blue of Mary's robe have almost all flaked off. Since true fresco must be painted on wet plaster, an artist can work on only a small portion of the wall where the plaster was freshly applied beforehand. A normal area for a day's work might be about one or two square yards, although the size depends on the complexity of the painting. Some days a fresco painter might complete only a head. In Giotto's fresco The Nativity, the breaks between one day's work and the next are visible on close inspection.

Fresco painting takes a lot of preparation because artists have to have the entire composition worked out before they get started on an individual patch of plaster. In addition to the usual preliminary drawings, fresco painter also makes cartoons-d on paper in the same scale as the mural-so that the essential lines of the composition can be traced and transferred to the wall.

Most artists produce easel paintings-painting that are painted on a portable wooden support- that are intended to be hung on a wall. Bernardo Daddi painted the three-part Madonna and Child (Jesus Christ and mother) for use during private prayer Originally, it may have rested on a table. After their prayers, the artwork owners could have folded the sides together to store it or carry it to another location.

When Daddi and other European artists in the fourteenth century wanted to create an easel painting, they usually worked on carefully prepared wooden panels. Wooden panels assembled from several planks and as large as ten to twelve feet high were common in that era. The wood had to be dried, aged, and sealed so that it would become an acceptable painting surface, as smooth as polished marble. Daddi mixed his pigments with egg yolk, a medium that is called egg tempera. If the panel received decent treatment over the years, a 650-year-old egg tempera such as Madonna and Child will have colors that are still preserved and fresh. Egg tempera is quite a permanent medium.

Egg tempera paint dries as fast as egg yolk dries on a breakfast plate and is just as stubborn to remove. An artist applies it in short strokes because of the rapid drying time, and by the time the artist comes back to the panel with the next brushstroke, the preceding one has already dried. Little blending is possible, and the only way to remove a mistake is by scraping it off. Every stroke has to be meticulously placed with small brushes.

Recognizing Social Play in Animals

Many animals engage in some type of social play-that is, playing with others including chasing and fighting. Three functions of social play have been proposed: Social play may (1) lead to the forging of long-lasting social bonds, (2) help develop much needed physical skills, such as those relating to fighting, hunting, and mating, and (3) aid in the development of cognitive skills. One cognitively related benefit of social play revolves around the idea of self-assessment. Here, animals use social play as means to monitor their developmental progress as compared to others. For example, in infant sable antelope (Hippotragus Niger), individuals prefer same-age- play partners. While this preference could be due to numerous factors, Kaci Thompson has hypothesized that it is primarily a function of infants attempting to choose play partners that provide them with a reasonable comparison from which to gauge their own development.

Since many of the behavior patterns seen during play are also common in other contexts-hunting, mating, dangerous aggressive contests-how do animals know they are playing and not engaged in the real activity? And even more to the point how do they communicate this information to each other? Bekoff has proposed three possible solutions to this important, but often overlooked, question. One way that animals may distinguish play from related activities is that the order and frequency of behavioral components of play is often quite different from that of the real activity. That is, when play behavior is compared with the adult functional behavior that it resembles, behavioral patterns during play are often exaggerated and misplaced. If young animals are able to distinguish these exaggerations and misorderings of behavioral patterns by, for example observing adults that are not involved in play, a relatively simple explanation exists for how animals know they are playing.

A second, somewhat related, means by which animals may be able to distinguish play from other activities is by the placement of play markers. These are also known as play signals and can serve both to initiate play and to indicate the desire to continue playing and to warn adults that the young are playing and not in danger of injury. In canids (the animal group that includes dogs and wolves), for example, biting and shaking are usually performed during dangerous activities such as fighting and predation. Yet, biting and shaking are also play behaviors of young canids. Bekoff found that play markers such as a bow (lowering the head) would precede biting and rapid side-to-side shaking of the head to indicate that they were not dangerous behaviors. The bow would communicate that this action should be viewed in a new context-that of play. Another play marker might be a particular kind of vocalization-for example, chirping in a rat, whistling in a mongoose, panting in a wolf or a chimpanzee before or during a play interaction. Or there might be a distinctive smell that indicated that the animals were engaged in play.

Play markers have also been found in primates such as the juvenile lowland gorilla. Juvenile lowland gorillas play with each other often, and play ranges from what Elisabetta Palagi and her colleagues call gentle play to rough play. Palagi's team discovered that when juvenile gorillas-particularly males-were involved with rough play, the play was often preceded by a facial gesture they call the play face. This facial gesture, which is not seen in other contexts, includes slightly lowered eyebrows and an open mouth. In addition to using this facial gesture during rough play, juvenile gorillas also displayed it when a play session was in a place that made escape (leaving)difficult -another context in which it may be important to signal to others that what is about to occur is play.

Yet another way by which young animals may be able to distinguish play from related behaviors is by role reversal, or self-handicapping, on the part of any older playmates they may have. In role reversal and self-handicapping, older individuals either allow subordinate younger animals to act as if they are dominant during play or the older animals perform some act (for example, an aggressive act) at a level clearly below that of which they are capable. Either of these provides younger playmates with the opportunity to recognize that they are involved in a play encounter.

Mesozoic Seed Dispersal

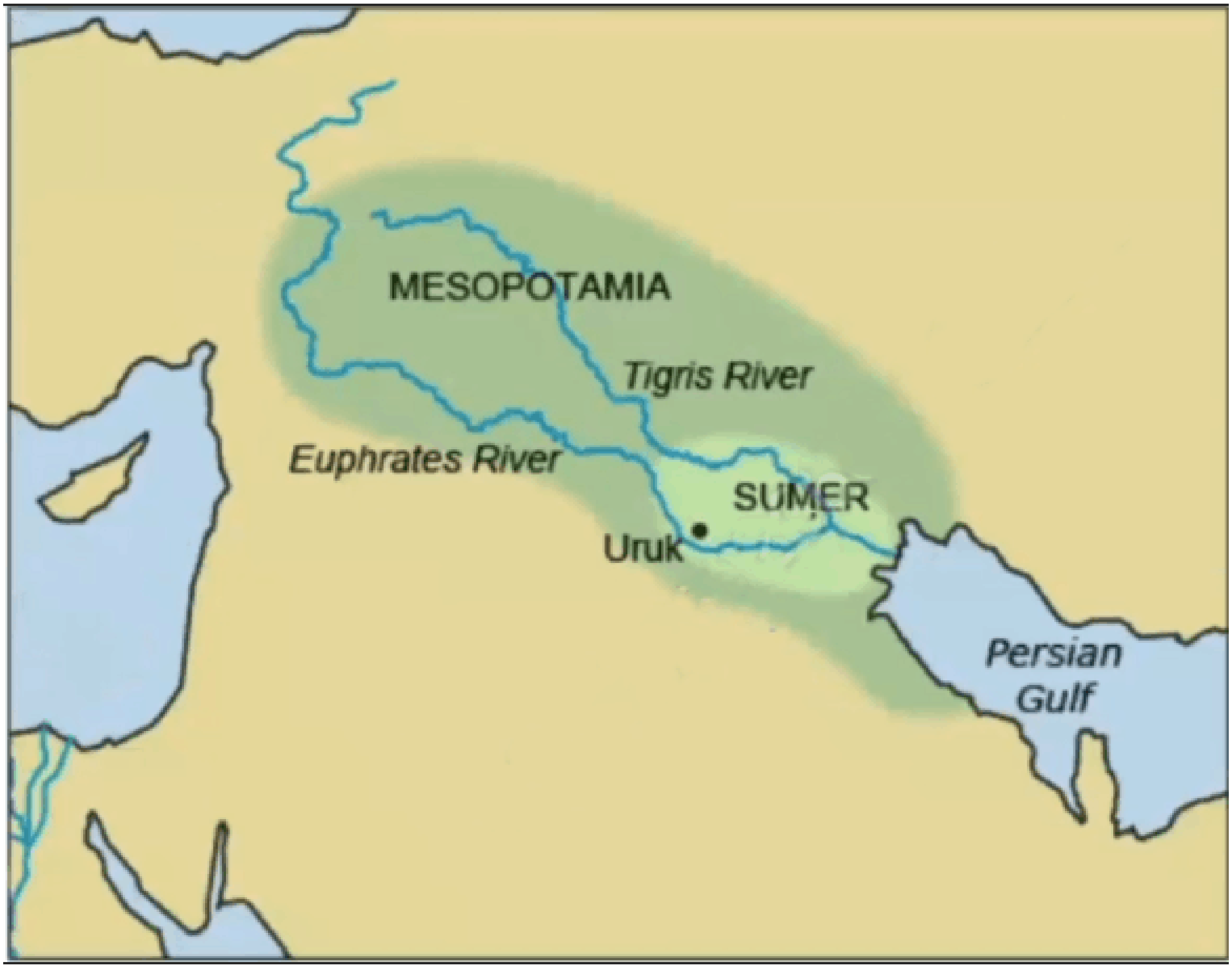
Many plants rely on wind to disperse(spread)their seeds. However, because seeds often fall into the wrong places. wind dispersal seems wasteful; it can work only in plants that produce great numbers of seeds. Wind-dispersed seeds must be light and small, so they cannot carry much energy for seedling growth. They have to germinate (grow from a seed) in relatively well-lit areas, where they can obtain energy from sunlight. Alternatively, a plant could have its seeds carried away by an animal and dropped into a good place for growth. Many animals can carry larger seeds, which can successfully germinate in darker places. Some animals visit plants to feed on pollen or nectar, and others eat parts of the plants. Others simply walk by the plant, brushing it as they pass. Small seeds may be picked up accidentally during such visits, especially if the seed has special hooks, burrs, or glues to help attach it to hairy or feathery visitor.

However, seed dispersal by animals is not cost-free. Many dispersers eat the seeds, passing only a few undamaged through their internal organs. So, there is a significant wastage of seeds, depending on a delicate balance between the seed coat and the teeth and stomach of the disperser. Too strong a seed coat, and the disperser will turn to easier food or germination will be too difficult; too weak a seed coat, and too many seeds will be destroyed. Some plants' seeds germinate well only if they are eaten by a particular disperser. Scientists believe animals first began dispersing seeds sometime during the Mesozoic era (252-65 million years ago). Seed dispersal by animals surely evolved after insect pollination. Insects of the Jurassic period (206-145 million years ago) may have become good pollinators, but they were too small to have been large-scale seed transporters. Jurassic reptiles (including dinosaurs) were large enough, but often had low food-processing rates, so any seeds they swallowed were exposed to digestive juices for a long time. Reptiles do not even have fur in which seeds can be entangled (though feathered therapod dinosaurs might have). Seeds were undoubtedly dispersed by dinosaurs to some extent, since the huge vegetarian dinosaurs ate great quantities of vegetation. But in spite of the size of the deposits of fertilizer (in the form of waste) that must have surrounded seeds passing through dinosaur, browsing dinosaurs (ones that eat grass, leaves, and shrubs) probably damaged and trampled plants more than they helped them. It is unlikely that any Mesozoic plant would have encouraged dinosaur browsing.

Transport over a long distance can take a seed beyond the range of its normal predators and diseases and can allow a plant to become very widespread provided that there are pollinators in its new habitat. As flowering plants adapted to seed dispersal by animals, they probably dispersed into new habitats much faster than other plants did. Other things being equal, we might expect a dramatic increase in the number of flowering plants found in the fossil record as they adapted Toward seed dispersal by animals rather than by wind.

There were few effective animal seed-transporters in the Jurassic period, and dinosaurs are unlikely candidates in the Cretaceous period (the last period of the Mesozoic era). The biologist Philip Regal suggested that birds and mammals prompted the distribution of flowering plants by aiding them in seed dispersal. Birds and mammals have feathers and fur in which seeds easily become entangled; seeds pass quickly through their small bodies with their high food-processing rates and are likely to be unharmed unless they have been deliberately chewed. Flowering plant seeds would have been especially suited to transport by vertebrate (back-boned) animals because of their extra protective coating. In contrast, seeds of conifers (plants with needle-like leaves) are usually small and light, designed to blow in the wind, and conifers depend on close clusters for pollination. Isolated conifers are likely to be unsuccessful reproducers, and additional transport would make little difference to their long-term success. However. many new types of flowering plants appeared in the Early and Middle Cretaceous period, when mammals and birds were still minor members of the ecosystem, so many scientists are unconvinced by this explanation of flowering plants' early success.

Challenges of Mesopotamian Agriculture



One of the world's first civilizations began in the area between the Tigris and Euphrates Rivers known as Mesopotamia (modern Iraq) where much of the fertile land was under cultivation by 4500 B.C. Sumer, in southern Mesopotamia, was dominated by eight major cities, including the city of Uruk, which had 50,000 inhabitants by 3000 B.C. But the irrigation that nourished Mesopotamian fields carried a hidden risk. Groundwater in semiarid regions usually contains a lot of dissolved salt. Where the water table is near the ground surface, as it is in river valleys and deltas, groundwater is moved up into the soil where it evaporates, leaving the salt behind in the ground. When evaporation rates are high. sustained irrigation can generate enough salt to eventually poison crops. While irrigation dramatically increases agricultural output, turning sunbaked floodplains into lush fields can sacrifice long-term crop yields for short-term harvests.

Preventing the buildup of salt in semiarid soils requires either irrigating in moderation or periodically leaving fields fallow(unplanted). In Mesopotamia, centuries of high productivity from irrigated land led to increased population density. which fueled demand for more intensive irrigation. Eventually, enough salt crystallized in the soil that further increases in agricultural production were not enough to feed the growing population.

The key problem for Sumerian agriculture was that the timing of river runoff did not coincide with the growing season for crops. Flow in the Tigris and Euphrates peaked in the spring, when the rivers filled with snowmelt from the mountains to the north. Discharge was lowest in the late summer and early fall, when new crops needed water the most Intensive agriculture required storing water through soaring summer temperatures. A lot of the water applied to the fields simply evaporated, pushing that much more salt into the soil.

Salinization was not the only hazard facing early agricultural societies. Keeping the irrigation ditches from becoming blocked by silt(mud) chief concern as extensive erosion from upland farming in the Armenian hills poured dirt into the Tigris and Euphrates Rivers. Conquered peoples were put to work pulling mud from the all-important ditches. Sacked and rebuilt repeatedly, Babylon was finally abandoned only when its fields became too difficult to water. Thousands of years later, piles of silt more than thirty feet high still line ancient irrigation ditches. On average, silt pouring out of the rivers into the Persian Gulf has created over a hundred feet of new land a year since Sumerian time. The ruins of the city of Ur. a once thriving seaport. now stand hundred and fifty miles inland.

As Sumer prospered, fields lay fallow for shorter periods because of the growing demand for food. By one estimate almost two-thirds of the thirty-five thousand square miles of arable land (land suitable for farming) in Mesopotamia were irrigated when the population peaked at around twenty million. The combination of a high load of dissolved salt in irrigation water, high temperatures during the irrigation season, and increasingly intensive cultivation pumped ever more salt into the soil.

Temple records from the Sumerian city-states inadvertently recorded agricultural deterioration as salt gradually poisoned the ground. Wheat, one of the major Sumerian crops is quite sensitive to the concentration of salt in the soil. The earliest harvest records, dating from about 3000 B.C., report equal amounts of wheat and barley in the region Overtime, the proportion of wheat recorded in Sumerian harvests fell and the proportion of barley rose. Around 2500 B.C., wheat accounted for less than a fifth of the harvest. After another five hundred years, wheat no longer grew in southern Mesopotamia.

Wheat production ended not long after all the region's arable land came under production. Previously, Sumerians irrigated new land to offset shrinking harvests from salty fields Once there was no new land to cultivate, Sumerian crop yields fell precipitously because increasing salinization meant that each year fewer crops could be grown on the shrinking amount of land that remained in production by 2000 B.C., crop yields were down by half. Clay tablets tell of the earth turning white in places as the rising layer of salt reached the surface.

Extinction Trends

In general, the diversity of life on Earth has been constant or increasing slightly over the past 200 million or 300 million years. The geologic record shows new species emerging and old ones dying away, resulting in broad range of forms over geologic time. On closer inspection, the geologic record also shows distinct short-term increases in the rate of extinction termed extinction events. One such event occurred 65 million years ago and marked the end of the Mesozoic Era, the age of reptiles. For years, the cause of this event puzzled scientists. However, in the980 s thin layer of unique sediment containing the element iridium was discovered in sedimentary bedrock whose age coincided with this event. Research showed that this sediment was created by an explosion on Earth's surface, probably caused by an asteroid impact. The explosion threw a huge mass of debris into the atmosphere, which, scientists reason reduced incoming solar radiation and cooled Earth. Plants declined and the food sources for many herbivores (plant-eating animals) apparently disappeared. followed shortly thereafter by a sequence of extinctions among predators.

Many other extinction events are also identifiable in the geologic record. Some occur at 26-million-year intervals, and scientists speculate that they may be associated with extraterrestrial impacts similar to the one at the end of the Mesozoic Era. Between these events, many smaller episodes of extinction are also apparent. The influence of humans on extinction patterns. however. does not appear until about 10.000 to 15.000 years ago.

The last glaciation on Earth reached its maximum 15,000 to 20.000 years ago. At this time the northern half of North America was covered with great sheets of glacial ice. A similarly large mass of ice-covered large part of northern Eurasia. Humans lived in the zone south of the ice border, where they survived by hunting and gathering. Among the animals hunted were large mammals such as mammoths and mastodons. which are now extinct. The extinction process appears to have taken place over several thousand years in the presence of humans as the ice sheets were melting and the bioclimatic environment was undergoing change. By 8,000 to 10,000 years ago, dozens of mammal species had disappeared. Scientists debate whether the extinctions were driven mainly by changing climate or mainly by human exploitation. No doubt these animals were under stress from a changing environment. but the evidence for population reduction by early hunters is also compelling. On balance, it appears certain that humans had a hand in the process. At the very least, hunting hastened the extinction process.

The next wave of extinction began about 10.000 years ago with the spread of agriculture and the growth of human populations. Before A.D. 1500 or so, much of the area taken up by sedentary agriculture in Europe and Asia was in subtropical and midlatitude environments. Many species were undoubtedly eliminated in these environments, but for most there is little or no record of their existence, especially for plants, insects, and smaller organisms. In the Mediterranean basin, on the other, where the early historical record is most complete there is sound evidence of widespread mammal extinction associated with the spread of early agriculture. Though less widely cultivated than the subtropics the wet tropics were not exempt from early agriculture. Large areas of monsoon forest and rain forest were destroyed in Asia and Africa by both crop farmers and herders. In the Americas, tropical rain forest and wetlands were also lost to agriculture. Recent evidence indicates that considerable areas of rain forest were cleared by the Aztec and Mayan civilizations in Central America. All told, the first several thousand years of agriculture undoubtedly forced the extinction of thousands of organisms, most of which we can never know about because they left no measurable record of their existence.

After A.D. 1500 and European colonization of the Americas and Africa, the rate of global extinction increased again. In North America, most of the midlatitude forests were eradicated. In the tropics, plantation agriculture, mining, and expanding subsistence farming (farming provides for the basic needs of the farmer) further reduced the tropical rain forest. Before the twentieth century, however, forest eradication was relatively minor by today's standards: beginning around 1950, rain forest destruction became a crisis that by 1975 had reached epidemic proportions.

Chemical and Biological Weathering of Rocks

Rocks can be broken into smaller pieces by a natural process known as weathering. Weathering refers to the chemical alteration and physical disintegration of rocks by the actions of air, water, and organisms. Two principal types of weathering are chemical weathering and biological weathering.

The minerals that rocks are made of are subject to alteration by chemical weathering. Some minerals, such as quartz, resist this alteration quite successfully, but others, such as the calcium carbonate of limestone, dissolve easily. In any rock made up of a combination of minerals, the chemical breakdown of one set of mineral grains leads to the disintegration of the whole mass. In granite (made chiefly of the minerals quartz and feldspar), the quartz resists chemical decay much more effectively than does the feldspar, which is chemically more reactive and weathers to become clay Often granite surfaces are heavily pitted (marked with many small holes or depressions). In such cases, the feldspar grains are likely to have been weathered to clay and blown or washed away. The quartz grains still remain, but they may eventually be loosened too. So even rock as hard as granite cannot withstand the weathering process forever.

Three kinds of mineral alteration dominate in chemical weathering: hydrolysis, oxidation, and carbonation. When minerals are moistened hydrolysis occurs, producing only a chemical alteration but expansion in volume as well. This expansion can contribute to the breakdown of rocks. Hydrolysis, it should be noted, is not simply a matter of moistening: it is a true chemical alteration, and minerals are transformed into other mineral compounds in the process. For example, feldspar hydrolysis yields a clay mineral (silica)in solution (that is, dissolved in water), and a carbonate or bicarbonate of potassium, sodium. or calcium in solution. The new minerals tend to be softer and weaker than their predecessors. In granite boulders, hydrolysis combines with other processes to cause the outer shells to flake off.

When minerals in rocks react with oxygen in the air, the chemical process is known as oxidation. We have plenty of evidence of this process in the reddish color of soils in many parts of the world and in the reddish-brown hue of layers exposed in such places as the Grand Canyon. The products of oxidation are compounds of iron and aluminum, which account for the reddish colors seen in so many rocks and soils. In tropical areas, oxidation is the dominant chemical-weathering process

Various circumstances may convert water into a mild acid solution, thereby increasing its effectiveness as a weathering agent. With a small amount of carbon dioxide. for instance. water forms carbonic acid, which in turn reacts with carbonate minerals such as limestone and dolomite (a harder relative of limestone). This form of chemical weathering, carbonation, is especially vigorous in humid areas, where limestone and dolomite formations are often deeply pitted and grooved, and where the evidence of solution and decay is prominent. This process even attacks limestone underground. contributing to the formation of caves and subterranean corridors. In arid areas, however, limestone and dolomite resist weathering much better, although they may show some evidence of carbonation at the surface

Biological weathering is the breakdown of rock caused by the actions of living organisms. This kind of weathering plays an important role in the formation of soils. It is through the breakdown of rocks and the accumulation of a layer of minerals that plants can grow--plants whose roots and other parts, in turn, contribute to the weathering processes. But it is likely that the role of plant roots in weathering is somewhat overestimated. The roots follow paths of least resistance and adapt to every small irregularity in the rock. Roots certainly keep cracks open once they have been formed. More importantly, however, are as of roots tend to collect decaying organic material that is involved in chemical-weathering- processes.

One of the most important aspects of biological weathering is the mixing of soil by burrowing animals and worms. Another interesting aspect is the action of lichens, a combination of algae and fungi that live on bare rock. Lichens draw minerals from the rock through an absorption process. The swelling and contraction of lichens as they alternately get wet and dry may also cause small particles of rock to fall off.

The Professionalization of Painting in Europe

Before the eleventh century most painters in Europe were monks (members of a religious group that lives in seclusion) and their work was exclusively religious. Such artists worked in a variety of art forms, including metalwork and manuscript illumination (the art of illustrating handwritten books), and they did not sign their work, as art was considered an opportunity for religious meditation not self-advancement. Between the eleventh and the fourteenth centuries, however, painting began being practiced by lay professionals, who had no connection to religious institutions. Inevitably, this had consequences both for working practices and for the art that was produced. These new painters followed a trade like any other. Like the woodworker, the potter, the baker and the weaver, the lay painter offered his technical proficiency for fee. He was by no means above asking the baker for use of his ovens to make charcoal, employed as a black pigment. Nor would the artist sneer at the cook, with whom he had many skills in common.

One of the consequences of this transition from monk to artisan was increasing specialization. Painters were painters, not to be confused with illuminators, dyers, or workers in wood and metal. Such distinctions were rigidly enforced by the guilds-powerful associations of workers within specific trades-that developed to safeguard the employment of tradespeople against competition and economic uncertainty, so that it would have been unthinkable for a painter to be called in to illuminate a book page. There were even fine distinctions drawn among painters. In fifteenth-century Spain one could find cialists in altarpiece painting (the painting of images for placement behind the altar of a church). fabric painting, and interior decorating. Associated with these divisions was a hierarchy of trades in which status tended to reflect the value of the materials. Goldsmiths were the most prestigious and powerful artisans, painters more humble, and woodworkers still more so. Guild restrictions prohibited the use of the most valuable pigments (coloring substances), such as ultramarine, for lowly purposes such as the painting of playing cards, carts, or parrot perches. Thus, these valuable pigments played role in establishing the social standing of painters: it was in their interests to use fine materials.

Yet the art of painting was still regarded as a mechanical process. There was good and bad, to be sure, but the task of the painter was to meet the specifications of a contract, not to give free expression to artistic inspiration. It was the employer-the patron-who made the decisions. Initially, the patrons of lay painters, when not church-affiliated, tended to be royal or aristocratic, and the painter could find himself affiliated to a court. But as a prosperous middle class emerged during the fourteenth and later centuries, the base of patronage broadened, and artists could find commissions (orders for artworks) among merchants.

Membership in a guild was attained by apprenticeship. Aspiring painters would serve a training period of typically between four and eight years in a workshop operated by a master craftsman, beginning as all apprentices do with the most menial of jobs, such as grinding pigments and making glue. Pigment grinding was a particularly arduous and time-consuming job, and painters might need to allow several days for this alone in their scheduling for a commission. To qualify as “master painter “ready to accept contracts, an apprentice would present “master piece “to the guild for approval. Strange, then, that this term has come to refer to an artist's most accomplished piece of work rather than his or her first attempt to gain recognition.

A workshop undertook a commission as group, much as a band of builders would (and still does) perform a job collectively. The master was responsible for the overall execution of the work, but the application of paint to surface was as much, if not more, the responsibility of his apprentices. Thus, the signatures that appear on fourteenth-and fifteenth-century art are simply trade stamps, denoting the name of the workshop's master. This traditional training placed little value on personal talent or inspiration: a person became a painter simply through hard work, dedication, and adherence to the master's demands. Skill was the product of diligence.

The Economics of Academic Tenure

In some countries, many universities use an employment system for teachers known as tenure. After a lengthy trial period, a faculty member whose performance meets with the approval of the senior members of the department and the administration of the institution may be awarded tenure. A tenured faculty member enjoys considerable job security for the rest of his or her working life and can only be fired for reasons of "moral turpitude" (bad or evil behavior) or "gross incompetence" or if the financial stability of the institution requires the elimination of an entire department or program.

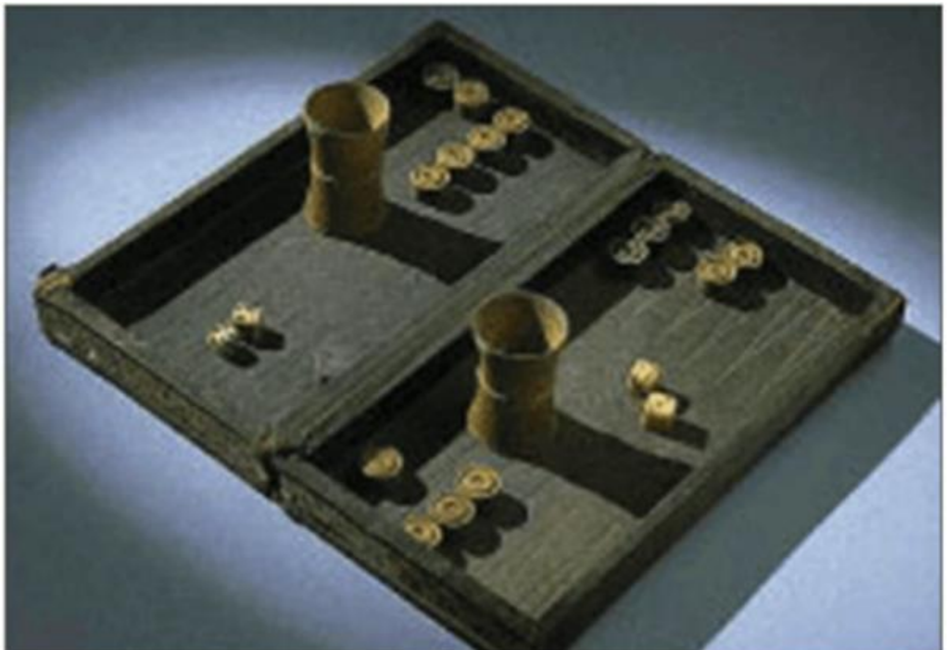
The high degree of job security enjoyed by tenured faculty members has been the source of complaints about the tenure system. One issue that has been raised by many, including legislators evaluating the finances and managerial practices of state universities in the United States, is that tenure shelters faculty from accountability for poor performance. Another argument is that tenure makes the university inefficient in responding to changing instructional demands. It is difficult to substitute computer engineering faculty for civil engineering faculty if most of the latter have tenure. In 1988, the Education Reform Act significantly softened" the tenure system in the United Kingdom, making it easier to fire individual faculty members for financial reasons. More recently, some universities in the United States have taken steps to give university administrators more control over tenured professors And, in general, American institutions of higher learning have ended up increasing the use of part-time and nontenured instructors over time. In 1992, just 48 percent of all instructors had tenure on were in a position that was expected to lead to tenure.

The traditional argument in favor of tenure is based on academic freedom, the freedom to investigate and teach any area of knowledge without restriction or interference. In this view, tenure protects faculty members from retaliation for voicing unpopular views. For example, a labor economist might not present a complete examination of the costs and benefits of worker unions it he or she feared that a rabidly anti-union university leader might seek to have the economist fired for speaking of the positive aspects of unions. In fact, the American Association of University Professors (AAUP), a group dedicated to protecting academic freedom, got its start in the wake of a 1901 decision by Stanford University to fire economics instructor Edward Ross at the insistence of the university's co-founder, Jane Stanford, who objected to his views on economics and other matters.

Going beyond academic freedom, the economics literature has recently turned to an emphasis on tenure as a labor-market institution that may have a positive payoff to universities through the incentives (motivation) it provides. For example, economist Lorne Carmichael's model of an academic department treats tenure as the means of providing incentives for incumbent (current) faculty to participate in identifying the best candidates for new positions. If incumbent faculty had to worry that more-able new additions to the department might replace them one day, they would be less inclined to make hiring decisions that were in the best interests of the university. Incumbents are much better positioned to judge the talents of potential new hires than is the university administration. Moreover, the long-term job security they gain through tenure gives incumbents an incentive to hire new faculty who might be more productive than the existing faculty in a department.

The economists Michael McPherson and Morton Shapiro have also emphasized the notion that tenure has a positive payoff for the university by aligning the self-interest of individual faculty members with the long-run interests of the institution. They see two valuable economic benefits from the tenure system beyond the incentive to hire and mentor more productive new faculty. First, job security allows tenured faculty the independence to perform credibly objective evaluations of students and other faculty. People outside the university who rely on the information provided by student grades or faculty reviews of papers or proposals can have greater confidence that these evaluations have not been colored by the faculty member's concern about job security. Second, tenure allows faculty to make long-run strategic decisions about educational policy and research even if these are in conflict with the short-run, career interests of administrators. The fact that tenured faculty often are viewed as obstacles to change by ambitious administrators looking to enhance their records for their next career move might well be a good thing for the long-run interests of the university.

Programming Computers to Play Games



Backgammon Set

© National Park Service Museum

Backgammon is the oldest board game in the world. It was first played in ancient Mesopotamia, starting around 3000 B.c. The rules of backgammon were codified in the seventeenth century, and the game has changed little since. The same can't be said about the players of the game. One of the best backgammon players in the world is now a software program. In the early 1990s. Gerald Tesauro, a computer programmer at IBM, began developing a new kind of artificial intelligence (AI). At the time, most AI programs relied on the brute computational power of microchips. This was the approach used by Deep Blue, the powerful set of IBM mainframes that managed to defeat chess grand master Garry Kasparoy in 1997. Deep Blue was capable of analyzing more than two hundred million possible chess moves per second, allowing it to consistently select the optimal chess strategy. (Kasparov's brain, on the other I hand. evaluated only about five moves per second.) But all this strategic firepower consumed a lot of energy: while playing chess. Deep Blue was a fire hazard and required specialized heat-dissipating equipment so that it didn't burst into flames. Kasparov, meanwhile, barely broke a sweat. That's because the human brain is a model of efficiency: even when it's deep in thought, the cortex consumes less energy than a lightbulb.

While the popular press was celebrating Deep Blue's stunning achievement-a machine had outwitted the greatest chess player in the world! -Tesauro was puzzled by its limitations. Here was a machine capable of thinking millions of times faster than its human opponent, and yet it had barely won the match. Tesauro realized that the problem with all conventional AI programs, even brilliant ones like Deep Blue's, was their rigidity. Most of Deep Blue's intelligence was derived from other chess grand masters, whose wisdom was painstakingly programmed into the machine. (IBM programmers also studied Kasparov's previous chess matches and engineered the software to exploit his recurring strategic mistakes.) But the machine itself was incapable of learning. Instead, it made decisions by predicting the probable outcomes of several million different chess moves. The move with the highest predicted value was what the computer ended up executing. For Deep Blue, the game of chess was just an endless series of math problems.

Of course, this sort of artificial intelligence isn't an accurate model of human cognition. Kasparov managed to compete on the same level as Deep Blue even though his mind had far less computational power. Tesauro's surprising insight was that Kasparov's neurons were effective because they had trained themselves. They had been refined by decades of experience to detect subtle spatial patterns on the chessboard. Unlike Deep Blue, which analyzed every possible move. Kasparov was. able to instantly isolate his best options and focus his mental energies on evaluating only the most useful strategic alternatives.

Tesauro set out to create an AI program that acted like Garry Kasparov. He chose backgammon as his model and named the program TD-Gammon. (The TD stands for temporal difference.) Deep Blue had been preprogrammed with chess acumen, but Tesauro's software began with absolutely zero knowledge. At first, its backgammon moves were entirely random. It lost every match and made stupid mistakes. But the computer didn't remain a novice for long; TD-Gammon was designed to learn from its own experience. Day and night, the software played backgammon against itself, patiently learning which moves were most effective After a few hundred thousand games of backgammon, TD-Gammon was able to defeat the best human players in the world.

How did the machine turn itself into an expert? Although the mathematical details of Tesauro's software are numbingly complex, the basic approach is simple. TD-Gammon generates a set of predictions about how the backgammon game will unfold. Unlike Deep Blue, the computer program doesn't investigate every possible permutation. Instead, it acts like Garry Kasparov and generates its predictions from its previous experiences. The software compares these predictions to what actually happens during the backgammon game. The ensuing discrepancies provide the substance of its education, and the software strives to continually decrease this error signal. As a result, its predictions constantly increase in accuracy, which means that its strategic decisions get more and more effective and intelligent.

Chondrites

Meteorites are the keys to understanding how Earth and the solar system formed because, although superficially many meteorites do not look significantly different from most Earth rocks close study shows that they are very different, with those differences holding clues to their distant origin. The most common type of meteorite-known collectively as the "chondrites-contains clues about the kinds of material that went into the construction of our planet.

The chondrites are just one of a variety of meteorite types, all ancient, but each with a different history-and each containing clues about how Earth and other planets formed and evolved. For example, the members of one group, the iron meteorites, are made up of solid iron metal, alloyed (or mixed) with a modest amount of nickel All the evidence indicates that these meteorites are similar to the metallic cores that inhabit the interiors of planets. In contrast to the iron meteorites. the chondrites are made up of a jumble of mineral grains, many of them familiar constituents of rocks on Earth, but also including pure iron metal-which does not occur in rocks on Earth-and small marble-like spherical objects called "chondrules"(it is from these that the chondrites take their name). The chaotic texture of chondrites indicates that they were formed in a process that randomly swept together their different components and cemented them together. The texture is also an instant clue to one of their most important characteristics: they have never been melted. Furthermore, mineral grains in the chondrites have provided the oldest ages ever measured by radiometric dating they date from the very earliest days of the solar system. These two properties have led the geochemists who work on these intriguing objects to conclude that chondrites bring us something we cannot find on Earth or among other meteorite varieties: an unprocessed sample of the original material from which Earth and our neighboring planets were made. They appear to be unaltered samples of the solid matter that was floating around in the solar system just as Earth was being born; there is strong evidence that they come from small asteroids that never grew big enough to heat up and melt, as larger objects did.

The primitive nature of the chondrites has made them very important for information about Earths overall chemical composition. You might wonder why these rare rocks from space should play such a key role when the Whole Earth Is right below our feet, ready for us to analyze. The answer, in brief, is that if we want to understand how Earth got to its present state, we have to know something about its initial, overall composition. But we only have access to rocks from our planets thin outermost skin-which is very different from the inaccessible interior. However, by using information from the chondrites as a reference point, and integrating that data with direct measurements on surface rocks and information about the interior obtained using remote sensing methods, geochemists have been able to work out models for Earths overall composition that satisfy already obtained independent evidence such as the density of our planet.

An important concept in formulating these models is that the composition of Earth and the other “terrestrial planets (the solar systems inner, rocky planets, Mercury, Venus, and Mars) depends on the proportions of the main minerals found in the chondrites that each planet incorporated, A good example is iron, which is so abundant and so heavy that the well-known density variations among the terrestrial planets can almost entirely be attributed to differences in their iron contents. Grains of iron metal are abundant in the chondrites, but different chondrites contain different amounts. The reason Earth is much denser than Mars, according to the models, is that its chondrite-like building blocks happened to contain more iron. Similarly, other differences among the terrestrial planets can be understood in terms of different planets incorporating different amounts of the various constituents of chondrites. This is undoubtedly an overly simplistic description of what actually happened, but that does not invalidate the chondrites as a good starting point for understanding the composition of Earth and other planets.

Greek Art in the Classical Age

Greek art is thought to have reached its peak during the Classical period in the fifth century B.C.E. Leading up to this period, the most common type of sculpture was the kouroi, which were life-size or larger marble statues of nude males that stood on sacred sites, often as grave markers, but also as offerings to the gods. With very stiff, straight poses (they evidently were modeled after Egyptian statues), it is clear that kouroi were not intended to look like real people. However, by the early fifth century, the style of Greek artwork changed. The transition is usually symbolized by the Kritios Boy, a marble statue found in the center of ancient Athens and attributed to Kritios, a sculptor active in Athens around 490-460 B.C.E. It is dated by experts to just before 480 B. C.E. and represents Callias a victor in the boys' footrace in an athletic competition. The changes from the traditional kouros are slight, but the boy is standing as a boy might actually stand, the right leg forward of the left, which bears the weight of the body so that the right can relax slightly not how artistic convention decrees a hero should pose. Yet this naturalness is achieved without the loss of an idealization (representation as perfect) of the human body. Here is, in the words of the art historian Kenneth Clark. "The first beautiful nude in art. "As John Boardman, an authority on Greek art, puts it: "This is a vital novelty in the history of ancient art-life deliberately observed, understood, and copied. After this all becomes possible."

There are a few clues as to why this revolution in art, from the stylized to the observed, took place. One is that bronze was becoming the most popular medium in which statues were being created. (It has been suggested that the Kritios Boy is a copy of a bronze original now lost.) The technical problems involved in casting and assembling bronze statues had been solved by the end of the sixth century B.C.E., as the earliest examples show. From the Classical period on bronze predominated in Greek sculpture, but as almost every statue was later melted down so its metals could be reused it is hard to guess this today. The few bronzes to survive (the Riace warriors, the Delphi charioteer and the majestic Zeus found in shipwreck off Cape Artemisium foremost among them) simply highlight what has been lost in quantity and quality. Bronze allowed far greater flexibility in modeling the process of building up a figure in bronze is totally different from cutting into marble. As a wonderful exhibition at the Royal Academy in London in 2012, Bronze, also showed bronze can be burnished (smoothed and shined) to produce a wide variety of aesthetic effects that pure white marble lacks.

The revolution also suggests a preoccupation with human form. While earlier Greek artists were focused on those few human beings who had become heroes, they now seemed concerned with the physical beauty of human beings as an end in itself. It is hard to see the Riace warriors without being aware of their intense sensuality. Yet within a few years this sensuality fades and is replaced by a greater concentration on the nature of the human body as an ideal. It was the sculptor Polycleitos, probably a native of Argos working from the fifth into the fourth century B. C.E., who allied aesthetics with mathematics when he suggested that the perfect human body was perfect precisely because it reflected ideal mathematical proportions that were capable of being discovered. One of his statues, the Doryphoros, or "spear bearer"(originally in bronze, but now known only through Roman copies in marble), was supposed to represent this ideal. If this approach was followed to its extreme, all statues would have had the same, perfect, proportions but the Greeks could not close their eyes to the variety of human experience. There always remained a tension in the art of the period between the abstract ideal of the human body and a particular body copied by the artist. This may be one reason for its aesthetic appeal.

Trade and Herring in Dutch Society

Although the people of the Netherlands were at war with their Spanish rulers, the Dutch economy prospered between 1588 and 1648, and trade was at the heart of this prosperity The expansion of trade directly and indirectly brought increased employment opportunities: toward the middle of the seventeenth century, the fleet of trading ships employed almost 30,000 people, and the handling of cargoes in the ports provided work for large numbers of people. In addition, there were those involved in the internal transport system, moving goods and people along the rivers and canals. Another important source of employment was provided by artisan production of various kinds; manufacturing was still labor-intensive-- literally handwork -with very limited inputs of nonanimal energy, with the exception of the large-scale use of wind power for sawmills. Also, the fisheries and secondary industries were important sources of employment. Although the import and re-export of goods through the staple-market (market for bulk commodities, especially raw materials) was paramount to the success of the trading economy, manufacturing and fisheries supplied valuable export products for foreign trade.

Herring are an edible ocean fish, and the herring fishery was important as a source of employment because it provided a crucial export item, as well as helping to feed the Dutch population. In terms of the numbers of boats and people employed, most if not all of the growth of the fishery seems to have already taken place before the revolt against Spanish rule began in 1568, with subsequent developments consisting chiefly of improvements in organization and co-ordination between the various branches of the industry. Dutch success in the fisheries was by no means certain ahead of time; the movement of the best herring fishing grounds to the

North Sea gave them the opportunity, but they were geographically no better placed to exploit this than many potential rivals indeed less so than the Scots and the English. What then accounts for the greater success of the Dutch?

Over the course of the fifteenth and sixteenth centuries. northern Netherlands fishing fleets had introduced a number of key innovations, which gave them a decisive edge. Specialized fishing boats came into use, and mother ships (supply ships) began to sail with the fleet to enable these boats to stay on the fishing grounds longer. Also, a new method of preserving herring was found in the fifteenth century, and curing began to take place at sea to improve the quality of the product. High quality was indeed the key to the success of the Dutch herring business, and concern to maintain this advantage over their competitors led to the use of the finest salt that could be found and to careful monitoring of the barreling storage process. It seems to have been agreed by contemporaries that Dutch herring was the best which provided the Dutch with a market even in Scandinavia with its own sources of the fish. In fact, herring made a significant contribution to the development of the Baltic trade providing an export item with a ready market there.

There may have been as many as 500 specialized boats employed at the peak of the herring fishery in the early seventeenth century. This in itself must have provided employment for 6,000 to 7,000 people, though only for part of the year, as the herring fishing was seasonal, starting in late June and ending in December or early January. However, if herring packers and the workers making the barrels in which the fish were packed are included, then the work provided directly by the fishery rises considerably. Moreover, the indirect boost to shipbuilding and sail and rope making is impossible to calculate but must have been significant and, if the transport to various markets by barge boat is taken into consideration, then the overall impact of the fishery on employment opportunities rises again.

There were other sea fisheries, especially cod, but the biggest growth in the early seventeenth century came in Arctic whaling, which rose from practically nothing to an important, if not always thriving, business. Initially this was open sea whaling, but from about the 1640s onward certain whales became increasingly difficult to find in open waters, which meant that whaling had to turn to the construction of specially strengthened ships that could follow the whales into the pack ice. The real expansion of Dutch whaling, however, belongs to the second half of the century.

Increasing Jellyfish Populations

Scientists have had a tough time trying to discover whether recent increases in jellyfish populations are really worth worrying about. On the one hand, jellyfish are known to proliferate rapidly in response to positive changes in prey abundance or environmental conditions such as water temperature and sunlight. The size of these "blooms" can vary from year to year. On the other hand, these population explosions are occurring in many places on a scale now widely viewed as unprecedented. In the Sea of Japan, for instance, Nomura's jellyfish are known to have drifted in from the south in large numbers three times during the twentieth century: in 1920 1958 and 1995. Beginning in 2002, however, they have turned up every summer but one, and in astonishingly high numbers. In 2005, one of the worst years, up to 500 million Nomura's jellyfish were reported to be drifting into the sea each day

Several factors have now been identified as possible contributions to the increased success of jellyfish worldwide. One of the leading suspects is the human exploitation of fish and other marine resources. something that has intensified in recent decades partly because of advances in large-scale seafood harvesting and processing techniques. Only a handful of species are thought to prey directly on jellyfish, and most of these predators-including giant sea turtles-are becoming increasingly rare. The main impact of overfishing, though, may stem from the reduction of filter-feeding fish such as sardines and anchovies, which eat the same food as jellyfish. In the southern Atlantic waters off Namibia. where overharvesting has resulted in the complete collapse of a once-thriving sardine fishery, unusually large numbers of jellyfish are now a permanent feature of the near-shore marine ecosystem.

At the same time, jellyfish seem to thrive under conditions that are becoming increasingly widespread because of human-associated activities. Although the theory is highly speculative and still under debate, global warming and acidification of the oceans a result of more carbon dioxide dissolving in the water-may be two such factors. Jellyfish love warm water, for one thing. And at least one study, from the North Sea, has reported finding a connection between greater jellyfish abundance and higher acid levels.

Another environmental change, and one that is more firmly linked to expanding jellyfish populations, is eutrophication, the process by which water becomes enriched with dissolved nutrients, resulting in more aquatic plant life and, subsequently, reduced levels of oxygen. Eutrophication occurs in near-shore ocean waters close to large human population centers and the mouths of large rivers. A strong correlation exists between blooms of algae and other plankton, caused by excessive nutrients from sewage and fertilizer runoff, and jellyfish eruptions. One example is the waters off the southern United States coast. Nutrient-enriched waters from the Mississippi River have created in the Gulf of Mexico a massive dead zone (ocean area with low oxygen levels).

While fish and other aquatic life-forms are finding survival increasingly challenging, jellyfish such as moon jellyfish and sea nettles are becoming increasingly numerous. This is not really surprising. A checklist of jellyfish traits-good survival rates during periods of starvation, rapid reproduction, diverse diet, ability to feed in murky water, capacity for surviving under low oxygen conditions typical of dead zones-reveals that these seemingly fragile organisms are actually tough life-forms ideally suited to survive in disturbed environments. According to the National Science Foundation, the main science-funding agency in the United States there are currently 400 known ocean dead zones where almost nothing lives except jellyfish.

Finally, like many other marine invasive species, jellyfish have benefited from the globalization of human trade. Many species are hardy enough to survive in the ballast water of ships, and jellyfish polyps can attach themselves to their hulls. Because of this, jellyfish species are being inadvertently introduced into new aquatic habitats. One of the best examples involves the warty comb jelly (Mnemiopsis leidyi), a species that likely made its way into the Black Sea for the first time in 1982. Since then, this highly adaptable organism has been thriving; its original nonnative range has expanded to include not just the Black Sea but also the Caspian, Baltic and North seas

Northwest Coast Art

The Pacific Northwest of North America produced stable Native American cultures. This was in great measure due to the abundance of resources and temperate climate, which figured prominently in the social and cultural fabric of the peoples. Fishing, hunting, and foraging produced ample means of sustenance, and therefore, the people of this region had no reason to cultivate crops or domesticate animals. Their cultures tended to share certain features, probably because they traded and warred with one another. Artistic motifs (recurring artistic elements) were alike, and they shared a common religion-shamanism (ancient belief systems centered on shamans. individuals who communicate with invisible forces or spirits), Although their gods and myths differed, the Northwest peoples all acknowledged the power of shamans to. contact the spirits of the forest and waters, to heal the sick, and to predict the future.

The rich forests of the area provided Native American artists with a wealth of materials for sculpting. Huge spruce and cedar trees abounded for many centuries, and when steel knives were obtained from fur traders and used as tools, Northwest-coast artists excelled in producing magnificent totem poles, carved posts for wooden dwellings, masks, rattles, and other objects. Carved wooden house posts not only supported the roof but also gave additional decoration to the interior. When living closer to the sea, artists had access to abalone shells that were used as inlays to give luster to their sculpted works.

Although many art objects from the area, such as masks, concern themselves with shamanistic religious rituals, a fair amount of art was secular. Like some African art, it was used to maintain the social fabric and bolster a ruler's power. For example, in the Tlingit group--a people who lived in the islands and bays of the upper reaches of the Pacific Northwest-communities were made up of a number of families, each of which had its own chief who inherited his rank from his mother. Carefully devised social customs obliged both men and women to marry outside their own clan. In this way a balance was established in which no one family achieved dominance. Nonetheless, chiefs competed fiercely with each other in displays of riches. Totem poles formed a part of this ostentation proclaiming prestige and family pride through genealogy-much like the coats of arms of the European aristocracy. Totem poles were carved from single tree trunks and often reached a height of 90 feet (27.4 meters). Probably originating as funerary (burial) monuments, by the nineteenth century they had become fixtures adorning the exteriors of chiefs' houses.

Eagles, beavers, and whales appear on the totem poles, in essence as symbolic crests that a chief inherited from his ancestors. The Tlingit did not worship these figures or have any supernatural relationship with them. They are governed by traditional artistic principles. One of these is that of bilateral symmetry-the designs on either side of the central axis are identical. Another specifies the transition from one motif to another. Each design must appear to grow from the one below. Thus, not only the overall vertical form of the totem pole, but also the design of its interior parts, leads the eye upward along its central axis.

As historical documents that record the wealth, social position, and relative importance of the person who paid for them, totem poles are quite similar to sculptural records from other cultures. The ultimate function of such art was probably to act as a gift. As the anthropologist Frederick J. Dockstader claimed, "The life goal of many ... involved the belief that the greatest value was to give away all of one's possessions." The actual working out of such a philosophy created some interesting scenarios. The more one gave away, the greater one's prestige. In turn, one's rival was more or less obliged to give back the same or more material wealth in order to prove greater disdain for possessions. As a result, totems and other goods were often burned, thrown into the sea, or otherwise destroyed.

Examining the Problem of Bycatch

A topic of increasing relevance to the conservation of marine life is bycatch-fish and other animals that are unintentionally caught in the process of fishing for a targeted population of fish. Bycatch is a common occurrence in longline fishing, which utilizes a long heavy fishing line with baited hooks placed at intervals, and in trawling, which utilizes a fishing net (trawl) that is dragged along the ocean floor or through the mid-ocean waters. Few fisheries employ gear that can catch one species to the exclusion of all others. Dolphins, whales, and turtles are frequently captured in nets set for tunas and billfishes, and seabirds and turtles are caught in longline sets. Because bycatch often goes unreported, it is difficult to accurately estimate its extent. Available data indicate that discarded biomass (organic matter from living things) amounts to 25-30 percent of official catch, or about 30 million metric tons.

The bycatch problem is particularly acute when trawl nets with small mesh sizes (smaller-than-average holes in the net material) are dragged along the bottom of the ocean in pursuit of groundfish or shrimp. Because of the small mesh size of the shrimp trawl nets, most of the fishes captured are either juveniles (young), smaller than legal size limits, or undesirable small species. Even larger mesh sizes do not prevent bycatch because once the net begins to fill with fish or shrimp, small individuals caught subsequently are trapped without ever encountering the mesh. In any case, these incidental captures are unmarketable and are usually shoveled back over the side of the vessel dead or dying.

The bycatch problem is complicated economically and ecologically. Bycatch is a liability to shrimp fishers, clogging the nets and increasing fuel costs because of increased drag (resistance) on the vessel. Sorting the catch requires time, leading to spoilage of harvested shrimp and reduced time for fishing. Ecologically, high mortality rates among juvenile fishes could contribute to population declines of recreational and commercial species. Evidence to this effect exists for Gulf of Mexico red snapper and Atlantic Coast weakfish. Because the near-shore areas where shrimp concentrate are also important nursery grounds for many fish species, shrimp trawling could have a profound impact on stock size.

Once the dead or dying bycatch is returned to the ecosystem, it is consumed by predators, detritivores (organisms that eat dead plant and animal matter), and decomposers (organisms that break down dead or decaying organic matter), which could have a positive effect on sport fish, seabird, crab, and even shrimp populations. Available evidence indicates that 40-60 percent of the 30 metric tons of catch discarded annually by commercial fishing vessels, and even more of the noncatch waste (organisms killed but never brought to the surface), does not lie unused on the bottom of the sea. It becomes available to midwater and ocean-bottom scavengers, transferring material into their food web and making energy available to foragers (organisms that search for food) that is normally tied up in ocean-bottom, deep-ocean, midwater, and open-ocean species.

Overfishing and overdiscarding may thus contribute to a syndrome known as "fishing down of food webs," whereby we eliminate apex (top) predators and large species while transforming the ocean into a simplified system increasingly dominated by microbes, jellyfish, ocean-bottom invertebrates, plankton, and planktivores. The strongest evidence for the fishing down phenomenon exists in global catch statistics that show alarming shifts in species composition from high-value, near-bottom species to lower-value, open-ocean species. In the last three decades of the twentieth century, the global fishing fleet doubled in size and technology advanced immeasurably. Despite increased effort and technology, total catch stabilized, but landing rates (rates at which species are caught) of the most valuable species fell by 25 percent.

Conservation organizations have condemned the obvious and extreme waste associated with bycatch. Public concern over high mortality rates of endangered marine turtles captured in shrimp trawls led to the development of turtle exclusion devices (TEDs) in the 1980s. TEDs were incorporated into the shrimp net design with the purpose of directing turtles out of nets without unacceptably reducing shrimp catches. Marine engineers and fishers also developed shrimp net designs that incorporate bycatch reduction devices (BRDs), taking advantage of behavioral differences between shrimp and fish, or between different fishes, in order to separate fishes.

Challenge of Dendrochronology

Dendrochronology is the technique of counting tree rings to determine a tree's age and measuring the width of these rings to determine characteristics of past climates. This might seem simple: each ring represents one year, and wider rings generally mean better growing conditions-plentiful rainfall, moderate temperatures, and so forth. But the seasonal growth of a particular tree is affected by factors other than the weather. Trees vary, one from another, just like people do. The genetic makeup of each individual tree is unique, so one particular tree may grow a bit more quickly that another. Highly local conditions can also change over time. It is easy enough to see that if part of the soil near a tree has been eroded, this will impact the tree's root system and limit its growth, at least until the situation stabilizes. Then again, an infestation of insects may affect a tree in one valley more than the same type of tree ten miles away. Or one tree may suddenly start to get a lot more sunlight when an old, big tree in the neighborhood finally falls. These kinds of factors produce significant variations among individual specimens, and that fact means that researchers need to average together samples from many specimens of a single tree species in one region over the same time period. Some dendrochronologists think that measuring an average of twenty-five to thirty tree-ring records in a locale is an essential first step in getting around the problem of individual variability. While it may be easy enough to find thirty samples in some locations for particular periods, it obviously becomes less and less likely the more ancient the wood samples are.

Another issue is more general. Trees that are fortunate enough to live on good soil and near local sources of groundwater often grow at steady rates. Such growth translates into attractive trees that are tall and well formed; they also have rings that are wide and quite uniform in thickness, but their uniform growth rings make them entirely useless when it comes to inferring anything about past weather patterns. That is why, instead of looking at superb botanical specimens, dendrochronologists focus their work on wood from trees that are living a tough life due to poor soil, steep slopes, the absence of local groundwater, or some other challenge. It is these "tortured" trees that are the most likely to grow very little during years of scarce rains or do poorly after a harsh winter and a late spring. What this means, of course, is that few trees in the woods are likely to be good samples for the scientist. Indeed, it may be quite a small fraction that yields useful ring patterns. Again, this increases the challenge of finding enough good samples to say with much certainty what past conditions were like.

Another factor of dendrochronology relates to wood itself. In the spring, a tree grows rapidly, creating new cells on the outside of its trunk and branches, just under the bark. These cells, called "earlywood" or "springwood," are large and have thin cell walls; both these factors contribute to making the wood relatively lightweight for its volume. In the summer, growth slows. Denser "latewood" is formed, creating the band that is relatively dark when you look at the end of a piece of lumber. But occasionally the sequence of a perfect pair of springwood and latewood does not hold up. If conditions-weather or disease-severely test a tree one year, it will not grow over all its surfaces. That may mean that a particular sample of wood taken by the dendrochronologist will have a missing ring in it, which will result in the scientist's inferences being off base by a year

A few trees also may trip up scientists by revealing a "false ring" made of latewood that's in the middle of springwood. These features, sometimes known as double rings, usually can be distinguished from true rings because the unusual dark ring is likely to change gradually rather than more abruptly into the springwood that lies on either side of the false latewood. It is not clear what creates such double rings, although people have speculated that unusual conditions during the middle of the growing season or even highly local issues might be the cause.

The Problem with Microplastics

Scientists think that about 10 percent of all plastic, which includes plastic bags and bottles, ends up in the ocean. The attributes that make plastic a useful material for a large number of products are its light weight and the strong chemical bonds in its internal structure, which make the material durable. Because plastic by itself is less dense than water, it floats along the ocean surface where it is continuously exposed to ultraviolet light from the Sun, which has the effect of loosening its chemical bonds. Ocean waves smash these weakened pieces of plastic against each other, and they are broken down into smaller and smaller pieces, eventually creating vast numbers of microplastics (pieces less than 5 millimeters across) that disperse throughout the water.

Due to the widespread increase in plastic production, there is six times as much plastic in the oceans as there was 40 years ago. Based on the amount of manufactured plastic, scientists have estimated that there are many trillion pieces of plastic in the ocean. To obtain a more precise idea about the amount, scientists conducted a study in 2010-2011 that involved sampling the ocean surface at 141 locations. After six months they concluded that only 7,000 to 35,000 tons of plastic were present on the ocean surface. It is widely believed, however, that tens of millions of tons of plastics have entered ocean waters. Moreover, the scientists primarily found larger pieces of plastic, which means that the quantity of microplastics found diverged from expectations to an even greater degree.

Scientists have attempted to explain the apparent disappearance of vast amounts of plastic from the surface of the ocean. Plastic is hydrophobic-it repels water, like oil does-which by itself contributes to its buoyancy (ability to float). However, this hydrophobic quality attracts single-celled organisms, such as diatoms and bacteria, which attach themselves to the plastic's surface and replicate. Over time the resulting biofilm (layer of organisms) alters the density of the plastic and reduces its degree of hydrophobia, increasing its chances of sinking. Organisms residing in the biofilm release certain chemicals that attract other organisms, including barnacles, aquatic insects, and algae, which make their homes on the plastic. This adds to the plastic's weight and further increases its chances of sinking. In one experiment scientists submerged plastic food bags for three weeks, measuring the amount of biofilm and buoyancy of the bags each week. The bags sank deeper during each week of the experiment. Severe coastal storms and vertical movement of water associated with differences in temperature or salinity may be another reason why microplastics move into deeper waters. Scientists have found as many as 40 pieces of microplastic per 50 milliliters of sediment in locations sampled along the ocean floor.

Some of the microplastics are likely ingested (eaten) by ocean organisms. Many marine invertebrates (animals lacking a backbone) are filter feeders-they draw water into their bodies and ingest tiny bits of food contained within it. These animals do not discriminate among food items and can easily take in plastics along with the rest of their meal. Thus, scientists have found microplastics in the bodies of marine organisms, including zooplankton, crabs fish, and marine worms. Although the plastic by itself can remain in or pass through the animals' bodies without significantly affecting them, many plastics come with pollutants left over from the manufacturing process or picked up as the plastics travel through the oceans, and these chemicals are often harmful to the animals.

Copepods-tiny, free-swimming organisms that eat algae-are among the filter feeders that have been found to ingest microplastics. But then, in an apparent attempt to avoid taking in even more of these materials, copepods switch to feeding on smaller algae, ones that are even tinier than microplastics. As a result of this diet change, copepods' energy intake can drop by as much as 40 percent. These copepods lay smaller eggs that are unable to hatch successfully. Microplastics may also get stuck in the antennae of copepods and other zooplankton, where they interfere with the organisms' ability to sense food in the surrounding waters, and also in copepods' joints, limiting their ability to move and catch prey.

The Rise of the Maya

A hundred and fifty years of painstaking archaeological inquiry allows us today to understand how the Maya emerged to transform the rain forest of Central America into a scene of urban civilization. By 1000 B.c. the Maya were settled agriculturalists growing a variety of crops in clearings in the forest, which they turned into villages They appear to have lived in a society of equals, without clear rulers or ceremonial centers. Then between 800 and 500 B.C, signs of a ruling elite within Maya society start to emerge in the form of elaborate burial monuments. At Los Mangales in the Salama Valley of highland Mexico, a chief was buried on a special mortuary platform accompanied by rich grave goods of jade and shell. Slightly later the ceremonial center at El Portón was built, involving the construction of earth terraces and platforms, on which were located altars and standing stones. one with a brief written inscription, although too damaged to be read today.

In the lowlands of Guatemala and the Yucatán Peninsula of Mexico, vast ceremonial centers appeared quite suddenly after 600 B.C. At Nakbe in northern Guatemala, the site was swiftly transformed from a modest village into a city with a major monumental structure at its heart. A massive platform was constructed on the ruins of the original settlement, on top of which were a series of terraced buildings up to 60 feet tall. Without any doubt we can see here the development of a more complex society. From 400 B.C. to Ad 250, major ceremonial centers developed in all parts of the Maya area, many carved out of the tropical rain forest that covered the southern lowlands of Guatemala. Belize, and Mexico. These cities were dominated by enormous terraced platforms, some forming giant temple-pyramids. Vast palaces of limestone masonry with vaulted rooms were constructed. set within architectural layouts that emphasized the most important buildings of a city, arranged around plazas with rows of standing stones lined up in front of them. A highly sophisticated art style emerged, seen in bas-reliefs, wall paintings, and beautiful pottery with multicolored fired decorations. Hieroglyphic writing became widespread, and inscriptions can be dated using the Maya Long Count, an elaborate but incredibly accurate calendarical system.

Earlier generations of archaeologists considered the great Maya cities to be purely ceremonial centers. In this view. the cities were occupied only by their peace-loving priestly rulers and their attendants except at great festivals. The British Mayanist Sir Eric Thompson, working at the Field Museum of Chicago, suggested that the Maya character encouraged the development of religious authority, and that their devoutness, discipline, and respect for authority would have facilitated the emergence of a theocracy.

This was not, however, an age of peace ruled over by unworldly priests living in solitude among the temples. In the last 25 years, a succession of remarkable breakthroughs has enabled us to read the Maya's hieroglyphic language. Whereas Thompson and others assumed that inscriptions outside temples concerned complex matters of astronomy and calendric systems that fascinated the priests, the translations now available show beyond any doubt that the cities were ruled by an aristocracy that was firmly secular (not based on religion) and indeed warlike in outlook. Hieroglyphic writing on monuments was mainly used to record the achievements of Maya rulers, especially in war. Victory stones were erected outside the temples, bearing the names of famous captives. Great monuments were liberally marked with the names and faces of the rulers who commissioned them.

Excavation has also played its part in overturning the established picture of Maya cities. Vital evidence that the cities were not just ceremonial centers has now been found at many lowland sites. On the outskirts of cities such as El Mirador are groups of low, rectangular mounds of earth and stone long ignored, but which archaeological investigations have now shown were occupied by small wooden houses, raised above the level of the summer flooding-humble dwellings that housed the ordinary inhabitants who served the aristocrats living in palaces at the heart of the city.

The Agricultural Revolution

During a relatively warm climatic period between 8000 and 6000 B.C. humans on the fringes of Mesopotamia began to shift from a hunting-gathering existence and started to control the animals and plants they would eat, thus initiating the Agricultural (or Neolithic Revolution. That it was a revolution there can be no dispute. It transformed the way human beings lived and shattered a tradition over two million years old. However, why the Agricultural Revolution occurred at this precise time is still largely a matter of conjecture. Why. for instance, did it not occur during one of the earlier interglacial periods (intervals of relatively warm climate that occurred periodically between ice ages over millions of years) when, presumably, the same conditions prevailed? It is difficult to find any uniformly satisfying answers. We know that agriculture developed more or less simultaneously in many different parts of the globe, so it is unlikely that it resulted from any single cause such as climatic change or population growth, although both have been offered as explanations. We also know that the move to agriculture was not always permanently successful. In some places it was tried for a while and then abandoned. It is even possible that certain plants and animals were domesticated more than once and by different peoples

Most modern explanations of the origins of agriculture tend to emphasize the role of microenvironments and long-standing human-plant and human-animal relationships. Such factors as changing climatic conditions, the presence of animals and plants that offered good potential for domestication, and the cultural and technological levels of achievement of the human populations present undoubtedly played important roles in the development of agriculture.

The key to understanding agriculture is the process known as domestication. Domestication was the essential technological breakthrough that allowed human beings to escape the age-old system of hunting and gathering and to control the production of food. rather than being at the mercy of what sustenance the terrain might offer at any given moment.

Domestication can be defined as a primitive form of genetic engineering in which certain plants and animals are brought under human control, their objectionable characteristics eliminated. their favorable ones enhanced. and in the case of animals. inducing them to reproduce in captivity. If wild animals cannot be induced to breed in captivity, they cannot be domesticated. Modern domesticated cattle, sheep, and pigs, for example, look only remotely like their leaner. meaner, and faster-moving ancestors. Domestication is best viewed as the creation of an artificial environment in which the chosen plants or animals come to exist exclusively. Left alone, domesticated species either die or revert to their original wild forms. Because herds. farms, orchards. and gardens are permanent, static entities, once they came into being, the old hunting-gathering forms of social organization had to be replaced

Hunter-gatherers place a low value on possessions and a high value on mobility. Always on the move, they carry only a few tools and weapons with them. Agriculture reverses this way of life. It cannot be practiced without a commitment to permanence and the accumulation of large amounts of material goods. Homes, villages and storage facilities must be constructed: fields cleared, divided, and fenced; herds built up and maintained; and tools fabricated. Constant effort is required to maintain all of these. Once settled, farmers may not move again for generations. Pastoralists (animal herders) are equally committed to their flocks and herds.

For practical purposes, hunting-gathering bands always remained small, in the range of 30 to 50 people. Larger groups would have been difficult to sustain in most environments; smaller groups could not reproduce themselves. Agriculture, by contrast knew no limits as far as population growth was concerned. Thus, where hunting-gathering bands restricted their numbers, agricultural communities tended to expand them. People could be put to work in the fields or gardens at an early age and at harvest time, when it was essential to maximize the number of people who could be mobilized. Overpopulation was solved by emigrating and opening up new land for cultivation. By about 6000 B.C., villages with populations in the thousands were common throughout the Middle East.

Christian Thomsen and Danish Artifacts

In 1807 the Danish Royal commission for the Preservation and Collection of Antiquities was established. It began to gather together a collection of antiquities (objects from the ancient past) from all over Denmark that soon became one of the largest and most representative in Europe. In 1816 the commission invited the scholar Christian Thomsen to classify and prepare this collection for exhibition. The main problem that Thomsen faced was how the diverse assortment of prehistoric material in the collection could be exhibited most effectively. He decided to proceed chronologically by subdividing the prehistoric period into successive ages of stone, bronze, and iron. The notion of successive ages of stone, bronze and iron was not merely speculation but a hypothesis for which there was already some evidence.

In attempting to sort the prehistoric material in the collection into three successive technological stages or eras, Thomsen faced a daunting task. He recognized that even for the stone and metal objects, automatic classification would not work. Bronze and stone artifacts had continued to be made in the Iron Age, just as stone tools had been used in the Bronze Age. The challenge was therefore to distinguish bronze artifacts made during the Iron Age from those made during the Bronze Age and to differentiate which stone tools had been made in each era. There also was the problem of assigning objects made of gold, silver, glass, and other substances to each period. Individual artifacts were no help in beginning this work. Yet the Danish national collection contained sets of artifacts that had been found in the same grave, collection, or other contexts and that could safely be assumed to have been buried at the same time. Thomsen called these "closed finds “and believed that, by comparing the various items from each such discovery and noting which types of artifacts occurred together and which never did, it would be possible for him to determine the sorts of artifacts that were characteristic of different periods.

Thomsen sorted and classified his artifacts into various use categories, such as knives, adzes, cooking vessels, safety pins, and necklaces. He further refined each category by distinguishing the artifacts according to the material from which they were made and their various shapes. Having in this way established a set of informal artifact types, he began to examine closed finds in order to determine which types did and did not occur together. He also examined the decorations on artifacts and found that these, too varied consistently from one closed find to another. On the basis of shape and decoration, it became possible for Thomsen to distinguish types of bronze artifacts that never occurred together with iron artifacts from ones that did occur with them. He also was able to demonstrate that large flint knives and spearpoints that had similar shapes to bronze ones had been made at the same time as bronze artifacts. Eventually, he succeeded in dividing the prehistoric artifacts in the collection into five distinct groups. Once these groups were established, he could assign single artifacts to each group on the basis of similarities in outward form and structure. Thomsen also studied the contexts in which artifacts had been found and discovered that these varied consistently from one group to another.

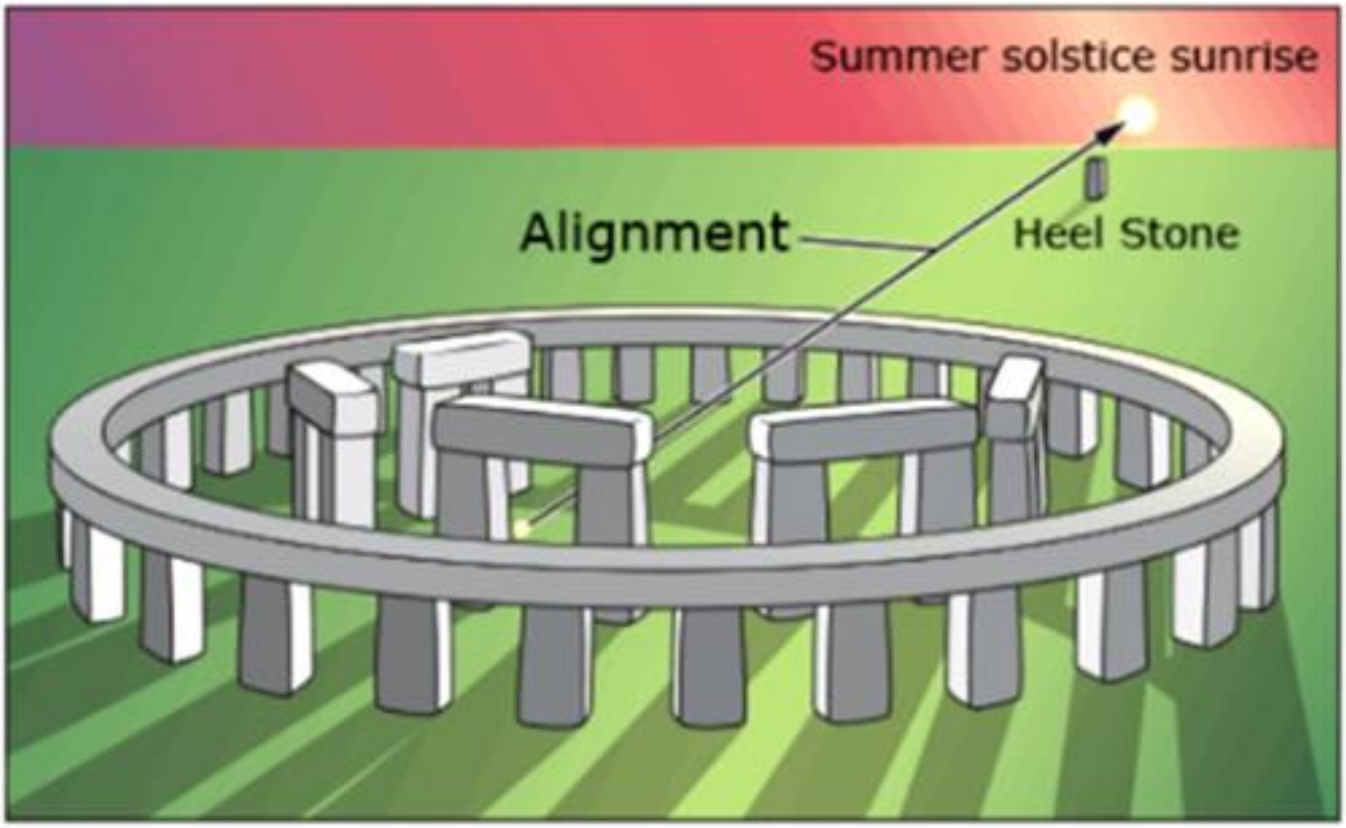
Thomsen then proceeded to order his groups into a historical sequence. He identified the simplest assemblages, which contained only chipped stone artifacts, as the remains of an early Stone Age. This material came invariably from small, simple sites. Next was a later Stone Age, which he described as the period when polished as well as chipped stone tools were manufactured and the first use was made of metal. At this time, the dead were buried, uncremated, in prehistoric tombs, accompanied by crude pottery vessels with incised decoration. In the full Bronze Age, both weapons and cutting tools were made of bronze, the dead were cremated and buried in urns under small tumulimounds), and artifacts were decorated with ring patterns. In the Iron Age, tools and weapons were made of tempered iron, whereas bronze continued to be used to manufacture ornaments and luxury goods. Thomsen divided the Iron Age into two stages, the earlier characterized by curvilinear serpent motifs and the later by more elaborate dragons and other fantastic animals.

Early Astronomy

As surviving records and artifacts make abundantly clear, many early cultures took a keen interest in the changing nighttime sky. But unlike today, the major driving force behind the development of astronomy in those early societies was probably neither scientific nor religious. Instead, it was decidedly practical. Seafarers needed to navigate their vessels, and farmers had to know when to plant their crops. In a real sense, then. human survival depended on knowledge of the heavens. The ability to predict the arrival of the seasons, as well as other astronomical events, was undoubtedly a highly prized, perhaps jealously guarded, skill.

The human brain's ability to perceive patterns in the stars led to the "invention" of constellations as convenient means of labeling regions of the nighttime sky. The realization that these patterns returned to the night sky at the same time each year met the need for a practical means of tracking the seasons. Widely separated cultures all over the world built elaborate structures to serve, at least in part, as primitive calendars, but often early experts on astronomy enshrined their knowledge in myth and ritual, sometimes turning sites used for astronomical observation into places for religious ceremonies.

Perhaps the best-known such site is Stonehenge located on Salisbury Plain in England. This ancient monument, which today is one of the most popular tourist attractions in Britain, dates from the Stone Age. Researchers believe it was an early astronomical observatory of sorts-not in the modern sense of the term (a place for making new observations and discoveries pertaining to the heavens) --but rather a kind of three-dimensional calendar or almanac, enabling its builders and their descendants to identify important dates by means of specific astronomical events. Its construction apparently spanned a period of about 17 centuries, beginning around 2800 B.C. Additions and modifications continued to about 1100 B.C, indicating its ongoing importance to the Stone Age and, later, Bronze Age people who built, maintained, and used Stonehenge. The largest stones weigh up to 50 tons and were transported from many miles away.



Many of the stones are aligned so that they point toward important astronomical events. For example, the line joining the center of the inner circle to the so-called heel stone, set off some distance from the rest of the structure, points in the direction of the rising Sun on the summer solstice (the longest day of the year). Other alignments are related to the rising and setting of the Sun and the Moon at other times of the year. The accurate alignments (within a degree or so) of the stones at Stonehenge were first noted in the eighteenth century, but it was only relatively recently-- in the second half of the twentieth century, in fact--that the scientific community began to credit Stone Age technology with the ability to carry out such a precise feat of engineering. While some of Stonehenge's purposes remain uncertain and controversial, the site's astronomical function seems well established. Although Stonehenge is the most impressive and the best preserved, other stone circles, found all over Europe, are believed to have performed similar functions.

The early Chinese also observed the heavens Their beliefs attached particular importance to omens" such as comets, which are seen in the night sky only occasionally, and "guest stars"-stars that appeared suddenly in the sky and then slowly faded away-and they kept careful and extensive records of such events. Twentieth-century astronomers still turn to Chinese records to obtain observational data recorded during the Dark Ages (roughly from the fifth to the tenth century AD.), when turmoil in Europe largely halted the progress of Western science. Perhaps the best-known guest star was one that appeared in AD.1054 and was visible in the daytime sky for many months. We now know that the event was actually a supernova-the explosion of a giant star-which scattered most of its mass into space. It left behind a remnant that is still detectable today, nine centuries later. The Chinese data are a prime source of historical information for supernova research.

The Switch to coal

In the United States and Great Britain, coal was not widely utilized until more easily available energy resources, such as wood were on the point of exhaustion. The shift to fossil fuels and the corresponding rise in energy consumption is illustrated by the transformation of the shipping industry in the nineteenth century. Until the mid-nineteenth century, the world's ships had been powered by wind and human power. The first river steamboats came into use around 1810-they first crossed the English Channel in 1821 and by 1839 had crossed the Atlantic. Technological developments, such as the introduction of iron and steel hulls (watertight bodies of ships) in the 1850s and 1860s, increased the effectiveness of sailing ships and the largest could still compete with the early, inefficient, and expensive steam-powered ships that operated from the 1840s. It was the development of the high-pressure steam boiler made from steel that transformed the situation. By the late 1860s, steamships could bring three times as much cargo from China to Europe in half the time taken by sailing ships. The amount of steam-powered shipping in the world rose from just 32,000 metric tons in 1831 to over three million metric tons by the mid-1870s, and then rose exponentially as sailing ships gradually died out and the steamship took over the world's merchant and naval fleets. Britain built a chain of coaling stations across the globe to sustain the worldwide deployment of the Royal Navy. Not only had the amounts of shipping in the world increased dramatically, its energy demands had increased even more.

The growing use of coal led to the rise of an important byproduct-the use of the waste gases to provide the first nonnatural source of lighting. Coal gas was first used to light a factory in Great Britain in 1807 and, six years later, a cotton mill in the United States. The advantage for the factory owners was that artificial lighting enabled far longer hours to be worked. Coal gas was cheap to use for street lighting once the high installation costs (laying gas pipes, known as mains, and installing new street lights) had been paid. It was cheaper than whale oil and available in much greater quantities. Street lighting could therefore spread on a much greater scale-by 1816 the first districts in London were lit by gas supplied from a central coal-burning plant through underground mains. By the late 1820s, gas lighting had been adopted in Boston, New York (which depended on imported British coal), and Berlin. The use of coal gas to light streets and houses and eventually for cooking spread through the industrialized world in the nineteenth century.

The peak of the world's dependence on coal for its energy came u in the first decades of the twentieth century. Coal's share of world energy consumption then fell from about 90 percent in 1900 to less than 25 percent in the early twenty-first century. However, coal production continued to increase-from about 760 million metric tons in 1900 to just over 5,000 million metric tons in 2000. The decline in the importance of coal occurred first in the United States because of its large oil reserves. In Europe the change came much later-in 1950 coal still provided over 80 percent of the continent's energy. Yet by 1970 the proportion had fallen to less than a third, as cheap imported oil replaced coal. Until the 1950s, Europe's railways still depended on coal-fired steam engines, as they had over a century earlier. Then, in the space of a couple of decades, they were replaced by diesel-powered locomotives and electrified systems. In 1900, Britain was the second-largest coal producer in the world, and the industry employed about 1,200,000 men. By the end of the twentieth century, output was at 10 percent the level of a century earlier and there were only 10.000 coal miners. Production also shifted from deep mines to open-pit mining, which was much more destructive environmentally (but cheaper)-in the United States two-thirds of coal production now comes from open-pit mines. However, coal is still the second-most-important energy source in the world. About 40 percent of the world's electricity still comes from coal-fired plants, and in many countries of the world it remains the primary fuel.

Why Did Agriculture Begin?

Transitions from a glacial to interglacial world occurred many times during the last two million years. Through all but the most recent glaciation, people moved in response to environmental changes rather than staying put and adapting to a new ecosystem. Then, after living on the move for more than a million years, they started to settle down and become farmers. What was so different when the glaciers melted this last time that caused people to adopt a new lifestyle?

Several explanations have been offered to account for this radical change. Some argue that the shift from a cool, wet glacial climate to less hospitable conditions put an environmental squeeze on early people in the Middle East. In this view, hunters began growing plants in order to survive when the climate warmed and herds of wild game dwindled. Others argue that agriculture evolved in response to an inevitable process of cultural evolution without any specific environmental pressure. Whatever the reasons, agriculture developed independently in Mesopotamia, northern China, and Mesoamerica.

For much of the last century, theories for the origin of agriculture emphasized the competing oasis and cultural evolution hypotheses. The oasis hypothesis held that the postglacial drying of the Middle East restricted edible plants, people, and other animals to well- watered flood plains. This forced proximity promoted social bonds, which eventually led to domestication. In contrast, the cultural evolution hypothesis holds that regional environmental change was unimportant in the gradual adoption of agriculture through an inevitable progression of social development. Unfortunately, neither hypothesis provides satisfying answers for why agriculture arose when and where it did.

A fundamental problem with the oasis theory is that the wild ancestors of our modern grains came to the Middle East from northern Africa at the end of the last glaciation. This means that the variety of food resources available to people in the Middle East was expanding at the time that agriculture arose-the opposite of the oasis theory. So, the story cannot be as simple as the idea that people, plants, and animals crowded into shrinking oases as the countryside dried. And because only certain people in the Middle East adopted agriculture, the cultural evolution hypothesis falls short. Agriculture was not simply an inevitable stage on the road from hunting and gathering to more advanced societies.

The transition to an agricultural society was a remarkable and puzzling behavioral adaptation. After the peak of the last glaciation, people herded gazelles in Syria and Israel. Subsisting on these herds required less effort than planting, weeding, and tending domesticated crops. Similarly, in Central America several hours spent gathering wild corn could provide food for a week. If agriculture was more difficult and time-consuming than hunting and gathering, why did people take it up in the first place?

Increasing population density provides an attractive explanation for the origin and spread of agriculture. When hunting and gathering groups grew beyond the capacity of their territory to support them, part of the group would split off and move to a new territory. Once there was no more productive territory to colonize, growing populations developed more intensive (and time-consuming) ways to extract a living from their environment. Such pressures favored groups that could produce food themselves to get more out of the land. In this view, agriculture can be understood as a natural behavioral response to increasing population.

Modern studies have shown that wild strains of wheat and barley can be readily cultivated with simple methods. Although this ease of cultivation suggests that agriculture could have originated many times in many places, genetic analyses show that modern strains of wheat, peas, and lentils all came from a small sample of wild varieties. Domestication of plants fundamental to our modern diet occurred in just a few places and times when people began to more intensively exploit what had until then been secondary resources.

Causes of Amphibian Declines

A striking feature of amphibian population declines, and what sets them apart from the general biodiversity crisis, is that they have occurred in some of the best-protected natural areas. For example, Sierra Nevada Yellow-legged Frogs have disappeared from wilderness areas high in the United States mountain range known as the Sierra Nevada. While remote locations such as these have escaped obvious habitat alteration and destruction, they are still being affected by human activities. Scientists have investigated five possible causes for the mysterious rapid amphibian declines that have occurred in such remote locations: (1) introduced nonnative species, (2) increased ultraviolet radiation, (3) disease, (4) climate change. and (5) toxic contaminants such as pesticides.

People introduce nonnative species for a variety of reasons, including for recreational fishing, for human food, and for biological control of pests. For example, as early as the 1800s, fish stocking of high-elevation lakes in the Sierra Nevada was done by horseback. Today, trout are raised in hatcheries and dropped by airplane into remote sites in mountain ranges throughout the West. Trout are heavy predators and will eat both tadpoles and adult frogs, so they could easily be the cause behind some declines. While introduced species such as trout have significant negative impacts on certain amphibian species, scientists are concluding that they are not the predominant cause of rapid declines.

Increased ultraviolet radiation was long thought to play a role in declines, and it has received much scientific attention. The release of ozone-destroying chemicals such as chlorofluorocarbons (CFCs), formerly used in refrigerators, has resulted in increased ultraviolet radiation reaching Earth's surface. Ultraviolet radiation can cause DNA damage and affect animal immune systems. However. the largest number of rapid declines has occurred in tropical mountain-slope forests where amphibians are sheltered from sunlight by the forest canopy. Thus, ultraviolet radiation is unlikely to be a primary cause of these declines.

In 1998 scientists discovered a previously unknown fungus that was associated with frog die-off in both Australia and Panama. The fungus, Batrachochytrium dendrobatidis, is in the chytrid family of fungi and is the only member known to be a pathogen (cause of disease) to vertebrates. The discovery of the chytrid fungus proved to be a turning point, and now disease, particularly that caused by the chytrid fungus, has emerged as the leading explanation for rapid amphibian-population declines. Chytrid fungus has now been found in more than 400 amphibian species and has been associated with population die-offs in North, Central, and South America, as well as in Africa, Europe, Australia, and New Zealand. While it is clear that chytrid fungus is the immediate cause of many rapid amphibian-population declines, there is still much debate about the disease. Is chytrid fungus a newly emerging disease, spreading and attacking defenseless amphibian hosts? Or have recent environmental changes facilitated disease outbreaks by a pathogen that has long been widespread?

Global climate change, especially warming trends, could be contributing to chytrid fungus outbreaks. Unusually hot spells may stress amphibians, which would weaken their immune systems and make them more susceptible to disease. Another possibility is that temperature shifts caused by climate change may create environments more favorable for the growth of chytrid fungus. In addition, climate change may cause drier conditions, forcing amphibians to congregate in smaller bodies of water, facilitating the transmission of disease. All these factors point to a possible connection between climate change and disease, but to date the necessary research has not been done to determine if any of these inks are actually operating.

Pesticides and other contaminants may facilitate disease also. Such chemicals can be carried long distances from where they are applied and often wind up in remote ecosystems. Studies have shown that contaminants can suppress immune systems in many organisms, including amphibians, which may lead to disease outbreaks. In California, researchers have found a strong association between the geographic pattern of population declines of a number of frog species and the pattern of pesticide applications. However, as with climate change, few studies have been conducted to establish a clear link among contaminants, disease, and amphibian declines. More research is clearly needed.

The Price Revolution

Unprecedented inflation, or the price revolution, swept through Europe in the sixteenth century. The main cause of the price revolution was the population growth during the late fifteenth and sixteenth centuries. The population of Europe almost doubled between 1460 and 1620. Until the middle of the seventeenth century, the number of mouths to feed outran the capacity of agriculture to supply basic foodstuffs, causing the vast majority of people to live close to subsistence (the minimum food necessary to live). Until food production could catch up with the increasing population, prices, especially those of the staple food bread, continued to rise

The other principal cause of the price revolution was probably the silver that flowed into Europe from the Americas via Spain, beginning in 1552. At some point, the influx of silver may have exceeded the necessary expansion of the money supply and may have begun contributing to the inflation. A key factor in the price revolution, then, was too many people with too much money chasing too few goods. The effects of the price revolution were momentous.

The price revolution had its greatest effect on farming. Food prices, which rose roughly twice as much as the prices of other goods, spurred ambitious farmers to take advantage of the situation and to produce for the expanding market; the opportunity for profit drove some farmers to work harder and manage their land better.

All over Europe, landlords held their properties in the form of manors. A particular type of rural society and economy had evolved on these manors in the fourteenth and fifteenth centuries. By the fifteenth century, much manor land was held by peasant tenants according to the terms of a tenure (relationship between tenants and landlords) known in England as copyhold. The tenants had certain hereditary rights to the land in return for the performance of certain services and the payment of certain fees to the landlord Principal among these rights was the use of the commons-the pasture, wood. and pond. For the copyholder, access to the commons often made the difference between subsistence and real want because the land tilled on the manor might not produce enough to support a family. Arable land was worked according to ancient custom. The land was divided into strips, and each peasant of the manor was traditionally assigned a certain number of strips This whole pattern of peasant tillage and rights in the commons was known as the open-field system. After changing little for centuries, it was met head-on by the incentives generated by the price revolution.

In England, landlords aggressively pursued the possibilities for profit resulting from the inflation of farm prices. This pursuit required far-reaching changes in ancient manorial agriculture, changes that are called enclosure. The open-field system was geared to providing subsistence for the local village and, as such, prevented large-scale farming for a distant market. In the open-field system, the commons could not be diverted to the production of crops for sale. Moreover, the division of the arable land into strips reserved for each peasant made it difficult to engage in profitable commercial agriculture

English landlords in the sixteenth century launched a two-pronged attack against the open-field system in an effort to transform their holdings into market-oriented, commercial ventures. First, they denied their tenant peasantry the use of the commons, depriving poor tenants of critically needed produce; then they changed the conditions of tenure from copyhold to leasehold. Whereas copyhold was heritable and fixed, leasehold was not. When a lease came up for renewal, the landlord could raise the rent beyond the tenant's capacity to pay. Both acts of the landlord forced peasants off the manor or into the landlord's employ as farm laborers. With tenants gone, fields could be incorporated into larger, more productive units. Landlords could hire labor at bargain prices because of the swelling population and the large supply of peasants forced off the land by enclosure. Subsistence farming gave way to commercial agriculture: the growing of a surplus for the marketplace. But rural poverty increased because of the mass evictions of tenant farmers.

Grinding Grain

It now seems that humans began to grind grain into flour earlier than was originally thought. Grinding stones have been found at African and Asian sites dating from 200,000-50,000 years ago It was presumed at first that these stones were used primarily to grind plant and animal materials, or minerals, to make pigments, rather than for the preparation of foodstuffs. However, new findings from the Middle East raise the intriguing possibility that some human groups may have been using grinding stones to process cereal grains, such as wheat, and maybe other types of edible plant, as early as the Middle Paleolithic Era (i.e., 50,000 years ago and earlier). But why is it such an advantage to grind cereal grains before eating them? The main reason is that grinding breaks down the hard, fibrous cereal grain to release the easily digestible starch granules contained within This served two purposes. Firstly, it enabled people to save enormously on the wear and tear of their teeth, compared to eating raw unprocessed grains. Unlike the teeth of grazing animals, human teeth do not continue to grow after childhood. Tooth wear due to a diet high in fiber and raw plants can result in the substantial erosion of molars (back teeth by early adulthood. People with worn or absent teeth faced starvation unless they could find alternative types of food that did not require chewing. Alternatively, they could try to find another way of grinding the fibrous plant material before eating it. Perhaps it was one of these incentives that led to the use of stone grinding tools for seed processing.

The development of grinding technology would have been socially advantageous to a human group. People would tend to keep their teeth for much longer, and thus older, more experienced individuals could have lived longer, despite the ultimate loss of their teeth. Such people could then serve their clan either as "grandparent" childcare givers or by acting as instruments for the innovation and transmission of oral culture. The latter role was a key adaptation in preliterate societies (societies without a writing system particularly in relation to strategies for food acquisition and technology in an era of considerable climatic flux. The remembered knowledge of how their grandparents dealt with the last arid period, including alternative food acquisition strategies, would have enabled such surviving elders to greatly enhance the ability of their clan to deal with such difficult situations.

Unfortunately, grinding seed to make flour could be a mixed blessing. Depending on the type of stone used, the prolonged and laborious process of grinding cereal grains would produce small chips of stone that could get into the flour. People eating the products of such flour every day would be repeatedly exposed to stone chips as they chewed their food, and eventually their teeth might become chipped and worn. This problem was partially alleviated many millennia later by the invention of pottery, which enabled a porridge to be made from grains mixed with water and boiled without grinding.

The second, and more immediate, reason for grinding cereal grains is that it enables us to produce a more attractive. sweeter tasting, more nutritious, and calorie-rich foodstuff. Rather than a hard, dry, indigestible, tooth-destroying- cereal grain, people could enjoy foods such as seed cakes, biscuits, and all the various forms of bread that we still relish so much today. Cereal grains that have been ground and processed into flour can be much more easily digested due to the higher surface area that is available for digestive enzymes (molecules that break down food). This means that not only the plentiful starches but also the grain proteins and the much less abundant micronutrients, are more easily absorbed from processed cereals. In the cold, dry climate of the Last Glacial Maximum, plants of the grass family, such as cereals. would have been a more reliable source of food than woodland plants (e.g., nuts and berries. Many of these woodland plants would have died out as the weather worsened, and edible animals would have also become increasingly unavailable as they migrated to warmer climates, leaving cereals as one of the few remaining options for the people who chose. or were obliged. to remain in the Middle East.

Machines and Manufacturing

The tremendous growth in European industry in the eighteenth and nineteenth centuries was the result of a number of changes, technology foremost among them. The accumulation and diffusion of technical knowledge necessary for manufacturing began in the countryside, where handicraft operations were gradually enlarged and mechanized. Often it was small-and medium-sized producers, and the occasional amateur experimenter in a barn, who were the inventors and innovators. Numerous small inventions, applied and diffused on both sides of the

Atlantic gradually built up a stock of technical knowledge and practice that was widely available.

Most famous of these inventors was James Watt (1739-1819) of Scotland who succeeded in making steam engines more efficient Steam engines burned coal to boil water, which condensed into steam that was used to drive mechanized devices. Watt devised a way to separate steam condensers from piston cylinders so that pistons could be kept hot. and therefore, running constantly. This set the stage for a fuel-efficient engine. Early prototypes of Watt's engine were used to pump water out of mines. After moving to Birmingham in 1774. Watt joined forces with the industrialist Matthew Boulton (1728-1809). who marketed the steam engine won an extension of the patent for another twenty-five years, and set up a special laboratory for Watt so that he could refine his device. Thus, technical exploration joined forces with the interests of business: collaboration between inventors and entrepreneurs was a sign of the times. Moreover. perhaps the most important aspect of Watt's engineering feat was that it was subject to a stream of improvement and adaptation. not just from Boulton and company, but from its competitors as well.

The steam engine, driven by burning coal, provided vastly increased, power and sparked a revolution in transportation. Steam-powered ships and railroads, built once inventors were able to construct lighter engines that required less coal to run. slashed the time and cost of long-distance travel. Steam power's diffusion accelerated when iron-making improved, allowing for the production of railroad track and cables used to hang suspension bridges. The first public rail line opened in 1830 in England between Manchester and Liverpool. During the next twenty years, railway mileage increased from less than 100 to almost 25,000 in England, France Russia. and the German-speaking countries. Steamships appeared in the 1780s in France. Britain, and the United States. and in 1807 Robert Fulton inaugurated the first commercially successful route between New York City and Albany on the Hudson River. A century of toying with boilers and pistons culminated in the radical reduction of distances. Moreover steam-powered engines also improved sugar refining, pottery making, and many other industrial processes. Mechanizing processes that would have taken much longer and been subject to human error if done by hand enabled manufacturers to make more products at cheaper cost.

Textile production was one of the areas that benefited from both technical changes and the consolidation of different stages of the work in a factory with new machinery, a single textile operator could handle many looms and spindles at once, and could produce bolts of cloth with stunning efficiency. Gone were the hand tools. the family traditions, and the loosely organized and dispersed systems of households producing cloth in their homes for local merchants to carry to markets. The material was also stronger, finer, and more uniform. Thanks to such innovations. British cotton output increased tenfold between 1770 and 1790. leading to a 90 percent decline in the price of cloth between 1782 and 1812.

Most raw cotton for the British cloth industry had come from colonial India until 1793, when the American inventor Eli Whitney (1765-1825) patented a device called a "cotton gin “that separated cottonseeds from fiber. Cotton farming quickly spread through the southern states- from South Carolina into Georgia. Alabama Mississippi, and Louisiana-as the United States came to produce more than 80 percent of the world cotton supply by the 1850s Thus, the American south became a supplier of raw cotton to Britain.

The British Economy in the Eighteenth Century

The British economy expanded significantly in the eighteenth century, particularly with the development of factory manufacturing. By the middle of the century, it had begun to alter the northern English landscape. "From the Establishment of Manufacturers, we see Hamlets swell into Villages, and Villages into Towns" exclaimed one gentleman in the 1770s. The production of manufactured goods doubled in the second half of the century. Cotton manufacturing led the way. from 1750 to 1770, British cotton exports doubled. The production of iron followed in importance, along with wool and woolen fabrics, linen, silk, copper, paper, cutlery, and the booming building trades. Coal was substituted for wood as fuel.

Despite its relatively small size, Britain had significant economic advantages over the other nations of Europe, helping to explain why the manufacturing revolution began in Britain. Unlike the German or Italian states, Britain was unified politically. People living in England spoke basically the same language. France and the Italian and German states still had internal tariffs that made trade more costly. whereas in Britain there were no internal tariffs once the union between England and Scotland had been achieved in 1707, creating Great Britain. The system of weights and measures in Britain had largely been standardized. Indeed, Great Britain was by far the wealthiest nation in the world. colonies in faraway parts of the world provided raw materials for manufacturing and markets for goods produced in Britain; for example, the amount of raw cotton imported from India increased by twenty times from 1750 to 1800.

England's stable banking and credit arrangements also contributed to England's advantage by facilitating the reinvestment of agricultural and commercial profits in manufacturing. London's banks particularly the giant Bank of England, were profitable and respected. Merchants and manufacturers accepted paper money and written orders for payment, or bills of exchange with confidence. Gentry, or those who owned land, invested in overseas trade expeditions and in manufacturing without the reticence of landowners on the European continent London's financial market could provide information twice a week on what investments were worth in Amsterdam and Paris. Joint-stock companies. which had begun in the late seventeenth century, offered investors shares in their businesses together with limited personal liability, which meant that in the case of a company's financial disaster, individual investors would be liable only to the extent of their investments.

Expanded demand for manufactured goods led to a dramatic improvement in Britain's transportation systems. A new process of road surfacing-using small pieces of stone mixed with tar-improved travel on the main routes. Major roads were extended and improved, as investors formed turnpike trusts, repairing the highways and turning a profit by charging a toll to travelers using them. In 1700 it took 50 hours to travel the 180 kilometers from Norwich to London by coach. by 1800 the journey could be achieved in 19 hours. Moreover, England's water transportation was unmatched in Europe, a gift of nature. Rich sources of coal and iron ore lay near waterways and could be transported with relative ease along the coast. No part of England stands more than 70 miles from the sea. Navigable rivers and canals built in the middle decades of the century also facilitated the transportation of raw materials and manufactured goods.

Finally, the British government offered business people more assistance than any Contnental rivals could anticipate from their own governments. The powerful Royal Navy protected the merchant fleet. which tripled during the first three-quarters of the century. Laws forced foreign merchants to ship export goods to Britain in British ships. Bowing to pressure from woolens producers. the British government in 1700 had imposed protective tariffs or imported silk and calico (printed cotton fabric). Agreements with the Dutch Republic and France in the late 1780s reduced trade tariffs. with those states, which helped British exports. Furthermore, the political influence of business people kept taxes low.

Yet the British government rarely interfered in operations of the economy in ways that businesses might have considered intrusive. Adam Smith (1723-1790), who emerged as the most important economic theorist of the time, rejected prevailing theories that prescribed more government control of the economy, instead extolling economic liberalism, that is relatively little government intervention.

Characteristics of Sixteenth-Century European Towns

Size, legal status, or presence of fortifications (walls or other defensive barriers) might seem natural criteria in defining the distinction between town and village in sixteenth-century Europe. Ultimately however, it was none of these, but rather "urban" functions that distinguished even the smallest towns from villages. Some villages could be relatively large. Some had their own walls. On the other hand, many towns were unfortified and lacked legal status as a town.

Early modern towns were multifunctional, whether they were large or small this was what they shared. and this in turn is what distinguished them most from rural settlements. Not all of these functions were economic, but the economic functions were the foundations upon which all others were constructed. Only newly established military towns, such as Palmanova in northeastern Italy or Neuf-Brisach on the Franco-Imperial border, and the occasional ecclesiastical center were an exception to this pattern. Without the money and demographic momentum generated by economic activity, towns were not selected as the location for administrative centers, law courts, capital cities, colleges and universities, cathedrals, or the sites for religious orders. They might be transformed in the process, as were Madrid, when it was chosen definitively by Philip II in 1561 as the capital of Spain, and Weilburg in Hesse, which was reconstructed for a similar purpose in the late seventeenth century by Count Johann Ernst of Nassau, but the preconditions for growth were already there.

Five economic functions distinguished towns from the countryside: their location as centers for exchange; the presence of artisans; occupational diversity; regular links with other centers of exchange; and influence over a hinterland, or rural area far from any town. These distinguishing characteristics were then reinforced by social and cultural indicators, such as more complex forms of government, the presence of a stratified society with an identifiable elite, the presence of processionals, such as lawyers or school teachers and of religious orders and educational establishments. As time passed, new cultural indicators were added, such as discussion groups and charitable societies.

By using these urban characteristics and indicators, it is possible to suggest that centers of all sizes in western Europe shared common urban identity. Consider two Italian urban centers: Venice and the Sicilian town of Gangi, which, while not quite at the opposite ends of the urban spectrum, are usually considered belong to totally different spheres. Gangi had a population of some 4,000 in the mid-sixteenth century. Two-thirds of its inhabitants were engaged in agriculture. The others were artisans, shopkeepers, merchants, servants, rentiers (people whose earnings are derived from rent on properties), and members of religious orders. At the height of its prosperity at the end of the sixteenth century, on the other hand, Venice had a population of 190,000. It was a major international trading center, supported by a substantial industrial sector, and was the capital of an extensive maritime and land-based empire. It had a significant number of merchants, renters, and professionals among its permanent population. As a point of departure for pilgrims and a point of arrival for tourists, it also functioned as a center of hospitality. Servants and others engaged in the business of food, drink, and hotels, swelled the already large numbers who worked in the households of the permanent residents. In spite of these substantial contrasts both Gangi and Venice shared the basic economic functions and the social and cultural indicators that have already been discussed. Both functioned, above all, as centers of exchange, which linked their rural hinterlands to long-distance trading networks. Gangi lay on the old Roman road between Palermo and Messina and was an important center for the export of livestock and wine. Venice's strategic location at the head of the Adriatic Sea enabled it to link the eastern and western Mediterranean with the producers and consumers of central Europe. It was also a center of exchange for goods emanating from northern Italy. Each town had its important public buildings and churches, a central square, and large houses belonging to an elite. Each had its cultural identity. Comparisons of similar kind could be made from any region of Europe.

Origins of Earth's Salty Oceans

Scientists have long been interested in discovering the origin of Earth's water and establishing why Earth's oceans are so salty There has been speculation that earliest Earth was so hot that no liquid water existed, and all light elements (such as hydrogen and oxygen) were rapidly stripped away from Earth by the solar wind (a stream of charged particles emitted by the Sun). If this were true, then the elements needed to form water on Earth would not have been freely available. As a consequence, it was proposed that collisions with icy comets or similar gas-and water-rich materials brought water to Earth after the planet had sufficiently cooled to retain it. This concept was supported by comparisons of the gas compositions of meteorites with those of rocks from beneath Earth's surface, notably using krypton and xenon, nonmetallic gases that do. not react with other materials. There certainly is enough ice in space to have supplied our water (and atmosphere) in this manner.

In July 2015, the space probe Philae, which landed on comet Churi, discovered not only ice and dust, but also 16 types of organic compounds, present not in a loose distribution but in discrete clumps. Suddenly, the idea gained lots of traction that comets brought not only water, but also the ingredients for life, even in ready-made clumps. Intriguingly, in October 2015 it was reported. that-as this comet slowly thaws-molecular oxygen (02) escapes in a constant and high proportion (1% to 10%) relative to water which suggests that the comet also contains a surprising amount of primordial (ancient) oxygen, which was incorporated during the comet's formation.

Other work favors an alternative explanation. This work found that the hydrogen isotope ratio (the proportion of different forms of hydrogen) of ice in comets may be different from that of water on Earth. It instead emphasizes that the chemical composition of water on Earth resembles that of the small percentage of water contained within rocky meteorites, and thus in asteroids, which are essentially very large meteorites. Thus, a theory was developed that the asteroids planetesimals. and protoplanets that clumped together to form Earth had carried enough water in their rock minerals to explain our oceans. It would have escaped from the planet's interior as steam, which in turn would have condensed into water at the surface and in the early atmosphere. Calculations indicate that this mechanism can also provide plenty of water to explain Earth's observed water content.

We have a more complete understanding of the origin of salt in our oceans. It represents an accumulation of dissolved minerals over tens of millions to hundreds of millions of years. These minerals were broken up and dissolved during chemical weathering We are all familiar with this process from limestone buildings that become pitted or smoothed by the action of water, wind, and weather; this is where the term weathering comes from. The key process at work is one of chemical reactions between the rock and the water, with an important role for gases that are dissolved in water, such as carbon dioxide or sulfur dioxide, since these make the water more corrosive. The chemical weathering reactions break up rock minerals into charged atoms or molecules, called ions, which are removed in solution by river water and groundwater. This is exactly what happens when you dissolve table salt in water: the mineral salt breaks down into sodium and chloride ions that are held in a solution.

The early atmosphere contained high levels of carbon dioxide, or CO2. This gas is easily dissolved in water, forming a mildly acid solution. In the Co2-rich early atmosphere, this resulted in a corrosive acid rain that was highly effective at chemically weathering rocks, and fresh volcanic rocks are especially easily weathered. The intense weathering released dissolved minerals in the form of ions into river water and groundwater. From early times onward, river and groundwater flow has transported the dissolved minerals to their final collection point, the ocean basins. Given the extremely slow input and removal of salts, it becomes clear that the oceans' vast store of salt has accumulated because the oceans have for ages been the end station for salt transport. Meanwhile water itself continually evaporates from the oceans-concentrating- its salts-and- the evaporated fresh water continues the weathering cycle.

Latin America in the Nineteenth Century

A series of wars that took place between 1808 and 1826 brought independence to most former colonies of Spain and Portugal in Latin America. After winning their independence, the new Latin American states began a long, uphill struggle to achieve economic and political stability. There faced immense obstacles, for independence was not accompanied by economic and social changes that could spur rapid Progress. Large estates, generally operated using primitive methods and highly exploited labor, continued to dominate economic life. Far from diminishing, the influence of the landed aristocracy (established upper social class) actually increased. This was the result of the leading military role it had played in the wars of independence and the passing of Spanish authority.

Economic life stagnated, for the anticipated large-scale influx of foreign capital did not materialize, and the European demand for Latin American staples remained far below expectations. Free trade brought increased commercial activity to the coasts, but this increase was offset by the near destruction of some local craft industries by cheap, factory-made European goods. The sluggish pace of economic activity and the relative absence of interregional trade and true national markets encouraged local self-sufficiency, isolation, political instability, and even chaos.

As a result of these adverse factors, the period from about 1820 to about 1870 was for many Latin American countries an age of violence and of alternate dictatorship and revolution. Its symbol was the caudillo (strongman), whose power was always based on force, no matter what kind of constitution the country had. Usually, the caudillo ruled with the aid of a coalition of lesser caudillos, each supreme in his region. Whatever their methods, the caudillos generally displayed some regard for republican (representative government) ideology and institutions. Political parties, bearing such labels as "conservative" and "liberal," were active in most of the new states. Conservatism drew most of its support from the great landowners and their urban allies. Liberalism typically attracted provincial landowners. professional people, and other groups that had enjoyed little power in the past and were dissatisfied with the existing order. As a rule, conservatives sought to retain many of the social arrangements of the colonial era and favored a highly centralized government. Liberals usually advocated a federal form of government (in which power is distributed between a central government and regional authorities), guarantees of individual rights, lay (nonreligious) control of education, and an end to special privileges for the clergy and military. Neither party displayed much interest in the problems of the native peasantry and other lower-class groups.

Beginning in about 1870, the accelerating tempo of the Industrial Revolution in Europe stimulated more rapid change in the Latin American economy and politics. European capital flowed into the area and was used to create the facilities needed to expand and modernize production and trade. The pace and degree of economic progress of the various countries were very uneven and depended largely on their geographic position and natural resources.

Extreme one sidedness was a feature of the new economic order in which one or two products became the basis of each country's prosperity, making these commodities highly vulnerable to fluctuations in world demand and price while other sectors of the economy remained stagnant.

The late nineteenth-century expansion was accompanied by a steady growth of foreign control over the natural and human-made resources of the region. Thus, by 1900 a new structure of dependency, or Colonialism, had arisen, called neocolonialism, with Great Britain and, later, the United States replacing Spain and Portugal as the dominant powers in the area.

The new economic order demanded peace and continuity in government, and after 1870 political conditions in Latin America did, in fact, grow more stable. Old party lines dissolved as conservatives adopted the dogma of science and progress, while liberals abandoned their concern with constitutional methods and civil liberties (protections for individuals against unjust government interference) in favor of an interest in material prosperity. The cycle of dictatorship and revolution continued in many lands, but the revolutions became less frequent and less devastating.

These major trends in the political and economic history of Latin America in the period extending from about 1820 to 1900 were accompanied by other changes in the Latin American way of life and culture-notably, the development of a powerful literature that often sought not only to mirror Latin American society but to change it.

The Meaning of Upper Paleolithic Art

The period beginning 40,000 years ago (the Upper Paleolithic) witnessed a marked increase in human artistic and symbolic expression. At about this time, a large number of statues carved from bone or stone begin to appear in the archaeological record, as do magnificent paintings of animals that were hunted and animals that were not, as well as other images on Cave walls and ceilings. It is difficult for modern viewers to remain unmoved by these images, but what did these works mean to their creators and why did they create them?

Some researchers regard Paleolithic artwork as part of a system of communication of ideas-a system that uses animals and geometric patterns as symbols, the specific meaning of which may be lost forever. Anthropologist Meg Conkey views the 1,200 bones engraved with abstract geometric patterns at Altamira Cave, Spain, as the identifying symbols-the "flags"-of different groups of people who came together at the cave during certain periods. Archaeologist Michael Joachim views the cave paintings of northern Spain and southern France (the so-called Franco-Cantabrian region) as symbols marking territory. Social stresses that accompanied the population influx into the region during the period beginning 25,000 years ago may have resulted in the need to mark territory with symbols of ownership. Painting animals-probably the most important resources of a territory-within a sacred place in the territory, like a cave, might have served to announce to intruders the rightful ownership of the surrounding lands. Archaeologist Clive Gamble views the small stone statues of female figures, known as Venus figurines, as a symbolic social glue, helping to maintain social connections between geographically distant groups through a common religion and art style.

More recently, researchers Patricia Rice and Ann Paterson have returned to a more economic perspective. Their statistical analysis of the numbers and kinds of animals seen on cave walls in the European Upper Paleolithic shows interesting correlations with the collections of animal remains found at habitation sites in Spain and France. Small, non-aggressive animals such as reindeer and red deer were important in the diet of the cave painters and seem to have been depicted on cave walls in proportion to their economic importance. In addition, animals whose remains are found less often at archaeological sites, but that were impressive, dangerous, and produced large quantities of meat when they were successfully hunted, were commonly included in the artwork as well. So, it would appear that cave painters wanted to depict animals that were important food sources. However, the relatively recently discovered Chauvet Cave contradicts this pattern, with its stunning depictions of animals not known to have been exploited for food by Paleolithic Europeans, including carnivores like lions, bears, and panthers, as well as woolly rhinoceroses.

A neuron-psychological approach has been applied by researchers J. D. Lewis-Williams and T. A. Dowson to explain at least some of the less naturalistic cave art. They note that there are six basic geometric forms that people who are placed into an altered state of consciousness (for example, through hypnosis) under experimental conditions report seeing: dots, wavy lines, zigzags, cross-hatching or grids, Concentric circles or U-shaped lines, and parallel lines. Interestingly, these geometric forms are precisely those seen in some ancient cave art dating to more than 30,000 years ago.

Lewis-Williams and Dowson's approach is cross-cultural-in other words, they surveyed a wide variety of historical and archaeological cultures, finding common images in artwork all over the world. Lewis-Williams and Dowson point out ethnographic records of shamans (priests) who, in an attempt to communicate with spirits or see into other worlds, fall into a trance like state by fasting, dancing, hyperventilating, going into isolation in absolute darkness, undergoing sleep deprivation, or even ingesting natural hallucinogens. When these shamans produce an artistic representation of what they have seen in their trances, they often include geometric shapes that are also seen in Upper Paleolithic artwork. These images from trances are not culturally controlled but result, in part, from the structure of the optic system itself and are therefore universal. Perhaps through sleep deprivation, staring at a flickering fire or the ingestion of drugs, ancient shamans or priests produced these images in their own optic systems. They then translated these images to cave walls as part of religious rituals.

The Roots of Economic Transformation in England

England was the first nation in Europe to develop a social structure that strongly supported the innovation and economic growth we associate with modern times. England's advantages were many, some of them deeply rooted in geography and history. This comparatively small realm contained an excellent balance of resources. The plain to the south and east, where traditional centers of English settlement concentrated, was fertile and productive. The uplands to the north and west possessed rich deposits of coal and Ron, and their streams had powered flour mills for hundreds of years. Proximity to the sea was another natural advantage. No part of the island kingdom was distant from the coast. At a time when water transport offered the sole economical means for moving bulky commodities, the sea brought coal close to iron, raw materials close to factories, and products close to markets. Above all, the sea gave Britain's merchants access to the much wider world beyond their shores.

Efficiency of transport was critical in setting the size of markets. During the eighteenth century, Britain witnessed a boom in the building of canals and turnpikes (roads that could be traveled for a fee). By 1815 the country possessed some 2,600miles of canals linking rivers, ports, and other towns. In addition, few institutional obstructions to the movement of goods existed. United under a strong monarchy, Britain was free of internal tariffs (payments for goods transported across a border), unlike prerevolutionary France, Germany, or Italy. English merchants everywhere counted in the same money, measured their goods by the same standards, and conducted their affairs under the protection of the common law. By contrast, in France local regions differed in their legal codes and in weights and measures, which complicated and slowed exchange. As the writer Voltaire sarcastically remarked, the traveler crossing France by coach changed laws as frequently as horses.

The English probably had the highest standard of living in Europe and generated strong consumer demand for manufactured goods. English society was less stratified (divided into groups based on status) than elsewhere in Europe, and the aristocracy was powerful but much smaller. Primogeniture-the right of the eldest son to inherit the family's land-was the rule both among the aristocratic members of the House of Lords and among the other land-owning classes. Left without lands, younger sons had to seek careers in other walks of life, and some turned toward commerce. They frequently obtained capital for their ventures from their landed fathers and elder brothers. English religious minorities, chiefly Calvinists and Quakers, formed another pool of potential businessmen; denied careers in government because of their religion, many turned their energies to business enterprises.

A high rate of reinvestment is very important to industrialization; reinvestment, in turn, depends on the skillful management of money by both individuals and public institutions. Here again, Britain enjoyed advantages. Early industrial enterprises could rely on Britain's growing banking system to meet their capital needs, a system which in the seventeenth century was taken over by the goldsmiths of London, who accepted and guarded deposits, extended loans, and provided other financial services. In the eighteenth century, banking services became available beyond London; the number of regional banks rose from 300 in 1780 to more than 700 by 1810. English businessmen were familiar with banknotes and other forms of commercial papers, and their confidence in paper money facilitated he recruitment and flow of capital.

The founding of the Bank of England in 1694 marked a distinctive period in the history of European finance. The bank took responsibility for managing England's public debt, sold shares to the public, and faithfully met the interest payments due to the shareholders with the help of government revenue, such as the customs duties efficiently collected on Britain's extensive foreign trade. When the government needed to borrow, it could turn to the Bank of England for assistance. This stability in government finances ensured a measure of stability for the entire money market and, most important, held down interest rates in both the public and private sectors. In general, since the late seventeenth century, England's government was sensitive to the interests of the business classes, who in turn had confidence in the government. Such close ties between money and power facilitated economic investment.

Understanding Insects through Fossils

Although it has been estimated that insects account for roughly one-third of all animal species alive today, insects are, on the whole. poorly represented in the available fossil record, where many species are known from just a single specimen, and a high proportion of fossil insects come from exceptional fossil deposits that are sporadically distributed in time and space. Nonetheless, about 40.000 species of insects have been described as fossils, with many more awaiting description. Foremost among insect-rich deposits are ambers in which complete external preservation of insects is routine. Amber is the fossilized resin of a few particular kinds of trees. Oozing out of the bark, this resin had the ability to trap and surround insects, as well as other small animals, protecting them from the normal processes of organic decay as it hardened into transparent, yellow or orange amber. The chemical process of "amberization" could take up to 10million years. During this time, it was common for amber initially buried in the soil to be washed out by rivers and redeposited in the sea.

Although the oldest amber comes from the Carboniferous period (360-290 million years ago) the great majority of amber deposits were formed between the start of the Cretaceous period (146 million years ago) and the present. They provide priceless windows on the insects and other small animals living at the time in the forests where amber-producing trees grew. Elsewhere in the fossil record, insects can be found in fine-grained sedimentary rocks, such as clays and silts deposited in freshwater lakes and sluggish rivers. Unlike the insects in amber, these fossils generally comprise only fragments. particularly of wings or wing cases, although more complete examples can be found, such as the dragonflies of the famous Jurassic Solnhofen Limestone of Bavaria.

The fossil record describes a multiplicity of insects that scientists have grouped according to their features. The most basic categorization of insects is into a primitive group without wings, called the Apterygota, and the winged Pterygota. Surviving apterygotes include the springtails and silverfish. They are now relatively rare, comprising less than 1 percent of all insect species. Pterygotes are divided into those with wings that cannot be folded, which are called the Palaeoptera, and a larger, more advanced group, the Neoptera, capable of folding their wings close to the body. Mayflies, dragonflies and damselflies are all palaeopteran insects. while the neopterans include locusts, butterflies, and wasps.

In spite of its imperfection, the fossil record holds a lot of useful information about the times of origination of insect groups that are alive today. Primitive wingless insects-the apterygotes-appear to have undergone an initial diversification during the Devonian period (416-359 million years ago)， possibly even the Silurian period (444-416 million years ago)。 Unfortunately, however, relatively few fossil insects of this age are known and there is a great need for further discoveries. The oldest known fossil insect is currently Rhyniognatha hirsti from the early Devonian fossils in Scotland. However, this species, preserved in sinter (mineral sediment) from an ancient hot water spring that was active between 400 and 412 million years ago, exhibits some advanced characteristic, implying that there are more primitive, older insects still to be discovered.

Fossil insects with preserved wings (Pterygota) first occur in the mid Carboniferous. The evolution of wings was accompanied by an increase in maximum body size. A remarkable dragonfly called Meganeura with a wingspan approaching 70 cm has been described from the late Carboniferous. This inhabitant of the forests is one of the largest insects ever to have lived the huge size of Meganeura has led to speculations about the composition of the atmosphere at the time, the powered flight of such a large insect perhaps demanding an atmosphere containing higher levels of oxygen than that of the present day. Unfortunately, the fragmentary insect fossil record sheds little light on the origin of flight, as the oldest winged insects already had fully formed wings.

Interplanetary Seeding

Some scientists believe that microbial life may be distributed across terrestrial planets by interplanetary rocks. Rocks are capable of carrying microbes from the surface of one planet, across hundreds of millions of miles of space, to neighboring planets. Each year, Earth is impacted by half a dozen half-kilogram or larger rocks from Mars. These rocks were blasted off Mars by large impacts and found their way to orbits that cross-Earth’s path, where they eventually collided with Earth as meteorites. Nearly 10 percent of the rocks blasted into space from Mars end up on Earth. All planets are impacted by interplanetary objects large and small over their entire lifetimes, and the larger impacts actually eject rocks into space and into orbit about the Sun.

A glance at the full Moon with binoculars shows long streaks, or rays, radiating from the crater Tycho, located near the bottom of the Moon as seen by observers in the Northern Hemisphere. The rays are produced by the fallback of impact debris (impact material) ejected from the crater, which is 100 kilometers in diameter. The rays can be traced nearly across the full observable side of the Moon. and such long "airborne" flight is evidence that some ejecta (ejected materials) were accelerated to near-orbital speed. Debris ejected to speeds higher than the escape speed(2.2 kilometers per second)did not fall back but flew into space It has long been appreciated that material could be ejected from the Moon by impacts, but only in the relatively recent past have we realized that whole rocks greater than10 kilograms in mass could be ejected from terrestrial planets and not be severely modified by the process It was formerly believed that the launch process would shock-melt or at least severely heat the ejected material. There was little expectation that rocks capable of carrying living microbes from planet to planet would survive the great violence of the launch. The discovery of lunar (Moon) rocks in Antarctica showed that this is possible.

There is also rare class of meteorites called SNCs. or Martian meteorites，" that are widely believed to be from Mars. The first suggestion that these odd meteorites might be Martian was greeted with considerable skepticism. The discovery of lunar meteorites changed this by proving that there actually was an adequate natural launch mechanism. The lunar meteorites could be positively identified. because rocks retrieved by the Apollo program. which brought astronauts to the Moon, showed that lunar samples have distinctive properties that distinguish them from terrestrial rocks and normal meteorites derived from asteroids (rocky objects that orbit the Sun and are smaller than planets). Positive linking of the SNC meteorites with a Martian origin was a more complex process. It included showing that gas trapped in glass in the meteorite matched the composition of the Martian atmosphere, as measured by the Viking spacecraft that landed on Mars in 1976. The general properties of the SNC meteorites revealed that they were basalts (dark-colored volcanic rocks) formed on a large, geologically active body that was definitely neither Earth nor the Moon. Because the atmosphere of Venus is too thick and its surface too young. Venus was also ruled out as a source.

The astounding discovery that meteorites from the Moon and Mars reach Earth has profound implications for the transport of life from one planet to another. Over Earth's lifetime, billions of football-size Martian rocks have landed on its surface. Some were sterilized by the heat of launch or by their long transit time in space. but some were not. Some Martian ejecta are only gently heated and reach Earth in only few months. This interplanetary transporter is capable of carrying microbial life from planet to planet. Like plants releasing seeds into the wind, or palms dropping coconuts into the ocean. planets with life could seed their neighbors. Perhaps, then, nearby terrestrial planets might contain microbial life with common origins. The seeding process would be most efficient for planets that have small velocities of escape (the minimum speed needed for an object to break free from a planet's gravitational attraction) and thin atmospheres. In this regard, Mars would be a more likely source than Earth or Venus.

Development of Mass Transportation in the United States

Before the development of cheap, reliable urban transportation in the United States (U.S.) , city size had been limited by the distance people could comfortably walk to work. In such "walking cities, “the radius rarely extended beyond a few miles, and people of all classes lived and worked close to one another By the end of the nineteenth century the walking city had metamorphose into the sprawling, segmented city. Economic function and income divided the city roughly into a concentric pattern. A central business district, or downtown core, where few lived, was ringed by a manufacturing and wholesaling district, where immigrants and the working lower-income classes jammed into tenements and subdivided old houses. Beyond them was stretch of lower-middle-class to middle-class row houses or apartment buildings, where skilled, clerical, and some factory workers resided. Then came the fringes of the city, where professionals, managers, and businessmen retreated to their spacious homes, leading comfortable lives and tending their manicured lawns and shrubs away from the stench, noise, dirt, and push-and-shove of the city.

New means of transportation made possible the new urban geography of economic integration and social segregation. As early as the 1820s and 1830s such eastern cities as Boston, New York, and Philadelphia had operated horse-dray public carriers called omnibuses. Omnibus systems Spread rapidly through the 1850s, but the slow, heavy vehicles accommodated few passengers and high fares limited ridership. By mid-century, steam-powered commuter railroads were pushing city boundaries outward for the affluent. The New York Harlem Railroad connected the sub urban village of Harlem with New York, and the Chicago& Milwaukee Railroad linked communities such as Evanston. Wilmette and Lake Forest with Chicago.

At the same time, many cities had switched to horse-drawn streetcars. Rails smoothed the ride, compared to the bouncing of omnibus wheels over cobblestones, and they made possible larger conveyances and more rapid movement. By establishing fixed routes, the horse railways promised riders predictability. They also broadened the compass of the city to five miles or more. With lower fares, the horse car systems attracted a middle-class clientele, who commuted to work downtown from newly developed residential areas, or suburbs, on city borderlands. By the 1880s, horse-drawn streetcars were carrying almost 190 million passengers annually in the United States. Horse cars, however, were not as clean, cheap, or reliable as needed. Technology soon replaced the horse cars with cable cars, pulled by steam-driven, underground wire rope clamped to the car. Introduced in San Francisco in1873， cable cars proved especially effective there and in such other hilly cities as Pittsburgh and Washington, D.C.

The most important advance in urban transit was the electric trolley. In 1888， on a twelve-mile track in hilly Richmond, Virginia, Frank J. Sprague, a naval engineer and former associate of Thomas Edison, demonstrated the practicality of powering streetcars with electrical currents transferred from overhead wires to the Vehicle by a four-wheeled spring device, or troller. A frenzy of construction and conversion to electricity followed, as city governments awarded generous trolley franchises to transit entrepreneurs. Trolleys rolled along at twice the speed of horse cars, and the US system of flat-fee fares (the same fare no matter how long the trip) with free transfers encouraged mass ridership. For ten cents one could escape the soot and smells of the city on daily commute or, for the poor, on a Sunday outing.

Mass ridership encouraged construction. In 1890 the nation had 5,700 miles of horse-drawn track and only 1,260 miles of electrified track; twelve years later, just 250 miles of horse-drawn track remained in service, while the total length of electrified track had grown to 22,000 miles. Transit companies tried to increase ridership by developing areas surrounding cities. By the 1890s, the Peoples Railroad trolley in St. Louis ran out to Grand Boulevard, a dirt road amid cornfields that the transit company Was transforming into an avenue worthy of its name. The company laid out private streets with fountains, and it planted shade trees and shrubs to lure prospective home builders and buyers. Everywhere in the urban United States, transit companies invested heavily in such suburban real estate development, thereby encouraging population dispersal.

Spider Web Decorations

© Howard Cheek/Dreamstime.com

An evolutionary puzzle is provided by orb-weaving spiders (weavers of round webs) that incorporate impressive zigzag lines of white ultraviolet silk into their webs. which would seem to make the trap more obvious to the prey that the spider needs to catch. Given that many insects can see ultraviolet light, one wonders how orb weavers could gain by alerting potential victims to the danger posed by their webs. But perhaps these web additions actually attract ultraviolet-reflecting bees that specialize in taking nectar from ultraviolet-reflecting flowers. The fact that bees constitute the large majority of prey of some spiders with ornamented webs supports this hypothesis.

On the other hand, it could be that spiders that have been catching many prey. simply because their webs are in a good spot, are the spiders that tend to add decorations to their webs. If this is true, then the correlation between web decorations and higher rates of prey capture need not mean that the decorations cause the increase in prey capture. Instead, perhaps high feeding rates provide the extra energy that the spiders need to spin their web decorations. which they do for some purpose other than prey attraction. This hypothesis can be evaluated by testing the prediction that spiders given an abundance of food should invest more in web decorations than spiders deprived of food. Indeed, in two species of orb-weaving spiders, females that received more food made significantly larger decorations, and in one species. Well-ed females were more likely to add decorations to their webs than were food-deprived individuals.

If one places an orb-weaving spider in a wooden frame and lets it build a web there, so that the web can be moved to a site chosen by the experimenter, then one can put two such webs side by side in a field. In this way, one can compare the rate of prey capture in an ornamented web with that in another web in the same spot whose decorations have been removed- -something that can be easily accomplished by cutting out the two web lines that support the decoration (two web lines are also cut from the other web, but not those that hold the decoration), Such an experiment controls for the effects of site productivity. Under these conditions, the decorated web catches about a third less prey than the undecorated one. Thus, web decorations involve a cost to foraging (obtaining food), not a benefit, under some conditions.

If it is true that decorating one's web carries a cost in lost calories, what counterbalancing benefit exists for this behavior? One possibility is that when passing birds see the bright decorations, they swerve to avoid colliding with the web and getting covered in sticky silk. If this is true, then in the experiment above. webs with decorations should have been less often damaged by birds than webs without any "keep away" signals. And in fact, the actual results from the experiment matched the predicted ones, with decorated webs damaged by birds 45 percent less often than undecorated ones.

Another benefit of web decorations might come from having the added silk camouflage (hide) the body of the web builder, making the spider less vulnerable to attack from its predators. Evidence in support of this explanation comes from a set of spiders whose females regularly incorporate a silky egg sac into their webs. The spider perches (sits) on this moderately conspicuous vertical tube of silk, eggs, and debris in ways that make her much less visible to human observers, especially because the spider's color pattern matches that of the egg sac so closely. Moreover, when the egg sac is experimentally removed, the spider replaces it with a vertical strip composed entirely of silk, on which she perches as if it were an eggs sac. Given that the egg sac seems to be used as a concealment aid. its replacement can be assumed to serve the same function.

More data on this point come from a study in which orb-weaving spiders that had built webs with or without extra silk decorations in an enclosed area were exposed to wasps. which hunt spiders and feed them to their offspring. In one experiment, only 32 percent of the spiders with web decorations were captured, as opposed to 68 percent of those without.

The Beginning of Planet Formation

The four innermost planets of our solar system-Mercury, Venus, Earth, and Mars-are terrestrial, or rocky, planets. The beginning of their creation process occurred when the cloud of leftover material from the formation of the Sun settled into disc around the young star. Most of the material in the cloud, like the material of the Sun itself, was in the form of hydrogen and helium. But there was a trace of dust, no more than 2 percent of the original material, in the form of particles as fine as the particles in smoke. Heat from the young Sun blew much of the gas away, but the rotation of the original cloud ensured that the dust settled into a disc around the young Sun-a protoplanetary disc like the ones seen around young stars today.

Within the disc, all the particles were moving in the same direction around the Sun, like runners going round a track. This meant that when they bumped into one another, they did so relatively gently, not in head-on collisions, giving the particles a chance to stick to one another. The tendency to stick may have been helped by electric forces produced by particles rubbing against one another, in the same way that you can make a child's balloon stick to the ceiling after rubbing it on a woolen sweater. Another important factor was turbulence in the gas, creating swirling structures like whirlwinds which gathered pieces of material together and gave them a chance to interact. Computer simulations show how objects as big as Ceres can form in this way-provided the particles can stick together.

Something else may also have helped the particles to stick together-something else that is special about the solar system. Studies of pieces of rock from meteorites show that the dusty disc around the young Sun contained tiny globules of material, known as chondrules, formed by melting at temperatures between 1,200 degrees Centigrade and 1,600 degrees centigrade.

Molten, or partly molten, blobs would be more sticky and encourage the buildup of larger lumps of stuff in the disc. But how did they get so hot? The most likely explanation is that the heat was released by radioactive elements that had been sprayed by a nearby star in the process of dying into the gas cloud from which the planets formed. One possibility is that a supernova occurred close to the cloud that became the Sun just before the Sun formed; it is even possible that the blast wave from this explosion triggered the collapse of the gas cloud that became the Sun and solar system. Supporting evidence for this idea comes from measurements of the proportions of various isotopes (different forms of an element) found in meteorites. Radioactive aluminium-26 seems to have been present in the proto-solar system from the beginning, but a pulse of iron-60 arrived about million years later. This matches what we know about the fate of a very large star, with more than 30times as much mass as the Sun. In the late stages of its life, the star first blows away much of the outer layers of material, which by then is relatively rich in aluminium-26, in a wind easily strong enough to cause any nearby gas cloud to collapse. The star only explodes at the very end of its life, showering the neighborhood with elements including iron-60.

There is a rival idea, developed in Barcelona by Josep Trigo-Rodriguez and colleagues, which suggests that the radioactive material was fed into the solar system as it was forming from much less massive star which came much closer to the Sun. The right proportion of isotopes could have come in the wind of material being blown away from a star with only six times as much mass as our Sun in the last stage s of its life. But the star would have to be very close to the Sun for this to happen-closer than 10 light-years-which makes such an event unlikely, statistically speaking.

Species Competition

Interspecific competition occurs when two or more species seek the same limited resource. In the 1930s, Russian biologist G. F. Gause devised a set of elegant laboratory experiments that provide the basis for our formal understanding of competition. Gause grew two different species of the single-celled Paramecium—P. aurelia and P. caudatum - -separately and together. Populations of both species always increased more rapidly when they were grown alone. When grown together, populations of both species grew more slowly. Eventually, P. aurelia totally displaced P. caudatum. The results of his experiments with Paramecium species, along with similar experiments he performed on other organisms, led Gause to form this postulate: two species that directly compete for essential resources cannot coexist; one species will eventually displace the other. This postulate has come to be known as the competitive exclusion principle.

An acre of tropical forest may include over 100 species of trees, all of which depend on the same soil, water, and nutrients. Freshwater lakes may have dozens of species of fish, all of which feed on the planktonic algae and animals suspended in the water. Indeed, two or more species of Paramecium may be found in the same lake. These and many other examples from ecological communities in nature seem to contradict Gause's principle. If two competing species cannot coexist in the laboratory, how are they able to coexist in natural settings? This question has been the basis for hundreds of ecological studies.

Ecologist G. Evelyn Hutchinson provided one of the most important explanations for the coexistence of competing organisms. He proposed that each species has a fundamental niche, the complete range of environmental conditions, such as requirements for temperature, food, and water, over which the species might possibly exist. Hutchinson noted, however, that few species actually grow and reproduce in all parts of this theoretical range. Rather, species usually exist only where they are able to compete effectively against other species. Hutchinson used the term realized niche to describe the range of conditions where a species actually occurs given the constraints of competition. Species whose fundamental niches overlap significantly are potential competitors. Hutchinson suggested that these potential competitors are able to coexist because they divide up the fundamental niche. Hutchinson called this division of resources niche differentiation.

Niche differentiation occurs among many different kinds of organisms. For example, five different species of warblers, small insect-eating birds, occur together in the evergreen forests of the United States. During nesting season, the primary food of all the warblers is caterpillars. Careful studies of the birds' feeding behavior reveal that each species competes most effectively in a different part of the forest's highest layer, and that is where each species can be found. The diverse grasses and herbs that grow in native prairies provide another example of niche differentiation. Above ground, these plants appear to be vying for the same space and resources. However, a careful mapping of root systems shows that different species are adapted to exploiting different portions of the soil. In addition, some species compete most effectively when growing in bright light, whereas others compete effectively when growing in the shade of taller plants.

Some of Gause's experiments support Hutchinson's niche differentiation hypothesis. Under any specific set of conditions-the same temperature, water availability, food source, etc. Gause's principle holds true. But if conditions change, competition among species may produce different winners and losers. Indeed, if waste products are periodically removed, the outcome of the competition between P. aurelia and P. caudatum is reversed and P. caudatum wins. Thus, in a complex environment where waste materials are collected in some places and not in others, these two species could coexist.

Time is required for one species to competitively displace another, and the competitive exclusion principle presumes that environmental conditions remain constant during that time. In nature, however, environments change from season to season and from year to year, SO conditions that are favorable to a particular species may not persist and environments that are constantly changing may allow competing species to coexist.

Origins of the Industrial Revolution

The Industrial Revolution, the wave of technological, economic and social changes that helped produce what we know as modern Europe, arose among back-country English cottage craftspeople in the early 1700s and fundamentally restructured industry. First, human hands were replaced by machines in the fashioning of finished products, rendering the word manufacturing ("made by hand technically obsolete. No longer would the weaver sit at a hand loom and painstakingly produce each piece of cloth. Instead, large mechanical looms were invented to do the job faster and more economically. Second, human power gave way to various forms of inanimate power. The machines were driven by water power, the burning of fossil fuels, and later by hydroelectricity and the energy of the atom. Men and women, once the proud producers of fine handmade goods, became tenders of machines. Within a century and a half of its beginnings, this economic revolution had greatly altered industrial activity.

The initial breakthrough came in the secondary, or manufacturing sector More exactly, it occurred in the British cotton textile industry, centered at that time in the district of Lancashire in western England At first the changes were modest and on a small scale Mechanical looms were invented, and flowing water, long used as a source of power by local grain millers, was harnessed to drive the looms During this stage, manufacturing industries remained largely rural, diffusing to sites where rushing streams could be found, especially waterfalls and rapids. Later in the eighteenth century the invention of the steam engine provided a better source of power, and a shift away from water-powered machines occurred In the United States, too, the first factories were textile plants.

Metallurgy was also affected. Traditionally, metal industries had been small-scale, rural enterprises, carried on in small forges (fireplaces where metals were heated and shaped) situated near ore deposits. Forests provided charcoal for the smelting process in which ores were melted and fused. The chemical changes that occurred in the making of steel remained mysterious even to the people who made steel, and much ritual superstition and ceremony were associated with steelmaking Techniques had changed little in 2, 500years. The Industrial Revolution radically altered all this. In the eighteenth century, a series of inventions by iron makers allowed the old traditions, techniques, and rituals of steelmaking to be swept away and replaced with scientific, large-scale industry. Coke, nearly pure carbon derived from high-grade coal, was substituted for charcoal in the smelting process. Large blast furnaces replaced the forge, and efficient rolling mills took the place of hammer and anvil Mass production of steel resulted, and the new industrial order was built of steel. Other manufacturing industries made similar transitions and entirely new types arose, such as machine-making.

Primary industries-those that gather or extract raw materials-were also revolutionized. The first to feel the effects of the new technology was coal mining. The adoption of the steam engine necessitated huge amounts of coal to fire the boilers and the conversion to coke in the smelting process further increased the demand for coal. Fortunately, Great Britain had large coal deposits new mining techniques and tools were invented, so that coal mining became a large-scale, mechanized activity. Coal, heavy and bulky. was difficult to transport. As a result, manufacturing industries began flocking to the coalfields in order to be near the supply. Similar modernization occurred in the mining of iron ore, copper, and other metals needed by rapidly growing industries.

The Industrial Revolution also affected the tertiary(service) sector, most notably in the form of rapid bulk transportation. The traditional wooden sailing ships gave way to steel vessels driven by steam engines, canals were built, and the British-invented railroad came on the scene. The principal stimulus that led to these transportation breakthroughs was the need to move raw materials and finished products from one place to another, both cheaply and quickly. The impact of the Industrial Revolution would have been minimized had not the distribution of goods and services also been improved, it is no accident that the British, creators of the Industrial Revolution, also invented the railroad, initiated the first large-scale canal construction and revolutionized the shipbuilding industry.

The Heavy Bombardment and Life on Earth

It is estimated that Earth formed around 4.5 billion years ago, yet life-even the simplest sort of microscopic life that must have started things off did not begin right away. Until about 3.9 billion years ago, Earth was subject to an intense barrage of large objects from space, a time called the "heavy bombardment." Radiometric dating of Moon rocks shows that most of the

Moon's visible impact craters, or holes formed on its surface from impacts, must have formed during this early period of the solar system's history. This is not surprising: According to our modern theory of solar system formation, the planets were built as larger and larger chunks of rock (sometimes mixed with metal or ice). or planetesimals, collided with one another. When a colliding planetesimal stuck to a growing planet, the planet got larger, increasing its gravity and allowing it to draw in even more planetesimals. Even after the planets had reached essentially their current sizes, there must still have been many planetesimals floating around; some of them still remain today, as the objects we call asteroids and comets. Those planetesimals that had orbits intersecting the orbits of the planets were doomed to eventual collisions. and most of those collisions must have occurred early in the solar system's history, when the number of planetesimals was still large. In other words, the heavy bombardment was the period of time during which impacts were most common, and the evidence from the Moon tells us that this period ended by about 3.9 billion years ago.

Some of the planetesimals were quite big. We have good reason to think that the Moon itself was created when a planetesimal the size of the planet Mars struck the young Earth within just 20 to 30 million years after Earth's formation. This "giant impact" is thought to have blasted rock from Earth's outer layers into space, where some of it settled into Earth's orbit and then was collected together by gravity to make the Moon.

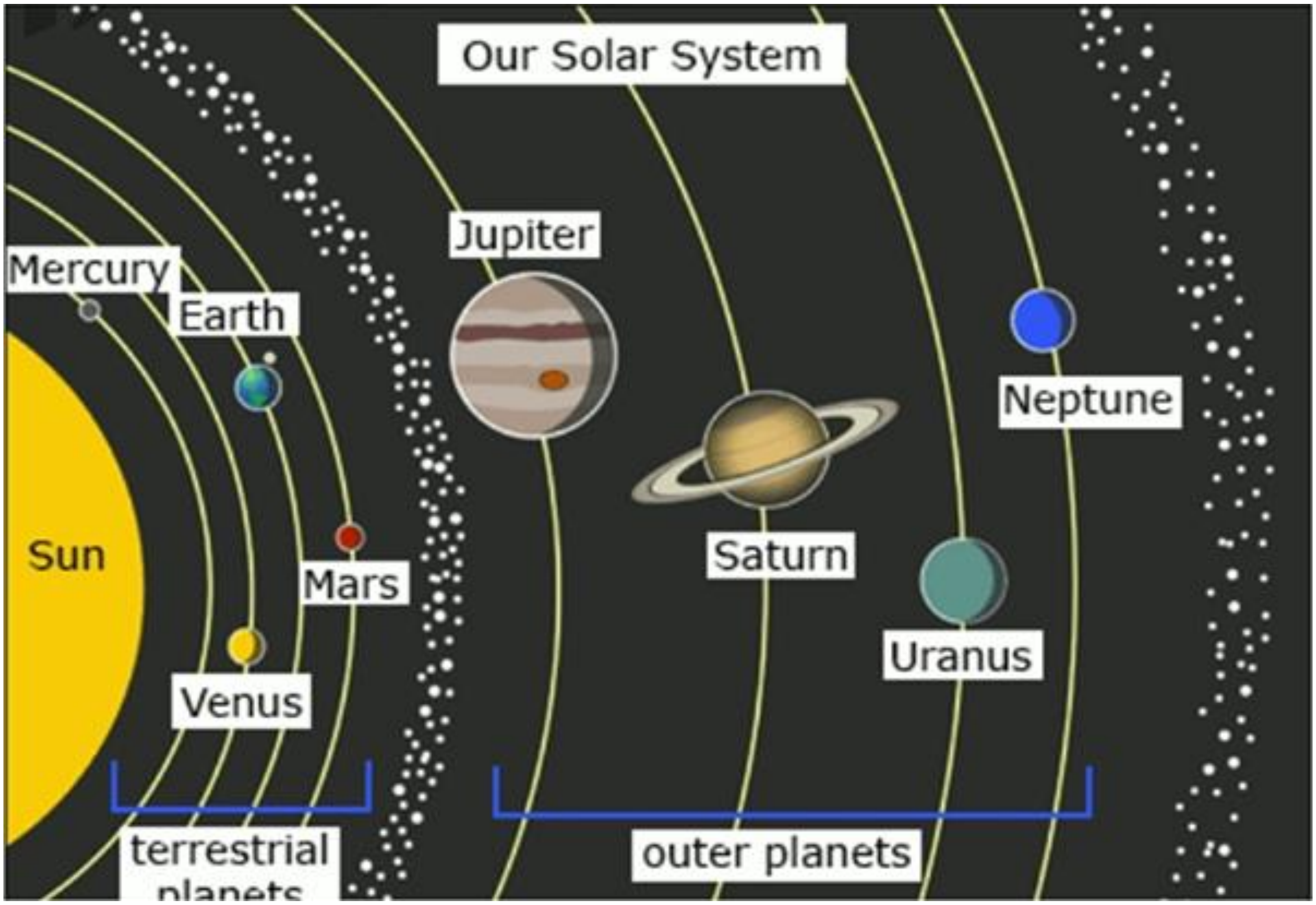
Once the Moon formed, it became a record of the continuing impacts, not only telling us when the heavy bombardment occurred but also telling us about the sizes of the impacting objects from the sizes of the craters they left. Because human spaceflight missions visited and brought back rocks from only six sites on the Moon, we have only incomplete data about lunar cratering. Nevertheless, these data point to two key ideas: First, while there were no more Mars-size impacts (fortunately!), the Moon continued to be pelted by objects tens of miles to a couple hundred miles across. Second, some of the largest impacts occurred as the heavy bombardment was ending, marking what many scientists now call the "late heavy bombardment." These large impacts created the smooth lunar maria (large, flat surface areas), that you can see easily with a pair of binoculars.

Because the heavy bombardment was a phenomenon of the solar system, it cannot have been unique to the Moon. This explains why we see craters on so many other planets arid moons. Earth, too, must have been frequently scarred by large impacts during the heavy bombardment. In fact, Earth should have been hit even more than the Moon, because our planet presents a bigger target and Earth's stronger gravity would have drawn in more objects and accelerated them to higher speeds by the time, they hit the ground. The only reason we do not see the craters from these impacts on Earth is that they were erased long ago by volcanic eruptions, erosion, and other geological processes that occur here but not on the Moon.

What does all this have to do with the origin of life? Calculations suggest that some of the larger impacts would have had a devastating effect on life. For example, the impact of an object larger than about 225 miles across would have released enough energy to completely vaporize the oceans and raise the global temperature to more than 3,000F. Such an impact probably would nave sterilized our planet, wiping out any life that existed when it occurred. Somewhat smaller impacts would have vaporized all but the deepest ocean water, killing off any life that was not either living near the ocean bottom or in rock deep underground. This means that life on Earth probably could not have begun until after this stage.

Planetary Formation

According to the condensation theory of planet formation, planets form out of a spinning disk of gas that surrounds a newborn star known as a protoplanetary disk. The disk and star both originate in a rotating, collapsing cloud of material, and this process of collapse produces different abundances of materials in the disk at different distances from the star. In the higher temperature regions, comparable to the region around the planet Mercury in our solar system, the only kinds of material that can condense from the gas to the solid state (in this case, microscopic dust grains) are metals. Farther out, about where Venus, Earth, and Mars are now, the gas temperatures are lower. At these distances and temperatures, rocky materials such as silicates can also begin to form dust grains. Even farther out, the temperature gets low enough for water ice to form, and even farther from the star, ices of other compounds such as ammonia and methane can condense but how do young planetary systems go from making dust grains to making planets?



"The answer to that question, "explains astronomer David Jewitt, "is a process called binary accretion, where collisions between pairs of objects let larger and larger structures get put together. Collisions between grains, which are small and sticky, quickly lead to the construction of pebbles. Collisions between pebbles lead to rocks. Collisions between rocks lead to boulders. Collisions between boulders lead to planetesimals, rocky bodies the size of asteroids (bodies that orbit between Mars and Jupiter and range in size up to about 1,000 kilometers). If this accretion process happens far enough from the star, significant quantities of ices will be included in the planetesimals. This is the likely origin of comets. Eventually, the objects are large enough for gravity to begin compressing and heating the planetesimal interiors. As planetesimals grow even larger, gravity pulls them into a spherical shape, and the heaviest elements sink to the center of the body. Iron and nickel will, in this way, form the dense metallic cores of the young planets Eventually, full-sized terrestrial planets, the rocky cores of gas giants (such as Jupiter and Saturn), and moons are formed through the accretion process.

The timescale for the accretion process depends on the density of gas and dust in the protoplanetary disk, because higher densities mean more frequent collisions. In the inner part of our pre-solar system disk, the density was high, and it took only100 million years to build the terrestrial planets. The formation of the outer planets is a more complicated story, and scientists are still not sure how the gas and ice giants formed. "There are basically two models for the formation of giant planets like Jupiter, says Jewitt. In the first model, an icy terrestrial planet grows by binary accretion up to a mass about 5-10 times that of Earth, at which point new process begins. "When that small core planet reaches the critical mass, "Jewitt explains, "it has enough gravity to start pulling in gas from the surrounding disk. You get very rapid flow of gas onto this core, taking the planet all the way up to Jupiter's or Saturn's mass." This idea is called the core accretion model. According to Jewitt among most solar system astronomers the core accretion model is the preferred idea for the formation of the large gas planets Jupiter and Saturn. The problem with the core accretion theory for these particular planets is that building up the rocky center takes an exceedingly long time. Near the end of the accretion phase, the disk begins losing its gas content when radiation from the Sun causes it to disperse. If the gas disperses before the core has a chance to reach its critical mass, the idea cannot work. That is why astronomers have developed a second theory, called the hydrodynamic instability model. This begins with an enormous disk of gas that collapses in on itself due to the influence of its own gravity. Just as the star and its disk were formed from a gravitationally unstable cloud, some astronomers claim that planets form from the gravitational collapse of gas within the disk itself. "Parts of the disk would just contract under their own gravity, says Jewitt. "The planets would form directly without needing a core.

The Columbian Exchange

When Christopher Columbus and his men became the first Europeans to arrive in the Americas in 1492, they set in motion a process of cultural, economic, and environmental exchange that had profound effects for the whole world. For the "old," Afro-Eurasian world, an important transformation included the addition of valuable new food items into the diets of people in such widely separated regions as Ireland, China, and sub-Saharan- Africa. For the "new" American world, the most significant transformation involved the deadly effects of Afro-Eurasian diseases among its peoples, which in turn helped pave the way for imperial conquest by Europeans. After two centuries of what is now known as the Columbian Exchange, both Old and New Worlds were transformed.

About 12,000 years ago, the land bridge linking the Americas with Afro-Eurasia across the Bering Strait disappeared as a result of rising sea levels. As a result, the peoples of the Americas were almost completely separated from Old World peoples until the Spaniards arrived in 1492. In other words, for twelve millennia the people as well as the plants and animals of the Americas developed in isolation from Afro-Eurasia according to the specific environmental conditions of the Americas.

Some human developments were similar: for example, agriculture and writing developed in the Americas just as they had done in Afro-Eurasia, as did the birth of hierarchical societies and patriarchies (societies headed by the eldest male. In other ways, however, developments in the Americas followed a different course from those of the Old World. For example, plant species such as corn, potatoes, and tobacco were unique to the Americas, and thus American peoples had a different set of choices regarding possible domesticated crops. Just as importantly, not long after humans arrived in the Americas, all of the large mammals that existed there became extinct. As a result, when humans in the Americas began to turn to settled agriculture, they did not have the choice to domesticate large mammals such as cattle or horses. Because of this, American societies did not develop innovations like the wheel, which proved unnecessary without large animals to pull carts or push plows. Furthermore, other animals familiar across much of Afro-Eurasia-including sheep, pigs, chickens, and goats-were nonexistent in the Americas. In fact, the only animals that could be domesticated available to American peoples were the dog, the turkey (in North America), the llama/alpaca and guinea pig (in the Andes region of South America), and the Muscovy duck (in the South American tropics).

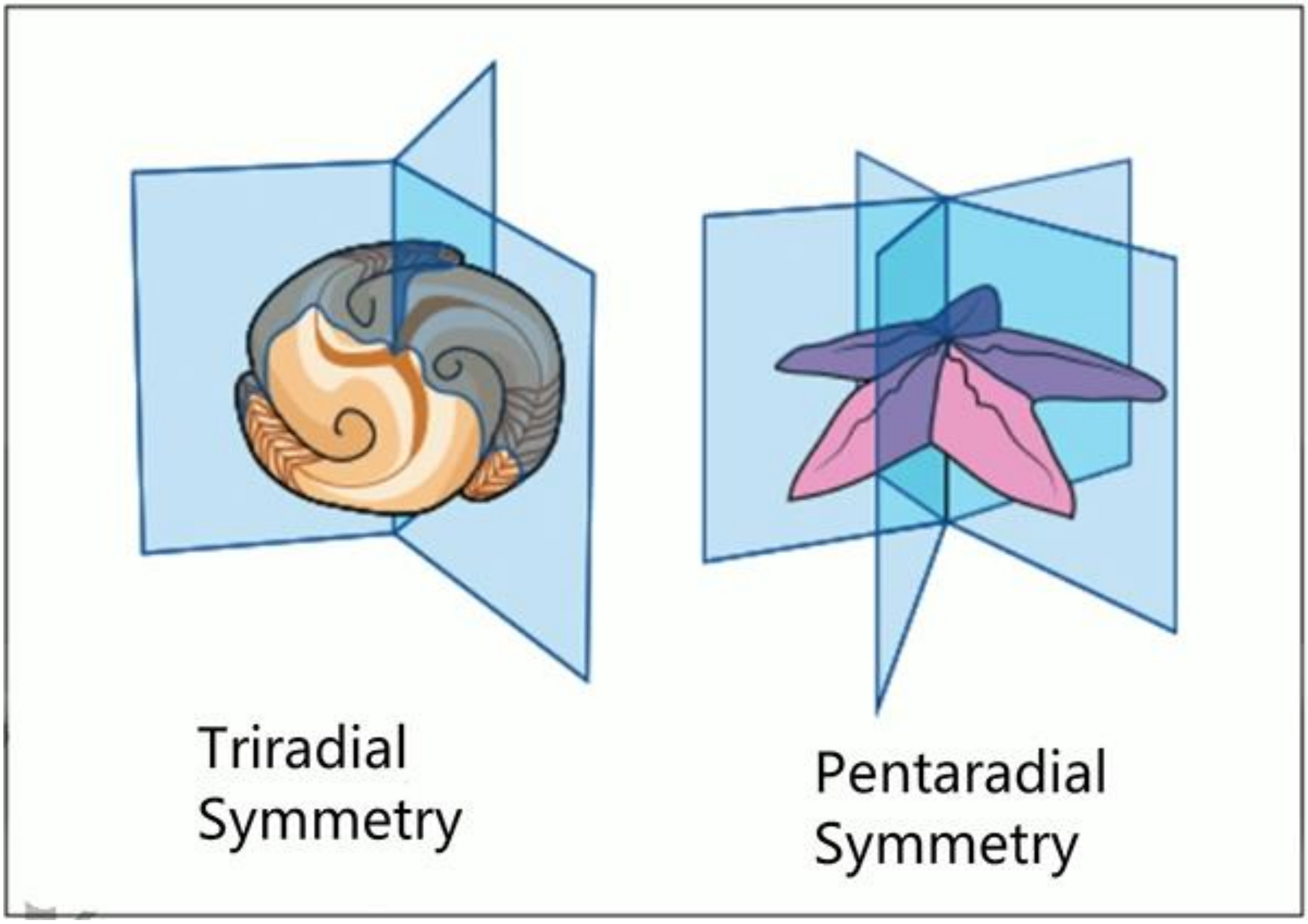
The absence of animals that could be domesticated is critical because in the Old World the domestication of animals such as sheep, chickens, goats, and cattle contributed greatly to the development of serious diseases such as influenza, smallpox, measles, and pertussis. In fact, many of these serious diseases resulted from microbes that were first present in herds of domesticated animals and then made the jump from their animal hosts to human hosts. The jump was made much easier because early Afro-Eurasian communities that relied on domesticated animals for food and labor tended to live in close proximity to their herds- even keeping them inside their houses for protection or warmth. Over time, a number of these microbial jumps occurred between domesticated animals and humans. In addition, because of the land connections across much of Afro-Eurasian, these diseases were able to spread to many Old-World populations While these diseases continued to be serious for the peoples of Afro-Eurasia, the human populations that survived them over the generations tended to pass on an acquired partial immunity to their offspring.

Without many animals that could be domesticated and living in enforced separation from Old World peoples, the people of the Americas remained freer from the type of contagious diseases that so often plagued the Old World. Some contagious diseases did exist, to be sure, and included dysentery, pneumonia, and possibly syphilis. But the “crowd” diseases of the Old World that flourished in crowds of large, centralized populations of humans did not exist, which meant that American peoples—when exposed to them—had no immunity whatsoever. Therefore, when Europeans carrying these diseases inadvertently spread them among American peoples, sickness spread rapidly, resulting in many deaths.

Echinoderm Evolution

The echinoderms are a phylum of invertebrate animals that includes starfish, sea urchins, and sand dollars. One feature that distinguishes echinoderms from other groups of organisms is their system of locomotion-the bottoms of echinoderms are covered with a number of tiny tube feet, which aid the echinoderm in feeding and slow movement on the ocean floor Like other invertebrates, echinoderms lack a backbone and have a fairly primitive nervous system. Unlike many invertebrates, the soft bodies of which are often poorly preserved as fossils, echinoderms have hard internal skeletons made of calcite. For this reason, echinoderms are well represented in the fossil record. In spite of their generally good preservation as fossils, there are several puzzles surrounding the evolution of echinoderms.

One major mystery surrounding echinoderms is how they originated. The first clearly recognizable echinoderms appeared around 540 million years ago during the Cambrian period; however, these first echinoderms were already fairly complex. Paleontologists have been unable to conclusively identify a simpler organism that could have been the ancestor of the echinoderms. One fossil that has been tentatively suggested as an ancestral echinoderm is Tribrachidium, which dates to the Ediacaran period just before the Cambrian. Tribrachidium is round, flat, disc-shaped fossil that was discovered in 1959. Some paleontologists have suggested that Tribrachidium could have been a primitive echinoderm, as it shares the radial body plan of modern echinoderms. A radial body plan is one in which an organism's body is composed of parts that are arranged symmetrically around the center like the petals of a flower. Since most known echinoderms have radial body plans, the radial symmetry of Tribrachidium has been viewed as suggestive of an evolutionary relationship. However, Tribrachidium has now been discounted as an ancestral echinoderm since it has triradial (three-point) symmetry instead of the pentaradial (five-point) symmetry found in adult echinoderms. In 1985 Tribrachidium was placed in its own phylum, Trilobozoa, a group with no known living descendants.



Another puzzle related to echinoderm evolution is how and why these organisms acquired their distinctive radial body plan. Among animals, all but the most primitive have bilateral symmetry-the left sides of their bodies are roughly mirror images of the right sides. Two main groups of animals have radial symmetry instead. One of these groups is jellyfish, which are extremely primitive and are assumed never to have evolved bilateral symmetry The other radially symmetrical group is the echinoderms, which are fairly sophisticated. In fact, biologists know that the echinoderms did once have bilateral symmetry because the free-swimming larvae(young)of echinoderms are bilaterally symmetrical. Once the larvae settle down on the ocean floor, they develop the radial symmetry of adult echinoderms. It is not clear why echinoderms do not retain bilateral symmetry as adults.

Paleontologists are always attempting to determine evolutionary relationships among organisms. One widely accepted hypothesis is that echinoderms are closely related to vertebrates-the group of animals that includes fish, reptiles, and mammals. Although adult echinoderms do not resemble vertebrates biologists have based this conclusion on DNA comparisons and embryological evidence. In most non-echinoderm invertebrates, the mouth forms from the firs opening in the embryo (an early stage of development), whereas in both echinoderm and vertebrate embryos the mouth develops from a secondary opening.

A group of animals called carpoids, which are only known through the fossil record, have been considered to be intermediate between echinoderms and chordates, a group that includes all vertebrates. Carpoids share several characteristics with vertebrates, including gill slits and complex nervous system. In fact, some experts have placed them near or at the beginning of chordate evolution. However, carpoids possess a skeleton of calcite plates identical to that of echinoderms. calcite skeleton is not present in any chordates. Furthermore, no ancestral genes for calcite skeleton development have been identified in any present-day chordates, which indicates that chordates and echinoderms most likely followed their own evolutionary paths and that carpoids are echinoderms. In 2012 scientists reported the discovery in southern Europe of fossils from the early middle Cambrian period; these fossils show adult echinoderms with a fully bilateral body plan. The animals represented by these fossils may be the earliest echinoderms from which all other echinoderms evolved.

Ancient Mapmaking

Reconstruction of Ptolemy's world map

Claudius Ptolemy, who lived from approximately 85 to 168 AD, was an ancient mapmaker whose works were rediscovered in Europe after being lost until the fifteenth century. He lived in Alexandria, Egypt, where he used Alexandria's famous library to compile existing knowledge of astronomy. geography, and astrology into three treatises. The astronomy and geography treatises had a long-lasting influence. but they both presented serious errors that went uncorrected for about 1,300 years. Ptolemy's astronomy treatise, Almagest, rejected the theory earlier proposed by Aristarchus (approximately 230 BC) that Earth revolves around the Sun. Ptolemy's geocentric idea-that Earth was the center of the universe -accepted the ideas of Aristotle and formed the main thesis of his treatise. When Ptolemy's works resurfaced in the fifteenth century. They were accepted as gems of ancient wisdom, and few had the nerve or the authority to challenge them. Likewise, any sixteenth century maps that altered the Ptolemy map were regarded with suspicion.

In his other influential treatise, Geographia, Ptolemy rejected the nearly correct computation of the distance around Earth-Earth's circumference-made by Eratosthenes in approximately 240

BC Rather, he chose an erroneous and much smaller distance (about 75 percent of the actual size). Ptolemy did not make any measurements himself, as Eratosthenes had done, but selectively compiled other information that was known at the time. The estimate he chose came from the Greek astronomer Poseidonius. Subsequently, however, his choices became known as Ptolemaic ideas and were considered I irrefutable. Also, Ptolemy assumed that the known world's land surface covered 180 degrees of longitude ranging from the Canary Islands in the west to the easternmost part of Asia (about 20 degrees of longitude too much). This error on his map showed the Atlantic Ocean much too narrow and connecting western Europe and east Asia, without the American continents in between. This remained the understanding of the world for 1.300 years.

The combination of these two errors-Earth's circumference too small and land area too large-encouraged mariners of the fifteenth century to assume that a relatively short voyage across the Atlantic Ocean would take them to Asia. Columbus was the first to promote an expedition on the basis of these errors. When Columbus reached land, he had traveled as far as he expected to travel to reach Asia and logically assumed that he had succeeded.

To his credit, Ptolemy's map introduced some excellent standard to mapmaking. despite the errors. Though he was not the first to use the idea of a gridded coordinate system, his method of showing attitude and longitude became a standard for future maps. Also, Ptolemy insisted that maps should be drawn to scale. Many maps of his time were distorted by enlarging the better-known places in order to include all the known information. Unfortunately, many mapmakers of his time failed to adopt his practical approach to scale and location.

Maps began to proliferate in the sixteenth century. Each voyage of exploration and discovery provided new information that had to be mapped. In 1507, a German mapmaker, Martin Waldseemoler, produced a map of the world, Universalis Cosmographia, which was the first to show Columbus's discovery as a separate continent. But he cautiously made the new continent very narrow-just a long. skinny island-rather than contradict Ptolemy's erroneous circumference of Earth By the middle of the sixteenth century, enough voyages had been made, including Magelan's trip around the world, that most mapmakers recognized that Earth's circumference as shown on Ptolemy's map was wrong.

In 1569, the Flemish mapmaker, Gerardus Mercator, made a map of the world showing all the known lands using his now famous innovative grid system of latitude and longitude. In 1570, Abraham Ortelus, a Belgian mapmaker, made the first known atlas of the world in an effort to compile the rapidly accumulating geographic knowledge. Although these maps still had some remaining traits of the Ptolemy map, they showed great improvements in detail and accuracy.

A major feature retained from the Ptolemy map was the presence of a very large continent in the Antarctic-large enough to counterbalance the weight of the Northern Hemisphere land. This belief was based on the Greek concept of symmetry, as well as the idea that Earth needed to be balanced to turn smoothly.

The Difference Threshold and Signal-detection Theory

How much weight must be added to or subtracted from a stack of books for the carrier to sense that the load is heavier or lighter? The just-noticeable difference (JND) is the smallest change in sensation that a person is able to detect 50 percent of the time. The difference threshold is a measure of the smallest increase or decrease in a physical stimulus that is required to produce the JND. A person holding a 5-pound load would notice the addition of a single pound, but a person already holding 100 pounds would not be able to sense an additional pound. More than 150 years ago, researcher Ernst Weber (1795-1878) observed that the JND for all the senses depends on a proportion or percent of change rather than a fixed amount of change. This observation became known as Weber's law. A weight being held must increase or decrease by 2 percent for the difference to be noticed. According to Weber's law, the greater the original stimulus, the more it must be increased or decreased for the difference to be noticeable.

The difference threshold is not the same for all the senses. very large (20 percent) difference is necessary for some changes in taste to be detected, for instance. In contrast, a difference is noticeable if a musical tone becomes slightly higher or lower in pitch by only about 0.33 percent. Nor are the difference thresholds for the various senses the same for all people. In fact, there are great individual differences, often arising from experience or expertise. Professional food tasters know if a particular item or batch is a little too sweet, even if its sweetness varies by only a fraction of the 20 percent usually necessary to detect changes in taste. Furthermore, people who have lost one sensory ability often gain greater sensitivity in the other sensory abilities. Actually, Weber's law best fits people with average sensitivities and sensory stimuli that are neither very strong nor very weak, for example, not as loud as thunder or as quiet as a faint whisper.

The classic methods in psychophysics for measuring difference thresholds focus exclusively on the physical stimulus-how strong or weak it is or how much the stimulus must change for the difference to be noticed. But even within the same individual, sensory capabilities are sharper and duller from time to time and under different conditions. Factors that affect a person's ability to detect a sensory signal are, in addition to the strength of the stimulus, the motivation to detect it, previous experience, expectation that it will occur, and alertness or level of fatigue.

In another approach, these factors are taken into account. According to signal-detection theory the detection of a sensory stimulus involves discriminating that stimulus from background noise (all the other stimuli present in the environment) and deciding whether the stimulus is actually present. But deciding that a stimulus is present in the first place depends partly on the probability that the stimulus will occur and partly on the potential gain or loss associated with deciding that it is present or absent Suppose you were given the description of a cousin you had never seen before and were asked to pick her up at the airport. Your task would be to scan a sea of faces for someone fitting the description and then to decide which of the several people who fit the description was actually your cousin. All the other faces and objects in your field of vision would be considered background noise. How sure you would have to be before you approached someone would depend on several factors-among them the embarrassment you might feel approaching the wrong person as opposed to the distress you would feel if you failed to find your cousin.

Signal-detection theory has special relevance to people in many occupations: air-traffic controllers, police officers, military personnel on guard duty, medical professionals, and poultry inspectors, to name a few. Whether these professionals detect certain stimuli can have important consequences for the health and welfare of vast numbers of people.

Bird Songs and Calls

Birds use song both in courtship and to define areas of territory. Both of these are communicative purposes: the bird is passing specific messages to other members of its species. Birds communicate for other reasons as well: a blackbird, for instance, will make a sharp "pink-pink" sound when there is a cat nearby, which warns other birds in the neighborhood of the danger.

William Thorpe (1961) studied the behavior of gannets in a colony containing many thousands of birds. Thorpe found that when a bird was returning to its nest, it would drift on an updraft of air from the bottom of the cliff upwards, calling as it went. When the bird on the nest heard its mate calling, it would call in reply, showing that each bird's call could act as an identification signal. Thorpe also calculated that a bird might have as many as fifteen or sixteen different kinds of calls, each serving a different function.

J. R. Krebs (1976) investigated how birds seem to sing more intensively in the early morning--the “dawn chorus”. By investigating what the birds actually did during each day, and how much time they spent on each activity, Krebs found that the dawn chorus serves a largely territorial function. The early morning is not a particularly good time for gathering food, because it is dark, so visibility is lower, and it is also cold, so many insects are still inactive. On the other hand, at this time many birds move around looking for living space, so establishing and defending a territory is necessary. Birdsong is not just territorial, of course. A bird's song can serve a dual purpose: it can be used to defend a territory and by indicating to a prospective mate that the singer has a territory to defend, can also attract a female bird.

P. J. B. Slater (1981) suggested that bird calls and birdsong are partly learned from other birds. He found that chaffinches which had been hand-reared and had not heard other wild birds made an entirely different kind of “chink" call from that of wild birds. In one case, Slater observed a laboratory chaffinch in a duet with wild sparrow outside the window of the laboratory. The chaffinch imitated the sparrow's "cheep" whenever the sparrow produced it. Slater concluded that learning through copying is an important part of the way in which birds acquire their songs. Slater also found that individual chaffinches can have up to five different types of song. Some of these are personal, sung by that bird alone. Others are shared by several birds. In some cases, too, Slater observed chaffinches singing songs which were almost identical to those sung by others, but with just a note or two different-possibly because the bird had made an error in copying the song from another.

Slater studied a population of 40 chaffinches on the Orkney Islands and found that among them they had seventeen different song types. So, it was not a matter each bird having its own individual songs--there was a considerable amount of sharing. Slater found that this sharing related to geographical distribution, but that the boundaries were not distinct enough for it to be accurately described as a dialect, or regional variety of a song. Instead, there was considerable overlap between the songs sung in one area and those sung in an adjoining one, but gradually the overlap would become less, until birds a long distance away from one another would be singing entirely different songs.

In 1970, Peter Marler proposed that birdsong and speech were directly comparable in certain key respects and that the study of birdsong might provide psychologists with some useful indicators as to the nature and development of speech in human beings. One of the parallels which Marler identified was the way that both humans and birds show a strong genetic predisposition to pick up and imitate certain sounds rather than others. Marler showed that young birds will learn the songs of their own species if they are played to them when young, but they will ignore songs of birds from other species. Similarly, young human beings are surrounded by all kinds of sounds and noises, but it is the human voice to which they listen most closely and human speech which they imitate.

Mass Production under China's First Emperor

Attempts to optimize the function, value, and appearance of mass-produced goods-goals associated with modern industrial design-date to as early as China's first emperor, Qin Shi Huang, who harnessed the power of design and mass production in the name of conquest. He is remembered for unifying by force China's warring states in 221 B.C. into the most powerful country the world had ever seen. He was also responsible for building the Great Wall of China to protect his new nation's borders from raiding Mongolian invaders. He is even better known, certainly in archaeological circles, for the Terracotta Army that was made to honor him and to protect him in the afterlife. This collection of ceramic funeral sculptures depicting the emperor's army is estimated to have included 8,000 life-sized statues of soldiers and 670 horses, as well as depictions of other members of his vast following, from officials and servants to musicians, singers, and acrobats. Although this army should probably be considered collections of statues rather than expressions of manufacturing design, the figures were essentially factory-pro-, using systemized methods of production similar to those used to make serially manufactured functional objects. The figures were also buried with weapons, and around 40,000 bronze spears, swords, crossbows, halberds, and staffs have been recovered by archaeologists so far.

Terracotta Army

As with the manufacture of the terracotta warriors themselves, the production of their accompanying weapons reveals that Emperor Qin's workforce was organized into highly efficient production teams that used standardization measures and quality-control procedures. The technical knowledge of these Chinese weapon-makers, however, was not confined to equipping the Terracotta Army with weapons-far from it. Recent research has revealed that some of the burial material was actually used in battle and bears the scars to prove it. Emperor Qin's craft workers perfected the art of bronze-making to such an extent that they were able to produce swords that are now considered some of the finest bronze weapons ever made. Because their raw material was of such a high grade, the emperor's craft workers could manufacture swords that were significantly longer than any that had gone before, which gave Qin's soldiers the enormous benefit of 30 percent greater reach and cutting power. Thus, the development of better-quality materials enabled the production of better-performing designs, which gave the first Chinese emperor's forces a decisive advantage; this is a recurring theme throughout the story of design-as more advanced materials are invented or discovered, their unique benefits are employed by designers to improve existing designs, thereby creating superior products.

Another reason that Qin was able to achieve his epic conquest of the warring states was that his workers produced weaponry to such precise specifications that their parts were completely interchangeable. If, for example, a section of a crossbow broke in battle, it could be replaced easily with an exactly replicated piece. Similarly, the arrows used by his soldiers had interchangeable shafts so that their arrowheads could be reused even if the shaft had broken. A high degree of manufacturing efficiency and standardization was bolstered by a culture of manufacturing accountability, with workers' output overseen by supervisors to ensure that no defective workmanship was ever allowed to creep into the production system-and if it did, the consequences were severe. Precision manufacturing and an exacting quality-control system were used not only for weaponry but also for weights and measures during Qin's reign, and this system of standardized production formed the basis upon which Chinese design and manufacturing would flourish over the succeeding centuries.

Over the next thousand years, Chinese craft workers used methods similar to these to design and manufacture ceramics and bronze wares that were technically far superior to their European equivalents. The delicate blue-and-white porcelain crockery imported from China into Europe in increasing quantities during the 1700s must have seemed the finest example of refinement and modernity when compared to the heavy and rather primitive earthenware pottery produced in Europe. A capacity for the precise replication of designs is still a defining characteristic of Chinese production, and it can surely be traced back to the first emperor's innovative implementation of rigorous design standards, which became incorporated into the nation's manufacturing culture.

Changes in the Art Market During the Late Nineteenth Century

Today's market for European art has its roots in the nineteenth century. In 1882 the passage of the Settled Land Act reformed inheritance law in Britain. As a consequence of industrialization and the availability in Europe of large quantities of cheap American wheat, the fortunes of British landowners were in decline. After bankruptcies began to mount, the inheritance laws were changed, making it possible for landowners to support their landholdings by selling their art treasures. Until 1882, estates were strictly settled, meaning that the head of the family did not actually own the land and estate goods, but rather was entrusted to preserve and protect the family property. This was meant to hinder large estates from being sold off in sections. The Settled Land Act made it possible to sell individual portions of an estate, such as works of art, in order to save the real estate (land and buildings). The art treasures of the British nobility began to flow into London auction houses, shoring up their dominant position in the art market.

Besides the passage of this law, the early 1880s marked the public acknowledgement of French Impressionism, a revolutionary art movement that emphasized subjects from everyday life and the accurate depiction of light and its changing qualities. Prior to this, Impressionist artists had been regarded as radical, immoral individuals who had declared war on beauty. Impressionist paintings were not accepted by the Salon de L'Academie des Beaux-Arts in Paris, which had been since 1830 the most important exhibition forum in the world and was historically the guardian of traditional standards of French painting. In reaction to the conservative tendencies of this institution, a group of Impressionist artists in Paris organized their own exhibition in 1874. Reviews of the exhibition were mixed. But in 1881, the Impressionist painter Edouard Manet was named a knight of the Legion of Honor; in 1882, Paul Cezanne was allowed to exhibit his work for the first time at the Salon.

With the advent of Impressionism, the modern art market was born. Since most collectors rejected the Impressionists, especially early on, the usual paths for distribution were closed to these artists, and they had to find other ways to sell their pictures. The rejection, however, meant that the artists were not dependent on the tastes of potential patrons (wealthy supporters), and hence they began to see themselves as autonomous masters. This new approach also awakened a newfound respect for the art dealer, since the dealer had access to artistic genius and could open up the path to new art and explain it to potential buyers. In the early twentieth century, the Cubist style (abstract art based on geometric forms) arose and representational art- art that represented objects and people realistically- became less significant. This development also strengthened the notion of the artist genius, as well as that of the art dealer as a go-between. On the one hand, the content of art became more complex, and on the other, it also became more difficult to understand and communicate.

Of course, artworks by the so-called old masters continued to be sold. The most successful dealer of such art was probably Joseph Duveen, who arrived in New York City at the age of seventeen in 1886. After a brief apprenticeship under his uncle, the art dealer Henry Duveen, Joseph opened a gallery (a privately owned establishment for displaying and selling artwork) in proximity to the Waldorf-Astoria Hotel, which was already the meeting place for the city's elite. Duveen's business strategy entailed the purchase of complete art collections from their owners. In this way, a buyer would be unable to calculate Duveen's wholesale price for an individual artwork and hence the percentage that he kept for himself. If he made purchases at auctions where the price, he paid would be public knowledge, he did not shrink from paying record prices, since this practice brought him a reputation and also raised the value of the pieces already in his inventory. He also stirred up enthusiasm for collecting among many American industrialists; these millionaires regarded the acquisition of European masterpieces as an opportunity to assure the immortality of their names.

Ocean and Atmosphere on Early Earth

Where did Earth's ocean water come from? Scientists agree that a large amount of water must have arrived during planet, accretion, which is the process of collision and sticking together of the material in orbit around the Sun by which Earth and the other planets in the solar system formed 4.6 billion years ago. Perhaps significant volumes were added during the period of heavy bombardment, during which Earth was being hit by a large number of comets (whose nuclei are composed mostly of ice) and other objects left over from the formation of the planets. The volume of water eventually found on Earth may be related to the formation of Earth's core (innermost layer). When the iron- and nickel-rich core formed, most of the water in the forming planet was consumed in oxidation processes whereby the oxygen component of water was used to make iron and nickel oxides. It is the residual water that make up the oceans. Perhaps that residual quantity was significantly enhanced by water carried by comets after Earth's initial formation, perhaps not .1n either case, the oceans reached approximately their present volume by 3.8 billion years ago. But this does not mean they were at their present area. Geologist Don Lowe has estimated that before 3 billion years ago, less than 5 percent of Earth's surface was land. Earth's atmosphere was also very different from that of today. There was no oxygen, and there was a great d.1 more carbon dioxide (Co_2)—perhaps 100 to 1,000 times as much as today.

Earth's surface temperature was higher than it is now because more heat was emanating from the interior and because of the warming generated by the extensive CO_2 and other greenhouse gases in the atmosphere. Greenhouse gases trap heat near the surface through the so-called greenhouse effect. Earth's internal generation of heat was an important factor; the Sun at this time was much fainter, a delivering perhaps a third less energy, than at the present time.

What would have happened if Earth had stayed a water world? Probably global temperatures would have remained high or even increased. For animal fife to form, the temperature had to drop from the levels acknowledged to have been characteristic of Archaean time (about 4 to 2.5 billion years ago). A drop in global temperature while the Sun was getting hotter required a drastic reduction of atmospheric CO2--a reduction of the greenhouse effect. Thus, there had to be some means of removing CO2.The most effective way to do this is through the formation of limestone, which uses CO2 as one of its building blocks and thus removes it from the atmosphere. But significant volumes of limestone form today only in shallow water; the most effective limestone formation occurs in depth of less than 20 feet (6 meters), In deeper water, high concentrations of dissolved CO2 slow or inhibit the chemical reactions that lead to limestone formation. There is evidence of deep-water, inorganic limestone formation in very old rocks on Earth, as demonstrated by geologist Jo. Grominger and his team. These studies thow. that early Earth's ocean may have been saturated in the compounds that can produce limestone and thus could have formed solid limestone in deeper water at that time, removing carbon dioxide from the atmosphere as a consequence. However, Grominger points out that occurrences of carbonate rocks such as limestone during the early Archaean--roughly the first billion years of Earth's existence—are rare. And this is only partly due to the rarity of rocks of this age. It looks as though the central mode of removing carbon dioxide from the atmosphere-the formation of carbonate rocks—seldom occurred.

To form limestone in significant volumes, then, shallow water is needed, but on a planet without continents, shallow water is in short supply. On Earth from perhaps 2.7 to 2.5 billion years ago there occurred a rapid buildup of continental areas, resulting in an increase in the land surface from perhaps 5 percent to about 30percent.Larger continents meant larger shallow-water regions, for the emergence of continents created the shallow areas near the continents as well as large inland seas and lakes.

Zebra Stripes

Zebras, horse like animals native to the grasslands of Africa, are known for their distinctive black and white stripes. Historically many scientists thought zebras' stripes served to camouflage (hide) them from predators, such as lions and hyenas. This assumption was based on the observation that other animals, such as tigers, have similar stripes that make them less visible. However, in recent years scientists have noted that zebras' environment and behavior are not well suited to camouflage by stripes. Tigers often inhabit heavily forested areas in which there, vertical stripes help them blend in with the surrounding trees, but zebras typically inhabit open grasslands. Unlike tigers, who use stealth to stalk their prey, zebras are herbivores that rarely hold still when threatened by predators; instead, they rely on their good eyesight to spot predators at a distance and flee at any signal of danger.

A zebra's stripes may help to protect it in other ways. Stripes may help zebras blend in with each other, rather than blend in with their environment. Zebras live in large herds, and they flee as a group when threatened. The dense pattern of moving zebra stripes may appear as a mass of confusing images, making it difficult for predators to target individual zebras. Catching a fleeing zebra requires a precisely timed final leap, and a zebra's stripes may interfere with predators' perception of distance. Because zebras have some ability to defend themselves by using their hind legs to kick at pursuing predators, at times powerfully enough to cause serious injury, a zebra's distinctive stripes may also serve as a Warning that encourages predators to seek less dangerous prey. However, none of these explanations is strongly supported by observation. Zebras are killed by lions about as frequently as other unstriped prey animals.

More recently, scientists have suggested that the primary purpose of stripes is to protect zebras from biting flies. The grasslands of Africa are home to a number of species of flies that feed on the blood of large mammals and can cause considerable damage through blood loss and the transmission of diseases. Laboratory tests have shown that biting flies are less likely to land on objects covered in black and white stripes than on solid black on white surfaces. It is not yet known why biting flies would avoid striped objects. but scientists have suggested that the explanation lies in the mechanisms of flies' vision, which is simple and apparently confused by stripes. The stripes may make it difficult for flies to detect the outline of a zebra's body or cause the flies to confuse a zebra for a collection of thin, vertical objects that do not resemble potential victims. Scientists have analyzed the stomach contents of wild biting flies and have found relatively little zebra blood.

If stripes are an effective means of protection against biting flies. then why have other animals not evolved stripes as well? One theory is that zebras are particularly vulnerable to flies because of their unusually short hair and developed stripes as an alternative protective measure. Horses, which are closely related to zebras, are indeed extremely susceptible to biting flies when imported to Africa. They are frequently infected by fly-borne parasitic infections which are often fatal even when veterinary treatment is provided. Horses do not develop immunity to the parasites and so can be repeatedly re-infected

Other puzzles remain. For instance, biting flies seem to be most discouraged by horizontal stripes, but the stripes on zebras are mostly vertical. More mysteriously, not all zebras are covered in stripes: the quagga, an extinct subspecies of zebra. had fainter stripes that were present only on the front half of its body. If stripes protect from flies, why would the quagga lack protection on half of its body? This cannot be explained by environmental factors. because quaggas' range overlapped with that of fully striped zebras. One possible explanation is that if they only have stripes on half of their body, quaggas can more easily distinguish between. their own and other species of zebra, making it easier for them to follow the solid-colored hindquarters when fleeing from predators. This behavioral hypothesis is unfortunately impossible to test, because scientists can only study the quagga through preserved museum specimens and a handful of nineteenth-century photographs.

The Behavior of Magma

Why do some volcanoes explode violently while others erupt gently? Why do some kinds of magma (molten rock) solidify within Earth to form hard rocks, and others rise all the way to the surface to erupt from volcanoes as lava? To answer these questions, we must consider the properties and behavior of magma. Once magma forms, it rises toward Earth's surface because it is less dense than surrounding rock. As it rises, two changes 0ccur. First, it cools as enters shallower and cooler levels of Earth. Second, pressure drops because the weight of overlying rock decreases. Cooling and decreasing pressure have opposite effects on magma: Cooling tends to solidify it, but decreasing pressure tends to keep it liquid.

Whether magma solidifies or remains liquid as it rises toward Earth's surface depends on the type of magma. Basaltic magma commonly rises to the surface to erupt from a volcano. In contrast, granitic magma usually solidifies within Earth's crust (outer layer). Granitic magma contains about 70 percent silica, whereas the silica content of basaltic magma is only about 50 percent. In addition, granitic magma generally contains up to 10 percent water, whereas basaltic magma contains only 1 to 2 percent water.

As in silicate minerals, silicate tetrahedral (four-faced solids, similar to triangular pyramids) in magma link together to form chains, sheets, and framework structures. They form long chains if silica is abundant in the magma but shorter chains if less silica is present. Because of its higher silica content, granitic magma contains longer chains than does basaltic magma. In granitic magma, the long chains become tangled, making the magma stiff or viscous. It rises slowly because of its viscosity and has ample time to solidify within the crust before reaching the surface. In contrast, basaltic magma, with its shorter silicate chains, is less viscous and flows easily. Because of its fluidity. It rises rapidly to erupt at Earth's surface.

A second difference is that granitic magma contains more water than basaltic magma. Water lowers the temperature at which magma solidifies. Thus, if dry granitic magma solidifies at 700C, the same. Magma with 10 percent water may remain liquid until its temperature drops below 600C. Water tends to escape as steam from hot magma, but deep in Earth's crust where granitic magma forms, high pressure prevents the water from escaping. As magma rises, pressure decreases and Water escapes. Because the magma loses water, its solidification temperature rises causing it to become hard rock. Thus, water loss causes rising granitic magma to Solidify within the crust. For this reason, most granitic magmas solidify at depths of 5 to 20 kilometers beneath Earth's surface. Because basaltic magmas have only 1 to 2 percent water to begin with, water loss is relatively unimportant. As a result of its comparatively low viscosity, rapidly rising basaltic magma remains liqudl1 the way to Earth's surface, and basal volcanoes are common.

In most cases, granitic magma solidifies within Earth's crust to form a rocky mass called a pluton. Many granite plutons are large, measuring tens of kilometers in diameter. To form a large pluton, a huge volume of granitic magma must rise through continental crust. But how can such a large mass of magma rise through solid rock? If you place oil and water in a jar, screw the lid on, and shake the jar, oil droplets disperse throughout the water. When you set the jar down, the droplets coalesce to form larger bubbles, which rise toward the surface, easily displacing the water as they ascend. Granitic magma rises in similar way. It forms near the base of continental crust, where surrounding rock is soft and flexible and behaves plastically, it flows and deforms without fracturing, because it is hot. As the magma rises, it shoulders aside the hot, plastic rock, which then slowly flows back to fill in behind the rising bubble. After a pluton forms, forces in the crust may push it upward, and erosion may expose parts of it at Earth's surface.

Transport of Food to Rome

During the reign of Emperor Augustus (63 B.C-14 A.D.)， the city of ancient Rome may have had as many as 1,200,000 inhabitants. With so many residents, the city needed an extremely large food supply, necessitating the development of an improved system for shipping food. Transportation of bulk goods by land was prohibitively expensive and slow. It simply was not practical to haul a wagonload of wheat very far, for example, since the animals needed to pull the wagon would quickly eat an amount of food equal to the relatively limited amount that could be carried in the wagon itself. The solution to this problem was to move goods by sea. Scholars have estimated that the cost of shipping grain from one end of the Mediterranean to the other was cheaper than hauling the same amount of grain 120ilometers overland. It Was perhaps 40 times more expensive to transport goods by land rather than by sea, and while it was somewhat more costly to move goods along a river than over the open ocean, river transport was still many times more efficient than land transport.

Since Italian resources were not sufficient to feed Rome. Romans naturally looked to the Roman-controlled areas that were closest to Rome and that had access to the sea. The first provinces that had surplus grain collected from them and transported to the city of Rome in large quantities were, logically enough, the nearby islands of Sicily and Sardinia. Once Rome had conquered the coast of North Africa, the foods from this region were rounded up and routed toward the city of Rome as well. By the first century A.D., Egypt and the coastal areas of Spain and Gaul had been added to this list. Despite its distance from Rome, Egypt in particular was an important source of grain. The safe arrival of the Egyptian grain fleet off the Coast of Italy was a major cause for celebration.

While necessary as the only practical way to transport enough food to Rome, this maritime traffic could also be problematic. Particularly during the winter, the Mediterrar Sea can produce violent storms that would have sunk ancient ships. Thus, the prime sailing season was restricted to only three or four months during the summer since most of the supplies for the city had to reach Rome's ports during this narrow window of opportunity or else face greatly multiplied chances of being caught in a storm and sinking. Natural dangers were not the only threat to shipping, however. Piracy in the ancient Mediterranean was rampant, and despite sporadic attempts at suppression, pirates had nearly free rein to prey upon merchant ships. The Romans were aware of challenges that had to be overcome to keep the city fed. The Roman historian Tacitus commented, "Italy relies upon external supplies, and the life of the Roman people is daily at the uncertain mercies of sea and storm" (Annals 3.54) 。The scale of the food and other supplies pouring into Rome demanded an appropriate infrastructure, both administrative and physical.

Rome is not located directly on the Mediterranean Sea but is about22 kilometers inland on the Tiber River. The mouth of the Tiber lacked a natural harbor where ships could be safely unloaded. The smallest ships could have traveled upriver directly to Rome, but medium or large freighters had to be off-loaded somewhere else. During the republic (509 to about29b.c.) many ships (including the huge Egyptian grain freighters) docked at the good harbor at Puteoli on the Bay of Naples. From here, their cargoes had to be either shifted to smaller watercraft for their trip to Rome or else hauled overland. At the mouth of the Tiber was located the port city of Ostia. The harbor facilities at Ostia remained rudimentary through the republic, and larger ships that docked there simply had to anchor offshore and have their cargoes transferred onto barges or small craft for the trip upriver. As Rome continued to grow and traffic increased, these harbor arrangements were clearly unacceptable, and in A.D.42， the emperor Claudius tackled this problem and began to construct substantial harbor works at Ostia. Just north of the city, he excavated out of the coastline an artificial harbor known as Portus, although it was not a wholly successful project. Rome at last got first-rate harbor when the emperor Trajan rebuilt Portus and added an inner harbor where ships could be completely safe.

Water and Life on Mars

The question of life on Mars depends heavily on the characteristics of its air and water. Mars has a relatively thin and dry atmosphere, with high percentage of carbon dioxide compared to Earth's atmosphere. Mars tilts about25°, which leads to seasons like those on Earth. Carbon dioxide freezes into the polar caps during the winter and evaporates into the air in the summer This causes a fluctuation in the density of the atmosphere of up to 25 percent over the year with so little atmosphere, the planet experiences a wide range of temperatures, heating up in the day and rapidly cooling at night. Temperatures on Mars have been measured between-225F and95°

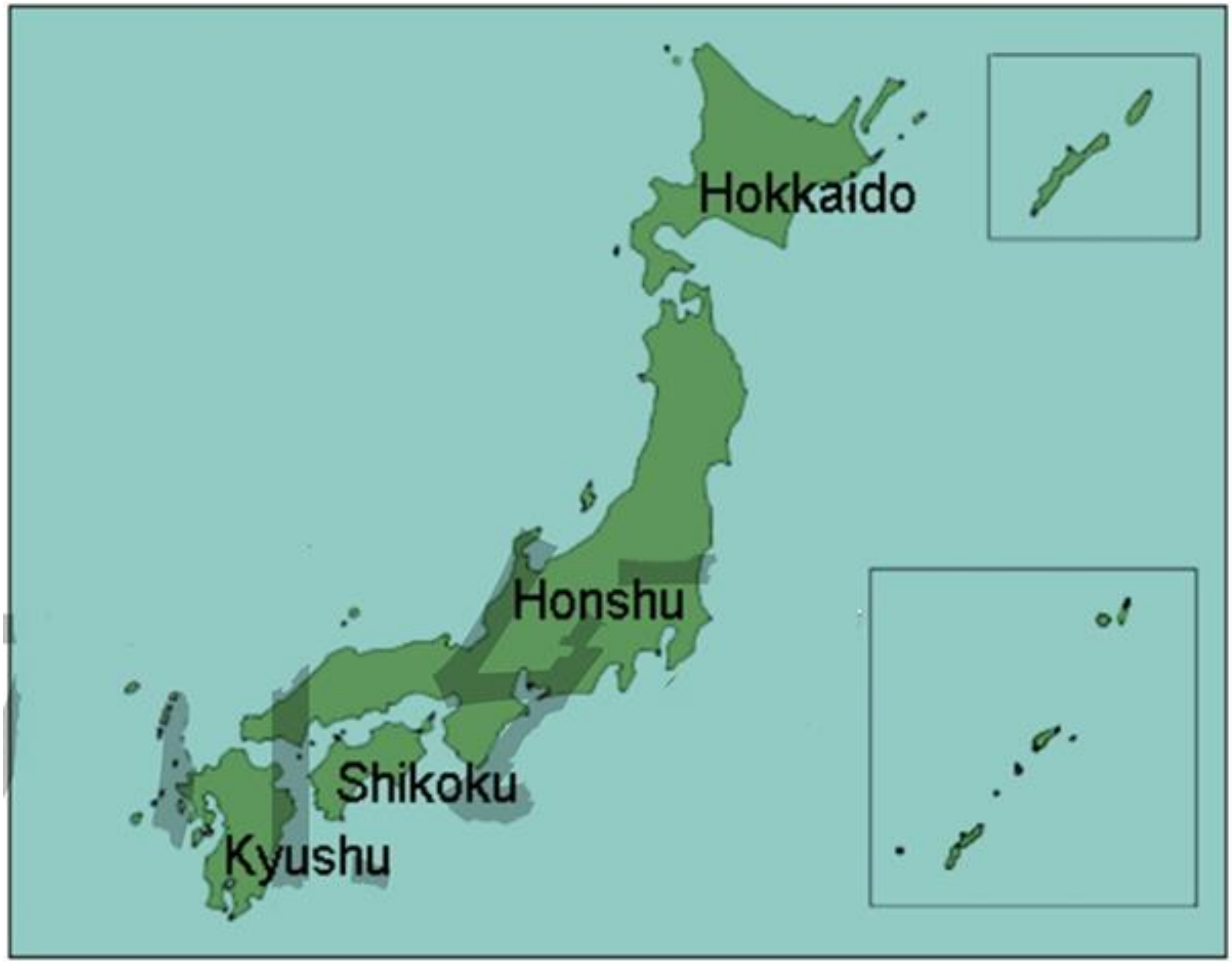
Scientists remain fascinated by the possibility of liquid water on Mars. Life as we know it requires water so it should be good marker for habitability. Current conditions on Mars suggest that liquid water could not occur on the surface. The low pressure and low temperature mean that ice would sublime-go straight from solid to gas-rather than melt. Water may be hiding in subsurface reservoirs, however, and stream forth for limited periods. In 2002 the Mars Odyssey orbiter detected large quantities of water ice near the surface of the southern hemisphere Channels on the surface suggest the presence of moving water in large quantities. In 2006 Mars Global Surveyor provided pictures of gullies (channels) on the surface that could only have been produced by liquid water flowing within the past seven years. Theories exist for how climate and erosion may reveal underground ice and cause sudden floods, but specific details are still a mystery. The water probably represents flash floods that quickly evaporate and may not provide an adequate medium for the development of life. If life arose in the distant past, though, the selfish floods may have been enough to maintain organisms adapted to such environments.

Astrobiologists continue to investigate whether the current situation-frozen water with occasional flash floods-was always the situation on Mars. Some surface channels on Mars could not have been produced by flash floods or geologic activity. The Opportunity rover, which landed on Mars in 2004. made two interesting discoveries in several craters on Mars. Rock layers on the sides of the craters look like the patterns inside sedimentary rocks on Earth. Sedimentary rocks form gradually at the bottom of bodies of water as particles settle down to the bottom. Sedimentary rocks on Mars would imply that a bod y of water was present for hundreds of years at some point in the past. In addition, Opportunity has found countless tiny spheres of the mineral hematite (a mineral composed of iron and oxygen) on the surface. These tiny spheres, or "blueberries" as they have come to be called, strongly resemble accretions (accumulations) found on Earth in highly acidic stream beds. Together with rippling patterns in the stone of the surface, the layers and blueberries make a compelling argument. Acidic, salty, flowing water must once have covered large regions of the plain where Opportunity landed. The Sun has been very slowly cooling off the past few billions of years. A warmer Sun in the past means a warmer Mars. Planetary scientists do not think the difference was enough to bring Mars up to Earth like temperatures, but it may have been enough to support liquid water in the warmer regions of the planet.

Overall, Mars looks like a good candidate for past life. Water, carbon, and energy were all present in useful amounts. If life arises quickly wherever resources allow, then Mars probably had life once. Unfortunately, we have no way to assess how common the origin of life might be. This is one of the questions scientists were asking when they explored Mars in the first place.

Astrobiologists Another import ant question to ask in the exploration of Mars has to disagree about evidence for life in the first billion years of Earth's history. Mars' history must be even more contentious. The most unambiguous evidence of life on Earth comes in the form of structural fossils, which probably did not start forming on Earth until life had been around for at least a billion years. If life did arise on Mars, it may not have been around long enough to leave that kind of evidence.

Jomon Pottery



The oldest known pottery in the world comes from Japan, and is known as Jomon, which means "cord marks, after its typical decorations made by impressing cords into the wet clay. The earliest Jomon pottery is dated to around 14,000 B.C., considerably earlier than any pottery produced in Europe and western Asia, the earliest of which dates to approximately 8,000 B.C. The oldest known pottery in the world comes from Japan, and is known as Jomon, which means "cord marks", after its typical decorations made by impressing cords into the wet clay. The earliest Jomon pottery is dated to around 14,000 B.C., considerably earlier than any pottery produced in Europe and western Asia, the earliest of which dates to approximately 8,000 B.C.

Why should the Jomon people have been so inventive? Why were they making pottery so much earlier than anywhere else in the world? Only pottery from China comes remotely close to it in date, and there it is explained by the requirements of rice cultivation. Melvin Aikens, an authority on the Jomon period, believes that Japanese pottery was invented to cook and store the produce of the thick broad-leaved woodlands that had already covered Kyushu, Japan's southernmost main island, by 13,000 B.C. The relationship is evident, he argues, from the simultaneous spread of broad-leaved woodlands and pottery into the northern islands of Japan, both appearing on the northernmost island of Hokkaido at around 7,000 B.C.

There are, however, two problems with this idea. First, there is no necessity for hunter-gatherers to have pottery when living in wooded environments-the inhabitants of other villages of this era, such as Ain Mallaha in western Asia at 12,500 B.C. and Star Carr in northern Europe at 9, 500 B.C., flourished by relying entirely on vessels made from bark, skins, wood, and stone. Rottery no doubt made life easier for those who did the cooking in the woodlands of Kyushu, and we know from food residues that pottery vessels had indeed been used to make vegetable, meat, and fish stews. But people could have easily survived without such vessels.

A second problem for Aikens' theory arose in 1999 when a new sample of pottery was found in northern Honshu (Japan's largest island). Radiocarbon dates on the residues stuck to the interior of the pot dated to 14, 500 B.C., pushing back the origin of pottery by at least another thousand years. At this date Honshu would have had no more than a sparse covering of pine trees. And so, the theory that Japanese pottery was invented to store and cook the produce of broad-leaved woodlands cannot be correct.

An archaeologist, Brian Hayden, has proposed an alternative explanation. It provides another example of his belief in social competition as the driving force of social change that he applied in explaining the origin of squash cultivation in Mexico. Hayden suggests that ceramic vessels have a number of important qualities that make them prestigious objects to own and ideal containers for serving food to guests. At the outset, the potter's art would have been difficult to master; clay had to be carefully selected, tempers (materials added to clay to reduce its plasticity) prepared, and construction and firing techniques explored practiced, and refined. Neighbors and other visitors would have been struck with the amount of labor and skill required to produce a pottery vessel. The display of novel forms with fancy decoration would have impressed them even more. Most striking of all might have been the dramatic smashing of vessels during feasts as an ostentatious display of wealth.

Theatrical smashing of pots may have occurred in the later Jomon period, as immense piles of broken pottery have been found. And as they come in astonishingly elaborate forms, there can be no doubt that many later Jomon vessels were primarily for display. They have spectacular rims modeled as licking flames or serpents winding around the vessel. Sometimes the decoration is so top-heavy that the pots can hardly stand alone. Lacquered (glossy) objects must have been very striking. But we must be cautious about applying such interpretations to the earliest and rather dull specimens of pottery. We currently know too little about the very earliest pottery makers of Japan to decide whether they had been more concerned with impressing their visitors or devising a means to cook vegetable stew. We do know, however, that by 9, 500B.C. many were living sedentary lives in permanent settlements. Although pottery had already been invented, the sedentary lifestyle must have been crucial in enabling ceramic technology to flourish.

The Origins of the Arctic Fox

The arctic fox lives in the far northern Arctic region with other well-known arctic animals like the caribou, musk ox, and polar bear. Scientists aren't sure exactly when the arctic fox first evolved. Arctic fox bone fragments from the Pleistocene epoch (2.6 million-11,700 years ago) are rare, and the ones that do exist are difficult to date with any degree of accuracy. The oldest confirmed arctic fox bone is believed to be 200,000 years old. There are other bones that may be 400,000 years old, but whether they belonged to an arctic fox or some other fox remains under debate. "It's still an open question, “says Love Dalen, a Swedish biologist who has been using genetics in an attempt to understand the evolution of the arctic fox. "Right now, the best estimate is that arctic foxes emerged somewhere between 200,000 and 400,000 years ago, presumably during one of the ice Ages."

For many years researchers believed the species arose in Europe, basing this conclusion on the location of those 2000-year-old fossils. Since the early 1980s, however, genetic comparisons between different fox species have revealed some surprising results. For example, research done by Robert Wayne at the University of California at Los Angeles has found that the arctic fox, despite its unique appearance and adaptations, is actually much more closely related to other foxes than had previously been thought. Its closest living relative, according to the genetic studies, is the North American swift fox.

The result suggests one of two scenarios. One is that arctic foxes gave rise to swift foxes, an evolutionary sequence that would depend on the arctic fox being much older than the fossil record currently indicates. The second possibility is that swift foxes gave rise to arctic foxes, likely during the height of a period of glaciation when tundra forming to the south of the ice sheets may have met the grassland habitat of the swift fox.

On the one hand, the latter scenario could hardly seem more implausible. Swift foxes, after all, live on the arid, sun-baked prairies and open desert, where temperatures can top 120°F (49°C). It's hard to imagine a greater environmental leap from this environment to the frozen tundra. To have acquired over the course of a few hundred thousand years all the adaptations necessary for survival in such an extreme environment suggests a rapid rate of evolution. And for the early species to have so rapidly conquered such a large territory in a polar region seems equally mystifying.

In other ways, however, this scenario works. As one of the smallest foxes, the swift fox can survive on very little food, a trait that would have helped its ancestors gradually stray farther and farther into the biologically barren tundra. And being a nocturnal hunter that seeks shelter in a den during the day, the swift fox is not overly adapted for life at high temperatures. Indeed, in the northern parts of their range, modern swift foxes have faced winter temperatures every bit as severe as those encountered in the Arctic.

From what researchers know about the evolution of other polar species, there seems to be a pattern of evolution in which species living in the biologically rich temperate zone give rise to animals that are more specialized for survival in the more extreme Arctic. Polar bears, for instance, are thought to have evolved from a northern population of brown bears that learned to hunt seals out on the sea ice. As for the rapid evolution and widespread dispersal, one must remember that the arctic fox is no ordinary species when it comes to both reproductive efficiency and migratory tendency; rapid genetic change and rapid migration would both appear to be well within the species' capabilities.

According to Love Dalen, most of the arctic species evolved during the Quaternary Period (the last 2.6 million years) and they all seem to have evolved from temperate species. In evolutionary terms, arctic species are quite young, including, presumably, the arctic fox. Dalen adds that if that is true, then the arctic fox would have evolved in North America, though he notes that there is no solid proof either way.

Deriving Scientific Facts from Theories

In science, a theory is an idea that. once confirmed by experiment or meticulous observation as certainty. becomes a scientific law. Facts are generally obtained through empirical research (through observation or experiment). Occasionally: however, facts that are not easy to obtain directly are deduced from theory. A good example is the work done in the 1840s on Saturn's rings by the great Scottish physicist James Clerk Maxwell. At the time, astronomers had observed three concentric rings about Saturn, all in the same plane. They knew that at least some regions of the rings must be quite thin, since in some areas the planet behind could be plainly seen. Aside from that, their structure was an intriguing mystery. Maxwell carried out a careful theoretical and highly mathematical treatment, and concluded that the rings could not be single solid or liquid units, since mechanical forces acting upon rings of such immense size would break them up. He suggested instead that the rings must be composed of vast numbers of individual solid particles rotating in separate concentric orbits at different speeds. He even drew some conclusions about how large the particles would be and how fast they would move.

For many decades there was no way of testing Maxwell's conclusion, but in the latter years of the twentieth century. observations-particularly those from Voyager spacecraft confirmed Maxwell's conclusions. The particles are composed of window impure ice, or at least are ice covered. Radar observations have even confirmed the range of masses and speeds predicted by Maxwell. This was a remarkable achievement on Maxwell's part. Or the basis of theory; he had arrived at factual information that could not be directly obtained for almost a century and a half.

Scientific work is best done with a theory in mind. It does not matter if the theory is proved wrong: it will nevertheless guide the initial experiments, which may lead to the correct theory. Paradoxically, people occasionally do much better with the wrong theory than with the right one, and there is one rather striking example of this. By the year 1900, physicists had become aware of the properties of radio waves, which they knew to be of the same character as light but with much longer wavelengths. Experiments had shown that radio waves (unlike 1ight) would pass through walls because of the long wavelengths, and that like light they travel in straight lines.

One of the people working on radio waves at the time was the Italian Guglielmo Marconi. Unlike almost everyone else working on them, he knew hardly any physics. (Hence, the award to him of the 909 Nobel Prize for physics amazed and horrified most physicists.) He did not really understand the properties of electromagnetic waves. and did not even realize that radio waves were of that type He said that his waves were Marconi waves, and that, instead of traveling in straight lines, they could be controlled by him to go to any point he chose. This was nonsense, but his ignorance of physics led him to try to send a signal across the Atlantic Ocean. No other physicist thought this was a reasonable thing to try -how could a wave, which has to travel in a straight line, possibly travel so far over the curved surface of the Atlantic? On December 12. 1901. Marconi's assistant sent signals across the Atlantic Ocean, from Cornwall, England, to St. John's, Newfoundland. where Marconi was apparently just able to receive them.

Physicists were quite correct in thinking that radio waves go in straight lines. How, then, can a radio signal go across the Atlantic, in view of Earth's curved surface? The most plausible answer is that they are reflected in some way. In 1902. the British physicist Oliver Heaviside and the American physicist Arthur Edwin Kennelly postulated the existence of a layer in the upper atmosphere in which they thought that molecules would be ionized. meaning they would carry electric charges. Such a layer would reflect the waves from the upper atmosphere and allow them to travel great distances over Earth's surface.

Industrial Activities in Britannia

As the Roman Empire expanded throughout Europe, it brought profound changes to newly conquered lands. One of the later territories to be absorbed as a province was Britain, in the first century A. D. There, with the arrival of the Roman army and its- relatively speaking-very rapid advance north, a number of new, or at least improved, technologies were introduced. Although ceramics had been made in Britain for several millennia, new kiln (furnace) technology allowed far-better-quality vessels to be made. Furthermore, Iron Age Britain had developed a specialized and highly artistic metalworking industry; with Roman technologies came the ability to mass-produce items. Consequently, while Iron Age communities had mastered many crafts and industries, what the Empire brought was the apparatus and the will to exploit these activities, with the initial impetus coming from the army.

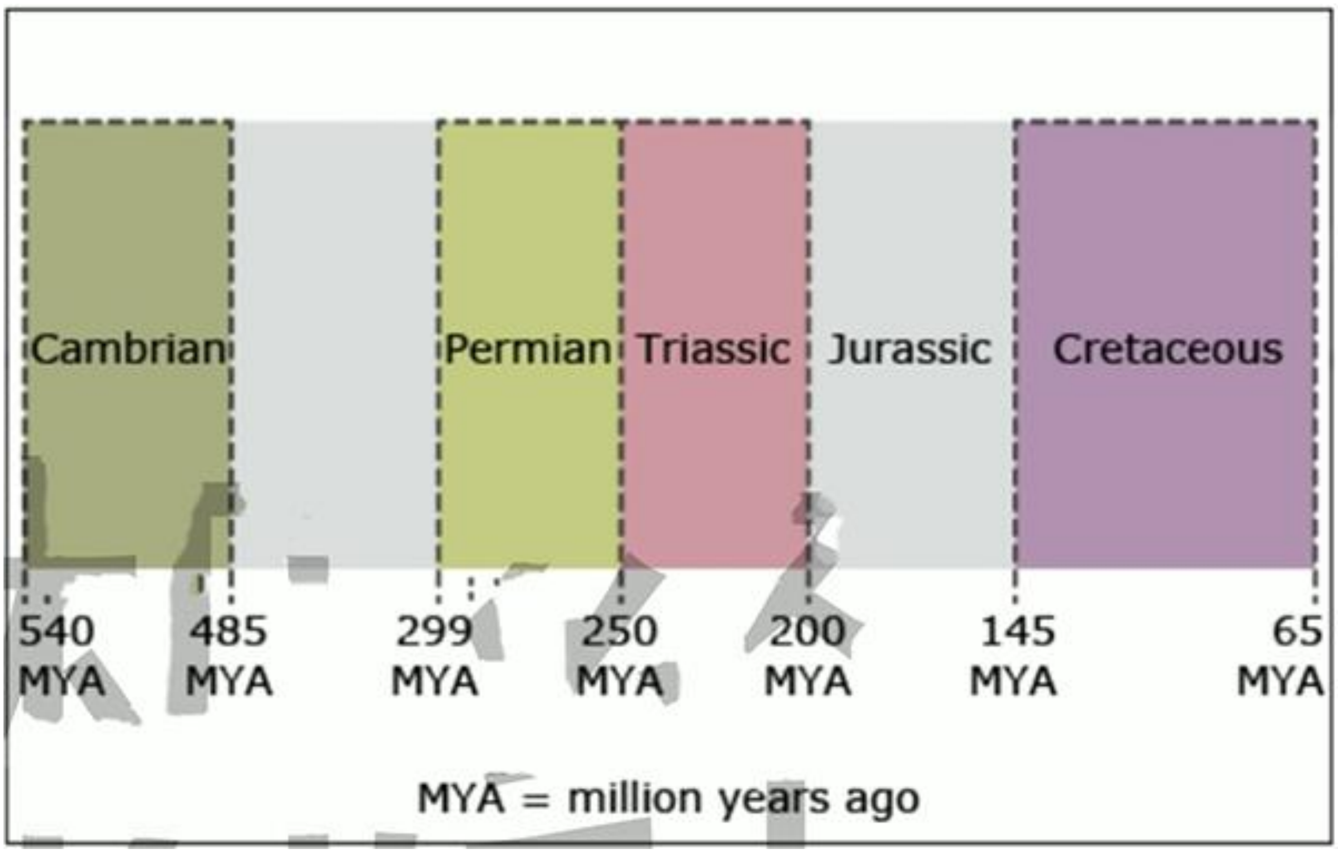
It has long been recognized that the army was responsible for many industrial activities early on in the life of the new province of Britannia, and this conformed to the well-established pattern of conquest and Romanization. While undertaking and controlling industrial production had obvious fiscal, political, and even social benefits, it is unlikely that these were the initial motives for the role the army took in manufacturing. First and foremost, the army was the largest consumer of goods in early Roman Britain, and what the legions required had to either be produced locally or more expensively imported from abroad. Consequently, it made far more sense for the army to produce as many of its own goods as possible, on one hand saving limited financial resources but also ensuring a controlled supply of items that were vital to the day-to-day existence of the legions.

Not only did the active role of the army in industrial production make strategic sense, it was an ideal way to promote the spread of new technologies. While it is well recognized that legions themselves were made up of people from all over the Empire, what is more often overlooked is that the army consisted of individuals with diverse skills drawn from across most of the known world. Glassmaking, among other expertise, was not originally native to Britain or any of her immediate neighbors, but with the arrival of the army, practiced experts were immediately placed at the heart of Britannia. It was therefore inevitable, rather than just a matter of chance, that glassmaking became so quickly established in the first century A.D.

This association of the army with industrial processes can be seen archaeologically, with many production sites being associated with forts. A typical example can be seen at Templeborough,South Yorkshire, where excavations just prior to the First World War revealed the remains of a small glass furnace in the vicus, or attached settlement, of the fort. Obviously, glassmaking was not the only industrial activity undertaken by the army, and it is therefore no surprise that furnaces have been found in areas of more general and mixed industrial production. Perhaps the best example of this can be seen just beyond the confines of the fort at Minstermen Warwickshire. A glass furnace dating to the second half of the second century, with associated working waste, was found to be operating among a group of pottery kilns. This pattern of mixed industrial quarters in close association to forts is mirrored at other sites across the Western Empire.

However, what is clear is that glassmaking was not being undertaken only at military sites. Indeed, there is probably more evidence from Britain of glassmaking being practiced in civilian settlements, and more specifically in towns. The growing urban populations of the first-and second-century province were a clear market for any products. And the positioning of furnaces as close as possible to these populations inevitably reduced transportation costs and the possibility of accidental breakage through carriage. The main difference with urban-centered production was that it was less likely to have been undertaken by the army directly, although since many towns were closely associated with forts, they still remained likely consumers.

Iridium and the Terminal Cretaceous Event



There have been a number of natural catastrophes in Earth's past that wiped out huge numbers of plant and animal species- -in effect, “resetting the clock" and requiring a new start for much of life on Earth. One such catastrophe took place at the end of the Permian era, some 251 million years ago. But perhaps the most famous one occurred about 65 million years ago at the end of the Cretaceous, causing the dinosaurs to go extinct (except for a small number that evolved into modern birds). Evidence that the terminal Cretaceous event was triggered by the impact of an asteroid with Earth began to emerge in the early 1980s, when Luis and Walter Alvarez discovered a thin layer of iridium in rock strata (layers) 65 million years old at widely separated sites around the world. Iridium is very rare on Earth's surface but is much more common in some kinds of meteorite. The two scientists calculated that the impact of such an object, about 10 kilometers across, could have spread the right amount of iridium around the globe in the cloud of debris (fragments) blasted out by the impact. The idea was greeted with skepticism at first, but more traces of debris from an impact occurring at the time of the death of the dinosaurs began to turn up as geologists searched for evidence for or against the hypothesis. To most people, the most decisive evidence came in the early 1990s, when a geological feature at Chicxulub in the Yucatan Peninsula of Mexico was identified as the remains of an impact crater in exactly the right place and with exactly the right age to connect it to the terminal Cretaceous event.

There remained one doubt. Iridium does exist within Earth, and it is brought to the surface by volcanic eruptions. It still seemed just possible that volcanic eruptions on a truly massive scale could have put the iridium layer in place, and such eruptions would, of course, be bad news for life on Earth. There was even a known volcanic event that could have been responsible. Huge eruptions at roughly the right time spread vast amounts of lava across what is now west-central India, forming a structure known as the Deccan Traps. This is one of the largest volcanic features on Earth, more than two kilometers thick and covering an area of more than a million square kilometers. But it is not a unique feature. The Siberian Traps were produced in one of the largest known volcanic events since the start of the Cambrian, which lasted for a million years and spanned the Permian-Triassic boundary, about 250 million years ago. This coincided with the terminal Permian extinction event, which killed some 90 percent of species existing at the time; in fact, the eruption of the Siberian Traps almost certainly caused this extinction.

So it is reasonable to implicate the formation of the Deccan Traps with the extinction at the end of the Cretaceous, and it is certain that such events have played a part in the history of life on Earth. And it also seems possible that life at the end of the Cretaceous was already under stress because of the environmental changes associated with the formation of the Deccan Traps when the meteorite struck. All of these possibilities encouraged a great deal of work over the best part of the next two decades, culminating in a meeting of 41 geologists, paleontologists, and other researchers who reviewed all the data and published their conclusions in the journal Science in 2010. They found that the volcanic activity lasted for about 1 .5 million years, much longer than the time span over which the famous iridium layer was deposited, and that the eruptions began about half a million years before the terminal Cretaceous event. During that time, ecosystems did not change dramatically, but they suddenly collapsed at the time of the Chicxulub event. In light of all the data, the team concluded that a large asteroid impact 65 million years ago in modern-day Mexico was the major cause of the mass extinctions.

The Evolution of Plant Roots

Roots are essential to the development of large plants because they provide a means of anchoring and maintaining an upright position. Most land plants are literally rooted to the spot. Roots also play a key role in water and nutrient acquisition. More significantly still, roots have a tremendous impact on the environment. They can break up rock, bind loose particles together, and prov idea conduit for the movement of water and dissolved minerals, all of which are essential to the development of soils.

In piecing together a fossil plant to form a conceptual whole, it is usually the rooting system that remains the final piece in the puzzle. It is often the case that roots are poorly studied or completely unknown. Although the fossil record of roots is therefore less complete than that of other plant organ systems, it is possible to discern some general trends. The earliest land plants, like modern mosses and liverworts, did not have well-developed root systems. These plants simply bore absorbing hairlike cells on stems and leaves that grew flat along the ground. From their fossils, some very early plants are known to have borne branches that appear to be specially modified for rooting. In other cases, roots were able to form from dormant buds on aerial stems. Fungi are also known to have played a key role in these early rooting systems, as they do in modern plants. Fungal symbionts—fungi that live in mutually beneficial relationships with another organism-have been recorded in the petrified plants of the 400million-year-old Rhynie Chert fossil site in Scotland, demonstrating a link with mycorrhizal fungi that goes back to the dawn of the land flora. These tiny, shallow rooting systems were adequate for small plants (30-50 centimeters tall), but larger organisms required something more substantial.

By the Late Devonian and Early Carboniferous eras (385 to 300 million years ago), an enormous variety of rooting structures had evolved. The evolution of large erect plants, and in particular trees, placed increasing demands upon the anchoring and supply functions of roots. These problems were solved mainly through the development of more extensive underground systems. The evolution of the cambium, the layer of living cells between wood and bark, enabled continuous perennial growth and long-term survival of roots in soils.

One important consequence of all this was that there was a progressive and massive increase in root biomass during the Devonian, which had an enormous impact on the development of soils. Prior to the Devonian, soils, if developed at all, are thought to have been predominantly thin and of microbial origin. By the Middle Devonian, soil penetration depths of roots were still shallow(less than 20 centimeters), but this increased to 1 meter or more as forests spread. The diversity of soils also increased. This change was brought about by root-induced weathering and mixing. By the end of the Devonian, there was an increase in soil clay content, structure, and differentiation into distinct layers—a development that correlated with increases in depth of root penetration. Soils with modern profiles(series of layers) are recognizable at this time.

The impact of roots on the environment extends beyond their immediate effects on the development of soils. The presence of roots in soils increases the natural weathering of calcium and magnesium silicate minerals. This apparently mundane fact turns out to have extremely important consequences for climate and temperature globally. Under natural circumstances, calcium and magnesium silicates react chemically with a dissolved form of the gas carbon dioxide (a process referred to as weathering), which comes from the atmosphere. This produces calcium and magnesium carbonates, which are transferred through the groundwater system to rivers and ultimately to the oceans, where they accumulate in the form of limestone and dolomite rock. Across the surface of the Earth, these chemical reactions occur on a vast scale, removing carbon dioxide gas from the atmosphere and locking it up as carbonate in rock formations. This reduces the so-called greenhouse effect, which leads to lower global temperatures. In other words, the widespread development of roots in land plants affected the chemistry of the atmosphere and the oceans, which, summed over millions of years, added up to changes in climate on a global scale.

Herbivores on the Serengeti Plain

Herbivores (plant- eating animals) living in the same environment often compete for food plants. But in some cases, herbivores may cooperate in the harvesting of plant matter. The grazing system of large herbivores on the Serengeti Plain of east Africa is an excellent illustration of how these animals may interact over their food supply. The Serengeti Plain contains the most spectacular concentration of large mammals found anywhere in the world. Approximately 1,000,000 wildebeests, 600,000 Thomson's gazelles, 200,000 zebras, and 65,000 buffalo occupy an area of 23,000 square kilometers (9,000 square miles), along with undetermined numbers of 20 other species of grazing animals.

The dominant grazers of the Serengeti Plain are migratory and respond to the growth of the grasses in a fixed sequence. First, zebras enter the long-grass communities and remove many of the longer stems. Zebras are followed by wildebeests, which migrate in very large herds and graze and trample upon the grasses until they are close to the ground. Wildebeests are, in turn, followed by Thomson's gazelles, which feed on the short grass during the dry season.

The various grazing species in the Serengeti system do not each select different species of grasses but instead select different parts of the grass plant during different seasons. Zebras eat mostly grass stems and sheaths and almost no grass leaves. Wildebeests eat more sheaths and leaves, and Thomson's gazelles eat grass sheaths and a large fraction of herbs not touched by the other two herbivores. These feeding differences have significant consequences for grazing species because grass stems are very low in protein and high in lignin, an organic substance that strengthens the cell walls of some plants. Grass leaves, on the other hand, are relatively high in protein and low in lignin, which means leaves provide more energy per gram of dry weight than stems. Herb leaves typically contain even more protein and energy than grass leaves. So, zebras seem to have the worst diet and Thomson's gazelles the best.

How can zebras cope with grass stems as the major part of their diet during the dry season? Most of the grazing animals in the Serengeti are ruminants, animals that have a specialized stomach containing bacteria and protozoa that break down the cellulose in the cell walls of plants. But the zebra is not a ruminant and is similar to the horse in having a simple stomach. Zebras survive by processing a much larger volume of plant material through their gut than ruminants do, perhaps roughly twice as much. So even though a zebra cannot extract all the protein and energy from the grass stems, it eats more and thus compensates by volume. Zebras also have an advantage of being larger than wildebeests and Thomson's gazelles, and larger animals need less energy and less protein per unit of weight than smaller animals do.

It would seem that competition for food would occur between wildebeests and Thomson's gazelles because they eat the same parts of the grass. Wildebeests have what appears to be a devastating effect on the grassland as they pass through in migration. Green biomass was reduced by 85 percent and average plant height by 56 percent on sample plots. By establishing fenced areas as grazing enclosures, one researcher was able to follow the subsequent changes both in grassland areas subject to wildebeest grazing and in areas protected from all grazing. Grazed areas recovered after the wildebeest migration had passed and produced a short, dense lawn of green grass leaves. As gazelles entered the area during the dry season, they concentrated their feeding on areas where wildebeests had previously grazed and they avoided areas of grassland that the wildebeest herd had missed.

Grass production was reduced by both wildebeests and gazelles, but no signs of competition were found. The Serengeti grazing populations instead show evidence of grazing facilitation, in which the feeding activity of herbivore species improves the food supply available to a second species. Heavy grazing by wildebeests prepares the grass community for subsequent exploitation by Thomson's gazelles in the same general way that zebra feeding improves wildebeest grazing.

British Agriculture

England was one of the first nations in the modern period to increase the efficiency of its agricultural production, making it possible to produce the same or more while using fewer workers. By the end of the seventeenth century, England was already in advance of most of continental Europe in agricultural productivity, with only about 60percent of its workers involved primarily in food production. Although the actual number of workers in agriculture continued to grow until the middle of the nineteenth century, the proportion declined steadily to about 36 percent at the beginning of the nineteenth century, to about22 percent in the mid-nineteenth century (when the absolute number was at its maximum), and to less than 10 percent at the beginning of the twentieth century.

The means by which England increased its agricultural productivity owed much to trial-and-error experimentation with new crops and new crop rotations. Turnips, clover, and other fodder crops(plants used to feed livestock) were introduced from the Netherlands in the sixteenth century, and became widely diffused in the seventeenth. Probably the most important agricultural innovation before scientific agriculture was introduced in the nineteenth century was the development of so-called convertible husbandry, involving the alternation of field crops with temporary pastures in place of permanent cultivated fields and pastures. This had the double advantage of restoring the fertility of the soil through improved rotation, including leguminous crops, and of maintaining a larger number of livestock, thus producing more manure for fertilizer as well as more meat, dairy produce, and wool. Many landowners and farmers also experimented with selective breeding of livestock.

An important condition for both the improved rotations and selective breeding was the enclosure (the conversion of common land into private plots) and consolidation of the fields. Under the traditional open field system, it was difficult, if not impossible, to obtain agreement among the many participants on the introduction of new crops or rotation; and with livestock grazing in common herds, it was equally difficult to manage selective breeding. The most famous enclosures were those carried out by acts of Parliament (the British governing body) between 1760 and 1815.Enclosure by private agreement, however, had been going on almost continually from the late Middle Ages; it was especially active in the late seventeenth and the first six decades of the eighteenth centuries. By that time more than half of England's arable land (land that can be farmed productively) had been enclosed.

The new agricultural landscape that emerged to replace villages surrounded by their open fields consisted of fields that were compact, consolidated, and closed-in by walls, fences, or hedges. Along with the processes of enclosure and technological improvement, a gradual tendency toward larger farms emerged, further boosting productivity. By 1851 about one-third of the cultivated acreage was in farms larger than 300 acres; farms smaller than 100 acres accounted for only 22 percent of the land. Even so, the occupants of the small farms outnumbered those of the others almost two to one. The reason for this is that the small farmers were owner-occupiers who farmed the land with the help of family labor; the larger farmers rented pieces of their property to others and hired landless agricultural workers. It used to be thought that the enclosures led to a loss of population in the countryside, but in fact the new techniques of cultivation associated with them actually increased the demand for labor. Not until the second half of the nineteenth century, with the introduction of such farm machinery as threshers, harvesters, and steam plows, did the absolute size of the agricultural labor force begin to decrease.

In the meantime, the increasing productivity of English agriculture enabled it to feed a burgeoning population at steadily rising standards of nutrition. For about a century, from 1660 to 1760,English farmers produced a surplus for export; after that period, the rate of population growth overtook the rate of increase of productivity. The relatively prosperous rural population provided a ready market for manufacturing goods, ranging from agricultural implements to such consumer products as clothing, pewter-ware, and porcelain.

The Documentary Film in the United States

In the United States, the nonfiction film was primarily defined and sustained by the travelogue, which was filmed in foreign lands and shown at lectures and sideshows to introduce audiences to different cultures and exotic locations. In 1904, at the St. Louis Exposition, George C. Hale's Tours and Scenes of the World was particularly successful but did not reach the mythic proportions of the film made from President Teddy Roosevelt's African safaris or Robert Scott's expedition to the South Pole. These kinds of travelogues appealed to the American public because they demonstrated a spirit of enterprise and adventure. This outlook underpins the Romantic tradition of filmmaking that begins with travelogues of the American West and comes to its fullest expression in the films of Robert Flaherty. It is he who most embodies the development of the documentary form as an objective tool of ethnography- -the scientific study of other cultures from a position "within" the community- -and anthropology.

His film Nanook of the North (1922), a study of Inuits of northern Canada, is acknowledged as one of the most influential films within the genre. It perhaps provides us with all the clues we require to define both the documentary and its acceptable limits. Flaherty's films, which have been called “authored" films, are made with a specific intent: not merely to record the lives of the Inuits but to recall and restage a former, more “primitive" era of Inuit life. This nostalgic intent only serves to mythologize Inuit life and to some extent remove it from its real context, thus calling into question some of the inherent principles that we may assume are crucial in determining documentary "truth."

Although Flaherty was an advocate of the use of lenses that could view the subject from a long distance so as not to affect unduly the behavior of the natives, and he filmed long, uninterrupted scenes at one time without stopping the camera instead of using complex editing, it is his intervention in the material that is most problematic when evaluating Nanook as a key documentary. Flaherty was not content merely to record events; he wanted to dramatize actuality by filming aspects of Inuit culture that he knew of from his earlier travels into the Hudson Bay area between 1910 and 1916. For example, he rebuilt igloos to accommodate camera equipment and organized parts of Inuit lifestyle to suit the technical requirements of filming under these conditions. In another of his documentaries, Moana (1926), Flaherty staged a ritual tattooing ceremony among the Samoan Islanders, recalling a practice that had not been carried out for many years. In Man of Aran (1935) shark hunts were also staged and did not characterize the contemporary existence of the Aran Islanders.

John Grierson, the British documentary maker, argues that Flaherty becomes intimate with the subject matter before he records it and thus, “He lives with his people till the story is told 'out of himself' and this enables him to 'make the primary distinction between a method which describes only the surface value of a subject and a method that more explosively reveals the reality of it."" This seems to legitimize Flaherty's approach because Nanook, Moana, and Man of Aran all succeed in revealing the practices of more “primitive “cultures- cultures which in Flaherty's view embody a certain kind of simple and romanticized social perfection.

Clearly then, Flaherty essentially uses actuality to illustrate dominant themes and interests that he is eager to explore. In some ways, Flaherty ignores the real social and political dimensions informing his subjects' lives and indeed does not engage with the darker side of human sensibility, preferring instead to prioritize larger, more mythic and universal topics. There is almost a nostalgic yearning in Flaherty's work to return to a simpler, more physical, preindustrial world, where humankind could pit itself against the natural world, slowly but surely harnessing its forces to positive ends. Families and communities are seen as stoic and noble in their endeavors, surviving often against terrible odds. Flaherty obviously manipulates his material and sums up one of the apparent ironies in creating documentary “truth" by suggesting that “Sometimes you have to lie. One often has to distort a thing to catch its true spirit."

The Decline of the Arctic Fox in Scandinavia

The arctic fox has lived in the Scandinavian countries of Norway, Finland, and Sweden since the last Ice Age, which ended more than10,000 years ago. Today, however, the arctic fox in Scandinavia is in severe trouble. Despite intense conservation efforts, its numbers have been radically declining. Why this most resourceful of species has been unable to rebound since hunting it was prohibited is not altogether clear.

One possibility is that the animals have gone through what biologists refer to as a population bottleneck. Such bottlenecks occur when the number of animals in a given population is reduced to the point where inbreeding enables the enhanced proliferation of genes that may be detrimental to a species' overall fitness (ability to survive). A related hypothesis is that the arctic foxes had lost their genetic advantage owing to the spread of genes from selectively bred foxes that escaped from farms. Captive-raised foxes are known to have escaped, and recent genetic studies from Norway have shown that such interbreeding between farm-bred foxes and foxes in the wild has indeed taken place. While these results are being viewed as potential future trouble, there is still no indication that they explain past difficulties. The number of crossover genes detected in the Norwegian study was small, suggesting that mixing between wild and escaped foxes is a recent occurrence.

Among changes in the environment that might be to blame, one of the earliest ideas relates to the absence of wolves, animals that for thousands of years held the role of top predator throughout Scandinavia but that disappeared from most of Norway and Sweden during the twentieth century after hundreds of years of persecution. There is no winter sea ice along the coastlines of Norway and western Russia, as there is throughout the Canadian Arctic and off the coast of Siberia, meaning the arctic foxes in this part of the world have no access to remains of polar bears' prey, an important source of food for arctic foxes. What they do have access to is reindeer. Since wolves are the only predator of reindeer in Scandinavia, it has been argued that the elimination of the wolf has led to a decline in the availability of a necessary winter food supply.

Scientists, however, have been unable to uncover any strong evidence to support the claim that arctic foxes are suffering because of the absence of large predators. One study even suggests arctic foxes do better in the absence of large predators because when animals such as reindeer die of natural causes, the foxes have the large carcasses all to themselves. Also casting doubt on the wolf's culpability is the fact that it had been hunted out of many alpine regions of Scandinavia during the mid-nineteenth century, long before the arctic fox entered into its decline. What's now known about wolf biology suggests the amount of meat and entrails (internal organs) left behind by these animals may not be sufficient to have a noticeable impact on the arctic fox population.

Today there is little enthusiasm among arctic fox scientists for the idea that a lack of wolves is to blame. Instead, most now believe the trouble stems from the increased presence of another canine carnivore- one that, in the eyes of the arctic fox at least, is far more ferocious: the red fox. The red fox is one of the most successful carnivores anywhere. It has evolved a repertoire of instincts that has enabled it to thrive in a world increasingly dominated by humans and human-altered landscapes.

For the most part, arctic foxes and red foxes occupy different environmental niches. Generally, the arctic fox keeps to the tundra, while the red fox is at home in a wide range of habitats throughout the northern hemisphere. In recent decades, however, there is evidence that red foxes are increasingly encroaching on what previously had been territory inhabited only by arctic foxes. During the nineteenth century, there were occasional red fox sightings on southern Baffin Island in Canada, but none of the animals were known to settle down and breed. In the last half of the twentieth century, however, they returned and became a permanent population there.

The Cooling of Early Earth

Earth formed just under 4.6 billion years ago, and for its first 50 to100 million years, it was a boiling ball of liquid rock without a permanent crust (hard outermost layer).An important event in Earth's earliest phase is known as the differentiation event, which completely changed the initially uniform composition of the planet. it happened around 4.5 billion years ago, when the planet had grown large enough for pressure to drive temperatures in the interior above1,000℃(degrees Celsius), the point at which rocks melt. Then, denser (metal-rich) materials sank to the center of the planet, and less dense (rocky) materials rose toward the surface. The sinking dense materials formed Earth's nickel-iron core, the planet's inner 3,500 kilometers or so. The lighter materials that rose up formed the less-dense rocky mantle, the planet's outer 2,900 kilometers.

The formation of Earth's core transformed conditions on Earth's surface. This is because it created the right conditions for development of the planet's magnetic field, which originates from movements in the outer layers of Earth's core. It had been known fora while that the magnetic field was already operational by about 3.5billion years ago, and very recent research has brought that back to before 4 billion years ago. The magnetic field is Earth's only real protection against the solar wind (charged particles from the Sun),which was stripping gases from the earliest atmosphere before the magnetic field had started up. Thus, the differentiation event is thought to have been critical for reducing the loss of light elements from the atmosphere to space. Without it, Earth might have ended up without hydrogen, and thus without water. And over time, many heavier gases would also have been stripped off by the solar wind. Mars is thought to have started out with a magnetic field but—being much smaller than Earth—to have cooled enough for its magnetic field to die at around 4 billion years ago. It subsequently lost almost all of its atmosphere and surface water. While this often-used explanation for retaining an atmosphere by presence of a magnetic field sounds plausible, some further thought suggests that things maybe a little more complicated. Venus has no magnetic field and is closer to the Sun yet has a very well-developed atmosphere. Venus and Earth have similar sizes and masses, while Mars is much smaller—hence, gravity may have been equally or more important for retaining an atmosphere than a magnetic field, especially when the gases concerned are heavier gases, like the dominant CO_2 (carbon dioxide) on Venus.

At Earth's position in the earliest solar system, temperatures of 250°C to 350°C would be expected, but the energy from the intense early impacting by objects from space had pushed temperatures up well above the melting temperature of rocks. Still, research has demonstrated that as early as about 4.4 billion years ago, Earth's surface had not only cooled sufficiently to form early crust (likely below 1,000°C) but even enough to allow for the presence of liquid water, which at modern atmospheric pressure would mean that temperatures had dropped below 100°C. However, at higher pressures this value is higher, and we don't really know how dense the early atmosphere was. So, 100°C is a lower estimate; true temperatures may have been double that value if the early atmosphere was very dense.

Now that clouds and rain had appeared on the scene, oceans began to develop. Large portions of the planet were covered with water by about 4 billion years ago. This is somewhat unexpected because, at this time, Earth’s fiery birth phase had rapidly settled down, and the Sun was only about 70 percent as strong as it is today. After their first bright ignition, stars like the Sun shine more weakly and then grow gradually more intense with age; we refer to this as the “faint young Sun." With only 70 percent of the modern energy coming in from the Sun, early Earth should have been covered by ice, not water. Somehow, Earth's early atmosphere must have retained heat more effectively than the modern atmosphere. Indeed, reconstructions of the early atmosphere's composition suggest high levels of CO2, water vapor, and methane, which would have caused efficient heat retention.

The Commercialization of Agriculture in the United States

Thomas Jefferson, a Founding Father of the United States, believed that farmers were the foundation of American democracy. To execute his plan for democracy, Jefferson proposed the United States Rectangular Land Survey- -familiarly known as the grid. Under the plan, surveyors were first sent to eastern Ohio with instructions to divide the land into boxes that would measure six miles square. Then they were instructed to divide these larger boxes into smaller ones, one-mile square, which were divided yet again into quarter sections measuring 160 acres each, considered to be the appropriate size for a single farm. In 1785 Congress passed the grid into law, and from that point on the same checkerboard pattern was etched across the West- -one of the most far-reaching attempts at rationalizing a landscape in world history.

The grid was the outward expression of a culture wedded not simply to democracy but to markets and exchange as well. It would aid in the rapid settlement of the country, turning millions of Americans into independent landowners, while at the same time transforming the land itself- its varied topography, soil, and water conditions- into a commodity, a uniform set of boxes easily bought and sold. But the grid was only the first step in the commercialization of Western farmlands.

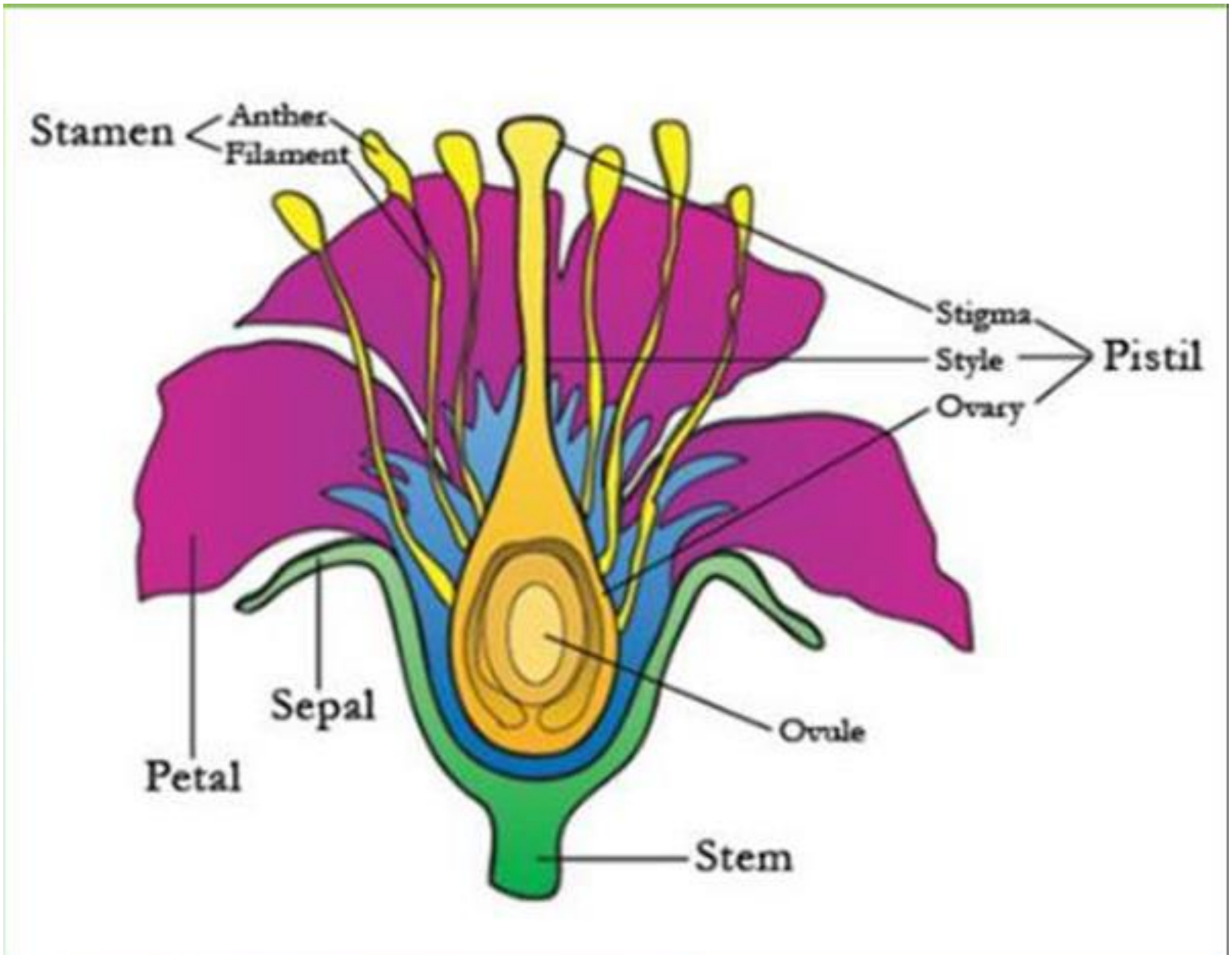
Once farmers purchased land, they needed to plow up the existing vegetation. The grasses that thrived on the organically rich, deep soil laid down by the glaciers thousands of years earlier were at first a challenge to cut. Wooden plows with edges made of iron proved virtually useless. The development and spread of the steel plow-invented in 1837 by John Deere, an llinois blacksmith- -made plowing successful. In place of the native vegetation, farmers planted corn and wheat, domesticated species of grass that grow best in a monocultural environment, that is, in fields by themselves. These crops tend to grow quickly, storing carbohydrates in their seeds. With bread constituting a major component of the American diet, wheat would eventually emerge as the West's major cash crop; acres and acres of some of the world's best agricultural land in states such as Ohio, Indiana, Ilinois, Iowa, and Kansas were plowed up and given over to the plant.

In the early years of settlement, farmers grew a variety of grains, including wheat, corn, oats, rye, and barley. Increasingly, however, farmers became more specialized, as commercial agriculture, aided by improved railroad transportation, proceeded apace. Much of the grain ended up in the Northeast, where, by the 1840s, population growth had outstripped the local farm economy's ability to provide. In effect, the West's surplus of soil wealth underwrote industrial development farther east.

The railroads not only delivered the products of the rich soils of the Western grasslands into the stomachs of Easterners, they also changed the meaning of the crops themselves. With waterborne transportation, farmers put their grain into sacks so they could easily be loaded into the irregularly shaped holds of steamboats. The advent of the railroads and steam-powered grain elevators (first developed in 1842) spurred farmers to eliminate the sack altogether. Now grain would move like a stream of water, making its journey to market with the aid of a mechanical device that loaded all the wheat from a particular area into one large grain car. Sacks had preserved the identity of each load of grain. With the new technology, however, grain from different farms was mixed together and sorted by grade. The Chicago Board of Trade (established in 1 848) divided wheat into three categories- spring, white winter, and red winter- -applying quality standards to each type. Wheat was turned into an abstract commodity, with ownership over the grain diverging from the physical product itself. By the 1860s, a futures market in grain had even emerged in Chicago. It was now possible to enter into a contract to purchase or sell grain at a particular price. What was being marketed here was not the physical grain itself so much as an abstraction, the right to trade something that may not even have been grown yet.

The Angiosperm Revolution

Of all the kinds of modern land plants one group dominates: the angiosperms, or flowering plants. With over 250,000 living species, they are the majority of plants of most habitats- except marine environments, which are still habitats for the more primitive algae. But angiosperms were a comparatively recent development in plant evolution. They arose in the mid-Mesozoic (approximately 200 million years ago to 145 million years ago), but by about 100 million years ago they had pushed the conifers (plants having cone-shaped reproductive structures rather than flowers for reproduction) into the background. Even earlier types of plants such as ferns are now restricted to certain wet habitats, and many of the dominant gymnosperms (plants with exposed seeds, such as conifers) of the early Mesozoic have now been largely replaced by angiosperms; the formerly dominant gymnosperms that did not become completely extinct now survive in comparatively few places.



Why were the angiosperms so successful? A major advantage they have over more primitive plants is their efficient mode of reproduction- -the flower and all of its complex reproductive mechanisms that ensure success. Instead of the inefficient wind-pollinated gymnosperm seed, which wastes a huge amount of pollen and is dependent on random breezes, angiosperms have evolved flowers specifically as devices to attract pollinators- -mainly insects (especially moths, butterflies, and bees) but also birds, bats, and other flying creatures. The pollinators ensure that the pollen is carried directly from one flower of the same species to another, which is more efficient than relying on the wind. This process of delivery is called cross-pollination. The reproductive cycle is highly modified: the ovules (egg-producing organs) are fully enclosed within protective covers called carpels, which form the core of the flower. The carpel protects the ovule from drying out, from fungal infection, and from predation by plant-eating insects.

Pollen-producing organs called stamens are surrounded by petals (which serve to attract the pollinator and guide it to the ovules in many cases) and an outer covering of sepals for protection. Typically, a pollinator gets pollen stuck onto it as it climbs into a flower, seeking the nectar that is generated to lure it. The ovule is usually pollinated by the sperm carried from a different flower, thus minimizing self-fertilization (but angiosperms can also self-fertilize if cross-pollination is not possible).

Once the pollen has been delivered, a pollen tube transports the sperm to the ovules. Here angiosperms have another advantage: double fertilization. The pollen carry two sperm nuclei, one of which fuses with the egg nucleus to form the embryo, the other of which fuses with two other nuclei to form a food supply for the embryo. This means that angiosperms don't need to invest a lot of energy creating food stores for each seed until it is fertilized (unlike gymnosperms, which create food even for infertile seeds).

The entire process of fertilization and producing an embryo takes place in only a few weeks or days, so angiosperms can sprout, flower, reproduce, and die in a single season if necessary. By contrast, most gymnosperms are slow to grow and reproduce (usually taking at least eighteen months between reproductive cycles) and cannot accomplish the entire process in a single season. For gymnosperms such as evergreen conifers to live in highly seasonal, cold-winter climates, they must be able to survive the cold and shut down much of their physiological systems during winter. Many angiosperms, on the other hand, are annuals-sprouting in the spring, flowering, and producing seeds that can survive until the next winter while the rest of the plant dies. This rapid reproduction enables them to quickly exploit habitats that other plants cannot.

Finally, angiosperms are known not only for their rapid growth rates but for their ability to grow back quickly after they have been munched by animals. Think of how quickly the grass grows back after you mow it (or an animal grazes it). By contrast, ferns cannot grow back so quickly after they have been heavily eaten, and often die if the damage is too great (such as when an animal eats the growing tip of the plant)- whereas many angiosperms can be eaten right down to their roots but grow back again.

Formation of the Solar System

Where did the solar system come from? This question has tantalized astronomers for centuries. While we do not yet have a wholly complete answer, a consensus has arisen about the most likely series of events that led to the present-day system of the Sun and planets.

A key piece of evidence about the origin of the solar system is that all the planets orbit the Sun in the same direction and in nearly the same plane. As long ago as the eighteenth century, German philosopher Immanuel Kant and French scientist Pierre-Simon deLaplace independently suggested that this state of affairs could not be a coincidence. They proposed that our entire solar system-the Sun as well as all of the planets, satellites, asteroids, and comets—formed from a vast, rotating cloud of gas and dust called the solar nebula. In the modern version of their theory, the solar nebula is thought to have had a mass somewhat greater than that of our present-day Sun.

Each part of the nebula exerted a gravitational attraction on the other parts, and these mutual gravitational pulls tended to make the nebula contract. As it contracted, the greatest concentration of matter occurred at the center of the nebula, forming a relatively dense region called the protosun, which eventually developed into the Sun. The planets formed from the much lesser amount of material in the outer regions of the solar nebula. Indeed, the mass of all the planets together is only 0.1 percent of the mass of the Sun.

When you drop a ball, the gravitational attraction of Earth makes the ball travel faster and faster as it falls; in the same way, material falling inward toward the protosun would have gained speed. As this fast-moving material ran into the protosun, the energy of the collision was converted into thermal energy, causing the temperature deep inside the solar nebula to climb. While the protosun's surface temperature stayed roughly constant, the temperature inside the protosun increased even more by means of further contraction. Eventually, after perhaps 100 million years had passed since the solar nebula first began to contract, the central temperature of the protosun reached a few million degrees, and nuclear reactions began in its interior. When this happened, the contraction stopped and a true star(the Sun) was born. Nuclear reactions in the interior of the present-day Sun are the source of all the energy that the Sun radiates into space.

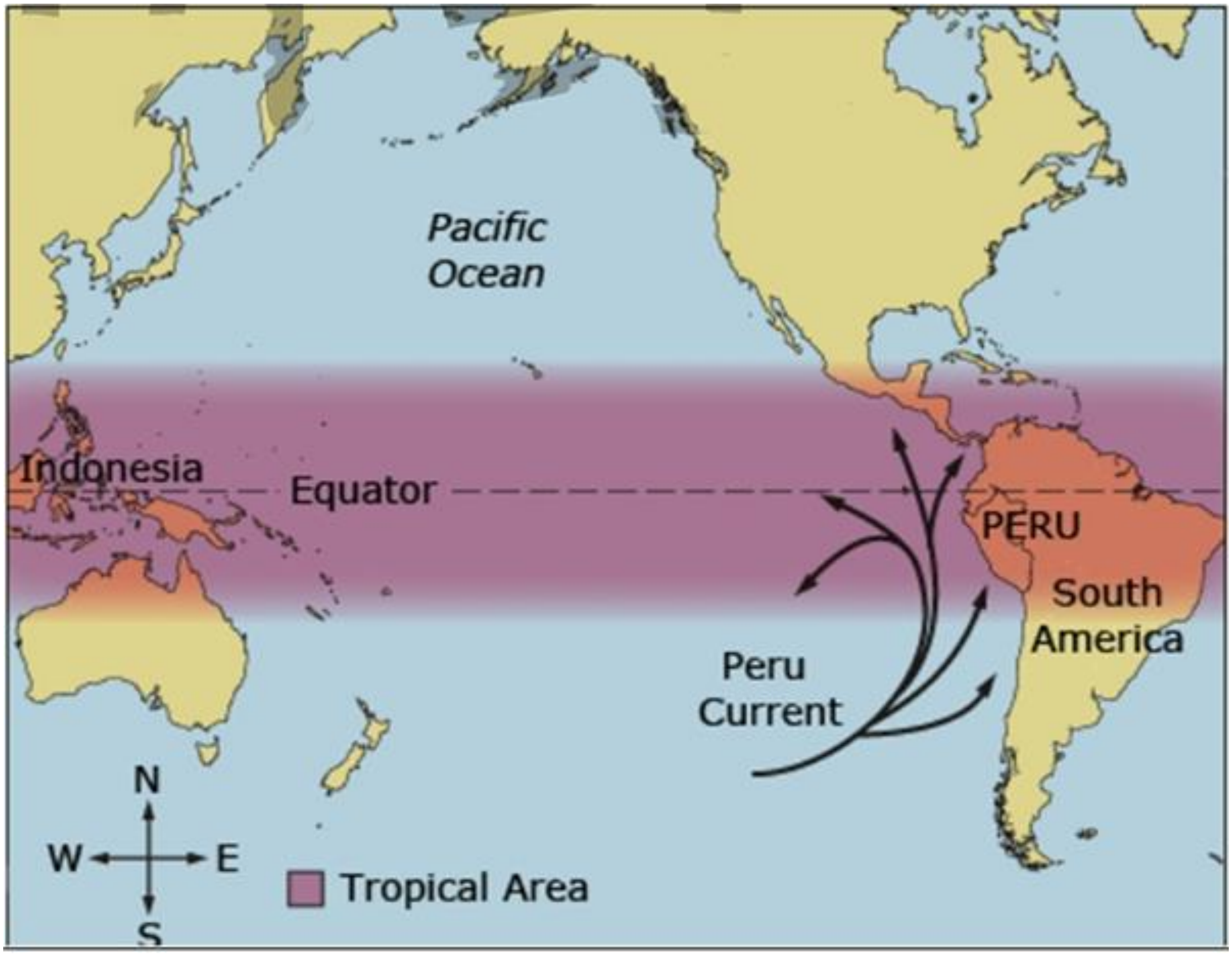
If the solar nebula had not been rotating at all, everything would have fallen directly into the protosun, leaving nothing behind to form the planets. Instead, the solar nebula must have had an overall slight rotation, which caused its evolution to follow a different path. As the slowly rotating nebula collapsed inward, it would naturally have tended to rotate faster. Figure skaters use this same phenomenon; when a spinning figure skater pulls her arms and legs inward, close to her body, the rate at which she spins automatically increases. As the solar nebula began to rotate more rapidly, it also tended to flatten out, just as a spinning ball of dough flattens out when it is spun rapidly by a pizza chef. The eventual result was a structure with a rotating flattened disk surrounding a newly formed Sun. The planets formed later from this disk, which explains why their orbits all lie in essentially the same plane and why they all orbit the Sun in the same direction.

Naturally, there were no humans to observe these processes taking place during the formation of our solar system. But astronomers have seen disks of material surrounding other stars that formed only recently. These are called protoplanetary disks, or proplyds, because it is thought that planets can eventually form from these disks around other stars. By studying these proplyds, astronomers are able to examine what our solar nebula may have been like some 5 billion years ago.

El Nino and the Southern Oscillation

Between the ocean surface and the atmosphere. there is an exchange of heat and moisture that depends, in part, on temperature differences between water and air. Even a relatively small change in surface ocean temperatures could modify atmospheric circulations and have far-reaching effects on global weather patterns.

Along the west coast of South America, where the cool Peru Current sweeps northward, southerly winds promote upwelling (rising to the surface and flowing outward) of cold. Nutrient- rich water that gives rise to large fish populations. especially anchovies. The abundance of fish supports a large population of seabirds whose droppings (ale guano) produce huge phosphate rich- deposits that support the fertilizer industry. Near the end of the calendar year, a warm current of nutrient-poor tropical water often moves southward, replacing the cold, nutrient-rich surface water.



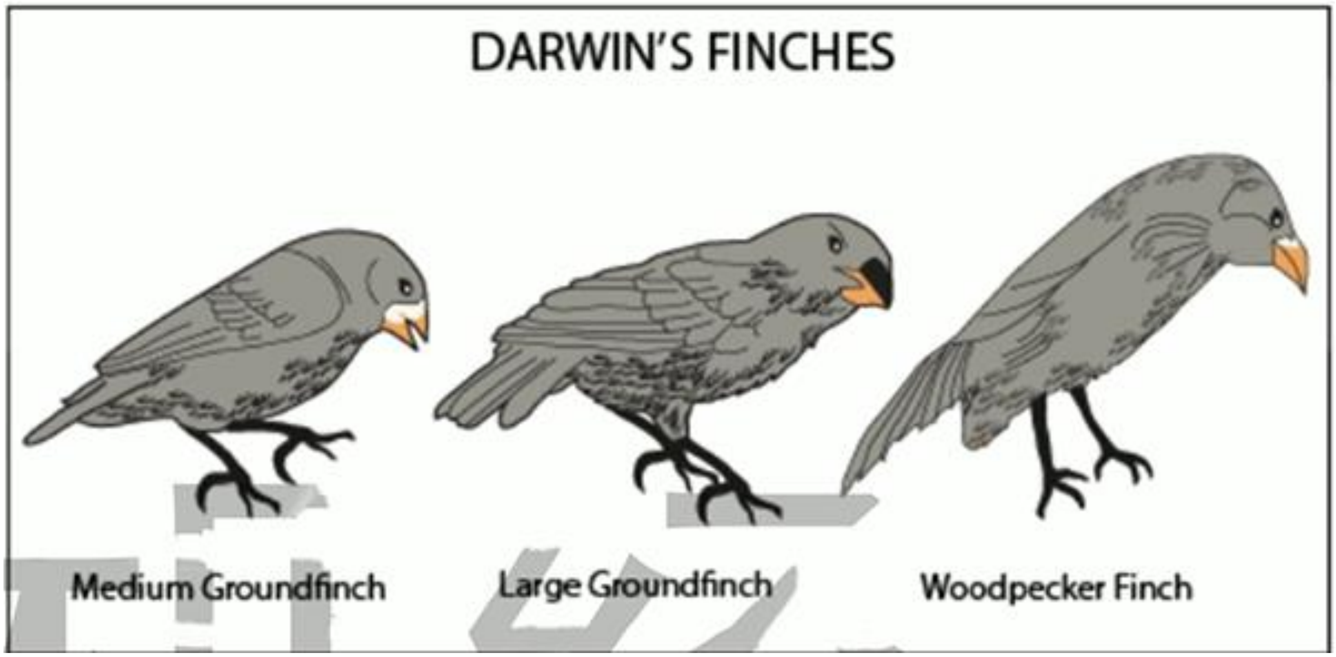
In most years, the warming lasts for only a few weeks to a month or more, after which weather patterns usually return to normal and fishing improves. However, when conditions last for many months, and a more extensive ocean warming occurs, the economic results can be catastrophic. This extremely warm episode, which occurs at irregular intervals of two to seven years and covers a large area of the tropical Pacific Ocean, is now referred to as a major El Nino event, or simply El Nino.

During a major El Nino event, large numbers of fish and marine plants may die. Dead fish and birds may litter the water and beaches of Peru; their decomposing bodies reduce the water's oxygen supply, which leads to the bacterial production of huge amounts of hydrogen sulfide. The El Nino of 1972-1973 reduced the annual Peruvian anchovy catch from 10.3 million metric tons in 1971 to 4.6 million metric tons in 1972. Since much of the harvest of this fish is converted into fish meal and exported for use in feeding livestock and poultry, the world's meal production in 1972 was greatly reduced. Countries such as the United States that rely on meal for animal feed had to use soybeans as an alternative This raised poultry prices in the United States by more than 40 percent.

Why does the ocean become so warm over the eastern tropical Pacific? Normally in the tropical Pacific Ocean, there are the trades-persistent winds that blow westward from a region of higher pressure over the eastern Pacific toward a region of lower pressure centered near Indonesia. The trades create upwelling that brings cold water to the surface. As this water moves westward, it is heated by sunlight and the atmosphere. Consequently, in the Pacific Ocean, surface water along the equator usually is cool in the east and warm in the west. In addition, the dragging of surface water by the trades raises the sea level in the western Pacific and lowers it in the eastern Pacific, which produces a thick layer of warm water over the tropical western Pacific Ocean and a weak ocean current (called the counter current) that flows slowly eastward toward South America.

Every few years, the surface atmospheric pressure patterns break down, as air pressure rises over the region of the western Pacific and falls over the eastern Pacific. This change in pressure weakens the trades, and, during strong pressure reversals, east winds are replaced by west winds. The west winds strengthen the counter current, causing warm water to head eastward toward South America over broad areas of the tropical Pacific. Toward the end of the warming period, which may last between one and two years, atmospheric pressure over the eastern Pacific reverses and begins to rise, whereas, over the western Pacific, it falls. This seesaw pattern of reversing surface air pressure at opposite ends of the Pacific Ocean is called the Southern Oscillation. Because the pressure reversals and ocean warming are more or less simultaneous, scientists call this phenomenon the El Nino/Southern Oscillation, or ENSO for short. Although most ENSO episodes follow similar evolution, each event has its own personality, differing in both strength and behavior.

Beaks of Darwin' Finches



In 1835, before he had developed his theory of evolution, Charles Darwin collected specimens of 13 previously unknown species of finches from the isolated Galapagos Islands. The Galapagos finches closely resembled a species of finches living on the mainland of South America, but each of the Galapagos species of finches had a differently shaped beak unique to it. His observations led Darwin to speculate that "from an original paucity of birds in this archipelago [the Galapagos Islands],one species has been taken and modified for different ends. “This is the essence of Darwin's theory of evolution by natural selection: Birds with a particular beak shape survived and reproduced because their beak made them well adapted for using a particular food source. In this way one original species that came to Galapagos from the mainland ultimately evolved into 13 new species.

The correspondence between the beaks of the 13 finch species and their food source immediately suggested to Darwin that evolution had shaped them. If his suggestion that the beak of an ancestral finch had been shaped by evolution is correct, then it ought to be possible to see the different species of finches acting out their evolutionary roles, each using their beaks to acquire their particular food specialty. The four species that crush seeds within their beaks, for example, should feed on different seeds, those with stouter beaks specializing in harder-to-crush seeds.

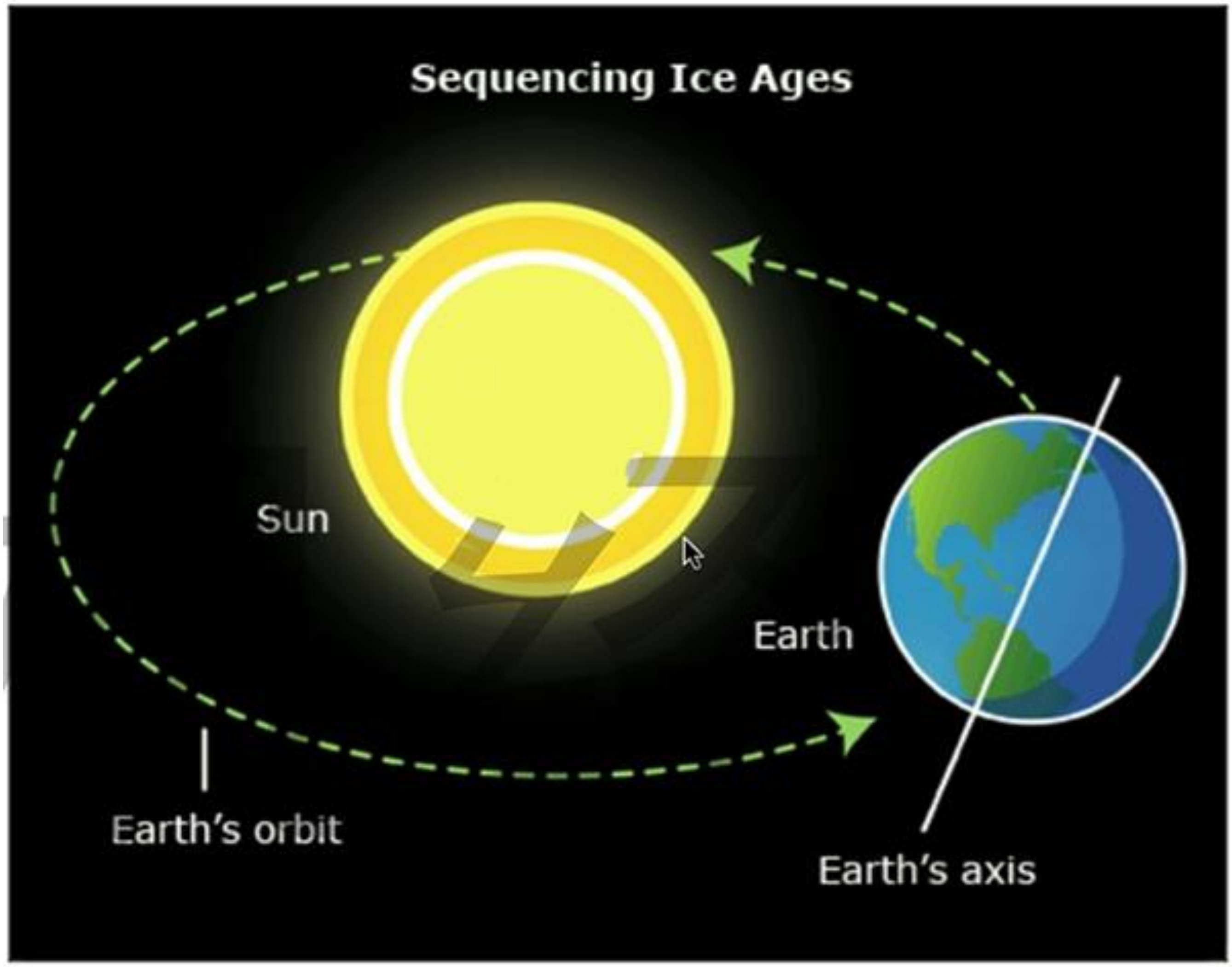
Many biologists visited the Galapagos after Darwin, but it was 100 years before any tried this key test of his hypothesis when the great naturalist David Lack finally set out to do this in 1938, observing the birds closely for a full five months, his observations seemed to contradict Darwin's proposal. Lack often observed many different species of finch feeding together on the same seeds. We now know that it was Lack's misfortune to study the birds during a wet year, when food was plentiful.' The finch's beak is of little importance in such flush times; small seeds are so abundant that birds of all species are able to get enough to eat.

The key to successfully testing Darwin's proposal that the beaks of Galapagos finches are adaptations to different food sources: proved to be patience. Starting in 1973, Peter and Rosemary Grant of Princeton University and generations of their students have studied the medium ground finch Geospiza fortis on a tiny island in the center of the Galapagos called Daphne Major. These finches feed preferentially on small, tender seeds produced in abundance by plants in wet years. The birds resort to larger, drier seeds, which are harder to crush, only when small seeds become depleted during long periods of dry weather and plants produce few seeds.

The Grants quantified beak shape among the medium ground finches of Daphne Major by carefully measuring beak depth(width of beak, from top to bottom, at its base) on individual birds. Measuring many birds every year, they were able to assemble a detailed portrait of evolution in action.' The Grants found that beak depth changed from one year to the next in a predictable fashion. During droughts, plants: produced few seeds and all available small seeds were quickly eaten, leaving large seeds as the remaining source of food. As a result, birds with large beaks survived better, because they were better able to break open these large seeds. Consequently, the average beak depth of birds in the population increased the next year, only to decrease again when wet seasons returned.

Could these changes in beak dimension reflect the action of natural selection? An alternative possibility might be that the changes in beak depth do not reflect changes in gene frequencies, but rather are simply a response to diet- -perhaps during lean times the birds become malnourished and then grow stouter beaks, for example. If this were the case, it would not be genetics but environment alone that influences beak size. To rule out this possibility, the Grants measured the relation of parent bill size to offspring beak size, examining many broods over several years. The depth of the beak was passed down faithfully from one generation to the next, regardless of environmental conditions, suggesting that the differences in beak size indeed reflected genetic differences.

Sequencing Ice Ages



Traditionally, climate scientists believed that Earth underwent four major periods of glaciation (being covered by ice sheets) with warm interglacials in between. But climate research has now shown that over the past 800,000 years Earth has seen a complex pattern of some twenty major climatic shifts, with temperatures alternating between very warm and intensely cold, featuring warm seas in northern Europe at one extreme and ice sheets covering vast areas of the globe at the other. The cause of these fluctuations lies in the complex relationship between Earth and the Sun. We are all familiar with the fact that Earth orbits the Sun and that it spins around its own axis, which is at an angle to the plane of its orbit. This causes some parts of Earth's surface to be nearer the Sun for periods of time, accounting for the differences between summer and winter. If all parts of this system were stable. Earth's climate would remain constant, but this is not so. First, Earth's orbit is not perfectly circular but is slightly elliptical, causing a variation on a 100,000-year cycle. Secondly, the axis of Earth Changes its tilt (angle by a fraction over a .41,000-year cycle; and thirdly, the planet . has a slight wobble (shaking movement) about its axis as it spins, setting up changes over a cycle of 23,000 years. The combination of all these factors (known as the Milankovitch cycles) creates very small changes in the Sun-Earth relationship that determine the expansion or contraction of the polar ice cap and thus the sequence of fluctuating ice ages.

The complexity of these changes has taken some time to unravel. The earliest work was done on glaciers in the Alpine region, and it was here that the four major stages of glaciation were identified, providing a useful regional sequence. More recently, over the last fifty years or so, work on deep-sea cores has pioneered a new way to study the phenomenon globally. The principle is quite simple: Deep-sea cores provide stratified sequences of accumulations of calcium carbonate derived from the shells of dead organisms. The calcium carbonate in these layers contains two different oxygen isotopes (oxygen of different atomic weights), 16o and 180, which the organisms extract from the atmosphere. Since it can be shown that cold conditions favor 180 over 160, by assessing the ratio of the two it is possible to arrive at a direct measurement relating to the temperature of the ocean at the time of deposition. From these measurements a system of twenty phases, known as marine isotope stages, can be distinguished, reflecting changes in Earth's climate. The method was extremely valuable in providing a relative sequence, but it needed to be calibrated to give approximate dates for the phases.

The breakthrough came when, in one of the cores, it was possible to identify a point at which a major reversal had occurred in Earth's magnetic field, the time when the North and South Poles took up their present positions. This same reversal has been recorded in rocks, where it could be dated to around 736.000 years ago using an absolute dating method. With this one point securely established, and assuming that the deep- sea sediments had accumulated at a standard rate, it has been possible to assign dates to the entire sequence. Another valuable dating method, which has been developed over the last thirty years, is based on cores bored out of the Greenland ice cap. The largest core is 3,000 meters deep and represents the buildup of the ice over a 110.000- year period. From the cores it is possible to measure annual increments, one year's winter ice being separated from the next year's by a fine dust layer formed during the summer melt. The temperatures prevailing at the time are estimated from the (16)0 to (18)0 ratios and from proportions of the windblown chemical particles The cores provide a very precise dated sequence of climatic events extending back from the present.

The Roman Road System of Britain

During its occupation of Britain (A.D. 43-410), the Roman Empire built an extensive network of roads. The Roman government recognized four categories of roads ranging in order from the most important to the least: public roads funded by the state; military roads built at the army's expense but also used by the public; local roads on which less engineering effort was expended; and private roads that were built and maintained by their owners. The latter two categories encompassed roads and tracks of varying quality. Before the Romans built roads, the ancient trackways of Britain had followed the natural terrain, seeking the easiest ground to traverse. Such tracks often detoured around marshy areas, hills, or ravines. Romans did not like to waste effort building long, meandering roads, so they ignored the older routes, preferring instead to move in the straightest line possible except where major obstacles in the landscape left no choice. Since Roman roads usually connected such new places of military importance as forts, towns, and administrative centers, the old trackways often did not take the desired direction. There were a few exceptions. Silchester was one of the pre-Roman centers of British activity reused by the Romans. There, the old native roads connected to the new road system.

Laying roads, the shortest distance by going through obstacles required a great investment of labor. If a hill stood in the way, earth was hauled away until the land was leveled. If a wetland needed to be crossed, earth was moved in to build a causeway (raised road). Construction was systematic. A dependable road required a solid foundation. Roman roads were constructed with thick layers of tightly packed stones and gravel sorted for size. The larger stones formed the bottom layer, while layers of progressively smaller gravel filled in and leveled the roadbed. The final result was well over 30 centimeters thick and resistant to the wear and tear of heavy traffic and severe weather. More important roads were elevated above the surrounding land surface and provided with ditches on both sides so that the road surface would never flood. Elevated roads were so well fabricated they did not collapse or become rutted even in the worst rainy seasons. Upright curb stones buried in a line along the sides reinforced the road surface, keeping it stable. These elevated roadbeds formed the original highways of Britain and were much used until modern times.

Construction methods varied according to the official status of the planned road, but certain techniques Were distinctly Roman. The careful layering and tight packing of selected gravels and other materials to form the roadbed is called metaling. Many different materials were used, depending on their availability and suitability to local subsoil conditions. Sand was added to keep the road from becoming too rigid and prone to cracking. In areas where iron ore was smelted into iron, the rock-hard chunks of slag, produced as a waste by-product during the smelting process, were added to the layering material. The iron content of the slag pieces had the added value of rusting over time. The rust would physically combine and harden with adjacent stones and sand to create an extremely solid, concrete-like mass. Heavily used roads were resurfaced from time to time. The final road metaling performed at the end of the fourth century sometimes resulted in a total road thickness of about a meter.

Roman roads in the fifth and sixth centuries continued to carry traffic, but roads do not choose their travelers. Historians speculate that the existing road system enabled a quick penetration of southeastern Britain by Germanic newcomers. The road system also allowed the entire population of Britain to move about in response to foreign threats and changing weather conditions. Townspeople left for the urban amenities of the European continent. Merchants sought markets and customers in safer jurisdictions, while farmers migrated to drier and warmer fields. The old Roman roads retained their importance in successive medieval centuries, too. Even where they did not remain in use as roads, the lines they imposed on the landscape often became boundaries for parishes and other significant administrative or private land units. The well-known Roman highway Watling Street in Northamptonshire now serves as the line for a number of modern administrative boundaries.

Group Foraging

Bluegill sunfish obtain food by flushing (forcing into the open)small aquatic insects from dense vegetation. Researcher Gary Mittelbach hypothesized that larger foraging (food searching)groups might flush out more prey and increase the average number of prey obtained per group member. He examined this hypothesis by experimentally manipulating the foraging group size of bluegill sunfish in a controlled laboratory setting. Mittelbach placed 300 small aquatic prey into a large aquarium containing juvenile bluegill sunfish. He examined the success of bluegills that were foraging alone, in pairs, and in groups (ranging from three to six bluegills). Mittelbach measured the number of prey captured per bluegill, and he found a positive relationship between foraging group size and individual foraging success (the larger the group, the greater the success) up to a group size of four fish. The increased feeding rate per individual was due to two factors. First, more prey were flushed when group size rose. Second, prey clumped together, so when one group member found prey, others swam over to this area and then often found food themselves.

While increased group size in bluegills does benefit each individual forager in its intake of food (its per capita foraging success), there is no evidence to suggest that bluegill foragers hunt in any coordinated or cooperative fashion. The manner in which bluegills increase foraging efficiency in larger groups (that is, by flushing) is only one means of increased foraging efficiency through group living. For example, in a number of species, increased group size reduces the amount of time that any given individual needs to devote to antipredator activities-often, but not always, increasing per capita foraging success.

The general relationship between group size and foraging success uncovered by Mittelbach has been found over and over in animal studies. For example, Scott Creel ran an analysis on foraging success and group size in seven species that hunt in groups. Overall, Creel found a strong positive relationship between per capita foraging success and group size.

Individuals may cooperate with one another when hunting in groups. For example, when wild dogs hunt down an animal, it is a coordinated effort, with different members of the hunting pack playing different roles in the hunt: flushing the prey, making the initial attack, killing the prey, and so forth. In such cases, it is useful to separate the effects of cooperation from that of group size per set. To see how animal behaviorists disentangle such effects, let us examine hunting behavior in chimps. Cooperative hunting, or the lack of it, has been examined in chimp populations in the Gombe preserve in Tanzania, the Ma hale Mountains in Tanzania, and the Tai National Park in the Ivory Coast. The most comprehensive studies of cooperative hunting among chimps are those of Christophe and Hedwige Boesch, who compared hunting patterns across the Gombe and Tai chimp populations. Major differences in hunting strategies emerged between these populations.

In Tai chimps, a positive relationship was found between hunting success and group size in what is called a nonadditive fashion. That is, adding more hunters to a group did not simply increase the amount of food by a fixed amount for each new hunter added. Rather, with each new hunter, all group members received more additional food than they did when the last new hunter was added to the group (up to a limit). In addition to these group-size effects, Christophe Boesch found evidence of cooperation in Tai chimp hunting behavior. Very complex, but subtle, social rules exist that regulate access to meat and assure hunters—that is, those that actually cooperate during the hunt—-greater success than those who fail to join a hunt.

The situation was quite different with Gombe chimps—there was no positive relationship between group size and hunting success in chimps from this population. That is, no group-size effect was uncovered in the context of obtaining food. In addition, unlike in the Tai population, behavioral rules limiting a nonhunter's access to prey were absent, and such individuals received as much food as those that hunted cooperatively. Part of the difference in hunting behavior in Tai and Gombe chimps may be due to the fact that the success rate for Gombe solo hunters was quite high compared with the individual success rate for chimps in the Tai population.

Nutritional Changes in Human History

Although we now think of an optimal diet as one that promotes longevity (long life), historically our optimal diet was whatever made us strong enough to have healthy children and then live long enough to help those children survive to do the same. The best measure of good nutrition is average height, which is about 80 percent genetic but is also strongly linked to childhood and adolescent calories and especially protein intake. Although big and strong people might out-hunt, out-gather, and even out-fight their smaller neighbors, they also need more calories. So, people faced with chronic or recurrent food shortages are better off if they are short and lean. The small bodies of less-well-nourished humans are not only an effect of their lower intake of calories, protein, and calcium but also protection against the likelihood of even lower intakes in the future. By these metrics, our nomadic hunter-gatherer ancestors were remarkably well nourished. Remains from the Paleolithic Stone Age (2.6 mya to 10,000 mya) show that men had an average height of about 179 cm (5 feet 10% inches) and an estimated weight of about 67 kg (148 pounds). The women were substantially smaller, averaging an estimated 157.5 cm (5 feet 2 inches) and 54 kg (119 pounds). After agriculture began, about 10,000 years ago, in the Neolithic era, humans actually got smaller. Archaeological evidence suggests that the average Neolithic man was about 165 cm (5 feet 5 inches) tall and weighed 63 kg (139 pounds), while women were an average of 150 cm (4 feet 11 inches) and weighed 45 kg (99 pounds).

Paleolithic men were bigger than modern European men through the end of the nineteenth century and were as big as they are today. And when the Europeans came to North America, as supposedly well-nourished invaders from the most technologically advanced countries in the world, the indigenous hunter-gatherer population of the American plains was far taller than they were. Why would that be the case? It all came down to nutrition. The plains diet of the hunter-gatherers centered around grass-fed wild bison, whose meat was less fatty than meat from modern corn-fed animals. And the plains tribes ate pretty much the whole animal. By comparison, even when agriculturalists' crops provided enough calories to meet their energy requirements- -which probably were as high if not higher than those of hunter-gatherers because farmers often worked every day from dawn till dusk- they didn't provide as much protein or calcium as the hunter-gatherers' diets did. Even nuts generally have more protein per ounce than grains, especially domesticated grains. This is why agriculturists also looked to legumes- -beans, peas, and even peanuts- for their protein, much as modern vegetarians do.

But it's a challenge to get enough protein from nonanimal sources, and it was even harder in the era before supermarkets.

Nutrition continues to correlate with human height today. The economist Richard Steckel compared the height of people living in Europe from the ninth through the nineteenth century by measuring the length of their thigh bones, or femurs, which account for about 25 percent of our height. He estimated that an average man in the early Middle Ages (5th-10th century A.D.) stood 174 cm (5 feet 8/z inches). But by the seventeenth and eighteenth centuries, when more people lived in cities, where there was less vitamin D (from exposure to sunlight) and fresh meat, the average man's height had declined to 167 cm (5 feet 5%4 inches)- a reduction of over 6 cm (2/z inches). And by the early twenty-first century, better nutrition explains why the average height of young men in affluent European countries increased to about 178 cm (5 feet 10 inches) on average.

In most societies, women are generally 10 to 13 cm (4 to 5 inches) shorter than men, regardless of the height of the men. In this context, the 21.5 cm (8%2 inch) difference in height between Paleolithic men and Paleolithic women was noticeably larger than the sex difference we've seen ever since. This large difference may have been because Paleolithic women didn't fully share in the protein bounty of the hunt, or perhaps became pregnant at a younger age, when they were still growing, and their growth was stunted by having to share their protein and calcium with a developing fetus.

Accounting for the High Density of Planet Mercury

Venus, Earth, and Mars are all thought to have formed from similar materials. Their iron cores take up similar fractions of the planets, and their silicate mantles- -the mantle is the layer of material between a planet's central core and its outer crust- -melt to produce similar kinds of volcanic rocks. By contrast, Mercury, the smallest and closest to the Sun of the four inner, or terrestrial, planets, may have been formed from a different bulk composition, or it may have formed under different conditions than the other terrestrial planets. The planet has a huge iron core, indicating either that the planet contains at least twice the iron content of the other terrestrial planets or that virtually all the iron from the bulk composition was pulled into its core, leaving its mantle almost iron-free. At the same time, the planet shows no recent volcanic activity, and so, unlike Venus, Earth, and Mars, its interior can be assumed to be stationary. The lack of volcanic activity is usually taken to mean that the planet has largely finished losing its internal heat. Mercury, however, has a small magnetic field. Magnetic fields are generally accepted to be caused by fluid movements in the mantle caused by heat loss. Somehow Mercury has created a balance that allows the generation of a magnetic field in the absence of mantle movements.

Mercury's average density is 341 pounds per cubic foot, just less than the densest planet, Earth, at 347 pounds per cubic foot. The planet's high density was first discovered by measuring the surprising acceleration of the Mariner mission toward Mercury in the grip of the planet's strangely strong gravity field. Earth is a much larger planet, and so its interior pressure is much higher, pressing the material into much denser forms. With the effects of pressure taken away, Mercury's density is actually much higher than Earth's, and it is therefore the densest planet in the solar system. The high density of the planet is thought to imply that it contains a lot of iron-core, by proportion, than any other planet. Iron is the most common dense inner solar system element, and it is likely to be responsible for high density in Mercury. If Mercury's extra density is due to iron content, then Mercury has a core that is 60 percent or more by mass, the largest core relative to planetary radius of any planet in the solar system. Mercury's core is probably 1,120 to 1,180 miles in radius, nearly 75 percent of the planet's diameter. This is at least twice the metal content of Venus, Earth, or Mars.

Current models for planetary formation suggest that any terrestrial planet should form with a large percentage of silicates, which would make up a thick mantle above the core, as they do on Venus, Earth, and Mars. Mercury may have lost most of its silicate mantle in some catastrophic event after its early formation. The simplest candidate for this event is a giant impact. If a large body struck Mercury more or less directly, then its iron would combine with Mercury's into an unusually large core, and much of Mercury's silicate mantle could have been lost to space. The impacting body may also have been made predominantly of metal, like an iron meteorite. In this scenario, the metal meteorite would combine with the early, small pro to Mercury, creating a planet with an unusually large iron core. Alternatively, the silicate mantle could have been removed by vaporization in a burst of heat from the early Sun. A third possibility is that there may have been strong compositional variation in the early solar system and that in fact Mercury formed with the iron-rich composition it now seems to possess. Finally, Mercury could have differentiated in a particularly oxygen-poor environment. Without oxygen, iron will preferentially form a metal and sink into the core rather than forming iron oxide and combining into silicate mantle materials. In this scenario, Mercury could have started with the same bulk composition and ended with its current structure without having endured a giant impact.

Niloufar Salehi

M.Sc., Energy Systems Engineer

PROFILE

Master's degree in Energy Systems Engineering with Chemical Engineering background, have 3 years successful experience researching at Sharif Energy Research Institute in waste management systems which led me to sustainable consumption, production and circular economy models and concepts, both as a team member and an independent researcher.

CONTACT INFO.

Bågvägen 29, 16756, Stockholm, Sweden

Niloufarsalehi.ns@gmail.com

+46-735576574

https://www.linkedin.com/in/niloufar-salehi-76580b4b/

Technical Skills

- **Programming:** MATLAB, Turbo Pascal
- **Chemical Engineering:** AspenOne Process Modeling (Aspen Plus), Aspen Energy Analyzer, HYSYS
- **Energy Systems Engineering**: ESM (Energy Systems Model), Sima-Pro, COMFAR III

Language

- **Persian:** Native
- **English:** Professional
- **French:** Fair

Education

Sharif University of Technology

M.Sc. In Energy Systems Engineering, Department of Energy Engineering

Sep. 2014 – Jan. 2017

Thesis

Multi-criteria Model Development to Evaluate and Optimize Energy Material Recovery from Municipal Solid Waste of Tehran

- Considering a combination of economic and environmental options simultaneously
- Analysis of Recycling Effect on Resource Recovery
- Using National Emission Factors Based on IPCC Guideline 2006

GPA: 17.78/20

Sharif University of Technology

B.Sc. In Chemical Engineering, Department of Chemical Engineering

Sep. 2009 – Sep. 2013

B.Sc. Project

CO2 Removal of Sour Gas by Tertiary Amines and Their Synthesis

Experience

Senior Researcher

Energy and Environment Group, Sharif Energy Research Institute (SERI), Tehran, Iran

Feb. 2015 – Feb. 2018

- Techno-economic Analysis of Anaerobic Digestion of Sewage Sludge of "Maragheh" City
- Involved in "Green Region Development" Project, "Shif" Island and "Varamin" City, Waste Management Sector
- Involved in "Determination of Iran's Commitments in Relation to Climate Change and Global Warming-Base on COP21" Project, The Academy of Science
- Member of Environmental Committee Aimed to Prioritize Greenhouse Gas Mitigation Technologies in Energy Sector
- Technical Advisor of Hybrid Bicycle Design and Construction Competition

English Teacher

Kish Language Institute

Apr. 2014 – Apr. 2015

- Teaching to adults and teenagers

Intern in Control Unit of Catalyst Transformation

Tehran Oil Refinery

July – Sep. 2012

Niloufar Salehi

Niloufarsalehi.ns@gmail.com

+46-735576574

Research Interests

- Climate Change
- Circular Economy
- Energy Systems Analysis & Simulation
- Life-Cycle Analysis
- Sustainable Consumption and Production
- Mathematical Programming and Optimization
- Bioconversion, Biofuel Production & Waste Recycling and Resource Recovery
- Multi-Criteria Decision-Making to Sustainable Environmental and Energy Planning

Publication

Exergy and Life Cycle-Based Analysis. In: Hussain C. (eds) Handbook of Environmental Materials Management. Springer, Cham, January 2018

Selected Course Projects

- Feasibility Study, Design and Techno-Economic Analysis of Combined Heat and Power Generation from Incineration of Sharif University's Solid Waste, Dr. Avami, Fall 2015
- Exergy Analysis of Environmental Pollution, Dr. Saboohi, Fall 2015
- Tehran's Municipal Solid Waste Management Assessment by Zero Waste Index, Dr. Roshandel, Spring 2015
- Conceptual Design of Acetaldehyde Production process, Dr. Avami, Spring 2015
- Thermodynamic Operation Assessment of Air Cooled Heat Exchangers, Dr. Sattari, Fall 2015
- Effect of LED Lamps Usage in Household Sector on Iran's Energy Balance, Dr. Saboohi, Fall 2015
- Thermal Design of Gas Heat Exchanger and Demethanizer Tower, Dr. Farhadi, Spring 2013
- Conceptual Design of Glass Production, Dr. Rashtchian, Spring 2013

References

- **Yadollah Saboohi**, My M.Sc. Supervisor, Professor of Energy Systems Engineering Department, Sharif University of Technology saboohi@sharif.edu
- **Alireza Bazargan,** Assistant Professor of Civil Engineering Department K.N. Toosi University of Technology, alirezabazargan@kntu.ac.ir

Graduate Courses

- Energy System Analysis, 18.1/20
- Advanced Mathematical Programming, 17.7/20
- Energy System Modelling, 17.3/20
- Conceptual Process Design, 17.3/20
- Electric Energy System Analysis, 19/20
- Advanced Energy Conversion, 17.2/20
- Energy Recovery from Wastes, 17.7/20
- Exergy, Energy System Analysis and Optimization, 19/20

Honors & Awards

- **Ranked 3rd** among M.Sc. Students in Energy Engineering, Sharif University of Technology, Tehran, Iran
- Member of "Behab" Team which won the **1st place** in Sharif Start-Up Trigger[4], Sharif Event on Water and Energy Nexus (Enerwat 2016), Tehran, Iran
- **Ranked 180** among 5,000 Participants in National Graduate Examination for Chemical Engineering, Iran
- **Ranked 1480** among 260,000 Participants in National Entrance Examination for Iranian Universities, Iran

Volunteer Experiences

Scientific Conferences Organizer (Dec. 2014 – Dec. 2015)
"Pouyesh" Scientific Group, Energy Engineering Department, Sharif University of Technology

Financial Manager (Apr. 2011 – Apr. 2012)
"Kimia" Scientific Group, Chemical Engineering Department, Sharif University of Technology

Student Executive Committee Member (Jan. – Feb. 2012) & (Oct. – Nov. 2011)

- 4th National Conference of Health, Safety and Environment, Chemical Engineering Department, Sharif University of Technology
- 20th National Food Conference, Chemical Engineering Department, Sharif University of Technology

Foreign Journal Editorial Member (Sep. 2011 – Dec. 2012)
"Farayand" Magazine, Chemical and Petroleum Engineering Scientific Journal, Student Premier Magazine in in 5th National Festival of "Harkat"

Career Advisor (Sep. 2011)
Educational Institute "Qalamchi"

Polar Dinosaurs

Once there was an idea that dinosaurs were cold-blooded and only thrived in the swamps and wetlands of tropical climes. But the more we look, the more we realize dinosaurs were found in as many different kinds of habitats as birds and mammals are today. Polar dinosaurs, probably warm-blooded and feathery, were thriving roughly 70- -100 million years ago in great polar forests, of which there is no modem equivalent. Fossils of such dinosaurs have been discovered in Siberia, specifically at the Kakanaut River on the Kamchatka Peninsula. The dinosaur remains found here over the years are mostly teeth, but also include some bones, and they reveal the presence of a variety of species. All of these remains are from the very end of the Cretaceous, 66- 68 million years ago, just before the mass extinction event that saw the extinction of the non-bird dinosaurs. "The material was fragmentary but showed that the Arctic dinosaurs were very diverse," says Pascal Godefroit, an expert on early birds and birdlike dinosaurs who has been involved in the work at the Kakanaut River.

The Kakanaut finds have been important in understanding the end-Cretaceous extinction event. This is in part because Kakanaut was within the Arctic Circle and just 1 ,600 kilometers from the North Pole at that time. Conditions then were warmer than today, but mean annual temperatures were still around 10 degrees Celsius, and there would have been frequent spells below freezing and many months of darkness. Dinosaur fossils found farther east across the Bering Sea in Alaska were left by creatures that endured similar climatic conditions, but some experts have argued that they migrated south in the winter, en masse, to avoid the coldest months. At Kakanaut, researchers found fragments of eggshell from hadrosaur and theropod dinosaurs, which suggested that these animals were breeding in polar regions and living there year-round.

Many scientists argue that the dinosaurs went extinct as a result of Earth's collision with a massive asteroid or comet about 66 million years ago that created the Chicxulub Crater on Mexico's Yucatan Peninsula. Other scientists have claimed that falling global temperatures had led to a decline in dinosaurs around the world in the period before the impact (collision). However, the rich fauna (animal life) found by Godefroit and his colleagues suggests that not only had a diverse ecosystem of dinosaurs persisted in the Arctic in the L ate Cretaceous, but also that they were thriving in very cold conditions. The herbivores here must have fed on plants, such as conifers, that remained green year-round and also taken advantage during summer months of a profusion of nutritious fresh growth thrown out by plants bathing in light 24 hours a day.

"For the first time we have firm evidence that these polar dinosaurs were able to reproduce and live in these relatively cold regions. There is no way of knowing for sure, but dinosaurs were probably warm- blooded just like modem birds, which are the direct descendants of dinosaurs," Godefroit told reporters when his findings were published in the German journal Naturwissenschaften in 2009. "The dinosaurs were incredibly diverse in polar regions- as diverse as they were in tropical regions. It was a big surprise for us."

Rather than dinosaurs slowly dying out due to climate change in the period before the impact, Godefroit believes that the discovery backs up the idea that dinosaurs were killed off in a rapid and brutal fashion by cataclysmic conditions that swept the world following the Chicxulub impact. Debris in the atmosphere may have blackened the skies for several years, killing off plants and destroying the food supply- -particularly as large herbivores (plant eaters), such as sauropod dinosaurs, required vast quantities of plant matter to fuel their massive bulk. Starved for meat, the flesh eaters would eventually have succumbed too, as herbivores disappeared.

It is likely that a combination of factors led to the demise of the non-bird dinosaurs, but the precise explanation remains a fascinating and enduring mystery. It had been thought that Siberia had a paucity of fossils in comparison to its southern neighbors Mongolia and China, but the recent discoveries suggest this may not be true.

Dating the Arrival of Humans in North America

Developed in the mid-twentieth century by Willard Libby and Jim Arnold, radiocarbon dating (used for dating organic materials) brought the chronology of North American prehistory into sharp focus. The method provided accurate ages for the deposits of organic material left behind by the last great ice sheets that covered the continent during the Pleistocene Ice Age that began approximately 1.6 million years ago. One such deposit was determined to be 11 ,400 years old. Radiocarbon was also used to date the traces (indications of their presence) left by America's earliest human inhabitants, enabling scientists to map out the movements of people and glaciers and to investigate the interrelationships between the two. Libby's first attempts to analyze an ancient North American site included what archaeologists call the Folsom culture.

Folsom is a small town in northeastern New Mexico. In 1926-27, arrowhead-like stone points were found there, mixed together with bones from a now-extinct type of bison, a discovery that caused great excitement because it placed humans in New Mexico during the last glacial period, when bison were abundant. Beyond that general observation, however, there was no way to date the site. Eventually, additional "Folsom" sites were discovered in other regions, all characterized by the same distinctive stone points. A few of these were in places that could be correlated with specific glacial deposits, which, through a fairly weak line of reasoning, were thought to be between 10,000 and 25,000 years old. Most workers favored the older end of the range.

Arnold and Libby included a charcoal sample linked to the Folsom culture in their first published list of radiocarbon dates. The result was a surprise: 4,283 +/- 250 years B.P. (before the present). This was clearly much younger than any of the earlier estimates suggested, and, if the date held up, it would mean that what appeared to be one of the oldest Native American cultures was actually quite recent. Although they were confident about their analysis procedures, Libby and Arnold were suspicious of the result and wondered if the sample had been contaminated with young carbon, or if there was some other difficulty, they were unaware of. In the end it turned out to be a classic case of improper sampling, and an example of the importance of careful field documentation. When the charcoal was collected in 1933 (it had been stored away from then until the analysis), it appeared to be lying within a soil layer that contained both animal bones and the distinctive Folsom stone points. But the unexpectedly young age prompted reexamination of the site, and it was determined that the charcoal came from a channel that cut into and through the older layers. Although it appeared to be at the same level as the ones and stone points, it was actually much younger. Once this problem was recognized and new samples from this and other sites were analyzed, it became clear that the most reliable Folsom ages fell in the range of 10,000 to 11 ,000 B.P.

However, it was also discovered that Folsom sites are not the oldest evidence for humans in North America. At some localities, slightly different varieties of stone hunting points occur; initially it was thought that these were simply regional variations, or perhaps weapons used for hunting different types of animals. But, in some places- notably at Clovis, New Mexico-they appear in layers that lie beneath the typical Folsom points. This indicated that they were older, and soon archaeologists began to distinguish between Clovis and Folsom cultures. Obviously, Clovis sites became another target for radiocarbon dating, and the results confirmed their antiquity. Clovis sites consistently gave dates that were a few hundred years older than those characterized by the Folsom artifacts, and there seemed to be little or no overlap between the two cultures.

With these results, the radiocarbon dates of both glacial deposits and archaeological sites in North America seemed to be painting a consistent picture. As the last severe glaciation of the Pleistocene Ice Age lost strength, early people spread into the United States. Clovis people were the first widespread hunters, making distinctive stone points for their weapons and hunting large game such as mammoths. Within a few hundred years, however, a new culture appeared, making smaller and finer stone points and apparently taking over from its Clovis predecessors as the dominant hunters in North America.

Industrialized Moths

When most people think of evolution, they imagine slow, incremental changes in a population of plants or animals. But evolutionary changes can also appear suddenly. This is illustrated by the most famous example of natural selection at work- the changing coloration of the peppered moth (Biston betularia). The story of this population of light-colored moths that turned dark and then light again in response to environmental change is an elegant example of Darwinian evolution at work. What happened to the peppered moth reminds us how easily populations can adapt to new conditions in times of environmental stress.

The peppered moth, easily recognizable by its distinctive off-white wings peppered with black specks, is common in England, where its preferred daytime environment is light-grey-lichen-covered trees. The species, which includes a variety of color gradations from dark to light, has always been popular with insect collectors because it's easy to preserve and because the specimens generally maintain their color after death. Before 1850, most of the specimens netted by English collectors were a speckled light grey-roughly the color of the lichens (organisms that often cover the bark on trees). But then the collectors noticed a change: increasingly, the moths they captured had dark wings, and many of them were almost totally black. British entomologists (scientists who study insects) were fascinated by this rapid change in the peppered moth population, especially when they realized that it was almost entirely restricted to polluted industrial areas. They began to realize that what they were seeing was the process of evolution through natural selection.

Because the light grey moths resembled the lichens on the trees, they were relatively safe from their main daytime predators, birds. But the darker moths were easy to see, and this disadvantage limited the percentage of genes for dark coloration in the peppered moth gene pool. When, in the mid-nineteenth century, industrial pollution in England began turning the lichen on the tree trunks black, the darker moths were suddenly protected while the light-colored moths became vulnerable. As more and more dark moths survived, the gene pool shifted toward darker coloration. Similar examples of industrial melanism, a darkening in color linked to a rise in industrialization, were identified in Europe, Japan, the United States, and Canada.

For about a century, dark peppered moths continued to dominate the populations. Then, in the 1950s, scientists noticed that the gene pool was rapidly shifting back toward the lighter coloration. As a result of antipollution legislation, the air was cleaner, the lichen on the tree trunks was becoming lighter, and so were the moths. In fact, this second shift in the gene pool was so radical that many entomologists believe the genes for dark coloration may disappear completely, and that soon the dark peppered moths will be extinct.

The story of peppered moth evolution is so simple that many evolutionary biologists and entomologists have wondered if it could possibly be true, and several of them have tested it. Although the results of their experiments have been reported in the popular press as falsifying Darwinian evolution, what they actually revealed both supports the main points of Darwinian evolution and emphasizes its complexity. The coloration of the peppered moth population definitely shifted in response to environmental factors, but almost every aspect of the process was more multifaceted than anyone thought. During the day peppered moths rest on a number of locations besides tree trunks. Moths' survival is less dependent on their coloring than was originally assumed. It also turns out that the changes in the peppered moth gene pool didn't parallel the changes in pollution levels as closely as had been reported- -migration, as well as bird predation, also had an effect on the ratio of dark to light forms. In short, far from illustrating the elegant simplicity of natural selection, the story of the peppered moth highlights its elegant complexity, and reminds us that even simple evolutionary changes aren't easy to trace.

Brick Technology in Mesopotamia

One of the earliest civilizations was that of Mesopotamia (part of the present-day Middle East), including Sumer and Assyria. Some of its buildings survive to this day. Sun-dried bricks made of mud and straw were its main building materials, as river mud was found in abundance along the Tigris and Euphrates Rivers. Here the scarcity of stone may have been an incentive to develop the technology of making oven-fired bricks to use as an alternative. To Strengthen walls made from sun-dried bricks, fired bricks began to be used as an outer protective skin for more important buildings like temples, palaces, and city walls and gates. Making fired bricks is an advanced pottery technique. I Fired bricks are solid masses of clay heated in ovens to temperatures of between 950 and 1,150 degrees Celsius, and a well-made fired brick is an extremely durable object. Like sun-dried bricks they were made in wooden molds, but for bricks with decorations special molds had to be made. Unlike the river mud used for sun-dried bricks, the clay for proper bricks needed to be carefully prepared, and the building of an oven, finding suitable fuel, and controlling oven temperatures required a professional level of skill and know-how. This is perhaps the reason why the use of fired bricks came in gradually over time.

The technical complexities involved in making fired bricks may explain why one of the first recorded instances of pieces of fired clay, used as a protective covering for mud-brick walls at the Sumerian city of Uruk in southern Mesopotamia in 3600- 3200 B.C., did not involve bricks but small clay cones. Uruk was one of Mesopotamia's first cities and had an important temple complex at its center known as the Eanna precinct. Here was found the Limestone Temple, which was connected to a second temple via a courtyard. In this courtyard was a terrace with massive pillars made of mud and bundled reeds. To protect them from weathering, thousands of nail-shaped ceramic cones were pushed into an outer bed of clay. The cones have flat tops and were painted red, white, or black and arranged in such a way that they formed geometric patterns. What is so significant about this is that here we find, perhaps for the first time, fired clay as a practical protection against the environmental effects of rain and wind, safeguarding the columns from deterioration as well as creating an ornamental surface decoration.

Over time fired bricks also came to be used in Uruk as a building material and for decorative purposes. An example is the temple of Inanna, the goddess of fertility, built by the king Karaindash at the end of the fourteenth century B.C. The exterior of the temple was constructed of fired bricks with alternating projections (structures extending from the temple) and recessions (hollow areas). In the recessions were brick statues of gods holding vessels from which flowed streams of water.

Assyrian kings used fired bricks for the construction of ziggurats (a series of rectangular terraces of decreasing size), palaces, and gateways. Some fine examples have survived from the city of Nimrud in northern Mesopotamia, where in the ninth century B.C. temples and palaces were decorated with fired bricks during the reigns of the Neo-Assyrian kings Ashumasirpal II (883- 859 B.C.) and Shalmaneser III (859- -840 B.C.). A glazed tile was found at Nimrud at one of the palaces, which shows a king attended by servants and bodyguards, and probably represents Ashumasirpal II, who rebuilt Nimrud during his reign. Similar compositions of Assyrian kings attended by their servants can also be found on stone-relief carvings of that time.

Bricks with cuneiform script provide important written information about the history of the period. Cuneiform is the oldest known form of writing and dates from around 3200 B.C. Text was imprinted with a small stick or reed pen into the soft clay, which was then left to harden in the sun, or sometimes fired to become a permanent record of the royal, religious, military, or economic history of the various Mesopotamian kingdoms. Cuneiform became the basis for all subsequent Sumerian, Akkadian, Assyrian, Babylonian, and Persian writing. It was frequently used to record royal deeds, as on the clay tablets excavated at the ziggurat attached to the temple of the god Ninurta, a prominent building in Nimrud.

The Early History of Jupiter

The Sun and all planets in the solar system formed from the same cloud of interstellar material known as the solar nebula. However, the size and composition of the four outer Jovian planets (Jupiter, Saturn, Uranus, and Neptune) radically differ from those of the four terrestrial planets, which are closest to the Sun (Mercury, Venus, Earth, and Mars). Whereas the inner planets are composed mainly of heavier elements, the Jovian planets are much larger and contain a much higher proportion of the light gases hydrogen and helium. Jupiter, with its mass of around 300 times that of Earth, is the largest planet in the solar system and is the Jovian planet closest to the Sun.

It is generally believed that Jupiter's formation began with the accretion (the coming together of material under the influence of gravitation) of a solid core. This core grew by collision and sticking of dust, ice, rocks, and larger bodies- -a process similar to the accretion of Earth. Jupiter, however, formed outside, or beyond, the “snow line," a special place in the solar system where water vapor condensed (solidified) to form ice grains, and the presence of “snow" in this region would enhance the density of solid matter and accelerate the accretion process. The mystery is why the proto-Jupiter (early Jupiter) grew so rapidly. Apparently, Jupiter grew to a mass of 15 Earths before Mars grew to 10 percent of an Earth mass. Planetary scientist David Stevenson has suggested that outward migration of water vapor and condensation at the “snow line" may have provided larger concentrations of solid matter at this location, thus speeding up the formation of the early Jupiter.

Jupiter 's growth to a giant planet began when the rock-ice core mass reached 15 Earth masses. At this mass, the gravity of the core can pull in and hold hydrogen and helium, the light gases that account for 99 percent of the mass of the nebula. When this gas accretion process begins, it is very dramatic because the rate of accretion of gas is proportional to the square of the mass already accreted. In other words, the bigger it gets, the faster it grows. If gas could be continually fed to it, it would consume the Universe in a relatively short time! What actually happens is that Jovian planet formation depletes its feeding zone (the area from which a forming planet can accumulate material) of matter, which in turn truncates planet formation. And although the general properties of this process might be modeled, it just seems to have been by chance that our Jupiter formed as it did.

Because Jupiter's gravity efficiently scatters bodies that approach it, it cleans our solar system of dangerous Earth orbit-crossing asteroids and comets (objects orbiting the Sun that are smaller than planets), thus having a beneficial influence on life on Earth. However, it appears that we have been quite lucky that the Jupiter in our solar system has maintained a stable orbit around the Sun. A Jupiter and a giant neighbor like Saturn (the second planet outside the “snow line" and the second largest in the solar system) are a potentially deadly couple that can lead to disastrous situations where a planetary system can literally be torn apart. Recently, it has become possible to use powerful computers to determine the stability of the orbits of Jupiter and Saturn over the lifetime of the planetary system. There are minor chaotic changes but no major changes, and the solar system, at least to a first approximation, is stable over its lifetime. However, this would not be the case if either Jupiter or Saturn were more massive or if the two were closer together. It would also be dangerous to have a third Jupiter-sized planet in a planetary system. In an unstable system the results can be catastrophic.

Gravitational perturbations (destabilizing interactions) among the planets can radically change orbits, making them noncircular. Although instability might start with just two planets, the effects would spread to them all, and the resulting repeated changes in distance between a planet and the central star would prevent any of them from having the persistence of conditions required for a stable atmosphere, ocean, and complex life, such as plants and animals.

The Economic and Cultural Impact of Plant Diseases

Plant diseases have affected people for as long as humans have grown crops. The ancient Romans even had a god, Robigus, whom they needed to placate in hopes of preventing wheat rust, a disease caused by a fungus that affects wheat, barley, and rye stems, which can cause a significant decrease in crop yield. Probably the most famous example of a plant disease was the late blight of potatoes, which devastated the potato crop in Europe and especially Ireland in the mid-nineteenth century. A large proportion of the people in Ireland were completely dependent on potatoes for subsistence, and as a result of the blight, Ireland lost about a quarter of its population to famine and emigration. It is now known that plant diseases are caused by the same types of organisms, or pathogens, that cause human and animal diseases, including bacteria, fungi, and viruses.

The discovery of evidence that plant diseases are caused by microscopic organisms was contemporary to discoveries substantiating the theory that infectious animal diseases are caused by microbes. In 1878, fire blight, a disease of pears and apples, was determined to be caused by a bacterium, a discovery that came only two years after the German physician Robert Koch had confirmed that anthrax, a disease of livestock, is caused by a bacterium. The first evidence that infection with a virus, a pathogen smaller than bacteria, can result in disease came from an experiment with the tobacco mosaic virus, which can infect tobacco as well as tomato plants. The experiment, which was performed in the 1890s, showed that sap from a diseased tobacco plant was still infectious after having been passed through a filter with pore sizes small enough to prevent passage of bacterial cells.

Some plant pathogens, such as the organism responsible for cedar-apple rust, must infect an intermediate, or alternate, host before they can complete their life cycle on the economically important host (in this case the apple). For this disease, the alternate host is red cedar, on which the fungus overwinters. Therefore, the elimination of red cedar trees was advocated to help reduce the incidence of the disease. These kinds of relationships with intermediate hosts are not unique to agents infecting plants. Both yellow fever and malaria require an incubation period of roughly ten days in the mosquitoes that transmit them before infection of new human hosts can occur.

Perhaps the most bizarre example of a plant disease affecting the economy of a country was the appearance of streaks or featherings of darker colors on a lighter-colored background in tulip flowers in Holland. These symptoms became known as color breaking. The bulbs of these tulips, which in reality were infected by a plant virus, became highly prized. Prices for these infected bulbs initially skyrocketed but then suddenly fell in 1637, resulting in the collapse of Holland's economy, which had become excessively dependent on tulip exports. The coffee rust outbreak and its aftermath in Ceylon (now Sri Lanka) in 1875 had a lasting cultural impact. Ceylon had come under British control about 50 years earlier, and most of the island consisted of coffee plantations. The coffee rust fungus destroyed the coffee plants, and the plantations switched to growing tea bushes. As a result, drinking tea became an integral part of British culture.

Prevention of plant diseases is achieved by several methods, such as chemical treatment and altering the plants' genes to make them disease resistant. The first chemical treatment was discovered in the nineteenth century when botanist Pierre-Marie- Alexis Millardet was studying an outbreak of downy mildew, a devastating disease of grapes, on vineyards in the Bordeaux region of France. During his investigation, he noted that, even in infected fields, vines closest to the road were not affected by mildew. He eventually learned that farmers had sprayed the vines near the road with a visible and bitter- tasting mixture of copper sulfate and lime so that passersby would not eat the grapes. Soon this mixture, known as the Bordeaux mixture, became a common means of preventing downy mildew.

Shoaling Behavior

A shoal is any group of fish that remain together for social reasons. A shoal may be a school (a coordinated group with synchronized movements) at some times and a less organized mass at others. Shoaling is of considerable interest because of its prevalence, and various hypotheses have been put forward to explain it: increased efficiency of movement in water (hydrodynamic efficiency), increased efficiency of food finding, reproductive success, and reduced risk of predation.

The idea that shoaling increases the efficiency of swimming applies mainly to schooling. This idea is very appealing, both because of the regular spacing that seems to characterize fish in schools and because fish in shoals tend to be uniform in size. To gain hydrodynamic advantages, however, each fish must maintain rather precise positioning within a school to use the hydrodynamic lift created by its neighbors. By and large, measurements of fish positions within schools in the laboratory have not found this to be true. Nevertheless, the regular spacing of fish in schools observed in the wild indicates that being in a school probably does have a hydrodynamic advantage- -at least for fish that are behind other fish. Some scientists argue that the leadership of a school is constantly changing, because while being immersed in a school incurs hydrodynamic advantages to individuals, leaders of schools are first to find food, which is also advantageous.

Schooling by predators increases the probability of their detecting a school of prey, just as shoaling in planktivores (fish that feed on plankton, the very small plants and animals in the water) increases the probability of detecting suitable patches of plankton. The reason for this is the presence of many searching eyes over a large area. Fish in shoals share information by monitoring each other's behavior. Feeding behavior in one fish quickly stimulates food-searching behavior in others. For planktivores, however, this advantage may be decreased by having to share the patch with many other fish: fish in the rear of a school are likely to encounter much lower densities of plankton than those in the front as well as higher concentrations of waste products from their schoolmates.

For fish that shoal throughout their lifetimes, little energy has to be expended in finding a mate when spawning time comes. Presumably, shoaling also allows fish to closely synchronize their reproductive cycles (through behavioral and hormonal cues). In addition, for fish that migrate long distances to breed, schooling may increase the accuracy of finding the way home, because the average direction sought by fish in a school is likely to be a better choice than that chosen by an individual migrating alone. For tunas, with habitual migration routes, this means that groups of individuals may stay together for long periods.

Shoaling reduces predation risk in two main interacting ways: by confusing predators and by providing safety in numbers. When predators attack, large shoals are particularly advantageous because of the reduced probability that any one individual will be eaten; in any given attack, a smaller percentage of a large shoal will be eaten compared to a small shoal. The confusion effect is based on the common observation that shoaling fish tend to get eaten mostly when they have been separated from the shoal; many attack strategies of predators can thus be best interpreted as efforts to break up shoals so that individuals can be picked off. Shoaling fish are the same size and silvery, so it is difficult for a visually oriented predator to pick an individual out of a mass of twisting, flashing fish and then have enough time to grab its prey before it disappears into the shoal. Shoaling fishes also are fairly uniform in size and appearance, a further factor confusing predators. Individuals that stand out (for example, because of heavy parasite infestations) are quickly picked off. Even when shoals are made of several species, individual fish are similar in size, although the species may remain segregated within the school. Thus, it has been noted that banded killifish, golden shiner, and white sucker shoal together, but that the killifish occupy the top of the shoal, the shiners the middle, and the suckers the bottom. These shoal positions reflect the preferred feeding areas of the three species (surface, mid-water, and bottom, respectively).

Pleistocene Overkill

Around 11,000 years ago, near the end of the Pleistocene era (2.6 million to 11,700 years ago), most megafauna- Hlarge, mostly herbivorous (plant-eating) land animals- -went extinct in North America. Their disappearance has been the topic of much debate, but considerable evidence supports a hypothesis known as “Pleistocene overkill." The idea is that, as humans spread across the continent, they preyed upon large herbivores, such as woolly mammoths, ground sloths, and tortoises, and wiped them out. As originally formulated by the American geoscientist Paul Martin, the large mammals were driven to extinction in a few hundred years in a massive, fast-moving event. A newer version of the hypothesis posits that the extinctions were more gradual, based on evidence that, in some areas, humans and large animals coexisted in the same habitats for long periods of time, despite hunting. However, the end result was the same: extinction of the megafauna. Large animals are more vulnerable to extinction than smaller ones because they cannot hide easily from human predators and because they reproduce quite slowly. It is possible that the large animals were also relatively unafraid of human beings, since they would have evolved for hundreds of thousands of years without humans present. In addition, there is some indication that a rapid shift in climate reduced the habitats of many of the giant herbivores, making them more vulnerable to human predation. Likewise, Australian biologist Tim Flannery suggests humans may have changed the environment through their actions, especially by increasing the frequency of fires.

Not unexpectedly, when the large herbivores disappeared, their natural predators, such as saber-toothed tigers and short-nosed bears, became extinct as well. The large scavenger birds, which had adapted to eating the remains of large animals, also followed them into extinction. The California condor may have held on because it had access to the carcasses of large marine mammals such as whales and sea lions, which did not go extinct at this time. The loss of these giant animals also impacted the diversity of smaller animals. Because abundant large animals (such as mammoths and tapirs) alter plant communities by the way they graze (feed on plants), their disappearance would have caused a shift in the plant communities, resulting in the extinction of many smaller species that depended on the habitats maintained by the large grazers. In fact, there existed a grassland ecosystem in Alaska called the mammoth steppe that disappeared entirely once the woolly mammoth went extinct in that region.

The idea of the Pleistocene overkill is quite controversial, yet the principal alternative to explain the rapid loss of all these giant animals is a drastic climate change that occurred when the last ice age ended near the end of the Pleistocene. Recent fossil data and archeological discoveries increasingly support the idea that the early native peoples were responsible for the extinction of many species. One of the early groups (but not the first) to colonize North America was the Clovis people. At Clovis archeological sites, researchers have found distinctive, beautifully made stone spearheads that would have been well-suited to killing large herbivores. These same Clovis spearheads have been found imbedded in the skeletons of large animals, which strongly suggests that these animals were hunted by the Clovis people. Clovis culture rapidly spread throughout North America, and then abruptly disappeared after about 300 years- -a disappearance that seems to coincide with the extinction of many (but not all) of the large animals of interior North America.

Additional support for the overkill idea comes from the growth rings of mammoth tusks at the time of the mammoths' extinction, which indicate that the animals were eating well and not experiencing the starvation that would normally accompany climate-driven extinction. Also, many of the extinct species of megafauna had already survived several other glacial climate cycles during the previous 100,000 years, and so presumably they could have survived one more. It is worth noting that a similar event occurred in Australia, with early human societies apparently wiping out many species of giant marsupials, as well as on islands throughout the world. If we accept some version of the Pleistocene overkill hypothesis, then we also have to accept the idea that early hunter-gatherer societies were capable of having large and permanent effects on their environment.

The Absence of Snakes in Ireland

According to legend, Ireland in northern Europe has no snakes because a fifth-century monk named Patrick drove the island's snakes into the sea. In reality, Ireland has no snakes because of the interplay between organisms and their physical environment.

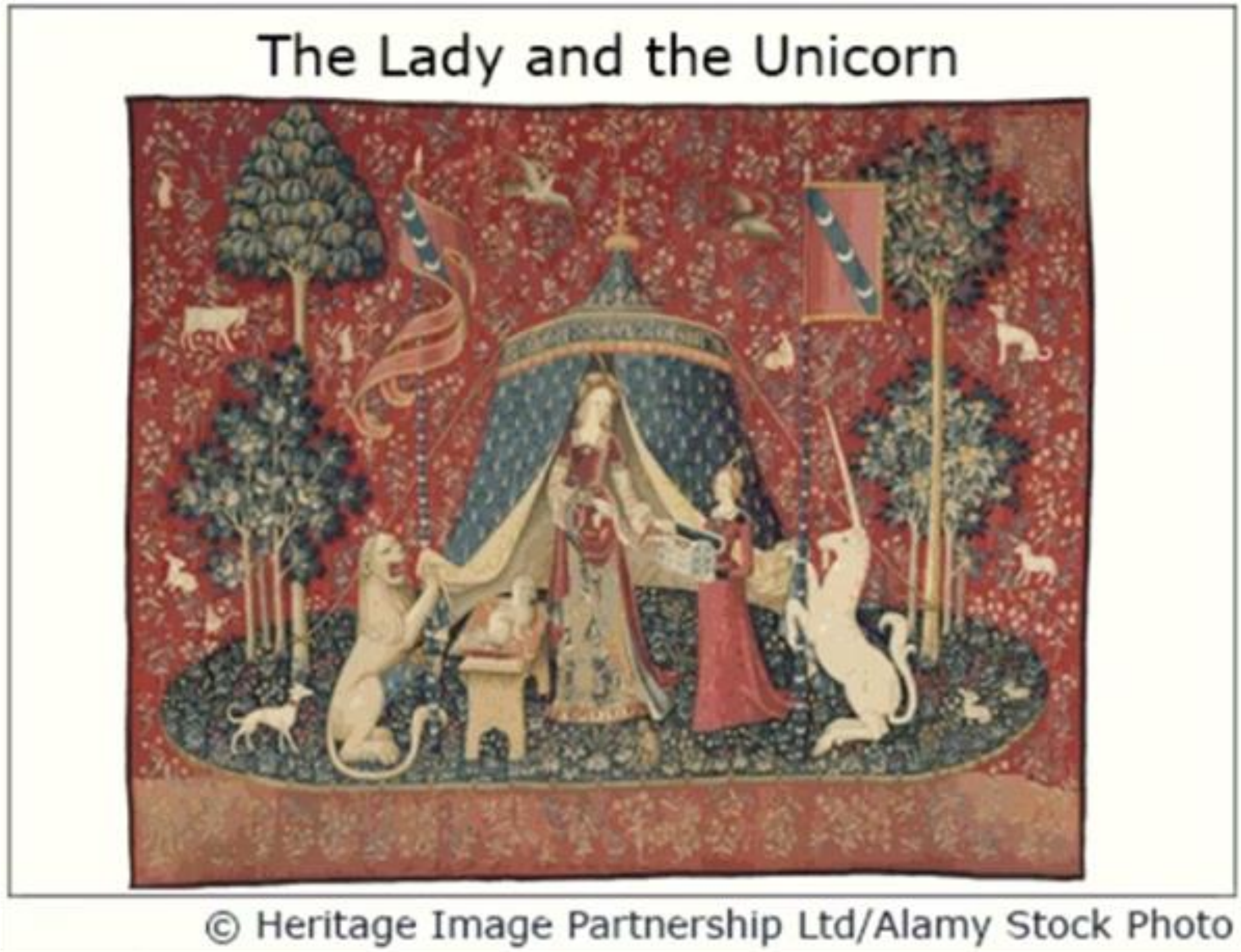
Snakes, like all reptiles and amphibians, are ectotherms (cold-blooded): their body temperature and functionality are dictated by the temperature of their external environment. Species diversity of reptiles decreases rapidly from the equator toward the higher latitudes. In North America, the number of lizard species is highest in the warm desert regions of the southwest and declines continuously as you move northward. The same pattern of species diversity is evident in Europe, where the number of reptile species declines markedly as you move from the warmer Mediterranean coast toward northern Europe and the British Isles. Although one snake species, the European adder (viper), is found above the Arctic Circle in Scandinavia, its unusually northern distribution comes at a cost. The adder has a very limited period of activity, often only three to four months a year. In addition, the species may take up to four years to attain sexual maturity, and females may breed only once every three or four years, using the intervening period to build up the fat reserves necessary to produce offspring.

The progressive decline in solar radiation and temperatures from the tropics to the poles not only reduces the abundance and diversity of reptiles but also has a direct influence on their body size. The reason for these patterns is that heat exchange occurs across the surface of a body, but warming usually occurs throughout the entire body's mass or volume. Large bodies, because of their low surface area to volume ratio, take longer to warm than smaller ones. This physical reality results in upper limits to the size of snakes (and other reptiles) depending on the distance of the snake's habitat from the tropics. All of the large snakes, such as the anaconda and python, are found within the tropic and subtropical regions. Other large reptiles--such as the iguanas, monitor lizards, and the crocodilians (alligators, caimans, and crocodiles)--are likewise limited in their distribution to the warm, a seasonal environments of the subtropics and tropics. The maximum body size for ectotherms declines as you move north and south from the equator. This pattern is the exact opposite of that observed for endotherms (warm-blooded animals), where average body size increases from the tropics to the poles, a pattern referred to as Bergmann's rule. The environmental constraint on the upper limit of body size for ectotherms is at the very heart of the current debate over whether the dinosaurs were cold- or warm-blooded. The fossil record places the dinosaurs well into the northern latitudes, with specimens found in Alaska and Siberia.

As for snakes, the fossil record reveals that the diversity of reptiles and amphibians (terrestrial ectotherms) has always been low in northern Europe. During the Pleistocene epoch (from 2.6 million to 11,700 years ago), a time when the temperature of Earth was considerably lower than that now observed, much of the Northern Hemisphere was covered by glaciers. Although certain refugia (small pockets of land not covered by ice) existed, the massive ice sheet virtually obliterated all life in northern Europe, including Ireland and Britain. As temperatures rose and the glaciers retreated during the Holocene epoch (11,700 years to the present), the region was recolonized by plants and animals, but the cool temperatures of northern Europe proved inhospitable to most snakes. Nevertheless, the island of Britain, similar in climate to Ireland, ended up with three species of snakes that survive there to this day.

Three species of snakes is a paltry sum, but nonetheless, it is more than Ireland has. Why the difference? After the retreat of the continental glaciers at the end of the last ice age, Britain had a land bridge connecting it with the mainland of Europe, whereas Ireland did not. The limited diversity of snakes on the mainland meant that the pool of species available to recolonize both Britain and Ireland was small in the first place. However, Ireland's lack of a land bridge with continental Europe coupled with the limited dispersal abilities of most snake species have to date prohibited their successful recolonization there.

Early European Tapestries



Tapestries, handwoven textiles made with different colored threads to produce designs or images, were highly popular in western and northern Europe between the thirteenth and seventeenth centuries. The nobility at the time usually lived in big, austere, drafty castles, and tapestries provided comfort and warmth. They could be hung on cold walls or used to cover doors and windows. Noble families often did not have a permanent home, instead moving periodically among several locations. Tapestries, which were expensive, were usually packed and moved each time, bringing with them a sense of luxury and familiarity.

Some of the most famous tapestries were woven from wool in Flanders (modern Belgium), a region well-placed for tapestry production. Many of the plants that supplied dyes grew there. The leaves of the woad plant were used to make blue, and madder root provided red. Yellow came from different materials, including onion skins and lemon peels, though most of these yellows faded quickly, thus also affecting the quality of green, which is a mixture of blue and yellow. High-quality wool was not produced in Flanders but was readily imported from nearby England.

Because of a long history of textile production, Flanders already boasted many skilled weavers and dyers when tapestry production began to increase, and this advantage grew in the late fifteenth century because of unrest in Flanders' southern neighbor, France. Originally, Paris had been a center of tapestry production, but the turmoil produced by the Hundred Years' War between England and France (1337- 1453) caused many textile workers to relocate to Flanders or to the city of Arras, which was then a part of the Duchy of Burgundy. Following the death of the duke of Burgundy in 1477, the French king Louis XI conquered the territories around Arras. Its inhabitants, still loyal to Burgundy, then expelled all the French who had lived in the city. In retaliation, Louis attacked the town and, in turn, expelled the inhabitants loyal to Burgundy, including the textile workers, many of whom fled to Flanders. Paris regained some of its strength in tapestry production in the seventeenth century, when Louis XIV established the famed Gobelins workshops there.

Tapestries were initially used mostly for warmth, but their visual aspect later gained in importance. Tapestries often told a story. Many early tapestries were hung in churches and cathedrals; these often-featured biblical scenes and were used for teaching or religious devotion. However, the use of tapestries in churches later declined with the rise of Gothic architecture. Because this new style emphasized light, churches had many more windows than before, reducing the space for tapestries. In homes, tapestries were still prized; they tended to show complex, varied scenes in vivid colors and provided visual unity to rooms. Historical or mythical scenes were popular, including depictions of heroes and heroines from Greek mythology and figures from European legend. Among the most famous tapestries is a set of six called The Lady and the Unicorn， now in the Cluny Museum in Paris. These tapestries, done in the intricate millefleurs design (with a background of many small flowers), all show an elegant lady accompanied by a unicorn and a lion. Five of them likely represented the senses, but the meaning of the sixth, bearing an inscription sometimes translated as “my sole desire," is debatable.

Tapestries were often used as symbols of power and success. A nobleman would sometimes commission a tapestry showing historical or mythical scenes that lent his family legitimacy or commemorating a battle his family had won. Because these tapestry pieces declared status, they were often finer than other kinds of tapestries, sometimes being made with gold or silver thread. One magnificent group of tapestries, called the Apocalypse of Angers , was commissioned by the duke of Anjou (France) in 1375. In Christian narratives, the Apocalypse is the final battle between good and evil, which is symbolized in the tapestry by various images, like battles between angels and beasts. Some of the angels are holding flags showing the Cross of Anjou, a symbol of the duke's territory. The tapestries were t probably designed to represent the duke's wealth as well as to reinforce his authority as a ruler.

Sexual Dimorphism in Lamprologus Callipterus

Lamprologus callipterus is one of over 300 species of cichlid (a freshwater fish) found only in Lake Tanganyika in Africa. Most African cichlids, including L.callipterus , exhibit sexual size dimorphism-the males are larger than the females. What is unusual about L. callipterus is the magnitude of the size difference. The key to understanding the sexual dimorphism of L.callipterus lies in the reproductive behavior of the species, particularly spawning (release of eggs by females and release of sperm by males) and brooding (protection of eggs). Female L.callipterus spawn and brood their eggs in empty snail shells on the lake bottom. In all other shell-brooding cichlids, both sexes enter the shell to spawn and thus both must be small enough to fit into the shells available on the lake bottom. In L. callipterus , however, only the females enter the shell to spawn and brood their eggs. The males remain outside, releasing their sperm only at the entrance to the shell. Consequently, female L. callipterus are small enough to fit into the shells, but the males, unconstrained by the need to enter shells, are much larger.

Spawning and brooding inside shells puts females in an evolutionary dilemma with respect to their body size. In cichlids, as in most fishes, the number of eggs spawned per batch (the fecundity) increases with female body size, and this favors females that grow large. In L. callipterus the correlation between female length and number of eggs spawned is as high as 0.95, which means that more than 90 percent of the variation in fecundity among females can be explained by differences in body length. Further, because fecundity is more a function of body volume than length, it increases disproportionally as body length increases. In one study, for example, an increase in female length from 3 centimeters to 4 centimeters was associated with an increase in fecundity from 59 to 134 eggs. In addition to this huge fecundity boost, larger females are able to exclude smaller females from access to the larger, preferred shells and so gain in the competition for spawning sites. All this suggests that female L. callipterus should be large, but this is where the dilemma comes in. No matter how beneficial large size might be, the scarcity of large shells ultimately limits the size that females attain in a given population. In laboratory experiments females actually adjust their growth rates so that their body size at maturity matches that of the available shells, and in the wild the correlation between shell size and female size explains more than 98 percent of the variation in female size among populations. The ultimate size of female L. callipterus is thus a compromise between growing as large as possible to maximize fecundity and remaining small enough to be able to find suitable shells for spawning and brooding their young.

Finding shells large enough for brooding is so critical for female L. callipterus that, when they are ready to spawn, they generally pay more attention to choosing shells than to choosing mates. Males have adopted a unique strategy to exploit this female choosiness. Rather than trying to attract females from afar with flamboyant displays, male L. callipterus entice females to mate with them by assembling piles of empty snail shells. They set up small territories, less than a meter in diameter, and each territory is centered on a patch of shells. In most areas of the lake these patches of shells are separated by stretches of sandy or rocky bottom devoid of shells. The males create this patchy distribution by picking up and carrying shells from as far away as 20 meters. In addition to clearing the surrounding area of abandoned shells, territorial males frequently steal shells from the shell piles of other males, sometimes with females and offspring inside. When this happens the female and her offspring usually disappear, and the empty shell becomes available for a new female in the territory of the male relocating the shell. The effect of this remarkable shell transporting is to create artificial clumps of shells surrounded by unsuitable stretches of lake bottom. When females are ready to spawn, they have to visit the shell piles of territorial males to find suitable shells, thus creating spawning opportunities for males.

Weather Forecasting

Because the many variables that affect weather are constantly changing, meteorologists have used high-speed computers to devise atmospheric models that describe the present state of the atmosphere. These are not physical models that paint a picture of a developing storm; they are, rather, mathematical models consisting of dozens of mathematical equations that describe how atmospheric pressure, winds, and moisture will change with time. Actually, the models do not fully represent the real atmosphere but are approximations formulated to retain the most important aspects of the atmosphere' s behavior.

Computer-generated forecast charts using these models to interpret recorded weather data are known as prognostic charts, or simply progs. At present, there are a variety of models, and hence progs, from which to choose. One model may work best in predicting the position of troughs (elongated areas of low barometric pressure) in the upper atmosphere. Another may forecast the position of surface lows quite well. And some models even forecast the state of the atmosphere 384 hours (16 days) into the future.

A good forecaster knows the idiosyncrasies of each model and carefully scrutinizes all the progs. The forecaster then makes a prediction based on the guidance of the computer. This prediction is a personalized practical interpretation of the weather situation given the local geographic features that influence the weather within the specific forecast area. Since the forecaster has access to hundreds of weather maps each day, why is it that forecasts are sometimes wrong?

Unfortunately, there are flaws inherent in the computer models that limit the accuracy of weather forecasts. Computer-forecast models idealize the real atmosphere, meaning that each model makes certain assumptions about the atmosphere that may be accurate for some weather situations and be way off for others, so a forecaster who bases a prediction on an inaccurate prog may find a forecast of rain and wind turning out to be a day of clear and colder weather.

Another forecasting problem arises because the majority of models are not global in their coverage, and errors are able to creep in along a regional model's boundaries. A model that predicts the weather for North America may not accurately treat weather systems that move in along its boundary from the Pacific Ocean. Obviously, a good global model of similar sophistication would require an incredible number of computations.

Even though many thousands of weather observations are taken worldwide each day, there are still regions where observations are sparse, particularly over the oceans and at higher latitudes. Temperature data recorded by satellites have helped to alleviate this problem, as the newest satellites are able to provide a more accurate profile of temperature for the computer models. Since the computer's forecast is only as good as the data fed into it and since observing stations may be so far apart that they miss certain weather features, a denser network of observations is needed, especially in remote areas, to ensure better forecasts in the future.

Another forecasting problem is that computer models cannot adequately interpret many of the factors that influence surface weather, such as the interactions of water, ice, and local terrain on weather systems. Some models do take large geographic features such as mountain chains and oceans into account, while ignoring smaller features such as hills and lakes. These smaller features can have a marked influence on the local weather. Given the effect of local terrain, as well as the impact of some of the other problems previously mentioned, computer forecasts presently do an inadequate job of predicting local weather conditions, such as surface temperatures, winds, and precipitation.

Even with a denser network of observing stations and near-perfect computer models, there are countless small, unpredictable atmospheric fluctuations that fall under the heading of chaos. These small disturbances, as well as small errors (uncertainties) in the data, generally amplify with time as the computer tries to project the weather further and further into the future. After a number of days, these initial imperfections (errors) tend to dominate, and the forecast shows little or no skill in predicting the behavior of the real atmosphere.

Class Structures in Postwar Europe

Rapid economic growth went a long way toward creating a new society in Europe after the Second World War. European society became more mobile and more democratic. Old class barriers relaxed, and class distinctions became fuzzier.

Changes in the structure of the middle class were particularly influential in the general drift toward a less rigid class structure. In the nineteenth and early twentieth centuries, the model for the middle class had been the independent, self-employed individual who owned a business or practiced a liberal profession such as law or medicine. Ownership of property- -very often inherited property- -and strong family ties had often been the keys to wealth and standing within the middle class. After 1945 this pattern declined drastically in Western Europe. A new breed of managers and experts replaced traditional property owners as the leaders of the middle class. Ability to serve the needs of a big organization largely replaced inherited property and family connections in determining an individual's social position in the middle and upper-middle classes. At the same time, the middle class grew massively and became harder to define.

There were several reasons for these developments. Rapid industrial and technological expansion created a powerful demand for technologists and managers in large corporations and government agencies. Moreover, the old propertied middle class lost control of many family-owned businesses, and many small businesses simply went out of existence as their former owners joined the ranks of salaried employees. Top managers and ranking civil servants therefore represented the model for a new middle class of salaried specialists. Well paid and highly trained, often with backgrounds in engineering or accounting, these experts increasingly came from all social classes, even the working class. Pragmatic and realistic, they were primarily concerned with efficiency and practical solutions to concrete problems. Managers and technocrats, of whom a small but growing number were women, could pass on the opportunity for all-important advanced education to their children, but only in rare instances could they pass on the positions they had attained. Thus, the new middle class, which was based largely on specialized skills and high levels of education, was more open, democratic, and insecure than the old propertied middle class.

The structure of the lower classes also became more flexible and open. There was a mass exodus from farms and the countryside, as one of the most traditional and least mobile groups in European society drastically declined. Meanwhile, the industrial working class ceased to expand, and job opportunities for white-collar and service employees grew rapidly. Such employees bore a greater resemblance to the new middle class of salaried specialists than to industrial workers, who were also better educated and more specialized.

European governments were reducing class tensions with a series of social security reforms. Many of these reforms-such as increased unemployment benefits and more extensive retirement pensions- simply strengthened social security measures first pioneered in Germany before the First World War. Other programs were new, such as comprehensive national health systems directed by the state. Most countries also introduced family allowances- direct government grants to parents to help them raise their children. These allowances helped many low-income families make ends meet. Most European governments also gave maternity grants and built inexpensive public housing for low-income families and individuals. These and other social reforms provided a humane level of well-being. Reforms also promoted greater equality because they were paid for in part by higher taxes on the rich.

The rising standard of living and the spread of standardized consumer goods also worked to level European society, as the percent of income spent on food and drink declined substantially. For example, the European automotive industry expanded phenomenally after lagging far behind the United States since the 1920s. In 1948 there were only 5 million cars in Western Europe, but in 1965 there were 44 million. Car ownership was democratized and came within the range of better-paid workers. Europeans took great pleasure in the products of the gadget revolution as well. Like Americans, Europeans filled their houses and apartments with washing machines, vacuum cleaners, refrigerators, dishwashers, radios, televisions, and stereos. The purchase of consumer goods was greatly facilitated by installment purchasing, which allowed people to buy on credit. With the expansion of social security safeguards, reducing the need to accumulate savings for hard times, ordinary people were increasingly willing to take on debt.

What Made Venetian Art Different

Venice was a major center for art in the Renaissance, yet it produced a style of art and architecture markedly different from that of other Italian cities like Rome or Florence. It owed this uniqueness to a thriving commercial empire: between the ninth and fifteenth centuries, Venice gained control over Mediterranean shipping and so became the most important market for goods from all over the known world. Because of the presence of traders and immigrants from far and wide and the experiences of its own merchants and diplomats, Venice benefitted from a huge range of cultural influences on its art and architecture.

Perhaps the biggest external influence on Venice was the East-Byzantium and the Islamic lands- seen especially in the architecture of its two most prominent buildings, Saint Mark's Basilica and the Doge's (Duke's) Palace. The basic interior plan of St. Mark's was derived from the sixth-century Church of the Apostles in Constantinople, and so, unlike most Italian churches of its time, it is in the form of a Greek cross, with four extensions or “arms" of equal length. The arrangement of domes above the arms is similar to that in Islamic architecture in Egypt. However, the architecture of St. Mark's is not a simple imitation of Eastern antecedents. Although the four arms of the church are of equal length, they vary in height and width, and the domes were raised and enlarged in the thirteenth century to make them more prominent in the city's skyline. The adjoining Doge's Palace, which was the political center of Renaissance Venice, is mostly an example of the Gothic style, which originated in northern Europe. Yet it too has Eastern elements. The details of its roof ornamentation, the patterns of its pink and white marble facing, and its pointed arches are reminiscent of major buildings in Egypt, one of Venice's most important trading partners.

St. Mark's interior has unusually fine examples of Byzantine ornamentation, especially its gold mosaics. Mosaics are decorations or images made up of small, colored pieces of glass, stone, or tile, called tesserae. Venetian tesserae were primarily made of glass, which was easily available from Venice's advanced glass workshops, like those on the island of Murano. An important effect of glass is that the color changes as the light in a room changes. This effect, combined with the curving surfaces of domes and vaults (arched ceilings), and the deliberate setting of tesserae at different angles to catch the light, produces a striking shimmering quality to St. Mark's. The heavy use of gold had long been established for religious images in Byzantium, but in Venice it was extended beyond the mosaics of St. Mark's to other artistic media. Textiles were often produced with gold thread. Painters such as Giovanni Bellini (C. 1430-1516) used gold in their paintings, and artists came from other regions of Europe to learn the techniques for using powdered gold. The Ca' d'Oro, a famous Venetian palace whose name means “House of Gold," even had substantial areas of gold on its exterior.

Eastern influences were not all that made Venetian art special. The city's commercial connections resulted in its being among the first to adopt new techniques of oil painting that were developed in Flanders (modern Belgium) in the late fifteenth century. Venice went on to become famous for its great oil painters, such as Veronese (1528- 1588), Giorgione (1477/1478- 1510), and Titian (C. 1488/1490- 1576). These painters adopted some stylistic elements of northern European painting, including the conventions of portrait painting and the use of mountains and forests in landscapes that typified German painting. At about the same time that the Flemish were making advances in oil painting techniques, Johannes Gutenberg (1395- 1468) was perfecting the technology of movable type in Germany. Venice quickly became one of the most important cities for book manufacturing in Europe. It had already produced a limited number of prints, but now the resulting upsurge in publishing created a greater need for illustrations, so there were increased opportunities for artists to produce woodcuts and engravings for books.

The Age of Sailing in Europe

Sailing began long before ships were capable of crossing entire oceans. Phoenicians are known to have sailed from the area of present-day Lebanon and Israel to the Atlantic Ocean and down the west coast of Africa over two thousand years ago. But the Phoenicians' boats (as well as those of other early Mediterranean sailors) were primitive in design and difficult to sail. These early galley ships had one mast and a single square sail, which meant they could sail well only downwind. Early sailors also lacked any reliable means of navigation on open seas. Once out of sight of land, sailors had nothing but the stars to guide them home, and without accurate timepieces and navigational tools, navigating by the night sky was a daunting and uncertain prospect. At that time, almost all travel by sea was within a single biogeographic province: from one end of the Mediterranean Sea to the other, or around the Baltic Sea, or from one South Pacific Island to the next.

“Blue water sailors"- -those willing to sail out of sight of land- -had to wait for several major developments in sailing technology before they could make open ocean voyages with any reasonable expectation of being able to return home. European sailors received an important tool when the compass (a Chinese invention) became available, probably in the tenth century A.D. Together with maps, the t compass allowed sailors to break away from coastal routes and set out across the open ocean without losing track of where home lay. But if they were to make long voyages safely and regularly, sailors also needed boats they could control in contrary winds. Single-masted boats with square sails hanging perpendicular to the boat cannot sail upwind efficiently because the sail cannot be adjusted over a wide range of angles to the wind. Technical improvements that began to address this problem included arranging the sail fore-and-aft (that is, with the sail parallel to the length of the ship) and using several sails on several masts or a triangular sail at the front of the boat. Replacement of the ancient steering oar at the rear of the ship with a much larger and more easily controllable rudder board also helped provide better control. Once these crucial innovations had been made around 1300 A.D., ship design evolved rapidly and sailors began to set their sights on more distant shores.

But even with compasses and improved ships, sailors still needed a working knowledge of the wind patterns that would carry them across the oceans- and back home. Between the 1330s and the 1520s, Portuguese and Spanish sailors discovered and mapped broad circles of wind that provided reliable routes across the Atlantic and back. A Portuguese sailor found the Canary Islands off the northwest coast of Africa in 1336 by following the northeast trade winds southwest from the Iberian Peninsula. Luck and insight eventually led the Portuguese to discover that the easiest way back was not to sail slowly and painfully against the wind up the African coast, but to sail west out to sea until they hit the westerlies, another group of winds that carried them northeast back to Europe. Other Portuguese and Spanish sailors mapped most of the rest of the great circulatory patterns of winds by the end of the sixteenth century, making possible the first controlled movement of large numbers of people and cargo back and forth between Eurasia (the combined continental landmass of Europe and Asia) and the Americas.

Several elements conspired to make Europe the place where this long series of inventions and discoveries- -compasses and star charts, ships that could sail into the wind, and knowledge of wind patterns- came together to create a power that could sail around the world. Europe's physical position on the globe was as critical a factor in its transoceanic explorations as were its technologically advanced ships. Compared to the Pacific, the Atlantic was a manageably sized ocean to cross on a regular basis, and the Americas (once discovered) were a much more profitable destination than the scattering of islands in the Pacific or the emptiness south of the Indian subcontinent. Perhaps even more important, the social and religious values that characterized many European cultures both impelled their transoceanic voyages and fueled the innovation that made these voyages possible.

Euglena: Ecosystem Engineers

In 1995 an extremophile, or organism that thrives in extreme environments, was discovered in the United States in a lake outside of Butte, Montana. The body of water where it lives, known as the Berkeley Pit, is a former open-pit copper mine that is slowly filling with groundwater that dissolves heavy metals like arsenic, cadmium, zinc, and aluminum from the mine's 2,000-foot-deep surface; the dissolved metals mix with the water to form a toxic (poisonous) solution. The Berkeley Pit is the largest hazardous waste site targeted for cleanup in the United States. As 2.6 million gallons of groundwater seep into the pit every day, the toxic brew edges ever closer to spilling into the Clark Fork River, which makes its way to the Columbia River and out to the ocean, representing a massive-scale ecological threat to the fisheries and communities in these areas.

Though the water in the pit is as acidic as lemon juice and considered inhospitable to life, a novel species has emerged and is slowly but surely changing the lake's toxic brew to one more habitable. An analytic chemist studying the lake noticed a clump of green slime floating in the lake and brought a sample to Dr. Grant Mitman of Montana Tech. Mitman brought his colleagues Don and Andrea Sterile into the lab, and they identified the slime as a colony of Euglena mutabilis , a single-celled organism that forms algae-like mats. How Euglena came to grow in the Berkeley Pit is unknown, although some attribute its spread to a flock of 350 snow geese that landed (and subsequently died) in the lake several months prior to the discovery of the organism. The geese may have carried reproductive material of the Euglena in their feces (waste matter).

Euglena are among the oldest organisms in the world, having survived since the time when conditions on Earth were not too dissimilar to those found in the Berkeley Pit- -ancient acidic oceans were full of heavy metals and other free- floating elements. These unique creatures exhibit traits of both plants and animals: they can photosynthesize (produce their own food using sunlight), but they can also move around in search of food. Their ability to photosynthesize produces oxygen. As a result of this oxygen combining with the dissolved iron in the water and by other means, the organisms cause iron to separate from the watery solution in the form of a solid, thereby creating stable substrates (surfaces) for other organisms to inhabit. In other words, Euglena are ecosystem engineers. They work to improve the environment for themselves and, in doing so, make their surroundings more habitable for other organisms. This type of bioengineering (engineering using biological organisms) fostered the development of all life on Earth, since the production of oxygen led to the proliferation of organisms that depend on oxygen, . and over the course of billions of years, the stable iron-based substrates eventually contributed to the formation of landmasses out of what was once a planet covered by water. Species like Euglena manufactured the current configuration of air, water, and land that makes this planet so uniquely hospitable to life today.

In the Berkeley Pit, Euglena are doing similar work. They thrive in the heavy-metal laden waters of the lake and are removing the iron, zinc, and cadmium out of solution, storing it in their bodies, and rendering the metals biologically inactive. When they die, their bodies and the metals' they contain are deposited in the sediments at the bottom of the lake. The chemistry of the Berkeley Pit is changing slowly but surely, and research is underway to encourage the population increase of Euglena and similar organisms to biologically treat the water before it spills into critical waterways. Since the discovery of Euglena , more than forty species of similar microorganisms have been discovered in the lake, several of them new to science. Many of these have found suitable habitat because of the pioneering work of Euglena , and they also serve to neutralize and repair the pit's toxic water. Species like Euglena are fascinating from an evolutionary perspective because they exhibit traits that not only confer advantages to their own species, but also create conditions that enable other types of life to thrive.

Plant and Animal Domestication

The domestication of plant and animal species in prehistoric times-the so-called Neolithic Revolution- was a vital innovation and has received a great deal of scrutiny. The domestication of a species ensured its reproductive success while providing humans with some type of necessity. Domesticated organisms are usually altered in various ways due to selective breeding, making them better adapted for interactions with humans.

Wild cereals have a very fragile stem, whereas domesticated ones have a tough stem. Under natural conditions, plants with fragile stems scatter their seed for themselves, whereas those with tough stems do not. When the grain stalks were harvested, their soft stems would shatter at the touch of a harvesting tool, and many of their seeds would be lost. Inevitably, though unintentionally, most of the seeds that people harvested would have been taken from the tough plants. Early domesticators probably also tended to select seed from plants having few husks (outer seed coverings) or none at all- eventually breeding them out- because husking prior to pounding the grains into meal or flour required extra labor. Many of the distinguishing characteristics of domesticated plants can be seen in remains from archaeological sites. Paleobotanists can often tell the fossil of a wild plant species from a domesticated one by studying the shape and size of various plant structures.

Domestication also produced changes in the skeletal structure of some animals. The horns of wild goats and sheep differ from those of their domesticated counterparts, and some types of domesticated sheep lack horns altogether. Similarly, the size of an animal or it sparts can vary with domestication, as seen in the smaller size of certain teeth of domesticated pigs compared to those of wild ones.

A study of age and sex ratios of animal remains at an archaeological site may indicate whether animal domestication was practiced. Investigators have determined that if the age and/or sex ratios at the site differ from those in wild herds, the imbalances are due to domestication. Archaeologists documented a sharp rise in the number of young male goats killed at 10,000-year-old sites in the Zagros Mountains of Iran. Evidently people were killing the young males for food and saving the females for breeding. Although such herd management does not prove that the goats were fully domesticated, it does indicate a step in that direction. Similarly, archaeological sites in the Andean highlands dating to around 6,300 years ago contain evidence that lamas were placed in enclosures, indicating the beginning of domestication.

Although it is tempting to think that a sudden flash of insight about the human ability to control plants and animals might have led ancient peoples to domestication, the evidence points us in different directions. There are several false ideas about the motivation for becoming food producers. First, contemporary foragers (people who search for food) show us that food production probably did not come about from sudden, unforeseen discoveries, such as that seeds can be planted and grown into plants, since these food foragers are perfectly aware of the role of seeds in plant growth, that plants grow better under certain conditions than others, and so forth. Jared Diamond aptly describes contemporary food foragers as “walking encyclopedias of natural history with individual names for as many as a thousand or more plant and animal species, and with detailed knowledge of those species' biological characteristics, distribution, and potential uses." In addition, food foragers frequently apply their knowledge to actively manage the resources on which they depend. Indigenous people living in northern Australia deliberately alter the runoff channels of creeks to flood extensive tracts of land, converting them into fields of wild grain.

Second, the switch from food foraging to food production does not free people from hard work. In fact, available ethnographic data indicate just the opposite. They show that farmers, by and large, work far longer hours compared to most food foragers. Finally, food production is not necessarily a more secure means of subsistence than food foraging. Seed crops in particular-of the sort originally domesticated in Southwest Asia, Central America, and the Andean highlands-are highly productive but not stable because of low species diversity. Without constant human attention, their productivity suffers.

The Atmospheric Greenhouse Effect

The atmospheric greenhouse effect is the result of electromagnetic energy from the Sun that is trapped in a planet's atmosphere. Atmospheric gases freely transmit visible- -short wavelength- solar energy, warming the planet's surface. The warmed surface tries to radiate the excess energy back into space, but because the planet is much colder than the Sun, it radiates at much longer, infrared (IR) wavelengths. But carbon dioxide and water vapor strongly absorb IR radiation, converting it to thermal (heat) energy. They subsequently re-radiate this thermal energy in all directions; some of the thermal energy continues into space, but much of it returns to the ground. The planetary surface receives thermal energy both from the Sun and from the atmosphere, and consequently it heats up.

Nowhere in the solar system is the atmospheric greenhouse effect more dramatic than on Venus. Its opaque cloud cover and massive atmosphere of carbon dioxide and sulfur compounds raises Venus's surface temperature by more than 400 K, to about 740 K (870'F). At such high temperatures, any remaining water and most carbon dioxide locked up in surface rocks would long ago have been driven into the atmosphere, further enhancing the atmospheric greenhouse effect.

Why does the atmospheric greenhouse effect make the composition of Earth's atmosphere so different from that of Venus? The answer lies in Earth's location in the solar system. Consider early Earth and early Venus, each having about the same mass, but with Venus orbiting somewhat closer to the Sun than Earth does. Volcanoes and comet impacts poured out large amounts of carbon dioxide and water vapor to form early atmospheres on both planets. Most of Earth's water quickly rained out of the atmosphere to fill vast ocean basins, but Venus was closer to the Sun, and its surface temperature was higher than Earth's, so most of the rainwater on Venus immediately re-evaporated. Venus was left with a surface containing very little liquid water and an atmosphere filled with water vapor. The continuing buildup of both water vapor and carbon dioxide in the atmosphere of Venus then led to a runaway (out of control) atmospheric greenhouse effect that drove up the surface temperature of the planet. Ultimately, the surface of Venus became so hot that no liquid water could survive there.

This early difference between a watery Earth and an arid Venus forever changed the ways that the two planets' atmospheres and surfaces evolved. On Earth, water erosion caused by rain and rivers continually exposed fresh minerals, which then reacted chemically with atmospheric carbon dioxide to form solid carbonates (minerals containing metal combined with oxygen and carbon). This reaction removed some of the atmospheric carbon dioxide, burying it within Earth's crust as a component of a carbonate rock called limestone. Later, the development of life in Earth's saltwater oceans accelerated the removal of atmospheric carbon dioxide. Tiny sea creatures built their protective shells of carbonates, and as they died, they built up massive beds of limestone on the ocean floors. As a result of water erosion and the various chemical reactions related to living organisms, all but a trace of Earth's total inventory of carbon dioxide is now tied up in limestone beds. Earth's particular location in the solar system seems to have spared it from the runaway atmospheric greenhouse effect. But what if Earth had formed a bit closer to the Sun? If all the carbon dioxide now in limestone beds had not been locked up by these reactions, Earth's atmospheric composition would resemble that of Venus or Mars.

Some scientists think that Venus once had as much water as Earth-as liquid oceans or as more water vapor than is measured today. If that's true, what happened to all the water? One possibility is that water molecules high in Venus' s atmosphere were broken apart into hydrogen and oxygen by solar ultraviolet radiation. Hydrogen atoms, being of very low mass, were quickly lost to space. Oxygen escaped more slowly, so some eventually migrated downward to the planet's surface, where it could have been removed from the atmosphere by combining with surface minerals. The Venus Express spacecraft has measured hydrogen, and some oxygen, escaping from the upper levels of Venus's atmosphere.

Megafauna Extinctions in Ancient Australia

In an effort to discover the connections in Australia's past between climate change, the vanishing of the large Ice Age animals (megafauna) there, and the arrival of humans, Gifford Miller turned to fossil eggshells he found in dune deposits along the ancient shoreline of a now vanished lake. The eggshells had been left by emus, ostrich- like flightless birds that still walk Australia's savannas and woodlands, and by Genyornis newtoni , an extinct bird species whose massive bones suggest each individual would have weighed about 550 pounds. Australia's acid soils and severe climate quickly leach all organic matter out of bone. But because eggshells have a different mineral structure than bone, the ancient shells retained traces of protein. This made it possible to date them using a technique called amino acid racemization. Results from a large set of samples- -1,200 dates collected from three different sites- -showed that the emu and Genyornis had coexisted for millennia. Then, about 45,000 50,000 years ago, Genyornis vanished.

The eggshells also offered clues to the big birds' diets. Grasses that thrive in hot habitats use a unique chemical pathway to capture the Sun's energy, a process that distorts the amount of the stable isotope carbon-13 they contain relative to most other plants. Miller and his colleagues compared the carbon isotope signatures of fossil emu and Genyornis eggshells.

The results show that the two bird species relied on different food sources. Grass must have been abundant right before the extinction event, because the emu of that era ate little else. The doomed Genyornis ate both shrubs and grass.

Two large, flightless birds living in the same habitat would have to develop different survival strategies, and different food sources, or constantly clash with each other. The emu is now, as it was during the Pleistocene (1.8 million years ago-11,700 years ago), a flexible eater. If necessary, it ate only grass, but it could also make do with shrubs and herbs. Genyornis included shrubs in its menu but seems to have been unable to get by without grass. About 45,000 years ago, the vegetation changed throughout the Lake Eyre basin. Emus shifted their tactics and began eating lots of shrubs. Genyornis could not adapt, and the species died out. More recently, Miller has analyzed the carbon isotope signatures of teeth fossils of the wombat (a pouched mammal). Like the emu, the wombat survived the ecological shift that came 45,000 years ago because it fed on shrubs, making do without grass.

According to Miller, the arrival of people, carrying fire sticks in their hands, changed everything. The harsh landscape of interior Australia does not hold well-preserved deposits of ancient pollen that might precisely track the changing vegetation. Based on the shifting diets of the herbivores he has studied, however, Miller can envision how and why the balance tipped. “Before people came, there must have been a savanna mixed with open woodland, the kind of habitat where grasses are abundant in years of good rainfall and the trees and shrubs sustain the animals when water is scarce," he explains. Frequent burning tends to favor grasses over shrubs and trees, yet at the moment of human arrival, emus and wombats stopped eating grass, and Genyornis simply vanished.

Miller argues that burning did promote one kind of grass growth, but that grass was spinifex, an unpleasant tasting species that now dominates the region. Spinifex leaves are heavily loaded with silica and are nutrient-poor. No native mammal eats it, and introduced cattle will take only the freshest, youngest shoots that sprout up after a fire. And spinifex loves fire: it grows in hummocks (low mounds) where live and dead blades intermix, laden with flammable resin, and it resprouts quickly after a burn. Australian plants had evolved with fire for millions of years. Before people came, the bush burned regularly at the end of the dry season, the time of many lightning strikes. But humans could light fires at any time of the year. In Miller's scenario, the increased frequency of fire wiped out many plants. Many ecosystems in the interior, he says, are adapted to burn every 20 to 50 years, the kind of frequency that occurs if lightning is the only ignition source. Torched frequently by Aboriginal people, such habitats would not have enough time to recover between burns.

The Port of Melaka

The port of Melaka, on the southwestern coast of the Malay Peninsula, was founded around the turn of the fourteenth century, and it continued the tradition of a previous regional power, Srivijaya, in ensuring the success of international trade. It owed its success to a number of factors. In the first place, it was generally able to guarantee the safety of its sea lanes. The rulers of Melaka, like those of Srivijaya, commanded the allegiance of various Orang Laut groups (nomadic sea peoples) who protected Melaka's clients and attacked ships going to rival ports. These safeguards (and the threat of attack for those who passed by Melaka) were an important element in the decision of traders to frequent the new settlement in preference to other ports in the region.

Secondly, Melaka was attractive to traders because of its commercial facilities. High priority was given to security within the town and to the protection of foreign merchants and their goods. For example, underground warehouses were constructed where stored goods would be less vulnerable to fire, damage, or theft. Such measures were necessary because departures, arrivals, and the exchange of goods were all governed by the monsoon winds (the seasonal wind of the Indian Ocean and southern Asia). Between December and March, the period of greatest activity, vessels reached Melaka from western Asia and the Far East; it was not until May, however, that ships from Java to the south and the eastern Indonesian archipelago (chain of islands) began to arrive. All traders, especially those from China and eastern Indonesia, had some time to wait before the change in monsoon winds made their homeward voyage possible. Secure storage facilities were therefore a significant factor in Melaka's ability to attract international clients.

Third, and most important, was Melaka's efficient legal and administrative machinery, which provided predictability essential for the long-term plans of foreign traders. The Undang-Undang Melaka, the first code of laws in the Malay world, devote considerable attention to the regulation of commercial matters. A separate codification of maritime laws concentrated specifically on matters concerned with sea-going trade, such as the collection of debts, shipboard crimes, and the duties of a captain and crew.

Melaka's administrative system also directly responded to the needs of a growing trading community. Four syahbandars , or harbor masters, were appointed, each one representing different ethnic groupings. Each syahbandar was required to oversee the affairs of his particular group; to manage the marketplace and the warehouse; to maintain a check on weights, measures, and coinage; and to adjudicate in any disputes between ship captains and merchants. The ruler of Melaka was the final arbiter who settled all quarrels between the different trading communities. Whenever a ship arrived in port, the captain reported to his particular syahbandar , who in turn referred him to Melaka's principal minister, the bendahara. The syahbandar then supplied elephants for the captain to transport his cargo to a warehouse assigned for the temporary storage of his goods. Before trading could be conducted, customs duties were paid in accordance with the value of the merchandise and the area from which the trader came. In addition, it was necessary to present gifts to the ruler, the bendahara , and the temenggung (the Melaka official principally involved in the collection of import and export duties), as well as to the appropriate syahbandar .

Melaka's reputation for security, a well-ordered government and a cosmopolitan and well-equipped marketplace all attest to the priority its rulers placed on creating the conditions for safe and profitable commerce. But these facilities alone would not have automatically attracted traders. The fundamental element in Melaka's success as a storage and distribution center was the dual role it played as the principal collecting point for spices such as cloves, nutmeg and mace from islands to the east and as an important redistribution center for Indian textiles. Indian cloth was carried mainly by Malay traders from Melaka to various parts of the archipelago and exchanged for spices, aromatic woods, sea products, and other exotic items highly prized by traders from both East and West. Without the spices from the eastern islands and the Indian cloth, Melaka would have been simply one of a number of other ports in the area specializing in a few local products.

Cave Artists

The earliest surviving cave and rock art was created in the Paleolithic period (40,000- 10,000 years ago) in locations that were not sites of human habitation. Some Paleolithic artwork was ephemeral—only six examples of open-air engravings survive—but the fine-art engravings and paintings that decorate caves were made to last and did last, in some cases for hundreds and even thousands of years, during which they were available for delight and use. Big cave galleries thus contained work done over a long period, available for comparison to people who had some notion of historical time and were developing a sense of their ancestry. The best Paleolithic art, especially the polychrome paintings, was the work of professionals.

We can say this with some confidence, for cave art at its best was difficult and expensive to produce. In the first place, it required lighting. Some 85 certain and 31 probable examples of Paleolithic lamps have survived, but less than one-third of them were found inside caves. The conjecture, therefore, is that artists usually worked by torchlight. Both lamps and torches consume animal fats in large quantities. Second, while it is true that some of the best cave paintings, especially at Altamira in Spain, were painted by artists standing up or in some cases lying down or squatting, others required elaborate scaffolding, no different in principle from that used by Renaissance artist Michelangelo when he painted the ceiling of the Sistine Chapel. Some of the paintings were done on a gigantic scale or at heights many feet above the cave floor. The sheer scale of the art is daunting. The big cave vault at Lascaux, known as the Picture Gallery, is over 1 00 feet long and 35 feet wide. Caves were especially chosen for their size as well as for their security. Niaux in the Pyrenees Mountains in Europe is over half a mile in length, and this is by no means unusual. The big caveat Rouffignac runs over 6 miles (more than 9 kilometers) into the mountain, and some of its huge collection of drawing-engravings are nearly 7 feet (over 2 meters) long.

Professional cave artists, then, needed not only platforms and scaffolding, whose existence at Lascaux, for instance, is betrayed by sockets cut into the walls, but assistants. They mixed the paints, some of which had to be used quickly before they dried; filled the lamps or held the torches; put up and secured the scaffolding; and made the brushes from twigs, feathers, leaves, and animal hairs, to the satisfaction of the master. These assistants probably graduated into masters themselves. It is not going too far to speak of a studio system as the basis of Paleolithic cave art. After all, there is positive evidence that art studios, where important works were fabricated and artists trained, existed in Egypt at least as far back as 3000 B.C. That still leaves a gap of 7 ,000 years from the end of the Paleolithic period, but the quality and consistency of the best painted work in caves, and the evidence of the time, expense, and skill required to produce them, does suggest that artists needed the collective support of something very like a studio. The probability is that the leading cave artists were important persons.

This brings us to quality. To modern eyes, accustomed to 5,000 years of continuous development in the depiction of living forms, the best of the Paleolithic paintings are magnificent. Indeed, seen in depth and in the total silence of the caves, the images are awesome. Human forms are rare and often quite unsophisticated, but the variety of animal forms is impressive. In the eight galleries of the great cave at Les Eyzies, there are multiple examples of mammoths, reindeer, horses, stags, bison, and wolves, as well as humanoids and abstractions or signs. These interlocking galleries, unfolding one by one, are meant to impress and they do. The art is ,both detailed and monumental, oscillating designedly between simplification and elaboration, between stasis and extreme dynamism. Some Paleolithic artists clearly understood both the anatomy of the animals they depicted and their principles of motion, the result of intense observation over many years and of a self-discipline in rendering that suggests a long apprenticeship and extensive study.

Breathing Inside an Egg

Tucked away inside its shell, a bird embryo has to breathe. But rather than using lungs to draw air in and push carbon dioxide and water vapor out, a bird embryo relies on “diffusion"- -the natural movement of gases- -much as do insects (which also lack lungs). In fact, both insects and eggs use tiny pores, or holes, and pore canals that connect the outside with their interior. For birds there are hundreds or thousands of tiny pores distributed all over the shell surface. The pores connect, via a narrow tube, the embryo's blood supply to the outside world. The number of pores per egg varies markedly between species, partly but not entirely related to the size of the egg. Since the pores are fairly straight and run vertically from the inner to the outer surface, their length is usually similar to the thickness of the shell. Generally, the number and size of pores determine how much and how fast oxygen diffuses into the egg. As well as taking away unwanted carbon dioxide, the pores allow water vapor to escape from the developing embryo. As the embryo grows it generates water, referred to as metabolic water, produced as a result of the metabolism of food. Different foods generate different amounts of metabolic water. Fat, for example, yields comparatively high amounts of metabolic water.

Inside an egg the developing chick generates plenty of metabolic water from the fat-rich yolk (yellow part of the egg) as it grows. This water has to be removed; otherwise, the embryo would drown in its own juices, so to speak, and it does this by allowing it to diffuse as water vapor through the pores in the shell. As a result, eggs lose weight during the course of incubation. What is remarkable is that, despite the huge variation across bird species in the size of eggs, the duration of incubation, and the relative size of the yolk, the loss of water between laying and hatching is always about 15 percent of the egg's initial weight. The water vapor lost during incubation ensures that the relative amount of water in the egg is the same in the chick at hatching as it was when the egg was laid. In other words, the composition of the newly laid egg has evolved through natural selection to ensure that the newly hatched chick has the right composition- -in terms of the amount of water in its tissues, too. This is achieved by adjusting-via natural selection- -the effective pore area such that all the metabolic water produced during development is eliminated before hatching. One consequence of this loss of water vapor is a space in the egg, roughly 15 percent of its volume, that becomes the air cell at the blunt, or flat, end of the egg and provides the amount of air needed by the chick just before it hatches.

The air cell is formed between the inner and outer shell membranes when the egg is laid. As the egg cools and its contents contract after leaving the female's body, air is drawn in through the pores and accumulates in a lens-shaped pocket at the blunt end of the egg. If you hold a hen's egg against a bright light, you can see the air cell. Furthermore, when you peel a hard-boiled egg to eat, the air cell's presence is revealed by the flattened area of white at the blunt end where the air cell has pressed down on the albumen (egg white). William Harvey in the 1600s was the first to think about the role of the air cell, dismissing the then-widespread belief that its position in the egg signaled the sex of the chick. As development proceeds, the air space increases in size, and it is for this reason that you can assess the age of an egg, or its stage of development, from how it floats in water: a very fresh egg with virtually no air cell sinks; an older egg floats.

Because gases behave differently under pressure, we might expect the size and number of pores to differ among birds breeding at different altitudes. Specifically, the loss of gases will be less at high altitudes. And this is confirmed by a comparison of birds breeding at different elevations: species breeding at high altitudes have fewer, smaller eggshell pores.

Polynesian Migration

Polynesia is a large grouping of islands scattered over the central and southern Pacific Ocean. The origins of the indigenous people of the Polynesian islands are the subject of some debate, as are the reasons for their migration from one island to another. In about 3000 B.C.E, people living in the Bismarck Archipelago (a chain of islands in the neighboring region of Melanesia) began making pottery, keeping domesticated dogs, pigs, and chickens, and growing vegetables. Their culture was known as the Lapita culture, and the people were likely ethnic Chinese. In about 1300 B.C.E., they began to spread eastward into Polynesia. Over the next 2,300 years, they brought their culture to several islands there, including New Zealand, where the Polynesian settlers became known as the Maori people. Their long voyages were made in canoes approximately 65 feet (20 meters) long. These canoes carried crews of at least five and up to 15, with a supply of vegetables, live chickens and pigs, and water stored in gourds that could be augmented by collecting rainwater in sailcloth.

Navigators memorized the routes to other islands and taught them to young men who were learning the skill of navigating, in the form of songs and drawings. Their method was based first on their knowledge of the direction and times of rising and setting of the most prominent stars and planets. Voyages began at dusk. The navigator set a course in relation to the direction of prominent landmarks that were still visible and of the stars, and during the night he would steer by the stars. During the day he would steer by the Sun. The condition of the sea and direction of the wind also provided valuable information. In the tropical Pacific, the prevailing winds blow from the northeast to the north of the equator and from the southeast to the south of the equator. Because they blow for most of the time, the winds produce a large swell, with waves that all move in the same direction. Navigators could steer by the direction of their canoe in relation to the swell, and they would tow a length of rope in the water behind the canoe. If a sudden wave or gust of wind blew the canoe to one side, the rope would not be affected and its line would record the direction they should steer. Small pennants (flags) tied to the canoe indicated the wind direction, which was also useful. In addition, Polynesian sailors knew the paths followed by migrating birds and whales. They were familiar with ocean currents, the cloud patterns that formed over distant islands, and they watched for floating plant debris that indicated land just over the horizon.

Polynesian navigation was remarkably skillful-but why were the Polynesians so restless? One possible clue can be found in the fact that despite living on islands surrounded by abundant stocks of fish, the people of the Cook Islands seldom eat fish. The seas around Rapa Nui support many species of edible fish, as well as lobsters and turtles. These have always been an important part of the Rapa Nui islanders' diet, and island traditions forbade fishing at certain times of year, which prevented overexploitation of the stocks. As recently as the early twentieth century, however, the islanders believed that the fish living in deep water farther from shore were poisonous, and they refused to touch them.

In 2009, Teina Rongo, a Cook Island marine biologist, aroused considerable interest among historians when he proposed that from time to time the fish on which Polynesians depended had turned poisonous and the islanders had suffered from a type of food poisoning called ciguatera. Robbed of their food supply, they had had no alternative but to move elsewhere. Ciguatera is caused by eating fish contaminated by a single-celled organism called Gambierdiscus toxicus . The poisoning is seldom fatal, although it can be; its symptoms include vomiting, blurred vision, a burning sensation on contact with a cold surface, and heartbeat irregularities. Common throughout the Tropics, this poisoning is most severe in people who have eaten carnivorous reef fish such as barracuda, snapper, and grouper. It may be that the waves of Polynesian migration were motivated by the need to find wholesome food.

The Ecological Roles of Birds

Birds provide numerous invaluable ecological services, and one of the most important is by scattering plant seeds. Seed dispersal enables plants to migrate into new territory and reduces competition with others of their own species, thereby increasing survival rates. In fact, if you enjoy spicy foods, you owe some appreciation to birds, who played a pivotal role in the evolution of chili peppers. In an example of evolutionary partnership, chili plants evolved in a way that offers birds a nutritious food source, while the plants benefited from the wide distribution of their seeds after passing through the birds' digestive systems. Capsaicin is the molecule in spicy peppers that creates the sensation of heat we feel when it binds to pain receptors in the mouth. On a human tongue, a solution of just ten parts per million creates a distinct sense of burning, but nerve receptors in birds do not register the heat of capsaicin even if they ingest as much as twenty thousand parts per million.

Bird digestive systems also do the seeds of the chili pepper no harm, because many birds with a fondness for them have a rather short, straight digestive pipe through which food transits quickly. Researchers have even found snails that had been eaten by a bird, passed through its digestive tract, cast out the other end, and were still alive. Moreover, chili peppers can be plucked readily from the stem only when they are ripe, which means only those seeds ready to germinate (begin growing from a seed into a plant) are swallowed by the bird. Catching the eye of birds helps pepper plants expand their range. If mammals ate chilies, they would digest and destroy them or deposit them fairly close to the plant. Birds, however, drop the undigested seeds in faraway places. The cradle of chili pepper civilization is in Brazil, in a lowland region dubbed “the nuclear area" because it has the largest number of wild varieties of chilies in the world. It is believed that the first wild chili peppers sprang to life here and were spread across much of the Western Hemisphere by birds.

Birds shape the plant world in myriad other fascinating ways. Some two thousand species inadvertently play the middleman in sexual relations between plants. White-winged doves that survive in the desert on cactus flower nectar carry pollen between saguaro cactus flowers, and hummingbirds seek out the nectar of a variety of wildflowers and domestic crops; some hummingbirds visit thousands of flowers in a day, and as they search for nectar, they carry millions of tiny grains of pollen stuck to their heads, beaks, and feathers, brushing flowers like flying paintbrushes.

There are places where birds and the pivotal roles they play have vanished. The extinction of the dodo, the three-foot-tall flightless bird of Mauritius, could well be the cause of the decline of the tambalacoque tree, also known as the dodo tree, which is highly valued for its timber (wood). Some plants have just a single animal partner that plays a key role in its life, and in the case of the peach- tree-like tambalacoque, the dodo's digestive tract may have been vital in paving the way for the tree's successful growth. As the seeds moved through the bird's system, fruit pulp was scrubbed of, reducing the risk that bacteria and fungi would kill the seeds before they germinated. These days, botanists growing tambalacoque trees pass their seeds through wild turkeys or even gem-polishing machines to roughen and clean them in preparation for planting.

Large enough flocks of birds can also be a factor in the weather. When millions of migratory seabirds arrive in the Arctic each summer, they bring a huge burst of guano (bird waste). As bacteria go to work on the guano and break it down, they emit ammonia. As sulfuric acid and water combine with the ammonia in the atmosphere, bigger particles are formed, which become the nuclei for cloud droplets and thus create cooling clouds. The number of cloud droplets near seabird colonies can be 50 percent more than in those areas without birds. Smaller droplets are also formed, which reflect sunlight and contribute to the cooling effect. When seabirds change their migratory patterns or disappear, experts believe they may contribute to changes in Arctic weather.

The Harappan Decline

The Harappan civilization flourished around the Indus River Valley on the western Indian subcontinent from about 2700 to 2000 B.C.E. The Harappans exploited the seasonal overflow from their rivers by capturing the water with dams and distributing it to crop-growing farms. They built cities of brick and traded widely, sending wood, fabrics, and other products to Mesopotamia, where other early civilizations thrived.

By around 1900 B.C E. the Harappan civilization was in decline, a result likely related to environmental degradation. Intensive plant growth, water-saturated land, and hot, dry summer climates cause rapid surface evaporation over time, drawing salts in the earth closer to the surface, where they slowly poison plant life through a process called salinization. Moreover, salt-permeated topsoil is easily blown away by passing winds, causing desertification (fertile land turning into desert). These conditions are further exacerbated where a hard layer of bedrock impervious to water lies not far beneath the surface, as is the case in the valley of the Indus. Such soil formations mean water tends even more to stay close to the surface (a high-water table), where the process of evaporation is greatest. Salinization is particularly deadly in regions experiencing a period of decreasing local rainfall, as the drier the air, the faster the evaporation, and hence the quicker the rise in salts to the surface. A shift eastward of seasonal rains may have been responsible for decreasing moisture on the western Indian subcontinent, encouraging a drying trend that became well established there and in Mesopotamia, where civilizations also began to decline around the same period.

Ironically , both the Harappan and Mesopotamian civilizations' management of their natural resources may have figured in this mutual decline. Much of the Harappan-Mesopotamian bulk trade was in Harappan wood. Harappan woodsmen may have stripped much of the Indus Valley of its local forests for this trade and to provide fuel for the ovens that dried the bricks used widely in Harappan construction. Mesopotamians had already deforested their lands, hence their need for outside suppliers of wood. The removal of forest cover, for any reason, increases flooding, because trees hold water in the soil. The making of burnt bricks and the trade in wood, combined with drought, was a prescription for disaster.

Dramatic earth movements have often altered the course of human history: earthquake-related tsunamis (large ocean waves) alone have eradicated cities from ancient Mycenae (Crete) to Port Royal (Jamaica) in the early modern period. The fate of Harappans was probably sealed by a series of earthquakes and other geologic events between 1900 and 1500 B.C.E. that uplifted land and shifted river beds so as to create even greater pooling of water in a time of flood, further increasing salinization. Increasingly unproductive land and declining Mesopotamian trade would have made it harder to fix the damage done by floods and earthquakes. Cities show signs of a slow dissolution of civic cohesion, including the rise of slums (low- quality residential areas) and a decline in the quality of crafts.

Where Life Arose

Life could not have come into being on Earth until after about 4 billion years ago, when bombardment by asteroids had subsided substantially. Even after Earth's surface had stabilized, certain conditions were required for the origin of life. Some researchers have concluded that precursor compounds, and ultimately life itself, arose in small bodies of water that were struck by lightning and turned into what is sometimes referred to as the "primordial soup “of organic compounds. The problem with that idea is that it would have required an atmosphere lacking oxygen, because even a small amount of oxygen would have oxidized, and thereby destroyed, chemical raw materials necessary for the production of essential organic compounds.

Knowing that photosynthesis produces the preponderance of oxygen in Earth's atmosphere today, scientists once assumed that the atmosphere lacked free oxygen before the origin of photosynthetic organisms. We now know, however, that ultraviolet light breaks down water vapor in Earth's upper atmosphere, slowly liberating oxygen, which spreads in small quantities throughout the atmosphere. This process would have contributed a small amount of oxygen to the ancient atmosphere-enough to destroy chemical compounds essential to life. Life must have originated not in a small body of water that was exposed to atmospheric oxygen but in some environment that was isolated from Earth's atmosphere. The most likely setting was a warm area beneath the seafloor in the vicinity of a mid-ocean ridge (submarine mountain range).

The heat that rises from Earth's mantle (the layer between the crust and the planet's central core)along mid-ocean ridges warms seawater that has percolated into the crust through pores and cracks. Because heating causes water to expand, and so reduces its density this water rises back to the ocean. In some areas it flows from the seafloor through large vents as columns of very hot water. Bacteria of many kinds inhabit the warm water of ridge environments, occupying pores, cracks, and vents. They live in a variety of ways, but most of them derive energy from the chemicals that the hot water has dissolved while moving through the ridge system. Some of these bacteria live in water warmer than 100 degrees Celsius, which remains in a liquid state because of the great pressure applied by the ocean above. Others live in lukewarm water farther from ridge axes. In general, these high-temperature bacteria may be inhabiting the kind of setting where life originated.

The principle that most warm-adapted bacteria put into practice to obtain energy is quite simple: they harness the energy of naturally occurring chemical reactions. Many of the chemical compounds that emerge from deep within mid-ocean ridges are not stable after the rising water in which they are dissolved cools and mixes with seawater; and so, they will enter into chemical reactions. Many such reactions do not occur quickly, however, and the bacteria take advantage of this situation. The bacteria consume the chemical compounds and simply allow chemical reactions that would have occurred in seawater to take place inside their cells. These reactions release energy. The bacteria harness the energy for their metabolism and excrete the chemical products of the reactions.

Some of the warm-adapted bacteria actively carry out chemical synthesis, but unlike photosynthetic organisms, they do not use light as an energy source. Other kinds of bacteria get energy by combining hydrogen or sulfur with oxygen in the way that higher organisms obtain energy by oxidizing sugars.

There is evidence that the warm-adapted bacteria that live in the vicinity of mid-ocean ridges are actually the most primitive living bacteria. Biologists have compared the genetic material, DNA and RNA of all kinds of bacteria to reconstruct their evolutionary relationships. Those features of DNA and RNA that are shared by many bacterial groups are regarded as primitive features inherited from very ancient ancestors. Traits shared by few bacterial groups are regarded as indicating more recent branchings of the evolutionary tree. Using this method, it turns out that the most primitive living bacteria (Archaebacteria)are adapted to warm habitats.

This finding suggests that bacteria evolved in such habitats, perhaps in the vicinity of mid-ocean ridges.

Porcelain in Seventeenth-and Eighteenth-Century England

Porcelain is a white ceramic that originated in China and that incorporates kaolin clay. When Asian porcelains first appeared in Europe, Europeans were delighted and mystified by their lightness, hardness, and translucence (allowing light to pass through).Starting in the seventeenth century, Britain's East India Company imported large quantities of goods from East Asia, with the volume of imported porcelain rising dramatically toward the end of that century and into the eighteenth. One reason for this increased importation of porcelain was the increasing popularity of the hot beverages coffee, chocolate, and tea. The earliest London coffeehouses date from the 1650s,and, though the first known advertisement for tea appeared in 1656,it was not imported by the East India Company for resale in Britain until 1678.The rising popularity of coffee drinking and tea drinking in Britain, and the vastly increased importation of Far East ceramics, are thus both phenomena that had their origins in high-class British social life during the last quarter of the seventeenth century. Being heavy and not susceptible to water damage, porcelain was packed underneath the very much lighter cargoes of tea brought by the East India Company. The histories of the two are thus closely intertwined, and the links extended from the act of importation to its sale-as porcelain dealers very often sold tea, coffee, and chocolate in addition to ceramics and glass-right through the final act of consumption.

Porcelain was resistant to the thermal shock of contact with boiling water, but unlike silver (of which late-seventeenth-and early-eighteenth-century teapots continued to be made)it did not get uncomfortably hot. And being of Asian origin, it no doubt seemed the appropriate material from which to drink the teas imported by the East India Company. As importation increased, some of the mystery surrounding its production was dispelled. Indeed, its preparation was described in two letters by Pere D'Entrecolles, a Jesuit missionary resident at the great ceramic center of Jingdezhen, China, in 1712 and 1722.These letters were published in Paris in the 1720s and subsequently appeared in a work on China published in French and English editions in the 1730s.

Although Asian wares had much to recommend them to seventeenth-and eighteenth-century Europeans, three characteristics in particular excited their admiration. Firstly, there were the physical qualities of the material itself, which was hard and white-bodied, and -most remarkable to Western eyes-could be translucent if thinly made. Secondly, the wares were technically superior to almost all of seventeenth-and eighteenth-century European ceramics, not allowing liquids to pass through and resistant to wear and tear. Finally, they came embellished with painted patterns in blue, or with brightly enameled decoration, the subjects and scenes of which must have appeared fascinating to Western eyes.

The late-seventeenth-century taste for such exotic porcelain wares was part of a wider fashion that embraced Asian goods in other materials, including Chinese and Japanese lacquerware (household goods coated with a protective material made from tree resins),silks and fans, and printed cottons and embroideries from the Indian subcontinent. From the fact that such goods were brought to the West by European East India companies, imported decorative work of this type was frequently called "Indian “during the seventeenth and eighteenth centuries since the geography and cultural diversity of Asia was very little understood by the European consumers in these years. Like the Indian cotton cloth known as calico, the majority of Chinese porcelains that found their way to the West during the eighteenth century were specially made for the Western market, very often after prototypes or design specifications supplied or laid down by European merchants. In the case of textiles, and to a lesser degree in ceramics, the designs with which they were embellished were also frequently based on Western reworkings of, or fantasies upon, Asian themes and decorative styles. The porcelain wares made to Western order and specifications were themselves copied and adapted in the ceramic factories of Europe, and some of these in due course made their way back to China, where they became the subject of further copying and variation.

Transitions in World Populations

Unless we know the size, density, age distribution, and reproductive capacity of a population, it is difficult to understand many aspects of its society, politics, and economy. In the contemporary world, most nations conduct periodic censuses to assess the present situation of their populations and to plan for the future. Before the mid-eighteenth century, when census taking became a regular procedure, population estimates and counts were sporadic and usually inaccurate. Thus, arriving at estimates of populations from the past, especially for non-literate societies, is a highly speculative exercise in which archaeological evidence and estimates of productive capacity of agricultural practices and technology are used. The earliest date for a population estimate with a margin of error less than 20 percent is probably 1750.

The history of human population can be divided into two basic periods: a long era-almost all of human history-of very slow growth, and a very short period-about 250 years from the mid-

1700s to the present-of very rapid growth. Before agriculture was developed, the hunting-and-gathering economies of the world's populations supported 5 to 10 million people, if modern studies of such populations can be used as a guide. After about 8000 B.C.E., when plants and animals were domesticated, the world's population began to increase more rapidly but still at a modest level. Although agriculture provided a more secure and larger food supply, population concentration in villages and towns would have made people more susceptible to disease and thus reduced their numbers. Some historians also believe that the settled agricultural life also led to intensified warfare (because of the struggle for land and water) and increasing social stratification within societies.

Still, the Neolithic revolution, the shift from hunting-and-gathering to settled agricultural life, stimulated population growth. It was the first major transition in the history of world population. One estimate, based on Roman and Chinese population counts and some informed guesses about the rest of the world, is an annual growth rate of about 0.36 per million. By 1 C.E. the world population may have been about 300 million people. It increased between 1 C.E., and 1750 c.E.to about 500 million people. We should bear in mind that during this period of general increase, there were always areas that suffered decline, sometimes drastic, because of wars, epidemics, or natural catastrophes, such as the disastrous decline of American Indian populations after contact with Europeans, caused by disease, conquest, and social disruption.

A second and extremely important transition took place between the mid-seventeenth and the mid-eighteenth centuries. Initially based on new food resources, this transition often is associated with the Industrial Revolution, when new sources of energy were harnessed. The growth rate greatly increased during this period in the countries most affected. Between 1750 and 1800,the world population grew at a rate of more than 4 percent a year to more than a billion people. By the mid-twentieth century, the world growth rate had tripled, and by 1990 the world population had risen to more than 5 billion.

This demographic transition took place first in Europe and is still more characteristic of the developed world. Most pre-modern agrarian economies were characterized by a balance between the annual number of births and deaths; both were high. Life expectancy usually was less than 35 years, and the high mortality was compensated by high fertility; that is, women had many children. Improvements in medicine, hygiene, diet, and the general standard of living contributed to a decrease in mortality in the eighteenth century. This allowed populations to begin to grow at a faster rate. By the nineteenth century in most of Western Europe, the decline in mortality was followed by a decline in fertility. In some countries such as France, these two transitions took place at about the same time, so population growth was limited. In much of Europe, however, the decline in fertility lagged behind the decrease in mortality, so there was a period of rapid population growth. Until the 1920s,population growth in Western Europe and the United States was higher than in the rest of the world, especially in the less industrialized countries. In recent times, that situation has been reversed.

The Jack Pine and Fire

The jack pine tree of Canada has several qualities that seem to equip it for a life with fire. As in many conifer trees, the branches nearer the ground tend to die off as the tree grows taller, not least because they find themselves without light. In the jack pine, however, these dead branches simply fall off. If the dead branches were allowed to persist, they would provide a "ladder" for the fire from the ground to the top. The physics of fire is in many ways counterintuitive. Crucially, a hot fire that burns itself out quickly can be less damaging than one that's somewhat cooler but lasts longer. Jack pine needles are high in resin (a flammable organic substance secreted by trees) and often low in water, especially in the droughts of spring and summer when fires are likely, and so they burn hot but quick. On much the same principle, the jack pine's bark is flaky. It picks up surface fires but then burns swiftly and does little harm. The stringy bark of eucalyptus trees in Australia, hanging loosely from the smooth trunk beneath, is protective in much the same way. In both cases the discarded bark prevents the fire from seriously harming the tree.

On the other hand, when jack pine bark accumulates on the ground (as it does if there is a long interval between fires),surface fires—particularly in spring and summer--can be very fierce. Then most trees of all kinds are killed. But the jack pine is typically the first to spring back. For a very hot fire in the summer burns both the leaf litter on the surface and the organic material in the soil itself, leaving a bare, mineral soil behind. Jack pines germinate well in such soil and, indeed, are inhibited by leaf litter. They like bright sunlight, too, and appreciate the open space.

By their fourth or fifth year many of the young jack pines are producing their first cones-which by tree standards is markedly young. Why are they so precocious: why not focus their precious energy on more growth rather than on reproduction? Forest fires often leave a lot of fuel behind, and sometimes a second fire quickly follows the first. It seems a good idea to scatter a few seeds before the possible subsequent fire.

But it is the cones and seeds of the jack pine that are adapted most impressively and specifically to fire. The cones are hard as iron, their scales tightly bound together with what could be called a "resinous glue." Many creatures attack cones; but only the American red squirrel will take on the jack pine cone, and even the red squirrel much prefers the easier, fleshier meat of spruce cones. The cones may persist on the trees for many years, and the seeds within them remain viable. In one study more than half the seeds from cones that were more than twenty years old were able to germinate. The cones do not open until there is a fire: it takes heat of 50C to melt the resin that locks the scales together. Then, they open like flowers. Thus, the seeds are not released until fire has cleared the ground of organic matter and of rivals and has created exactly the conditions they need. The output is prodigious. After a fire in the taiga (the northernmost forest, which then gives way to tundra),the burned ground may be scattered with twelve million jack pine trees per acre.

Although the cone responds to fire, and only to fire, it is remarkably fire resistant. It has been found that the seeds inside would survive for thirty seconds even when the cone was exposed to as much as 900C.It has also been shown that the cone does not respond simply to the presence of fire, like some crude unmonitored mechanical device: rather, as it is heated, it releases resin from within. Which oozes to the surface and creates a gentle, lamp-like flame around the cone which lasts for about a minute and a half. All in all, it seems that, once ignited, the cone is programmed to provide a flame for the right amount of time to open the cone.

Ziggurats in Mesopotamia

Ziggurat © Wikimedia Commons/Hardnfast

Like the pyramids of Egypt, the towers called "ziggurats" built in ancient Mesopotamia were monumental symbols of a great civilization. Ziggurats consisted of several levels or platforms of diminishing sizes with exterior stairways or ramps leading to their summits. New platforms were built on top of older ones over many centuries, leaving earlier stages buried under later enlargements. The Egyptian pyramids, however, were never meant to be climbed, and new pyramids were not built over the remains of older ones.

Unlike in Egypt, there was no stone for building projects in Mesopotamia, but mud-available everywhere-was used to produce bricks. Despite the availability of this cheap building material, ziggurats were always considerably smaller than pyramids, possibly due to a lack of manpower and wealth. The central core of most ziggurats consisted of unbaked, sun-dried bricks covered with a thick outer shell of baked bricks. Water occasionally leaked into the interior of a ziggurat, causing its unbaked core to expand and crumble. All kings of Mesopotamia had to face the constant task of rebuilding mud-brick structures; ziggurats, as well as other structures, rarely lasted a century without major renovation.

Why, then, were ziggurats not built entirely out of baked bricks instead of just their exterior walls? If they had been, they could have withstood the ravages of time, and their kings would not have been obliged to repair them every few years. As is so often the case, environmental factors may have dictated the quantity of burnt mud bricks used in large structures. There were very few trees in Mesopotamia, and the Mesopotamians lacked the necessary fuel to bake the millions of bricks required for large structures. Most of the wood and straw fuel available was used for cooking fires in private homes and could not be spared for brick-making. Another factor that contributed to the demise of ziggurats was the size of the bricks used to make them. Mud bricks are smaller and lighter than the great stones used in pyramid construction and, long after they were abandoned, peasants in search of easily available building material found ziggurats to be a convenient source of bricks for constructing houses and other domestic buildings.

The purpose of Egyptian pyramids is clear: they were tombs for their deceased kings. But if ziggurats were not tombs, then what was their purpose? Early explorers naively thought that they were used by Mesopotamian priests to escape the mosquitoes. Some maintain that the first small ziggurats were simply raised platforms where the village grain supply could be kept dry during the annual flood. As early as the fourth millennium B.c., temples were built on raised earth and mud-brick mounds, and ziggurats may have been a further development of this type of construction. The most widely accepted explanation is that ziggurats were meant to be climbed. Ziggurats always had several stairways leading to their summits, and it seems clear that their primary purpose was to elevate the priests closer to the realm of the gods in the heavens. In the city of Sippar, the ziggurat was called "The Staircase to Holy Heaven, and offerings were made to the gods from a small temple at the summit of the ziggurat. In this way, ziggurats formed an important spiritual link between people on Earth and the sacred realm of the gods in the heavens.

In the early days of research on Mesopotamia, it was claimed that ziggurats were built as celestial observatories where astronomers could have studied the stars without city buildings obstructing their view. It is likely that on some occasions celestial observers climbed to the top of ziggurats to observe the night sky, recite prayers to the gods of the night, and make offerings to the celestial gods. It should be kept in mind that, since the moon and planets would still appear to be the same size, climbing a few meters to the top of a ziggurat would not give an observer a significantly closer or better view of celestial objects. It is doubtful that ziggurats would have been of much use to astronomers and calendar-makers, but they would have elevated priests and celestial observers into the higher spiritual realm that was such an important element of their religious world.

Bat Diets

One of the most dramatic measures of the diversity of bats is the variety of food they consume. Although some 600 species eat insects as the main dietary staple, others live on fruit, nectar and pollen, fish, frogs, birds, small mammals, blood, and even other bats. Most of this diversity occurs in the tropics, although many bats from temperate regions vary their diets by eating a wide range of insects.

What any bat eats is determined by two important factors: the need for enough energy to keep the body going and the need for essential chemicals to maintain it. Just as an automobile requires fuel to move and lubricants to keep the engine running smoothly, animals require protein, carbohydrates, or fat for energy along with vitamins and minerals to stay healthy. Since it is essential for bats to consume enough calories to make flight, reproduction, growth, and other bodily functions possible, it is easy to see why such variety is necessary in their diets. Species that feed mainly on plant products require protein, which can be obtained by eating animals (insects),pollen (which provides only moderate amounts of protein),or large quantities of fruit (a very limited source of protein).Bats feeding on insects appear to obtain a balanced diet from this source of nutrition.

To obtain the energy and chemicals they need, bats consume vast quantities of food. Small insectivorous species (species that eat mainly insects)eat at least 30 percent of their body weight each night they are active, and in nursing mothers the amount may exceed 50 percent. These figures appear to apply to all bats, regardless of diet. It may not be impressive to learn that a little brown bat eats three grams of insects on a summer night, but to find 150 mosquitoes in its stomach certainly is, especially when you realize they are not an entire night's ration and the bat probably caught them in less than fifteen minutes. The most impressive statistic of insect consumption comes from Texas, where it is estimated that a local population of Mexican free-tailed bats eats slightly more than 6,000 tons of insects each summer.

The basic design of a bat imposes certain restrictions on the range of food available to it. Bats determine the direction and distance of objects in their environment by emitting high-pitched sounds and interpreting their echoes to find their location-the term for this is echolocation. Because they can echolocate, New World leaf-nosed bats that feed mainly on fruit are able to catch insects to supplement their diet with protein. Flying foxes (a species of bat found mainly in Indonesia and Malaysia)and their relatives, however, do not echolocate and must obtain their protein from something other than insects. Recent studies in the Ivory Coast by the Canadian biologist Donald Thomas suggest that several species of the herbivorous bat family pteropodidae get their protein from the fruit that composes the main part of their diet. The levels of protein in the fruit are low, but this is countered by the consumption of large quantities of fruit and by enzymes in the digestive tract that efficiently extract what little protein is available. Size is also a factor in determining the diet of a bat; a 3- gram butterfly bat has fewer prey species from which to choose than a 40-gram,large slit-faced bat. As a rule, smaller bats are almost entirely insectivorous, and larger species include larger prey in their diet, readily switching to small vertebrates such as fish, birds, and frogs. The smallest bats that feed on plant material are nectar-and pollen-feeders. Larger species more often feed on fruit but may also supplement their diet with nectar and pollen.

The food selected by bats also depends on where they feed. Flying bats chasing flying insects will not catch scorpions that do not fly, but they often snatch spiders ballooning on pieces of web. Bats feeding on stationary or terrestrial prey often catch resting insects that are able to fly. Bats concentrating their feeding activity over water catch more aquatic insects than those feeding high over the forest, and species hunting over water have opportunities to catch fish not available to high-flying species that visit the water only to drink.

Continuous Script and Oral Culture in Europe

Today people commonly read in silence and in private, but that was not the case in Europe during the periods of ancient Greece (800-146 B.C.E.),ancient Rome (753 B.C.E.-476 C.E.),and the centuries that followed. In ancient Greek times, reading was an oral (spoken)practice--one reflected in writing itself. Greek texts were written in a continuous script (without spaces between words)and with minimal punctuation; this both required and rewarded sounding them aloud. Continuous script could not have developed without the Greek introduction of letters for vowels, which allowed readers to identify syllables and hold them in memory as the eye moved across the text.

Though it seems awkward to us now, continuous script was not a natural construction, but a choice, as demonstrated by the fact that the Romans discarded their own punctuation in the first century in favor of the Greek model. It established literacy (ability to read and write)as the domain of a cultured elite, who either studied from a young age to master the skills appropriate for reading each individual text or employed a professional reader, or lector, for the task. It also facilitated a culture of shared inquiry, in which challenging texts were read aloud in groups as an incentive for debate. In ancient Greece, literature was primarily a social activity, with audiences gathering for performances of epic poetry and drama. Epic poems bear the hallmarks of this orality: they rely on repetition, formulaic images, meter, and rhyme as mnemonic (memory)aids to the performer. The term used to describe performances of such works, rhapsody, means "to stitch together-suggesting the extent to which oral composition relies on weaving familiar lines.

The great thinkers of ancient Greece, in fact, mistrusted writing as a technology that would destroy the oral arts of debate and storytelling on which they based their sense of the world, of philosophy, and of time and space. In Plato's dramatic play Phaedrus, the philosopher Socrates looks down on the written word for separating ideas from their source, citing Egyptian king Thamus as the first to voice this concern when he received the gift of writing from the god Thoth. Transcription (writing speech down), Socrates fears, is an aid that will both interfere with memory and trap philosophical thought in ambiguity, leaving interpretation in the hands of the reader. Texts, after all, can circulate without their author, thus preventing one from explaining or defending them. Despite these fears, the very writing Plato used to record his works proved instrumental in the development of ancient Greek oratory (art of speech making). As scholar Walter Ong points out in Orality and Literacy, his study of the ways writing technologies restructure consciousness, the written word enabled Greek scholars to transcribe and codify effective rhetorical (speaking)strategies. It also vastly increased human vocabulary, since we no longer had to rely on memory to hold all of language for immediate use. Writing, in fact, allowed rhetoric to flourish.

For the kind of silent reading, we now experience to take hold, reading would have to change its context and text its form. It would have to become a more private experience, which means literacy would have to extend beyond the elite and monastic (religious) communities. Texts, too, would need to become more legible, with standardized punctuation and word spaces so that the mumbling of readers sounding out text, common through the sixth century, could disappear. And libraries designed for quiet, contemplative reading could then develop to serve this new readership.

British scribes, like those who crafted the Book of Kells (around 800 C.E.),played a key role in making text more accessible. They wrote in Latin, and because it was a second language, and one more challenging to sound out in continuous script, they introduced several changes to improve its legibility, including word separation (around 675 C.E.),additional punctuation, and simplified letterforms. Still, it took nearly four hundred years for these small innovations to spread. The translation of Arabic scientific writing into Latin in tenth-century Europe likely played a vital role in solidifying word separation, since it was inherent in the language (because unlike Greek and Latin, it is written in consonants).Translators kept Arabic word separation when rendering these texts in Latin, in part because it made the complex technical prose significantly more comprehensible.

Tern Hunting of Fish

Tern

Terns are a group of bird species that hunt fish close to the water surface. Fish have adaptations to avoid being eaten, including cryptic coloration (most fish are dark along the back, even if they are silvery on the sides),habitually swimming at depths below those at which they are in danger from above, and taking evasive action if they see a bird diving toward them. Terns have their own cryptic coloration, most having white or light grey underparts so that they are difficult to see from below the surface, and they dive vertically toward the fish at the highest possible speeds.

Terns also have to learn to adjust for parallax: sunlight changes direction at the air-water interface, which makes fish appear always to be at shallower depths than they really are. This is an especially difficult problem when the water surface is wavy, which makes the image of the fish move around horizontally as well as vertically-as one can see by looking down from a bridge or dock into rippled water. Terns must learn by trial and error to compensate for parallax and dive accurately toward a target that appears to be moving unpredictably. It has been shown that terns' success in catching fish declines with increasing wind speed, presumably because increasing waviness of the surface makes the fish increasingly hard to locate. At the other extreme, terns' fishing success declines again at very low wind speeds, either because the terns have more difficulty hovering to locate the fish in still air, or because the fish can see the terns more easily when the surface is smooth. Except in the most favorable circumstances, terns' fishing success is usually quite low-typically only one fish caught for every three or four dives, with many attempts broken off even before the tern hits the water.

At most times and in most places, terns are unable to catch fish because the fish are swimming deeper below the surface than the terns can dive. Terns' ability to catch fish usually depends on factors that bring the fish toward the surface. Many fishes follow vertical movements of their invertebrate prey, or zooplankton (tiny water- based animals),but these prey habitually come to the surface at night when terns are usually unable to catch fish. Sand eels, one of the main prey species for terns, come to the surface to spawn (lay eggs), and several other fishes spawn in shallow water near the shore. These spawning events provide good opportunities for terns to catch fish, but they are usually localized and transitory. Fish larvae and juvenile fish are more likely to swim near the surface than adult fish, but these prey are often too small to be good food sources for terns.

An important factor that brings larger fish close to the surface is vertical movements of the water caused by currents running over obstacles or through narrow passages. Terns regularly concentrate over tide “races” such as that at Dungeness, England, where currents running around narrow strips of land generate turbulent eddies (circular currents)that bring fish to the surface, or at gaps in barrier beaches such as those of the north Norfolk coast (also in England), where tidal currents running in and out twice each day produce similar eddies. Other sites suitable for tern feeding are where tidal currents run around islands or over shallow rocks, reefs, or sandbars, bringing fish close to the surface at predictable times of the tidal cycle. Roseate terns in particular feed regularly over such places, and in some areas travel long distances to feed at sites where fish can be caught predictably.

The most widespread factor making fish come toward the surface, however, is predatory fish chasing them from below. In many coastal areas, flocks of terns gather over schools (groups)of predatory fish and follow them as they feed, diving frantically in the short intervals when the prey fish are forced close to the surface by the predators pursuing them from below. Many tropical terns habitually follow tuna, bonito, or other predatory fish that hunt in schools. The sooty tern is thought to be dependent for much of the year on tuna and the dolphins that accompany them, ranging widely over tropical oceans and coming together wherever the tuna come near the surface to feed.

Chaco Canyon

Archaeologists have discovered that the Anasazi civilization of present-day southwestern United States flourished even after environmental problems had reduced crop production and virtually eliminated timber supplies in Chaco Canyon, the center of the Anasazi population. Despite these problems, or because of the solutions the Anasazi found to them, the canyon's population continued to increase, particularly during a big spurt (sudden and brief increase)of construction that began in A.D.1029.Such spurts went on especially during wet decades, when more rain meant more food, more people, and more need for buildings. A dense population is attested not only by the famous great houses (such as Pueblo Bonito)spaced about a mile apart on the north side of Chaco Canyon, but also by holes drilled into the northern cliff face to support roof beams, indicating a continuous line of residences at the base of the cliffs between the great houses, and by the remains of hundreds of small settlements on the south side of the canyon. The size of the canyon's total population remains unknown and much debated. Many archaeologists think that it was less than 5,000 and that those enormous buildings had few permanent occupants except priests and were just visited seasonally by peasants at the time of rituals. Other archaeologists note that Pueblo Bonito, which is just one of the large houses at Chaco Canyon, by itself was a building of 600 rooms and that all those post holes suggest dwellings for much of the length of the canyon, thus implying a population much greater than 5,000. Such debates about estimated population sizes arise frequently in archaeology.

Whatever the number, this dense population could no longer support itself but was subsidized by outlying satellite settlements constructed in similar architectural styles and joined to Chaco Canyon by a radiating regional network of hundreds of miles of roads that are still visible today. Those outliers had dams to catch rain, which fell unpredictably and very patchily: a thunderstorm might produce abundant rain in one desert area and no rain in another area just a mile away. The dams meant that when a particular area was fortunate enough to receive a rainstorm, much of the rainwater became stored behind the dam, and people living there could quickly plant crops, irrigate, and grow a huge surplus of food at that area in that year. The surplus could then feed people living at all the other outliers that did not happen to receive rain then.

Chaco Canyon became a black hole into which goods were imported but from which nothing tangible was exported. Into Chaco Canyon came tens of thousands of big trees for construction; pottery (all late-period pottery in Chaco Canyon was imported, probably because exhaustion of local firewood supplies precluded firing pots within the canyon itself);stone of good quality for making stone tools; turquoise from other areas of New Mexico for making ornaments; and macaws (parrots),shell jewelry, and copper bells from the Hohokam and from Mexico, as luxury goods. Even food had to be imported from places as far as 50 to 60 miles away, as shown by a recent study tracing the origins of corncobs excavated from Pueblo Bonito.

Chaco society turned into a mini-empire, divided between a well-fed elite living in luxury and a less well-fed peasantry doing the work and raising the food. The road system and the regional extent of standardized architecture testify to the large size of the area over which the economy and culture of Chaco and its outliers were regionally integrated.

Why would outlying settlements have supported the Chaco center, dutifully delivering timber, pottery, stone, turquoise, and food without receiving anything material in return? The answer is probably the same as the reason why outlying areas of Italy and Britain today support cities such as Rome and London, which also produce no timber or food but serve as political and religious centers. Like the modern Italians and British, Chacoans were now irreversibly committed to living in a complex, interdependent society. They could no longer revert to their original condition of self-supporting mobile little groups, because the trees in the canyon were gone, irrigation was impossible, and the growing population had filled up the region and left no unoccupied suitable areas to which to move.

Energy Distribution in Plants

Annuals are plants that go through their entire life cycle within a single year. Annuals begin their life cycles in the spring when seeds that survived the winter germinate (begin to grow).In regions with distinct dry and wet seasons, germination occurs with the onset of the rainy season. Because it has only one growing season, an annual has to distribute its photosynthates (energy-rich molecules)first to leaves. Leaves, in turn, become involved in photosynthesis, which replenishes the supply of photosynthates and increases overall plant biomass. At the time of flowering, the plant decreases the amount of energy distributed to leaves and diverts most of its photosynthate to reproduction. For example, in the sunflower, the biomass of leaves declines from approximately 60 percent of the total plant weight during the period of growth to 10 to 20 percent by the time the seeds are ripe. When in bloom, the sunflower distributes 90 percent of its photosynthate to the flower head and the remainder to the leaves, stem, and roots.

Perennial plants maintain a vegetative structure over several years. Once established, they distribute their energy in a very different manner from annuals. Before perennials expend any energy on reproduction, they divert photosynthate to the roots. This distribution to roots is in excess of that required for the development of roots for the uptake of nutrients and water from the soil. In some species, such as the skunk cabbage, the roots develop into large storage organs. Energy stored in the roots makes up a reserve upon which the plants draw when they begin growth the following growing season. When they are ready to flower, perennials divert energy from storage to the production of flowers and fruit. As the flowers fade and the fruits ripen, the plant once more sends photosynthate to the roots to build up the reserves it will need for the following spring.

Trees and woody shrubs live a long time, which greatly influences the manner in which they distribute energy. Early in life, leaves make up more than one-half of their biomass; however, as trees age, they accumulate more woody growth. Trunks and stems become thicker and heavier, and the ratio of leaves to woody tissue changes. Eventually leaves account for only 1 to 5 percent of the total mass of the tree. The production system (the leaf mass)that supplies the energy is considerably less than the rest of biomass it supports. Thus, as the woody plant grows, much of the energy goes into support and maintenance, which increases as the plant ages.

When deciduous trees (trees that lose their foliage in winter) produce leaves again in the spring, they expend up to one-third of their reserve energy on the growth and expansion of leaves. This expenditure is repaid as the leaves carry out photosynthesis during the spring and summer. After leaves, trees give preference to flowers; then tissues that transport nutrients and water, new leaf buds, and deposits of starch in roots and bark; and finally, new flower buds.

Evergreen trees have a somewhat different approach. Many have fine, sharp-pointed leaves called “needles.” Because the photosynthetic tissues in these needles can function year-round when temperature and moisture conditions permit, they do not need to draw on root reserves for new growth in the spring. They can afford to wait until later in the growing season to produce new growth. Then evergreens can draw upon energy built up earlier in the spring. For the same reason, new growth develops rapidly and matures within a few weeks.

Reproduction and vegetative growth compete for energy allotments. If photosynthesis is limited, vegetative growth gets first claim. Because the energy that reproduction demands is high-up to 15 percent in pines,20 percent in deciduous trees, and 35 percent or more in fruit trees-trees can afford an abundance of fruit only periodically, once every two to three years in deciduous trees and two to six years in evergreens.

The proportionate distribution of net production to above-ground and below-ground biomass tells much about different ecosystems. Low light conditions favor the distribution of energy to the production of leaves and stems at the expense of roots. A reduction in water or nutrient availability favors the distribution of energy to the roots.

The Early American Economy

In the sixteenth century, settlers from England, a part of Britain, began to colonize North America. These colonies remained under British control until the American Revolution, which occurred at the end of the eighteenth century. The economic theory of mercantilism that was dominant from the sixteenth to the eighteenth century held that colonies should benefit the mother country by helping her become self-sufficient and wealthy. Beginning in the 1650s,the British Parliament passed a series of laws relating to the American colonies that were designed to advance those goals. Colonists were required to buy goods only from English merchants using English ships operated by English crews. The laws stated that certain products, such as tobacco, rice, and indigo, could be sold only in England and nowhere else. On the other hand, colonial agricultural products that competed with those produced by English farmers, such as wheat or meat, could not be sold in England.

The mercantile system was designed to profit the colonizers-not the colonies-and Americans grumbled about some aspects of it and evaded some of its regulations. But on balance, the colonies probably benefited from it. For one thing, it gave colonial producers of protected goods a monopoly in the British market, and English economic energy and strength maintained a vigorous resale trade in American commodities on the European continent. Also, the producers of some commodities received the inducement of a cash subsidy, called a bounty, that profited them immensely. Finally, all American trade benefited from a strong British commercial and financial infrastructure and from the physical protection provided by the powerful British navy, which controlled the North Atlantic. The British connection was especially helpful in the second third of the eighteenth century, when English and European prosperity and population growth led to increased demand for American products. The British even lowered their barriers to American foodstuffs, creating a new market for colonial farmers.

This prosperity had a generally positive, if uneven, effect on American standards of living. As productive people enjoying extensive use of a rich resource base and taking advantage of growing market opportunities, the colonists probably lived as well as any people anywhere. For the colonies as a whole, per capita wealth on the eve of the Revolution was only slightly lower than in England, and in some places-—Pennsylvania, for example-was actually higher.

As a consequence, Americans were progressively better fed, clothed, and housed. They drank coffee, tea, and rum, flavored their food with sugar and spices, and wore clothes constructed from imported textiles. They enjoyed luxuries their parents and grandparents had been unable to acquire, such as glass windows, China, silver candlesticks, and lace. Especially attractive to colonial consumers were luxury foods for the table, an indication that they were entertaining guests and perhaps that women were playing an important role in purchasing decisions.

Prosperity fueled expectations among colonists and prospective colonists. Immigrants poured in to share the American bounty, not only from England but from Ireland, Scotland, and Germany. Farmers bought and cleared more land and hired labor. Planters went into debt for luxuries, assuming that the upward economic trend minimized their risk.

The problem with discussing broad trends in economies or the economic behavior of people in the aggregate is that it leads us to lose sight of important qualifications and distinctions. The first and perhaps the most important qualification that must be made involves farmers' behavior and intent. Virtually all colonial farmers-including even most substantial planters-practiced a safety-first agriculture that was modified only slowly and incompletely by growing commercial opportunities. The essential goal of most farmers was not to fill a market demand but to provide a sufficiency for their families and dependents. It was their first priority to feed their families from their own flocks and fields, clothe their families with thread spun and cloth woven at home, and warm their families from their own woodlots. Self-sufficiency is not the same as subsistence, however, and most farmers produced some sort of surplus for market most of the time. Market involvement was necessary because few farmers could produce enough of everything their families consumed. Hence, one might sell surplus wheat to purchase another' s surplus apples or to buy a cooking pot. Moreover, farmers needed a surplus to generate profits to pay taxes, and they wanted a surplus in order to buy those luxuries that would make their lives more attractive.

The Mycenaean Collapse

The Mycenaean culture of ancient Greece, with major centers at Mycenae and Pylos on the Greek mainland and Knossus on the island of Crete in the Aegean Sea, was at its height from

1400 B.C.E. to 1200 B.C.E. With growing cities, thriving trade, and a prosperous economy. Around 1200 B.C.E., however, the Mycenaean world showed signs of great trouble, and by 1100 B.C.E.it was gone. Its palaces were destroyed, many of its cities were abandoned, and its art, patterns of life, and system of writing were buried and forgotten.

What happened? Some recent scholars, noting evidence that the Aegean Island of Thira (modern Santorini)suffered a massive volcanic explosion in the middle to late second millennium B.C.E., have suggested that this natural disaster was responsible. According to one version of the theory, the explosion occurred around 1400 B.c.., blackening and poisoning the air for many miles around and sending a monstrous tidal wave that destroyed the great palace at Knossus and, with it, Minoan culture. According to another version, the explosion took place about 1200 B.C.E., destroying the Bronze Age culture throughout the Aegean Sea region. This second version conveniently accounts for the end of both the Mycenaean and Minoan civilizations in a single blow, but the evidence does not support it. The Mycenaean towns were not destroyed all at once; many fell around 1200 B.C.E., but some flourished for another century, and the Athens of the period was never destroyed or abandoned. No theory of natural disaster can account for this pattern, leaving us to seek less dramatic explanations for the end of Mycenaean civilization.

Some scholars have suggested that sea raiders destroyed Pylos and, perhaps, other sites on the mainland. The Greeks themselves believed in a legend that told of the Dorians, a rude people from the north who spoke a Greek dialect different from that of the Mycenaean peoples. According to the legend, the Dorians joined with one of the Greek tribes, the Heraclidae, in an attack on the southern Greek peninsula of Peloponnese, which was repulsed. One hundred years later they returned and gained full control. Recent historians have identified this legend of the return of the Heraclidae with a Dorian invasion.

Archaeology has not provided material evidence of whether there was a single Dorian invasion or a series of them, and it is impossible to say yet with any certainty what happened at the end of the Bronze Age in the Aegean Sea region. The chances are good, however, that Mycenaean civilization ended gradually over the century between 1200 B.C.E. and 1100 B.C.E. Its end may have been the result of internal conflicts among the Mycenaean kings combined with continuous pressure from outsiders, who raided, infiltrated, and eventually dominated Greece and its neighboring islands. There is reason to believe that Mycenaean society suffered internal weaknesses because of its organization around the centralized control of military force and agricultural production. This rigid organization may have deprived it of flexibility and vitality, leaving it vulnerable to outside challengers.

The immediate effects of the Dorian invasion were disastrous for the inhabitants of the Mycenaean world. The palaces and the kings and bureaucrats who managed them were destroyed. The wealth and organization that had supported the artists and merchants were likewise swept away by a barbarous people who did not have the knowledge or social organization to maintain them. Many villages were abandoned and never resettled. Some of their inhabitants probably turned to a nomadic life, and many perished.

Another result of the invasion was the spread of Greek people eastward from the mainland to the Aegean islands and the coast of Asia Minor. The Dorians themselves, after occupying most of the Peloponnese, swept across the Aegean Sea region to occupy the southern islands and the southern part of the Anatolian coast. These migrations made the Aegean Sea a Greek lake. Trade with the old civilizations of the Near East, however, was virtually ended by the fall of the Minoan and Mycenaean civilizations, nor was there much internal trade among the different parts of Greece. The Greeks were forced to turn inward, and each community was left largely to maintain itself.

Examining the Diets of Prehistoric People

One of the most important pieces of information about a prehistoric people is the nature of their diet. This information is especially relevant when trying to answer questions related to the origins and development of agriculture. Archaeologists can approach diet in a number of ways. They can study diet indirectly by figuring out what people might have eaten based on what was available in their natural environment. For example, deer, duck, turkey, and fish are known to have been available in what is now the northeastern United States for about 7,000 years. A hunting-and-gathering people in this area are likely to have used such resources at one time or another. The weakness of this method of reasoning is that we cannot know with certainty which of the available foods were most important in the diet.

On the other hand, archaeologists can approach the question of diet more directly if there has been good preservation. In many instances, food remains themselves are still present in archaeological sites. Archaeologists can study the fireplaces and garbage heaps of the people who lived at a site and recover food material if it has been preserved. Such remains as bone, seeds, and nuts are often fragile and fragmentary, however, making it difficult to get them out of the ground and back to the lab for identification and analysis. In many cases, an archaeologist takes the entire feature, including all soil, back to the laboratory, instead of attempting to separate the dry soil matrix from the fragile archaeological remains in the field. In the lab, through a number of different procedures collectively called flotation, the archaeologist uses liquid to do the job of separation, taking advantage of the fact that soil and rock will not float in some liquids, whereas organic remains will.

The next task in the reconstruction of a prehistoric diet is the identification of the species of plant or animal represented by the remains. This task can be difficult because of the fragmentary nature of such remains and the changes in them as they decay. In some cases, no precise identification can be made-the piece of bone is too small or the seed too broken up to tell what it is with any degree of confidence. But by comparing the fragment to items in a “library" of bones, nuts, and seeds, archaeologists can often identify many of the dietary remains found at a site.

The examination of the animal remains found at a site is called faunal analysis. Here, the species represented, their seabags at death, health, and physical characteristics are identified. Knowing the species of the remains as well as their age, sex, and health status can provide insight into the hunting practices of prehistoric people. Were the people hunting large numbers of herd animals in group hunts, or were their prey solitary creatures that could be hunted by individual hunters? Were the ancient people able to kill large animals in their prime or only those that were very young, very old, or sick? Also, because many animals give birth to their young seasonally, knowing the age at death of a young animal helps determine the season of the hunt. For example, the North American white-tailed deer usually gives birth to its young in May or June. If the bones of a deer found at an archaeological site indicate an age of nine months, the animal must have been killed, and the site was most probably occupied, in February or March.

Information about diet can sometimes be gathered from an animal's teeth and bones. For example, in mammals teeth provide clues to overall food sources. Carnivores such as dogs and cats, for example, have slicing teeth adapted to meat eating. Humans have a more generalized dentition, built for chewing a variety of foods. Certain wear patterns on the teeth, examined microscopically, can reveal whether the diet was made up of soft foods such as fruits or more abrasive foods such as roots and tubers. Prehistoric bones can be analyzed to determine what proportion of their chemical composition is strontium and calcium; this information can be used to determine whether plants or meat made up the bulk of the diet of certain populations.

The Development of Factories

One of the most enduring images of the eighteenth and nineteenth century British Industrial Revolution is that of the large factory, filled with workers laboring amid massive machinery driven by either water or steam power. Mechanized factory production evolved out of forms of industry that had emerged only during the early modern period (1500-1750).In the Middle Ages virtually all industry in Europe was undertaken by skilled craftsmen belonging to urban guilds, and these artisans, working either by themselves or with the assistance of apprentices or journeymen, produced everything from candlesticks and hats to oxcarts and beds. During the early modern period the urban craftsman's shop gave way to two different types of industrial workplaces, the rural cottage and the large handicraft workshop. Both of these served as halfway houses, or transitional stages, to the large factory.

Beginning in the sixteenth century, entrepreneurs began employing families in the countryside to spin and weave cloth and make nails and cutlery. By locating industry in the countryside, the entrepreneurs were able to escape the regulations imposed by the guilds regarding employment and the price of finished products. They also paid lower wages, because the rural workers, who also received an income from farming, were willing to work for less than the residents of towns. Another attraction of rural industry was that all the members of the family, including children, participated in the process. In this “domestic system” a capitalist entrepreneur provided the workers with the raw materials and sometimes the tools they needed. He later paid them a fixed rate for each finished product. The entrepreneur was also responsible for having the finished cloth dyed and for marketing the commodities in regional towns.

Rural household industry was widespread not only in certain regions of Britain but also in most European countries. In the late eighteenth century, it gradually gave way to the factory system. The great attraction of factory production was mechanization, which became cost-efficient only when it was introduced in a central industrial workplace. In factories, moreover, the entrepreneur could reduce the cost of labor and transportation, exercise tighter control over the quality of goods, and increase productivity by concentrating workers in one location. Temporary labor shortages sometimes made the transition from rural industry to factory production imperative.

The second type of industrial workplace that emerged during the early modern period was the large handicraft workshop. Usually located in the towns and cities, rather than in the countryside, these workshops employed relatively small numbers of people with different skills who worked collectively on the manufacture of a variety of items, such as pottery and munitions. The owner of the workshop supplied the raw materials, paid the workers' wages, and gained a profit from selling the finished products.

The large handicraft workshop made possible a division of labor- the assignment of one stage of production to each worker or group of workers. The effect of the division of labor on productivity was evident even in the manufacture of simple items such as buttons and pins. In The Wealth of Nations (1776),the economist Adam Smith (1723- 1790)used a pin factory in London to illustrate how the division of labor could increase per capita productivity from no more than twenty pins a day to the astonishing total of 4,800.

Like the cottages engaged in rural industry, the large handicraft workshop eventually gave way to the mechanized factory. The main difference between the workshop and the factory was that the factory did not require a body of skilled workers. When production became mechanized, the worker's job was simply to tend to the machinery. The only skill factory workers needed was manual dexterity to operate the machinery. Only those workers who made industrial machinery remained craftsmen or skilled workers in the traditional sense of the word.

With the advent of mechanization, factory owners gained much tighter control over the entire production process. Indeed, they began to enforce an unprecedented discipline among their workers, who had to accommodate themselves to the boredom of repetitive work and a timetable set by the machines. Craftsmen who had been accustomed to working at their own pace now had to adjust to an entirely new and demanding schedule.

Causes of Glacial Ages

Any theory that attempts to explain the causes of glacial ages must successfully address two basic questions. First, what causes the onset of glacial conditions? For ice sheets to have formed, average temperatures must have been somewhat lower than at present and perhaps substantially lower than throughout much of geologic time. For that reason, a successful explanation would have to account for the gradual cooling that finally leads to glacial conditions. The second question is: What caused the alternation of glacial and interglacial stages that have been documented for the Pleistocene epoch (1.6 million to 10,000 years ago)?Whereas the first question deals with long-term trends in temperature that occur on a scale of millions of years, this second question relates to much shorter-term changes.

Probably the most attractive proposal for explaining why extensive glaciations have occurred only a few times in the geologic past comes from the theory of plate tectonics. Not only does this theory provide geologists with explanations about many previously misunderstood processes and features, it also provides a possible explanation for some previously unexplainable climatic changes, including the onset of glacial conditions. Since glaciers can form only on land, we know that landmasses must exist somewhere in the higher latitudes before an ice age can commence. Many believe that ice ages have only occurred when Earth's shifting crustal plates carried the continents from tropical latitudes to more poleward positions.

Glacial features in present-day Africa, Australia, South America, and India indicate that these regions experienced an ice age near the end of the Paleozoic era, about 250 million years ago. For many years this puzzled scientists. Was the climate in these relatively tropical latitudes once like it is today in Greenland and Antarctica? Why did glaciers not form in North America and Eurasia? Until the plate tectonics theory was formulated and proven, there was no reasonable explanation. Today scientists realize that the areas containing these ancient glacial features were joined together as a single supercontinent called Pangaea that was located at high latitudes far to the south of their present positions. Later, this landmass broke apart and its pieces, each moving on a different plate, drifted toward their present locations. It is now believed that during the geologic past, plate movements accounted for many dramatic climatic changes as landmasses shifted in relation to one another and moved to different latitudinal positions. Changes in oceanic circulation also must have occurred, altering the transport of heat and moisture and, consequently, the climate as well. Since the rate of plate movement is very slow, on the order of a few centimeters per year, appreciable changes in the positions of the continents occur only over great spans of geologic time.

Since climatic changes brought about by moving plates are extremely gradual, the plate tectonics theory cannot explain the alternation of glacial and interglacial climates that occurred during the Pleistocene epoch. Therefore, we must look to some other triggering mechanism that may cause climatic change on a scale of thousands rather than millions of years. Today many scientists believe or strongly suspect that the climatic oscillations that characterized the Pleistocene may be linked to variations in Earth's orbit. This hypothesis was first developed and advocated by the Yugoslavian scientist Milutin Milankovitch and is based on the premise that variation in incoming solar radiation is a principal factor in controlling Earth's climate. Milankovitch formulated a comprehensive mathematical model based on the following elements: variations in the shape (eccentricity)of Earth's orbit around the Sun; changes in obliquity, that is, changes in the angle that Earth's axis makes with the plane of Earth's orbit; and the unsteady wobbling of Earth's axis, called precession.

Using these three factors Milankovitch calculated variations in the seasonal timing of the receipt of solar energy and the corresponding surface temperature of Earth over a long period of time in an attempt to correlate these changes with the climatic fluctuations of the Pleistocene. In explaining climatic changes that result from these three variables, it should be noted that they cause little or no variation in the total amount of solar energy annually reaching the ground. Instead, their impact is felt because they change the degree of contrast between seasons. Somewhat milder winters in the middle to high latitudes means greater snowfall totals while cooler summers would bring a reduction in snowmelt.

Imitation in Child Development

When does imitation become possible? How important is this ability in the learning of infants and children? Andrew Meltzoff and Keith Moore argue that even newborns and very young infants imitate a variety of responses, including tongue protrusion, mouth opening, and possibly even facial expressions portraying such emotions as happiness, sadness, and surprise. Although some investigators have been unable to replicate these results, many others, including the authors of one study involving infants from Nepal, report a high degree of imitative competence in newborns.

Even more controversial is what imitative behaviors mean. The influential Swiss psychologist Jean Piaget, for example, claimed infants younger than eight to twelve months could imitate someone else's behavior only when able to see themselves making these responses. Because babies do not ordinarily view their own faces, imitative facial gestures would be impossible, according to Piaget, until after about a year in age, when symbolic capacities (abilities to use a symbol, an object, or a word to stand for something)emerge. From this perspective, then, the facial gestures that infants younger than a year make in response to a model's facial expression are stereotyped, rigid responses set off by or linked to limited forms of stimulation rather than imitations of what the child has seen. For example, perhaps tongue protrusion by a model arouses the infant, which in turn promotes a sucking response that naturally invokes tongue protrusion from the infant. If this is the case, infants could be responding to just a few types of stimuli and producing a kind of reflexive (automatic)motor activity that is not really a form of imitation.

Meltzoff and Moore counter that very young infants imitate a variety of responses, modify their imitations to increasingly match the modeled behavior over time, and exhibit their imitations primarily to other people and not to inanimate objects. These arguments contradict the view that such behaviors are simply a fixed pattern of reflexive actions. They propose instead that infants imitate in order to prolong interacting with others. In fact, babies as young as six weeks will imitate behaviors of a model up to twenty-four hours later. Such imitation is a way to continue the earlier imitative communication with the model. If this interpretation is correct, imitation has an important social-communicative function and is one of the earliest games babies play with other people.

Between six and twelve months of age, infants display far more frequent and precise imitations, matching a wide range of modeled behaviors. Piaget and others believed that deferred imitation, the ability to imitate well after some activity has been demonstrated, is not possible until about eighteen to twenty- four months of age. Piaget believed deferred imitation, along with pretend play and the emergence of language, marks an important transition from one stage of thinking to the next and provides one of the first major pieces of evidence for symbolic capacities. However, as we have already seen, Meltzoff and Moore claim infants as young as six weeks can reproduce a model's behavior a day after seeing it.

Deferred imitation involving actions associated with objects can also be observed far earlier than Piaget claimed. For example, six-month-olds will remove a mitten from a puppet's hand, shake it, and try to put it back on the puppet after observing this sequence of actions performed by a model twenty-four hours earlier. Moreover, toddlers as young as fourteen months who see a peer pulling, pushing, poking, and inserting toys in the laboratory or at a day care center will reproduce the behaviors in their own homes as much as two days later when given the same toys. The capacity for deferred imitation, then, appears to exist much sooner than previously assumed. In fact, the results are consistent with research on memory showing that infants younger than one year can recognize stimuli hours and even days later. The observances of imitation at very young ages are important from a social learning perspective, providing clear and compelling evidence that infants, as well as older children, learn many new behaviors by observing others.

Birds and Food Shortage

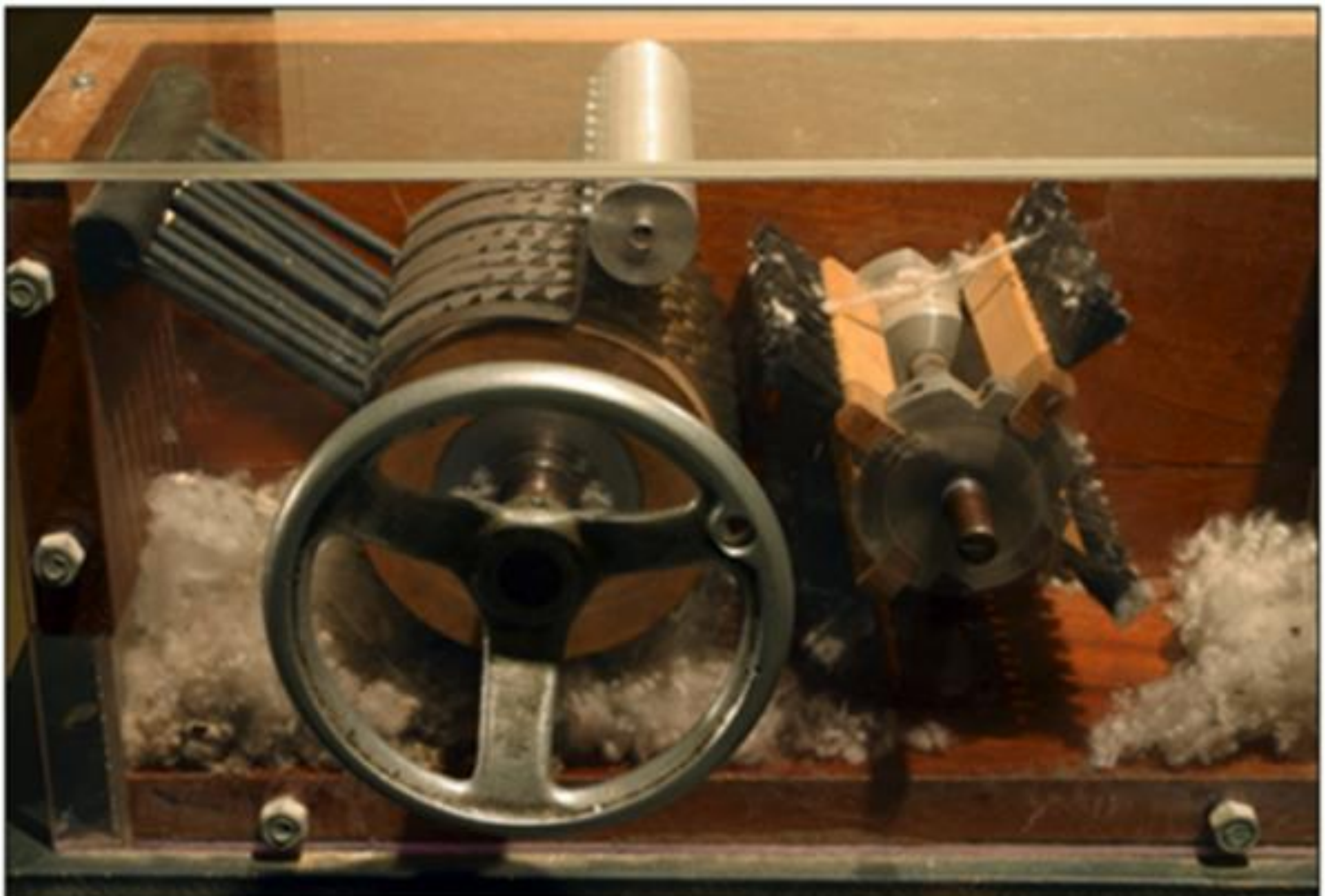
The term “food shortage” has been used for food that is inadequate in quantity, quality, or availability. Whatever the abundance of food in the environment, if it is for some reason unavailable, bird species in either inland or coastal areas may starve. Food sources like intertidal invertebrates retreat deeper in the mud in cold weather, putting them beyond the reach of shorebirds, while many soil-dwelling invertebrates move deeper into the ground during drought. Such behaviors illustrate the general points that only a proportion of potential food items may be accessible to birds at any one time, and that this proportion may change through time in response to other conditions. Secondly, if food is plentiful but of low nutrient content or digestibility, a bird may be unable to process enough each day to maintain its weight. This is particularly true of herbivores (plant eaters)eating fibrous, or coarse, vegetation, and food shortage may arise partly from the structure of a bird's inner parts, or gut, which limits the rate at which food can be processed. In general, among birds, species differences in gut structure correspond with dietary differences, and in species whose diet changes seasonally, gut structure may change accordingly. The need for change in gut structure can make some species vulnerable at times of rapid shift in diet, as when snowfall forces birds suddenly from one food type to another.

Many bird species cushion the effects of food shortage by storage, either of body fat or of food itself. In small birds, fat stores are accumulated mainly for use overnight, and few species can last more than a day or two without feeding. Larger birds, with their relatively lower metabolic (energy)demands, can store relatively more energy per unit of body weight. As a result, larger birds can rely on internal reserves for longer than small ones, more than two weeks in large waterfowl. Many birds store some body fat and protein in preparation for breeding, extreme examples being Arctic-nesting geese, in which the females produce and incubate their eggs almost entirely on reserves accumulated in wintering or migration areas. For some weeks, then, they are protected against the need to feed on their breeding grounds. Most migratory birds accumulate body fat for the journey, the amount depending on the duration of unbroken flight.

Species that store food items outside the body may retrieve them hours, days, or even months later, depending partly on the type of food. Some meat eaters cache (store)prey for consumption later in the day or in the following days, while some seed eaters may store food in autumn to last through the winter or even year-round. Long- term storage is frequent among various birds, such as tits, nuthatches, and woodpeckers, reaching its extreme in certain jays and nutcrackers. In some such species, stored food can provide more than 80 percent of the winter diet, and can also feed the adults into the next breeding season, influencing the numbers of young raised. In good seed years, huge amounts of seed can be stored. Crucial aspects of hoarding behavior are that individuals should guard their stores or should hide them (to reduce theft)and remember where they are. Some species store items individually, each in a separate place, and in experiments have shown incredible feats of memory. But none of this behavior alters the dependence of such birds on their food supplies; it simply enables them to use their foods more effectively, taking advantage of temporary gluts (oversupply)to lessen later shortages. It does not prevent them being limited by shortages in years of poor seed crops.

Yet other birds can cushion themselves against temporary food shortages by becoming torpid, meaning they achieve an inactive state in which body temperature and heart, respiratory, and metabolic rates are greatly depressed, thereby conserving energy. In some small species, this happens in some degree every night, but it is especially marked in some small hummingbirds, such as Anna's hummingbird of western North America. Some other species, such as the common swift, can remain torpid for several days at a time when cold weather renders their insect prey inactive, and at least one species, the common poorwill, can remain torpid for several weeks, in a state akin to hibernation in mammals. These adaptations help the species concerned to last through difficult periods, whether hours, days, or weeks, but they cannot eliminate altogether the risk of starvation.

Cotton Ginning and Interchangeable Parts: The Legacy of Eli Whitney



Eli Whitney's Cotton Gin

© Courtesy of Wikimedia commons/Tom Murphy VII

Cotton is a soft fiber that grows around the seeds of the cotton plant. It was first cultivated in the Indus Valley about 6,000 years ago and then spread to Egypt, Persia, China, and the Mediterranean. For thousands of years, the cultivation of cotton was very labor-intensive. Many steps were involved in preparing cotton to be used in a garment, and many machines were invented to make cotton more valuable as a general-purpose textile material.

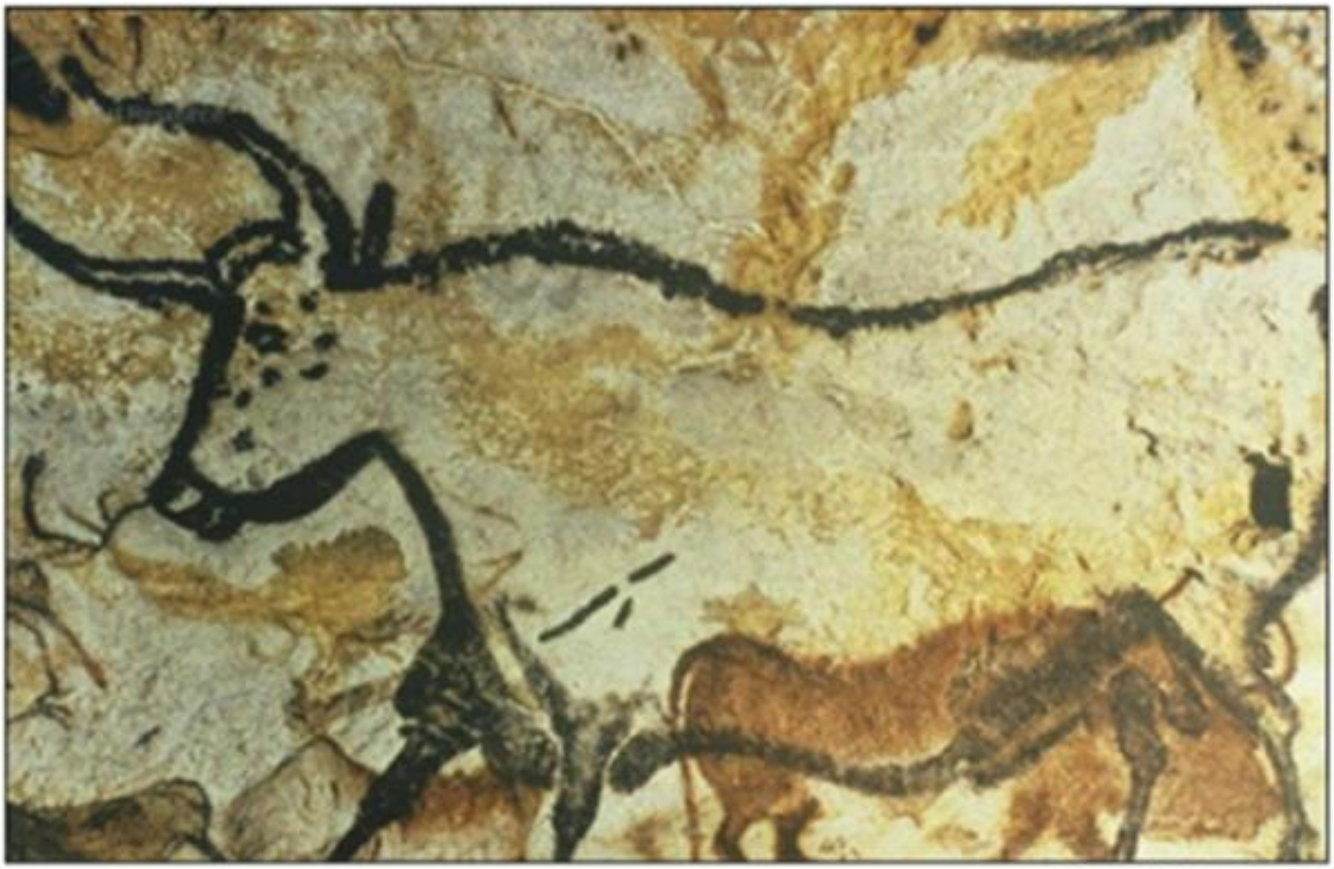
The process of separating cotton fibers from seeds is called ginning, which is depicted as far back as the fifth century in paintings in the Ajanta Caves in India. Egyptian cotton has the longest fibers and is used to make premium fabrics. In the American South, the high quality long-staple cotton can be grown only in a narrow band along the Carolina and Georgia coast and is called Sea Island cotton. The interior areas of the American South can only grow Upland short- staple cotton. The shorter-length fibers mean reduced yarn and cloth quality, and they have "fuzzy" seeds because the fibers are tightly attached to the entire seed surface. This strong attachment of fibers to seeds makes it difficult to remove the fibers without damage, which made the process expensive as it was time-consuming and labor- intensive. These time-consuming tasks were done by slaves in the American southern states during the 1800s.

The modern cotton gin was invented by the American Eli Whitney (1765-1825),who graduated from Yale University in 1789 and is known for two great inventions: the cotton Gin, which transformed the southern economy, and interchangeable parts in machinery, which led to the mass-production method of manufacturing. When he graduated, he was short of funds and went to Georgia to seek his fortune in plantations. He was introduced to the problem of finding a better way to gin the Upland short-fiber cotton, which led him to invent the cotton gin in 1794.Whitney's cotton gin made the mechanical separation of fiber from seeds much more efficient. His machine had spiked saw teeth mounted on a revolving cylinder in a box, which was turned by a crank. The teeth pulled the cotton fiber through small slotted openings that were too small for the seeds to pass; thus, the lint was separated from the seeds. Then a rotating brush removed the fibrous lint from the teeth, and the seeds fell into a hopper. Whitney's machine could separate up to 50 pounds of cleaned cotton daily, making cotton production profitable for the southern states.

Unfortunately for Whitney, his cotton gin was extremely simple and easy to copy, so he failed to profit from his invention. Soon after, imitations of his machine appeared, and his 1794 patent for the cotton gin could not be upheld in court until 1807.Instead,he decided to make money by going into the ginning business, and manufactured and installed cotton gins through Georgia and the southern states. However, his ginning plants were resented by the planters, who thought that he charged too much, and hence they found ways to circumvent his patent.

In later years, Eli Whitney cultivated social and political connections through his status as a Yale alumnus and through his marriage. Although he had never made a gun in his life, through those connections he won a contract from the War Department in 1798 to make and deliver 10,000 muskets (firearms used by soldiers before the invention of the rifle)for the army in 1800.Treasury Secretary Wolcott sent him a "foreign pamphlet on arms manufacturing techniques, and Whitney began to talk about interchangeable parts. This concept was a departure from the usual method of having an individual master machinist make each part of the musket and fit them together, which is slow and painstaking. Whitney's method was to have many less skilled machinists specialize in making only one part per person to highly specified dimensions, so that the parts would be interchangeable and could then be assembled. This is the heart of the mass-production method later made more famous by Henry Ford in the manufacture of the Model T automobile. Whitney ultimately delivered on the army contract in 1809,and he spent the rest of his life promoting interchangeability.

Interpreting Prehistoric Cave Art



© Art Resource

The Upper Paleolithic period began about 45,000 B.c. It is from this period, in several caves located in Europe, that archaeologists have discovered remarkable examples of prehistoric art. One of the early interpretations of cave art was that it was “art for the sake of art," much as we today might go to a museum to see the skills of artists. This is a Western-culture-centered interpretation, however, and some would argue that there might be other, context-specific interpretations that are more suitable. The location of this art deep within dark caves, in which small flickering lamps would only reveal small portions of painted walls, also does not seem to fit with an interpretation of prehistoric “art galleries."

Another early interpretation was based on the fact that most of the images are animals, and the majority of these, such as horses and bison, were hunted for food. This viewpoint became known as the "hunting magic" explanation. The drawing of the animals was interpreted as a way to magically ensure that an upcoming hunt would be successful. The hunting-magic interpretation has much appeal because we know that hunting was an important part of daily life during this period. And, in direct contrast to the art-for-art's-sake explanation, the practice of these “rituals” deep within caves suggests that not everyone participated. There are many nonhunting images in the caves, however, that do not support this explanation, such as geometric shapes and human figures.

In an effort to include all the types of images in Upper Paleolithic cave art in a comprehensive interpretation, some researchers turned to aspects of how the human mind works during altered states of consciousness and what images the mind “sees” during different phases of altered states. This explanation is called “entoptic phenomena,” and it argues that all modern human brains experience the same sets of visual images in the same progression. For example, during the first stage it is common to see geometric patterns; during the second stage, the brain begins to associate various geometric designs with real objects; and during the third stage, the brain sees actual animals, people, and monsters. Altered states can be achieved in many ways-drugs, intense dancing, sitting in absolute darkness as in a deep cave-and the entoptic-phenomena interpretation argues that cave-art images represent the "visions “seen during these experiences.

Other explanations focus more on the use of cave art as a form of communication. That is, its presence and the types of images were used to establish social identities and perhaps as territorial markers. Communication as an explanation is based on identifying different styles that represent different groups of people and is a key element at aggregation sites, places where these people came together at certain times. Communication also is seen as adaptive because it enhances the survival of groups using social networks or alliances. The abundance of cave art in France and Spain, particularly during the Late Upper Paleolithic, is thought to be one outcome of the dense packing of people as they moved south to escape the harshest conditions of the extreme cold of the glacial maximum of the last ice age. Art was used to form alliances and thus resolve disputes about resources between groups who could not easily move away because of the close presence of many groups and the inhospitable nature of more northern areas in Europe.

In recent years, accumulating evidence suggests that Upper Paleolithic people created wall art not only in relatively inaccessible caves, but also in the rock shelters where they lived. Like Later Stone Age rock shelters in Africa, those of the Upper Paleolithic in Europe were exposed to sun, rain, and other weather, and most of the art present in these locales has long since disappeared. The traces we do have, however, suggest that wall art was a much more common feature of daily life than the art present deep in caves might suggest. And, if wall art was much more widespread and was typical of people's living sites, then single-cause explanations for wall art, especially those that are based in part on its inaccessibility in deep caves, seem less likely to be accurate models for all Upper Paleolithic wall art.

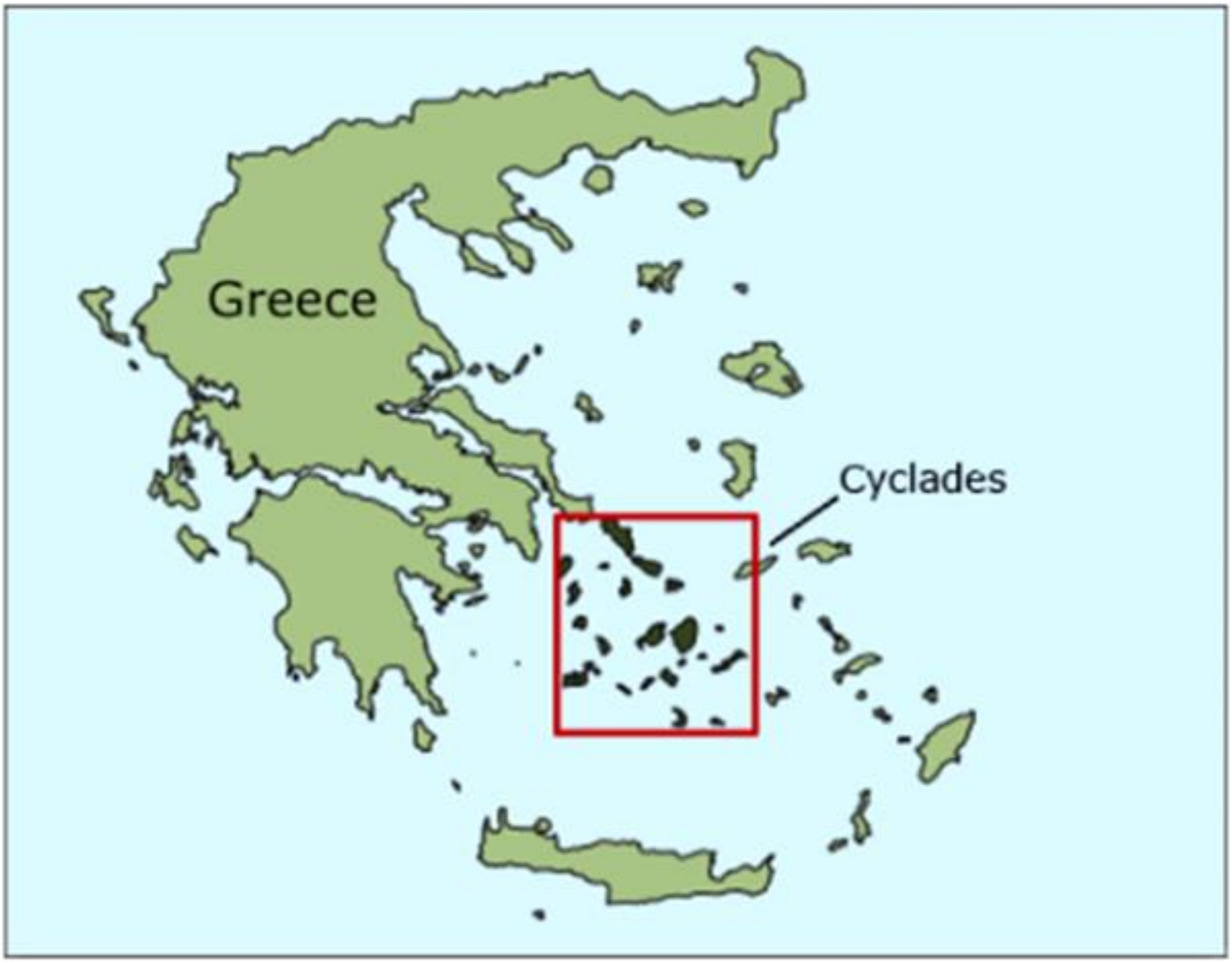
Sea Ice

Sea ice covers an area of almost 16 million square kilometers of the Arctic seas during its maximum extent in winter. Each summer the pack ice (Arctic ice cover)melts and shrinks in area, and, at a minimum, extends over about 9 million square kilometers. The sea ice is an important part of the climate of the Arctic, and indeed of the whole world, because it has a large effect on the way the surface of the Arctic is heated. Sea ice is highly reflective-we see this by the need to wear sunglasses when walking over snow or ice. Water, on the other hand, appears rather dark by comparison. Where sea ice covers the Arctic seas, much radiation from the Sun is reflected back into the atmosphere. However, where open water is present, most of that radiation is absorbed and acts to heat the water. As the water heats up, more ice can be melted, and so on. This is called a positive feedback effect and is one of the key reasons why scientists are monitoring both the extent and the thickness of Arctic Sea ice today. If sea ice covers a progressively smaller part of the Arctic seas in a world that may be affected by greenhouse gases, then this positive feedback could be critical to the rate at which our planet warms up.

Another important aspect of the formation of sea ice is that when water freezes, a large proportion of its salt content is rejected (forced out into the surrounding water). Indeed, it is possible to melt old sea ice and drink the water. However, the seawater produced in areas where sea ice forms is both very cold and also of high salinity (salt content).The presence of so much salt produces some of the densest water found in the world's oceans, and this water sinks to the seafloor as a result. In fact, the very dense waters produced by sea ice freezing in the Arctic are a vital factor in driving the circulation of the oceans, because these waters flow to the deepest parts of the ocean floor. Some areas of the Arctic seas are places of intense sea ice production, where upwelling (the movement of above-freezing water from the lower depths of the ocean to the surface)or strong winds consistently keep the sea open year-round. One example is the North Water at the head of Baffin Bay. New sea ice can, therefore, continue to form throughout the winter in such open-water areas, which are known as polynyas, from the Russian language.

Sea ice also presents a number of hazards to human activity in the waters around the Arctic islands. In the heroic era of nineteenth-and early-twentieth-century exploration of the Arctic archipelagos (groups of islands),the presence of sea ice in many passages between islands made travel by sea difficult and dangerous. The British navy, for example, sent several tens of ships to the Canadian Arctic during the first half of the nineteenth century in an attempt to find a navigable passage connecting the north Atlantic and Pacific Oceans, referred to as the Northwest Passage, as a route for trade with Asia. However, this period of exploration coincided with some of the coldest weather of the last few thousand years, which is known as the Little Ice Age. The ice conditions encountered by many of these ships were appalling. A number of vessels became trapped in sea ice as the short summers came to an end, and several were crushed when pressure built up in the sea ice as it was driven by strong winds. The great dangers inherent in these early explorations of the islands and passages in, for example, the Canadian north are exemplified by the remarkable story of Sir John Franklin and his crews of the ships Erebus and Terror, who disappeared in the late 1840s and whose fate has never been finally established. By contrast, the Norwegian explorer Roald Amundsen was the first to navigate a ship through the Northwest Passage, and he did so only in 1903-06,when the climate had warmed up after the Little Ice Age and sea ice conditions had become less severe.

The Early Cyclades



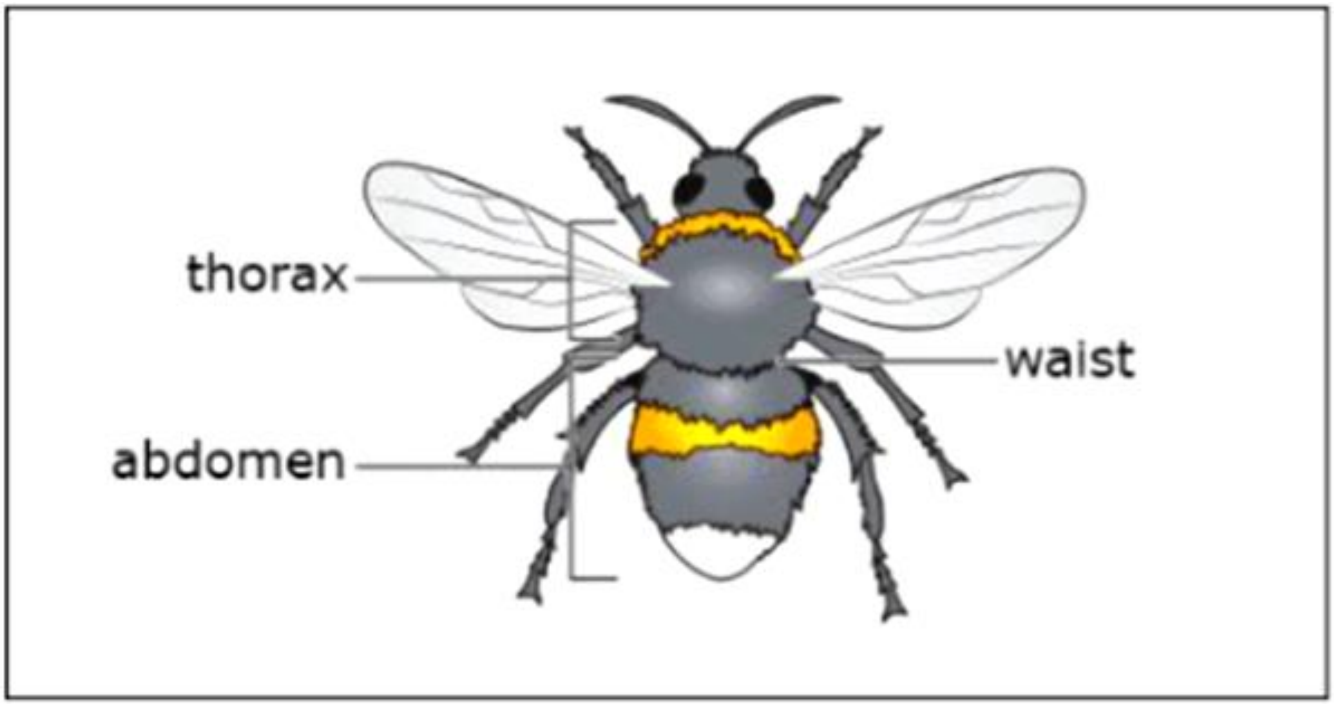
The Cyclades, a group of islands in the Aegean Sea to the east of mainland Greece, eventually played an important part in Greek civilization, though at first, they were sparsely populated. Radiocarbon dates from excavations throughout the islands tell us that farmers had settled on Naxos as early as the fifth millennium B.C.E. Between 4000 and 3000 B.C.E., farming communities flourished on Andros, relatively close to the mainland, as far out as Paros, and possibly on Mykonos, in the heart of the Cyclades. As far as one can tell, the early settlers avoided smaller islands, with their tiny patches of arable soil and often sparse water supplies. Milos was rich in obsidian, a volcanic glass used for tools and jewelry, and the island was certainly visited for fishing and fine-grained stone, but the first date for permanent occupation remains an open question. Throughout the Cyclades, the founding settlers seem to have preferred larger and medium-sized islands, even if they were far from the mainland. This speaks volumes about the seaworthiness of what by now must have been fairly large boats. Quite apart from anything else, such vessels would have had to transport sheep and goats across open water for as long as two days.

Why did people colonize islands far out in the Aegean? One theory argues that regular visits in search of obsidian, or for tuna fishing in otherwise inaccessible waters, might have led in due time to permanent settlement. Archaeological finds on the mainland do not support the obsidian hypothesis, for early obsidian finds are merely fragments, apparently carried back haphazardly. It was only much later, some centuries before 3000 B.C.E., that much greater quantities of obsidian arrived on the mainland, deliberately prepared in a variety of ways, arriving as cargoes that included carefully hewed blocks of the volcanic glass. Whether greater loads resulted from colonization or simply from an increase in the tempo of exploitation and visitation is a matter for discussion. Nor is fishing a particularly compelling reason for colonization. Fish were never more than a dietary supplement in these waters, where marine life was far from abundant. Tuna migrations move rapidly and unpredictably, usually heading south in the fall. Laborious voyages in search of such an unreliable food source were not worth the trouble or the risk, given the difficulty of predicting them.

Living permanently on the Cyclades was never easy, for survival depended on diversifying one's crops among carefully selected microenvironments. Barley, wheat, and pulses (a plant group including peas and beans)were staples, while goats and sheep thrived on rugged island slopes. Olives and vines became important crops later, when they were widely traded commodities, but almost certainly not in earlier millennia. The uncertainty of the Aegean agricultural year meant that storage of foods of all kinds was of great importance, a process that must have involved sharing among neighboring communities and perhaps even between one island and another. Unlike mainland farmers, islanders could not fall back on a cushion of reliable wild plant foods or game like fallow deer. The realities of daily life meant that most island communities were small and widely dispersed over generally rugged island terrain. Populations were tiny, to the point that people in the Cyclades were a scarce resource. Few islands attained self-sufficient populations and marriage networks of about three hundred to five hundred people. There were many more people living in a single early Mesopotamian city such as Uruk, one of the earliest urban settlements in the world in the fourth millennium B.C.E., than in the entire Cyclades. Inevitably, then, there was a high degree of interdependence among the islands, for exchange of food and other commodities and mobility were the only long-term survival strategies.

This constant movement touched every facet of island existence- marriage and kin ties, planting and harvest, food storage, the need for toolmaking stone and other commodities as prosaic as grinding stones, movements triggered by drought, excessive rainfall, and other short-and long-term climatic events, chance migrations of tuna shoals, and all the complex demands of ceremonial life and ritual observance. Seafaring operated here within ever-changing lattices of interaction that linked people over considerable distances and across often turbulent seas.

Bumblebee Heat Regulation



While flying, bumblebees have a body temperature that is generally well above that of the air around them, and tends to be constant at about 35 C.Keeping warm is harder the smaller you are. Small creatures, like bees, have a vast surface area relative to their volume, and lose heat incredibly quickly. Yet bumblebees can keep themselves warm even when the surrounding air is 30C cooler than their body temperature. Scientist Bernd Heinrich found that the answer to bees' ability to retain heat is in two parts: keeping heat in, and generating it in the first place. Keeping warm is helped if you have insulation, and of course bumblebees have furry coats. The vital thing for a bumblebee is to keep its thorax (the middle section)warm, because this is where the flight muscles are; unless the thorax is warm enough, the muscles cannot contract sufficiently fast, and the bee cannot take off. The temperature of the abdomen does not matter much in flight. The abdomen and thorax in a bumblebee are connected by a very narrow waist, and the front part of the abdomen contains a sac of air, so heat loss from the thorax to the abdomen is minimal (air is a poor conductor).A bumblebee's heat is generated by the contractions of the flight muscles. In flight a bumblebee flaps its wings 200 times per second. This generates a lot of heat, but of course comes at a cost: bumblebee flight is enormously expensive in terms of the energy that it uses. Consequently, bumblebees have to eat almost continually to keep warm.

A bumblebee's dense furry coat has obvious advantages, but it can create problems when the weather is warm. Bumblebees cannot help but produce lots of heat when they fly, which can be difficult to get rid of if the air temperature is also high. This is probably why bumblebees are not common in Mediterranean countries. If their body temperature exceeds 44 C they will die; as they approach this lethal limit, their metabolism collapses and they become unable to fly. It is noticeable that those species of bumblebee that occur in warmer climates tend to have shorter fur, while those from high latitudes and altitudes tend to be very large and very furry. Hence the huge size and long shaggy coat of the world's largest bumblebee, Bombus dahlbomii, which inhabits the high Andes of South America.

Bumblebees do have a trick to help them lose heat in hot weather. They normally keep their abdomen at a much lower temperature than their thorax, with the narrow waist connecting the two acting as a barrier. If the thorax starts to get too hot, the abdomen starts rhythmically to contract, sending surges of cool blood into the thorax and sucking back waves of hot blood. This heats the abdomen and so raises the surface area from which heat can be lost. Nonetheless, on very hot days in summer bumblebees will tend to stop seeking food as noon approaches and recommence in the cool of the evening.

Pumping heat from the thorax to the abdomen can also serve a very different purpose. The northernmost social insect in the world is a bumblebee known as Bombus polaris which lives well within the Arctic Circle: large and unusually hairy, it can exist in regions where, even in the height of summer, air temperatures rarely exceed 5C. Unlike other bumblebees, the Bombus polaris queens maintain a stable, high abdominal temperature (greater than 30 C)by pumping hot blood from their thorax to the abdomen. This enables them to develop eggs within their ovaries quickly, which is important in the very short Arctic summer. As the workers and males have no eggs to develop, their abdomens are substantially cooler.

The larger the bee, the easier it is for it to keep warm but the more prone it is to overheating in hot weather. This may explain why queen bees are so much larger than workers, for they begin flying earlier, in the spring when the weather is cold. It may also explain why the worker bees that leave the nest to gather food tend to be larger, on average, than those that stay in the nest to look after the brood.

Dams and Flood Control

Water traveling through a dam

There are both advantages and disadvantages to building dams on rivers. On the one hand, dams can be beneficial to human populations. They provide reserves of fresh water, a commodity crucial to life, both directly as drinking water and indirectly because irrigating farmland provides crops for food. Moreover, the energy of water flowing through dams can be harnessed to provide electricity. The location of dams on rivers also enables flooding to be controlled. Thus, dams can provide protection from the extremes of the local climate. Heavy rains in headwater areas, which might otherwise lead to downstream flooding, are impounded in the reservoir(water storage area) behind a dam. By the same token, during dry periods that might otherwise produce drought, water can be released from the reservoir to downstream areas.

On the other hand, introducing a dam into a river system will alter the structure of the river. Sediment will be deposited upstream of the dam and in the reservoir (reservoirs are inevitably only transient features and will ultimately fill up with sediment). Downstream, increased erosion will occur. Landscapes and ecosystems are submerged beneath the new reservoir. The dam itself represents a potential hazard should it ever fail. The collapse of the St. Francis Dam in southern California in 1928 was the consequence of building it against a weak rock support only two years earlier.

Government policies in the United States regarding the construction of dams have shifted considerably in the past century. The early part of the twentieth century saw an enormous boom in dam construction, particularly in the arid West. In recent years this trend has been halted and even reversed in some cases, as plans to tear down dams and restore stream systems to their natural state are under consideration. Any discussion of the construction of a dam across a "wild" river usually pits conservationists against power companies and farmers. One side stresses the need to conserve for future generations; the other stresses the needs of the modern population and economy. The example of the Trinity River Dam in northern California serves to illustrate many of the issues at work here.

Prior to the construction of a dam on the Trinity River near Mount Shasta in northern California, the river was a typical youthful mountain stream; its bed was rocky and strewn with boulders, and only minor brush (woody plants) grew close to the river edge. It was a clear, cold stream with various game fish including a large resident population of trout and annual spawning runs of steelhead and salmon. The stream bed was kept free of sediment and brush by vigorous floods derived from the spring thaw in the high mountains nearby. This mountainous region was a wilderness, with little industry except logging. Downstream, however, was a growing population and economy. Energy needs of the area grew, and the spring floods became more than just a nuisance-they were a threat to safety. A dam was built to control the floods, conserve the water for later use in irrigation, and act as a hydroelectric facility.

After the construction of the dam, the spring floods were contained by the new Trinity Lake. Control of the river meant that regular water delivery could be provided, and irrigation for farms and ranches downstream was a boon to agriculture. Power generated from the hydroelectric plant is completely nonpolluting and provides a substantial proportion of the increasing energy needs of the area. But since there are no spring floods to clean the river of sediment, weeds, and brush, instead of a clear, fast-moving, cold stream, the Trinity River has become sediment laden, slower moving, warmer, and less hospitable to the prized game fish that formerly populated the river. There are no longer any salmon or steelhead runs up the Trinity River.

Trinity Lake, formed behind the dam, is a popular recreational area that attracts many people for boating, water skiing, and fishing. Agriculture and residents of the cities and towns downstream benefit from the reliable supply of water and power. The lake has an excellent resident population of trout, and because of the good fishing, the surrounding forest supports a growing population of osprey-a hawk that many conservationists considered on the brink of extinction.

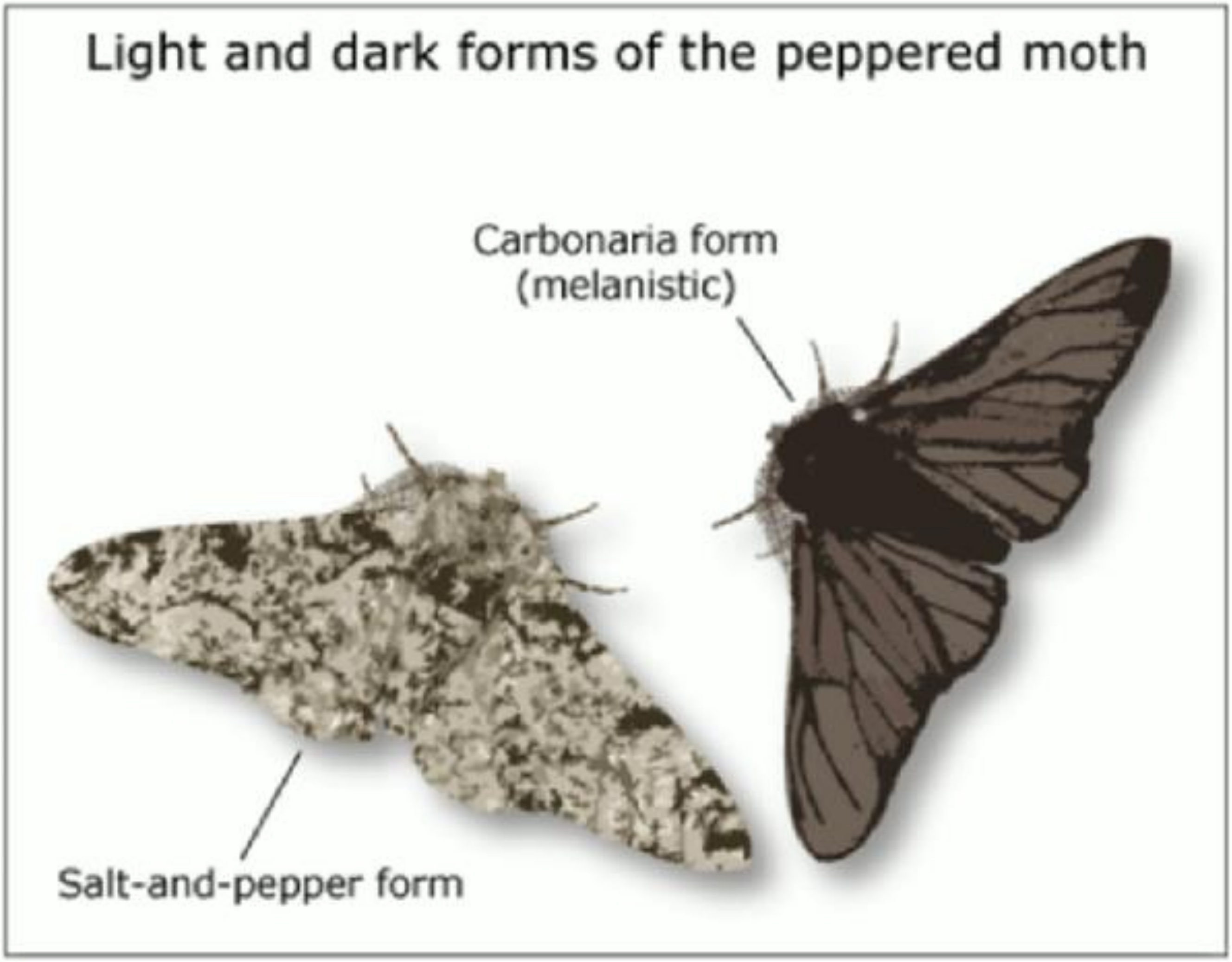
The Development of Land Flora

Today's most primitive land plants, the liverworts and hornworts, have patterns in their DNA that indicate a connection to a group of complex green algae that are found only in freshwater, which suggests that land plants arose in freshwater lakes and streams or from river deltas-not from the ocean shore. But what might have caused a plant to adapt to the harshness of a land environment? The most reasonable scenario suggests that early land plants lived along streams or in ponds that dried up during part of the year. Their first adaptation was the spore, a reproductive cell that could survive this dry period. Carried by the wind, the microscopic spore could be scattered across wide areas, germinating (starting to grow)—if lucky -in a distant pool of quiet water or moist mud. Such spores are the first evidence of plants living on land.

Earlier, blue-green bacteria probably lived on damp soil and in shallow pools over hundreds of millions of years. They were probably joined by green algae and fungi to form puddles of green glop in moist depressions and along the edges of ponds and streams. These lineages (lines of descent) may have populated land surfaces for many millions of years before more complex land plants arose. And they, too, had to develop sporelike stages to survive drying out and achieve dispersal. Unfortunately, because they did not possess resistant tissues, these early forms of land flora left no trace in the fossil record. The first "true" land plants were probably flat liverwort- like plants with little leaves only one cell layer thick. Again, these first land plants are remembered in the rocks only by their spores. Being able to withstand the physical stresses of drying and rehydration, the tough spore wall is also resistant to decomposition. Thus, spores became an important element in the fossil record. Algal and fungal spores are quite different from these. These early spores are first recorded in the rocks dated at around 470 million years ago, and they are usually found as groups of four (called tetrads). By about 430 million years ago, spore tetrads declined in abundance and separate individual spores with a Y-shaped (trilete) configuration on one side became common.

About 410 million years ago, with plant spores diversifying, the rocks preserved the first fragmentary evidence of larger plants. It was not until more complex land plants had evolved desiccation-resistant surfaces and a strong erect structure that a really three-dimensional terrestrial vegetation came into being. Waxy surfaces were a major innovation in the advent of larger terrestrial plants. In addition to reducing water loss through outer surfaces, a waxy layer protected against ultraviolet radiation, microbial attack, and corrosive chemicals. Unfortunately, such protection also presents a problem: how can a plant absorb carbon dioxide from the air if it's covered with wax? Moist interior cell surfaces must be exposed to air in order to absorb carbon dioxide for photosynthesis (the process through which plants obtain energy by using sunlight, carbon dioxide, and water). Inevitably, exposing moist cells to air will also result in water loss by evaporation from within the plant. And because carbon dioxide concentrations are low in air, a great deal of water will be lost in the process of absorbing carbon dioxide. (This is why plants require a lot more water to survive than do animals with a similar volume of active tissue.) The solution to this plant problem came in the form of special little pores called stomates on the surfaces of the plant: pores that could be opened for gas exchange, and closed when conditions became too dry. The other major adaptation for life on land was a 'plumbing system" to conduct water from the roots to the parts of the plant active in photosynthesis. The vascular/plumbing system includes thick-walled cells that quickly become dead, empty, and open at the ends. Laid out end-to-end these empty cells resemble a system of pipes, facilitating the flow of water through the plant. These same dead thick-walled cells would also serve as architectural support. With a vascular system and stomates in place, the earliest erect plants were able to draw water up to the aerial parts of the plant and provide strength in the face of wind and weather.

Evolution in Peppered Moths



Since Darwin proposed the theory of evolution, a vast literature has emerged that supports the precepts of the theory and increases our understanding of the mechanisms of evolution. One of the most striking cases of evolution in action is that of industrial melanism, or darkening of the body's tissues, in the peppered moth of England. Early in the nineteenth century, occasional dark (melanistic) specimens of the common peppered moth were collected. Over the next 100 years, this darker form, referred to as carbonaria, became increasingly common in forests near heavily industrialized regions of England, which is why the phenomenon is often referred to as industrial melanism. In places without factories and other heavy industry, the lighter-colored, salt-and-pepper form of the moth prevailed. This phenomenon aroused considerable interest among geneticists, who showed by cross-mating light and dark forms that melanism is an inherited trait determined by a single dominant gene. Because the melanistic trait is an inherited characteristic, its spread reflected genetic changes (evolution) in the population.

Peppered moths inhabit dense woods and rest on tree trunks during the day. Where melanistic individuals had become common, the environment must somehow have been altered so as to give dark forms a survival advantage over light forms. It seemed reasonable to suppose that natural selection-the process by which organisms best suited to survival in their environment achieve greater reproductive success, allowing advantageous genetic characteristics to pass on to future generations-had led to the replacement of the typical light individuals with darker carbonaria individuals. To test this hypothesis, the English biologist H. B. D. Kettlewell captured a sample of moths of both forms, marked each individual with a dot of cellulose, and then released them back into the woods. The mark was placed on the underside of the wing so that it would not attract the attention of predators to a moth resting on a tree trunk. Some days later Kettlewell captured more moths by attracting them to a mercury-vapor lamp in the center of the woods or to caged virgin females at the edge of the woods. (Only males could be used in the study because females are attracted neither to lights nor to virgin females.)This type of study is referred to as a mark-recapture study.

In one such mark-recapture experiment, Kettlewell marked and released 201 lighter moths and 601 darker moths in a wooded area near industrial Birmingham, England. The results indicated that a greater percentage of the darker form survived over the course of the experiment. A similar experiment in a nonindustrial area revealed higher survival by the lighter salt-and-pepper form of the moth.

The specific agent of selection was easily identified. Kettlewell reasoned that in industrial areas pollution had darkened the trunks of trees so much that typical moths stood out against them and were readily found by predators. Any unusual dark forms were better camouflaged against darkened tree trunks, and their coloration made them more likely to survive. Eventually, the different survival rates of dark and light forms would lead to changes in their relative frequency in the population. To test this idea, Kettlewell placed equal numbers of light and dark forms on tree trunks in polluted and unpolluted woods and watched them carefully from a concealed spot some distance away. He quickly discovered that several species of birds regularly searched tree trunks for moths and other insects, and that these birds more readily found a moth that contrasted with its background than one that resembled the bark it clung to. These data were consistent with the results of the mark-recapture experiments. Together they clearly demonstrated the operation of natural selection, which over a long period resulted in genetic changes in populations of the peppered moth in polluted areas.

One of the most gratifying aspects of the peppered moth story is that with the advent of new smoke-control programs and the return of forests to a cleaner state, frequencies of melanistic moths have decreased. In the area around the industrial center of Kirby in northwestern England, for example, the darker carbonaria form decreased from more than 90 percent of the population to about 30 percent over a period of twenty years.

United States Literature and Art in the Nineteenth Century

At the beginning of the nineteenth century, the United States was a provincial culture, still looking to Britain for values, standards, literature, and art, despite all the rapid improvements in communication- -for example, the great growth in the number of magazines and newspapers that occurred in the United States following independence from Great Britain. In a famous essay in the Edinburgh Review in 1820, Sydney Smith sarcastically inquired, "In the four quarters of the globe, who reads an American book? Or goes to an American play? Or looks at an American picture or statue?"

Yet interest in the arts and the cultural institutions that sustain them was rapidly growing in America. To be sure, cultural institutions were difficult to create in the South, mainly because the population was so widely dispersed; and in the West, dominated in this era by pioneers, emphasis was on the practical rather than on the literary or artistic. But many eastern seaboard cities were actively building the cultural foundation that would nurture a distinctively American art and literature. During these years, Philadelphia's American Philosophical Society, founded by Benjamin Franklin in 1743, boasted a distinguished roster of scientists, including Thomas Jefferson, concurrently its president and president of the United States. Culturally, Boston ran, a close second to Philadelphia, founding the Boston Athenaeum (1807), an impressive library and reading room containing‘ 'works of learning and science in all languages, particularly such rare and expensive publications as are not generally to be found in this country," The North American Review, which became the most important and long- lasting intellectual magazine in the country, was published in Boston. Devoted to keeping its readers in touch with European intellectual developments, it had a circulation of 3,000 in 1826, about the same as similar British journals.

Of the eastern cities, New York produced the first widely recognized American writers. In 1809 Washington Irving published America's first great book of comic literature- -A History of New York- -a humorous account of life in New York both in colonial times and in Irving's own day. James Fenimore Cooper's Leatherstocking novels (of which The Last of the Mohicans published in 1826, is the best known) achieved wide success in both America and Europe. In these novels, Cooper established the American experience of westward expansion as a serious and distinctive American theme.

It was New England, however, that saw itself as the home of American cultural independence from Europe, a claim voiced in 1837 by Ralph Waldo Emerson in a lecture at Harvard: "Our day of dependence, our long apprenticeship to the learning of other lands, draws to a close," Emerson announced. Rather than draw on British models, American artists, Emerson argued, should seek inspiration in the ordinary details of daily American life. Immensely popular, Emerson gave more than 1 ,500 lectures in 20 states between 1833 and 1860. "The American Scholar," his most famous lecture, carried a message of cultural self-sufficiency that Americans were eager to hear.

Artists in this period were as successful as writers in finding American themes. Thomas Cole, who came to the United States from England in1818, found great inspiration in the American landscape. Cole founded the Hudson River school of America painting, whose romantic renderings of New York landscapes were frankly nationalistic. Other painters- -realists such as Karl Bodmer and George Catlin— drew on the dramatic landscapes and peoples of the American West. Their art was an important contribution to the American sense of the land and to the nation's identity. Catlin, driven by a need to document Native American life, spent eight years among the tribes along the upper Missouri River. Then he assembled his collection of more than 500 paintings and toured the country in an effort to increase the public's awareness and appreciation of Native American lifeways. George Caleb Bingham, an accomplished genre painter, produced somewhat tidied-up scenes of real-life American workers, such as flatboat men on the Missouri River. All these painters found much to record and celebrate in American life. Ironically, the inspiration for the most prevalent theme, the American wilderness, was profoundly endangered by the rapid western settlement of which the nation was so proud. Despite such contradictions, the early nineteenth century was the period in which writers and painters found the national themes that defined the first distinctively American literature and art.

Development of Intensive Agriculture

Anthropologists and scientists in other fields have long been concerned with the origins of intensive farming and early civilization, with the social and economic structure of rural society, and with the strategies that have enabled diverse populations to adapt to environmental and other problems. Agricultural intensification involves increasing the yield from labor or land. This implies that due to the complex ways in which land and labor are interrelated, there is more than one route to intensification. Generally, the perspective of most American or European economists is to concentrate on labor. Agricultural history is usually described in terms of progress in labor-saving technology- -the plow, seed drills, cultivators, and the like- -because the economies of the Western world experienced labor shortages over much of their histories. Technology, of course, not only may increase food production by allowing the same labor force to cultivate more land but also may free up labor for other endeavors. Thus Australia, Canada, and the United States produce massive amounts of food with a relatively small rural labor force.

Production can also be intensified by increasing the productivity of land without reducing the labor requirements; that is, expanding production by using an existing labor force or even a larger one. This can be important, for example, where there are few alternative sources for employment, as is often the case in densely populated developing countries. Irrigation and the introduction of new crop strains are well-known examples of this form of intensification. Water control may allow for multiple harvests of a particular crop; new plant strains may also increase productivity without new capital input needed to reduce labor requirements.

One of the most ancient (and still important) ways land productivity can be increased is through water management. So diverse are the ways in which moisture can be controlled that it is somewhat misleading to refer to them all as irrigation." Simply adding small stones to fields, as did some ancient peoples of North America, can enhance the field's ability to retain moisture. In this sense, irrigation, or, at least, "moisture control," is as early as agriculture itself. Archaeological evidence in the Middle East indicates that simple systems of water control predate the rise of large agrarian states with their concomitantly dense populations. Populations near rivers or marshes would simply capture or divert annual floodwaters or runoff from rains. The term "irrigation" refers to actually transporting water to the field and then managing its direct application and subsequent drainage, since drainage is important in order to maintain a salt-free soil base.

The earliest known large-scale system of irrigation in Egypt appears with the emergence of one government throughout that land in 5100 B.C. We know that even fairly complicated irrigation systems can be managed by local farmers, although the potential for interfamilial or intercommunity conflict is substantial. People who share a common water resource may have very different interests in its use. As a consequence, we see a widespread pattern of large-scale irrigation systems that come to be run by special managers, with a corresponding lessening of control by households or even by local communities. Centralized decision making facilitates the mobilization of large workforces, allocation of water, conflict resolution, and storage of surpluses.

In the emerging prehistoric states of Mesopotamia, this managerial role was first assumed by religious leaders and only later by secular rulers. It is interesting to note a similar pattern in large-scale irrigation systems in the southwest United States that were managed by the Mormon Church (a religious organization) in the nineteenth century. Making water control feasible required centralized control by a committed bureaucracy willing to use resources to sustain building and rebuilding dams and canals beyond the ability of local communities.

The main impetus for irrigation, in most places, is simply the need to have water available in areas where rainfall is unpredictable-not necessarily a wish to increase the average yield of a unit of land. But with the advent of irrigation, slight differences in the productivity of different pieces of land become greatly magnified. Fields that lend themselves to irrigation -fields close to the water source or that drain well- -produce far more than those less suited to irrigation.

Why Did Nonavian Dinosaurs Become Extinct?

Data from diverse sources indicate that the Late Cretaceous (100 to 65 million years ago) climate was milder than today's because of shallow seas covering the continents. At the end of the Cretaceous, the geological record shows that these seaways retreated from the continents back into the major ocean basins. No one knows why. Over a period of about 100,000 years, while the seas pulled back, climates around the world became dramatically more extreme: warmer days, cooler nights; hotter summers, colder winters. Perhaps nonavian dinosaurs (that is, all the dinosaurs except those belonging to the Ave, or bird, family) could not tolerate these extreme temperature changes and became extinct.

If true, though, why did cold-blooded animals, such as snakes, lizards, turtles, and crocodiles survive the freezing winters and torrid summers? Those animals are at the mercy of the climate to maintain a livable body temperature. It's hard to understand why they would not be affected, whereas nonavian dinosaurs were left too crippled to cope, especially if some dinosaurs were warm-blooded. Critics also point out that the shallow seaways had retreated from and advanced on the continents numerous times during the Mesozoic, SO why did the nonavian dinosaurs survive the climatic changes associated with the earlier fluctuations but not with this one? Although initially appealing, the hypothesis of a simple climatic change related to sea levels is insufficient to explain all the data.

Volcanism has also been implicated in dinosaur extinction. The end of the Cretaceous coincided with a great increase in volcanic activity throughout the world. Lava flooded large areas of India, and explosive eruptions in the South Atlantic and the midwestern United States hurled ash over much of the globe. These eruptions could have spewed great quantities of poisonous gases into the atmosphere, causing acid rain and more acidic waters in the surface layers of the ocean. Over the short term, a cooling would result from the airborne volcanic ash, which would cut off sunlight. Climatic changes from increased volcanism may have caused the extinction of many nonavian dinosaurs, but they do not satisfactorily explain the selective patterns of extinction in the fossil record.

Dissatisfaction with conventional explanations for nonavian dinosaur extinctions eventually led to a key observation that, in turn, has fueled a decade of vigorous and often vitriolic debate. Many plants and animals disappear abruptly as one moves from older layers of rock documenting the end of the Cretaceous up into younger rocks representing the beginning of the Cenozoic. Between the last layer representing the end of the Cretaceous and the first layer representing the start of the Cenozoic, there is often a thin bed of clay. Scientists felt that they could get an idea of how long it took to deposit this 1 cm of clay by looking at the concentration of the element iridium (Ir) in it.

Iridium is no longer common at Earth's surface. Because it usually exists in a metallic state, it was preferentially incorporated into Earth's core as the planet cooled and consolidated. Iridium is found in high concentrations in some meteorites, in which the solar system's original chemical composition is preserved. Even today, microscopic meteorites continually bombard Earth, filling on both land and sea. By measuring how many of these meteorites fall to Earth over a given period of time, scientists can estimate how long it might have taken to deposit the observed amount of iridium in the boundary clay. These calculations suggest that a period of about one million years would have been required. On the basis of other evidence, however, scientists know the deposition of the boundary clay could not have lasted one million years, so the unusually high concentration of iridium seemed to require a special explanation.

Consequently, scientists hypothesized that a large asteroid, about 10 to 15 km across, collided with Earth, and the resulting fallout created the boundary clay. Their calculations show that the impact kicked up a dust cloud that cut off sunlight for several months, inhibiting photosynthesis in plants, decreased surface temperatures on continents to below freezing; caused extreme episodes of acid rain; and significantly raised long-term global temperatures through the greenhouse effect. This disruption of the food chain and climate would have eradicated the nonavian dinosaurs and other organisms in less than 50 years.

Infant-Directed Speech

Infant-directed speech is a style of speech directed toward infants. This type of speech pattern was previously called motherese, because it was assumed that it applied only to mothers. However, that assumption was wrong, and the gender-neutral term infant- directed speech is now used more frequently.

Infant-directed speech is characterized by short, simple sentences. Pitch (the highness or lowness of a sound) becomes generally higher, the range of frequencies (essentially, the difference between the highest and the lowest pitch used) increases, and intonation (the rise and fall in pitch) is more varied. There is also repetition of words, and topics are restricted to items that are assumed to be comprehensible to infants, such as concrete objects in the baby's environment. Sometimes infant-directed speech includes amusing sounds that are not even words, imitating the prelinguistic speech of infants. In other cases, it has little formal structure but is similar to the kind of telegraphic speech that infants use as they develop their own language skills.

Infant-directed speech changes as children become older. Around the end of the first year, infant-directed speech takes on more adultlike qualities. Sentences become longer and more complex, although individual words are still spoken slowly and deliberately. Pitch is also used to focus attention on particularly important words.

Infant-directed speech plays an important role in infants' acquisition of language. Newborns respond to such speech more readily than they do to regular language, a fact that suggests that they may be particularly receptive to it. Furthermore, some research suggests that unusually extensive exposure to infant-directed speech early in life is related to the comparatively early appearance of first words and earlier linguistic competence in other areas.

Is infant-directed speech similar across cultures? In some respects, it clearly is. According to a growing body of research, there are basic similarities across cultures in the nature of infant-directed speech. Consider, for instance, a comparison of the major characteristics of speech directed at infants used by native speakers of English and Spanish. Of the ten most frequent features, six are common to both: exaggerated intonation, high pitch, lengthened vowels, repetition, lower volume, and instructional emphasis (that is, heavy stress on certain key words, such as emphasizing the word "ball" in the sentence, "No, that's not a ball."). Similarly, mothers in the United States, Sweden, and Russia all exaggerate and elongate the pronunciation of the three vowel sounds of "ee," "ah," and "oh" when speaking to infants in similar ways, despite differences in the languages in which the sounds are used. Even deaf mothers use a form of infant-directed speech: When communicating with their infants, deaf mothers use sign language at a significantly slower tempo than when communicating with adults, and they frequently repeat the signs.

The cross-cultural similarities in infant-directed speech are so great, in fact, that they appear in some facets of language specific to particular types of interactions. For instance, evidence comparing American English, German, and Mandarin Chinese speakers shows that in each of the languages, pitch rises when a mother is attempting to get an infant's attention or produce a response, while pitch falls when she is trying to calm an infant. Why do we find such similarities across very different languages? One hypothesis is that the characteristics of infant-directed speech activate innate responses in infants. As we have noted, infants seem to prefer infant-directed speech over adult-directed speech, suggesting that their perceptual systems may be more responsive to such characteristics. Another explanation is that infant-directed speech facilitates language development, providing cues as to the meaning of speech before infants have developed the capacity to understand the meanings of words.

Despite the similarities in the style of infant-directed speech across diverse cultures, there are some important cultural differences in the quantity of speech that infants hear from their parents. For example, although the Gusii of Kenya care for their infants in an extremely close, physical way, they speak to them less than American parents do.

A Mutualistic Fungus of Tall Fescue Grass

Mutualism is an interaction between two species in which both benefit. A classic example of this relationship is between tall fescue grass and a fungus named Epichloé coenophiala . That fungus is completely internal and grows intercellularly (between the cells) in the above-ground portion of the grass. The grass supplies all the nutritional needs of the fungus. Infected plants exhibit no external symptoms of the fungus and no disease symptoms. In contrast, pathogenic (disease-causing) fungi result in symptoms that include wilt, leaf spots, deformed plant structures, or death of plant tissue. Pathogenic fungi often invade plant host cells and absorb the cells' contents (thereby eventually killing the cells) or produce chemicals that kill them outright.

Tall fescue is extensively used as both a pasture grass and a turf or lawn grass. It is a hardy, vigorously growing grass but has been associated with various health problems in cattle and horses. Beef cattle often show reduced weight gains when grazing on tall fescue pastures, and dairy cattle experience lower milk yields. Female horses may be more likely to lose their offspring during pregnancy and may not produce milk following a live birth. Reports of such adverse effects resulted in microscopic investigation of leaf and stem tissue that revealed the presence of the Epichloe coenophiala fungus. Surveys of infected pastures found a positive correlation between the percentage of infected plants and the severity of animal symptoms. The fungus in the grass was found to produce various alkaloids (toxic nitrogen-containing organic molecules). One type of alkaloid (ergot alkaloids) contributes to the livestock's symptoms and reduced grazing, whereas another type (loline alkaloids) makes the grass more resistant to insects. Interestingly, the insect-deterring alkaloids are not produced when the fungus is grown in the laboratory apart from the plant. Because these chemicals defend the grass as well as the fungus from consumption by both livestock and insects, the mutualistic relationship has been termed defensive mutualism.

Because the Epichloe coenophiala fungus is not external on the plant, the usual mechanism of spread by fungi does not take place. Typical fungi form spores (reproductive units consisting of one or more cells that are similar to the seeds of plants). Fungal spores spread through the air to susceptible plants and usually require an extended period of moist conditions for their survival and germination (the beginning of growth of a spore or seed), and for the subsequent infection of the host plant. The tall fescue fungus, in contrast, penetrates the flower head and infects the plant embryos in the developing seeds. Therefore, seeds collected from infected plants result in infected seedlings following germination. As the plant grows, the fungus grows along with it by division and stretching of the hyphae (threadlike structures that make up the body of a fungus) at the base of the grass, where plant cell division also occurs. Thus, growth of the fungus is concurrent with growth of the plant, which ensures that new stems and leaves of the grass are infected with the fungus. Fungus-free tall fescue plants can result if infected seed is stored for several months before planting. The fungus usually dies out during most long-term storage conditions.

The alkaloids produced by the fungus also affect microorganisms in the immediate environment of the plant. Nondisease-causing bacteria, recovered from the surfaces of fungus-infected tall fescue plants, can use loline alkaloids, released on the leaves, as a source of nutrients. The population size of these bacteria is about eightfold greater on infected plants than on fungus-free tall fescue plants or on plants colonized by related fungi incapable of producing loline. Thus, many components of the fungus-infected tall fescue community are impacted by the fungus: the grass itself, animals, insects, and microbes.

Although the fungus of tall fescue is detrimental to livestock, it is considered beneficial when grass is grown for other purposes. Insect pests are numerous in turf (short, thick grass and the soil it grows in), and tall fescue's increased resistance to them is advantageous when grass is grown for house lawns or sports fields. It may be possible to develop pasture varieties of tall fescue that are infected with this fungus for the purpose of insect resistance, but with the deletion of the fungal genes for the toxins that harm livestock.

How Old is the Continental Crust?

Although we now know that the Earth formed roughly 4.5 billion years ago, the next 600 million years of Earth history is essentially blank. The oldest rock samples yet identified on Earth(from the Northwest Territories in Canada) have an age slightly greater than 3.9 billion years. These rocks have undergone strong metamorphism (changes to the form of the existing rock due to heat and pressure), and it is difficult to tell much about their origin. However, they are not very different from many other typical continental rocks that are much younger. Thus, we know that there were at least some fragments of continental crust in existence 3.9 billion years ago.

The question of when the first continents formed is one that has long intrigued geologists, because it is evident that the continental crust has grown and evolved over geologic time. It is probable that there were small continents even before the creation of the 3.9-billion- year-old rocks. The clues that lead to this conclusion are rare and minute and difficult to discover. Where would one look for such evidence? The answer provides a good example of how geologists often work: by using the present as a window into the past. We know that the products of erosion accumulate on the edge of continents today, and there is no reason to think that the situation was different in the past. Even the earliest continents must have had beaches. There is a chance that if some of those very old sediments have been preserved, they might contain mineral grains from even older continents.

Geologists have thus sifted through some of the oldest known sandstones, which were probably originally laid down along the shorelines of ancient continents, in search of mineral grains that are especially resistant to destruction during weathering and transport. One such search was in a 3.6-billion-year-old sandstone in Western Australia. Some of the grains in these rocks are much older than the sandstone itself and have apparently survived multiple cycles of erosion, deposition, consolidation into solid rock, uplift, and re- erosion. Geologists have found that a few grains of the weathering- resistant mineral zircon from these sandstones have ages in the range of 4.1 to almost 4.3 billion years.

Zircon crystals are small but common constituents of many rocks. The weathering and erosion processes that break down their parent rocks have little effect on the inert zircon crystals. Large, transparent zircon crystals are often sold as gemstones. But the ones that are most useful to geologists are the small zircon grains that are carried long distances in streams or even by wind. These become tracers of the ultimate source for the sedimentary materials in which they now reside.

As their name implies, zircons are rich in the element zirconium. Fortunately, they also incorporate considerable amounts of uranium when they form, and the radioactive decay of uranium produces isotopes of lead that can be measured to determine the age of the grain. Modern techniques are so sensitive that even the minute amount of lead in a single zircon grain can be measured precisely and its age determined. Because these ancient zircons are single grains rather than rock fragments, it is difficult to say much about the types of rock from which they were originally eroded. However, zircons are common in continental rocks such as granite but virtually absent from the ubiquitous basalts of the ocean floor. The inference is that these grains must have come from continental rocks, and if this is indeed the case, the existence of continents can be extended back to almost 4.3 billion years, only a few hundred million years after the formation of the Earth.

Even if the Earth's crust began to form very early, there are several possible reasons why no fragments of continental rock are preserved from approximately the first 600 million years of our planet's existence. One is that for much of this period, the Earth was experiencing heavy bombardment from space as the residual material from the planet's formation process was gathered in by the Earth's gravity. A second is that the early Earth was very hot. Vigorous circulatory motion of materials in the hot Earth may simply have destroyed much of the early formed crust.

Animals and Forests

Many species in the deer family, particularly white-tailed deer and moose in North America and red deer in Europe, browse(feed on) immature trees such as seedlings and saplings and in so doing curtail the regeneration of trees. In Scotland high numbers of red deer pose a particular problem for the regeneration of Scots pine forests, some of which have not seen substantial regeneration for 300 years. Many smaller mammals, including mice, squirrels, voles (small, mouse-like animals), and rabbits, also regularly eat tree seeds and seedlings, sufficient to alter the course of woodland development.

These animals have a preference for particular tree species and can therefore potentially change woodland composition and future development. Moreover, some tree species can tolerate repeated browsing, but others cannot. Many broad-leaved tree saplings, for example, will readily resprout if the leading stem is browsed, but conifers(trees with needlelike leaves) generally lack this ability. This is significant in northern zones, where voles more than make up for the comparative absence of large animals. In northern Europe, vole outbreaks occur on a three- to four- year cycle (and somewhat more irregularly in North America), during which populations multiply several hundredfold. As voracious feeders, they rapidly consume their preferred plant roots and bulbs and then switch to seedlings of pines, spruces, and other conifers. This can have substantial economic impact for commercial forestry, and in Finland alone the cost of seedling losses to voles has been estimated at €20 million during such outbreaks.

Browsing mammals feed on trees, leaves, and shoots and consequently suppress tree growth. Conversely, grazers feed on grasses and can enhance tree establishment. This is most obviously played out in the African plains. By suppressing trees in African dry forests, elephants, giraffes, and other large browsers facilitate the spread and growth of grasses. This creates a substantial fuel load (amount of burnable material), and the resulting intense fires reduce woody vegetation further and prevent tree regeneration. Grazers such as zebras and wildebeests reverse this process by cropping the grasses, which reduces fire intensity. Trees reestablish, and the resulting forest shade suppresses the grasses. The change of African savannas from woodland to grassland and back again is therefore maintained by different kinds of herbivores(plant eaters).

Yet it is predators that have the key role in triggering the change from woodlands to grasslands and back. Trees provide predators with cover within which they can stalk and ambush herbivores. Herbivores therefore exist within a "landscape of fear” in which they avoid wooded areas in favor of more open grasslands where they can more easily spot predators. The concentration of grazing herbivores in grasslands is therefore brought about by predators, and this serves as the main driver of conversion to woodland. Elephants, by contrast, are more or less immune to predators on account of their bulk and have no aversion to wooded areas. As large browsers they feed on leaves, breaking branches and even knocking over trees to get to them, thereby initiating a return to grassland.

No matter the destruction done by mammalian herbivores, it pales into insignificance compared to that caused by insects. Since 1990 some 30 billion conifers from Alaska to Mexico have been killed by tiny bark beetles. Individually, bark beetles are smaller than a grain of rice, but when conditions are right, their populations explode in epidemics that can last a decade or more. In recent decades a few bark beetle species, such as the mountain pine beetle in North America and the European spruce bark beetle, have been particularly destructive of coniferous forests. The increased incidence of summer droughts in these regions has rendered many trees vulnerable to attack, while milder winters have facilitated the growth of bark beetle populations. Healthy trees defend themselves by drowning the tiny pine beetles in resin, a sticky substance produced by some trees. Female beetles therefore target vulnerable drought-stressed trees, and once located they release a pheromone (chemical signal) to attract other beetles. The tree responds with sticky resin and poisonous gases, but sheer force of numbers eventually overcomes a tree's defenses. The beetles, which might now number several thousand, lay their eggs under the bark, and the larvae (young) feed on the living tissue, which ultimately kills the trees.

Painting in the Dutch Golden Age

The seventeenth century in the Netherlands is often called its Golden Age, not least because of its painting, featuring such masters as Johannes Vermeer, Jacob van Ruysdael, and Frans Hals. Many factors allowed the Dutch (Netherlandish) artistic scene to flourish, but two of the most important were innovations in the stylistic treatment of paintings and in the process of producing them. In the early 1600s Esaias van de Velde began to paint landscapes with a simplified color scheme, fewer people, and a lower horizon, creating a sense of space that helped make the image more attractive. Painter Jan Porcellis and others modified their seascapes in a similar way, using simpler compositions, as well as a more monochromatic approach with less varied and less vivid colors-now known as the "tonal" approach. This helped create an interest in pictures of everyday life, moving away from historical or religious subjects that had long been dominant.

The art historian Michael Montias presented some of the inventions by van de Velde, Porcellis, and others as innovations in process. By combining a swifter painting technique with simpler compositions (fewer figures and objects)and more restricted colors, Porcellis and van de Velde are considered to have set in motion a trend for producing cheaper paintings that could penetrate a broader market and thus attract new groups of potential purchasers. By using more sky, more shade, and less crowding in their pictures, painters effectively reduced the amount of labor they needed to invest in the painting. Such specialized and "painterly" works took much less time to complete than their meticulously executed counterparts, and since labor costs were the prime determinant of production costs, this had a dramatic impact on the price of paintings. Montias asserted that the works of the realistic "tonal" school of landscape painting, for instance, brought substantially lower prices than those of their predecessors(typically 15 to 30 florins, versus 70 to 100 florins for the older works).

At roughly the same time, Dutch publishers cut back on production costs by reducing the size of books. It has been suggested that painters also applied this strategy, but as of yet there is no quantitative evidence to convincingly corroborate this. Fortunately, the few quantitative studies that exist on the size of Dutch paintings provide some clues. Historian Advander Woude's analysis of the average size of paintings in the s Dutch Rijksmuseum produced by Dutch painters indicates a gradual decline in the size of paintings over time. On average, painters born between 1550 and 1599 produced larger paintings than the groups born in 1600-1649 and 1650-1699. His data, however, are not extensive enough to show exactly when this trend set in. Another clue can be found in the distribution of paintings by subject matter. The sizes of traditional subjects of religion, mythology, and history paintings produced by the 1550-1649 group were on average significantly larger than landscapes, figure painting, or still lifes--paintings of inanimate objects, such as a bowl of fruit. We can combine this finding, which is based on a limited sample of paintings, with the relative distribution of subjects in probate inventories (court documents created for inheritance purposes)in seventeenth-century Amsterdam(a Dutch city) and collected by Montias. He compared inventories from the periods 1620-1649 and 1650-1679 and found that the share of landscapes, figure paintings, and still lifes increased, while the share of history paintings dropped significantly. This suggests that an increasingly large share of paintings in Amsterdam inventories were of smaller than traditional size.

By offering quality paintings for reasonable prices, the new generation of painters unlocked the demand for paintings. Whether the primary motives of the trendsetters were artistic or economic, the consequences were unambiguous as productivity increased and paintings could be offered for lower prices without losing out on quality or necessarily threatening painters' profits. The middle-income groups that previously could only afford copies or prints were now able to own new and original paintings by living masters. Because the artistic novelties did not replace or exclude other subjects, styles, and techniques, the range of paintings on offer expanded dramatically.

Strengths of the Byzantine Empire

In 330 AD., Constantinople (modern-day Istanbul) became the new capital of the Roman Empire. Historians traditionally consider the destruction of the city of Rome by foreign invaders in 476 AD.to mark the end of the Roman Empire. However, Constantinople continued to be the capital of another large empire, called the Byzantine Empire, for many centuries after the fall of

Rome. By the seventh century, the Byzantine Empire faced enormous challenges. Not only did it have to fight off frequent attacks from neighboring peoples, there was great instability internally as well with many emperors coming and going. And yet the empire survived.

What gave it its extraordinary resilience was partly the strength of its deeply rooted administrative system and partly its geographical position. By the mid-seventh century the territory was divided into a number of administrative units or provinces known as "themes," each with its own military detachment under its own commander. The soldiers were allotted land and thus had a real incentive to protect their home district. The land grants were large enough to support the soldiers' families as well as tenants and hired librarian this way at least some of the financial burden of the army was met by its own labor. The military domination of the themes also led to a shrinking of the civil service (the class of professional government employees),which saved further costs. While the system had many advantages, not least in providing a degree of stability spread across the empire and making it difficult for anyone commander to assume control of the whole army, there was understandable reluctance among the sedentary forces to respond to the emperor's demands if this meant fighting far from home. The problem was to some extent countered by the creation of six army units stationed in and around the capital. Although the soldiers in these units, too, were landholders, they were trained to be a mobile force capable of being deployed quickly either as a field army or in defense of the emperor and the city.

The other great strength of the Byzantine Empire was its unique geographical position sitting at the crossroads linking Asia with Europe and the Mediterranean with the Black Sea. While there can be little doubt that the volume of trade overall diminished and the number of coins in circulation fell, the large and moderately wealthy urban populations still required a basic range of commodities as well as luxuries when they could afford them. One marketing innovation of this period was the provision of provincial warehouses. Their function seems to have been to supply arms and equipment for soldiers to buy using the produce of their farms. The goods brought in from the countryside could then be further sold into the regional market system.

At the elite level two of the most sought-after products were silk and purple dye made from murex, a mollusk found in the eastern Mediterranean. Silken garments, particularly those dyed purple, were the privilege of the elite and were used extensively in the royal court and by religious leaders in the Church throughout the Roman empire, both east and west. As early as the sixth century. the emperor Justinian had made silk production the sole right of the Byzantine Empire. There were huge profits to be made, not least in meeting the demands of the western Roman Church. Silks and the rare purple garment could also be used as diplomatic gifts. In 705 Justinian II rewarded the Bulgar ruler with gifts of silk cloth and purple leather for helping him to regain the throne. That he also licensed him to trade in controlled Byzantine goods is a fair indication that trade was still essential to the fragile Byzantine state.

That said, there can be no doubt that the empire suffered a dramatic economic contraction during the seventh and eighth centuries, and the population declined significantly, partly as the result of a declining birth rate in times of stress and partly from repeated outbreaks of disease, which had begun in the sixth century and returned many times until the middle of the eighth century. The declining population and economic downturn were responsible for many changes. Once-great towns like Athens shrank to a half or a quarter of their original size, but nonetheless urban life continued.

The Story of Film Editing

Many filmmakers regard editing as the single most important creative step in determining the look and shape of the finished film. A good editor can save a film that has been directed in a mediocre fashion, and poor editing can damage the work of even the finest director.

Editing is the work of joining shots (scenes recorded by the camera, usually on film) to assemble the finished film. The editor selects the best shots from the large amount of film footage (raw, unedited film) provided by the director and the cinematographer, assembles these in order, and connects them using a variety of optical transitions. In theory, the process of editing begins with the completion of filming, or cinematography. In practice, however, the editor may begin consultations with the producer and director and may even begin cutting the film, taking scenes and putting them together, while principal filming is being completed. Most editors, however, will not watch the process of filming or view the locations where the film is shot. This allows them to view the footage unhampered by knowledge about the actual conditions that existed in front of the camera and to visualize with greater freedom various ways of combining the shots.

Over the years, the basics of editing have remained relatively constant. The first task is to assemble a first version or rough cut, which is done by eliminating all of the unusable footage containing technical or performance errors. These may include out-of-focus shots or shots containing unstable camera movement, lines where an actor has made mistakes in delivery, inaudible sound recordings, or lighting problems. Once all of this footage has been removed, the editor then assembles the remaining footage in a narrative sequence. This rough assembly will be pruned, refined, and polished to yield the final cut. The final cut is the completed product of an editor's work. It includes the complete assembly and timings of all shots in the film's finished form. It is in going from the rough cut to the final cut that the real art and magic in editing lies.

With the introduction of computers, editors began to use a nonlinear, computer-based editing system. Film footage was converted to videotape, which was then digitized and stored on computer disk, giving an editor instantaneous access to any frame or single image, shot, or edited sequence distributed anywhere in the existing footage. The editor would then decide which footage to work on by using notes that described the characteristics, strengths, and flaws of particular shots. Prior to the 1990s, when the film industry adopted digital editing systems, editors worked directly with the actual film itself and had to search manually through all of the footage to find a desired shot or segment. This older approach was a linear system: the editor could only search for one shot at a time and had to do so by viewing footage sequentially, from beginning to end.

Digital systems made editing a much faster process, and the complex control they gave editors over the digitized footage helps explain why so many films had many more cuts and shot transitions than in earlier decades. Many shots were only a few frames long, less than a second of screen time.

While digital systems gave editors greater control over their footage and increased their ability to manipulate it in ever more elaborate ways, these systems had disadvantages. Unlike editors who worked in linear systems, those using digital systems did not view a film image but rather an electronic image on a small monitor screen, which was degraded in quality, with poor resolution. This could bias editors toward close-ups because they would look better on the monitor than long shots. Furthermore, because the monitor's image was a poor guide to the visual qualities of the actual film images, editors were forced to rely more heavily on their notes about the footage. It has been argued that editors using linear systems got to know their footage better because they had to manually search through all of it to find what they needed. The editor working on a digital system would not access footage that the notes had excluded. With recent advances in digital technology, even these disadvantages have been addressed.

The Medieval European Recovery

For centuries after the Roman Empire's fall in the fifth century c.E., economic activity in western Europe was negligible compared to that of civilizations to the east. Agricultural production suffered from repeated invasions, which seriously disrupted the economy and society. Moreover, the decay of urban centers resulted in decreased industrial production and trade. By the tenth century, however, political stability served as a foundation for economic recovery, and western Europeans began to participate indirectly in the larger trading world of the Eastern Hemisphere.

With the establishment of the Frankish kingdom and later the Carolingian Empire between the sixth and the ninth centuries in what is now part of France and Germany, the European center of gravity shifted from the Mediterranean to more northern lands, particularly France. But the agricultural tools and techniques inherited from the classical Mediterranean world did not transfer very well. In light, well-drained Mediterranean soils, cultivators used small wooden plows that basically broke the surface of the soil and disrupted weeds. This type of plow made little headway in the heavy, moist soils of the north. About the sixth century a more serviceable plow became available: a heavy tool equipped with iron tips that dug into the earth and turned the soil so as to aerate it more thoroughly and break up the root networks of weeds. The northern plow was more expensive than the light Mediterranean plow, and it required cultivators to harness much more energy to pull it through heavy soils. As a result, it was slow to come into wide use. Beginning about the eighth century, however, the heavy northern plow contributed to increased agricultural production.

As the heavy plow spread throughout western Europe during the ninth and tenth centuries, cultivators took several additional steps that increased agricultural production. Peasants cleared new lands for cultivation, and they also constructed watermills, which enabled them to take advantage of a ready and renewable source of inanimate energy, thus freeing human and animal energy for other work. Moreover, they experimented with new systems of rotating crops (systems for alternating the particular crop grown in a particular field). As a result, peasants could cultivate land more intensively than before, as key soil nutrients were less likely to be depleted.

The agricultural surplus of medieval Europe was sufficient to sustain feudal lords and their subjects, but not substantial enough to support cities with large populations of artisans, craft workers, merchants, and professionals. Whereas cities had thrived and trade had linked all regions of the Roman Empire, early medieval Europe was almost an entirely rural society that engaged in little commerce. Manors and local communities produced most of the manufactured goods that they needed, including textiles and heavy tools, and they provided both the materials and the labor for construction and other large-scale projects. Towns were few and sparsely populated, and they served as economic centers for the areas immediately surrounding them rather than integrating the economic activities of distant regions.

By the tenth century, however, political stability and increased agricultural production had begun to stimulate trade and the development of urban settlements in western Europe. This revived trade took place in the Mediterranean, the North Sea, and the Baltic Sea. Merchants from the coastal cities of Italy regularly traded with Muslims in Sicily and Tunisia, who linked Europe indirectly with the larger Islamic world of communication and exchange. Even more prominent was trade in the North Sea and Baltic Sea, where Scandinavian seafarers played the role of trader as well as raider.


Scandinavia
North Sea
Baltic Sea
Germany
France
Italy
Sicily
Tunisia
Mediterranean Sea

In c.E. 200, before the Roman Empire began to experience serious difficulties, the European population had stood at about thirty-six million. It fell sharply over the next four centuries, to thirty-one million in c.E. 400 and twenty-six million in c.E. 600-a decline that reflected both the ravages of epidemic diseases and the unsettled conditions of the early Middle Ages. Then, gradually, the population recovered, edging up to twenty-nine million in c.E. 800 and thirty-two million in c.E. 900. By c.E. 1000 the European population once again amounted to thirty-six million-the level it had reached eight centuries earlier. By this time, western Europe was poised for extraordinary economic and demographic expansion that occurred in the late Middle Ages.

Abandoning Hunting and Gathering

For much of human history, people survived by hunting and gathering, but in the Neolithic period, starting around 10,000 B.C.E., this was gradually given up in favor of growing grains and other plants or domesticating animals (pastoralism).Why did early societies in so many parts of the world gradually abandon a way of life based on food gathering? Some theories assume that people were drawn to food production by its obvious advantages. For example, it has recently been suggested that people settled in what is now the Middle East so they could grow enough grains to ensure themselves a ready supply of beer. Beer drinking is frequently depicted in ancient Middle Eastern art and can be dated to as early as 3500 B.C.E.

However, most researchers today believe that climate change drove people to abandon hunting and gathering in favor of pastoralism and agriculture. So great was the global warming that ended the great Ice Age that geologists give the next era which began around 9000 B.C.E. A new name: the Holocene. Scientists have also found evidence that temperate lands were exceptionally warm between 6000 and 2000 B.C.E., the era when people in so many parts of the world adopted agriculture. The precise nature of the crisis probably varied. In the Middle East taking up food production may have been a response to shortages of wild food caused by a dry spell or population growth. Elsewhere, a warmer, wetter climate could have promoted rapid forest growth in former grasslands, reducing the supplies of game and wild grains.

Additional support for an ecological explanation comes from the fact that in many drier parts of the world, where wild food remained abundant, agriculture was not adopted. The inhabitants of Australia continued to rely exclusively on foraging until recent centuries, as did some peoples in all the other continents. Many Amerindians in the arid grasslands from Alaska to the Gulf of Mexico hunted bison, while in the Pacific Northwest others took up salmon fishing. Abundant supplies of fish, shellfish, and aquatic animals permitted food gatherers east of the Mississippi

River in North America to become increasingly sedentary. In the equatorial rain forest and in the southern part of Africa, conditions favored retention of the older ways. The reindeer-based societies of northern Eurasia were also unaffected by the spread of farming.

Whatever the causes, the effects of the gradual adoption of food production in most parts of the world were momentous. A hundred thousand years ago, there were fewer than 2 million people, and their range was largely confined to the temperate and tropical regions of Africa and Eurasia. During the last glacial epoch, between 32,000 and 13,000 years ago, human populations may have fallen even lower. Then, as the glaciers retreated, people moved into new land and adopted agriculture. Their numbers gradually rose to 10 million by 5000 B.C.E., and mushroomed to between 50 million and 100 million by 1000 B.C.E. This increase brought important changes to social and cultural life.

The evidence that an ecological crisis may have driven people to food production has led researchers to reexamine the assumption that people in agricultural societies were better off than foragers. Modern studies demonstrate that food producers have to work much harder and for much longer periods than do food gatherers. In return for modest harvests, early farmers needed to put in long days of arduous labor clearing and cultivating the land. Pastoralists had to guard their herds from wild predators, guide them to fresh pastures, and tend to their many needs. There is also evidence that even though the food supply of early farmers was more secure than that of food-gathering peoples and pastoralists, the farmers ‘diet was less varied and nutritious. Skeletal remains show that on average Neolithic farmers were shorter than earlier food-gathering peoples and more likely to die at an earlier age from contagious diseases. People in permanent settlements were more exposed to diseases from water contaminated by human waste, to disease- bearing vermin and insects that infested their bodies and homes, and to new diseases that migrated from their domesticated animals (especially pigs and cattle).

Climate Change in Early Urban Centers

During the third millennium B.C.E., the first urban centers in Egypt, Mesopotamia, Iran, Central Asia, and South Asia flourished and grew in complexity and wealth in a wet and cool climate. This smooth development was sharply, if not universally, interrupted beginning around 2200 B.C.E. Both archaeological and written records agree that across Afro-Eurasia, most of the urban, rural, and pastoral societies underwent radical change. Those watered by major rivers were destabilized, while the settled communities on the highland plateaus virtually disappeared.

After a brief hiatus, some recovered, completely reorganized, and used new technologies to manage agriculture and water. The causes of this radical change have been the focus of much interest. After four decades of research by climate specialists working together with archaeologists, a consensus has emerged that climate change toward a warmer and drier environment contributed to this disruption. Whether this was caused solely by human activity, in particular agriculture on a large scale, or was also related to cosmic causes such as the rotation of Earth's axis away from the Sun, is still a hotly debated topic. It was likely a combination of factors.

The urban centers dependent on the three major river systems in Egypt, Mesopotamia, and the Indus Valley all experienced disruption. In Egypt, the hieroglyphic inscriptions tell us that the Nile no longer flooded over its banks to replenish the fields with fresh soil and with water for crops. Social and political chaos followed for more than a century. In southern Mesopotamia, rivers changed course, disrupting settlement patterns and taking fields out of cultivation. Other fields were poisoned by salts brought on through overcultivation and irrigation without fallow periods (periods when nothing is planted). Fierce competition for water and land put pressure on the central authority. To the east and west, pastoralists faced with shrinking pasture for their flocks, pressed in on the river valleys, disrupting the already challenged social and political structure of the densely urban centers.

In northern Mesopotamia, the responses to the challenges of dryness were more varied. Some centers were able to weather the crisis by changing strategies of food production and distribution. Some fell victim to internal warfare, while others, on the rainfall margin, were abandoned. When the region was settled again, society was differently organized. The population did not drastically decrease, but rather it distributed across the landscape more evenly in smaller settlements that required less water and food. It appears that a similar solution was found by communities to the east on the Iranian plateau, where the inhabitants of the huge urban center of Shahr-i-Sokhta abruptly left the city and settled in small communities across the oasis landscape.

The solutions found by people living in the cities of the Indus Valley also varied. Some cities, like Harappa, saw their population decrease rapidly. It seems that the bed of the river shifted, threatening the settlement and its hinterland. Another city, Mohenjo Daro, on the other hand, continued to be occupied for another several centuries, although the large civic structures fell out of use, replaced by more modest structures. And to the south, on the Gujarat Peninsula, the population and the number of settlements increased. They abandoned wheat as a crop, instead cultivating a kind of drought-enduring millet that originated in West Africa. Apparently, conditions there became even more hospitable, allowing farming and fishing communities to flourish well into the second millennium B.C.E.

The evidence for this widespread phenomenon of climate change at the end of the third millennium B.C.E. is complex and contradictory. This is not surprising because every culture and each community naturally had an individual response to environmental and other challenges. Those with perennial sources of freshwater were less threatened than those in marginal zones, where only a slight decrease in rainfall can mean failed crops and herds. Likewise, certain types of social and political institutions were resilient and introduced innovations that allowed them to adapt, while others were too rigid or shortsighted to find local solutions. A feature of human culture is its remarkable ability to adapt rapidly. When faced with challenges, resilience, creativity, and ingenuity lead to cultural innovation and change.

Commedia dell'Arte

"Harlequin" by Maurice Sand

In contrast to opera, which is chiefly a musical form, another equally popular type of entertainment in Renaissance Italy was pure theater. This is commedia dell'arte , which is Italian for "comedy of art" and means "play of professional artists" (as opposed to the kind of scholarly play performed by nonprofessionals).

Commedia companies usually consisted of ten performers-seven men and three women, though the number sometimes varied. They were traveling troupes, possibly the successors of Greek and Roman mimes, or clowns. Although there were instances when commedia performers staged serious forms of drama, they usually staged comedies, and through the years the term "commedia dell'arte" has come to be associated primarily with comedy.

Commedia thrived in Italy over a considerable period of time, from 1550 to 1750. It was not a written literary form but, rather, consisted of improvised presentations. Scenarios-short scripts without dialogue-were written by members of a company, and these scripts provided plot outlines; in other words, the performers had no set text but invented the words and actions as they went along. Over a thousand such scenarios survive from the Italian Renaissance. Using these outlines, actors would create the dialogue and would be expected to move the action along through improvisation.

The conventions of commedia dell'arte made the actors' task simpler than its improvisatory nature would suggest. For one thing, commedia actors played the same stock characters throughout most of their careers. Among the popular comic figures were a lecherous, miserly old Venetian, Pantalone ; a foolish pedant who was always involved in his neighbor's affairs, Dottore ; a cowardly, braggart soldier, Capitano; and sometimes foolish, sometimes sly servants known as zanni. Arlecchino, or Harlequin was the most popular of the comic servants. Commedia scenarios also included serious young lovers whose romances were often blocked by Pantalone and Dottore. Since the performers fused their own personalities with those of the characters, improvisation was easier. And since the performers worked together, playing the same characters for extended periods of time, they became adept at creating comic interaction on the spur of the moment.

Improvisation was also made easier because all the commedia characters employed standard lazzi- repeated bits of physical comic activity or "business." Capitano, for example, would get entangled with his sword. (Twentieth-century film and television comics had their own lazzi; the great American film clowns-Laurel and Hardy and the Marx Brothers, for example-had pieces of physical business that they repeated in all their performances.)

In addition, commedia actors used conventional entrance and exit speeches as well as prepared musical duets. Surviving from the Renaissance are manuscripts put together by commedia actors that contain jokes, comic business, and repeated scenes and speeches. (These books were referred to by many different names, but the most common term for them was zibaldoni.)

Costuming also facilitated improvisation. Commedia characters all wore traditional costumes, such as Harlequin's patchwork jacket and Dottore's academic robe, so that audiences could recognize them immediately. Usually, each character would wear the same outfit, and its exaggerated details reflected his or her comic personality. A significant addition to Harlequin's costume was the slapstick, a wooden sword used in comic fight scenes. Sometimes the slapstick consisted of two thin slats of wood, one on top of the other; when a performer was struck with it, the effect was greatly exaggerated by the sound of the two pieces of wood smacking together. Today we use the term "slapstick" for comedies emphasizing physical horseplay or roughness. Masks, covering either the whole face or part of the face, were an essential element of commedia costumes. Pantalone's mask, for example, always had a huge hooked nose. The young lovers, however, did not wear masks.

Commedia dell'arte was enormously popular with audiences. One measure of its success was its popularity outside Italy, particularly in France.

The most successful commedia companies were often organized by families and chose names that were meant to characterize them: I Gelosi (The Zealous), I Fideli (The Faithful), I Confidenti (The Confident), and I Accesi (The Inspired). Most companies were based on a profit- sharing plan: a company's members shared in its profits as well as its expenses and losses. Commedia performers were flexible; they could perform in town squares, in unused theater spaces, in the homes of wealthy merchants, or at royal courts.

Habitability of Planets

Magnetic fields deflect (turn away) charged particles. Earth has a magnetic field generated by the movement of liquid in its core. One reason that Earth's magnetic field is considered important for Earth's habitability (ability to support life) is that it helps protect the planet from bombardment by cosmic rays. Cosmic rays are energetic charged particles that originate from space, either from the Sun or from farther away in the galaxy. Solar cosmic rays are relatively low-energy particles that are released in a stream of charged particles (the "solar wind") from the upper atmosphere of the Sun. Galactic cosmic rays are high-energy particles and gamma rays that originate from outside the solar system. Prolonged exposure to either of these types of radiation can create health risks, including cancer-a serious consideration for astronauts out in space.

Although Earth's magnetic field helps protect the planet from cosmic rays, that protection is only partial: the magnetic field barely deflects the most energetic cosmic rays that have energies of many gigaelectron volts. What then shields Earth's surface from high-energy cosmic rays? The answer is the atmosphere itself. Most cosmic rays, including those falling at the poles, are absorbed by the atmosphere at altitudes of 80 km or more. The high-energy ones create cascades of secondary particles, and some of these do indeed make it down to the surface. By the time they get there, however, they have lost most of their force, and so they account for less than 10 percent of the radiation exposure that an average person receives during the course of a year.

Planetary magnetic fields have another effect that may actually be more important in keeping a planet habitable. Earth's magnetic field prevents the solar wind from interacting directly with its atmosphere. (This statement is equivalent to saying that it protects Earth from solar cosmic rays.) Mars, which lacks an intrinsic magnetic field, does not have such protection, and consequently its atmosphere may have been sputtered away by the solar wind. "Sputtering" is the term used to describe the process whereby collisions between solar wind particles and electrically charged atoms in a planet's upper atmosphere can lead to loss of atmospheric gases. Based partly on this observation, scientists Peter Ward and Donald Brownlee suggest that planets that lack magnetic fields are unlikely to be habitable.

This argument, though, overlooks something obvious: Venus, which also has no intrinsic magnetic field, has a very dense atmosphere —100 times thicker than Earth's. Furthermore, Venus is closer to the Sun than either Earth or Mars, and hence it is exposed to a denser, more energetic solar wind. If solar wind sputtering is so effective, then why didn't Venus lose its atmosphere? The answer probably has to do with planetary size. Venus is roughly the same mass as Earth, and so its upper atmosphere is less extended than that of Mars. That makes it harder for the solar wind to strip away atmospheric gases. Both Venus and Mars do have induced magnetic fields (fields generated in their upper atmospheres by their interaction with solar wind), and in Venus's case, this induced field is evidently sufficient to protect its atmosphere. Mars's problem is the combination of its lack of an intrinsic magnetic field and its small size, which is clearly not conducive to planetary habitability.

This brings us to the issue of planetary size itself. Mars's small mass (1/9 of Earth's mass) also allowed the planet to cool more quickly. Without enough internal heat to sustain volcanism, it was unable to recycle carbonates (consisting of one carbon and three oxygen atoms) back into CO2 (carbon dioxide, a gas that traps heat at the planet's surface). Mars appears to have already cooled off by about 3.8 billion years ago. No one knows exactly how big a planet needs to be to stay volcanically active for 4.5 billion years, as Earth has done, but one can guess that it needs to be 1/3 of Earth's mass or more. A higher mass should also give a planet a better chance of sustaining a magnetic field because its core should take longer to solidify.

Early Metallurgy

The earliest metal tools were made by cold hammering copper into simple artifacts. Such objects were fairly common in Near Eastern villages by 6000 B c. Eventually, some people began to melt the copper. They may have achieved sufficiently high temperatures with previously established methods used for firing clay pottery in kilns (ovens). The copper was usually melted into shapes within the furnace itself. Copper metallurgy was widespread in the eastern Mediterranean by about 4000 B c. European smiths (metalworkers)were working copper in the Balkans as early as 3500 B.C. In the Americas, copper working achieved a high degree of refinement in coastal Peru among the Moche and Chimu cultures, and also in western Mesoamerica. In North America, the Archaic peoples of Lake Superior exploited the native deposits of copper ore(rock containing copper) on the southern shores of the lake, and the metal was widely traded across the eastern continent and cold hammered into artifacts from Archaic(650-490 B.C)to Woodlands (1000 B.C.-A.D. 1000) times.

The real explosion in copper metallurgy took place midway through the fourth millennium B.C., when southwest Asian smiths discovered that they could improve the properties of copper by alloying(combining) it with a second metal such as arsenic, lead, or tin. Perhaps the first alloys came about when smiths tried to produce different colors and textures in ornaments. But they soon realized the advantages of alloys that led to stronger, harder, and more easily worked artifacts. Most early bronze-an alloy of copper and tin-contains about 5 to 10 percent alloy material; 10 percent is optimum for hardness. An explosion in metallurgical technology occurred during the third millennium BC, perhaps in part resulting from the spread of writing. The use of tin alloying may have stimulated much trading activity, for the metal is relatively rare, especially in the eastern Mediterranean region. Bronze working was developed to a high pitch in northern China after 2000 B.C.

Smiths during the Bronze Age (3000-1200 B.C.)certainly knew about iron, but it was considered of little apparent use. They knew where to find iron ore (rock containing iron) and how to fashion iron objects by hammering and heating. But the crucial process in iron production is carburization, in which iron is converted into steel. The result is a much harder object, far tougher than bronze tools. When an iron object is carburized, it is heated in close contact with charcoal(a form of carbon) for a considerable period of time. The solubility of carbon in iron is very low at room temperature but increases dramatically at temperatures above 910C, which could easily be achieved with charcoal and a good Bronze Age bellows-a device used to supply air. It was this technological development that led to the widespread adoption of iron technologies in the eastern Mediterranean area at least by 1000 B.C.

Iron tools are found occasionally in some sites as early as 3000 B.C., but such artifacts were rare until around 1200 B.C., when the first iron weapons in eastern Mediterranean tombs appeared. The new metal was slow to catch on, partly because of the difficulty of extracting it from its ore. Its widespread adoption may coincide with a period of disruption in eastern Mediterranean trade routes as a result of the collapse of several major kingdoms, including that of the Hittites, after 1200 B c Deprived of tin needed to make bronze, the metalworkers turned to a much more readily available substitute-iron. It was soon in use even for utilitarian tools and was first established on a large scale in continental Europe in the seventh century B c by people of the western and central European Hallstatt culture.

Iron ore is much more abundant in nature than copper ore is, being readily obtainable from surface rock formations and wetland deposits. Once its potential was realized, it became much more widely used, and stone and bronze were relegated to subsidiary, often ornamental, uses. It made available abundant supplies of tools with tough cutting edges for agriculture. Moreover, iron tools could be lighter than bronze ones, and they could hold a sharper edge. With iron tools, clearing forests became easier, and people achieved even greater mastery over their environment.

Essentials of Chinese Art

One important effect of recent research is that scholars now question ideas of essential, permanent characteristics of Chinese art. Many long-held conceptions turn out to be based on the tastes and values of ruling and literate(mostly male)elite patrons and audiences of certain periods. Often left out of such accounts are features of popular, religious, or craft arts and the cultural activities and values of women. There is also now an increasing awareness of the interactions and cross borrowings between China and other cultures. But even while the ideas of essentialism are questioned, we can still ask about distinctiveness. In what ways were Chinese arts special and noteworthy within the broad history of world art? A few long-term characteristics stand out, although, as important as these culturally distinctive elements were in the history of Chinese art, they still leave out much of significance and interest.

China is unique among the great cultures of the world in the degree to which its civilization was identified with its craft products. For the ancient Greeks and Romans, China was Seres, "the land of silk." In early modern Europe, the word "China" was as likely to refer to porcelain tableware as to the tableware's country of origin. Indeed, Chinese productivity and technical excellence in many crafts were unrivaled for centuries or even millennia. Ceramic production and the carving of the hard stones known collectively as jade are part of the earliest horizons of Chinese cultures in the Neolithic period (ending 2000 B.C.E.), and the products of these activities have been made continuously in large numbers- sometimes approaching an industrial scale-down to the present. Chinese silks were prized luxury items for both domestic and international consumption for nearly two millennia. Chinese lacquerware factories employed an early version of mass production more than 2,000 years ago. Bronze vessel casting, gold and silver metalworking, wood-block printing of books and pictures, and hardwood furniture making are among many other media with long histories of large-scale, technically distinguished production. Far from being mere cultural embellishments, such craft industries were major economic and social forces.

Thus the use of such terms as "craft arts," "craft skill," and "craft production" in the context of Chinese production refers to major traditions of highly accomplished technique and organization and should not convey the negative evaluations often implied by distinctions between fine arts and crafts in the modern West. Critical traditions in China do indicate distinctions between literate arts-especially those such as calligraphy, painting, and architecture that acquired an accompanying theoretical, historical, and critical literature-and art forms that depended more on specialized skills and sometimes on laborious or cooperative production. Such distinctions were fluid over time, however, as antique ceramics were prized and inscribed by imperial collectors, ancient ritual bronze vessels attracted the interest of antiquarian scholars, and individual potters, carvers, and metalworkers achieved independent renown.

Ritual performances of respect for ancestors are a Chinese cultural feature with roots that extend at least as far back as the Shang dynasty (1766-1122 B.C.E.) and continue down to the present. Among the artistic products directly associated with ancestor veneration are monuments of tomb architecture and memorial sculpture, as well as a wide array of ritual and grave goods: bronze offering vessels, along with tomb furnishings of stone and ceramic sculpture, metalwork, lacquer, and jade. More generally, the importance of ancestral cult practices contributed to a strong historical consciousness in Chinese culture and to such artistic by-products as antiquarian tastes, retrospective calligraphy and painting styles, and an art-historical literature.

The Chinese written language has been a unifying cultural force since the Shang period, facilitating communication across temporal and geographical divides. At the time, writing was in some ways exclusionary, since it was limited to literate elites who were always a distinct minority of the population until the recent past. Nevertheless, writing has been a powerful vehicle for forging cultural identity, and the high status accorded the written text, at times verging on the divine or magical, lent prestige to the arts of calligraphy, inscribed monumental vertical stone slabs called stelae, printed and illustrated books, and paintings inscribed with written texts.

Atwater Food Energy Value System

The Atwater system of ascertaining energy values in foods was developed by Wilbur Olin Atwater, a college chemistry professor, around the end of the nineteenth century. Atwater knew that food contained three main substances the body uses for energy: protein, fat, and carbohydrate. Using a simple laboratory device called a bomb calorimeter, he recorded how much heat was released when typical proteins, fats, and carbohydrates were completely burned. He found that there was not a lot of variation among the different types of each item. For example, all proteins tended to produce a little more than four kilocalories of heat per gram (kcal/gram).

After that, Atwater needed to know two more things. First was how much of the major macronutrients-protein, fat, and carbohydrate-a food contains. Fat was easy, because unlike protein and carbohydrates, fat dissolves in ether (a liquid used to dissolve other substances). So, Atwater chopped foods finely, shook them up with ether, and weighed how much material was dissolved in the ether. That gave him a food's fat content (or, more strictly, lipid content: lipids include both fats, which are solid at room temperature, and oils which are liquid). The same method is used today. Protein was harder to index because no test identifies proteins in general. However, Atwater knew that about 16 percent of the weight of an average protein was nitrogen. So, he found a way to measure the amount of nitrogen, which gave him the concentration of protein.

Carbohydrates were the hardest. There was no test then, nor is there now, for identifying the concentration of carbohydrates in general. But Atwater knew the main organic matter in foods was the three big items, protein, fat, and carbohydrate. He also knew how to calculate the total amount of organic matter. He simply burned the food completely, leaving only the mineral ash that did not burn and was therefore the inorganic part. Knowing how much organic matter the food contained and how much fat and protein it held, he obtained the amount of carbohydrate by subtraction: the weight of the carbohydrate was what was left when the weights of the fat, protein, and mineral ash had been subtracted from the total weight of the original food item.

Atwater was thus able to estimate the amount of protein, lipid, and carbohydrate in his food. The second piece of information he needed was how much of the food a person eats is digested, as opposed to being passed through the body unused. This required him to analyze the feces (waste matter) of people who were eating precisely measured diets, which he duly did. He was then able to estimate, for each of the three nutrients, how much of what was eaten was also digested. Once again, he found that there was little variation within the categories of protein, fat, and carbohydrate, so he assumed the variation could be ignored.

The chemist now had what he wanted. He knew how much energy each of the three big types of macronutrient contained, how much of each macronutrient was present in the food, and how much of it was used in the body. Ignoring variation within each type of macronutrient he proposed the convention that still dominates the food industry and government standards. By taking into account the proportion of the food that he found was not digested, which was rarely more than 10 percent, he claimed that on average proteins and carbohydrates each yield four kcal/gram, while lipids yield nine kcal/gram. These are known as Atwater's general factors.

This simple and convenient system forms the basis of the Atwater convention and is essentially what the United States Department of Agriculture's National Nutrient Database and McCance and Widdowson's The Composition of Foods use to produce their tables of nutrient composition. But nutritionists have long recognized important limitations in the Atwater convention, so it has been modified in various ways, such as by making the general factors more specific in 1955 when the Atwater specific-factor system was introduced to take advantage of a half century of nutritional biochemistry research. For example, the energy value of different types of protein is known to vary: egg protein produces 4.36 kcal/gram, whereas brown rice protein produces 3.41 kcal/gram. An exhaustive list of such variants has been compiled.

Egyptian Hieroglyphs

Sumer and Egypt were two of the world's earliest civilizations and the sites of two of the oldest known writing systems. Egyptian hieroglyphs came into existence early in the third millennium BCE, a little after Sumerian cuneiform, and were probably invented under the influence of the latter. There are many instances of early Sumer-Egyptian relations. but there is no direct evidence for the transfer of writing, and a credible argument can also be made for the independent development of writing in Egypt. Like cuneiform, hieroglyphs consisted of pictograms-pictorial symbols that initially stood for objects-some of which over time became phonetic signs standing for sounds. Unlike the symbols in Sumerian writing, which eventually lost their resemblance to the original pictograms, Egyptian hieroglyphs retained their recognizable form for a very long time. Even after the hieroglyphs became phonetic signs. they were recognizable as things: a man with hands raised, a crown, a hawk. Eventually the Egyptians evolved determinants, signs placed next to a hieroglyph to show whether it served as a phonetic symbol or as a pictogram. But neither hieroglyphic script nor cuneiform evolved into a fully phonetic form-an alphabet.

Indeed, Sumerian never had the chance to develop an alphabet. It was replaced by Akkadian, the language of Sumer's conquerors, before its development was complete. Hieroglyphs, on the other hand, existed for thousands of years without losing their character as pictures. Probably this can be explained by the Egyptian attitude towards writing. For the Egyptians. writing brought immortality. It was a magic form in which the lines themselves carried life and power. Some hieroglyphs were too powerful to be carved in a place considered to have magic power; they could only be written in a less powerful area, so they wouldn't bring unwanted forces into existence. The name of a king, carved in hieroglyph on a monument or statue, gave him a presence that went on past his death. To deface the carved name of a king was to kill him eternally.

This attitude towards writing preserved the pictorial form of hieroglyphs, since the pictures themselves were thought to have such power. In fact, far from being phonetic, hieroglyphs were designed to be indecipherable unless you possessed the key to their meaning. The Egyptian priests guarded this knowledge in order to keep this tool in their own hands. Ever since, the mastery of writing and reading has been an act of power. As a matter of fact, hieroglyphs were so far from intuitive that the ability to read them began to fade even as Egypt still existed as a nation.

Hieroglyphs could preserve their magical and mysterious nature only because the Egyptians invented a new and easier script for day-to-day use. Hieratic script was a simplified version of hieroglyphic writing, with the careful pictorial signs reduced to a few quickly dashed lines. Hieratic script became the preferred handwriting for business matters, bureaucrats, and administrators. Its existence depended on another Egyptian invention: paper. No matter how simple the lines were, they could not be written quickly on clay.

Clay had been the traditional writing material of both the Sumerians and the Egyptians for centuries. It was plentiful and reusable. The writing on a smooth-surfaced clay tablet that had been dried in the sun would last for years; but simply dampen the surface of the tablet, and the writing could be smoothed and altered. to correct or change a record. Records that had to be protected from tampering could be baked instead, fixing the marks into a permanent, unalterable archive. But clay tablets were heavy, awkward to store, and difficult to carry from place to place, severely limiting the amount of writing in any message. Sometime around 3000 BCE, an Egyptian scribe realized that the papyrus (a thick paper made from the papyrus plant) used as a building material in Egyptian houses could also serve as a writing surface. With a brush and ink, hieratic script could be laid down very rapidly on papyrus. Of course. papyrus was also much simpler to carry. And while hieroglyphs continued to be carved on the stone walls of tombs and on monuments and statues, letters and other important documents were written on papyrus.

The Industrial Revolution in Britain

In Britain, the eighteenth century was marked with a series of inventions that brought new uses to known energy sources (coal) and new machines to improve efficiencies(steam engines) and enable other new inventions (water pumps and railroads). Funding the inventions and financially supporting inventors and inventions through several trials required money. The eighteenth century was marked with a flow of capital (money or wealth) from Britain's colonies and from global trade into Western Europe. The flow of capital into Western Europe enabled investors to fund inventors and to perfect inventions. For example, James Watt is credited with improving the steam engine by creating a separate chamber to house the steam and by improving the engine's moving parts, getting them to perform correctly. The invention did not happen overnight: a series of attempts over a few decades finally worked when Watt partnered with toymaker and metalworker Matthew Boulton. Boulton, who had inherited money, financed the final trials and errors that made Watt's steam engine functional and reliable.

Innovations in iron manufacturing enabled the production of the steam engine and other products made of iron. In Coalbrookdale, England, in 1709, ironworker Abraham Darby found a way to smelt iron (remove the oxygen from rock containing iron). By burning coal in a vacuum-like environment, the English already knew they could cook off the impurities, leaving behind coke, the high-carbon portion of coal. Darby put iron ore and coke in a blast furnace, and then pushed air into the furnace, a combination that allowed the furnace to burn at a much higher temperature than wood charcoal or coal allowed. Mixing the iron ore with limestone (to attract impurities) and water and smelting it with coke enabled ironworkers to pour melted iron ore into molds (instead of shaping it with heat and hammers), making cast iron. The use of molds allowed more consistency in iron parts and increased production of iron components.

The steam engine alone had dramatic effects on production. It was used to pump water out of coal mines, enabling coal workers to reach deeper coal seams; to power spinning wheels that spun 100- plus spools of thread at a time; to power dozens of looms in a factory all at once; and to create a new mode of transportation, the railroad. The first railroad in England was opened in 1825. In 1830, Manchester (a center of textile manufacturing) was connected by rail to the nearby port of Liverpool, and in the next several decades thousands of miles of iron tracks were laid.

Before the railroad connected places and reduced the transportation costs of coal, manufacturing needed to be located close to coalfields. Manufacturing plants also needed to be connected to ports where raw materials could arrive and finished products could depart. In the late 1700s, plants were usually connected to ports by broad canal or river systems. In Britain, densely populated and heavily urbanized industrial regions developed near the coalfields. In the early 1800s, as the innovations of Britain's Industrial Revolution diffused into mainland Europe, the same set of locational criteria for industrialized zones applied: nearby coalfields and connection via water to a port.

Once the railroad was well established, some manufacturing moved to or grew in existing urban areas with large markets, such as London and Paris. London was an attractive site for industry because of its port location on the Thames River and more importantly because of its major role in the flow of regional and global capital. By locating in London, an industry was at the pulse of Britain's global influence. Paris was already continental Europe's greatest city, but like London, it did not have coal or iron deposits in its immediate vicinity. When a railroad system was added to the existing network of road and waterway connections in Paris, it strengthened the city's position as the largest local market for manufactured products for hundreds of miles. Paris attracted major industries, and the city, long a center for the manufacture of luxury items, experienced substantial growth in such industries as metallurgy and chemical manufacturing. These urban centers became, and remain, important industrial complexes not because of coalfields' proximity but because of the centers' commercial and political connectivity to the rest of the world.

The Chytrid Fungus Epidemic

The scientific community first suspected that frogs were dying out in the 1980s. Several cases of sudden disappearances of frogs from places where they were formerly common were noticed, and worryingly. these were places where the habitat was protected and in good health in other respects. Furthermore, these extinctions occurred in widely separated places, among them. liter frogs in Puerto Rico, stream-breeding frogs in Australia, western toads in Colorado's Rocky Mountains, and frogs and toads from the mountainous rain forests of Costa Rica and Panama.

When scientists working on amphibians around the world began to exchange notes on their experiences, it became clear that the amphibian declines in pure and widespread habitats were occurring much more quickly than they could explain in terms of habitat destruction, pollution, or climate change even though these factors have undoubtedly had some effect on frog populations in some places, and may even have caused local extinctions. A fungal disease was discovered in amphibians in the early 1990s. The fungus belongs to a group called the chytrid fungi and, although there are over 1,000 different forms of chytrid fungi, this was the only species known to become parasites on a vertebrate host. In 1999 it was named Batrachochytrium dendrobatidis, or Bd for short. It is now known that Bd only lives on amphibian skin, and often causes the death of frogs. Importantly. Bd infects all species of amphibians, and so it remains a threat even after one or more species has disappeared from a region. The researchers reached the conclusion that Bd was the main cause of amphibian declines in pure habitats and set out to investigate it.

There were several factors at work. Frogs at higher elevations are more likely to be affected than those at lower ones, indicating that the fungus is stronger at the cooler temperatures experienced in mountains than the warmer conditions lower down. It seems that the optimal temperature for the fungus is 170-25°C (62.50-770F), which is also, unfortunately, the temperature at which frog diversity tends to be greatest. Frogs restricted to cool mountains, especially stream-breeders, are the most affected. Higher temperatures reduced the activity of the fungus and can even be used to cure infected frogs under laboratory conditions. Species that tend to aggregate, under rocks, for example, are more likely to catch Bd, and periods of stress, perhaps as a result of unusually warm or dry weather, may also trigger outbreaks (and may also increase the tendency of the frogs to aggregate). Researchers discovered that although all amphibians have bacteria and other microbes on their skin that help them to fight off diseases, including fungal infections. The effectiveness of this defense appears to vary from one species to another. Partially immune species can act as reservoirs of infection, or carriers, allowing the fungus to spread throughout a community of frogs. They also found that Bd can survive away from its hosts, at least for limited periods of time, and can be collected from wet rocks on which frogs were previously sitting, for instance.

So far, nobody has definitely established where Bd came from. One theory that is gaining favor among experts is that it has always existed in parts of sub-Saharan Africa, as the fungus was discovered on a museum specimen of an African clawed erogenous fakery, collected in West Africa as far back as 1933. It was also found in Xenopus laevis collected in Uganda in 1934 and Xenopus gilli from the Cape regions of South Africa in 1982. Furthermore, the fungus appears not to be fatal to frogs from the southern half of Africa, suggesting that they had acquired immunity to it over a long period of time. Certainly, Bd was present on all clawed frogs collected in the region in 1973 with no apparent ill-effects. Clawed frogs were transported around the world in the middle years of the twentieth century for testing and, just as some of the frogs were accidentally introduced to new regions, this too could have been an opportunity for the fungus to spread. Regardless of its origins, Bd has spread to much of the world since the 1980s.

The Effects of the Tambora Volcano Eruption on the Atmosphere

In 1815 the Tambora volcano in Indonesia erupted and sent millions of tons of ash and sulfuric acid into the atmosphere. Tiny drops of liquid ash and sulfuric acid formed an aerosol cloud that spread throughout the winter of 1815-1816, carried by winds until it covered the Earth, cooling global temperatures by reflecting and scattering sunlight. Although the cloud reflected only 0.5 to 1 percent of the incoming energy, it reduced the Northern Hemisphere average temperature in 1816 by about 1 degree Celsius, a seemingly small amount of cooling that had a considerable impact on global weather patterns, with devastating consequences for agriculture on both sides of the Atlantic Ocean

Ironically, however, the effects of Tambora's aerosol cloud could have been far worse if the eruption had been slightly weaker. While immense in size and scope, Tambora's aerosol cloud was not particularly efficient at reflecting sunlight. Stronger volcanic eruptions tend to eject more sulfur dioxide into the stratosphere than do weaker eruptions, and they lead to more sulfuric acid droplets within the same volume of atmospheric gases. A greater number of droplets increases the chance that droplets will meet and collide, forming larger droplets that will be removed more quickly from the stratosphere by gravity. A single, larger droplet also has less total surface area than two smaller droplets, and so is less effective at scattering sunlight.

There is therefore a balance to be struck between eruptions that are too weak to penetrate the stratosphere-and so produce a small, short-lived amount of cooling-and eruptions that produce large, less effective sulfuric acid droplets. By measuring the remnants of Tambora's aerosol cloud in material taken from ice sheets and lake sediments, modern scientists have determined that the climatic consequences-while undoubtedly devastating-could have been far worse if the particles had been roughly half their size.

Unlike the sudden drop in temperatures in Indonesia that occurred immediately after the eruption of Mount Tambora, the planetwide cooling was a gradual process that took up to a year to be fully realized. While air temperatures can, and frequently do, change rapidly in response to variations in solar energy, soil and ocean temperatures adjust much more slowly. The land and sea possess considerable capacity to store heat, while the atmosphere has practically no storage capacity. When the atmosphere is cooler than the land and sea, heat will flow from these reservoirs back into the air; but since the air cannot store heat for long, much of this is soon lost to space. If, on the other hand, the atmosphere is warmer, some of that excess heat will be stored in soil and water until a balance is reached

As Tambora's stratospheric aerosol cloud began to cool air temperatures by subtly reducing the amount of solar energy reaching Earth, the land and oceans would have resisted this cooling by transferring stored heat into the atmosphere and cooling themselves as a result. By early 1816 the land, ocean, and atmosphere were shifting toward a new balance of energies, largely as a result of the solar-dimming effect of the aerosol cloud. The adjustment cooled first air, then land, and finally ocean temperatures across the globe. Using information from tree rings the width of each ring is related to the growing conditions (mostly temperature and rain or snowfall) that year-climatologists have determined that 1816 was the second-coldest year in the Northern Hemisphere since 1400, surpassed only by 1601 following the eruption of the Huaynaputina volcano in Peru.

In the meantime, the aerosol cloud had produced other noticeable optical phenomena, most notably a series of spectacular red, purple, and orange sunsets in London in the summer and autumn of 1815. Sunsets typically appear yellow, orange, or red because atmospheric gases scatter blue light more effectively than they do light of other colors, skewing the visible-light spectrum toward red. The effect is even more pronounced when the Sun is low on the horizon, since its light must pass through a thicker layer of the atmosphere to reach the ground, resulting in less blue light and more red light. Stratospheric ash, dust, and soot particles from volcanic eruptions enhance this atmospheric scattering effect leading to brilliant red sunsets.

Iron, Steam, and Factories in Eighteenth-Century Europe

Industrial production in eighteenth-century Europe was facilitated by the use of a new energy source, coal. Traditionally, charcoal had fueled the smelting of iron-the melting process used to extract iron from iron ore. Britain, however, ran low on wood-the source of charcoal-before other European countries did and needed an alternative fuel. There was plenty of coal, but it contained impurities, particularly sulfur, that contaminated the materials with which it came into contact. In 1708 the British iron master Abraham Darby discovered that coal in a blast furnace could smelt iron without these attending complications. His discovery triggered the iron industry's use of coal. In 1777 the introduction of a steam engine to operate the blast furnace considerably increased efficiency. In 1783 a steam engine was used to drive a forge hammer to shape the iron: three years later steam-driven rollers flattened the iron into sheets. With these innovations, the output of the British iron industry doubled between 1788 and 1796 and again in the following eight years.

The greater supply of iron stimulated other changes. Relatively cheap and durable iron machines replaced wooden machines, which wore out rapidly. The new machines opened the door to further advances. Improvements in manufacturing methods and techniques led to the production of ever-larger amounts of goods, usually at lower prices. Industrial change started with cotton, but breakthroughs in the use of iron and coal continued and sustained these changes.

Before the age of industry, the basic sources of power were humans, animals, wind, and water. Humans and animals were limited in their capacities to drive the large mills needed to grind grain or cut wood. Wind was unreliable because it was not constant Water-driven mills depended on the seasons-streams dried up in the summer and froze in the winter. And water mills could be placed only where a downward flow of water was strong enough to drive a mill. Clearly the infant industries needed a power source that was constant and not confined to riverbanks. The steam engine invented and improved on in Britain. met that need and stoked Britain's industrial growth. As late as the 1860s. people. animals, and wind- and water-operated machines still supplied more than half of the energy needs of manufacturing in Great Britain and the United States. But the steam engine was clearly the power source of the future.

The steam engine was first used to pump water out of coal mines. As mining shafts were dug ever deeper through groundwater, drainage became a critical factor. In 1712 Thomas Newcomen invented a steam-operated water pump. Its use spread rapidly. The first steam engine in the Americas was a Newcomen engine installed in New Jersey in 1753. James Watt improved on the Newcomen engine considerably, making it twice as efficient in energy output. Eventually, by developing a separate condenser, Watt devised an engine that enabled the steam engine to power a variety of machines. Thus, steam engines could operate mills that had previously been powered by water or wind. The high-pressure steam engine was even more powerful and energy efficient. The use of steam engines spread in Britain and then to the rest of Europe and to the United States.

The steam engine centralized the workplace. With the machine as a central power source, it became practical and commonplace to organize work in a factory. Locating a manufacturing plant where it was most convenient eliminated the expense of transporting raw materials to be worked on at a natural but fixed power source such as a waterfall. The central factory also reinforced work discipline. These factories were large buildings without decoration. sometimes inspired by military architecture and therefore resembling soldiers housing. With the introduction of blast furnaces and other heat producing manufacturing methods. the tall factory chimney became a common sight on the industrial landscape.

The steam engine powered a dramatic growth in production. It increased the force of blast furnaces and the mechanical power of machinery used to forge iron and to produce equipment for spinning and weaving. Assisted by machines. workers were enormously more productive than when they depended solely on hand-operated tools.

Formation of Earth's Atmosphere and Oceans

Early in its history, Earth was molten and had neither atmosphere nor oceans. However, as the gradually cooling surface began to solidify into rock, intense heat trapped below Earth's outer crust had to get out somehow. The result was surely volcanoes, geysers, earthquakes, and a variety of other geological events that literally blew off steam and pent-up heat through cracks in the surface. “Outgassing” of this sort happens even today: though at only a few locations on Earth and rather infrequently at that. But several billion years ago, this type of geological activity was surely more widespread and frequent. Scrutiny of modern volcanoes shows that lots of steamy water vapor, carbon dioxide, and nitrogen would have then undoubtedly emerged, along with vast quantities of ash and dust. Smaller amounts of hydrogen, oxygen, and other gases doubtless accompanied these early planetary eruptions. Calculations imply that over the course of Earth’s history enough gas was exhaled through fissures from Earth’s interior to create much of our current atmosphere. The rest presumably came from comets and meteorites that could have scattered large quantities of matter on young Earth. Even today, some 40 thousand tons of extraterrestrial matter falls to Earth each year almost all of it burning in the air or splashing in the ocean.

The origin of our present atmosphere is therefore a combination of terrestrial outgassing and interplanetary assault (further changed by later biological events). In truth, Earth's atmosphere is perhaps still adjusting, as present-day volcanoes occasionally sputter gas and heat amid incoming debris arriving from space. Today’s atmosphere was not, however, derived directly from a mixture of interstellar gases The composition of Earth's atmosphere thus differs considerably from the average cosmic abundance of the elements.

Since the atmosphere and oceans of planet Earth are so closely linked, they almost certainly originated, at least partly, from the same sources—our planet's interior and materials from space. As regards the ocean, geologists argue that, as the surface cooled sufficiently, the first store of liquid water pooled as water vapor condensed. After all, steam is the main component of volcanically vented matter, and the hydrated rocks (mostly silicates, with water trapped inside) that today make up Earth's mantle (the layer immediately below the crust) store several times more water than in all the seas combined. But outgassing from within may not be the whole story since our planet’s water has chemical subtleties implying that some of it may have come from beyond.

Debate swirls among geologists concerning the rate and timing of ocean formation. We are unsure whether Earth’s mantle outgassed the global seas all at once early in our planet’s history— known among geologists as the “big burp" theory. Or perhaps the seas took some time to form; having secreted from Earth’s interior in a series of volcanic events that occurred more gradually. Researchers argue that some of the waters of Earth have resulted from a rain of water-rich comets and meteorites that collided with our planet in great numbers during its first billion years A minority of researchers go so far as to argue that all of Earth’s water originated in this way, but this extreme view does not fit with the fact that the makeup of water on Earth does not match that trapped in interplanetary bodies. Three comets that recently bypassed Earth—Halley's in 1986, Hyakutake in 1996, and Hale-Bopp in 1997—all emitted radiation that revealed a higher concentration of water with heavy hydrogen atoms than is found in Earth’s oceans.

Most likely—as with many other aspects of the cosmic evolutionary story that are neither black nor white, neither solely this nor clearly that—Earth's large bodies of seawater emerged from both within and without, and then, as the rate of outgassing and bombardment declined, a global recycling system began to operate. Water locked in rocks was expelled back into the ocean whenever the rocks were heated, such as those near volcanoes. To be sure: most of today's seawater is thought to have been recycled many times, perhaps as frequently as every ten million years. Recently, water has been directly observed emanating from certain underwater vents Whether this is “juvenile” water still originating directly from Earth's mantle and incrementally adding to the world's supply or merely existing seawater cycling through the vents is not yet known.

A Transportation Transformation

In the northeastern United States, from Independence (1776) until the Civil War (1861-1865), state and local governments, rather than the federal (national) government, were the sources of most initiatives to improve transportation, although private entrepreneurs were the principal builders of toll roads in the early nineteenth century. All toll roads required payment for use, but few toll roads were profitable, and improvements in water transportation made greater contributions to commerce and economy. The use of steamships on rivers from the second and third decades of the nineteenth century reduced the cost and increased the speed of transporting goods up such major rivers as the Hudson, Ohio Mississippi, and Missouri and contributed to the growth of cities such as Albany, St. Louis, Cincinnati, Louisville, and Pittsburgh. Since there were over twenty thousand miles of navigable rivers in the pre-Civil War United States, the fuller use of natural inland waterways was a major step in integrating the new nation. Steamships soon lowered transport rates on large lakes, along the sea coasts, and in international trade.

Failing to obtain federal aid, New York State, with help from British investors, financed the 363-mile Erie Canal. The canal opened in 1825 and linked Buffalo on Lake Erie with Troy and Albany on the Hudson River and thus with the Atlantic Ocean. The Erie Canal lowered transportation costs from Buffalo to New York City by 90 percent and established New York City as the premier United States port. The canal that linked the Ohio river with Lake Erie in 1833 was another example of connecting major waterways and large regions with each other. Nearly four thousand miles of canals were operating by the 1850s, with three-quarters of the financing coming from state and local governments, borrowed mostly from British investors.

The principal canal networks connected the Northeastern Seaboard with the Upper Midwest. The low cost of canal and steamboat shipping made it feasible to convey even bulky, low-priced goods long distances. This put northeastern and Mid-Atlantic grain farmers at a competitive disadvantage in respect to grain from the rich soils of the Ohio River valley and beyond. The beleaguered farmers had to switch to dairying or vegetable and fruit crops to remain viable, but instead they often left farming or migrated west New England farmers were most likely to migrate, and recurrent sightings of stone walls in deep New England forests are modern-day reminders of those abandoned farms.

The canals were successful, but then came the railroads. The first railroad in the United States-the Baltimore and Ohio-opened in 1830 with a locomotive imported from Great Britain. Thirty years later, the United States boasted over thirty thousand miles of tracks with a total expenditure five times greater than that invested in canal construction. Like the canals, most railroad lines were located in the northern regions of the nation and facilitated east-west transport, and Chicago, with four thousand miles of track converging there by 1860, became one of the major transportation hubs. Rail competition led to the closure of many transportation canals and the cancellation of new canal projects.

Railroads were more flexible than canals because they could move goods and people over a variety of terrains with fewer problems posed by mountains, water levels, freezing weather, and trans shipments. They reduced overland travel time by about thirty times, and it was the greater speed that made rail travel attractive to people. But railroad freight (shipped goods) revenue exceeded passenger revenue by the end of the 1850s, also the point when the volume of freight sent by rail surpassed that of canals. River transport continued to be widely used

The telegraph was a spectacular change in communication. It complemented the expansion of the railroads by enabling railroads to coordinate and monitor schedules. The railroad companies often allowed a telegraph company to string lines along railroads' rights of way in exchange for discounted service fees.

There were over fifty telegraph companies in the early 1850s when a group of Rochester, New York. businessmen began consolidating small companies into what became Western Union (a famous telegraph company). This consolidation led to better efficiencies and improved service, facilitating the increased use of telegraph service for money orders, business news, newspaper reporting, personal messages, and conduct of war. Western Union completed the first coast-to-coast line across the nation in 1861, leading the United States Post Office to cancel the Pony Express (mail delivery by riders on horseback) when the Civil War began.

Communication: Bumblebees Versus Honeybees

The waggle dance of the honeybee has been frequently described and much studied, and it is one of the most complex systems in insect communication. In contrast, it has long been assumed that for bumblebees the search for food is essentially a solitary endeavor, that workers do not communicate with each other about good sources of forage, so that each individual has to learn for itself which flowers provide reward. Indeed, it has been known for many years that bumblebees (of a range of species)are unable to recruit nest mates to particular places. However, it has become apparent that bumblebee foragers (bees that look for food) do communicate indicating that they have found an available food source, but they do not indicate a specific location. In an elegantly simple experiment, Dornhaus and Chittka demonstrated that on their return to the nest successful foragers of B. terrestris (a species of bumblebee stimulate other workers to forage and communicate to them the scent of the food source that they have located. The returning forager runs around on the surface of the nest in an excited manner, frequently bumping into nest mates and buzzing her wings(very similar behavior occurs in some stingless bees). This stimulates workers to leave the nest and search for the source of the scent. This communication system is less complex than that of honeybees, for the new recruits do not appear to be given any positional information about the location of the food source. It would appear that there Is some sort of pheromone signal(a chemical factor that triggers a social response in members of the same species)released by the returning forager, for activity in adjacent nests is stimulated unless air flow between the nests is prevented.

Why do bumblebees not communicate positional information? Domhaus and Chittka argue that conveying the location of food sources may be less important to bumblebees than to honeybees evolved in tropical ecosystems where they rely mainly on flowering trees, which grow close to one another and may be several kilometers from the nest and therefore difficult to locate in the temperate habitats in which bumblebees probably evolved, the plants on which they feed are generally more scattered. There is nothing to be gained in recruiting workers to a specific small patch that one bee can adequately exploit single-handedly. However, communication about the types of flowers that are providing rewards will allow the colony to rapidly encourage feeding on a rewarding plant species when it comes into flower, and so keep track of the changing seasonal availability of different species.

A second possibility is that bumblebees may forage over shorter distances than honeybees, perhaps as a result of their smaller colony size, thus rendering communication about the precise location of forage less important.

There is also a cost to conveying location. Honeybee recruits can take over an hour to decide where to forage when presented with just two returned foragers advertising different locations. They also take a long time to find the food source that is being advertised, for the locational information is not precise. Thus, in bumblebees it may be that the costs of conveying this information outweigh any gains.

Communication also occurs between bees while foraging. Both bumblebees and honeybees can distinguish between rewarding and nonrewarding flowers of the same species without sampling the reward available. When doing so, they are often observed to hover by and "inspect each flower, accepting some and rejecting others. In some circumstances the bees may be directly assessing the reward levels, or perhaps examining correlates of reward such as flower size and symmetry. However, there is now clear evidence that perhaps the most important cue used by bees to decide whether to probe or reject a flower is scent marks left by bees on previous visits. Such marks may increase foraging efficiency by reducing the time spent handling flowers that have recently been emptied by another bee. Bumblebees, honeybees, and carpenter bees leave short-lived repellent marks on flowers that they visit, and members of the same species use these to discriminate between visited and unvisited flowers.

Natufians and the Domestication of Grain

It is known that the Natufian people used sickles to cut and harvest wild grains in the Mediterranean region as early as 12,500 B.c. But did this practice lead to domesticated grains? Wild grains are brittle (easily broken) and the seeds burst forth when ripe, whereas nonbrittle "domesticated" grains remain on the plant until beaten off. In wild grains, there are one or two nonbrittle mutants to every 2 to 4 million brittle individuals. If ripe grain is cut with a sickle and taken to a village to dry and be beaten out, a higher percentage of the nonbrittle mutants is harvested since many of the brittle seeds are lost. If this grain is then planted, this ratio will increase each harvest

Gordon Hillman, an archaeologist, and Stuart Davies, a biologist from the University of Wales, have used their knowledge of plant genetics and ancient gathering techniques-much of it acquired by experimentation-to estimate how long the change from wild to domestic strains would take. By using computer simulation, they showed that in ideal circumstances as little as twenty cycles of harvesting and replanting in new patches could have transformed a wild, brittle type of wheat into the domesticated nonbrittle variant. Under more realistic conditions, 200 to 250 years is the most likely period of transition.

The archaeological evidence makes it clear that this transition did not happen during the Natufian Period (about 12,500 to 10,200 years ago). There are microscopic differences between the shape of the grain from domestic cereals and wild varieties, and although cereal grains are rare in the Natufian archaeological record, all of those known are clearly from wild cereals. Not for another millennium at least do we encounter the first domesticated grains-from the settlements of Abu Hureyra and Tell Aswad in Syria and Jericho in Palestine. So, the Natufian people appear to have cut the wild cereal stands with their sickles for as much as 3,000 years without the evolutionary leap from brittle to nonbrittle plants.

There seems to be a very simple explanation for this, identified in some remarkable research by Romana Unger-Hamilton during the 1980s while she worked at the Institute of Archaeology in London. Under the guidance of Gordon Hillman, she spent many months replicating the Natufian style of harvesting wild cereals. Using identical sickles made with bone handles and flint blades, she cut stands of wild wheat and barley on the slopes of Mount Carmel, around the Sea of Galilee and in southern Turkey in a series of controlled experiments. The blades were then microscopically examined for signs of "sickle-gloss" (traces of wear which cause a blade to shine)-the texture, location, and intensity of gloss will vary with different types of cereals and at different stages of ripeness.

Unger-Hamilton found that the sickle-gloss on the true Natufian blades was most similar to that on the blades she used to harvest cereals that were not yet ripe. In that state, the brittle plants would have shed only a little of their grain, so that it would be collected from the nonbrittle variants in virtually the same minute proportion as from those plants within the stand. So even if the Natufian people were planting seed to generate new stands of wild cereals, nonbrittle variants were unable to become dominant. Harvesting the unripe ears was perfectly sensible as it avoided the loss of most of the grain from the brittle plants, which would have already been shed to the ground.

Another factor probably prevented the emergence of domesticated cereals among the Natufians: their sedentary lifestyle. Patricia Anderson, of the Jales Research Institute in Paris, undertook a similar program of research to that of Romana Unger-Hamilton and confirmed many of her results. She also found that when wild stands are cut with sickles, even when still in a green stage, the grain that falls to the ground is quite sufficient to provide for the next year's crop. So, the Natufians would have only needed to plant if they were beginning a brand-new plot of cereals-otherwise they could have relied upon growback at the existing stands. Even if the grain collected by the Natufian people did have a higher proportion of the nonbrittle variants, unless they were creating new plots of cereals in new places, these variants would never have had the opportunity to become the dominant form.

Egypt in Early World History

The origins of Ancient Egypt represent some of the first stances of complex human societies. Indeed, the Nile Valley is the perfect place to start a state-level society. The annual flood brings fertile silts from the Ethiopian highlands to the plains along the Nile River Valley, allowing agriculture to continue indefinitely without overtaxing the soil. Evidence suggests that the inhabitants of the upper Nile region(that is the "upstream " or southern regions of Egypt)were harvesting wild barley as early as 10,000 B.C.E. Perhaps as early as 5000 B.C.E., permanent agricultural settlements had been established both in the upper and lower Nile regions. These settlements Were similar in some ways to other sites found in what is now known as Egypt's Western Desert. Pottery found at the early Nile sites is of a style also found across the West African Sahel and North African region. This evidence suggests that the early Nile settlers may have come to the river valley to escape a savanna (grassland)that was beginning to make the transition to desert. Still during the period up to the third millennium B CE, the region around the Nile Valley was well enough watered by rain to support some degree of agriculture, so it is probably more accurate to think of the drying of the Sahara as a series of gentle pushes rather than as a single dramatic shove out of the region.

During the period from 3600 to 3300 B.E, the settlements along the upper Nile appear to have begun the process of creating what anthropologists call complex societies by developing a specialization of labor and moving toward more hierarchical political and economic systems. Two of these early Nile states have been excavated and examined in some detail: Nagada and Hierakonpolis Nagada was located just north of where the city of Thebes would later be established, and Hierakonpolis lay just to the south. The inhabitants of both proto-states undertook a process of environmental modification. They cleared trees and heavy growth to expand the area of cultivated land and also built dykes and canals to control the flow of water.

These efforts were successful enough that by around 3500 B.C.E. they had expanded the amount of arable land fourfold and led to a population explosion of up to 1,000 percent-perhaps reaching a population density o 1,000 persons per square kilometer. Such an increase is a testimony both to their ingenuity and to the bounty of the Nile Further, the major towns of Nagada and Hierakonpolis were walled suggesting that the inhabitants were sometimes forced to defend their newly created wealth from their neighbors.

Both towns show increasing economic activity by the middle of the fourth millennium BCE Hierakonpolis seems to have developed an economic specialization in the manufacture of a fine ware pottery known to archaeologists as "Plum Red", and examples of this pottery were exported throughout the Nile Valley. The town was also the home of one of the world’s first large-scale breweries producing an estimated 1, 200 liters of beer a day. Such economic specialization led to social differentiation. Rather than a relatively egalitarian society with very few differences in wealth from individual to individual or family to family, now some people were rich and others were not. Within the towns. a wide variation in the size of homes suggests growing social distance between the rich and poor. Cemeteries found in the Nagada region during this period show an increasing social stratification. Some tombs include luxury items such as gold jewelry, gems, and finely decorated pottery. The wealthy tombs tended to be clustered together in a certain part of the cemetery, maybe as family groups or maybe just to be separate from the poor, even in death.

Even in this period before the unification of the Nile, there was considerable contact with other regions. Although the archaeological record for the lower Nile is less complete, it is clear that there was some degree of exchange along the Nile Valley. This was no doubt aided by the fact that whereas the current runs north, the prevailing winds blow south, allowing for relatively easy round-trips.

Uniformitarianism and Earth's Cycles

During the seventeenth and eighteenth centuries, before geology became the scientific discipline it is today, people believed that all Earth's features had been produced by a few great catastrophes. Those catastrophes were thought to be so huge that they could not be explained by ordinary processes but must have supernatural causes. This concept came to be called catastrophism. The catastrophes were thought to be gigantic and sudden, and they were also thought to have occurred relatively recently.

During the late eighteenth century the concept of catastrophism was reexamined, compared with geologic evidence, and found wanting. The person who assembled much of the evidence and proposed a counter theory was James Hutton. a Scottish physician and gentleman farmer. who was intrigued by what he saw in the environment around him. He observed the slow but steady effects of erosion: the transport of rock particles by running water and their ultimate deposition in the sea. He reasoned that mountains must slowly but surely be eroded away. that rocks must form from the debris of erosion, and that those rocks in turn must be slowly thrust up to form mountains. Hutton didn't know the source of the energy that caused mountains to be thrust up, but he argued that everything moved slowly along in a continuous, repetitive cycle.

Hutton's ideas evolved into what we now call the principle of uniformitarianism, which states that the same external and internal processes that we recognize in action today have been operating throughout Earth's history. We can examine any rock, however old and compare its characteristics with those of similar rocks that are forming today in a particular environment. We can then infer that the ancient rock very likely formed in the same sort of environment. For example, in many deserts today we can see gigantic sand dunes formed from sand grains transported by the wind. Because of the way they form, the dunes have a distinctive internal structure. Using the principle of uniformitarianism, we can safely infer that any rock composed of cemented grains of sand and having the same distinctive internal structure as modern dunes is the remains of an ancient dune.

Geologists since Hutton's time have explained Earth's features in a logical manner by using the principle of uniformitarianism. But in so doing they have made an outstanding discovery—Earth is incredibly old. An enormously long time is needed to erode a mountain range, or for huge quantities of sand and mud to be transported by streams, deposited in the ocean, and cemented into new rocks, and for the new rocks to be deformed and uplifted to form a new mountain. Yet, slow though it is, the cycle of erosion formation of new rock, uplift, and more erosion has been repeated many times during Earth's long history.

During the nineteenth century, geologists tried to estimate the duration of the rock cycle by estimating the thickness of all the sediments that have been laid down through geologic time. They assumed that the principle of uniformitarianism applied to the rate sat which processes occur as well as to the processes themselves and hence that rates of deposition of sediment have always been constant and equal to today's rates. Thus, they thought, it would be a simple calculation to estimate the time needed to produce all the sediments. The results, we now know, were greatly in error. One of the reasons for the error was the assumption of constancy of geologic rates.

The more we learn about Earth's history and the more accurately we determine the timing of past events through radiometric dating (using rates of decay of naturally occurring radioactive atoms to determine the ages of rocks), the clearer it becomes that rock cycle rates have not always been the same. Some rates were once more rapid, others much slower. This means that the relative importance of different geologic processes has probably differed in the past. For example, just because glaciation is an important process today, we cannot assume that it has been equally important throughout geologic time. But we can assume that when glaciation did affect Earth in geologically remote times the processes and effects were the same as those, we observe in glaciated regions today.

The Kingdom of Meroě

The kingdom of Meroě was one of the most important states of ancient Africa. Though overshadowed by Egypt to its north, it was a major regional trading power and had one of the most advanced ironworking industries in the world. Its roots date from around 1100 B.C., when Egypt began to decline and pulled out of its southern region of Nubia (modern Sudan). Local Nubian rulers then established their own kingdom, known as Kush, which became strong enough to invade and rule Egypt for a short while, withdrawing after being defeated by the Assyrians in 670 B.C. About 120 years later, Kush shifted its capital to the southern town of Meroë. This move may have been made under the threat of Egyptian invasion, but there were many other advantages to the southerly location, where the kingdom (often called Meroě in its new location) flourished.

One advantage was its natural resources, particularly iron ore and the hardwood timber needed for making charcoal, which was essential for iron production. Meroě was determined to promote the production and use of iron, especially given its rulers' memory of the devastating Assyrian defeat, when they had no defense against iron weapons. It is clear from archaeological evidence that Meroě in fact achieved a high level of iron production: there are large numbers of metal-waste heaps to be found in the region, as well as furnace remains and locally produced iron weapons. Iron production also gave Meroě tools such as hoes for farming and axes to cut timber and clear agricultural land. On the other hand, intensive iron production helped sow the seeds of Meroě's eventual downfall, since the need for charcoal resulted in overexploitation of timber, causing a degradation of the environment.

Meroě's southerly position provided it with extensive tropical rains in the summer, so that it could rely on a productive agriculture. This had not been possible in the north, where the cultivable floodplains on either side of the River Nile were relatively narrow, yet where the relative lack of rain meant that farming depended on those plains, nourished mainly by the Nile's floods. In Meroě, the rains allowed farmers to grow tropical cereals like sorghum and millet, and to cultivate fields far away from the river. However, it was also to Meroě's advantage to find ways to exploit drier lands within its territory. One way was to adopt the Egyptian water wheel, or saqia, as an aid to irrigation, especially for cotton, permitting Meroë to be a major producer of textiles. In dry plains to the east, where cattle and other livestock could graze, the Meroites improved pastoral production by building large earthen dams across seasonal streams, which provided water for the livestock during the long dry season. They may also have practiced recessional irrigation, in which crops are planted in the wet soil left behind as a body of water shrinks in the dry season.

Its location was ideal for the kingdom to serve as a trading center for the northerly flow of sub-Saharan goods like ivory, leopard skins, ebony, and gold to the economic centers of the Greek and Roman Mediterranean world, and for fine metalwork, glasswork, and other manufactured and luxury goods going south. Not only was trade to the north possible, but Meroě's relative proximity to the Red Sea to the east allowed a connection to commercial networks linking India, the Far East, and the Mediterranean. These networks increased in importance during the centuries of Greek and Roman rule in Egypt (starting in the fourth century B.C.). However, Meroě's dependence on trade may also have been a major factor in its decline, starting in the second century A.D. It was at this time that harsh Roman rule in Egypt impoverished that country, so that it was no longer able to afford goods from Meroě. Around the same time, desert nomads gained power and made the trade route between Meroě and Egypt more and more dangerous. Finally, the rising new Axumite kingdom, to Meroě's southeast, had its own port on the Red Sea, taking away much of Meroě's eastward trade.

Physiological Adaptations of Tuna

Tuna fish not only swim fast, but they can also cover great distances over short periods of time or during long migrations. At night, tuna are known to cover some 15 kilometers during offshore excursions (possibly for feeding).For swimming at such great speeds and over long distances, the tuna has evolved a muscle structure and circulatory system that athletes can only dream of.

In general, fish move by a series of muscle contractions that begin at the head and progress through the body toward the thruster, the tail. Two types of muscle, white and red, are used in swimming and are arranged along the flanks (sides)of the fish. White muscle contracts quickly and is used for bursts of high-speed swimming. It operates anaerobically (without oxygen),generating lactic acid as a by-product (similar to the lactic acid that can build up in humans after prolonged vigorous exercise and produce fatigue or muscle cramps).Red muscle contracts more slowly, is used for slow, continuous swimming, and operates through an aerobic (with oxygen)metabolic pathway. Because red muscle does not produce lactic acid, it can continue to contract almost indefinitely with the necessary fuel (oxygen and glucose).In most fish, the bulk of the muscle is white, with only a small lateral strip of red. The tuna, however, is equipped with a much larger proportion of red muscle than the average fish. This allows the tuna to cruise at high speeds aerobically, without lactic acid buildup. White muscle is used to generate additional thrust to go from cruising to extremely high speed. To sustain such athletic skill, the tuna requires an efficient means of supplying blood and oxygen to its hard-working muscles and a way of releasing the heat generated during muscle contractions.

Fish, like other animals, need oxygen to respire and to fuel their working muscles. Dissolved oxygen is some 30 times as dilute in the sea as in the air. Fish use their gills to extract oxygen from the water by pumping water in through their mouths, then over and out through the gills. The pumping is done by repetitively opening and closing the mouth, thus sucking or gulping the water in and pushing it past the gills. The tuna's skull and jaw are rigid to enhance swimming speed, so it cannot physically pump water in. Instead, it breathes by means of ram ventilation As a tuna swims, its mouth stays open and water is literally rammed in and over its extraordinarily large gills. Like some of the other fast-swimming fish, the tuna must constantly move or else it will suffocate.

Tuna and some sharks have evolved a specialized circulatory system that uses excess heat production to stay warm in their often-chilly medium. Fish are typically cold-blooded; their body temperature is regulated by the surrounding water. But the tuna has a special internal heat exchanger made up of intermingling arteries and veins. Cool, oxygen-rich blood flows from the gills to the tissues through arteries and a network of small blood vessels. Running parallel to and intermingling with the arterial network (the vessels carrying blood from the gills to the organs)is the venous system. In the venous system, warm, oxygen-depleted blood is carried from the organs and tissues back to the gills. Because the veins and arteries are intermingling and in close contact, heat is readily transferred from the warm blood in the veins to the cold blood in the arteries. With elegant physiological simplicity, this countercurrent heat exchange uses the tuna's own internal heat from muscle contractions to warm incoming blood. In most other fish, heat is lost directly from the gills; but in the tuna, there is little heat loss, and by the time the blood reaches the tuna's interior, it is nearly as warm as the internal body tissues. This exchange is so effective that a tuna's core temperature has been measured at 10 to 20 degrees Celsius warmer than the surrounding water. Scientists suspect that elevated body temperatures help keep the tuna's metabolism high for food processing, allow its red muscle to contract more quickly, and may enhance lactic-acid breakdown. Tuna may also be able to partially shut down the heat exchanger if they become too warm, or they may make excursions into deeper, colder water to avoid overheating.

The Carboniferous Atmosphere

An explosion of new animal forms took place about 530 million years ago, and from that era onward, oxygen built up rapidly in the atmosphere. As oxygen levels climbed, the new ecosystems grew in stature and complexity until lush, swampy forests had developed. Thus began the Carboniferous period (roughly 360 to 300 million years ago), named for the deep beds of coal—a form of carbon—that formed from the remains of trees and other organic matter during this time. We know a lot about life in the Carboniferous world because of the many fossils found in coal seams (layers). In 1979, miners working in the town of Bolsover, England, stumbled upon the fossil of an enormous dragonfly, with a wingspan of more than half a meter. Carboniferous scorpions were a meter long, and mayflies had 40-centimeter wingspans. Plants were huge too. The lycopods, which we know today as tiny club mosses, reached heights of 50 meters. All these life forms were giants by today's standards, so what was going on? A hint at the answer comes from the way the animals breathed. Insects do not have lungs. Instead, their bodies are riddled with tiny tubes called trachea that let air diffuse in and out passively. This puts a size constraint on insects because the passive breathing system cannot deliver oxygen quickly to deep tissues. In today's atmosphere, a dragonfly the size of the Bolsover specimen would not be able to beat its wings fast enough to fly because its flight muscles would be deprived of oxygen. To get airborne, it must have flown through air much richer in oxygen than ours.

Calculations suggest the Carboniferous air was 30 to 35 percent oxygen, as compared to 21 percent today. The explanation for this high level of oxygen lies in the coal itself. Coal forms when dead plant material doesn't fully decompose, instead becoming buried and then compressed over vast spans of time. By 375 million years ago, the swampy forests of the Carboniferous were already laying down the first coal seams at a prodigious rate, and they continued doing so for millions of years. In fact, about 90 percent of the world's coal stems from this time. The Carboniferous plants must have been piling up in the ground when they died, perhaps because the standing water of the swamps protected them from decay—still (unmoving) water lacks oxygen that is needed in order for decomposition to occur. Or perhaps bacteria had not yet evolved the ability to digest the tough fibrous material (lignin) in wood. And because the dead plants weren't being fully recycled, there was an imbalance in the carbon and oxygen cycles. The trees were taking in carbon dioxide and giving off oxygen through the process of photosynthesis, but the reverse process wasn't happening—the carbon was staying trapped in the plants, and the oxygen wasn't being used up by the process of decay. So as forests flourished, producing more and more oxygen, the atmosphere's oxygen level rose. Life had created an atmosphere in which it could make a leap of evolution.

The Carboniferous was a green world of mosses, vines, horsetails, and ferns, and vast areas of equatorial land were given over to forest swamps. But the oxygen-rich atmosphere brought something else. This is the time when charcoal first appears in the fossil record, and charcoal results from organic matter exposed to high temperatures, which tells us that the atmosphere now had sufficient oxygen to support fire. From around 365 million years ago, ferocious wildfires swept across the land and they were fires like no other, for in air that was 30 percent oxygen, even the damp swamps would have burned. Today, studies of Carboniferous charcoal reveal not only the great intensity of the fires but the nature of the plants that were consumed by them. This, along with fossil evidence, reveals that the plants had adapted to become resistant to fire, with thick bark, high crowns, and long tubers that were protected deep in the soil. Destructive as the wildfires that swept the planet must have been, they were also part of the mechanism that kept the atmosphere in balance, because they would have consumed oxygen and prevented levels from rising too high while also, by burning the forests, injecting carbon dioxide back into the air.

Escape Mechanisms of Ypsolopha Dentella

Y. dentella larva © neil hardwick/Shutterstock.com

Ypsolopha dentella (Y.dentella)is a moth that lives in the northeastern United States. It appears as a larva (the immature form that differs entirely from the adult moth)early in the spring, feeding on the young leaves of honeysuckle plants. Y.dentella larvae are well camouflaged. Predominantly green, they match the color of the leaves they eat. As additional defenses, the larvae also have the ability to squirm and to jump off a leaf and hang in the air using a silken thread. Squirming, or wiggling, is an escape mechanism they share with many small caterpillars, but Y.dentella manifests this behavior in the extreme. Release a Y.dentella larva on a smooth surface and poke it gently with a toothpick, and it will propel itself over the surface by twisting and turning with such speed that it will literally vanish from view. The response may last only seconds, but that is long enough to allow the larva to distance itself from the scene.

When a larva has wiggled its way to the edge of a leaf and can go no farther, it jumps, plunging into space, suspended by a slender silken thread that it squeezes from a small spigot that projects from just beneath the mouth. The spigot bears the openings of two large sac-like glands that take up a considerable portion of the larva's body cavity and are bulging with the viscous liquid from which the silk is made. Jumping does not require that the larva take time to anchor the silken thread to the edge of the leaf before jumping. The silken thread is fastened to the larva's cocoon, and the larva needs only to let out a piece of silk long enough to cover the distance to the leaf edge. It is therefore connected to home base at all times and free to squirm away and take a plunge at a moment's notice.

When they plunge, the larvae drop some 10 to 20 centimeters or more, a considerable distance relative to a body length of 15 millimeters or less. In essence their plunging is another vanishing act, a trick that in combination with squirming enables the larvae to avoid trouble by leaving the scene. When the larvae plunge, they drop virtually instantaneously. The rate at which they are able to convert liquid glandular precursor into silken strand is therefore quite remarkable.

The suspended larva, as a rule, does not remain hanging from its thread for long. Within minutes, it begins the slow process of climbing up the thread to return to its cocoon. To ascend, the larva pulls itself up along the cord, a short distance at a time, using its mouthparts to exert the pull and to tuck away the loops of slack reeled in as it proceeds upward. The process is elegant. The entire thread is stuffed into a holding space between the third pair of legs. When the larva completes its climb, it relinquishes the package, leaving it stuck to the leaf margin the moment it is back at home base. In so doing, the larva is essentially squandering a resource. Other silk producers are more conservation oriented. Spiders, for instance, when they take down their webs, eat the silk.

Jumping is doubtless effective against many predators, including ants and jumping spiders, but it comes at a price. Aside from the cost of the silk itself, which the larva needs to conserve for eventual cocoon production, jumping takes time. Although the larvae usually ascend within minutes, they sometimes remain suspended for over an hour. Potentially this wait could cut significantly into feeding time and therefore have a retarding effect on development. Whether it does depend on how often the larvae are attacked under normal circumstances, which has not been determined. But there is no question that predation pressure could be high at times upon Y. dentella larvae, particularly from ants.

One intriguing possibility is that jumping protects Y.dentella larvae against predation by wasps. No data on Y.dentella itself are available, but other caterpillars have been shown to avoid wasp attacks by resorting to squirming behavior.

The Medieval Agricultural Revolution

After around 900 A.D. the North Atlantic region began a 300-year spell of elevated temperature known as the medieval warm period, which affected agriculture both in North America and in Europe- including a burst of productivity that has sometimes been called a revolution. Although some have questioned whether the term "agricultural revolution” is appropriate for western Europe during this era, it is clear that dramatic changes in food production coincided with the medieval warm period. The central innovation was the three-field system of crop rotation, which began to take hold in much of northern Europe and Britain during the tenth and eleventh centuries. Although historians are not as certain as they once were about either the extent of its use or its novelty, the three-field system replaced in many areas an earlier agricultural model that Germanic peoples of the northern parts of western Europe had inherited from the Roman Empire. Many farmers had formerly used two sets of fields; one was cultivated and the other left unplanted, and the cultivation would alternate from year to year. The main crop had been wheat, sometimes joined by plots of rye or other grains, planted in the autumn and harvested in the spring.

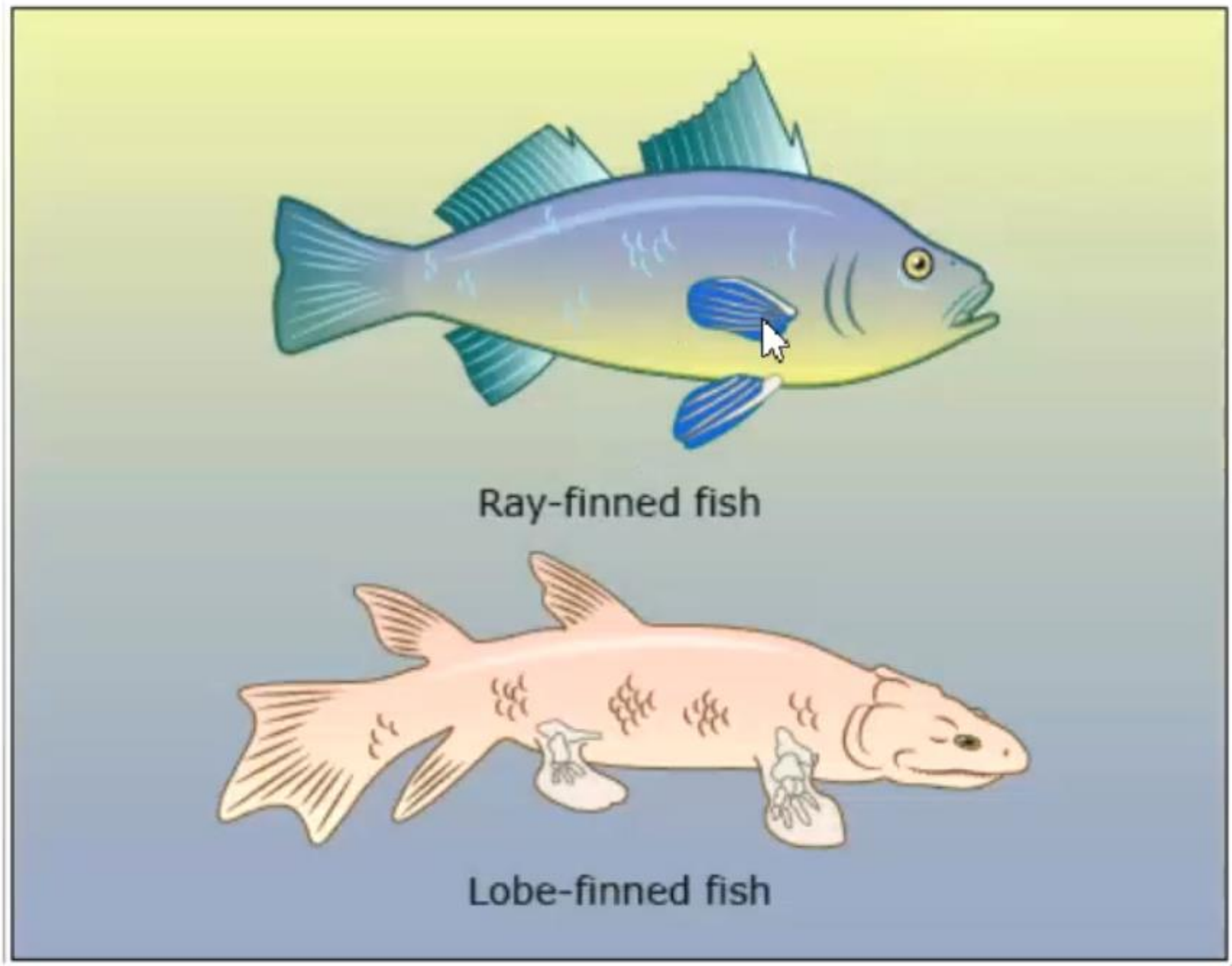
The new three-field pattern took advantage of climatic changes during the warm period to add summer crops to the ancient mix. The two-field system had always been better adapted to the wet, mild winters and dry summers of the ancient Mediterranean region than to the very different environments north of the Alps and the Pyrenees. The three-field system was not only better adapted to northern conditions but substantially more productive than the Mediterranean model. For any given place, the three-field revolution increased the amount of land under annual cultivation by a third; a community that formerly cultivated,say,300 of its 600 acres annually now farmed 400.The transition may not have been quite as earthshaking as the replacement of the crops first grown by Native Americans in eastern North America with a regime of corn, beans, and squash, but it was dramatic nonetheless. On both continents, new agricultural patterns transformed life in temperate zones.

In northwestern Europe, more was involved than new crops and a new organization of the landscape. A novel kind of yoke allowed horses to be employed to pull heavy loads, augmenting the slower and less powerful oxen that still did much of the work. Horsepower was particularly useful in pulling improved heavy, wheeled plows, which were now frequently fitted with moldboards (curved iron plates)that turned as well as dug the soil. As these innovations spread, simple wooden tools called harrows also began to replace dragged thornbushes in the chore of breaking up and smoothing freshly plowed ground. An even simpler device-two pieces of wood attached together to make a flail-improved the process of separating the seed from the plant.

None of these changes occurred overnight, and all built upon precedents that were hundreds of years old. Nonetheless, the cumulative impact was striking. Vast areas either abandoned since the decline of Rome or never before cultivated were put into production. More important, the three-field system diversified the diet of ordinary people. Among the new summer crops were oats, barley, and a variety of legumes, including vetches, peas, lentils, and beans. The expanded availability of legumes (apparently grown previously, if at all, mostly as garden vegetables)was a major nutritional improvement to a diet formerly composed largely of daily bread. Western Europeans and North Americans discovered the virtues of legumes at almost exactly the same time.

Most European households kept gardens, which, like the gathered vegetation of North America, contributed essential nutrients for at least part of the year. The extent to which this phenomenon was novel is unclear, and, even for the elite, the yield was what the most careful student of early medieval nutrition calls "a somewhat monotonous diet heavily oriented to members of the cabbage and onion families.” still, the nutritional gains from western Europe's agricultural revolution were substantial. As the system took full hold, in the eleventh through fourteenth centuries, the region's population seemed to have roughly tripled.

From Water to Land



Fish were the earliest vertebrates (animals with backbones), able to live only in water. During the Devonian period (416-358 million years ago), a group of lobe-finned fish started a transition from water to land that ultimately led to the evolution of tetrapods-four-limbed animals including amphibians, reptiles, birds, and mammals (some tetrapods, like snakes, have later evolved to be limbless). Unlike ray-finned fish-the most common types of fish found today-lobe-finned fish have bones supporting some of their fins. This arrangement made it possible for fins to evolve into primitive limbs. It is likely that the main anatomical transition from fins to limbs was completed in an aquatic environment, where the limbs did not have to support the entire weight of the body due to the water's buoyancy (the upward force of water on a submerged object). The primitive fin-limbs probably first developed flexible joints, helping them to bend and push the fish forward as it crawled along in the shallows or negotiated thick waterweeds. The fin-limbs grew stronger as they developed simple knee, ankle, elbow, and wrist joints. Eventually, the primitive fish-tetrapod would have been able to crawl out of the water, where the lack of buoyancy would drive the evolution of limbs to support the weight of the animal's body. In turn, this would favor a stronger internal skeleton.

One such skeletal adaptation was the development of a rigid backbone, with sliding and overlapping processes to link adjacent vertebrae, while providing the necessary overall flexibility. In addition to securing support for the body, the hind limbs became the actual driving force for tetrapod locomotion. This required strong hip bones to anchor these rear limbs to the backbone. The fore limbs became the steering force, and so shoulder bones- which attach to the skull in fish-became detached in tetrapods, providing them with a flexible neck. Muscular changes accompanied these skeletal adaptations, promoting strong limb function and linking the hip and shoulder bones to the backbone.

Another requirement for the evolution of tetrapods, together with legs that work on land, is the ability to breathe air on land. Primitive air sacs existed in ray- finned and lobe-finned fish long before the first tetrapods appeared. In ray-finned fish, these air sacs were used as swim bladders to maintain buoyancy. In lobe-finned fish of the Devonian period, however, the air sacs had already evolved into air-breathing lungs. Lobe- finned fish, such as living lungfish, breathe by gulping air into their mouths at the surface of the water. When they subsequently dive, the increasing water pressure bearing onto them forces the air in the mouth down into the lungs. The process reverses as the fish rises to the surface again, forcing stale air out of the lungs and mouth as the fish gulps in a fresh mouthful. Since this basic mechanism for respiration already existed in Devonian lobe-fins, the earliest tetrapods were already well on their way to the evolutionary stage of breathing on land.

The transition from fins to limbs was only one part of the evolutionary journey for terrestrial tetrapods. Many other anatomical changes must have taken place to help early tetrapods cope with life on the land. For example, the evolutionary change that resulted in a flexible neck increased the mobility of the head, helping them pick up food instead of grabbing items that passed by in the water. No one knows exactly which evolutionary forces prompted the transition from fins to limbs. The first tetrapods were probably still fully aquatic animals that lived in shallow waters. Some experts think they may have used their primitive limbs to stalk prey through the dense vegetation that lined the bottom of rivers and swamps. Others believe that environmental pressures, such as an increasingly dry climate, prompted the transition from water to land. As the heat dried up the shallow waters, primitive limbs may have helped the earliest tetrapods crawl between pools of water. Yet another hypothesis is that they were chased out of the water by bigger predatory fish. Any tetrapod that could survive out of water would have found itself in a new paradise, with an abundance of plant and insect foods and a notable absence of dangerous predators.

Economic Growth in Europe's Smaller Countries

Much has been written about the economic growth of England, France, and Germany in the nineteenth century, but the economies of a number of smaller European countries also performed impressively in this period, especially those of Belgium, Switzerland, the Scandinavian countries (Sweden, Norway, and Denmark), and, to a lesser degree, the Netherlands. Still, these countries suffered from various handicaps, the first being their geographic smallness: their home markets were narrow, and they were relatively poor in mineral resources. Belgium alone had coal (and in plenty), so it was a leader in the application of electricity to industry from the 1870s onward, but Switzerland, Sweden, and Norway had a rich waterpower potential. In the latter three countries, much of the territory was mountainous or arctic in climate and therefore scarcely populated. In the early nineteenth century, some of these countries were quite poor. The Netherlands, meanwhile, retained high per capita incomes but suffered from a difficult transition from its seventeenth-century golden age of commerce to industrial capitalism.

On the other hand, these small countries developed their educational systems quite early, and their workforces were literate and skilled, while wages (except in the Netherlands) were low. Thanks to the accumulation of valuable knowledge and skills among the workforce, the abundance of either local or foreign capital (money and other assets), and other advantages, such as a quiet political life-with only one short war for each country, except Sweden, which had none-they succeeded in overcoming the obstacles to industrialization. Most of them had an industrial tradition, but any growth of industry had to be driven by exports, and they became heavily involved in international trade. This was particularly true in the case of Switzerland, even though it alone was landlocked and surrounded by neighbors that put up barriers to trade; up to 1914, it was the world leader in exports of manufactured goods per capita. These countries benefited from the demand by the large advanced countries of Europe-especially Britain-for their manufactures and primary products, but they also exported to distant markets like Latin America. The adoption of free trade (allowing trading across boundaries without government interference) by Britain and the general liberalization of trade after 1860 were beneficial, but in order to export, these countries needed to be competitive, and they achieved efficiency by a strategy of specialization adapted to their resources.

Belgium experienced the most complete industrial revolution on the European continent, with an emphasis on heavy industry and large exports of coal and semifinished and finished metal products. In Switzerland, on the other hand, there were mainly light industries making high-quality goods: luxury cottons, watches, machinery (at first for the textile industries, with the manufacture of electrical equipment becoming important later), fine synthetic chemicals such as dyes and drugs (for which the Swiss held second rank, far behind Germany), and processed foods.

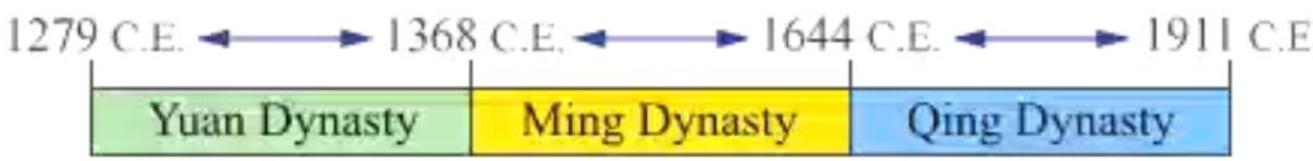
Sweden was rather a latecomer. For a long time, its exports were mainly of primary and semifinished products-bar iron, cut timber, paper pulp, and iron ore. From the 1880s onward, however, it developed a high- tech engineering industry, often on the base of Swedish inventions. This allowed it to become a large-scale exporter of various specialized products such as ball bearings, telephones, cream separators, and capital goods (machines for making products). Norway specialized in exports of fish, lumber, pulp, and paper, and in shipping services. Its industry was small, but electrochemistry was introduced in the 1900s with the help of French and Swedish capital.

Denmark's main export was initially grain, but in the face of competition its farmers reconverted to animal products: bacon, butter, cheese, and eggs-mainly for British breakfast tables. Danish agricultural productivity became the highest in Europe, making the country proof that industrialization is not the only road to wealth. The Netherlands also benefited from high-productivity agriculture and from its exports. Industry there, which had suffered in the wars of the early nineteenth century, had a slow but steady recovery, with large-scale industry emerging in the late nineteenth century.

Each country thus had its own specialized products and services. Their growth rates were fast. From 1870 to 1913, the Scandinavian countries outperformed the rest of Europe in the growth rates of individuals and achieved a spectacular convergence with the richest countries, especially in standards of living. Belgium and Switzerland had been "rich" for some time.

Chinese Salt Production and Government Power

By the year 2000 B.C.E. techniques had been developed in China for the production of salt. Archaeological evidence from northwestern China suggests that local inhabitants were by then using the technique of boiling water from saltwater lakes to extract its salt. In the region of Shansi and Shensi provinces in northern China, salt was sometimes even collected from evaporated lakebeds in the dry season. Inhabitants also learned to divert water from salt lakes into shallow holding pools, allowing the water to evaporate over a period of months. The natural salts would then be collected. Deposits of several million round-bottomed ceramic pots found in Sichuan Province in southwestern China suggest an organized system of salt processing by boiling during the same period. Scholars believe that the salt was then traded to the salt-poor regions of Hubei and Hunan. Similarly, evidence suggests that sometime prior to 2000 B.C.E. salt also began to be extracted directly from seawater. Over time, the coastal regions became the center of Chinese salt production, eventually providing roughly 80 percent of China's annual salt output. Given the increasing scope and value of salt in China, the control of its trade became a perpetual goal of the Chinese government and administration. The first formal attempt to establish the state's monopoly (exclusive control) over the production and distribution of salt goes back to the Eastern Zhou Dynasty (770-221 B.C.E.). The creation of this monopoly appealed to those in power because it provided the funds necessary to create a formidable standing army.



By the time of the Ming (c.E. 1368-1644) and Qing (C.E. 1644-1911) Dynasties, the government had developed intricate systems for the control of salt production and distribution. Beginning in 1370, the Ming government administered the salt monopoly through the "certificate system." Under this arrangement, each salt-producing household was required to provide an annual quota of salt to the government. In return the household was provided with a fixed payment to cover wages and expenses. The salt was then transported to one of over two hundred official salt depots (storage centers), which were located within fifteen salt administration regions. Salt merchants, for their part, received certificates giving them the right to transport and market salt from the government depots, but only after they had delivered grain to troops (military) serving on the northern frontier. This replaced the previous system, used by the Yuan Dynasty (c.E. 1279-1368), wherein merchants bid for the right to distribute salt. Thus, the Ming government guaranteed a supply of food for troops through the monopoly on salt. Salt was then delivered to local retailers and sold to the public. In the late Ming period, beginning in 1631, a new process known as the "syndicate system," which eventually created a hereditary class of salt merchants, placed salt distribution into the hands of a small group of families. This system lasted until 1831.

While both the certificate and the syndicate systems directed considerable amounts of money into the government treasury, they had significant drawbacks. First, the mandatory system of wholesale depots run by the government meant that salt had to be transported long distances even when there was a local market. Thus, salt might be produced in a region, transported to a depot far away, and then returned in the hands of licensed merchants to be sold to the public by local retailers. Such a process greatly elevated the cost of salt to the common buyer, and created a substantial demand for smuggling and even the raiding of vehicles transporting salt. It is estimated that anywhere from one-quarter to one-half of the salt consumed during the Ming and Qing periods was manufactured and sold on the illegal black market outside of state control, despite the continued efforts of the state to restrict such activities. The state derived considerable revenue from the control of salt, but it did so at the expense of the workers, who not only had to provide labor to the government, but who also paid elevated prices for a daily necessity of life. In turn, the people resisted the government's control over salt by creating an underground (unofficial) economy in smuggled and stolen salt. In effect, the state encouraged the populace to undertake an illegal trade in salt.

Temporary Pools

Temporary pools are freshwater habitats that retain water for only three to four months of the year or even shorter periods: typically for some eight or nine months they are dry basins without surface water. At the latitude of Canada's southern Ontario, for example, the basin of a typical temporary pool fills with water when snow melts in March and April. Because temporary pools (by definition)have neither a surface inlet nor outlet, the water steadily recedes through evaporation and seepage into the surrounding soil, disappearing altogether from the surface in June or July. Most of the animal residents one sees in temporary pools are aquatic insects such as caddis flies, whose larvae (wingless, immature insects)are sometimes the most conspicuous animals in these communities.

At least two advantages to living in this type of habitat can be recognized. One involves predation. Fish are major predators of aquatic insects and other small freshwater creatures; however, fish, at least in temperate regions over much of North America, are unable to survive the dry phase, and hence are eliminated from temporary pools. Predatory insects such as aquatic beetles and bugs and dragonfly nymphs do live in temporary pools, but there are far fewer species, and therefore a less efficient predator population than in permanent waters. Moreover, populations of the invertebrate predators in temporary pools have to develop anew every year from colonizing adults or from drought-resistant eggs. All of this means that the impact of predation is reduced for aquatic animals that have adaptations to survive the dry phase of temporary pools.

A second advantage is concerned with nutrition and growth. Temporary pools, especially in wooded sites, are fueled primarily by the energy and nutrients stored in fallen leaves and other decomposing plant detritus (nonliving matter that originates from plants).Furthermore, the dry phase of temporary pools usually supports a community of specialized herbaceous plants that can tolerate springtime flooding but grow rapidly in the moist, organically enriched soil after surface water disappears. These plants contribute to the detrital base every year. Although the food web is based mainly on detritus, the processes of decomposition in temporary pools are somewhat different from those in the detritus- based communities of streams and the shore zones of lakes. Detritus decomposes faster when exposed to air during dry periods and has a higher protein content upon flooding in spring than when submerged continuously in permanent water. The reason for this difference is that the terrestrial fungi causing the decomposition of plant detritus in the absence of surface water flourish where oxygen is not limited. In permanent waters, where detrital materials are covered by water, oxygen is not plentiful and decomposition is brought about by aquatic fungi and bacteria, which do not grow as abundantly as terrestrial fungi. Colonizing terrestrial fungi contribute much of the protein of decomposing plant materials, and in feeding experiments limnephilid caddis fly larvae of temporary pools showed a preference for decaying leaves with higher protein levels. Consequently, rapid development, which is so important to animals in transient waters, would be enhanced by the protein-rich detritus as well as by the higher summer temperatures of small bodies of water.

These principles have led to some practical applications. An increase in productivity following a dry phase underlies the practice in wetland management of alternately draining and flooding wetlands. Aquatic birds such as ducks feed on larval caddis flies and other aquatic invertebrates and also on plants; the increased nutrient levels resulting from exposure of more of the bottom detritus to air and terrestrial fungi enhance productivity throughout the food web. By the seventeenth century, fish farmers in Europe had learned that periodically leaving ponds dry and empty of fish yielded higher productivity in controlled ponds of carp fish than did constant water levels. Through manipulation of water levels, the ponds were wintered as dry basins and periodically over a full year as well-practices that led to increased production of fish. In retrospect then, it appears that after centuries of experimentation to increase productivity in marshes and fish ponds, humans succeeded in duplicating the natural cycle of temporary pools.

Seasonal Bird Populations

Humans have hunted birds for their meat and harvested their eggs since earliest times, and therefore humans have an interest in understanding the seasonal movements of bird populations. Birds are regarded as resident when they spend all year in the same area, and migratory if they move seasonally between the breeding and nonbreeding areas. Most birds of the tropics and some temperate species(species that live in areas with moderate temperatures) are resident, but many mid- latitude species, and most which breed in high latitudes, are migratory. The most common reason for an annual migration is to allow the bird to occupy a part of the world that provides food at only one time of year. In continental North America and Eurasia, insects are available only in summer in northern forested regions, so insect-eating birds migrate south in winter. Other reasons for migration include the weather, which becomes too cold or too wet at certain times of year, and competition from other species for food or nest sites. Nearly all migratory species are from the northern hemisphere, partly because seasons there are more distinct than in the southern hemisphere. Current migration patterns of northern hemisphere birds have evolved only in the past 10,000 to 5,000 years, since the end of the last Ice Age. They are thought to have been preceded in the Pleistocene (2,500,000 to 11,700 years ago)by shorter migrations because the temperate climatic zone at that time was more compressed geographically.

Many of the waterfowl today breed in the tundra of northern Eurasia, northern Canada and Alaska, Iceland, and Greenland, where the vegetation and insect life can support the birds during what is quite a brief breeding season. Geese, which breed up to the edge of the Arctic, migrate south in winter to escape frozen water; they remain in the northern hemisphere, but in milder regions. For example, the brant(brent goose)in North America migrates from Alaska and northern Canada to the Pacific and the Atlantic coast, and in Eurasia it migrates from Siberia to the coasts around western Europe. Many of the plovers follow a similar strategy, breeding inland and north, but returning to more southerly latitudes and to coastal locations in winter. Birds of these families are most easily caught on migration or in winter. Migration in seabirds takes a different form. Many come to land to breed but spend the rest of their life cycle at sea.

Even birds that do not migrate change their social behavior in winter. Many of the species that congregate and breed colonially (in large groups) become solitary in winter and occupy a single territory. Others congregate in flocks at that time. They also become more mobile, and both are adaptations to less predictable food supplies: a tree that may be full of berries one winter may have few or none the next. Hunters often target flocks rather than solitary birds, so hunting of those species that congregate is often a winter activity. Some species may be both migratory and sedentary. The woodcock, for instance, has a sedentary British population, which is enhanced in winter by migrants from Scandinavia.

From the point of view of humans who live in one place, certain birds are present in one season only, either in summer to breed or in winter; or they are passage migrants, appearing for a short time in spring and autumn during temporary stops to feed and rest while en route between summer and winter territories. Migratory birds can be a reliable seasonal source of food for people because they return each year at a predictable time but then leave, thus preventing numbers from being reduced too much by predation. In the main, migration routes are fairly fixed. For instance, many ducks follow the shoreline, and other species follow major rivers. Some avoid crossing wide expanses of water, so they concentrate at narrow crossing points such as Gibraltar at the entrance to the Mediterranean and Falsterbo at the mouth of the Baltic. Migrating birds store fat reserves under the skin, an adaptation to the journey to be taken, so many are at their most desirable from the point of view of human hunters as they start the migration.

Hunting the American Bison

© Photo by Jim Carr, U.S. Fish and Wildlife Service

When nonnative settlers arrived on the flat, treeless North American plains in the nineteenth century, American bison-a large herd animal-thrived despite hunting by Native American peoples. Scholars once assumed that Native American cultural values and hunting practices were designed to limit demands on natural resources. Native Americans harvested only around half a million bison every year for domestic consumption (a sustainable figure, given that it was such a successful species). Almost all parts of the animal had value. Most organs could be used for food, skins for shelter and clothing, and bones for tools. The Blackfeet (a Native American tribe) obtained more than 100 specific items from the bison that sustained daily life. But in emphasizing Native American ingenuity in turning a variety of bison parts into useful products, scholars overlooked considerable evidence of waste and excessive killing.

In recent years, the role played by Native Americans in the near extinction of the bison has been more closely studied. Archaeological evidence has revealed that traditional methods of hunting, such as forcing bison off cliffs, could be remarkably wasteful. For instance, one such site in southern Colorado contained the bones of some 200 bison (males and females, adults and juveniles), but over 40 animals at the bottom of the pile had either not been touched or been only partly butchered. But by the late eighteenth century, the adoption of the horse and the rifle had made Plains Indians (Native Americans who lived in the plains)more effective and selective hunters. They preferred to kill female bison because their meat was juicier and their skins were easier to work. After 1700, almost all Plains Indians had become involved in commercialized hunting, exchanging bison tongues and skins for European trade goods and horses. Native Americans began to slaughter the bison for their tongues and skins alone, leaving the rest of the body to rot. Between the 1830s and the 1860s, as demand increased, they supplied firms such as John Jacob Astor's American Fur Company with over 100,000 bison robes annually. In combination with environmental factors, especially lack of rain, the Plains Indians' preference for killing females (thus limiting the herds' reproductive capacity), the competition for grazing on prairie grasses from their horses, and the pressures of hunting both for their own use and for the market began to speed the decline in bison numbers.

Contrary to the old belief, there were elements in Native American belief systems that were also destructive to the bison. The Plains Indians believed that when the great bison herds dispersed during the winter months, they went to grasslands underground, reemerging every spring in inexhaustible swarms "like bees from a hive." If bison herds did not reappear in expected numbers, it was because many had not yet left their underground prairies. Such a notion undoubtedly discouraged the conservation of what was a diminishing resource, as Plains people could still hunt in enormous quantity, secure in the knowledge that there were always more animals grazing down below. Today, few historians would accept uncritically the idealized image of the environmentally friendly Native American who consciously pursued a lifestyle that left few traces on the land. But the pressure that Plains Indians exerted on the bison were slight in comparison with that exerted by Euro-American market hunters.

The bison population collapsed rapidly from the mid- nineteenth century. The demand for bison skins for the tanning industry, which turned animal skin into leather, was the primary cause. In 1850, when tanners still mostly used the skins of cows, tanning was already the fifth-largest industry in the United States, and it grew in value by supplying markets at home and overseas with a variety of leather products-most importantly, the heavy belting that powered machinery in mills and factories. Demand outstripped the supply of cowskins in the United States, forcing tanners to import them from Argentina and elsewhere in Latin America. But in 1870 a process for transforming bison skin into tough, elastic leather was discovered by tanners in Philadelphia, making bison hide an ideal material for use as industrial belting. Thousands of Euro-American commercial hunters flooded onto the plains, killing around 2 million bison for their skins annually between 1870 and the early 1880s. At the same time, the extension of the railroads to the plains made transporting bison skins to tanneries both quick and inexpensive.

In the Darkest Depths

Although scientists in the early 1800s were unable to explore the deepest regions of the ocean, they Knew that sunlight could not penetrate depths greater than 275 meters. Without sunlight, there can be no photosynthesis (the process by which plants convert sunlight into energy), and therefore no plants or algae to serve as food for other organisms. Since ocean depths can exceed 10,000 meters, scientists hypothesized that the deepest areas of the ocean could not support life.

Throughout the nineteenth century, scientists exploring the ocean collected organisms from ever greater ocean depths. In an 1873 expedition, researchers aboard the ship HMS Challenger dragged an open-sided box suspended from the ship across the floor of the Atlantic Ocean. This box -known as a dredge- samples the sea floor in different parts of the ocean at depths of up to 4,57 meters. The scientists were astonished to discover nearly 5,000 previously unknown species. When it became clear that life flourished at depths beyond the penetration of light, scientists were forced to reject their earlier hypothesis that no life existed in the deep ocean waters. After discovering this rich abundance of deep-sea life, scientists were faced with the need to understand how it could exist. The lack of light suggested that deep-sea organisms were somehow sustained by energy that did not come from photosynthesis on the ocean floor. Scientists had observed that the surface waters of the ocean produced a steady descent of tiny particles that were produced by the death and decomposition of organisms living there. These particles are known as "marine snow". Large organisms such as whales also occasionally died and fell to the ocean floor. Scientists hypothesized that marine snow and the remains of large organisms provided the energy needed to sustain organisms in the depths of the ocean.

In the 1970s, scientists were finally able to send submersibles -small manned submarines- to take a first-hand look at the deepest ocean areas. Their discoveries were striking. They not only confirmed that much of the ocean floor supported living organisms but that areas near openings in the floor of the ocean, which later came to be known as hydrothermal Vents, contained a great diversity of deep-sea species. Hydrothermal vents release plumes of hot water with high concentrations of sulfur compounds and other mineral nutrients. A tremendous diversity of species surrounded these hydrothermal vents. Indeed, the total amount of life at these depths rivaled that seen in some of the most diverse places on Earth. It became clear that the amount of energy contained in organic matter resting on the sea floor was not sufficient to support such a diverse and abundant set of life forms. That earlier hypothesis now had to be rejected.

How could so much life exist at the bottom of the ocean? That this life existed near the hydrothermal vents suggested that the vents were somehow responsible. Scientists had known for a long time that some species of bacteria could obtain their energy from chemicals rather than from the Sun. The bacteria use the energy in chemical bonds, combined with carbon dioxide (CO2), to produce organic compounds -a process known as chemosynthesis- similar to the way that plants and algae use the energy of the Sun and CO2, to produce organic compounds through photosynthesis. Based on this knowledge, scientists hypothesized that the hot vents, which release water With dissolved hydrogen sulfide gas and other chemicals, provided a source of energy for bacteria and that these bacteria could be consumed by the other organisms living around the vents.

After several years of investigations, scientists found that the immediate area around the hot vents contained a group of organisms known as tube worms. These animals have no digestive system, but possess specialized organs that house vast numbers of bacteria that live in a symbiotic (mutually beneficial) relationship with the tube worms. The tube worms capture the sulfide gases and CO2; from the surrounding water and pass these compounds to the bacteria, which then use the sulfide gases and CO2 to produce organic compounds. Some of these organic compounds are passed on the tube worms, which use them as food. These bacteria also represent a food source for many of the other animals that live near the vents. In turn, these bacteria-consuming animals can be consumed by larger animals.

Early Native American Cultures

In the seventeenth century, the gulf separating European colonists in North America from those who were native to America was defined not only by their material cultures, but also by how they viewed their relationship to the environment and how they defined social relations in their communities.

Regarding the soil as a resource to be exploited for human benefit, Europeans believed that land should be privately possessed. Individual ownership of property became a fundamental concept, and an extensive institutional apparatus grew up to support it. Fences symbolized private property, inheritance became the mechanism for transmitting land from one generation to another within the same family, and courts gained the power to settle property disputes. Property was the basis not only of sustenance, but also of independence, material wealth, political status, and personal identity. The social structure in Europe directly mirrored patterns of land ownership, with a land-wealthy elite at the apex of the social pyramid and a mass of individuals lacking property forming the broad base.

Native Americans also had concepts of property, and tribes recognized territorial boundaries. But they believed that land was invested with sacred qualities and should be held in common. Communal ownership sharply limited social layers in most Native American communities. Accustomed to wide disparities of wealth, Europeans often found this remarkable. Not all Europeans were wealthy, competitive individuals. The majority were peasants scratching a subsistence living from the soil, living in kin-centered villages with little contact with the outside world, and exchanging goods and labor through barter. But in Europe's urban centers, a wealth-conscious, ambitious individual who valued and sought wider choices and greater opportunities to enhance personal status was coming to the fore. In contrast, Native American traditions stressed the group rather than the individual. Holding land and other resources in common, Native American societies were usually more egalitarian and their members more concerned with personal courage than personal wealth.

European colonizers in North America also found the matrilineal (female-based) organization of many tribal societies odd and unfamiliar. Contrary to European practice, family membership among the Iroquois, for example, was determined through the female line. A typical family consisted of an old woman, her daughters with their husbands and children, and her unmarried granddaughters and grandsons. When a son or grandson married, he moved from this female-headed household to one headed by the matriarch of his wife's family. Divorce was also the woman's prerogative. If she desired it, she merely set her husband's possessions outside the door to their dwelling. European women, with rare exceptions, were entirely excluded from political affairs. By contrast, in Native American villages, again to take the Iroquois example, designated men sat in a circle to deliberate and make decisions, but the senior women of the village stood behind them, lobbying and instructing. The village chiefs were male, but they were named to their positions by the elder women of their clans. If they moved too far from the will of the women who appointed them, these chiefs were removed.

The role of women in the tribal economy reinforced the sharing of power between male and female. Men were responsible for hunting, fishing, and clearing land, but women controlled the cultivation, harvest, and distribution of food. They were responsible for probably three-quarters of their family's nutritional needs. When the men were away on hunting expeditions, women directed village life.

In economic relations, Europeans and Native Americans differed in ways that sometimes led to misunderstanding and conflict. Over vast stretches of the continent, Native Americans had been involved in trading networks for centuries before Europeans arrived. This experience made it easy for them to engage in trade with arriving Europeans and incorporate new metal and glass trade items into their culture. But trade for Native American peoples was not only a way of acquiring desirable goods. It was also a way of preserving interdependence and equilibrium between individuals and communities. This purpose displayed itself in elaborate ceremonies of gift giving that preceded the exchange of goods. For Europeans, trade was an economic matter with the additional benefit of building good will between two parties, but the social and spiritual functions of trade were sharply limited in comparison with Native Americans.

The Waterfall Environment

Waterfall

Waterfalls can act as barriers that pose a number of challenges to plants and animals. There is evidence that waterfalls can so isolate the upper parts of rivers that distinct varieties or even unique species of fish may evolve in the river on each side of the waterfall. Waterfalls commonly form obstacles to migration but, as the well-known example of salmon testifies, some creatures are able to overcome low waterfalls and rapids. particularly where the descent is broken into separate waterfalls forming a natural staircase. While no fish can swim against falling water, many jump effectively. Atlantic salmon can clear over three meters. To enable it to jump higher, the salmon leaps from the peak of the standing wave that commonly forms near the foot of a waterfall. Here there is an upward component in the turbulence which assists the fish in its leap. To facilitate the movement of fish upstream, fish passes, which are artificial stepped channels bypassing waterfalls, have been built on many rivers, while in some places the waterfalls themselves have been deliberately modified or destroyed for this purpose.

When descending rivers, fish may have no difficulty dealing with waterfalls and rapids, particularly where the drop is small enough for them or their parents to have ascended. Here water acts as a cushion and lessens the effect of the fall, helping prevent damage. Waterfalls can be destructive of aquatic life, however. Some researchers have found that many species of plankton, minute water animals and plants usually living in colonies, are eliminated by waterfalls and rapids. Others have shown that some species can survive the descent of steep rapids and even major waterfalls.

Swimming and jumping are not the only means by which fish can overcome the barriers imposed by rapids and waterfalls. Species with suckers (organs that help an animal cling to a surface) can often ascend wet vertical rock faces beside or beneath falling water in stream courses. Some fish belonging to the large family of gobies have the ability to climb waterfalls in this way. Other fish that have the ability to overcome waterfall barriers include young eels, which make use of wet areas beside the stream to bypass falling water.

While waterfalls often act as barriers to movement and form obstacles in the stream that separate parts of the stream where life can be readily sustained, they provide a habitat to which some species are well adapted. There are plants and other organisms that survive and even flourish in the fast-flowing, turbulent water of waterfalls and rapids. Its waters well supplied with oxygen by numerous falls, the tumbling stream provides an ideal environment for the production of chemical energy by living organisms in the form of plants such as algae, so long as there is adequate sunlight for photosynthesis. Particularly important are the small plants which grow closely attached to other plants, rocks, stones, and dead branches in the stream. These form a dense, slimy surface of algae or mosses, which are the most important photosynthetic organisms in running streams. The highly productive level at which living material is generated is due largely to the rapid current. This prevents the development of a layer of nutrient- and carbon dioxide-depleted water that would otherwise build up around the plants, as they absorb nutrients and use carbon dioxide for photosynthesis. Such a layer would reduce photosynthesis. Within the mass of algae and other plants lives a very rich collection of small animals, such as water mites and insect larvae. Clinging to solid holds in the tumbling flow and in the splash zones of waterfalls, these communities of plants and tiny animals flourish, providing food for the rest of the inhabitants of the stream. Many of the latter, however are unable to live in the fast-moving, turbulent waters of rapids and waterfalls, except for some species that pass through them for short periods on their way to more favorable environments. Indeed, the animals that are commonly found in and around fast-moving streams and waterfalls normally take advantage of any available protection from the powerful currents, finding refuge behind or beneath rocks, or in gentle currents and quiet pools outside the main flow.

Easter Island's Statues

© istockphoto.com/tankbmb

Easter Island, a small, isolated island in the South Pacific Ocean, is ringed by monumental statues known as moai. Researchers have long wondered how the islanders were able to construct and transport the statues using only the resources available to them. The statues range from 6 feet to almost 33 feet high, but there is a standard style and shape: a long human head and torso with prominent chin and stretched earlobes, arms held tightly at the sides and hands resting on the stomach. To some of the statues were added eyes of red and white stone and coral and red stone pukao (topknots) on top of the head, which could represent hair or the red feather headdresses seen by European visitors in the early eighteenth century. Some 230 of the statues were once set upright on platforms, some with multiple statues in a row

There are questions about how the statues were carved, how they were moved, and how they were put in place. Locating the source of the stone used to make nearly all the statues is no great achievement, for it forms an impressive monument in itself. The stone quarry inside the old volcano at Rano Raraku is an extraordinary sight, with hundreds of carved-out niches left behind after the finished statues were removed and nearly 400 mostly uncompleted examples.

As for carving the stone, the surface of the volcanic rock is quite hard when it is weathered. Yet once this surface crust has been broken through, the rock beneath is only a little harder than chalk and can easily be shaped, helped along by softening it with water. This difference between the surface and the inside of the rock led some early explorers to conclude mistakenly that the statues had a hard pebble coating and a soft interior of clay. The tools used to carve and free the statues from the ground were undoubtedly the pointed picks of hard stone discarded in vast numbers at the quarry. In a well-known experiment, Thor Heyerdahl, leader of the 1955 Norwegian archaeological expedition to Easter Island, arranged to carve the outline of a statue at Rano Raraku. Six men hammered away with stone picks for three days, wetting the rock as they worked, at the end of which they had produced the outline of a statue some 16 feet long From this, Heyerdahl estimated that six men could have carved the whole statue in about a year.

Once the giant statues had been freed from the rock, some of them were transported to their eventual resting places on platforms up to six miles away, along the tracks that radiate out from Rano Raraku, although the larger the statue, the shorter the distance it was moved. This was not necessarily because of their weight but more likely because of the fragility of the carved statues. The largest statue to be transported is that known as Paro, a giant 32 feet tall and over 80tons in weight, which was shifted some four miles across rough country.

The early explorers who assumed that the island had always been treeless were completely baffled as to how the statues could have been transported without the help of wooden levers and rollers. However, archaeologists have been able to show that the landscape of Easter Island was once very different. By analyzing the pollen deposited by vegetation in the three lake beds on the island, they have drawn up a picture of the changing environment, confirming the idea that the island was once forested, with palm pollen being the dominant type. Therefore, trees or ropes could have been used to move the statues.

By attaching ropes to the head and base of a 13-foot statue, Thor Heyerdahl used a fifteen-man crew to move the statue by turning it on its base when it was tilted forward. They moved it only a few yards but damaged the base in the process. Geologist Charles Love placed a statue on a small wooden platform and ran it across wooden rollers. Twenty-five men managed to move the statue 150 feet in only two minutes, but misplacing the rollers caused the statue to come crashing down

Mire Biology

Mires are wetland ecosystems comprising bogs, marshes, and similarity swampy areas. They provide ideal conditions for the formation of the partially decomposed vegetation known as peat. One aspect of the biology of mires that has long puzzled ecologists is why so many of the plants, particularity the small woody shrub species, appear to be so well adapted to drought. Although some peatlands become dry on their surfaces during particularity hot and dry summers, drought is not usually long lasting, nor is it usually severe. Yet many of the dwarf shrubs of peatlands, especially those in the temperate zone, have small and leathery rolled, evergreen leaves, which are more often associated with the vegetation of dry conditions, such as chaparral and Mediterranean scrub. These structural characteristics associated with drought in plants are termed "xeromorphic".

Courtesy of National Park Service

A Labrador Tea Plant

In the arctic mires and many northern temperate peatlands, the black crowberry is a common member of the surface vegetation. This low-lying shrub is evergreen and has shiny, almost cylindrical leaves. On close inspection it can be seen that the leaves are actually tightly rolled in upon themselves, leaving only a very narrow, white-colored strip on the lower side where a thin gap leads to the delicate undersurface of the leaf. The upper part of the leaf is glossy with the presence of wax, which protects its surface cells from desiccation (drying up). The Labrador tea has large and leathery, evergreen leaves, which are not tightly rolled (although they do curl inward at their edges) but have dense masses of hair on the leaf undersides that prevent air movement and thus minimize the plant's evaporation and water loss. The leatherleaf, another woody shrub of peatlands, has quite narrow evergreen leaves whose undersides are covered with small scales that serve a similar purpose of reducing evaporation.

In fact, almost all of the small, woody plants of temperate peatlands exhibit some degree of adaptation to drought, and this type of structure appears, at first sight, most inappropriate for such wet habitats. Plant ecologists and physiologists have developed several theories to account for the phenomenon, and it is entirely probable that a number of different causes lie behind this surprising aspect of mire vegetation. Perhaps the most obvious explanation is that the plants of the high latitude peatlands experience physiological drought, which means that although water may be present in abundance in the environment, it remains unavailable to the plant, which therefore suffers drought. The water is unavailable because of its very low temperature (or even its frozen state) in winter. Some experiments have shown that plants reduce their water uptake when the soil they are grown in is cooled, but this is not universally the case. It could be argued that under cold conditions the transpiration rate (the rate at which the leaves lose water vapor through pores on their surfaces) would be low anyway, so reducing evaporation in winter is not a problem. The idea that physiological drought can account for the xeromorphic characteristics of wetland plants is no longer strongly favored.

An alternative explanation, currently receiving considerable support from experimental work, is that reduced transpiration from leaves can cut down the quantity of toxins absorbed from the wetland soils. The main toxins proposed are iron and manganese. These elements are needed in small quantities by plants but in excess can prove harmful. Both elements are present in mire waters, especially under acid conditions when they become more soluble. The toxicity problem is overcome in part by the diffusion of oxygen from wetland plant roots. Oxygen passes from the atmosphere, through the plant, and leaks from the roots, where it intercepts iron and manganese, forming oxides that precipitate (condense) in the soil and therefore do not enter the plant. A fast rate of transpiration, however, can result in a greater inward movement of these toxins, leading to their accumulation in the leaves and causing death in sensitive species. In one very informative experiment, a sensitive species, bell heather, was sprayed with silicone to prevent transpiration, following which there was less uptake of iron, and the plant survived. So, the xeromorphic adaptations of these wetland species may simply be a means of protecting themselves from accumulating unwanted metals in their tissues.

Modern Inuit Commercial Arts

The Inuit are a group of culturally similar peoples who live in some areas of the Arctic. Before the nineteenth century, European explorers and other early nonnative visitors in the Arctic collected souvenirs of their travels to Inuit communities. But the commercial market for Inuit art likely began in the latter half of the nineteenth century with the large collecting expeditions mounted by, and on behalf of, museums in the United States, Canada, and Europe. Inuit were encouraged to sell the sacred and everyday objects which they supposedly no longer needed in modern industrial societies and which ethnologists and museum curators assumed would disappear. In Alaska, the arrival of large numbers of people looking for gold(1899-1909)produced a steady market for Inuit arts in North Alaska. Both on their own and with the encouragement of Christian missionaries and government administrators, Inuit in many different regions produced replicas and miniatures of traditional items to sell to the growing number of nonnative visitors. Contemporary Inuit commercial arts developed out of these early marketing activities.

The art forms most often associated with Inuit, such as tupilak figurines, baleen baskets, and caribou-skin masks, are all recent forms produced for sale rather than for domestic use. For the most part, they originated as tourist arts. It is a mistake, however, to discount their authenticity on that basis. Nor is it correct to assume that the quality is poor. Artists of all cultures and ethnic groups make their art for the market, and Inuit artists are no exception. Today, art is just one component of the mixed economies of Inuit communities. Through the materials employed and the motifs and symbols presented, Inuit art is very much representative of contemporary Inuit culture.

Tupilak figurines are small, grotesquely shaped sculptures carved of animal tooth or bone by Greenlanders. Meant to represent agents of malicious magic or supernatural power, tupilaks were first produced as commercial art in the 1930s. According to Greenlandic anthropologist Robert Petersen, the sculptures that are sold as tupilaks did not exist prior to that time. Rather, tupilaks existed in the imagination and were brought into being through stories. Physical models of tupilaks were carved for the benefit of Danes and other Europeans who could not otherwise understand what these evil beings looked like. European purchasers proved to be most interested in the most grotesque and bizarre images, and thus tupilak figures became more monstrous and alarming in response to the market.

Baskets of baleen, the plastic-like substance with which bowhead whales filter plankton(small organisms)from seawater, are an art form of the Inupiat (one of the Inuit peoples)dating from around 1915. The first baleen baskets were produced in Barrow, Alaska, at the instigation of American whaler-turned-trader Charles Brower. These small, lidded baskets are woven of coiled baleen and topped with a carved ivory piece, usually in the shape of an animal such as a seal. Disks of ivory are used as starter pieces for attaching the first coils of the base and lid. Brower is reported to have asked an Inupiat named Kinguktuk to make baskets similar to willow-root baskets from baleen that Brower might give as gifts. This art form did not attract other weavers until the 1930s when the fur market, which had replaced commercial whaling as an income generator for the Inupiat, collapsed. Basket making then provided an alternate source of income.

The caribou-skin masks from the village of Anaktuvuk Pass, Alaska, originated in an entirely different way. Under the influence of Christian missionaries, Inupiat stopped making masks for religious purposes prior to 1900, but it is also the case that the Inupiat residents of Anaktuvuk Pass did not use masks historically. Caribou- skin masks are a modern creation unrelated to any traditional Inuit religious practices, but they were initially created by a pair of Inupiat hunters for local use. Two Inupiat hunters made the first caribou-skin masks in 1951 as part of costumes they wore to the community Christmas celebration. As part of the holiday festivities, performers disguised themselves and clowned to amuse fellow residents. Several years later, a market for caribou- skin masks developed, and other residents of Anaktuvuk Pass began to experiment with mask-making techniques.

Forest Management in the Blue Mountains

The Blue Mountains of the western United States stretch from southeast Oregon to northwest Washington State. When Euro-Americans first came to the Blue Mountains in the early nineteenth century, they found a land of open forests full of large ponderosa pines-fire-resistant. insect-resistant forests that had been growing for centuries. But after less than ninety years of management by federal government foresters, the great ponderosa pines vanished. In their place were thickets of dying fir trees that were first attacked by defoliating insects and then devastated by intense fires.

In part, the landscape changed for straightforward ecological reasons. When foresters suppressed fires in open, semi-arid disturbance-prone forests dominated by ponderosa pine, firs grew faster than pines in the resultant shade-a change that left firs dominating the forests. Heavy grazing (animals' feeding on grass) which eliminated the grasses that had previously suppressed tree regeneration and kept forests relatively open, also contributed to the change. High grading-a logging method that removed old pine while leaving firs behind-further encouraged the replacement of open pine forests with dense fir stands. When droughts later hit, firs growing on dry sites succumbed to insect epidemics

But the real story is much more complex than this, since changes in the land are never just ecological changes; people made the decisions that led to these ecological changes, and they made those decisions for a complex set of motives. Federal foresters came to the Blue Mountains with the best of intentions: to save the forest from the scourges of industrial logging, fire, and decay. When they looked at the Blue Mountains, they saw two things: a "human" landscape in need of being saved because it had been ravaged by companies and the profit motive and also a 'natural" landscape that they felt needed saving because it had become decadent, wasteful, and inefficient. Not only were federal foresters going to rescue the grand old western forests from those who exploited timber resources, they were going to make them better. Using the best possible science of the day, foresters felt they were going to make the best possible forests for America in the twentieth century.

Industrial logging had been underway for less than a decade, and in that short time enormous changes were already beginning. Government foresters firmly believed that these industrial cutting practices produced sterile lands and, in particular, those practices destroyed the vegetation cover that protected the water supply in the arid west. Young trees were critical to the future of the forest, and industry seemed to have sacrificed them for short-term profits. Scientific forestry, the foresters felt, would change everything

Two major interrelated tenets of scientific forestry developed First, foresters felt they should encourage the growth of young trees by suppressing fire; and second, foresters felt they should replace old growth with regulated, rapidly growing forests. Fire seemed to threaten the forests by killing young trees, and since foresters were certain that young trees were the future of the forest, fire was clearly the enemy. The foresters thus decided that to protect the pine forests and the water supply, they needed to keep out fire and encourage reproduction. The federal government's Forest Service was convinced that the more young pines they had, the more marketable pine would necessarily follow.

Although it was clear to early foresters that suppressing fire would lead to dense thickets of young trees, they felt that this would be a good thing, for reasons that show how their cultural beliefs affected their ecological reasoning. The early twentieth-century United States was a culture that glorified competition. It was commonly believed that competition would produce stronger individuals by eliminating the weak in favor of the mighty. This feeling was encouraged by emerging industrial capitalism and then projected onto the landscape. Foresters reasoned that dense stands of young trees would lead to intense competition for light and water, and that competition in the forest economy, just as in the industrial economy, would create vigorous individuals. Without competition, weaklings would result-or so the foresters reasoned The opposite turned out to be true, unfortunately. Western conifers --trees with cones and needlelike leaves-do not thin themselves; without fires to thin them, what resulted were not a few big trees but thickets of stunted trees all the same age.

The Horse and the Camel in Early China

During the Han Dynasty (206 B.C.E.-C.E. 220),China began to trade with lands to its west over the Silk Road, a set of trade routes stretching all the way to the Mediterranean Sea. While some goods reached China from as far away as Rome, many valuable items originated in neighboring Central Asia. Perhaps the most consequential Central Asian goods were able to travel into China under their own power: horses and camels. China required a constant supply of horses for its cavalry, having learned a painful lesson in the importance of mounted horsemen from the Xiongnu, a Central Asian people whose superior horsemanship had once allowed them to invade western China. While it was difficult for the Chinese to breed their own horses, possibly due to the lack of calcium in Chinese soils and water, Central Asia had ample grasslands that were well-suited for producing horses of the highest quality. The Chinese thus traded the silk produced by their silkworms for Central Asian horses, sending tens of thousands of bolts of silk westward each year. Much of it went to the Xiongnu court, where this and other Chinese luxury goods were highly valued.

In some cases, horses arrived not as objects of trade but as spoils of war. In the early days of the Silk Road, some of the most prized horses were those from the Ferghana valley in present-day Uzbekistan. In 102 B.C.E. the Emperor Wu sent an enormous military expedition to Ferghana, capturing over 3,000 horses and forcing the defeated kingdom to deliver two additional horses to China every year. Historians have traditionally believed that these special horses were acquired for their superiority in the cavalry. Yet one historian, Arthur Waley, has maintained that the Chinese rulers desired Ferghana horses for spiritual, rather than military, reasons. At the time of Emperor Wu's expedition, many Chinese believed in the existence of water dragons that appeared in the form of special, divine horses and often retained their dragon wings. The association of the Ferghana horses with these imaginary beasts gave them a special status.

The unprecedented prosperity of the Chinese Tang Dynasty in the 7th century led to greater levels of Chinese trade along the Silk Road and a dramatic increase in the number of foreign merchants, religious pilgrims, and other visitors to China. The large number of rich Persian merchants is suggested by the fact that the proverbial phrase "a poor Persian" was used by some Chinese to indicate an inherent contradiction. Tang China became famous for its openness to new products, ideas, and fashions. From Persia, for example, came the game of polo, in which players on horseback attempted to strike a ball through a goal using long wooden sticks. Polo, which was played by both men and women, was so popular for a time that the imperial palace had a field dedicated exclusively to the sport. Some types of horses were better suited to polo than others, and in c.E. 717 special polo ponies were imported from the kingdom of Khotan in what is now Xinjiang province. Overall, the imperial stables held some 40,000 horses for games and war.

Horses were not the only Central Asian product to figure in the cosmopolitan Tang Dynasty society. Perhaps the most representative import was the camel, which had been used there for transporting goods for thousands of years and was crucial to the functioning of the Silk Road. Aside from the ability to withstand long periods of time without drinking water, camels could survive on the scrub and thorn bushes found in the arid regions of the caravan routes, and they could transport huge loads of up to 500 pounds(about 227 kilograms). The demand for camels in Tang China grew to enormous proportions; they were used not only for transportation, but also for their valuable hair, which was woven into cloth, and sometimes for meat. Many of the colorful ceramic statuettes produced during the Tang depict camels carrying riders with beards and large noses, details that seemed exotic to the Chinese. Camels remained a common sight in some parts of China all the way into the 20th century.

Silver in North America in the Late Eighteenth Century

Silver and silver objects had multiple meanings in late-eighteenth-century North America. Insights about social attitudes and the buying habits of consumers come partly from the records kept by silversmiths such as Paul Revere, the most politically active, artistically innovative, and successful silversmith of the period. His customers came from a wide range of socioeconomic groups—artisans, middle-income merchants, the business elite, and government appointees. Revere produced more low-end goods—eating utensils, shoe or knee buckles, buttons, and harness fittings—than high-end. The elite preferred to buy more elaborate silver goods from England or other European countries. Much of his success after the American Revolution (1776-1783), however, lay in his ability to respond to demands for specific kinds of more elaborate silver objects, in particular sugar bowls, creamers, and teapots. One expert notes that after the Revolutionary War, Revere dominated silver teapot production, taking advantage of a return to popularity of a beverage that earlier symbolized a resistance to English taxes and the consumption of which had thus diminished immediately prior to and during the war.

Before the war, inhabitants of the American colonies had preferred to purchase silver as single pieces, thus slowly acquiring complete tea sets; after the war the purchase of larger tea services as a complete set became popular. Revere's production of such en suite tea services was facilitated by the availability of rolled sheet silver, which was a more economical method of producing silver vessels, in particular fluted teapots—which have grooves or ridges—like the one at the center of the set he made for a wealthy Boston merchant and his wife, John and Mehitable Templeman, in 1792. The rolled sheet metal also made it easier to achieve the fluting and straight edges that mark these objects. Their clean lines and finely engraved designs of draped fabric and tassels recall neoclassical forms (inspired by ancient Greek and

Roman art) that were then popular in much of Europe and associated with newly emerging republics on both sides of the Atlantic. These neoclassical attributes, found on all manner of goods and buildings in the immediate post-Revolution period, were the hallmarks of what became known as the Federalist style. Revere also engraved Templeman's initials—JMT—on the spoon handles, stands, and sugar urn, reinforcing the association between these objects and their owners, and serving as a kind of insurance, making it easier to track down the owner if the item was stolen.

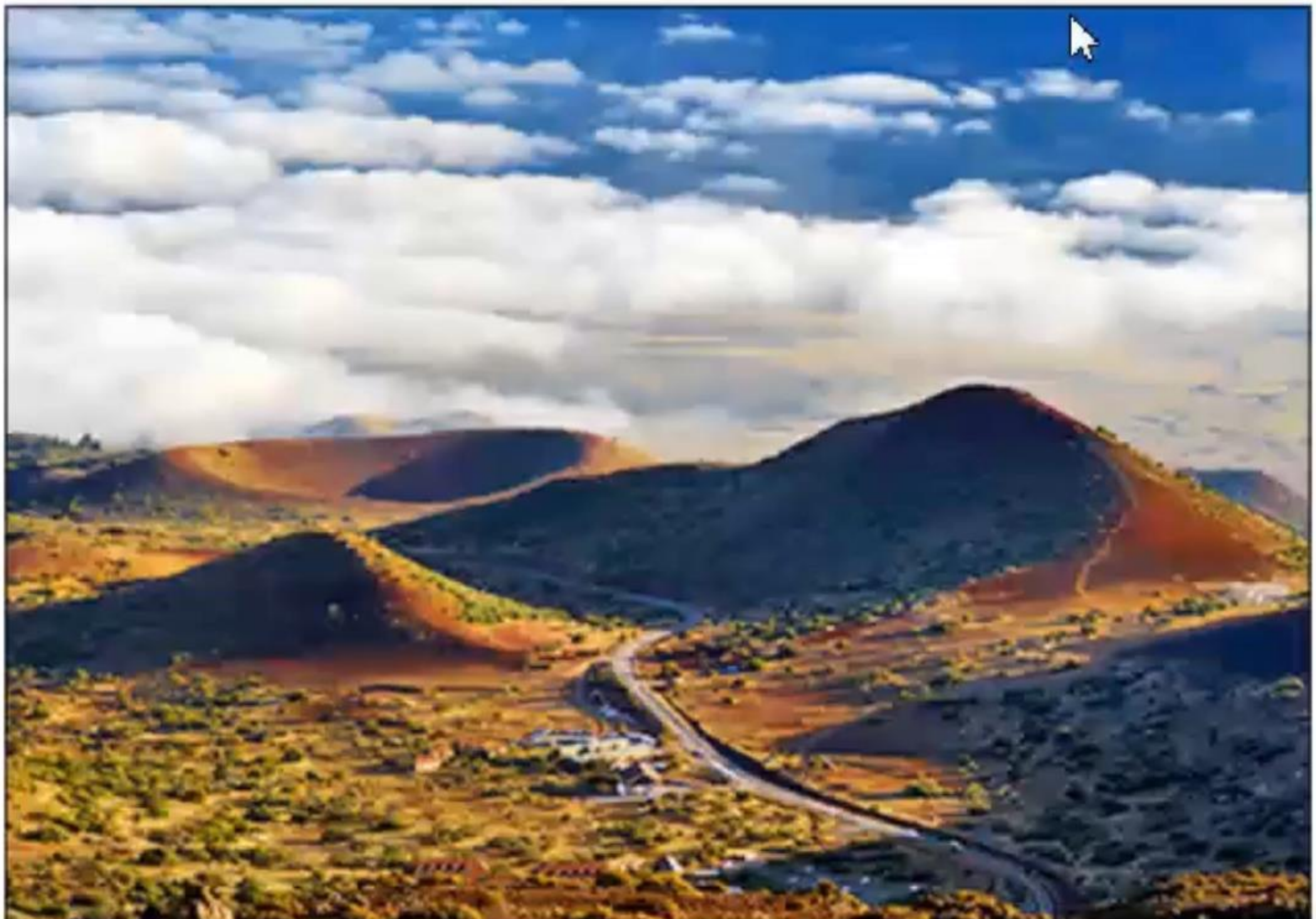
Much of the silver in the homes of the well-to-do in the eighteenth century took the form of coins that were often melted down and transformed into domestic objects, and sometimes remelted to pay off debts or to fund revolutionary armies. This transference from craft to currency and back suggests the particular place that silver held in colonial and post-Revolutionary America. According to historian Richard Lyman Bushman, silver was "a powerful material for establishing identity and configuring hierarchical relationships... Everywhere," silver was used to command assent, to assert authority, or to claim respect. Silver's power, notes Bushman, lay in the fact that it could signify three things at once: money; beauty, which was associated with divinity; and rank. In an era of dramatic fluctuations in the value of paper currency, silver maintained its value.

Yet the sheen of silver was tarnished by both its origins and its association with thievery and deceit. Much American silver originated in mines in Mexico and Peru and was extracted through a system of forced indigenous labor marked by long hours and appalling conditions; conditions that were remarked upon in political treatises and literary tracts available to American colonists. Many merchants, through their participation in the slave trade, then acquired the coins created from this silver. Once acquired, owners of such silver had to worry about thieves who recognized the wealth and status associated with this valuable metal and who were not averse to counterfeiting it as well as stealing it. Thus, as Bushman writes, "silver figured in a complex and ambiguous narrative of power." Whether as coinage or as teapots, it represented both elite status and a more democratic consumer revolution, as well as the ugly underside of the economic system that made both possible.

Measuring Climate Change on Mountains

Climate change has been linked to greenhouse gases in Earth's atmosphere that allow sunlight to reach the planet's surface while trapping the infrared energy that it emits. While the most abundant greenhouse gas is water vapor, human activities have only a small direct influence on its overall concentration, and its residence time (the time it stays in the atmosphere) is only nine days. The other major greenhouse gases are carbon dioxide and methane, and their concentrations in Earth's atmosphere have been increasing since the mid-eighteenth century, due particularly to the burning of fossil fuels, the clearing of forests, and other changes in land use. Molecules of carbon dioxide and methane stay in the atmosphere for years after they are emitted-their residence times are between 5 and 200 years (carbon dioxide) and 12 years (methane)-so human activities have long-term effects.

Evidence of these effects can often be found on mountains. For instance, the clearest direct evidence that concentrations of carbon dioxide are increasing around the world comes from continuous measurements taken since March 1958 near the summit of Mauna Loa, Hawaii, at an altitude of 3,397 meters. The site was chosen by the United States Weather Bureau after it was unable to find anywhere in the continental United States where the air was clean enough. The atmospheric scientist Charles Keeling's original reason for monitoring carbon dioxide was to better understand the daily cycle of its concentration in the atmosphere. However, within a few years, he had identified a seasonal cycle-because vegetation in the Northern Hemisphere absorbs carbon dioxide in summer and releases it in winter-and also discovered that, each year, concentrations were higher. Since 1958, the concentration of carbon dioxide at Mauna Loa has increased from 317 to over 400 parts per million. There is now a global consensus that Earth's climate is changing because concentrations of carbon dioxide and other greenhouse gases are increasing as a result of human activities.



Mauna Loa

© MNStudio/Dreamstime.com

Mountains are not only where the primary cause of climate change has been measured, but they are where crucial evidence of its effects can be seen. Two of the most compelling types of evidence are the melting of glaciers and the fact that plants are growing at ever higher elevations on mountain summits. An international program of glacier monitoring began in 1894, as it was hoped that long-term observations would provide insights into processes of climate change, such as the occurrence of ice ages. Initially, the program focused on measuring the lengths of glaciers. Subsequently, the monitoring program was expanded to include the measurement of the mass balance of glaciers-the difference between the inputs of snowfall and the losses, mainly by melting-and the use of images from satellites. The great benefit of these images is that they provide complete coverage of areas that are often not easily accessible, such as many glaciated areas. Since 1986, the resulting standardized data on changes in glaciers, provided by scientists in over thirty countries, has been compiled by the World Glacier Monitoring Service. These data show that the world's glaciers are shrinking-and at an accelerating rate. While the primary reason is a warming atmosphere, an additional cause, also of human origin, is the deposition of fine particles of black carbon, or soot, on the surface of glaciers. This comes mainly from engines and the burning of coal and other biofuels. Because soot absorbs more radiation than ice, its presence increases rates of glacier melt. This phenomenon occurred in the European Alps in the nineteenth century, so this was the region where the rate of glacier melting was greatest at the time, and it is currently happening in the Himalaya.

While long-term measurements of carbon dioxide and glaciers were planned, finding evidence of alpine plants growing at higher elevations was largely a lucky accident. Early in the twentieth century, the Swiss botanist Josias Braun-Blanquet climbed many peaks in the Swiss Alps. While on their summits, he recorded the highest specimens of each plant species. In the early 1990s, two young Austrian botanists, Michael Gottfried and Harald Pauli, returned to the peaks that Braun-Blanquet had climbed and repeated his measurements. They found a statistically significant increase in the number of species on the summits. These changes could then be correlated to changes in regional climates.

Cool Early Earth Hypothesis

The oldest Earth rocks known to date, found in Quebec, Canada in 2008， were determined to be about 4.3 billion years old. However, the oldest known materials of the solar system-Moon rocks and meteorites-are around 4.55 billion years old. Why the difference? Earth's surface is constantly being recycled and remodeled due to the motion of tectonic plates. The Moon, by contrast, has a dead surface with no such movement. Meteorites that formed with the early solar system, like carbonaceous chondrites, have been unaltered since they cooled, so they give the oldest dates of all.

These are the dates for the oldest rocks on Earth, but they are not the oldest Earth materials known. That distinction goes to a handful of zircon sand grains from a much younger sandstone found in the Jack Hills of Western Australia. Each individual grain can be dated by uranium-lead methods, so they give a range of ages. But the oldest grains of all give an age of 4.404 billion years, at least 100 million years older than the 4.3-billion-year-old materials from Quebec. Thus, the current record holder for the oldest material from Earth (that is, not a Moon rock or meteorite) is 4.4 billion years. These sand grain dates put us closer and closer to the age of Moon rocks and meteorites, but we still have a gap of about 200 million years. This is about the same time span as the beginning of the Age of Dinosaurs (Late Triassic) until today, so it is not a trivial amount of time.

But those same tiny zircon sand grains held even more surprises. Not only did they give the oldest known dates, but when scientists analyzed the tiny bubbles of gases trapped inside them, they found evidence of the early atmosphere from over 4 billion years ago. These bubbles had oxygen isotopes (variants) in them that suggested Earth had liquid water on its surface as early as 4.4 billion years ago! This is surprising because when Earth formed 4.55 billion years ago, it was in a molten state (liquefied by heat). Prior to this discovery geologists had always assumed that Earth took a long time to cool from its molten state. Most thought that Earth took about 700 million years to cool down below the boiling point of water (100C), because that was the age of the oldest sedimentary rocks that would have been produced by running water (the Isua Supracrustals from Greenland, which are 3.8 billion years old). But the Jack Hills zircons turn that assumption inside out. If the zircons truly indicate the presence of liquid water on Earth 4.4 billion years ago, then it took only 200 million years for Earth to cool from its molten state to a condition that was below the boiling point of water. This also suggests that there were not as many meteorite impacts during this time interval, or the oceans would have been vaporized over and over again. Taken together, these data suggest what is now called the "cool early Earth hypothesis."

So where did this early Earth water come from? Traditionally, geologists thought that it was water trapped inside Earth's mantle (middle layer)when it cooled, gradually escaping through volcanoes in a process called degassing. But lately, chemical analyses of extraterrestrial objects match the chemistry of Earth's oceans(especially carbonaceous chondrite meteorites). This suggests that there was a lot of water trapped in the debris of the early solar system (of which the chondrites are remnants), which represent the material from which Earth formed. The same is true of Moon rocks, which do not have much water in them today, but apparently, they were pretty wet when the solar system formed. If this is so, then Earth was born with its water already present as it cooled and condensed. It only required its surface temperature to drop below 100C for that water to form the first oceans. One thing we can rule out is comets. Although comets are often called "dirty snowballs” because they are made mostly of dust and water ice, chemical analyses of four comets now show that their geochemistry is very different from Earth's water. Thus, the popular idea that comets impacted early Earth and melted to form its oceans can be dismissed.

The Columbia Exchange

When Christopher Columbus and his men became the first Europeans to arrive in the Americas in 1492, they set in motion a process of cultural, economic, and environmental exchange that had profound effects for the whole world. For the "old," Afro-Eurasian world, an important transformation included the addition of valuable new food items into the diets of people in such widely separated regions as Ireland, China, and sub-Saharan Africa. For the "new." American world, the most significant transformation involved the deadly effects of Afro-Eurasian diseases among its peoples, which in turn helped pave the way for imperial conquest by Europeans. After two centuries of what is now known as the Columbian Exchange, both Old and New Worlds were transformed.

About 12,000 years ago, the land bridge linking the Americas with Afro-Eurasia across the Bering Strait disappeared as a result of rising sea levels. As a result, the peoples of the Americas were almost completely separated from Old World peoples until the Spaniards arrived in 1492. In other words, for twelve millennia the people as well as the plants and animals of the Americas developed in isolation from Afro-Eurasia according to the specific environmental conditions of the Americas.

Some human developments were similar: for example, agriculture and writing developed in the Americas just as they had done in Afro-Eurasia, as did the birth of hierarchical societies and patriarchies (societies headed by the eldest male). In other ways, however, developments in the Americas followed a different course from those of the Old World. For example, plant species such as corn, potatoes, and tobacco were unique to the Americas, and thus American peoples had a different set of choices regarding possible domesticated crops. Just as importantly, not long after humans arrived in the Americas, all of the large mammals that existed there became extinct. As a result, when humans in the Americas began to turn to settled agriculture, they did not have the choice to domesticate large mammals such as cattle or horses. Because of this, American societies did not develop innovations like the wheel, which proved unnecessary without large animals to pull carts or push plows. Furthermore, other animals familiar across much of Afro-Eurasia-including sheep, pigs, chickens, and goats-were nonexistent in the Americas. In fact, the only animals that could be domesticated available to American peoples were the dog, the turkey (in North America), the llama/alpaca and guinea pig (in the Andes region of South America), and the Muscovy duck (in the South American tropics).

The absence of animals that could be domesticated is critical because in the Old World the domestication of animals such as sheep, chickens, goats, and cattle contributed greatly to the development of serious diseases such as influenza, smallpox, measles, and pertussis, In fact, many of these serious diseases resulted from microbes that were first present in herds of domesticated animals and then made the jump from their animal hosts to human hosts. The jump was made much easier because early Afro-Eurasian communities that relied on domesticated animals for food and labor tended to live in close proximity to their herds-even keeping them inside their houses for protection or warmth. Over time. a number of these microbial jumps occurred between domesticated animals and humans. In addition, because of the land connections across much of Afro-Eurasia, these diseases were able to spread to many Old-World populations. While these diseases continued to be serious for the peoples of Afro-Eurasia, the human populations that survived them over the generations tended to pass on an acquired partial immunity to their offspring

Without many animals that could be domesticated and living in enforced separation from Old World peoples, the people of the Americas remained freer from the type of contagious diseases that so often plagued the Old World. Some contagious diseases did exist, to be sure, and included dysentery, pneumonia, and possibly syphilis. But the "crowd" diseases of the Old World that flourished in crowds of large, centralized populations of humans did not exist, which meant that American peoples-when exposed to them-had no immunity whatsoever. Therefore, when Europeans carrying these diseases inadvertently spread them among American peoples, sickness spread rapidly, resulting in many deaths.

Monochrome Landscape Painting in Medieval Japan

During Japan's Muromachi period (1336-1573 A.D.), an important art form was monochrome painting (painting using one color), done in the manner evolved several centuries earlier by artists of the Song dynasty of China. Like their Japanese counterparts of this later age, the Song monochrome artists painted a variety of subjects, including Zen Buddhist religious officials, folk deities, and flowers and birds. But their primary interest lay in landscapes (known in Japanese as Sansui, or pictures of mountains and water). And indeed, Song monochrome landscapes are among the more striking works of Chinese art. They are, moreover, perhaps the most supremely moving tributes of any people to the grandeur and vastness of nature.

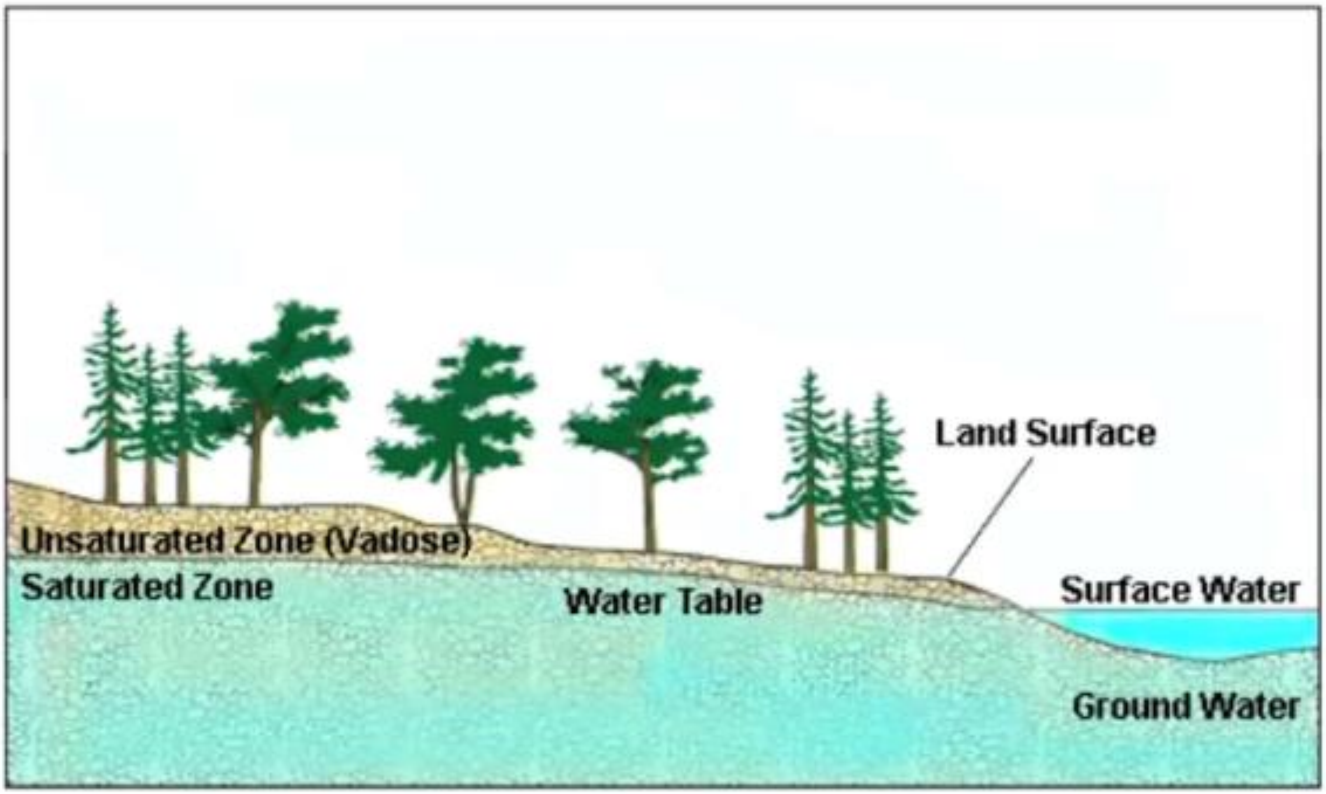
The Song masters did not attempt to reproduce nature as it really was; rather, they employed bold and even daring brushwork to capture in stylized outline misty scenes of forests, jagged cliffs, waterfalls, and awesome mountains (the most distant of which seem to be on the point of vanishing into space). Human figures sketched into these landscapes are usually ant-like in size. We see them, insignificant figures engulfed by the universe, as lone travelers moving slowly along mountain trails or as solitary beings seated in pavilion-like huts nestled on the sides of towering peaks. Song monochrome painting appealed particularly to the medieval Japanese because its medium of black ink was so compatible with the tastes of the age. In the first phase of painting in the Song manner during the fourteenth century, Japanese artists devoted themselves primarily to portrait and figure work, but in the fifteenth century they turned increasingly to landscapes.

Among the greatest masters of monochromatic ink work of the fifteenth century was Shubun. Although Shubun, who was active during the second quarter of the century, is reputed to have painted many different subjects in a variety of media, the only extant works attributed to him are landscapes, mostly on folding screens and sliding doors. A typical Shubun landscape is “visionary" in that it is a depiction, derived wholly from imagination, of a scene set in China. Like that of other Japanese artists of his time, Shubun's work is also impressionistic, since space is not clearly differentiated (that is, it is difficult to judge the relative depths of the various sections of a painting) and mountains, cliffs, and other pictorial elements often appear to be suspended or not properly integrated with the rest of the landscape. By contrast, Song-style landscapes by Chinese artists are notable for the care with which they are constructed: Foregrounds, middle distances, and backgrounds are clearly distinguishable, and all parts of the picture “fit together" into a coherent reproduction, even if stylized, of a view of nature.

Thus there appears to have been a fundamental difference in the approach to landscape between the Song-style Chinese artist and such Japanese painters as Shubun, a difference that seems to consist in the fact that the Chinese artist was as much concerned with presenting a philosophical idea as with representing visual qualities. The Chinese artist sought to portray in nature the kind of harmony and overall agreement of parts that ideally ought to prevail in human society. In other words, the Chinese artist tried to make a social statement; and the greater sense of structure and depth he could incorporate into his landscapes, the greater the philosophy of his work.

The Japanese, on the other hand, have never dealt with nature in their art in the universalistic sense of trying to discern any grand order or structure; much less have they tried to associate the ideal of order in human society with the harmonies of nature. Rather, they have most characteristically depicted nature -in their poetry, painting, and other arts- in specific distinct glimpses. The Chinese Song-style master may have admired a mountain, for example, for its enduring, fixed quality, but the typical Japanese artist (of the fifteenth century or any other age) has been more interested in a mountain for its changing aspects: for example, how it looks when covered with snow or when partly obscured by mists or clouds.

The vadose Zone and its Structure



When rain falls on dry land, it doesn't become part of the groundwater immediately. First it must penetrate the unsaturated ground (the ground that is not soaked with water and still has space to absorb more water). The water table, or water level, marks the end of the unsaturated zone and the beginning of the saturated (water soaked) zone. Reaching the water table may take considerable time, and not all of the incoming water necessarily gets as far as the water table; usually, after a temporary stay in the unsaturated zone, some of the water returns to the atmosphere.

Water in the unsaturated zone typically is moving, up and down and horizontally; it is seldom at rest. The unsaturated zone has two other names: sometimes it is called the zone of aeration, because its pores usually contain some air, and sometimes the vadose zone (from the Latin vadosus, meaning shallow). From this point on, we will use the term vadose zone; its brevity makes up for its being a jargon term, and it is more accurate, too; the vadose zone is not unsaturated always and everywhere, as we will see. Nor is it a single, homogeneous zone: in most places it has three distinct layers. The topmost layer is the soil; below it is the intermediate zone, consisting of the subsoil together with any solid bedrock that happens to lie between the subsoil and the water table; and at the bottom, in contact with the water table, is the capillary fringe, a layer into which groundwater soaks upward, like water into a sponge.

The soil is the most familiar, and certainly the most complex, of these layers. The distinction between soil and subsoil is that the former contains living organisms larger than bacteria, while the latter does not. The soil is the layer occupied by the roots of plants; it is also the home of myriads of mostly unnoticed soil organisms. This abundance of living material affects the soil's structure, its color, its chemistry, and the way water flows through it. Indeed, when water descends from soil into subsoil, it encounters entirely different conditions.

The thickness of the soil varies enormously: in some places it is no more than a thin layer of dust on bare rock; in others, it may consist of peat beds, areas made up of partly decomposed plant materials, 4 or 5 meters thick. Soil is continually subject to two opposing forces: dead organic material, accumulating ceaselessly, continually builds it up; at the same time never-ending erosion, as water flows through and over it, continually washes it away. If the soil thickness remains unchanged for long periods, as it does in many places, the reason is that these two processes balance each other.

The intermediate zone is the zone of loose mineral sediments, and sometimes bedrock, sandwiched between the soil and the capillary fringe. In places where the soil is deep or the water table shallow, there may be no intermediate zone; then all that lies between the surface and the capillary fringe is true soil.

The two upper layers of the vadose zone together form a temporary holding ground for water on its way to some other destination: some flows down to join the groundwater, some flows laterally to adjacent streams and lakes, and some is transported back into the atmosphere by the action of plants

The lowermost of the three layers of the vadose zone is the capillary fringe. It is not necessarily different in texture from the material above and below it, but it is permanently wet, saturated with water sucked from the groundwater beneath. The height to which water will rise above the water table because of this spongelike suction depends on the texture of the material holding the water The water moves up to fill the pores and interstices (spaces) of this material and the tiny channels that link them. It is constantly being pulled downward by gravity at the same time as it is being pulled upward by the suction force of surface tension; the level at which these opposing forces balance determines the thickness of the capillary fringe.

The Chemical Defenses of Trees

Woody plants produce a great variety of chemical compounds to provide protection against other plants, diseases, and herbivores (plant-eating animals) big and small. These chemicals can be found in just about any part of the plant (for example, the waxy surface of apple leaves contains toxins to repel certain insects). They can also be emitted into the air in considerable quantities, which explains why pinewoods have a strong fragrance and the eucalyptus-dominated bushlands of Australia have a blue haze. These chemicals seep out through the bark or the holes in a leaf (the stomata) or ooze out of special glands, often on teeth around the leaf edge (as found, for example, in pines, alder, and willows) usually when the leaf is young but sometimes (as in certain willows) even late in life.

Some chemical defenses are highly toxic and will kill attackers in small doses. Known as "qualitative" defenses, they work in low concentrations. These include alkaloids present in plants such as the tansy ragwort, which is notorious for killing cattle and horses when ingested. Other chemicals work by building up in the herbivore. Referred to as "quantitative" defenses, they become more effective the more they accumulate in the animal's body. Classic examples are phenolic resins in creosote bushes and tannins found in a number of plants. Tannin is a "protein precipitant which gradually makes food less digestible, so herbivores end up starving and having stunted growth. These often have the effect of causing the herbivore to move elsewhere. It is possible, for example, that the endangered British red squirrel is declining not through competition with the introduced North American gray squirrel, as commonly thought, but because of acorns. Acorns (nuts from oak trees) contain digestion inhibitors that gray squirrels can disarm but red squirrels cannot. Red squirrels do well in conifer tree plantations, feeding on the more nutritious pine seeds, and where there are no oaks to give the gray squirrels a competitive edge.

Plants are not immune to the effects of chemical defenses. Spanish moss, which hangs from trees in the southern United States and in South America, never grows on pines, presumably because of the resins pine trees secrete. Perhaps the best-known example of antiplant chemical defenses was first reported in the first century AD. by Pliny the Elder, who wrote that "the shadow of walnut trees is poison to all plants within its compass." Walnuts and related trees contain the chemical juglone, which, seeping from the roots. Reduces the germination of competitors, stunts their growth, and even kills nearby plants, resulting in open-canopied walnut groves having very little growing under them. Tomatoes, apples, rhododendrons, and roses are very susceptible, but many grasses, vegetables, and Virginia creeper will happily live under walnuts.

Defensive chemicals are costly for the plant to produce and store; many are toxic to the plant producing them. Oaks may put up to 15 percent of their energy production into chemical defenses (which explains why stressed trees are most prone to attack: they have less energy available to produce defenses). Thus, although there may be large pools of defenses available where attacks are common and ferocious, it makes evolutionary sense in lesser situations to produce the chemicals in earnest only when they are needed. Oaks (and a number of other trees) manage with low concentrations of tannins in their leaves, but once part of the canopy is attacked, the tree will produce tannins in large quantity More than that, the tannins released into the air are detected by surrounding oaks and they in turn will produce more tannins in preparation for the onslaught. In a similar way, willow trees in Alaska that had their leaves eaten by snowshoe hares produce shoots with lower nutritional concentration and higher levels of lignin and deterrent phenols, thus rendering them less palatable. These "induced" defenses may have an effect for some time: birch trees can remain unpalatable for up to three years after being eaten. Expensive defenses are saved not only when they are not needed but also where. The North American aspen has been shown to produce fewer tannins in trees growing on fertile soils: it is cheaper to replace lost material than to defend it. Similarly, more is invested in defenses where herbivores and diseases are more common.

The Porcelain Industry at Jingdezhen

Porcelain is a ceramic material that is prized for its translucence, hardness, and smoothness. Porcelain was first produced in China, perhaps as early as C.E. 100–200. By the Tang period (C.E. 618–960), porcelain objects were being exported from China to the Islamic world. Over the centuries, manufacture of porcelain for both domestic consumption and export grew. The large scale of production for export in the Yuan period (1279–1368) is indicated by a shipwrecked transport vessel bound for Japan that was found off the Korean coast in 1976. It was loaded with some seventeen thousand pieces of export wares (goods to be sold). Included among the porcelains were many Longquan celadon (a particularly beautiful type of green porcelain from the period) and white or bluish-white porcelains from Jingdezhen.

The Yuan period was pivotal in the history of porcelain for several reasons. There was a definitive concentration of production at the Jingdezhen kilns (ovens) in the southeastern province of Jiangxi. This region was favored by abundant concentrations of the necessary special clays, as well as forests for wood to fire the kilns and a relatively central location well connected for water transport and distribution. The Yuan rulers established an official agency to supervise ceramic production at Jingdezhen in the thirteenth century, and it continued as the official governmental porcelain center under the succeeding Ming (1368–1644) and Qing (1644–1911) dynasties. In addition to official kilns that supplied wares for palace use and for diplomatic gifts or awards to officials, the Jingdezhen region had many private kilns that produced wares for broad domestic consumption and for export.

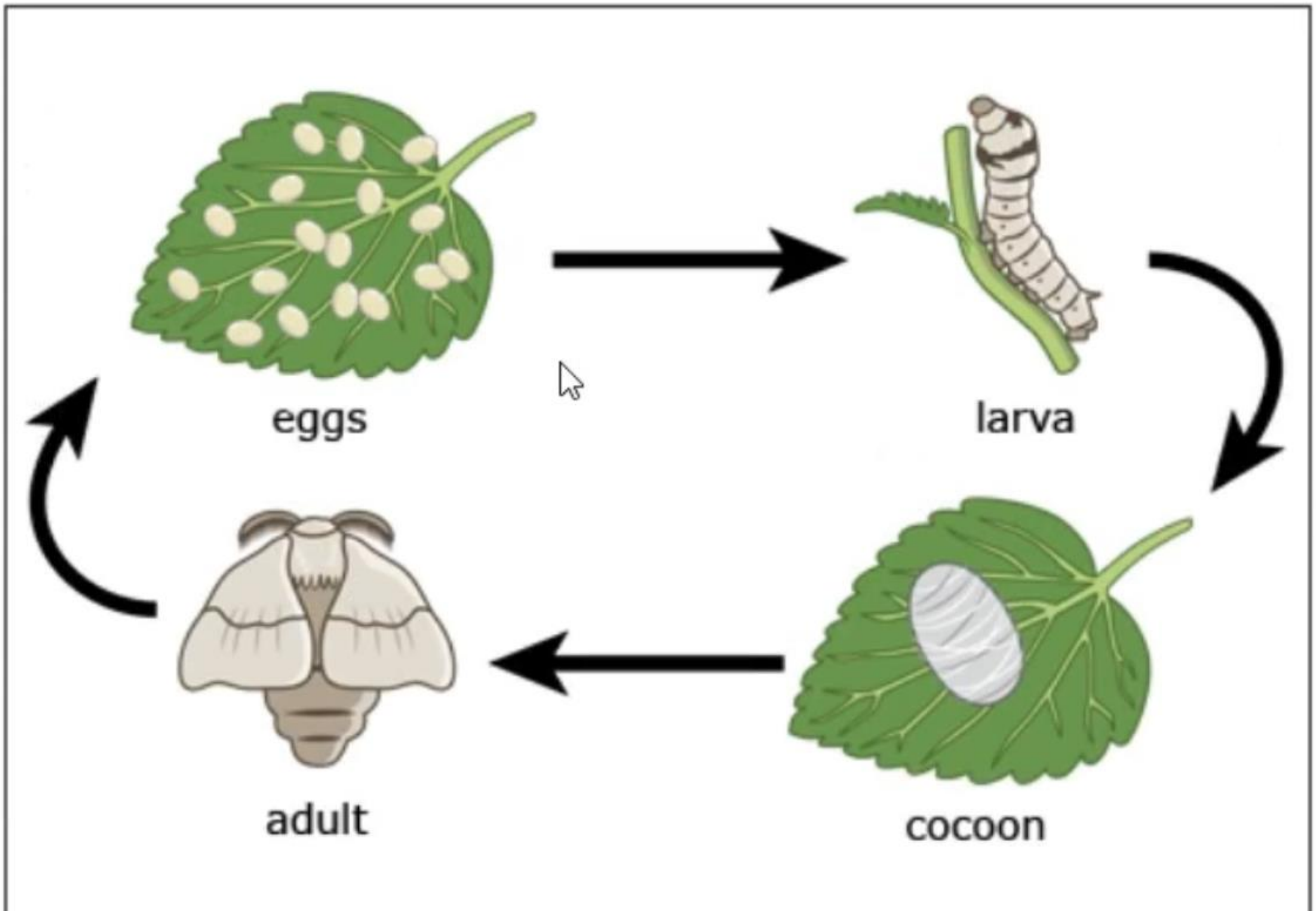
The concentration of resources, artisan expertise, and imperial prestige at Jingdezhen led to the development of a porcelain industry of national and international scale. Specialization of labor and rationalization of manufacture, the process of improving the means and methods of production to improve efficiency in mass production, characterized porcelain production as it did later industries (and some earlier productions, such as lacquerware in the Han period, from 206 B.C.E. to C.E. 220). By the mid-sixteenth century, porcelain production involved more than ten thousand craftsmen in the region, and a single piece might undergo seventy-two processes before completion. The result was a centralized and sophisticated craft industry of vast scale that supplied markets ranging from the imperial households to export markets in Southeast Asia and, eventually, Europe and America.

In terms of technique and style, the Yuan period was important for introducing painted decoration in which copper-red and, most famously, cobalt-blue pigments were used to paint the porcelain and then covered over with a transparent protective glaze, known as underglazing. Although the previous Song period(C.E. 960–1279) had favored wares decorated in a single color with pictorial designs limited to carved or molded patterns, earlier precedents existed for lively pictorial designs brushed on ceramics. Ink-like iron oxide or colored enamel pigments were used on Cizhou types , and painted designs appeared on the still earlier Changsha wares of the Tang period. These were not porcelains but less refined stoneware, fired at lower temperatures, and Yuan-period Jingdezhen saw the first large-scale combination of porcelain technique with underglaze painting. The taste for underglaze blue-and-white porcelains may have been both stimulated and enabled by the international connections of the Mongol rulers of the Yuan regime. The best sources for the cobalt used for the blue ornamentation came from Western Asia and so were more easily supplied during the period of Mongol domination of Asia, with its open trade routes. Much early blue-and-white production seems to have been aimed at export markets in the Middle and Near East and Southeast Asia. Some of the design of these wares may have aimed at replacing cruder painted Near Eastern wares with more durable and refined porcelains decorated in similar style, and much of the demand that stimulated production of blue-and-white wares may have originated from Arab clients in West Asia.

One of the great advantages of underglaze-painted decoration was its adaptability to a wide range of tastes and markets. Skilled painting specialists produced lively designs of decorative flowers and plants, dragons, ducks, fish, or figural scenes drawn from popular drama, as the occasion demanded; in addition, designs in Islamic script or incorporating Tibetan Buddhist motifs were produced in large numbers for those markets and recipients.

The Silk Makers

The silkworm-the larva (immature form) of the domestic silk moth-produces silk: a fine, continuous protein fiber.



Life Cycle of a Silkworm Moth

A very large number of invertebrates can make some type of silken thread, including moths and butterflies, but also spiders and grasshoppers. These animals create a variety of silk fibers, which they use for a variety of purposes, such as making webs, creating nests, and spinning cocoons. Variations in silk production and use appear to be one of the most important factors in giving rise to the immense diversification of insect species, which often emerged because there were so many different ways to use so many different types of silk to establish unique niches (ecological roles).

Evolutionary biologists estimate that the ability to make silk in the ancestors to present-day Lepidoptera (the order of insects that includes more than 180,000 species of butterflies and moths) first evolved some 250 million years ago. For silkworms and the larvae of other moths and butterflies, silk became a tool for undergoing metamorphosis. Metamorphosis is the extraordinary process during which larvae, such as caterpillars or worms, change into adult forms, such as flying insects or spiders. The worms weave silk cocoons as a sort of second egg inside of which they essentially consume their previous body in order to grow a new winged one. The cocoon is both a receptacle for this complex biochemical process and a means of protecting the defenseless insect from the environment and predators while it is metamorphosing. The silk fibers render the cocoon less palatable while also making it more difficult for would-be predators to get at the tasty protein-rich prize inside.

Nonetheless, many animals are still happy to eat the vulnerable worms during metamorphosis, raising the question of why any organism would have evolved such a risky behavior in the first place. It seems counteradaptive if not downright suicidal. Evolutionary biologists are not entirely sure, though since some 45-60 percent of all the animals on the planet undergo metamorphosis, it must confer some useful advantage. The strongest current theory suggests that metamorphosis permits insects to more fully utilize available food sources in their niche while avoiding intergenerational competition. In the case of silk moths, the larvae grow rapidly while eating the leaves of mulberry trees, but the adult moth form feeds solely on the nectar of flowers. This permits any given environmental niche to support both more worms and moths, conferring an evolutionary advantage. Metamorphosis also allows the wingless larval form to burrow into plants, fungi, and meat, which would likely harm the delicate wings of the subsequent adults. Since their larvae are such efficient feeding machines, some adult insects even have enough stored energy that they do not have to eat at all during their short life spans, freeing them to concentrate all their energies on the more important tasks of mating and egg laying.

It is equally fortuitous that at least some silkworms are even capable of coevolving with humans in a way that led to the animal partnership that we call domestication. In the case of silkworms, it led to silk farming, a practice first developed by the Chinese, who established a lucrative transcontinental silk trade near the end of the first millennium B.c.E. Domestication is always a two-way street in which another species must have the inherent capacity to tolerate and cooperate with humans. Silkworms' ability to tolerate domestication becomes evident if we compare them to spiders. Spiders actually produce a superior silken fiber, yet modern attempts at domesticating spiders have largely failed because they are very territorial creatures and will often eat one another if forced to live in close proximity. Indeed, spiders' need to make silk as adults-rather than just during a larval stage as with silkworms-may explain why they do not coexist well with other members of their own species. As is the case with mammals like wolves and humans, in many insects social behaviors are linked to juvenile hormones that make them more cooperative than aggressive. However, in spiders these juvenile hormones just happen to interfere with the biochemical synthesis of silk. So, for spiders to enjoy the many adaptive advantages that come from a lifetime of silk making, it appears they had to give up the many potential benefits of social cooperation.

Methane Hydrate as a Potential Fuel

At the bottom of the world's oceans lies a potential source of energy that could provide society with fuel well into the next century. These deposits are the product of ancient seafloor bacteria, and they store twice as much carbon as all the world's other fossil fuel reserves combined. This potential fossil fuel is not oil or coal, or even natural gas as we know it, but methane gas locked in crystals of ice. Resembling sparkling snow, this peculiar ice, known as methane hydrate, burns if ignited, leaving behind a pool of water. Methane ice forms under pressure and expands to more than 160 times its original volume when it is brought to the surface.

First discovered in 1970, methane hydrate has been found within just a few hundred kilometers of almost every coastline, as well as in permafrost regions on land, such as those of the United States state of Alaska. And it has been found in enormous quantities. Off the east coast of the United States near the Carolinas, for example, the United States Geological Survey, a scientific agency that studies natural resources, has discovered two deposits of methane hydrate, each covering about 3,139 square kilometers. Together they are estimated to contain over 37 trillion cubic meters of methane gas, or more than 50 times the amount of natural gas consumed in the United States in 2012. Is methane hydrate the fuel of the future? The sheer volume and richness of methane hydrate deposits make them a strong candidate for development as an energy resource. However, the challenges are enormous.

Although methane hydrate forms in layers that could potentially be mined, gaining access to the deposits beneath the seabed poses huge technical problems. In addition to the problems of mining it, the hydrate is stable only at the icy temperatures and crushing pressures of the deep ocean and starts to break down long before it reaches the surface; thus, preventing the gas from escaping as the hydrate is recovered adds to the challenge of recovery.

Like conventional natural gas, the layers of methane hydrate could also be tapped by drilling. Several oil companies and government agencies are actively researching the possibility of such an endeavor. Because the deposits are fa deeper than most underwater oil and gas fields, special deep-water drilling vessels would have to be constructed. Nevertheless, drilling is at least a possible option, although the methane ice is under so much pressure, the challenge is akin to bursting a balloon and trying to capture all the escaping gas. One potential solution would be to expel the methane by pumping hot water or steam into the deposit through one drill hole and extracting the expelled methane through another. But once recovered, the methane would still have to be brought ashore, and this would pose an additional challenge.

Methane hydrate would provide a "cleaner" source of energy than oil and coal because methane releases less than half the amount of carbon dioxide (a greenhouse gas that traps heat in the atmosphere)into the atmosphere when it is burned and so contributes less significantly to climate change. However, its exploitation is not without potentially serious environmental consequences. It is estimated that the methane trapped in methane hydrate amounts to more than 3,000 times the volume of methane in the atmosphere, where it forms a potent greenhouse gas that is ten times more effective at trapping heat than carbon dioxide. The global warming effect of methane accidentally released into the atmosphere is consequently ten times more serious than that of carbon dioxide. Breakdown of methane hydrate at the base of hydrate layers is also known to have triggered massive landslides on the ocean floor, adding a further hazard to the production of this resource.

Although the technological challenges facing the successful exploitation of methane hydrate are great, they are not insurmountable. Indeed, both Japan and China have projects aimed at commercial-scale extraction, and the oil industry is field-testing the possibility of extracting methane. We may therefore live to witness the exploitation of this resource as other reserves of fossil fuel dwindle and methane hydrate becomes an economically attractive alternative.

The River Nile in Ancient Egypt

The special character of the Nile, which made it central to ancient Egyptian culture, was its annual inundation (flooding). During June the river began to rise, and a quantity of green water appeared. The color is said to have resulted from the brief period of reproduction of myriad minute organisms. During August the Nile rose rapidly and assumed a muddy red color created by the rich red earth brought into its waters by its tributaries. The Nile continued to rise until mid-September, then remained at that level for two or three weeks. In October it rose again slightly, then began to fall gradually until May, when it reached its lowest level.

The Nile has created a convex floodplain. In convex floodplains, sediments(clays and silts)are deposited by flood waters, making the land nearest the river have the highest elevation. The convex floodplain is marked by natural levees that form elevated barriers immediately adjacent to the river. These levees rise a few meters above the seasonally inundated lowlands. When the Nile floods, the water covers most of the low-lying land up to the edge of the desert. When the floods subside, the waters are trapped behind the levees and prevented from returning to the river. The benefit of such topography is obvious: the water can be used where it stands or can be channeled to other areas as dictated by agricultural needs.

Ancient records, those preserved both in texts and in the visible evidence on ancient devices for measuring water levels called nilometers, indicate that a flood of six meters was perilously low and that one of nine meters was high enough to cause damage to crops and villages. A flood of seven to eight meters was ideal in that low- lying areas and basins throughout the whole valley would be flooded up to the edge of the rising ground of the desert, but towns, villages, and dikes that served as paths and water barriers remained above the water level.

The ancient Egyptians fully understood the extent to which their lives and prosperity depended on the unfailing regularity of the inundation. The occasional low flood and consequent shortage of food were enough to cause much anxiety among the populace at the beginning of each flood season. Ancient Egyptians, therefore, never became completely confident about the annual inundation and its gifts, even though it usually brought a layer of fresh, rich silt and waters for irrigation that made agriculture in the Nile Valley relatively easy. The generally predictable crops and resulting surplus freed a significant segment of the population from agricultural labor, allowing for the development of nonfarming occupations, such as full-time craftspeople, bureaucrats, and priests.

The importance of the Nile to Egyptian civilization is reflected in the role that it played in religion and the myths that revolved around the river. The Nile was known in antiquity by the

Egyptian name Iteru, meaning "great river.” The personification of the inundation was a god named Hapy, who was associated with fertility and regeneration. The ancient Egyptians had various conceptions of the origin of the inundation. Some texts relate that it began in a cavern at Philae, while others credit the site Gebel Silsila (about 100 kilometers to the north)as the source. It was believed that veneration of the gods associated with these sites in the Aswan area could ensure a sufficient inundation. The Famine Stela, a text carved on rocks at Sehel near Philae, records a famine that was averted by donations of land and goods to the Temple of Khnum at Aswan. This text was formerly thought to date from the reign of Djoser(2687-2667 B.c.), but in reality, it dates to the Ptolemaic period some 2,500 years later.

The Faint Young Sun Paradox

Billions of years ago, in the first part of what is called the Archean period, our Sun was much weaker than it is today, emitting only about seventy percent of its current light and heat. This means that Earth should have been a much colder place, covered in ice. Yet there is good evidence that it was not as cold as we would expect. This is called the faint young Sun paradox. Something must have provided Earth with an atmosphere that retained the Sun's heat. It was initially hypothesized that this atmosphere most likely included carbon dioxide, a greenhouse gas that induces this effect.

This question has been debated for the best part of half a century, as the puzzling non-iciness of early Archean times continued to emerge from the evidence of the rocks. The atmosphere, we know, had little or no oxygen in it, as layers of sediment of that age deposited by rivers include grains of minerals such as pyrite and uraninite that, in today's oxygen-rich climate, would quickly oxidize and rust away. Among the ingredients of the atmosphere, many have suggested high levels of carbon dioxide from sources such as volcanic eruptions. However, extremely high levels of this gas would be needed to compensate for the faint early Sun-perhaps at about a hundred times current atmospheric levels. How can we tell if an ancient atmosphere had a lot of carbon dioxide or a little?

One way is to look at layers thought to represent ancient land surfaces-soils. Some minerals, such as siderite (iron carbonate), need high levels of carbon dioxide in order to form. So, if we find siderite in fossil soil, can we infer that carbon dioxide levels were high at the time of its formation? Not necessarily, because it depends on what other minerals were there to influence the reactions, and on the microenvironment. Even at a small depth within a soil, the carbon dioxide levels may be very much higher than in the air above, because of the activity of soil bacteria. And even moderately preserved Archean soils are exceedingly rare. The tentative studies made so far mostly suggest levels twenty or so times higher than today-but not a hundred times higher, and so insufficient to overcome the faint young Sun effect on its own.

So what other additional greenhouse gases might there have been? Methane (CH_4) is a good candidate. This simplest of the hydrocarbon gases is about twenty-five times as powerful a greenhouse gas as carbon dioxide. Virtually all methane on Earth today is biologically produced, much of it by the action of microbes called methanogens. It can, however, form without the involvement of life, such as by the reaction of water with certain types of iron- and magnesium-rich igneous rock. In today's atmosphere methane is oxidized to carbon dioxide within a few years. But in a low-oxygen world such as in the Archean it could have survived for many years in the atmosphere. So that is one possibility. There are other candidates too-ammonia, for instance. This is also a greenhouse gas, but it is easily broken down to nitrogen when exposed to ultraviolet light. Now the best way to keep out ammonia-destroying ultraviolet light is to have a global sunscreen made of ozone: to make ozone, a planet needs to have significant amounts of oxygen in its atmosphere, and that state was yet a billion years in the future.

Was the Archean world, perhaps, simply hazy? A haze (cloud) of organic-rich particles might take on the function of ozone and block out significant amounts of ultraviolet light. The trouble is, it can also block out the warmth-giving sunlight, as the early studies on this possibility indicated. Those early studies, though, simply treated the haze particles as simple spheres. On that mightiest of Saturn's moons, Titan, there is an organic haze-but the particles seem on closer examination to be anything but spheres, rather forming highly complex, fluffy collections of material. Shaped in this, way, they are not much of a barrier to incoming light-but they can still absorb ultraviolet radiation. So, if early Archean Earth was in that respect, at least, Titan-like, it might have harbored significant amounts of ammonia.

How the First People Got to the Interior of North America

Most researchers believe that the first people to come to North America came from Asia to what is now Alaska across a land bridge called Beringia, which was exposed as a result of low sea levels during the Ice Age. Evidence in the heart of North America points to a settlement time of around 14,000 years B.P. (before present). But there is some disagreement about which route these settlers took to the interior of a still uninhabited North America from Alaska. Until recently, everyone agreed that the first settlers traveled southward by land through the interior between two great ice sheets, the Cordilleran and the Laurentide; but in recent years, a growing number of people have favored the idea of a coastal route, triggering a major debate.

In the 1950s, Canadian geologists reported that Cordilleran ice did not fuse with the western margins of the Laurentide, creating between them an ice-free corridor sometimes called the Mackenzie Corridor. The Mackenzie Corridor became, in the minds of some, a kind of superhighway to the south, where mammoths and people traveled. This appealing picture is a geological myth, for the boundaries of the ice sheets are slowly being mapped, revealing only a partial corridor, and one that was very inhospitable indeed. With rugged land, mazes of meltwater lakes, crumbling rocks, the scantiest of vegetation, and few animals hunted for food, there was little incentive to travel through this ice-free zone at the height of the last Ice Age. If human beings did travel south, the most likely times for them to do so would have been either before the last glacial maximum (the period when ice sheets were at their maximum extension) beginning 25,000 years ago, or soon after 15,000 years ago, as the ice sheets began to retreat. Recent research has suggested that the corridor may have been quite wide by 11,000 years ago.

If the ice-free corridor was an inhospitable and late migration route to the south, what about other paths? Recent evidence suggests that long stretches of northern coastline from the Kuril Islands off Japan to Beringia and south through the Aleutians (a chain of volcanic islands in the northern Pacific Ocean) were more extensive 13,000 years ago than was once suspected. Did the Pacific coast provide a route into the heart of North America? Advocates of coastal settlement believe that fishing and sea mammal hunting, as well as a sophisticated maritime tradition, were well established in northeastern Asia during the late Ice Age, and that small groups of coastal people traversed the ice strewn shoreline of the Beringia land bridge, then hunted and fished their way southward through the fjords, bays, and more sheltered waterways of the Alaskan coast.

Such a theory is entirely plausible, but its advocates have to live with a reality that any sites likely to document such migrations are buried under 300 feet (90 meters) of rising seawater. Furthermore, there is absolutely no archaeological evidence for sophisticated sea mammal and maritime technology in the form of skin boats (umiaks) or skin kayaks in the far north before about 3000 years ago, when the Bering Strait region became a major center for sea mammal hunting. Maritime adaptations are considerably earlier along the Aleutian chain but certainly much later than a supposed, and perhaps conservative, crossing date of 14,000 years ago. No one has examined the difficulties that sailors would have encountered crossing from Siberia to Alaska in small crafts. Apart from ice flows and pack ice, both of which may have been thicker during a period of warming, the rowers would have had to develop the clothing and boats to deal with very cold water and hypothermia(low body temperature), as well as often savage weather conditions, even in midsummer.

The manual British Admiralty Ocean Passages for the World describes these waters in summer as wet, fog- ridden for days on end, and subject to frequent, severe westerly winds, conditions that challenge even today's well-equipped fishing boats and freighters. Even in more southerly areas like southeastern Alaska, conditions can deteriorate rapidly, with strong headwinds that are notoriously difficult to row against. Southeast Alaska has plenty of sheltered bays and inlets, but the same cannot be said of the exposed coasts of the Gulf of Alaska, or of the shoreline south of Cape Flattery on the Olympic Peninsula, the Pacific route to lands far to the south.

Martian Volcanoes

Earth is not the only planet to have volcanoes; in fact, Mars though all of its volcanoes have long been inactive, has the two largest ones in the solar system: Olympus Mons and Alba Patera. Mars has spectacular lava fields, large craters formed by a volcanic explosion, and shields (broad, low volcanoes shaped like shallow domes).

The most striking characteristic of Martian volcanoes is in fact their great size and longevity. The giant shields of Tharsis and to a lesser extent the volcanoes of Elysium and Hellas pack in many times more lava than the largest volcanoes on Earth, and yet Mars is a smaller planet. The reason lies in the planets' internal makeup. Because it is larger and hotter, Earth has a nearly molten (hot liquid) mantle that is overlain by a very thin rock crust-less than 10 kilometers thick under the ocean basins. The hot mantle churns over in great loops, and rips the overlying rigid crust into great slabs, called tectonic plates, that shuffle around the globe. In those places where deep-seated "hot spots" send plumes of molten material to the surface, the magma (the trapped lava) starts piling up on the crust to build a volcano, but because the crust is drifting, the volcano eventually breaks away from its source and goes extinct. In its place, a new volcano starts to grow, so that a hot spot under a mobile plate creates a string of medium-sized volcanoes rather than one giant shield.

On the smaller, faster-cooling Mars, the crust has solidified and thickened to the extent that it remains stationary with respect to the underlying hot spots. When a plume of magma erupts at the surface, it will build a single volcano in one place for as long as the supply lasts-typically hundreds of millions of years. Not surprisingly, then, the volume of one Martian shield like Olympus Mons or Alba Patera is comparable to that of an entire chain of volcanoes on Earth, such as the Hawaiian Emperor chain that stretches across the Pacific and spans nearly 100 million years of hot-spot activity.

Another characteristic of Martian volcanoes is the large size of their craters and the long run-out distances of their lava flows. This is principally due to the lower gravity on Mars: 38 percent, or about one third, of Earth's gravity. Gravity controls to some extent how solid rock behaves under stress. When the crust stretches and breaks in a low gravity field, as it does on Mars, the fissures that open up are wider than on Earth and can funnel larger amounts of magma toward the surface.

Magma chambers grow correspondingly larger and when they empty out and collapse, they yield larger craters. Eruption rates also tend to be higher on Mars, since greater quantities of lava can flow out of the wider fissures. These larger volumes guarantee a better retention of heat. Because it stays hot and molten for a longer period of time, lava on Mars travels greater distances than it does on Earth before cooling and slowing to a halt.

Explosive eruptions are also affected by the unique set of conditions that exist on Mars. The atmospheric pressure is so low that any gas bubbles trapped in the magma will undergo tremendous expansion upon reaching the surface. As a result, it will take lesser amounts of dissolved gases in Martian (versus terrestrial)magma for the mixture to foam and shoot out explosively from the vent. On Earth, the confining pressure of the atmosphere requires that magma contains close to 1 percent (by weight) of light gases for it to spray upward as a lava fountain. About 3 to 4 percent is needed for the bubbles to grow large enough to blow the magma to shreds and create an ash cloud. On Mars, thresholds for such disruptive behavior are much lower: respectively 0.03 and 0.2 percent gases by weight. Twenty to thirty times less gas is needed on Mars to achieve similar results.

Characteristics of Tropical Rain Forests

Almost all the action in a rain forest (not just photosynthesis but also flowering, fruiting, predation, and herbivory) happens high in the canopy, the leafy green roof of the forest. Apart from the trees, the vegetation is largely composed of plant forms that reach up into the canopy by climbing the trees (for example, vines) or root in the damp, upper branches (epiphytes). The epiphytes depend on the sparse resources of mineral nutrients that they extract from crevices and pockets of the organic matter of soil found on the tree branches. The plants and animals of the canopy are not easy to study; even to gain access to the flowers in order to identify the species of tree is difficult without the erection of tree walks. It is a measure of the problems of doing research in rain forests that botanists have trained monkeys to collect and throw down flowers and a research team has used hot air balloons to move over the canopy and work in it.

Most species of animals and plants in tropical rain forests are active throughout the year, though the plants may flower and ripen fruit in sequence. In Trinidad, for example, the forest contains at least 18 closely related trees in the genus Miconia whose combined fruiting seasons extend throughout the year; this contrasts with the situation in temperate latitudes.

Dramatically high species richness is the norm for tropical rain forests, and communities rarely, if ever, become dominated by one or a few species-a very different situation from the low biodiversity of northern coniferous forests. This raises some fundamental questions. First, what is it about the evolutionary history of tropical rain forests that has allowed such diversity to evolve? Part of the answer relates to the comparative stability of patches of rain forest during the ice ages. It is thought that during these periods, drought forced tropical rain forests to contract into islands in a sea of savanna (flat grassland), and these expanded and coalesced again as wetter periods returned. This would have promoted genetic isolation of populations, allowing individual species to become well established before joining other species when rain forests grew again. This phenomenon of isolation is so important for a variety of species to develop. We may also ask why it is that among the diversity of species in tropical rain forests, a few have not dominated and suppressed the rest in a struggle for existence. At least part of the answer is that populations of specialized pathogens and herbivores develop near mature trees and attack new recruits of the same species nearby. Thus, the chance that a new seedling will survive can be expected to increase with its distance from a mature tree of the same species, reducing the likelihood of dominance by one or a few species in the forest.

The diversity of rain forest trees provides for a corresponding diversity of resources for herbivores. A variety of fresh, young leaves are available throughout the year, and a constant procession of seed and fruit production provides reliable food for specialists such as fruit-eating bats. Moreover, a diversity of flowers, such as epiphytic orchids with their specialized pollinating mechanisms, requires a parallel, specialized diversity of pollinating insects. Rain forests are the center of diversity for ants-43 species have been recorded on a single tree in a Peruvian rain forest. And there is even more diversity among the beetles; Erwin (1982) estimated that there are 18,000 species of beetles in 1 hectare (10,000 square meters) of Panamanian rain forest compared with only 24,000 in the whole of the United States and Canada.

There is intense biological activity in the soil of tropical rain forests. Leaves that fall to the ground decompose faster than in any other ecological community. As a result, the soil surface is almost bare. The mineral nutrients in fallen leaves in rain forests are rapidly released and are unlikely to be removed by natural processes. In other ecological communities, rains may carry nutrients well below the upper soil levels at which roots can recover them. By contrast, the mineral nutrients in a rain forest are safe from loss to the lower soil layers and almost all are in fact held by the plants themselves.

Economic Reasoning

Economics begins with the assumption that human actions are rational-intended for maximum self-benefit. For example, people pay more for a larger box of the same cereal than for a smaller one. Yet people sometimes seem to act irrationally. When a celebrity endorses a product, sales increase although the endorsement appears to convey no information about product quality. Nonetheless, economists stubbornly insist on finding rational explanations for all behavior. Why? Imagine a physicist who encounters his first helium-filled balloon. which flies up rather that down when released near the ground, a blatant challenge to the laws of gravity. Two courses are open to him. He can say. "Well. the laws of gravity have exceptions." Or he can say, "Let me see if there is any way to explain this strange phenomenon without abandoning basic principles of physics." If he takes the latter course, and if he is sufficiently clever. he will eventually discover the properties of objects that are lighter than air and recognize that their behavior is in perfect harmony with existing theories of gravity. In the process he will not only learn about helium-filled balloons; he will also come to a deeper understanding of how gravity works.

Now it might very well be that there are real exceptions to the laws of gravity, and that our physicist will one day encounter one. If he insists on looking for a good explanation without abandoning his theories, he will fail. If there are enough such failures, new theories will eventually arise to supplant the existing ones. Nevertheless, the wise course of action, at least initially, is to see whether surprising facts can be reconciled with existing theories. The attempt itself is a good mental exercise for the scientist, and there are sometimes surprising successes. Moreover, if we are too quick to abandon our most successful theories, we will soon be left with nothing at all.

Much primitive agriculture shares a strange common feature. There are very few large plots of land: instead, each farmer owns several small plots scattered around the village. (This pattern was endemic in medieval England and exists today in less developed parts of the world.) Historians have long debated the reasons for this scattering, which is believed to be the source of much inefficiency. Perhaps it arises from inheritance and marriage: At each generation, the family plot is subdivided among the heirs, so that plots become tiny; marriages then bring widely scattered plots into the same family. This explanation suffers because it seems to assume a form of irrationality: Why don't the villagers periodically exchange plots among themselves to consolidate their holdings?

Inevitably. this problem attracted the attention of the economist and historian Deirdre McCloskey. whose instinct for constructing ingenious economic explanations is unsurpassed. Instead of asking "What social institutions led to such irrational behavior?" McCloskey asked, "Why is this behavior rational?" Careful study led her to conclude that it is rational because it is a form of insurance. A farmer with one large plot is liable to be completely ruined in the event of a localized flood. By scattering his holdings, the farmer gives up some potential income in exchange for a guarantee that he will not be wiped out by a local disaster. This behavior is not even exotic. Every modern insured homeowner does the same thing. One way to test McCloskey's theory is to ask whether the insurance "premiums" (that is, the amount of production that is sacrificed by scattering) are commensurate with the amount of protection being "purchased." using as a standard the premiums that people are willing to pay in more conventional insurance markets, and by this measure it holds up well. A serious criticism is this: If medieval peasants wanted insurance, why didn't they buy and sell insurance policies, as we do today? A simple answer is that nobody had yet figured out how to do it. But some economists demand more exact answers to this objection, and they are right. Theories should be tested to their limits.

Facial Expressions

Even without the more elaborate forms of nonverbal communication like sign language, gestures, or miming humans can actually communicate a great deal using only their bodies. Much of this, such as the posture of a person who is sad, is unintentional, but we can give deliberate signals with our bodies as well. Invading someone's space can be a way of intentionally signaling aggression. Other species also use body language, which was probably one of the earliest forms of communication between animals. However, unlike many other animals, humans also have an impressive ability to communicate things using only our faces. Once again, this can be intentional or unintentional. Our facial expression is almost constantly advertising our current emotional state and can even betray us by revealing feelings that we attempt to hide. We can also add important nuance to verbal communication using our faces. Given the right context, a simple glance at a friend can often communicate a great deal.

Many of the cues and signals that come from our facial expressions are universal among all of the people in our species. This indicates that this form of communication is much more innate (inborn rather than learned) than the recent innovation of spoken language. For example, the languages spoken among the hunter-gatherer tribes of New Guinea have absolutely nothing in common with English or any other Indo-European language. (Most of them have nothing in common with each other!) Yet, a smile means exactly the same in the rain forests of New Guinea as it does in the streets of New York City. This speaks to the unconscious, genetic, and inborn nature of facial expressions because if facial expressions were learned, like language, we would expect divergence among the cultures of the world. Instead, we see striking universality. Further proof of this is that people born blind display the same facial expressions as their sighted neighbors. They could not have learned the expressions by seeing them in others. It seems clear that facial expressions and their meanings are almost completely built into the human brain.

Many animal species also communicate with each other using facial expressions. This appears largely limited to mammals for two reasons. First, invertebrates, fish, amphibians, and reptiles, even if they live in communities. show only the simplest hints of interactive social dynamics and cooperation. When it comes to cooperation and communication in the animal world, birds and mammals are the stars of the show. Second, facial expressions require elaborate overlapping musculature in the facial region, which birds do not really have. There are forty-three muscles in the average human face. Many of them are strange, as skeletal muscles go, because they do not connect two bones in order to allow skeletal locomotion. Instead, they are loosely connected to the skull on one end and skin tissue on the other. In other words, their only purpose is to squeeze and stretch the skin of the face. This sort of thing does not exist anywhere else in the body. Much of our face musculature exists purely so that we can make facial expressions.

This is why mammals have the monopoly on facial expressions. We are the only animal group that has both the social-cooperative nature to use communication in the first place and the necessary muscles to accomplish that communication using our faces. One might be tempted to think that mammals evolved our elaborate facial muscles for the purpose of communication, since that seems like pretty much all we use them for, but it turns out that our face muscles perform an even more basic and essential function: suckling (drinking milk from the mother's breast). By studying very old mammal fossils, paleontologists have discovered that a muscle group that was originally located in the throat was co-opted and relocated to the face of early mammals and resulted in much-improved suckling from the then-recent mammalian innovation: the breast. With the new facial muscles and the highly social context of nursing the needed components were in place for the invention of facial expressions as a means of silent communication by early mammals, who must have lived in constant fear of predation by the dominant animals of their day: the dinosaurs.

Greece Emerges from the Dark Ages

The Mycenaean Greek civilization, which featured cities and a thriving economy, dominated the Aegean and parts of the Mediterranean beginning about 1600 B.C.E. But around 1100 B.C.E., Greek society entered a period of decline known as the Dark Ages that lasted for roughly three centuries. A new Greek culture emerged in the eighth century-the conventional start date is 776 B.C.E.-marking the beginning of the Archaic Age, which brought a revival of Greek trade and culture as the Greeks again became one of the major economic and political forces in the Mediterranean world. This happened for several reasons. For one thing, the Dark Ages had seen a constant population increase, and Greece, with its great expanses of rocky and hilly soil, simply could not grow enough agricultural staples to support a large population. Methods for dealing with the extra mouths were needed.

One response was to import food. As a consequence, Greek trade and manufacturing expanded. In exchange for olive oil, fine pottery, silver, and slaves, the Greeks acquired grain in Egypt and in the lands on the Black Sea coast. The expansion of Greek commerce brought the Greeks into conflict with the current Mediterranean trading power, the Phoenicians. The large,30-oared Greek warships, known as triakonters, of the ninth and eighth centuries B.C.E., were replaced by fleets of iron-beaked ships known as pentakonters, which were 90 feet long and had 50 oars. The clumsy, two-decked Phoenician warships were no match for the more maneuverable Greek ships, and the Greeks soon wrested control of much of Mediterranean trade from their Phoenician competitors, who at this time also were under attack by the Assyrians.

The revival of Greek trade also had other consequences. The Greeks brought back culture as well as grain, and as a result Greek culture evolved based largely on borrowings from the much more ancient and sophisticated cultures of the Near East. From the Phoenicians they borrowed the alphabet, for they now had just as great a need to keep records. In order to adapt the Phoenician alphabet, which had been designed to write Semitic languages, to write their own language, the Greeks added new letters representing vowels to make it more clear which word was meant.

In the late seventh century B.C.E., the Greeks of Ionia adopted coinage from the Lydians who lived in Turkey. In the early sixth century, coinage spread to the mainland, with the island city of Aegina issuing staters (coins of a standard weight)bearing a turtle on one side and a simple punch mark on the other. Around 570 B.C.E., Corinth issued coins bearing an image of Pegasus, the winged horse thought to have alighted at Corinth to drink from a sacred well, and about 525 B.C.E. Athens began issuing its famous "Owls," which bore the head of the goddess

Athena on one side and Athena's owl on the other A multitude of other Greek cities also issued coins as a means of asserting their status and independence. The coins of important trading cities such as Aegina, Corinth, and Athens became standard currency in the Mediterranean world because of their extensive circulation. One problem, however, was that the Greeks could not agree on standard weights for all their coins, and since trading cities attempted to carve out their own spheres of economic influence by favoring the use of coins with their own weights, this resulted in a complicated mathematical calculation if, say, someone wanted to use Corinthian coins to pay for something with a price expressed in Athenian coins.

The Greeks also assimilated ideas about artistic expression from the East. During the Dark Ages, artistic designs had consisted of simple geometric patterns. The Archaic Age introduced a multitude of Eastern styles. Greek sculpture assumed a very Egyptian look, and, in the Orientalizing style, at its height around 750 B.C.E., Greek pottery depicted many Eastern motifs, such as lions, bulls, and sphinxes. But Greek artists soon developed their own idiosyncratic style by incorporating into their designs scenes drawn from their own myths and legends. Greek potters competed with each other to perfect different manufacturing techniques and artistic styles, and they soon were the acknowledged masters of pottery making in the Mediterranean.

Animal Communication

Communication among animals has risks, as when an animal that makes a noise reveals itself to a predator. But it also has important advantages. Communication serves animals' needs to survive and reproduce and their ability to project their genes into future generations by aiding close relatives. When a male cowbird sings his song, he is attending to his need to attract a female cowbird and get his genes into the next generation. Worker honeybees, which are honeybees that do not reproduce, perform dances to communicate to other bees in the hive, the location of a food source, with specific paths, speeds, and angles indicating the distance and direction of the food. By imparting this information, a worker bee is aiding the survival of her sisters in the hive and the further reproductive efforts of her mother (this is the only way worker bees can help spread their genes).

Animals do not only communicate with their relatives. Prey also communicate with predators, as when a gazelle stots (jumps up suddenly in the air) to communicate to the lion stalking it that he has been spotted and thus is unlikely to catch the gazelle. They should both save themselves the bother of a pointless chase that will just waste their time and energy. Sometimes predators listen in on the communications of their prey (as when dolphins detect the sounds made by fish). In discussing animal communication, it is customary to identify a signaler and the recipient of a signal. This does not imply that either individual is consciously aware or even unconsciously intends to send or receive the signal. An older view construed communication as a cooperative activity in which a sender and receiver collaborated to get a message across. It is now understood that the picture can be much more complex. Receivers may eavesdrop on signals that the signaler would rather they did not pick up. Bats, for example, listen in on the mating calls of male frogs in order to locate and attack them.

Some species are sensitive to the presence of potential eavesdroppers and modify their signals accordingly. Monkeys that raid farmers' crops have quiet alarm calls, known as "conspiratorial whispers" that they use to warn each other of the arrival of the farmer. These calls are clear enough that the monkeys hear each other but quiet enough that the farmer does not detect them. In other situations, signalers compete to get their message across with ever louder, larger, and brighter signals. It is thought that the large bright tail of the peacock evolved in this way as a form of competition among male birds to impress the females.

The tail of the peacock is an example of an honest signal (one that is intended to provide accurate information). The size and bright coloration of an impressive male's feathers indicate health, requiring adequate nutritious food, energy and preening (grooming) to maintain. Peacocks do not have the resources to fake their tails. Furthermore, having bright, flashy feathers or a long train that might slow you down can come at the cost of being more susceptible to predators. A male that is fit enough to survive despite this potential handicap must surely be a prize for a lucky female.

The deep notes of frogs' and toads' mating calls are also honest signals of their owner's size. The laws of physics impose a necessary relationship between the depth of a tone and the size of the body that produces it (a big body means a deep call). While this is generally true, a recent study showed that males of one species of green frog, Rana clamitans, may sometimes lower the dominant frequency of their calls during territorial contests in order to make the calls sound deeper and the frog seem larger than normal. In other words, not all signals are honest. Examples of dishonest signals include the behavior of mimics, which copy the behavior of another individual. For example, two species of fireflies emit similar patterns of flashes of light. Females of one species, the Photinus species, use their characteristic flashing pattern to attract mates. Females of the other species, the Photuris species, mimic the flashing pattern of Photinus females so that they can lure Photinus males and eat them.

The Choreography of Merce Cunningham

Choreographer Merce Cunningham was an experimentalist who, in his final years, was almost routinely hailed as the world's greatest living choreographer. Cunningham productions -many of them collaborations with composer John Cage- aroused much controversy. Three aspects of Cunningham's choreographic theories were especially provocative: his use of chance and indeterminacy (not precisely determining the completion of his works but leaving some choices up to the performers); his treatment of stage space; and his tendency to regard the components of a dance production as independent entities.

Wishing his dances to possess some of the unpredictability of life itself, Cunningham began to make choreographic use of chance. For Cunningham, chance did not mean chaos, and his dances were not improvisations or free-for-alls. Typically, he prepared in advance a multitude of movement possibilities-more than he may actually have needed in a work-and then decided which sequences (groups of movements) he would use by some simple device, such as flipping a coin. Or he would choreograph works in which the episodes might be performed in any order. Since Cunningham prepared so much, one might ask why he bothered with chance at all. He would reply that chance can reveal to a choreographer ways of combining movements that the rational, conscious mind might not have otherwise thought of on its own. All people are, to some extent, prisoners of their mental habits. But Cunningham believed that through chance, choreographers could free themselves from habit and discover attractive new movement sequences.

A theatrical preoccupation with indeterminacy is perhaps to be taken for granted in an age of mechanical formats, such as films, recordings, and television. One characteristic of the mechanical is fixity: once something is on film, it will stay that way forever (or at least until the film decays). But even when live performers are trying to speak the same words, sing the same music, or dance the same steps, every live performance differs, if only slightly, from every other live performance. Therefore, it can be argued that in his concern for indeterminacy, Cunningham was calling attention to the indeterminacy inherent in all live theater.

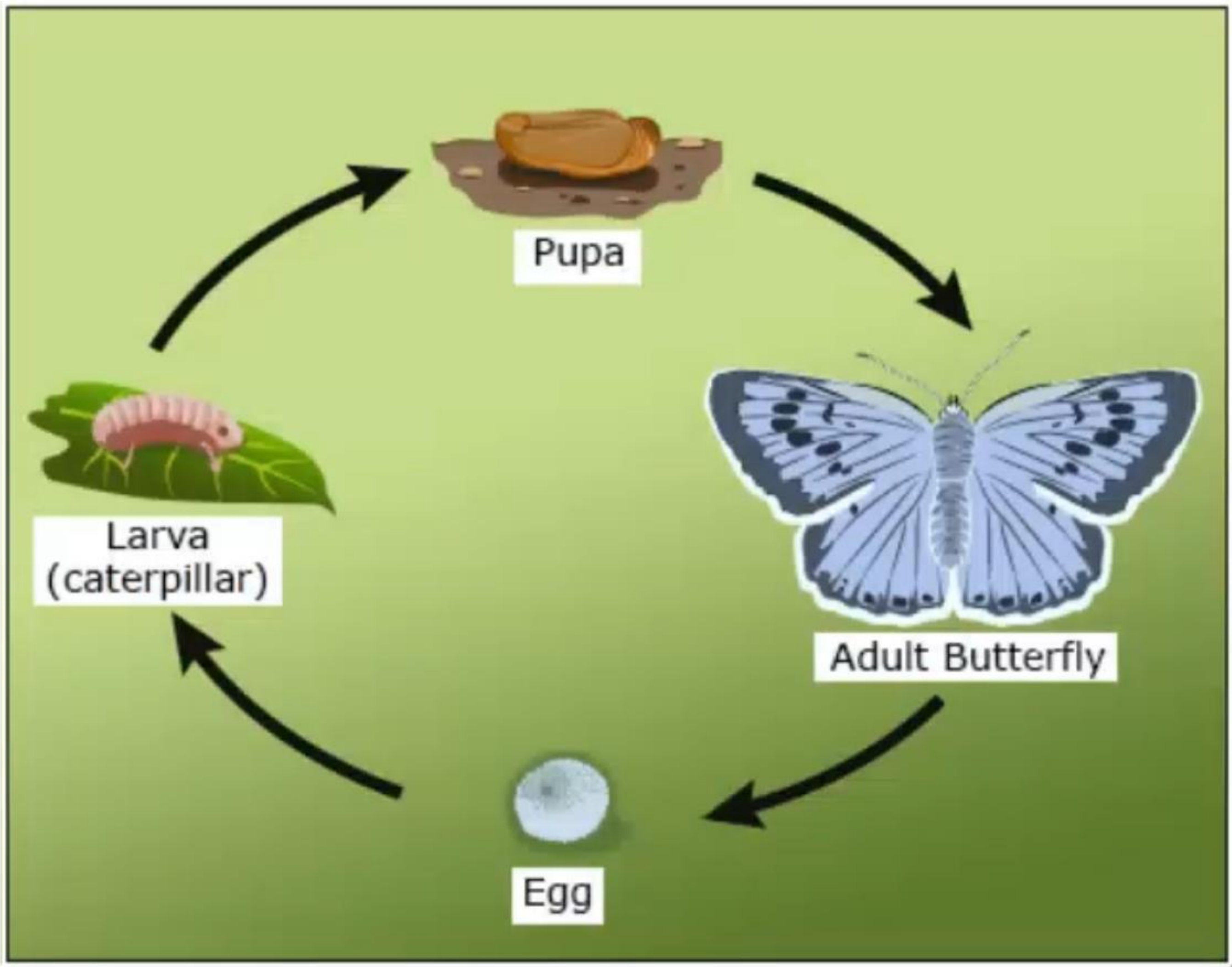
A second characteristic of Cunningham's choreographic approach was his treatment of stage space. Whereas many dances are structured around a central focus (a ballerina or danseur, or a hero or heroine, who may be framed by an ensemble) Cunningham gave equal importance to all parts of the stage. Corners and sides were as important as center stage, and many things happened simultaneously in different stage areas. Instead of being choreographically guided toward a single point, the spectator's eye was left free to roam as it pleased across a field of activity.

Finally, Cunningham regarded the components of a production (movement, music, decor) as coexisting independent entities. The dancers' steps were not phrased to coincide with the musical phrases; the scenery did not illustrate the choreography. Dance, music, and scenery simply occurred in the same space and time. The most radical manifestations of Cunningham's belief in the independence of theatrical elements occurred in what he termed "Events". These 90-minute productions, performed without intermission, consisted of choreographic sequences from dances already in the Cunningham repertoire. Yet the sequences were not presented to the audience as detached excerpts. Instead, they were woven together and performed to musical scores other than those for the works from which they derived. Thus, Events presented old movements in a new context, and the change of context often drastically altered the effect of the movements. Through his Events, Cunningham emphasized that everything is subject to change.

Each of Cunningham's dances managed to have its own special mood. Although there is nothing really tropical about Rainforest, the choreography is sensuous, and Andy Warhol's decor of floating silver pillows somehow seems appropriate for such a lush creation. Quartet, which despite its title, is a dance for five (one outsider keeps trying in vain to join a group of four people) can be interpreted as an image of social exile or the gulf between generations. Cunningham's dances consist of many different elements, but like objects in a landscape, they may cohere to produce an unmistakable atmosphere.

The Decline of the Large Blue Butterfly in England

The large blue butterfly undergoes a four-phase life cycle: egg, larva, pupa, and adult.



Adults lay eggs on plants that provide food for the larvae, known as caterpillars. When the caterpillar is fully grown, it enters the pupa stage, from which it emerges as a winged adult. The population of the large blue declined greatly in southern England in the late nineteenth century; this was attributed to poor weather. Final extinction of the species in England in 1979 followed two successive hot, dry breeding seasons. Since the beautiful butterfly is highly sought after by collectors, excessive collecting was presumed to have caused the long-term decline that made the species vulnerable to deteriorating climate.

In an attempt to protect the butterfly and its habitat, a reserve was established in the 1930s that excluded both collectors and domestic livestock, but this did not stop the decline. Evidently, habitat had changed through time, including a reduction in wild thyme, which provides the food for the early development period of the large blue's caterpillar. Moreover, the loss of grass-eating rabbits (through disease) and the exclusion of grass-eating cattle and sheep from the reserve harmed short-turf grasses, which require periodic grazing to flourish. Short woody plants (shrubs) replaced the short-turf grasses that had provided an optimal environment for the large blue. Thyme survived; however, the butterflies continued to decline to extinction in Britain.

A more complex story has since been revealed by research associated with the reintroduction of the large blue to England from continental Europe. The larvae of the large blue butterfly in England and on the European continent prey on colonies of Myrmica ant species. Larval large blues must enter a Myrmica nest to feed on larval ants. Similar predatory behavior and/or tricking ants into feeding them as if they were the ants' own offspring are features of the natural history of various butterfly groups. After hatching from an egg laid on the larval plant food, the large blue's caterpillar feeds on thyme flowers until it reaches the final larval phase, around August. At dusk, the caterpillar drops to the ground from its birth plant, where it waits motionless until a Myrmica ant finds it. The worker ant observes the larva for an extended period, perhaps more than an hour, during which the ant eats a sugary substance secreted from the caterpillar's nectar organs. At some point the caterpillar becomes swollen and adopts a posture that seems to convince the ant that it is dealing with an escaped young ant, and the caterpillar is carried into the ant nest. Until this stage, the caterpillar's growth has been modest, but in the ant nest the caterpillar becomes predatory on ant eggs and larvae and grows rapidly. The caterpillar spends the winter in the nest and, nine to ten months after entering the nest, pupates (passes through the pupal stage)in early summer of the following year. The caterpillar requires an average of 230 immature ants for successful pupation. It apparently escapes predation by the ants by secreting surface chemicals that mimic those of the ant offspring, and probably receives special treatment in the colony by producing sounds that mimic those of the queen ant. The adult butterfly emerges in the summer and departs rapidly from the nest before the ants can identify it as an intruder.

Adoption and incorporation into the ant colony turns out to be the critical stage in the life history. The complex system involves the 'correct' ant, Myrmica sabuleti, being present, and this in turn depends on the appropriate microclimate associated with short turf grass. Longer grass causes cooler near-soil microclimate conditions, favoring other Myrmica species, including Myrmica scabrinodes, which may displace Myrmica sabuleti. Although caterpillars associate apparently indiscriminately with any Myrmica species, caterpillar survivorship differs dramatically; with M. sabuleti approximately 15 percent survive, but an unsustainable reduction to less than 2 percent survivorship occurs with M. scabrinodes. For large blue populations to succeed, more than 50 percent of the adoption by ants must be by M. sabuleti. Other factors affecting survivorship are that the ant colony must provide enough larvae to feed the caterpillar well and that it must lie within 2 meters of the birth thyme plant. Such nests are associated with newly burnt grasslands, which are rapidly colonized by M. sabuleti.

The Nile River Valley

Though the verdant green course of the Nile Valley appears to be the very antithesis of the bleached Saharan sands through which it flows, these two contrasting features have together constituted a primary force in the development of agriculture and settled human society in Africa. The Sahara acted as a pump, drawing people from surrounding regions into its watered environments during the good times and driving them out again as conditions deteriorated (though not necessarily returning them to their point of origin). The Nile, for its part, was a refuge for people retreating from the desert and then a reservoir from which the desert region was repopulated as conditions improved.

The Nile is commonly believed to have been a conduit along which the principles of food-crop cultivation moved from Egypt into more southern regions of Africa along with the wheat, barley, peas, and lentils initially domesticated some 9,000 years ago in Southwest Asia. It is true that the ancient Egyptian civilizations founded on the Nile over 5,000 years ago were sustained by the exceptional productivity of those crops, but they were a comparatively recent introduction, arriving long after the cultivation of indigenous African plants had begun farther south.

Contrary to expectations, the earliest evidence of a shift toward a critical dependence on food production as opposed to food gathering comes not from the floodplain of the Nile but from sites in what is now the empty and waterless Sahara. The development was complex, involving the domestication of plants and livestock, technological innovation, the establishment of villages, and an increasing level of social interdependence.

The Sahara Desert had been widely inhabited until the last glacial maximum, when conditions of increasing aridity drove people (and practically all animals) out of its previously productive wooded grassland and savanna environments. Essentially nomadic—though exhibiting a tendency to settle at lakesides and other sources of food and water —the groups moving to the north and south continued to follow their established hunting and gathering way of life. The groups that moved east into the Nile Valley, however, adopted a distinctly sedentary lifestyle. Indeed, they had no choice. The narrow strips of green floodplain flanking the Nile were bounded by waterless desert, and movement along the riverside plains was restricted by the presence of competing groups.

The Nile along which people congregated 18,000 years ago was a quite different river from that which flows through Egypt today. It was much smaller and flowed more slowly through a tangle of braided channels across a wide elevated floodplain rather than along a single massive stream. The river carried a heavy sediment load, and the silts deposited along its length steadily raised the level of the river and its floodplain.

In effect, the Nile Valley at that time was an elongated oasis: a sharply defined area of inhabitable territory beyond the boundaries of which human survival was impossible. The oasis extended all the way from the Sudan to what is now Cairo—a distance of over 800 kilometers—but it was nowhere more than a few kilometers wide. Its food resources were varied and nutritious. Quantities and availability, however, were always subject to seasonal variation and the vagaries of the Nile's annual flood. During the critical two to four weeks when the flood was at its height each year, people and animals were driven out of the Nile Valley and crowded into the narrow band of inhabitable land that lay between the floodwaters and the arid desert beyond.

People hunted large mammals for meat and hides, but environmental and territorial constraints meant that only small numbers of a limited range of species were available. Indeed, large mammals constitute a very small proportion of the faunal remains found at archaeological sites, and only three species are represented: the hartebeest, the dorcas gazelle, and a type of wild cattle called the aurochs. Birds—particularly coots and migratory geese and ducks—were caught regularly. Fish, however, were by far the most important source of protein—primarily catfish. For carbohydrates, the Nile Valley offered its inhabitants a considerable variety and seasonal abundance. Twenty-five different seeds, fruits, and soft vegetable tissues have been distinguished among archaeological remains.

Cloth Manufacturing in England

By the fourteenth century, the European cloth industry, which had previously been centered mainly in Flanders (northern Belgium), Italy, and Holland, transferred to England, where fewer property taxes and market restrictions made greater profits for cloth makers. In England, the industry's production methods yielded improved cloth and changed the economy.

A technique called fulling was widely used in England and caused manufacturers to relocate from urban areas of England to different rural villages all over the country. The process of fulling involves submerging a cloth in water (usually containing a natural clay detergent called fuller's earth) and beating it vigorously. Proper fulling will shrink the cloth, making the fabric tighter and stronger, and make the surface smoother and softer. Three traditional methods of fulling were used: a submerged cloth was beaten with the feet, with the hands, or with clubs. One of these is depicted in an ancient wall painting in Pompeii of a man standing in a trough containing water and pounding a cloth with his feet. These traditional methods were still used in Flanders, Italy, and, for a time, England. But then, in the eleventh or twelfth century (historians disagree on the precise date), a new method was introduced: two wooden hammers were attached to a drum and a crank was turned to raise and drop the hammers on the cloth. The real breakthrough came when this device was hooked to a water mill (one probably constructed to grind grain). As a result, a single operator overseeing a series of hammers could perform the work that had previously required a crew of fullers-and he could do it much more quickly, too.

This is why the historian Eleanora Carus-Wilson observed that the invention of the fulling mill "was as decisive an event [for the woolen industry] as were the mechanization of spinning and weaving in the eighteenth century.' Whether fulling mills developed in the eleventh or twelfth century, they were so common in the thirteenth century that they revolutionized the English industry, leaving the Continent (the rest of Europe) far behind. Fulling gave English cloth a significant advantage on the international market. With many fewer fulling mills, cloth makers on the Continent could full only some of their cloths, which entailed a large sacrifice in quality.

The ascendance of the fulling mill helps explain the English woolen industry's marked preference for villages and rural areas on good streams. Such locations had several additional advantages. Moving water was useful for dyers, who needed to rinse excess dye from their cloths. Moreover, locating in rural areas permitted firms to escape the repressive regulations imposed by guilds (professional organizations of practitioners of the trades), to avoid the higher taxes of towns and cities, and to pay lower wages (the cost of living was lower in rural areas than in cities).

Why didn't the woolen industry on the Continent similarly disperse to small towns and villages? Because in Continental Europe only the cities provided enough freedom and property rights to sustain industry. On the European Continent, the rule of the nobility prevailed in the countryside, and everyone had to fear the local lord's greed. But in England, freedom and security prevailed throughout the country, and medieval English industrialists did not need to concentrate in crowded expensive, dirty cities-many devoid of water power - as their counterparts in Flanders, in Holland, along the Rhine, and in Italy were forced to do. As a result, the English woolen industry was remarkably decentralized.

Finally, dispersion and relatively unimpeded capitalism may have contributed to the international dominance of English woolens by encouraging the production of more fashionable and attractive products. As the historian A. R. Bridbury put it, to explain the success of the English woolen industry, it is not enough to cite better wools or lower prices; what should be stressed is "art and skill... the exotic dyeing of these cloths and... the subtle blending of design and color in their creation... the search for making cloth which would be more fashionable internationally." In textile centers on the Continent, the guilds-very traditional in their outlook-often limited experimentation with colors and designs, and originality nearly always suffers when creative people are crowded together and fully aware of one another's work. England's dispersed woolen industry produced greater variations in styles and quality.

Origins of the Deciduous Forests

Forests dominated by deciduous trees (trees with leaves that fall off at the end of the growing season) are found around the world. But they were not always so widespread, and there are unanswered questions about their evolution. The ancestors of deciduous trees can be traced back to the Cretaceous Period, 113-65 million years ago, when dinosaurs ruled the land and the continents occupied different positions from today. At this time, extensive deciduous forests existed only at very high latitudes, near the north and south poles.

Alaska was closer to the North Pole during the late Cretaceous than it is today, but the climate was milder, reflecting higher temperatures for the entire planet. This can be deduced from the shape of fossil leaves. There is a remarkably consistent relationship between leaf shape and mean annual temperature for modern plant species. For example, the proportion of species with toothed leaves consistently increases as one moves from warm to cold climates. Based on statistical analyses of a large number of such leaf characteristics, temperatures in the Cretaceous Arctic generally hovered above freezing during the winter. This is confirmed by the lack of geological evidence of permafrost, a layer of earth that is permanently frozen. However, fossilized tree trunks at Ellesmere Island in northern Canada display damaged sections (growth interruptions) consistent with damage caused by severe frost indicating that occasional frosts occurred. Also, although the winters were generally mild, they were-like modern Arctic winters-very dark. Because of the tilt of Earth's axis, very little sunlight reaches high northern latitudes for several months each year, so trees and shrubs of the ancient Arctic could not photosynthesize(produce chemical energy from sunlight) for several weeks or even months depending on the latitude. This is apparent from growth rings from fossilized wood of the period; the annual growth rings end abruptly, evidence of a rapid interruption of photosynthesis and plant growth with the onset of winter darkness.

The combination of mild temperatures and dim light during the winter may explain why so many Cretaceous trees shed their leaves at the end of the growing season. This would help reduce water loss and energy use during an extended period of darkness. Because the winters were relatively warm, leaves on an evergreen tree, which does not lose its leaves, would continue to undergo respiration (breathing), resulting in a steady loss of energy during a period when photosynthesis was not possible and energy stores could not be replenished. Thus, the food reserves of the plant would be expended to keep the leaves alive. In contrast, deciduous trees dropped their leaves and entered a period of deep dormancy during which they lost relatively little energy. When the Sun reappeared in the spring, new leaves sprang out to take advantage of the short- but almost continuously sunny-growing season.

This explanation for the prevalence of deciduous trees in the ancient Arctic is so logical that it was not rigorously tested for many years. In the early 2000s, however, this hypothesis was undermined by experiments with both evergreen and deciduous species belonging to plant groups that grew in the Arctic or Antarctic forests of the Cretaceous. A team of researchers compared growth rates of young trees for two types of deciduous trees-ginkgo and dawn redwood-and two types of evergreen trees-southern beech and coast redwood-in greenhouses with a simulated (imitated) Cretaceous Arctic winter (dim light and mild temperatures). Surprisingly, the deciduous trees lost more of their food reserves(as measured by the amount of carbon in the trees)during the simulated winter than did the evergreen trees. This is due to the immediate and massive loss of carbon as leaves are shed, which offsets the carbon saved by winter dormancy. If these modern trees responded in a similar way to their ancient relatives, then it is perplexing that evergreen trees did not predominate in Arctic forests. Deciduous trees may have an advantage during the summer rather than during the winter, however, because their leaves could grow and photosynthesize more rapidly than evergreen leaves during the brief growing season.

Proto-industrialization in Europe

The wealthy merchants of the early modern era (approximately 1500-1800) took commodities (products) made by others and shipped them to distant markets. Or they supervised manufacturing enterprises, especially textiles, which worked on the domestic or "putting-out" system. In this system, the merchant acquired raw materials, distributed them to country workers who labored in their own cottages, and collected the finished goods for sale. Employing men, women, and children, this system flourished into the eighteenth century. It constituted a "proto-industrial" phase of European manufacturing, featuring rural production and city-directed exchange.

Technological innovation also contributed to the growth of the European economy. By 1500, Europe already had put machines to work in productive ways. Waterwheels and windmills helped grind grain, brew ale, pump water, make paper, saw wood, and treat textiles. The circulation of goods was facilitated by the building of canal networks, which in turn was made possible by the invention of locks (gates on rivers or canals that change the water level, allowing boats to pass obstacles). Over the next centuries, other new machines and industrial processes improved textile and glass manufacturing, coal mining, and iron production.

All this growth, however, had a disadvantage: rapid inflation. The injection of new quantities of bullion (ingots or gold coins) into the European economy, the deliberate devaluation of existing currencies, and the steady climb of the European population caused an explosion of values and prices. Prices rose for goods, while wages and rents lagged, with regional variations: in Spain, prices more than tripled in the century before 1600; in England, the prices of basic goods rose nearly sixfold during the same period. The pattern of steep price increases affected different groups differently - it allowed merchants to accumulate capital for investment in more manufacture and trade, while it had a very negative impact on peasants and urban workers.

The forward wave of commercial economy favored population increase, which in turn supported the economic boom. The European population generally increased until about 1600, when it surpassed for the first time the pre-bubonic plague level of 73 million. During the seventeenth century, population growth slowed again, owing in part to the disastrous effect of the Thirty Years' War, which reduced the population of central Europe by 30-40 percent. The marriage age increased in many regions, showing that people were adopting a familiar strategy to reduce the birth rate. Spain and Italy fell behind the north Atlantic nations, and France's rate of growth fell behind England's. In the eighteenth century, the population began again to increase as mortality rates declined. After a last outbreak of the bubonic plague in Marseilles in 1720, the bubonic plague left Europe (perhaps because of unfavorable shifts in the ecology of the black rat, the critical carrier of the deadly infectious flea). The cycle of hunger receded too, and the proto-industrial cottage industries, which allowed young persons to make a profit from their labor without delaying marriage, drove the numbers upward.

Despite economic growth, the steady upward pressure on prices and the increasing population brought hardship, especially to the rural poor. Grain prices rose by factors of three, four, even six, while the average price of manufactured goods merely doubled. Bread was expensive or scarce, while landowners strove to extract more labor from their workers. The real wages of agricultural workers fell catastrophically, and the force of hunger, briefly subdued in the aftermath of the bubonic plague, returned. The numbers of the landless increased, and beggars proliferated.

With economic growth came urbanization. Merchants concentrated in cities, where goods changed hands, where banks held funds, where artisans labored, and where consumers purchased. In the early modern centuries, towns became cities, and cities expanded. For the first time, many European cities grew to exceed the largest cities previously known in the Western world (London, Paris, Naples, and Milan). Not only did large cities grow larger, but a greater proportion of Europeans lived in cities and towns; even country workers often migrated to the city for part of their lives before returning to their villages.

Mate Choice in Birds

Male birds are often strikingly different in plumage and size from their female counterparts. Charles Darwin concluded that exaggerated sexual differences such as the tail of a peacock or the displays of the wild turkey evolve as a result of what he called sexual selection - namely, contests among males for mates and female preferences for particular males. As potential male reproductive success increases, so does the value of the characteristics - large size, fancy plumage, intricate songs, and striking displays - that are responsible for the success. The resulting evolutionary process of sexual selection leads to differences between the sexes in size and ornamentation. Darwin's insights into the evolutionary role of sexual selection are now largely confirmed, but the effects of competition among males, female choice, and resources other than mates connect in even more intricate ways than Darwin proposed.

Mutual assessment of prospective partners is a vital aspect of the early stages of courtship and pair formation. The ornaments and displays favored and maintained by sexual selection are those that reliably reflect the superior condition of certain males, enabling females to select the best possible mates. For example, house finch females prefer brightly colored males, which have better survival rates and are better family providers. The familiar flight displays of male bobolinks over lush fields advertise their condition. Females favor males that display longer. Such males have larger fat reserves and consequently raise more young than their neighbors do.

Why do females choose males with more elaborate plumage or displays? The good-genes hypotheses propose that exaggerated male plumage and courtship displays truthfully signal genetic or physiological superiority. Females should recognize superior males and select them to sire offspring. What aspects of genetic or physiological superiority might exaggerated courtship displays serve to indicate? One possibility would be a male's superior survival skill. For example, the enormous tail of a peacock might actually be a handicap during flight or escape. So would be bright colors that attract predators. Males that survive to display such handicaps would have superior stamina or abilities to escape predators. Evolution would tend to favor bigger and bolder badges of this so-called handicap superiority if females preferred to mate with the males that bore such badges.

Another application of the good-genes hypotheses asserts that ornamented plumage provides an index to a male's health, particularly its resistance to pathogens and parasites. Females could detect disease-prone males by the lower quality of their display plumage or by their reduced display stamina. Strong evidence now exists for this hypothesis. Among their many effects, parasites reduce the brightness of ultraviolet coloration of bird feathers. The glittering blue plumage of male satin bowerbirds, for example, has a single wavelength peak in the ultraviolet. The visual intensity of this peak predicts the male's level of infection by blood parasites, because ultraviolet shine decreases with increasing infection.

Other evidence for the use of a bird's appearance as an index to its health comes from studies of red junglefowl, which are the ancestors of domestic chickens. Marlene Zuk and her colleagues first established that hens of the red junglefowl mated more quickly with roosters bearing large, fleshy, red combs on their heads. The hens use comb size as an index to the health of a potential mate. Comb size is strongly affected by the level of the hormone testosterone, which, in turn, affects the bird's physical condition. Intestinal worms reduce comb size, with the result that hens prefer roosters without worms over those that are infected.

The large sizes and conspicuous plumage favored in reproductive displays may be disadvantages in other regards. Large size itself requires greater energy expenditure. There is some evidence that large male red- winged blackbirds are at a disadvantage because they must sacrifice display time for feeding. Among species of North American blackbirds, males that are much larger than females tend to suffer greater mortality as nestlings. Similarly, because they grow twice as fast as females to reach their full adult size by the end of their first summer, male western capercaillie, a huge species of European grouse, are more vulnerable than females to starvation when food is scarce. And the same bright colors that announce a male's presence to potential rivals or mates may also attract predators.

Theatrical Staging

Staging, the most obvious function of a play's director essentially involves positioning the actors on the stage or set and moving them about in a theatrically effective manner. The basic architecture of staging is called "blocking", which refers to the timing and placement of a character's entrances, exits, rises, stage crosses, and other major movements of all sorts. The blocking pattern that results from the interaction of characters in motion provides the framework of an overall staging; it is also the physical foundation of an actor's performance (many actors have difficulty memorizing their lines until they know the blocking that will be associated with them).

The director may block a play either by preplanning the movements ("preblocking") on paper or by allowing the actors to improvise movement (or create it on their own without preparation) on a rehearsal set and then settling on a blocking pattern sometime before the first performance before an audience. Often a combination of these methods is employed, with a director favoring one method or the other depending on the specific demands of the play, the rehearsal schedule, rapport with the acting company, or the extent of the director's own preparation. Complex or stylized plays and settings and short rehearsal periods usually dictate a great deal of preblocking; simple domestic plays and experienced acting ensembles are often accorded more room for improvisation. Each method can produce highly commendable results in the right hands and at the right time. Both, however, can present serious problems if misapplied or ineptly handled.

For the most part, the blocking of a play is hidden in the play's action; it tends to be effective insofar as it is not noticed and insofar as it simply brings other values into play and focuses the audience's attention on significant aspects of the drama. By physically enhancing the dramatic action and lending variety to the play's visual presentation, a good blocking pattern can play a large role in creating theatrical life and excitement. But beyond this, there are moments when inspired blocking choices can create astonishing theatrical effects-effects that are not hidden at all but are so surprising and shocking that they compel intense consideration of specific dramatic moments and their implications. Such a dramatic effect was achieved, for example, by director Peter Brook in his celebrated 1962 production of King Lear, when Paul Scofield, as Lear, suddenly rose to his feet and, with one violent sweep of his arm, overturned the huge oak dining table at which he had been seated, sending pewter mugs crashing to the floor as he raged at his daughter Goneril's treachery. This stunning action led to a reevaluation of the characters of both Lear and Goneril and of the relationship between this tempestuous and sporadically vulgar father and his socially ambitious daughter.

Some plays require specialized blocking for certain scenes (for duels, for example, or dances). Such scenes demand more than basic blocking and are frequently directed by specialists, such as dueling masters or choreographers, working with the director. These specialized situations are not at all rare in the theater (almost every Western play that was written before the nineteenth century includes a duel or a dance or both) and the ability to stage an effective fight scene or choreographic interlude (or at least to supervise the staging of one) is certainly a requisite for any director who aspires to work beyond the strictly realistic theater.

"Business" is a theater term that refers to the small-scale movements a character performs within the larger pattern of entrances and crosses and exits. Mixing a drink, answering a telephone, adjusting a tie, shaking hands, fiddling with a pencil winking an eye, and drumming on a tabletop are all "bits of business" that can lend a character credibility, depth, and fascination. Much of the stage business in a performance is originated by the actor (usually spontaneously over the course of rehearsal) although it may be stimulated by a directorial suggestion or request. The director ultimately must select from among the rehearsal inventions and determine what business will become a part of the finished performance; when this determination is made, bits of business become part of the blocking plan.

Pollen Gathering by Animals

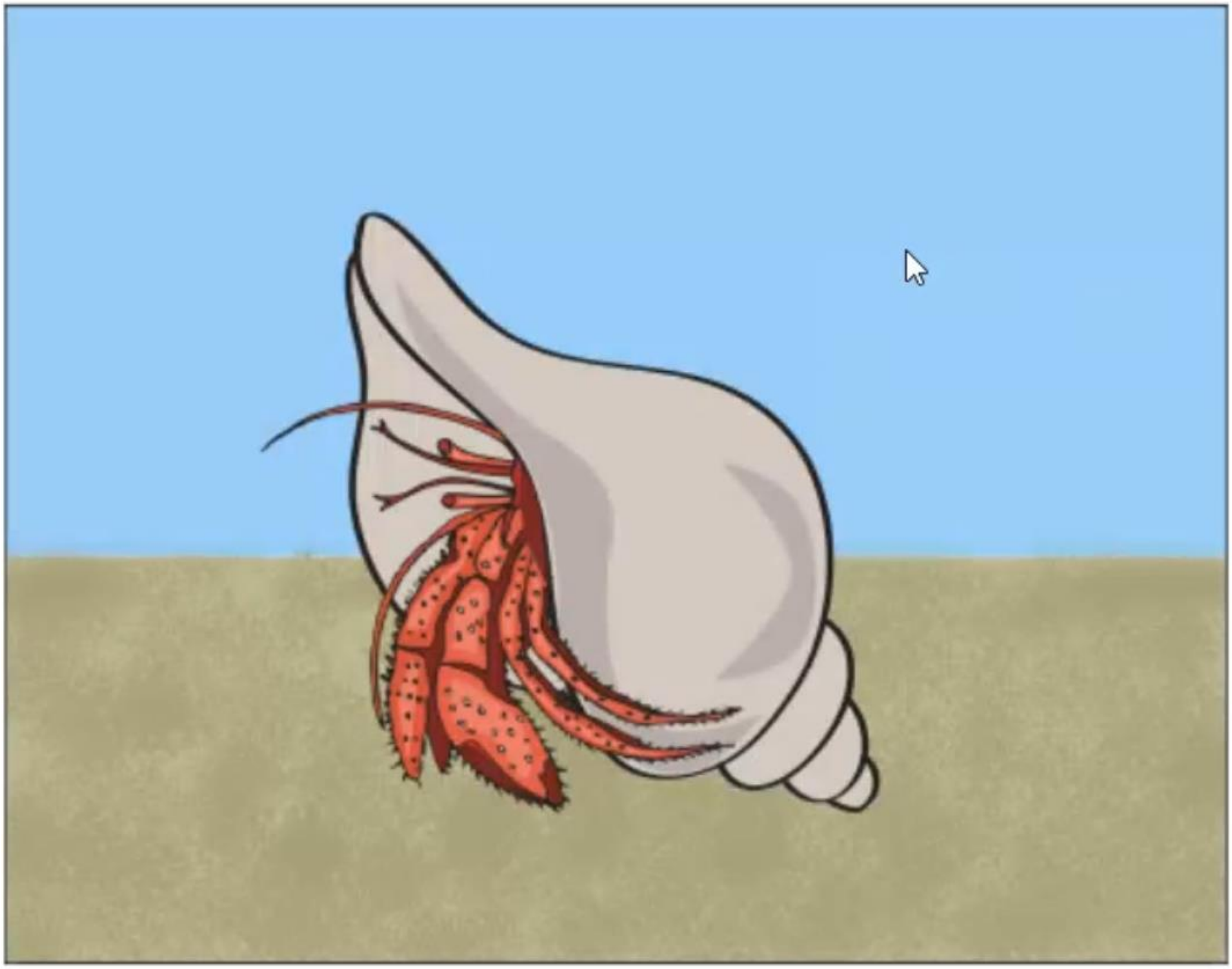
Pollen contains a seed plant's male reproductive cells, and in flowering plants it is produced by anthers, part of the male reproductive organs. Pollination in flowering plants is the process whereby pollen is transported from a flower's anthers to the female reproductive organ (stigma), enabling reproduction. This is sometimes accomplished by wind or water, but pollen is more commonly transported by animals that visit flowers, especially bees, butterflies, moths, and other insects, but also including some birds reptiles, and mammals. It is generally assumed that most pollen is transferred passively (without deliberate effort) onto a visiting animal's body when physical contact is made. Pollen adheres to the surfaces of visitors because of its naturally sticky properties. Its adhesion is often enhanced by the characteristics of the animal's surfaces: branched hairs for bees, scaled surfaces for butterflies, and moths, and fur or feathers in vertebrates.

There are some indications that what is termed passive gathering can be augmented by electrostatic attraction between pollen grains and animals' bodies. Plants are inherently slightly negatively charged in warm, still air, the magnitude of the charge field depending on the plant's composition and architecture and on weather conditions. The charge is greater at sharp tips and is therefore often higher in flowers than elsewhere. Hence it is possible that pollen grains acquire negative charges within the flower. In contrast, insects, and especially hairy bees, tend to acquire a positive surface charge as they fly. The charge difference might aid the transfer of pollen onto visiting insects. In short, even without direct contact, pollen grains could be attracted onto a closeup bee. Pollen might also be unloaded from bees onto dry stigmas in reverse fashion (the pollen having acquired a positive charge while on the flying insect), especially where stigmas stick out from flowers and thus concentrate charge. Small electrostatic charges could also lead a visiting insect to leave an electrical "footprint" on flowers, detectable to subsequent visitors for at least several minutes and thus potentially allowing avoidance of recently depleted flowers.

Rather few kinds of pollinators eat pollen and so gather it deliberately, but those that do are important and numerically dominant kinds. Syrphids (a type of fly that gathers in high numbers) may eat pollen directly from the anthers. A few bees also gather pollen directly with their mouthparts, eating some immediately and carrying the rest back to the nest in their crop (a part of the foregut that functions like an expandable sack). In such cases, only the pollen that is accidentally scattered during consumption (or that passively adheres to other parts of the insect) is of any use for pollinating. Most bees normally do not eat pollen at the flower but instead collect it-with their legs and mouthparts-by rubbing the underside of their body across a flower, or by using various comblike or rake-like structures. In these cases, the pollen is then packed away on or into specific collecting areas of the insect's body, often with the addition of some moistening flower nectar, a sugary liquid. Hence there may be substantial interruptions to food gathering while a bee lands, grooms (removes pollen from its hair), and packages its pollen, although more advanced bees require only a brief hover between flower probings to achieve grooming and packaging.

Although active pollen gathering may be highly beneficial to a flower-visiting animal, there are problems from the plant's point of view. First, much of the pollen is hidden away in sites on the visitor where it will never contact a stigma. Therefore, plant evolution might act to place as much pollen as possible in sites on the visitor that are hard to groom. With suitable arrangement of floral parts, there is then a good chance that the stigma of another flower of the same species will receive some pollen from the insect. Second, many insect pollinators, especially ants, have special glands that produce small quantities of substances that protect their nests against fungal and bacterial growth but that can also kill pollen. Third, when saliva and nectar are added to the pollen gathered by bees, the pollen may suffer damaging biochemical change, since bees do not "want" the pollen to germinate (develop) once deposited in the nest. For such reasons, usually the longer pollen adheres to a visitor, the more its viability is compromised.

Hermit Crabs in Snail Shells



Hermit crabs depend on a properly fitting snail shell for protection from predators. Crabs in shells that are too small grow more slowly, are less tolerant of air exposure and drying out, and are more likely to be eaten by a predator than those that can withdraw completely into their shell. When a crab has outgrown its snail shell, it has to look for a bigger one that fits properly. They locate shells by smell, either when the preceding owner (the snail) dies and begins to decay or by detecting calcium, the major component of snail shells. The crabs do not generally kill snails to find new shells, although they may fight with other hermit crabs over empty shells, especially when empty shells are scarce. After locating a new shell, the crab investigates its surface and internal size by rolling it over and exploring it with its claws and walking legs. If it looks like a good fit, the crab withdraws its abdomen from its old shell and inserts it into the new one so quickly that it can be difficult to see. Switching shells is risky-the crab could be attacked by a predator or lose one or both shells to other hermit crabs.

Crabs rarely abandon their current shells without a new home lined up. However, if a crab has been buried in the sand during a storm or flooding event, it is likely to abandon its shell to return to the surface. Although abandoning its shell increases its chances of surviving such a burial, this behavior also increases the risk of predation and of being buried or injured in another flooding event.

Particular species of hermits tend to prefer certain species of snail shells, but size is the critical factor. Occupying larger and heavier shells gives them greater protection, but it takes more energy to lug a large house around. Shells often become covered with other organisms, such as barnacles or algae, that make them heavier and more difficult to carry. Some hermit species live in shells that get entirely covered by bryozoan colonies that eventually dissolve the snail shell, leaving the crab with a house composed entirely of the bryozoans. As the bryozoan colony grows, so too do the crabs, so these crabs have no need to change their homes.

Hermit crabs with poor-fitting shells are chemically attracted to dying gastropods (snails and related animals) and other hermit crabs, where a shell may become available. In the Mediterranean species Clibanarius erythropus, gastropod predation sites attract dozens of hermit crabs. Researcher Elena Tricarico and colleagues observed that these aggregations function as shell exchange markets: the first crab to arrive takes the empty shell, and a chain of shell exchanges among the crabs follows. They found that simulated snail predation sites quickly attracted a larger number of hermit crabs than other types of sites; therefore, aggregation is the most efficient tactic for this species to acquire new shells. In Belize, the land hermit Coenobita clypeatus has an even more organized shell exchange in what is called a "synchronous vacancy chain" by Randi Rotjan and colleagues, who discovered this unique behavior. When a large, empty shell becomes available, many crabs gather around it, which can take hours. As they gather, the crabs arrange themselves into a line of decreasing size, starting with the largest crab holding onto the empty shell. As though choreographed, the crabs begin shell swapping, one after the other, a smaller crab climbing into a new shell right after it is vacated by the slightly larger crab ahead of it. What makes the synchronous chain possible is that smaller crabs linger near a too-large shell, perhaps attracting others, waiting until a bigger crab comes along, which increases their chances of getting a good-fitting hand-me-down.

Most hermit crabs and most snails are right-handed and spiral clockwise. Some species, however, are left-handed and spiral counterclockwise, and they need to find snail shells that are coiled that way. Since the majority of snail species are right-handed, left-handed hermits may have trouble finding left-handed snails. Adults of the left-handed Petrochirus diogenes, the largest hermit crab in the Caribbean, are often found in shells of the queen conch Eustrombus gigas. As an exception to the general rule, they will attack and eat the conch, thus obtaining both a meal and a shell.

Impressionists and Their Paints

Camille Pissarro's *Boulevard Montmartre*

© Courtesy of Wikimedia Commons/Hermitage

By the mid-nineteenth century, the provision of artists materials in Europe had become a regular trade with its own well-defined distribution channels. Previously, pigments (substances that give paints their color) tended to be categorized either with imported spices or with drugs (since some artists' materials also had pharmaceutical uses), and so they were sold by grocers who dealt also in foods and medications. But as paint manufacturing became increasingly a matter of chemical synthesis(production) rather than the grinding of naturally-occurring pigment, it was transformed into an industry, and most retailers did little more than package the ready-made paints from manufacturers.

Another major innovation in paint supplies was the collapsible metal tube, invented in 1841 by an American portrait painter named John Rand. The tin tubes replaced containers made from the bladders (hollow organs) of pigs, in which oil paints had previously been stored, and made paints far less liable to dry out in their packaging. This was particularly significant for the Impressionist painters- artists who represented the effect of light on objects with short brush strokes of color and preferred working outdoors rather than in the studio; Auguste Renoir, a painter and one of the founding members of the Impressionist school remarked that "without paints in tubes, there would have been no Cezanne, no Monet, no Sisley or Pissarro, nothing of what the journalists were later to call Impressionism."

This commercialization of artists' supplies contributed to the distancing of the painter from his or her materials- the beginnings of that problematic attitude toward the primary substance of paint that was ultimately to impel some artists of the twentieth century to experiment with the use of house paints. J.-F. L. Merimee, a French chemist and painter, complained in 1830 that painters were no longer able to distinguish good materials from bad. And there certainly were bad materials around. To the paint merchant, the bottom line was profit, and sellers often cared little for the quality or long-term stability of their colors. These people, said Merimee, "had a stronger feeling toward their own immediate profit than any regard to the preservation of pictures." They used binding materials that gave the paints a long shelf life, but at the cost of creating poor drying properties on the canvas. Because the pigment was the most expensive component of the paint, the color-makers would be inclined to minimize the amounts used. But as all oils turn slightly yellow as they dry, paints with a high oil-to- pigment ratio discolor more markedly. And to keep the paint stiff while lowering the pigment content, some manufacturers added wax, which resulted in a sticky material that was liable to crack. Some retailers diluted their pigments with extenders-inert materials such as chalk or gypsum that merely made the colored material go further. There were even cases of deliberate falsification, of one pigment being passed off as another more expensive variety or surreptitiously mixed with it. The ambiguous relation of some paint names to the pigments they contained only heightened the temptation for duplicity.

As a protection against such practices, many artists strove to establish a good relationship with a particular color merchant, relying on him to control paint quality. Impressionist painters Pissarro and Cezanne, and later Vincent van Gogh, used Julien Tanguy, who ran a small shop on the rue Clauzel in Montmartre from 1874. The Danish painter Johan Rohde described it as a "paltry little shop, poorer than the most miserable second-hand dealers in Adelgade [in Copenhagen]. There were stacks of pictures-no doubt payment for materials-and very valuable things among them," which drew many visitors to the shop. The wild van Gogh was befriended by Tanguy when the artist came to Paris in 1886, and he twice painted the merchant's portrait, despite complaining on several occasions about his wares. Tanguy ground his own colors, and one of the advantages of such a supplier was that he could provide materials to order: van Gogh, for instance, made specific requests for coarsely ground pigments. Among the Impressionists only Edgar Degas showed much curiosity about what was actually in his colors. His notebooks contain many chemical recipes and technical notes that show a very limited knowledge of chemistry.

Can Io and Europa Support Life?

Some scientists think that two of Jupiter's four large ("Galilean") moons, Io and Europa, may be candidates for hosting life-in part because both of them have two important preconditions, water and an energy source. Io is similar to the rocky inner planets (such as Earth and Mars) in its composition, having an iron core and large amounts of silicon. Proximity to Jupiter makes it far more geologically dynamic than any other planet or moon. Tidal forces- caused by the gravitational effects of Jupiter and other moons-deform Io as it passes through its orbit. The process heats the moon, and constant volcanism results as the surface crust fails to keep in the sloshing magma (liquefied rock) underneath. The surface can change up to a hundred meters between high and low tide in some places. Other than Earth, Io is the only object in the solar system known to have currently active volcanoes. Eruptions wipe out all evidence of meteoric impacts; no craters can be seen on the surface.

Io seems, however, to have too much energy to make a good habitat for life. In addition to volcanism, the moon gets energy from the intense magnetic field of Jupiter. As the moon passes through this field, tremendous amounts of energy are generated. The forces strip nearly a ton of mass from the surface every second, creating a cloud of charged ions around the moon. Life as we know it would quickly dissolve on the surface of Io.

The second nearest of the Galilean satellites is Europa. In many respects it resembles Io, but its composition is radically different. Europa possesses large quantities of water, either as ice or as a liquid. Massive tides warm Europa, though they are only about one-tenth as strong as the ones affecting Io. On this moon, the tides probably maintain an extensive ocean under the sea ice. Several lines of evidence point to liquid water. Planetary scientists were surprised to discover that very few large craters exist on the surface. Some process must be resurfacing the planet on a regular basis, leading many to believe that liquid water spills out. Second, the

Galileo spacecraft detected a magnetic field induced by Jupiter. Some conducting material must be present around the inside of the planet. Liquid water with salts seems the most likely solution. Finally, the mass and size of Europa are consistent with an ice-and-water mixture.

Unfortunately, Europa exploration will prove difficult. Engineers are working on plans to reach the subsurface ocean, but currently there are no workable plans to place a drill on the surface that could tunnel through many kilometers of ice. Looking at all the evidence, planetary scientists believe that Europa consists of a rocky core accounting for less than one-half its depth. Over this core rest many kilometers of liquid water and ice. Different models give different depths of surface ice over the ocean, and further studies of composition, salinity, and surface features will certainly lead to a better understanding, but a large saltwater ocean seems probable. Tidal forces from interactions with Jupiter and other moons would keep the water liquid despite the subfreezing exterior temperature of -170° Celsius. Craters would be eliminated whenever meteors pierced the surface ice or tidal forces formed giant cracks. Saltwater from underneath would spread out and freeze, making a new top layer in the area around the event.

The importance of liquid water to life causes astrobiologists to be very excited about Europa. Liquid water appears to be rare other than on Earth, but here we have an ocean of it. Nonetheless, a number of other considerations are important. Carbon might not be difficult to find, as long as some form of internal circulation rotates heavier metals from the rocky core out into the water. Energy might be more difficult, though. Solar radiation hits the icy surface where life is unlikely to survive. Energy would have to be generated from some other source under the ice, possibly volcanic or magnetic processes. Some astrobiologists believe that Europa could have magma vents (surface openings through which materials escape) like the deep-sea vents at the midoceanic ridges on Earth. Such vents could provide both carbon and energy for an ecosystem.

Agriculture in Classical China

In classical China (1000 B.C.E. - 500 C.E.), farmers and farming enjoyed greater standing than in most other civilizations. In the official Chinese social hierarchy, farmers came after the scholar-bureaucrats in prestige. Such was the prestige of farming in classical China that even scholars tried to keep their hand in agriculture, engaging when they could in plowing or other agrarian activities. After the farmers came artisans and, lastly, merchants, actors, and other groups presumed to be disreputable. In theory, peasants could become government officials; during the Han dynasty (202 B.C.E. - 220 C.E.), discrimination against traders increased, even leading to a prohibition against merchants becoming state officials. Traders were, however, typically much wealthier than the vast majority of the peasants, bringing about contradictions between wealth and social prestige. The lack of prestige in commerce did make the wealthy prefer to invest in land rather than in trading ventures, leading to increased landlordism.

The majority of the population were either peasants who owned their own land or permanent workers who cultivated plots that they did not own but that remained in the same family for many generations. This pattern had emerged in the preclassical period under the Zhou dynasty (1046 - 256 B.C.E.). The Zhou formed China's feudal period, in which political unity was an ideal rarely, if ever, achieved. Instead, the relatively weak central government and the rival kingdoms gave away land to aristocrats in exchange for armaments, horses, and other military needs. These aristocrats had to supply loyalty and military services to their benefactors. In return for military protection, peasants paid a certain percentage of their crops to the aristocrats, in what is called a manorial system. The landlords restricted the movement of the rural workers; after all, it was the peasants' work that made the land produce, and peasants' movement elsewhere meant the land, less productive, could yield less to the aristocracy.

The classical period in China was punctuated by a social revolution. In the Qin dynasty (221 - 202 B.C.E.), the emperor decreed the liberty of all manorial peasants, giving them the land, they had worked for generations. During the Qin period, this decree probably did not mean much less actual work, for it was during this dynasty that the emperor built much of what we know of as the Great Wall. The authoritarian methods to recruit workers brought about widespread discontent and, ironically, led to the Qin emperor's overthrow. The abolition of the manorial system was permanent, however, and led to higher population growth, as well as spontaneous migration to the fertile and less densely populated lands in southern China. The consequences of these actions had lasting importance for rural labor patterns. In the first place, instead of millet and wheat, peasants increasingly began to cultivate rice, which grew well in the hot southern climate. In addition, population growth diminished the size of the plots of households, especially in the densely settled eastern portion of the empire. By the time of the Han dynasty, the standard landholding of one farming family with seven members was slightly less than 5 acres. The small plot size meant that Chinese peasants had to engage in intensive farming. They had to use deep tilling (digging soil to prepare for planting) techniques and irrigation and to invest a much greater amount of labor into their plots, with frequent weeding, fertilizing, careful thinning, and the like. In the south, where winters came late or were mild, cultivators frequently tried to plant and harvest more than one crop annually or to plant a staple crop such as millet in between mulberry trees.

This intense activity was not enough in years of droughts, overabundant rainfall, or other natural disasters. Peasants needed to generate other income to cover their taxes and pay for other goods that they might want to acquire. The most important subsidiary activity was weaving, done mainly by women, as the ancient Chinese refrain 'man as tiller, woman as weaver' suggests. Indeed, the Han considered agriculture and weaving to be complementary, a distinctive trait of Chinese rural life. Even when they were sitting at their doorsteps conversing with their neighbors, women spun fibers or kept busy in other ways. It is possible that weaving brought in almost as much income as farming.

The First Settlers in Ancient Amazonia

Archaeologists believe that humans first arrived in North America at least 16,000 years ago and that by 11,000 years ago, these migratory hunter-gatherers (people that hunt for and collect food in the wild) had spread throughout North and South America. For some time, archaeologists believed that the region of Amazonia (the Amazon rainforest in South America) was not occupied until much later. However, it has since been proven that hunter-gatherers first entered Amazonia at least 11,200 years ago.


North America
Atlantic Ocean
Oyapok
River
Amazon
River
Salinas
Ecuador
São Marcos
Bay
Taperinha
Pacific
Ocean
South
America
Amazon
rainforest

Currently, some controversy surrounds the dates when hunter-gatherers in Amazonia became sedentary (settled in one location). In many parts of the world, from Japan to Peru, the first societies to enjoy a sedentary or semisedentary way of life relied on aquatic (water-based)resources. It was not different in Amazonia, a region known for its bountiful rivers and a 932-mile-long stretch of shoreline cut by bays, deltas, and straits, where we find one of the most extensive mangrove areas on the planet. This Atlantic landscape- which begins at the mouth of the Oyapok River and the Sao Marcos Bay, marking the eastern limits of the Amazon rainforest-was settled by humans at least 5,500 years ago, thanks to the rich biodiversity (wide variety of plant and animal species) available in the region year- round. There, native communities would find shellfish, crab, fish of various sizes, and small game, as well as forest fruits, nuts, and seeds in exceptional abundance. The daily consumption of shellfish produced huge amounts of discarded shell in the habitation sites, which is why such sites are called 'shell middens.' Why didn't people find the sea earlier? Well, they may have. It is possible that many earlier sites are now underwater due to the sinking of the continental shelf(continental part closest to the ocean) during the Holocene epoch(11,700 years ago to present). A half-submerged site recently found in Salinas, a beach town, reinforces this hypothesis. Moreover, shell midden populations were also found in riverine (river) settings at Taperinha-a site located inland at a farm at the lower Amazon River -which indicates an aquatic-based sedentary way of life by 7,000 years before present, long before the spread of crop cultivation.

Unfortunately, our ability to study those first river-and-shore-adapted peoples is hampered by the poor state of preservation and looting (theft) at most of the important sites belonging to this period. Because shells contain calcium carbonate, shell midden sites were mined at the turn of the nineteenth century and have become virtually flattened after decades of exploitation. In Taperinha, however, a six- meter-deep shell mound deposit excavated by archaeologists provided evidence of a way of life based on freshwater shellfish and other animal species, and also pointed to the production of stone artifacts and ceramics (pots and other articles made from baked clay) at an early time. The dates determined for the use of baked clay in the production of cooking vessels are the oldest for the entire continent, challenging the prevalent theory at the time, which accounted for the arrival of ceramics in the Americas by diffusion across the Pacific Ocean from Japan. According to this diffusionist model, the technology for the production of ceramics would have arrived at the northern South American coast(in the area of present-day Ecuador) by 4,000 years before present, where it was readily adopted, and then spread throughout the continent. This hypothesis was based both on the similarities between the two ceramic industries and on the assumption that ceramic technology was invented only once, later traveling around the globe.

Taperinha's well-preserved deposits, however, contained ceramic pieces 3,000 years earlier than Ecuador. It was previously thought that the ceramic culture of Valdivia in Ecuador was the oldest aquatic-based culture in South America. But the Taperinha deposits proved that ceramics were produced for the first time in the Americas in the heart of the Amazon rainforest, although ceramics did not constitute the bulk of cultural remains for that period, as it would in the millennia to come. Taperinha's deposits also yielded stone implements and utensils, as well as grinding and cooking stones. Turtles and shellfish, widely used in the diet, also provided shells that were used as raw materials for domestic utensils.

Waking Up Coming Out of Hibernation

There are animals in most temperate regions in the north that spend approximately six months in hibernation-from October to March. Within the Arctic Circle this period can be extended to nine months for some species. Elsewhere, such as the desert regions of the southwestern United States, droughts may result in the normal periods of estivation (inactivity during hot or dry periods) and hibernation running continuously so that animals like the spadefoot toads may be inactive for eighteen months. But dormant (inactive) hibernators do not sleep solidly for their whole term as once believed, and even those that sleep deeply awaken quite regularly. Why an animal should periodically arouse (wake up) for several hours, or perhaps a whole day, during its long hibernation is unclear and has been the subject of much speculation.

Arousal, which entails heat production, is a drain on an animal's precious stored reserves of fat as it raises its body temperature, and there must be a very worthwhile reason for doing it. Possibly life in such a reduced state-called metabolic depression- damages the animal's nerve cells and affects its ability to develop immune responses as defense against infection and disease. Arousal, even for just a few hours, restores the metabolic processes and perhaps allows the animal to recover from the stress of metabolic depression, giving it an opportunity to really sleep and rejuvenate its body, which does not happen when it is dormant. It is also believed that some animals may need to restart their metabolism in order to excrete waste, and it is possible that physiological imbalances occurring during hibernation, such as the depletion of blood glucose, could be corrected during their brief return to normal temperature.

There is also a safety mechanism that functions if the temperature of the hibernator's shelter drops below a mammal's set point, the minimum temperature at which an animal can exist without freezing. The hibernating animal arouses from its sleep through shivering and violently contracting its muscles, which increases its heart rate and respiration, and thus its oxygen intake and body temperature. In the case of the external food storers, such as the chipmunk, depleted food reserves in its body are the trigger to arouse and feed and thus replenish its energy.

In the early stages of arousal, the front of an animal's body-the part containing the essential vital organs, the heart and brain-is heated faster than the posterior and the extremities. Blood is directed to the vital organs and especially to the brown fat tissue that occurs along the neck and between the shoulder blades of mammals, indicating its importance in providing heat to power the arousal process. Vasodilation, or widening of the blood vessels, occurs as mammals arouse from hibernation, and arousal time depends on body size. Large hibernators warm up slowly, at the rate of about 1.8F (1°C) per ten minutes, whereas tiny creatures warm up much faster-at the rate of 1.8°F(1°C) per minute. Heat production warms the body at a much faster rate than the cooling- down process at the start of hibernation, and animals therefore arouse more quickly from the state of hibernation than they enter it.

Emergence from hibernation is controlled by the same factors as the entry weeks or months earlier-temperature, amount of daily exposure to light, or body rhythms. The increasing temperatures of springtime are the major factor initiating the arousal and emergence for many species. Terrestrial amphibians such as the wood frog and spring peeper, which hibernate under leaves, are the first to appear as the snow begins to melt, whereas aquatic hibernators like the leopard frog and bullfrog must await the melting of a large part of their pond's ice cover before awakening. Biological rhythms (daily rhythms to many physiological functions and activities, such as sleep and body temperature) appear to play a greater role in those that sleep deep underground, below several feet of frozen soil, for it seems unlikely that they are influenced by the slightly warmer temperatures or longer days above ground, and they often emerge to find that there is still considerable snow cover. There is rarely food immediately available for animals when they emerge, and it is important that they have enough internal reserves left to sustain them over this essential part of their life cycle.

Why Did Social stratification Emerge

Without exception, modern industrial and postindustrial societies are socially stratified-that is, they contain social groups such as families, classes, or ethnic groups that have unequal access to important advantages such as economic resources, power, and prestige. Based on archaeological evidence, it seems the emergence of social stratification was connected with the advent of agriculture roughly 10,000 years ago. Until then, all human societies depended entirely on food they hunted, gathered, and/or fished, and so anthropologists are reasonably sure that higher levels of stratification emerged relatively recently in human history.

Archaeological sites dating before 8,000 years ago do not show extensive evidence of inequality. Houses do not appear to vary much in size or content, and different communities of the same culture are similar in size. Signs of inequality appear first in the Middle East about 2,000 years after agriculture emerged in that region. Inequality in burial suggests inequality in life. Particularly telling are unequal child burials. It is unlikely that children could achieve high status by their own achievements. So, when archaeologists find statues and ornaments only in some children's tombs, as at the 7,500-year-old site of Tell es-Sawwan, the grave goods suggest that those children belonged to a higher-ranking family or a higher class.

Why did social stratification develop in the first place? Some scholars stress the importance of surplus production that resulted from increased agricultural activity. Others stress the degree to which wealth can be transmitted across generations. With regard to surpluses, the cultural anthropologist Marshall Sahlins suggested that surpluses would result in greater scope and complexity of the system of distributing goods, enhancing the status of chiefs (leaders) as agents for redistributing goods. Gradually, this would give the chiefs more control over resources and ultimately more power. The comparative sociologist Gerhard Lenski, too, argued that production of a surplus is the stimulus in the development of stratification, but he focused primarily on the conflict that arises over control of that surplus. Lenski concluded that the distribution of the surplus will be determined on the basis of power. Thus, inequalities in power promote unequal access to economic resources and simultaneously give rise to inequalities in privilege and prestige. A broader argument is that a surplus may lead to some advantages of one subgroup over another, such as more people to support a stronger military force, or more knowledge that could lead to the development of specialized, productive technology.

The surplus theories of Sahlins and Lenski do not really address why people would produce surpluses or why redistributors or leaders will want, or be able, to acquire greater control over resources. Sahlins later amended his theory to suggest the reverse-that leaders may encourage the development of a surplus to enhance their prestige. But even if that were so, prestige enhancement is not the same as wealth enhancement. After all, the redistributors or leaders in many nonindustrial societies do not have greater wealth than others, and custom seems to keep things that way. One suggestion is that, as long as followers have mobility, they can reject leaders they do not like by moving away from them. But when people start to make more permanent investments in land or technology (such as irrigation systems), they are more likely to put up with a leader's elevated status in exchange for protection. Another suggestion is that access to economic resources becomes unequal only when there is population pressure-increased competition for resources resulting from population growth. Such pressure may be what induces redistributors to try to keep more land and other resources for themselves and their families.

The anthropologist C. K. Meek offered a modern example of how population pressure in northern Nigeria may have led to economic stratification. At one time, a tribal member could obtain the right to use land by asking permission of the chief and presenting him with a token gift in recognition of his higher status. But, by 1921, the reduction in the amount of available land had led to a system under which applicants offered the chief large payments for scarce land. As a result of these payments, farms came to be regarded as private property, and unequal access to such property became institutionalized.

Thermoregulation in Marine Mammals

Any organism, in terms of regulating its internal body temperature, can be either ectothermic (cold-blooded) or endothermic (warm-blooded). In an ectothermic organism, its internal body temperature and thus metabolic rate largely varies with and is controlled at the surface by the external environment. In an endothermic organism, its internal body temperature is maintained at a constant temperature, usually between 37° and 40C by internal metabolic heat production.

The marine environment offers temperatures that are relatively more stable than those on land because water has a high specific heat capacity, which means substantial amounts of energy need to be absorbed or lost before the temperature of water changes. Therefore, ectothermic species tend to be more common in the ocean than on land. However, the metabolic rates of ectotherms change by a factor of 2 for every 10°C temperature change. Thus, the rate at which metabolic reactions occur for ectotherms is substantially reduced as temperatures drop. This gives endothermic species a big advantage over ectothermic prey or competitors in cooler water temperatures. That marine mammals are endothermic explains, in part, why they are so abundant and successful in polar and subpolar regions. However, being endothermic does have a cost, because these species require more energy to support a high metabolic rate, which means they must consume relatively more food to survive.

The thermal conductivity of water is 25 times greater than that of air. Although water temperatures in the marine environment, as noted above, are relatively stable compared with those of the terrestrial environment, marine mammals do have difficulties, nonetheless, with heat loss to the surrounding environment. This is especially true for the many marine mammals that occupy cool waters, where the temperature gradient between their internal body temperature and the surrounding seawater is high, and thus heat loss is greater. Thus, heat loss is greatest for animals with a high surface area (relative to their volume), a body surface that conducts heat well, and a large difference between their internal body temperature and the surrounding water temperature. As a result, marine mammals are able to minimize heat loss by increasing body size, reducing conductance, or transference, by increasing insulation, or, in some cases, avoiding low water temperatures. For example, sea otters experience heat loss because they are relatively small and their surface-area-to volume ratio is high. One way to decrease surface area is to have small appendages. Allen's rule, an ecological observation made by Joel Allen in 1877, states that endotherms from colder climates usually have shorter limbs (or appendages) than the equivalent animals from warmer climates because this reduces their relative surface area. Sea otters have relatively blunt muzzles (noses) and shorter tails than their otter counterparts in warmer climates.

Sea Otter

© Xfkirsten/Dreamstime.com

Although sea otters are one of the smallest of the marine mammals, they can weigh up to 45 kg and be over 1.5 meters in length, making them the heaviest and largest species in their group. Therefore, they have a relatively smaller surface area to volume ratio than other otters. This is an example of another ecological rule: Bergmann's rule (after Christian Bergmann, who published the rule in 1847) states that animals within a specific group tend to get larger as the latitude increases (and decreases in temperature). For example, Arctic-dwelling polar bears and sea otters are the largest species of bear and otters, respectively.

Heat loss is also minimized by reducing the conductance (increasing insulation) of the exterior surface. Sea otters possess a thick fur layer and have one of the densest furs of all mammals, with up to 150,000 hairs per square centimeter. This fur layer traps air (air has a lower conductance than water) and therefore insulates the otter well. However, when a sea otter dives, water pressure forces air out of their fur and they lose much of their insulation. Salt from seawater also clogs their fur, making hairs stick together. Thus, much of a sea otter's behavior is dedicated to grooming, cleaning fur, and "fluffing it up" so air can get between the fur hairs. Polar bears face a similar problem; although their fur is an effective insulator on land, when they enter the water, their fur becomes waterlogged and loses much of its insulating properties. However, polar bears have a substantial layer of adipose, or fatty, tissue.

Irrigation and Early Civilizations

For many analysts of early civilizations, the search for an explanation of how these cultures evolved in highly similar yet distinctive ways logically should begin with a consideration of these similarities and differences. What did these early civilizations share in the way of climate, ecology, demographics, or any other factors, that might explain why it was Mesopotamia, Egypt, the Indus Valley, China, Mesoamerica, and Andean South America that gave rise to the first great states and civilizations? If you knew nothing about the archaeology of early complex cultures and began to compare them, chances are you would be struck by the same facts that impressed Julian Steward, V. G. Childe, and other early scholars who studied the problem of cultural complexity origins. Evidently, most of these cultures developed in similar physical environments and were based on similar economies.

Perhaps the most obvious common denominator of ancient complex societies was extensive irrigation systems. Even today aerial photographs of Mesopotamia, Peru, and most other areas of early state formation clearly show the massive remnants of these ancient structures, and similar constructions were built by early “chiefdoms" in such places as Hawaii and southwestern North America. This led some scholars to conclude that the construction and operation of extensive irrigation systems were at the heart of the origins of complex societies. A particularly influential proponent of this view was Karl Wittfogel, whose Oriental Despotism is a detailed excursion into comparative history and sociological analysis.

Wittfogel notes that the limiting factors on agriculture are soil conditions, temperature, and the availability of water. Of these, water is the most easily manipulated, but its weight and physical characteristics impose limitations on this manipulation. To divert water to agricultural fields requires canal systems, dams, and drainage constructions that can only be built efficiently with organized mass labor; and irrigation systems require enormous investments of labor and resources to operate, clean, and maintain them. In addition, these systems necessitate complex administration and communication, because crucial decisions have to be made about water allocation, construction and repairs, and crop harvesting and storage. Thus, a complex irrigation system under ancient conditions required cooperation and centralized hierarchical decision-making institutions.

Irrigation systems also have the intrinsic capacity to create another element in the process of the evolution of complex societies: wealth and status differences. Fields closer to main rivers are better drained and more easily irrigated, and possess a higher natural fertility; thus, control of such lands would create immediate wealth differences. Correspondingly, wealth and status would most likely accrue to the elites of the decision-making hierarchies. Wittfogel concludes that irrigation-based agriculture has many other effects on a society. It encourages the development of writing and calendrical systems so that records can be kept of periods of annual flooding, agricultural production statistics, the amounts of products in storage, and the allocation of water. Construction of roads, palaces, and temples would also be encouraged, because the mobilization of labor for the canal works would be generalized to these other endeavors very easily, and roads would contribute to the movement of agricultural produce and to the communication required for efficient operation of the systems. The construction of temples and palaces would also serve to reinforce the position of the hierarchy. The creation of standing armies and defensive works would also likely follow, because irrigation systems are extremely valuable but not very portable, and they are easily damaged by neglect or intentional destruction.

Wittfogel's hydraulic hypothesis still has some currency, but there seem to be logical and empirical problems with his ideas as a general model of the origins of cultural complexity. Simple societies in several parts of the world have been observed operating extensive irrigation works with no perceptible despotic administrative systems or rapid increases in social complexity. More damaging to Wittfogel's hypothesis is the scarcity of archaeological evidence of complex irrigation systems dating to before, or to the same time as, the appearance of monumental architecture, urbanism, and other reflections of increasing cultural complexity in Southwest Asia and perhaps other areas where complex societies appeared independently and early.

Nonetheless, the difficulties of dating irrigation canals, inadequate archeological samples, and other deficiencies of evidence mean we cannot conclude that irrigation was unimportant in virtually any case of early cultural complexity.

Desert Adaptation in Kangaroo Rats

Courtesy of Wikimedia Commons/ Danny Chia

The desert kangaroo rat of western North America has adapted wonderfully to its dry environment, and in fact it rarely goes thirsty. Mammals lose water in three ways: when they breathe, sweat, or urinate. Water is lost through breathing, which mammals do to acquire the oxygen necessary for carrying out the chemical functions within cells that sustain life and for getting rid of carbon dioxide, a waste product. Water is lost in the process because it is a physiological requirement that oxygen be absorbed across a wet membrane. The insides of mammalian lungs have a nice wet membrane, and when they breathe out, some water is exhaled as water vapor. Mammals generally sweat to cool their bodies down because evaporation of water removes heat. Animals urinate to remove nitrogenous waste from their bodies because nitrogen is a toxic by-product of digesting food, particularly proteins. If a mammal did not urinate (or if its kidneys shut down), it would quickly build up too much nitrogen, its blood would become toxic, and it would die.

If we look to the kangaroo rat, however, we can see that evolution has equipped this animal with a host of amazing adaptations to help it conserve precious water in its desert environment. First, we turn to the kangaroo rat's nose, through which it breathes. The rat's nasal passages are set up in such a way that it is able to conserve and recover some of the water from its breath. In these passages, warm, moist air leaving the lungs comes into contact with cooler, drier air entering the lungs and loses some of its warmth to this cooler air. Thus, the breath leaving the lungs cools as it nears the outside, allowing water vapor in the breath to condense (change to a liquid) in the nasal passages since cool air holds less water than warm air. This relatively simple mechanism allows the kangaroo rat to recover somewhere between 50 percent and 80 percent of the water in its breath, making it very water-efficient

Another astonishing fact about this kangaroo rat is that it does not drink water. In the deserts where these rodents live, they can't rely on free water because they rarely encounter it. If they don't drink water, why don't they die of thirst, you might ask. One way they get water is through their food. When they break down food during digestion, one of the by-products is water. For demonstration purposes, we will represent all food using the simple sugar glucose, which has the chemical formula CC6 HH12 OO6. Digestion of glucose requires oxygen (O_2) and breaks this sugar into carbon dioxide(CO_2) and water (H_2O), releasing energy. For every molecule of glucose digested, we get six water molecules. This process is not unique to kangaroo rats; it happens when we digest food as well, but these rats are more efficient at generating and using this water. In addition, kangaroo rats are seed eaters and they will often cache, or bury, their seeds underground, where the temperature is significantly cooler. These seeds are bone-dry when collected at the desert surface, but when they are put into cool underground caches, they act like little sponges, sucking up some of the moisture from the surrounding soil. So, kangaroo rats also get some water from the way they handle and process food.

When we look at water loss through sweating, we find that kangaroo rats have solved this problem by staying in their burrows, limiting the time they spend out in the hot Sun. In fact, they don't sweat much because they run around at night, when it is significantly cooler and there is no sunlight to dry them out. Finally, in terms of water loss through urine, kangaroo rats have a specialized kidney that is five times more efficient than that of humans at removing water from their nitrogenous waste. So efficient are the kidneys that the kangaroo rat's urine is excreted more as a paste than a liquid, saving it large volumes of water. With all of these physiological and behavioral adaptations, the kangaroo rat is a model of water conservation, which allows it to live in the harsh, dry desert.

Late Classic Maya Agriculture

The great Maya civilization of Mesoamerica(what is now Mexico and Central America) underwent a precipitous decline in its Late Classic period(600-900 A.D.), and though many different explanations have been proposed for this collapse, Maya agriculture certainly played a role. While Maya agriculture could be remarkably productive, it also suffered from strong inherent risks and limitations.

Despite the fact that their farming practices were technologically underdeveloped compared to those of Old World(European, African, and Asian)civilizations, Maya farmers enjoyed several advantages, especially where population densities were low and plots of land could be developed and then abandoned for more fertile plots when the first land was no longer productive(a technique called shifting cultivation). One advantage derived from the very simplicity of their agricultural tasks. Because they could make the modest tools needed to work their fields from locally available materials, Maya farmers did not have to purchase metal implements made by specialists, or buy, rent, and maintain expensive animals to attach to plows or wagons, like Old World peasants. Necessary agricultural labor could be provided by domestic households, so it was unnecessary to hire farmhands (though illness or death of workers could cause severe problems for the domestic workforce). Even large projects, such as the hillside terraces and drained fields that many archaeologists believe supported vast Classic period populations, could be built with domestic labor augmented, when necessary, by the help of neighbors or relatives. In other words, the basic costs of agricultural production were low, and the necessary skills and knowledge were widely available.

Another set of advantages derived from the nature of the principal staple crop-maize. On decent soils watered by adequate rainfall, maize typically produced larger yields per unit of cultivated land-perhaps twice as much-than Old World small grains such as wheat, barley, or oats grown by medieval European farmers. And because of the way these latter grains were sown, anywhere from one-sixth to one-third of the European annual crop had to be reserved for seed (except where irrigation was used). For maize the ratio of seed input to output was much more favorable, and because there were few domestic animals to feed, virtually all of the harvest could be consumed by humans.

On the other hand, there were some severe limitations, one of which was caused by the history of land use itself. By the eighth century many of the most favorable parts of the Maya Lowlands had already been cultivated for hundreds of years. The population explosion of Late Classic times took place on local landscapes that no longer were natural in any sense of the word, because they had already long been humanized(degraded)by generations of use. Heavy episodes of deforestation and erosion have been documented for several regions as early as Preclassic times (2000 B.C.-200 A.D.), and there is strong evidence that some centers and regions experienced cycles of growth and abandonment long before the collapse. Even when overall populations were low, early farmers packed into any small region could obviously damage their environment severely. Early on, mobility seems to have been one of their solutions-people left for better lands elsewhere.

Another limitation involved technology. Without metal tools, complex machines, and sources of animal energy, Maya farmers could cultivate only very small amounts of land and they also had to schedule their work carefully to take into account the constraints of the seasons. Even where good land was abundant, Maya producers could accordingly generate only small per capita surpluses. Working diligently during a good year, a healthy farmer might cultivate 2-2.5 hectares(5-6.2 acres), which would yield roughly 1,000 kilograms per hectare(890 pounds per acre) of maize. If this land were of sufficient quality, it would yield twice as much food-around 2,000 kilograms(4,400 pounds)-as his family ate annually, after subtracting that amount reserved for next year's seed stock and lost in storage. In other words, each family, if it had good land and worked hard, might generate only enough surplus food to support one other family of similar size-as opposed to perhaps five to six times the amount of food a family needed in the case of some Old-World farmers.

Food Caching by Scrub Jays

© Thomas W. Woodruff/Shutterstock.com

The western scrub jay is a highly intelligent bird, and biologists have conducted a number of experiments on its cognitive abilities based on its propensity to cache (hide) food such as acorns, seeds, or worms for later consumption. Since they do not migrate for the winter, scrub jays must prepare in advance, storing food in caches (hiding places) to get through the cold months when less food is naturally available. In addition to this predictable seasonal change, however, scrub jays have to contend with two other factors that require their knowledge of the world to be updated more rapidly: first, different kinds of food tend to perish (spoil and become uneatable) at different rates; and, second, other scrub jays may find and eat their cache (hidden food) before they do.

In a series of complex and very clever experiments, scrub jays have been shown to remember what kind of food they have hidden where, and, more impressively, when they did so. When given the option of recovering either previously cached wax worms(their favorite food, but one that perishes rapidly) or peanuts(which are less preferred, but last longer), scrub jays would attempt to recover the wax worms if the delay between caching and recovery was short (4 hours) but would recover peanuts when the delay was long (120 hours). Wax worms are still tasty after 4 hours, so it makes sense to go for them after a short delay, but after 120 hours they have gone bad and become inedible (uneatable) and there is no point in recovering them. It seems, then, that the scrub jays have some means of keeping track of what they have cached, where they did so, and also when they did so, with the result that they do not waste effort recovering bad food, or, conversely, they do not spend time and effort recovering less preferred nonperishable foods, like peanuts, when there are more exciting and tasty foods like wax worms to be had.

In addition to recovering their own caches, scrub jays are also very adept at pilfering (stealing) the stores of other scrub jays. To counter such pilfering, jays frequently recache their food if there are competitor birds around that have had the opportunity to observe where food has been stored. In another series of experiments, scrub jays were able to keep track of which food caches had been seen by competitor birds, and even to remember exactly which specific individual had seen them hide their cache. In one experiment, for example, the birds were given access to food and allowed to cache in private or in the presence of an observer bird. After a three-hour delay, the birds were then allowed to recover their caches, and to recache the food in a new tray if they chose to do so. Jays were much more likely to recover their caches and hide them in the new tray if they had been observed initially than if they had cached in private. This suggests they were sensitive to the social context in which they had originally cached: food was moved to a new spot only if there was a risk that it could be pilfered by another bird.

Scrub jays also engage in various kinds of cache-protection strategies at the time of caching: if they can be seen by another bird as they cache, scrub jays are much more likely to choose caching sites in shady areas, or those that are less well lit(presumably because this makes it harder for a competitor to see exactly where they are caching), and they will also choose a more distant site over a closer one. Further, they are more likely to move a food item around multiple times between different cache sites before finally settling on one, making it difficult for a competitor to keep track of exactly where a food item ends up. Most intriguingly of all, it appears that these recaching tactics are dependent on a scrub jay's previous experience of pilfering: the only birds to recache their food after being observed are those that have previously taken food from other birds' caches. Birds with no experience of raiding another bird's cache sites show no tendency to recache their own food if another bird is witness to their caching.

When Did Humans First Colonize the Americas?

Nineteenth-century archaeologists struggled to estimate ages of American prehistoric sites and artifacts. Without radiocarbon isotope and other laboratory technologies to calculate years elapsed, some mistook stones obtained from old quarries (places where stone is dug out of a large hole in the ground) for million-year-old Early Paleolithic artifacts; others went to the opposite extreme, insisting that sites and artifacts couldn't be more than 2,000 years old. The breakthrough came in 1927 when excavation of a New Mexico site from which bison bones were eroding revealed a finely crafted stone weapon point (blade) between bison ribs. Calls went out to a paleontologist at the American Museum in New York, an archaeologist at the Smithsonian in Washington, and the most respected Southwestern United States archaeologist. The three authorities agreed that the large bison bones belonged to a species that became extinct about ten thousand years ago, and that the stone artifacts had been used to kill and butcher the bison. Therefore, there were people in America that long ago, hunting huge animals soon to become extinct.

Soon after "Folsom points," named after the village near the site with bison bones, became accepted as evidence of late-Pleistocene migration of humans into America, a somewhat similar style of stone weapon blade was discovered in another New Mexico site, near the town of Clovis. This type of stone blade is larger than the Folsom type, and the characteristic long, vertical handle groove (a narrow line cut into the surface) extends only about a third of the way up the blade, whereas the Folsom type has the groove nearly up to the tip. The two types share expert flint-knapping (chipping) skill, making the artifacts beautiful and easily noticed. At Blackwater Draw near Clovis, excavation demonstrated that Clovis points were in a layer lower than the layer with Folsom points, indicating that Clovis technology was older than Folsom technology. Clovis but not Folsom points have been found with mammoths, another clue to Folsom being later, after mammoths became extinct in America. Clovis-style points have been found over most of North America, even more in the East than in the West. Proponents of the Clovis First theory claim that this style of blade was ideally suited for hunting mammoths and other large animals and was used by the first wave of migrants moving south into the unpopulated Americas from Beringia, the land bridge between present-day Siberia and Alaska. Opponents of the Clovis First hypothesis interpret Clovis as a technology and style that spread through already-populated North America.

The answer to the debate hinges on whether there is any evidence of settlers in the Americas before Clovis. Clovis First archaeologists demanded copious evidence for earlier settlers, against those thinking there is an uncountable number of earlier sites destroyed or not yet discovered. Lack of evidence in archaeology should not be construed to prove that something never existed. Evidence for humans in America earlier than the Clovis horizon is thin-thin, but there. Skeptical archaeologists, citing abundant data, argue that sites alleged to predate Clovis may be incorrectly dated, or the simpler artifacts recovered might come from camps used by Clovis-equipped hunters who took their weapons with them when they moved on. Opponents of the "Clovis First" hypothesis say, of course, that the evidence is thin, because there have been many more years for sites to be buried or destroyed, and furthermore, first colonizers would have been less numerous than their descendants, so their sites would be less numerous. Most important, we don't have a clear idea of what pre-Clovis technology looked like. What do you look for to find a pre-Clovis settlement?

A newer twist in the Clovis First debate is the assertion that Clovis technology looks like the stone tools of Eurasian Upper Paleolithic Solutrean culture, most famous for Paleolithic cave art. This culture predates Clovis by several thousand years. Proponents of the Solutrean theory see hunters paddling along the North Atlantic ice front spearing seals and fish and then settling throughout eastern North America, modifying their tool technology eventually into Clovis blades.

Honeybee Juvenile Hormone

The activities of honeybee hives are highly coordinated. Honeybees live in colonies made up of a single queen and up to 80,000 workers. The tasks of a worker usually change dramatically with age, a phenomenon known as age polyethism. A typical honeybee worker newly emerged as an adult, winged insect will live for approximately six weeks and during this time moves through an orderly progression of tasks: Young adults (4 to 12 days old) tend the queen and the pupae (pre-adult insects); middle-aged bees (13 to 20 days old) perform various tasks associated with nest maintenance and food storage; and the oldest bees (over 20 days of age) concentrate on foraging (searching for food) outside the nest. In general, young workers mostly stay inside the hive and tend to nursing and housekeeping, and older workers often leave the hive to forage. Keep in mind, however, that the actual changes in the duties performed by a worker usually involve only changes in the frequencies of particular types of behaviors with age, rather than the addition or loss of entire categories of behaviors. Also, the sequence is flexible, allowing duties to be reassigned if the hive's needs change.

Age polyethism in worker honeybees is regulated by juvenile hormone (JH); levels of JH increase with age and mediate changes in work habits. These levels vary with age because of changes in the rate of its synthesis by the corpora allata, a pair of glands near the brain. Not only are the naturally occurring levels of JH correlated with a worker's change in duties but also experimental manipulations of JH levels have resulted in the expected changes in behavior. Treating newly emerged worker bees with a chemical similar to JH speeds up behavioral maturation. On the other hand, removing the corpora allata delays but does not prevent the development of a worker into a forager. But that delay can be eliminated by juvenile-hormone treatments.

How does JH cause an adult worker's behavior to change so dramatically? At least part of the answer is that JH affects brain structure. The most striking change in behavior during the usual progression of duties occurs when a worker becomes a forager. Unlike younger hive bees, foragers take long-distance flights; learn rapidly and form long-term memories; use the sun as a compass for orientation; and communicate to other foragers the distance to and direction of a food source. As a honeybee matures, there is a 20 percent increase in the size of the insect brain structures known as mushroom bodies. The mushroom bodies are important in learning, memory, and the processing of sensory information. The increase in volume is due to increases in the number of branches on the neurons, which increase the number of synapses (points at which nerve impulses pass from one neuron to another), thereby boosting the brain's ability to process information. Thus, these changes in the mushroom bodies may prepare a worker to learn to navigate during foraging flights and to find its way back to the hive.

Juvenile hormone, not experience, is responsible for the growth of the mushroom bodies. This was demonstrated by treating one-day- old workers with a chemical that mimics the effects of JH to cause them to begin foraging sooner than usual. Normally, young workers take orientation flights to learn the lay of the land before they begin to forage. But some of the treated bees were prevented from leaving the hive for orientation flights by a large tag attached to their backs. Although the bees were different ages, the mushroom bodies of treated bees grew to about the size of those of untreated workers that were beginning to forage. Furthermore, the growth of the mushroom bodies was equivalent in treated bees with and without foraging experience.

Hormonal regulation of the age-based division of labor in honeybees is not a rigid system of control. Instead, it is a flexible process that permits ongoing adjustments in the proportions of workers involved in any one task. The rise in JH may be modulated due to environmental conditions such as food availability or age structure of the colony. For example, workers from starved colonies begin foraging a few days earlier than workers from well-fed colonies.

The Origins of Life on Earth

As early as the 1920s, some scientists recognized that Earth's early atmosphere should have been oxygen-free(oxygen is a product of life on Earth); they also hypothesized that, in this oxygen- free environment, sunlight-fueled chemical reactions could have led to the spontaneous creation of organic molecules, molecules like those found in living systems. This hypothesis was tested in the 1950s in a famous experiment known as the Miller-Urey experiment.

The original Miller-Urey experiment used small glass flasks to simulate chemical conditions they thought represented those on early Earth. One flask was partially filled with water to represent the sea and heated to produce water vapor(water in the form of gas). Gaseous methane and ammonia were added and mixed with the water vapor to represent the atmosphere. These gases flowed into a second flask, where electric sparks provided energy for chemical reactions. Below this flask, the gas was cooled so that it could condense to represent rain and then was cycled back into the water flask. The water soon began to turn a murky brown, and a chemical analysis(performed after letting the experiment run for a week) showed that it contained many amino acids, the building blocks of life, and other organic molecules. Thus, the experiment appeared to support the idea that organic compounds would have formed spontaneously and in vast quantities on early Earth, covering our planet and filling at least the top layers of the oceans with a rich "organic soup" containing all the building blocks of life.

However, other scientists soon realized that the methane and ammonia gas mixture used by Miller and Urey probably was not representative of Earth's early atmosphere and the significance of their results was called into question. That the relevance of the Miller- Urey experiment is an even hotter scientific controversy today demonstrates the way that science often progresses unsteadily.

The major issue concerns exactly what the early atmosphere was made of. When analyzed in more detail, the Miller-Urey experiment shows that hydrogen is a key ingredient needed to form large quantities of organic compounds. In contrast, oxygen in any form, even when tied up in carbon dioxide, tends to suppress the formation of organic compounds. Today, we assume that Earth's early atmosphere was produced by volcanic outgassing(release of trapped gas). Outgassing may have occurred at a somewhat higher rate in the distant past than it does today, but we expect the gases to have been similar: largely water, carbon dioxide, nitrogen, and methane, along with sulfur-bearing gases and hydrogen. The water condensed to form the oceans, leaving these other gases as potential ingredients for the early atmosphere. If there was a lot of hydrogen relative to carbon dioxide, then the Miller-Urey results might still be valid, suggesting that our young planet did indeed have a rich organic soup. However, if there was little hydrogen, then the oxygen in carbon dioxide would have kept the amount of organic material far lower than the Miller-Urey experiment suggests.

Prior to 2005,it was generally assumed that hydrogen would have escaped from the early atmosphere at the same rate that it escapes today, which is rapid enough that little hydrogen would have been present at any time. However, new calculations, still subject to verification, suggest that conditions in Earth's early upper atmosphere could have slowed the escape considerably, allowing hydrogen to make up as much as thirty percent of Earth's early atmosphere. Until this issue is settled, we will not know whether spontaneous chemical reactions would have turned Earth's surface and oceans into a rich organic laboratory or left them with only small amounts of locally produced organic compounds.

We know of at least two other potential sources of organic molecules on early Earth. The first of these potential sources is chemical reactions near deep-sea vents. As these undersea volcanoes heat the surrounding water, a variety of chemical reactions can occur between the water and the minerals released from the vents. These chemical reactions would have occurred spontaneously in the conditions thought to have prevailed in the early oceans, and they should have resulted in the production of the same types of organic molecules thought to have been necessary for the origin of life. The other additional source of organic molecules may have been material from space. Analysis of meteorites shows that they often contain organic molecules, including complex molecules such as amino acids.

The Royal Palace of Foumban

The royal palace of Foumban, built from 1912 to 1922, is an architectural marvel and a mystery at the center of the former African kingdom of Bamum. Built from brick, its architectural style was unprecedented among indigenous buildings in the Grassfields (the prairies of western Cameroon)when it was constructed. Scholars seeking to write the history of the Grassfields continue to puzzle over the influences that led King Njoya (ca. 1885 to 1933) to conceive of such a structure.

Their task is made more difficult because no writing systems were in wide use in the region until the late 1920s. Few written documents illuminate the history of the area, and historians have been required to look to other sources, such as art and architecture. Palace complexes, which can be found in almost every kingdom in the Grassfields, required a great amount of effort, material, and labor to build, and so can provide important historical information. Their design and construction reflect deliberate choices in style, the use of materials, and the incorporation of decorative elements that can provide a window into past events and developments. The use of glass beads from centra Europe to decorate the walls of the throne room of the royal palace in the kingdom of Bali, not far from Foumban, shows that the kings of Bali were able to acquire rare and expensive materials from abroad. This fact alone, of course, does not resolve whether the beads were obtained by trade, as a gift, or by conquering another kingdom that possessed them. Such questions can only be answered, if at all, by assessing additional evidence.

In the case of the Foumban palace, many architectural historians believe King Njoya modeled his building on the 1905 stone mansion of the German colonial governor, Jesko von Puttkamer (1855-1917). The governor's mansion is in the town of Buea, only a hundred miles or so away from Foumban, and King Njoya visited Buea in 1908 to meet with a later German governor, Theodor Seitz (1863-1949). The trip to Buea was the king's only lengthy visit outside the Bamum kingdom before the Foumban palace was built, and it is likely the visit had a strong effect on the young monarch. During his stay in Buea, it is certain the king saw the governor's mansion, and he might have even stayed in it. In many respects the two buildings appear similar. Both are multistory structures, with symmetrical windows and turrets. They are made of similar materials and both buildings were originally covered with lime-based plaster to create a uniform and smooth exterior surface.

These building materials differentiate the Foumban palace sharply from other Grassfields palace complexes and even the palace complex built in the 1860s by King Njoya's father, King Nsangu, which burned in the first years of the twentieth century. The palace in the neighboring kingdom of Bandjoun is an example of typical Grassfields palace architecture. Though much larger and more complex than the houses of ordinary people, it was not built with brick but exclusively with less durable vegetable materials such as the immense ribs and leaves of raphia palm trees and the tall grasses from which the Grassfields take their name

More recent scholarship has cast doubt on how much Njoya drew on the Buea mansion when planning his own palace. Some scholars have pointed out that even before 1908, European missionaries in the Bamum Kingdom and elsewhere in the Grassfields had built churches and homes using brick and stone, suggesting a closer source of inspiration for the Foumban palace than Buea. Other scholars have suggested that Njoya's inspiration should not be sought exclusively in European architecture in Africa, but rather in other African indigenous architectural traditions. They argue that the Foumban palace bears a closer relationship to palaces in the kingdoms beyond the Grassfields in northern Cameroon and Nigeria. These palaces were also made of brick, and they possess architectural elements, such as arcades and vast, pillared rooms much like the central throne room of the Foumban palace. Although King Njoya never visited these northern palaces himself, his ambassadors did. They brought back a number of cultural innovations from their trips north, such as the use of horses in warfare, that were eagerly adopted by King Njoya and his court.

Temperate Plant Phenology

Phenology is the timing of growth and reproductive activity within a year, and it can vary greatly among plant species, populations, and even individuals. In temperate forests (characterized by moderate temperatures), many forest-floor plants expand their leaves and flower before the trees that form the forest's upper layer (the canopy) begin leaf expansion. The consequence is that these plants do most of their growing and reproducing in temperatures that are far colder than those experienced by the canopy trees during the same life stages. Many studies have shown that this timing is actually advantageous to the forest-floor plants. Once the leaves of the trees fill in the open spaces, very little sunlight sometimes as little as 1 percent-gets through to the forest floor. Thus, the forest-floor plants' growth and reproduction would be even more limited by light availability than they are by cool temperatures.

In most temperate species, it appears that temperature and photoperiod (day length) are the main factors determining plant phenologies. It is important to realize that temperature plays its role in a particular way. It is generally the sum of temperatures experienced over some period of time and not the temperatures on a particular day that determines the timing of leaf expansion. In cold years, then, leaf expansion is delayed. Plants, like other organisms, often use photoperiod as a reliable predictor of the average temperature. If they relied exclusively on temperature, a warm spell in midwinter would typically cause many plants to expand their leaves. By using temperature as a cue, plants can respond to an early spring by expanding their leaves early, but because they also use photoperiod as a cue, this response is limited. These plants, in other words, respond to environmental fluctuations, but they do so cautiously.

As the example of temperate forest-floor plants suggests, there are at least two main types of factors that act on plant phenologies: non-biological factors limiting growth-such as the timing of killing frosts (frosts cold enough to kill plants) or seasonal droughts-often determine the beginning or end of growth episodes (or both), but biological factors-in particular, competition for light or water-are also quite important for many species.

This observation raises an interesting question: why do the canopy trees in temperate forests wait so long before expanding their leaves? Why don't they use the early spring to add to their growth, as the forest-floor plants do? There are a number of selective forces affecting the timing of leaf expansion. First, the canopy is elevated. Temperatures in the treetops can be considerably colder than those at ground level. Thus, leaf expansion for trees occurs later on the calendar. but it may not really be much later in terms of average temperature. Second, late frosts are a common occurrence. Leaves on trees are much more vulnerable to frost damage than are those on forest-floor plants, because the latter are partly sheltered by the trees and because the ground-level temperatures are higher. Third, all enzymes (substances that facilitate chemical reactions) used for plants metabolism have different defined temperature ranges over which they can operate, and they are most efficient at particular temperatures. Enzymes adapted for peak functioning at warm temperatures are unlikely to be efficient in the early spring; it is possible that earlier leaf expansion might reduce the total year's growth of the trees, not increase it. Finally, many temperate-zone trees are wind-pollinated, and pollination in most of these species occurs while the trees are leafless. The presence of leaves earlier in the season would be likely to limit pollen transfer.

The phenologies of temperate forest-floor plants are actually more complex than we have implied so far. Most of the plants in northeastern North America have phenologies like those described here, but there are others that are able to capture and use light at other times. Some species can use light flecks on the forest floor to grow during the summer months. Still others use the additional light available in autumn, when some canopy species have begun shedding their leaves.

Classical Greek Tragedy

In 534 B.c.E. the city-state of Athens in what is today Greece added performances of tragedies to its annual festival honoring the god Dionysus. In ancient Greece, a tragedy was a serious drama that explored themes concerning human nature, often depicting a conflict between the hero and a superior force (such as destiny) and having a sorrowful or disastrous conclusion. Like the famous Olympic Games, theater for many years was a contest, at first among playwrights for the best set of tragedies, but later for best comedy and best actor, among other competitions. At first, the actor and playwright were the same person. The early tragic playwrights, such as Thespis, performed their own plays, which featured one actor and a chorus (a group of singers/dancers who interacted with and responded to the actor). Later a second and then a third actor were added, but all of the surviving tragedies were written for three actors who performed multiple roles with a chorus whose size is usually estimated at twelve to fifteen. The chorus, however, may have begun with fifty performers, since that was the size of the traditional dithyramb (a choric presentation sung and danced in homage to Dionysus). The dithyramb is older than tragedy and is claimed to be a source for it.

All actors and chorus members were strictly male(even though many characters and choruses represented females), and they performed in a large, typically circular performance space called an orchestra(dancing place). The chorus entered and exited through the parados (roofed passageway). At some point a space called a skene was added behind the orchestra for changing masks and costumes, and the audience of many thousands watched the action from a large theatron (seeing place) on a hillside to which seats were eventually added. All parts of the theater, though temporary at first for each Dionysian festival, were set in stone permanently by the fourth century B.C.E. Although a few theaters held only a few thousand spectators, most of the surviving theaters are so large that they remind many people of stadiums. The surviving architecture creates a dilemma for modern historians, since nearly all of the surviving plays are from the fifth century B.c.E. but surviving Greek theaters are from the fourth century B.c.E. or later. Therefore, because the theaters and plays were created in different eras, the theatrical spaces that still exist may not be good indications of the staging that was originally used for the plays.

At the Dionysian festival, Greek tragedies, once fully developed, were performed by three actors in three groups of three plays by three competing playwrights. After all the performances, a group of judges, chosen from the ten tribes of Athens, selected the set of three plays they found the best. Sometimes a playwright told one story through all three plays (a trilogy that can be read as one three-part play today). Other playwrights presented three different stories in their three tragedies. All Greek tragedies that we now possess except one are based on myths that were ancient even when the plays premiered. The famous stories were familiar to many in the audience, but it is clear that playwrights retold them in a variety of ways. The story of Electra, a princess who conspired with her brother to murder her mother and the mother's lover, survives in tragedies by Aeschylus (525-456 B.C.E.), Sophocles (496-406 B.C.E.), and Euripides (480-406 B.C.E.), and each tragedy has a very different focus and point of view. These three playwrights won numerous Dionysian contests, and some of their plays continue to inspire us in modern revivals.

The only surviving trilogy of plays, The Oresteia (458 B.C.E.) by Aeschylus, is still performed, translated, and adapted and has proved to be one of the most enduring dramatic presentations in theater history. The three plays that make up The Oresteia (one of which dramatizes the story of Electra)explore personal vengeance and the creation of a standard for criminal justice. Aeschylus also wrote the only Greek tragedy we have that was based on events in the playwright's own time rather than mythology: The Persians (472 B.C.E.)examines with sympathy enemies of Greece whom the playwright had fought a few years earlier.

Carbon Dioxide Levels in the Ocean

The concentration of carbon dioxide gas (CO2) is rising in the atmosphere, and some of this CO2 is taken up by ocean surface waters. To determine how much CO2 has ended up in the ocean, oceanographers have made measurements at nearly ten thousand sites. The results show that carbon content has increased everywhere in the ocean. These results contradict the claim that the atmospheric increase in CO2 is a result of CO2 being released by the ocean, similar to what happened at the end of the ice ages (long periods of reduction in the temperature of Earth's surface and atmosphere). The carbon added to the climate system from fossil fuels has increased CO2 levels in both the atmosphere and the ocean.

The fact that the ocean has taken up so much carbon from the atmosphere sounds like a blessing. If this were not the case, atmospheric CO2 concentration would already be much higher and climate would have warmed more. But scientists are now realizing that this ocean CO2 uptake is in itself a huge problem: it makes the ocean waters more acidic. This is very simple chemistry-CO2 dissolved in water makes carbonic acid. The effect can easily be calculated, and long-term measurement stations are showing it directly. Acidity is measured in units called pH-the lower the pH value, the more acidic the water is. Since preindustrial times, the pH value of the ocean surface has already dropped by 0.1 units. This corresponds to a 30 percent increase in the number of the chemically aggressive hydrogen ions (electrically charged particles) in the water.

The reason excessive acidity is very important is its effect on marine life. Many marine organisms build hard shells that are made of calcium carbonate (CaCO3), which comes in two forms: calcite and aragonite. Familiar sea shells are made from CaCO3, but more importantly so are the shells of the main microscopic plankton species that form the basis of the marine food web. And coral reefs, which host the most beautiful and diverse marine ecosystems, are also built from calcium carbonate. This marine life can thrive only when there is an overabundance of calcium carbonate in seawater. Put more technically, calcium carbonate needs to be supersaturated, so that these organisms can extract it from the water to incorporate it into their shells. There is a well-defined chemical threshold where calcium carbonate switches from being supersaturated to undersaturated and by adding CO2 and making seawater more acidic, we are pushing the ocean toward that threshold.

In fact, this threshold can be seen for real in the ocean: while the surface waters are supersaturated, the deep waters are undersaturated in calcium carbonate for natural reasons. The dividing line (called the saturation horizon) is at different depths depending on location, varying between 200 meters depth in high latitudes and 3,500 meters depth in parts of the Atlantic. What happens to the shells of dead plankton that sink down below the saturation horizon? They dissolve, as undersaturated water is corrosive to these shells. That's why the saturation horizon is often compared to the snow line in the mountains, with the snowflakes being the microscopic shells constantly drifting down from the biologically productive surface waters. Where the ocean bottom is above this line, the shells remain intact and form white sediments that are analyzed by scientists to give information about past climates. The white chalk cliffs of Dover, England, are made from this type of calcareous sediment that was uplifted above the sea surface.

White Cliffs of Dover

Measurements show that this saturation horizon is moving upward in the water column, consistent with the uptake of anthropogenic (human caused) CO_2 by the ocean. This raises concerns for marine life-particularly in the Southern Ocean, where the saturation horizon is already very shallow at only 200 meters depth. Laboratory experiments where plankton were grown in water enriched with CO2 show that their growth is impaired. According to model simulations, by the year 2050 the changes in ocean chemistry will start to harm marine life if the atmospheric CO2 concentration continues to rise. For coral reefs, the IPCC (Intergovernmental Panel on Climate Change) report concludes that the impacts of acidification may be severe-corals produce aragonite, the form of calcium carbonate that is more susceptible to acidification.

Prosperity in British North America

The British colonies of North America developed at a remarkable pace throughout the later colonial period. Colonial trade with Britain grew by 700 percent in the period 1689-1760, while the population increased from 250,000 to around 1.5 million. At the same time, per capita (per person) income rose by at least 0.5 percent a year in real terms, so that the standard of living for most of the population improved by between 50 and 100 percent. The American colonies were one of the first societies to escape the cycle in which increased resources merely stimulated population growth and ultimate decline in per capita income (the total national income divided by the number of people in the nation). The colonies never experienced the kind of severe food shortages which afflicted most European societies through the nineteenth century.

One indication of colonial well-being was the reaction of visitors. In the seventeenth century, it was the wild appearance of the continent that drew comment, whereas by the 1750s, visitors were impressed by the general prosperity of the inhabitants. Most people had land, and there seemed to be little poverty or unemployment. As a British officer commented, “Everybody has property and everybody knows it.” The general progress was well symbolized by the rise of Philadelphia, a town that did not exist in 1682, but which by 1760 was on a par with Dublin, Edinburgh, or Bristol.

The attractions of living in America were constantly emphasized by Benjamin Franklin. He pointed out how much better off the inhabitants of America were compared with their counterparts in the British Isles. The same points were made by Thomas Hutchinson in his History of Massachusetts. He wrote, "Property is more equally distributed in the colonies... especially those to the northward of Maryland than in any nation in Europe." The reason, of course, was the availability of land.

More recently, historians have also been paying attention to other types of evidence. Documents such as wills, for example, tend to support the view that living standards were improving. Whereas in the seventeenth century, most inheritances consisted of only a few tools, some rough furniture, and livestock, eighteenth-century inventories reveal a much richer standard of living, including China rather than coarse pottery, silverware, furniture, clocks, warming pans, and other household items designed for decoration or comfort rather than mere survival. After 1700, these increasingly included what earlier would have been considered inessential luxury items: tea, coffee, French wines, Indian silks and calicos, glass, porcelain, and even furniture. Previously, colonials had made their own, found substitutes, or done without. The first sixty years of the eighteenth century witnessed a consumer boom, which seems to suggest a standard of living 20 percent higher than in Britain, for the middle classes at least.

How was this prosperity achieved? One answer was greater efficiency. Although historians have generally condemned the American colonial farmer for wasteful land usage, improvements did occur through greater knowledge of the climate and terrain. Some modest technological improvements occurred in manufacturing and commerce, though less so in farming. Most colonials were still using tools similar to those of their seventeenth-century ancestors, though implements like plows were now used more widely. In addition, the colonial period was generally one in which prices remained stable or actually fell as production methods improved and transport became cheaper, giving settlers greater purchasing power. Throughout the period, the colonists continued to enjoy that almost unique combination of abundant land, cheap food, and unlimited fuel.

Lastly, the growth in living standards was ironically fueled by the consumer boom itself. Until the end of the sixteenth century, only the upper classes were expected to live opulently as a sign of their rank and a means whereby the social order could be maintained. Ordinary people, it was assumed, would merely be corrupted by such consumption into idleness and vice. By the end of the seventeenth century, such attitudes had changed in favor of personal improvement. Now it was argued that many of these same luxuries were items of refinement, which could be defended on moral grounds as leading to propriety. This change in attitudes had two consequences. One was a surge in demand. The other was an increase in production, as people sought to improve their standard of living.

Debating the Agricultural Diet

An affluent society is one in which resources are ordinarily not scarce, and most society members have achieved a general level of economic well-being. The earliest humans were hunter-gatherers, meaning that they subsisted on wild plants and animals. During the 1960s, anthropologist Marshall Sahlins famously described hunter-gatherer groups as "the original affluent society." Based on empirical data from some of the few remaining hunter-gatherer groups, Sahlins argued that hunter-gatherers worked less and had more leisure time than their contemporaries in either traditional agricultural economies or industrialized economies. Although they had few possessions, they were not poor; as Sahlins stated, "Poverty is a social status. As such, it is the invention of civilization" (social organization characterized in part by the unequal distribution of resources and status).

In the dietary realm, Sahlins noted that large numbers of people in agriculture-based societies went to bed hungry each night, while hunter-gatherers had a steadier and more varied food supply. This food supply was based on intensive knowledge of the seasonal availability of plant and animal foods in the environment. Historically, such knowledge was not obvious to all "civilized" observers of hunter-gatherers, although some did note that hunter-gatherers managed surprisingly well despite being "handicapped" (as these observers described it) by a decided lack of civilized knowledge.

The varied diet of hunter-gatherers stands in stark contrast to the monotonous diet of many agricultural populations. In some cases, reliance on a single food crop can lead to diseases resulting from vitamin deficiency. Populations in which corn is the principal food crop can be susceptible to high rates of pellagra, a disease caused by a deficiency in the B vitamin niacin. The symptoms of pellagra can be quite unpleasant, including a distinctive rash, diarrhea, and even mental disturbances. Some cultures traditionally treated corn with an alkali, which released niacin from the hull of the corn, thus reducing the likelihood of developing pellagra. An over-dependence on polished rice can lead to beriberi, a disease of the nervous system caused by a lack of vitamin B1 (thiamine).

More generally, traditional agricultural diets, because they are less varied, are not as good as hunter-gatherer diets at providing all of the specific nutrients that our bodies need to thrive. On the other hand, they clearly provide enough to allow people to survive and reproduce, increasing population numbers. Many archaeological studies have been done comparing the skeletal health of groups before and after the advent of agriculture in the same location. In almost all cases, the agricultural group shows indications of nutritionally based bone or tooth stress not present in the hunter-gatherer comparison population. In addition, agricultural groups, because of their higher population densities, are more likely to be exposed to infectious diseases. If domestic animals are present, then there is also a risk of disease being transmitted from animals to humans.

Things have changed considerably in developed countries with the shift from traditional to modern industrialized agriculture. Food is relatively cheap and abundant, and the supply is buffered from the effects of seasonality and poor yields. Initially, combined with medical advances to control infectious disease, modern agriculture supported the growth of healthier and larger populations; people were also individually healthier and of bigger physical size. In the mid-twentieth century, however, it became apparent that there were groups that did not do so well with this modern Western diet. For example, when some Native American and Pacific Islander populations gave up their traditional agriculture-based diets (which also included fishing and hunting) for a Western diet, they became highly susceptible to developing obesity and all of the medical conditions, especially diabetes, associated with it. Geneticist James Neel came up with the idea that non-European populations were at risk for these diseases precisely because they had not been previously exposed to such a flush nutritional environment. He argued that in European populations, metabolically efficient genotypes—genotypes associated with a high risk of developing obesity and diabetes—have been eliminated over time because European populations lived in a rich nutritional environment. Neel was probably wrong in thinking that premodern European diets were particularly abundant, so his scenario for the historical elimination of "thrifty" genotypes in those populations is probably wrong also. Nonetheless, the high susceptibility of some recently Westernized groups to developing obesity and diabetes was and is very real, for whatever reason.

Diving for Food by Birds

Seabirds are physiologically adapted to dive underwater for food, a strategy that necessarily poses problems. An obvious problem is that when underwater, the bird cannot breathe and must rely on its accumulated oxygen stores. Diving adaptations have been most extensively studied in emperor penguins, which can remain underwater for twenty or more minutes. During the first five minutes or so, an emperor penguin's body processes continue to function as they do when the animal is breathing air, with lactate building in the muscles as the bird expends oxygen from them. After about five minutes, however, lactate begins appearing in the blood, at which point the penguin shifts to anaerobic metabolism—cell function without oxygen—in which nonessential organs are shut down, and blood is sent only to selected tissues.

Another key adaptation to diving is a reduction in heart rate underwater. Detected via attached electrocardiogram recorders, the heart rate of a resting emperor penguin is around 70 beats per minute. This value roughly doubles to around 140 beats per minute immediately before the dive. If the dive is long, heart rate drops off dramatically and may reach as low as three beats per minute. Just before the penguin surfaces, the rate accelerates. It can be around 200 beats per minute when the penguin surfaces and can breathe once more to replenish its oxygen stores.

Remembering that even in tropical areas, the water temperature below 200 meters is probably no higher than 5°C, a further physiological problem faced by seabirds underwater is potentially that of cold. Penguins and auks have tight plumage (feathers) that retains air close to the skin. This assists heat retention, albeit by creating buoyancy (tendency to float) that hinders the downward dive. The situation is different in cormorants and shags. Their plumage is notoriously wettable. Think of the classic pose of a perched cormorant hanging out its wings to dry after a spell of swimming. If the water has reached the skin, the cormorant will have lost more heat than another seabird whose skin remains dry.


Cormorant perched on a rock

Courtesy of JJ Harrison/Wikimedia Commons

In fact, the paradox is more apparent than real. The great cormorant is one of the most widespread of all seabirds. Its breeding range extends from Australia via Southeast Asia to Europe and Greenland. Whether in northern France or Greenland, it requires less food per day than other seabirds of comparable weight. The birds are evidently not leaking heat and do not need to take in extra food. How they retain heat became evident when European researchers looked at the plumage more closely. All four subspecies studied, living in sub-Arctic to subtropical climes, retained an insulating air layer in their plumage, which was, however, much thinner than for other species of diving birds. Detailed examination of the plumage showed that each cormorant body feather has a loose, instantaneously wet outer section and a highly waterproof inner portion.

Because of air trapped in the plumage, plus that in the lungs and air sacs (lung extensions), seabirds are positively buoyant when first submerging. Their natural tendency is to float back upward. Therefore, they have to work hard to reach greater depths. As the birds reach those greater depths, the air compresses, and eventually a depth is reached where they are neutrally buoyant; that is, they will neither sink nor rise. This depth depends on the quantity of air in the lungs and air sacs at submergence; the more air to be compressed, the greater the depth of neutral buoyancy. In fact, penguins anticipating diving deeper take in more air before submerging, which then allows a longer dive and increases the depth at which they are neutrally buoyant. Therefore, rather neatly, there is a close correlation between diving depth and duration.

Having, in ornithologist Rory Wilson's evocative descriptions, pedaled downward, the bird faces an ascent that is relatively easy. This is simply because the compressed air expands at shallower depths, increasing positive buoyancy and serving to thrust the bird upward at an ever-faster speed. For example, Magellanic penguins beginning their ascent from 50 meters initially ascend at a little more than half a meter per second. When they have reached 20 meters, the ascent rate has topped a meter per second.

Dorset Culture

In the Canadian Arctic, the Pre-Dorset culture evolved into the Dorset culture by about 800 B.c.E. The Dorset culture, which was first recognized by Diamond Jenness in 1925， was named after Cape Dorset, where some of the first artifacts were discovered. Archaeologists originally believed that the Dorset culture was a new migration of people from Alaska, but this is no longer widely accepted. Understanding the transition from the Pre-Dorset to the Dorset culture is complicated because the rate of change was not equal in all areas of the Arctic. Starting about 1500 B.C.E., a cooling trend began that lasted until about 1 c.E. The impact in the northern Arctic was that there were fewer land animals. While some Pre-Dorset people migrated south in search of a warmer environment, others did not migrate, were unable to adapt to the changes in the climate, and died of exposure or starvation. However, some developed the Dorset culture that allowed for survival in a colder environment.

Thus the Dorset culture emerged between 800 and 500 B.C.E., during a time of climatic cooling, and was marked by a change in material culture including the addition of snow knives, presumably used to construct snow houses, and also of soapstone lamps, which were used inside snow houses for heat and light. The cooler climate resulted in an increase in the amount and duration of winter sea ice, which in turn resulted in an increased emphasis on hunting sea mammals, especially seals. Unlike the Pre- Dorset people who focused on terrestrial hunting, the Dorset people appear to have done inland hunting of tundra animals primarily in the summer months. The bow and arrow apparently disappeared during the Dorset period, as did the use of the drill. The Pre-Dorset people had used the drill to make holes in household items and weapons, whereas the Dorset laboriously cut or carved holes in items. Much of the tundra area in the Canadian Arctic occupied by the Pre-Dorset people was abandoned by the Dorset people as they moved to coastal areas.

The Dorset culture was adapted to the harsh Arctic environment and survived for more than a thousand years. However, there were two factors that contributed to its disappearance. The first was a warming trend that started about 900 c.E. The Dorset people had evolved a culture that was highly dependent on hunting sea mammals from the ice, and the warming trend changed not only the amount and duration of sea ice but also the distribution of sea mammals and the hunting techniques needed to obtain them. Although a thousand years earlier the Dorset people made the cultural changes necessary to survive when the climate became colder, they did not make the cultural changes necessary to survive in a warmer environment.

The second factor that contributed to their disappearance was the arrival from the west of a new culture, the Thule, which was named after a site in northern Greenland where their remains have been found. It is not known if the Dorset gradually died out or if they were absorbed or annihilated by the Thule. The Thule culture was fully developed in northern coastal Alaska by 1000 c.E. and had a highly specialized and effective technology for hunting large sea mammals. As the climate became warmer, previously ice-filled areas in the eastern Arctic opened up, and as sea mammals migrated east, toward the area inhabited by the Dorset, the Thule people followed. The migration was very rapid, and within 200 years they had spread from Alaska to Greenland. In the eastern Arctic, the Thule people are referred to as Neo-Eskimos since they shared no cultural relationship with the Dorset who inhabited the Canadian Arctic when they arrived.

It is generally believed that the Dorset and Thule peoples had some interaction. The Thule adopted several aspects of Dorset culture, such as snow houses and soapstone lamps, and Thule legends discuss contact in the Canadian Arctic with the "Tunit," also spelled Tuniit, who are believed to be the Dorset. While there is no direct evidence of violence between the two groups, the Dorset people were probably pushed to peripheral resource areas of the eastern Arctic since the Thule were better organized and had weapons, such as the bow and arrow, which helped them control the better hunting areas.

Life in the Desert

In the austere and inhospitable environment of the desert, each inhabitant must be highly specialized in order to survive. In spite of the common notion that a desert is deserted of life, there are actually many living organisms. The principal limiting factor for successful animal and plant species is, obviously, the dearth of water, and some of the most significant adaptations have evolved in solution to this basic problem.

In the desert, many of the plants are annuals, lasting only one season. Some are ephemerals, which arise only after the rare rains and rapidly conclude their cycle before the new drought. They have no need of any particular specialization, other than opportunism and rapid growth. The perennial plants that regularly grow year after year, in contrast, must be much more able to adjust. Normally, they have leathery leaves covered with waxy films to reduce the transpiration and evaporation of internal liquids. Many plants have reduced leaf surfaces, or they have transmuted them into sharp thorns. Some species, such as mosses and lichens, curl their leaves during the dry periods. Other plants have fleshy tissues in which they concentrate and conserve great quantities of reserve water. Still other plants grow only in the proximity of sources or basins of water dispersed in the desert (in oases)or around riverbeds where underground water flows. These plants have extremely long roots, which reach down to the moisture coming from the permanent water table. This is the case for the well-known date palm of the oases in the Sahara and the Middle East. When the water layer is too distant, however, other species live by spreading their roots along the surface in order to enjoy the moisture that condenses during the night but is insufficient to penetrate deeply into the soil.

Adaptations that are equally remarkable can be seen among the animals, which, in order to survive, have no fewer problems to resolve than do the plants. Heat and lack of water are again factors limiting the possibilities of life: the factors that must be addressed by the choices and strategies of each individual organism.

To these factors is often also added the scarcity of food, which is partially resolved by the low density of animals in the desert. Similarly, to what happens during the winters of the temperate latitudes, when many species hibernate, desert animals reduce their metabolism during the summer by becoming less active, often in the shelter of rocks or underground. The same can occur in the case of prolonged drought. This seasonal behavior recalls what also happens on the scale of a single day: activities are stopped during the hottest hours, and everyone seeks shelter in the shade. Large-dimensioned species, such as hoofed or carnivorous mammals, which are endowed with relative facility of movement, as flying insects and birds are also, often undertake nomadic movements or actual migrations to find more-hospitable areas during periods that are particularly unfavorable because of high temperatures or drought.

There are many adaptations in form and function to the torrid climate of the desert. For the purpose of more easily dispersing the heat, animals in hot desert regions have smaller bodily dimensions than similar species living in colder areas. The opposite is true for the appendages, which are more developed and sometimes moist in animals living in hot areas, in order to cool off more easily. This is the case for the enormous ears of the jackrabbit- the hare of the Sonora Desert in America. Many desert animals are capable of tolerating long periods without drinking. They aid themselves in this regard by producing urine that is highly concentrated, thus reducing their loss of water. Some desert mice, alternatively, conserve the water that would be lost in respiration by making it pass through a series of intricate ducts that open into the nasal sinuses, where the air loses moisture as it cools.

How the solar system Was Formed

Any model of the origin of the solar system must explain why all the planets orbit the Sun in the same direction and in nearly the same plane. One model that was devised explicitly to address this issue was the tidal hypothesis, proposed in the early 1900s. Two nearby planets, stars, or galaxies exert varying gravitational, or tidal, forces on each other that cause the objects to elongate. In the tidal hypothesis, another star happened to pass close by the Sun, and the star's tidal forces drew a long filament (threadlike formation of matter) out of the Sun. The filament material then went into orbit around the Sun, and all of it naturally orbited in the same direction and in the same plane. From this filament of material, the planets condensed. However, it was shown in the 1930s that tidal forces strong enough to pull a filament out of the Sun would cause the filament to disperse before it could condense into planets.

An entirely different model is now thought to describe the most likely series of events that led to our present solar system. The central idea of this model dates to the late 1700s, when the German philosopher Immanuel Kant and the French scientist Pierre-Simon Laplace turned their attention to the manner in which the planets orbit the Sun. Both concluded that the arrangement of the orbits—all in the same direction and in nearly the same plane—could not be a mere coincidence. To explain the orbits, Kant and Laplace independently proposed that our entire solar system, including the Sun as well as all of its planets and satellites, formed from a vast, flat, rotating cloud of gas and dust called the solar nebula. This model is called the nebular hypothesis.

The consensus among today's astronomers is that Kant and Laplace were exactly right. In the modern version of the nebular hypothesis, at the outset, the solar nebula had a mass greater than that of our present-day Sun. Because each part of the nebula exerted a gravitational attraction on the other parts, matter at the perimeter was pulled toward the center, whereas matter in the center, because it was pulled equally from each side, did not move at all, and thus the nebula contracted with the greatest concentration of matter occurring in a central region called the protosun. As its name suggests, this part of the solar nebula eventually developed into the Sun. The planets formed from the much sparser material in the outer regions of the solar nebula. As a result, the mass of all the planets together is only 0.1 percent of the Sun's mass.

When you drop a ball, the gravitational attraction of Earth makes the ball fall faster and faster as it falls; in the same way, material falling inward toward the protosun gained speed. As this fast-moving material ran into the protosun, the energy of the collision was converted into thermal energy, causing the temperature deep inside the protosun to climb. This process, in which the gravitational energy of a contracting gas cloud is converted into thermal energy, is called Kelvin-Helmholtz contraction, after the nineteenth-century physicists who first described it.

As the newly created protosun continued to contract and become denser, its temperature continued to climb as well. After about 100,000 years, the protosun's surface temperature stabilized at about 6,000 Kelvins, but the temperature in its interior kept increasing to ever-higher values as the central regions of the protosun became denser and denser. Eventually, after perhaps 10 million years had passed since the solar nebula first began to contract, the matter at the center of the protosun reached a density about 100 times greater than water, and a temperature of a few million Kelvins. Under these extreme conditions, nuclear reactions that convert hydrogen into helium began in the protosun's interior. When this happened, the energy released by these reactions stopped the contraction, and a large amount of light energy was radiated into space. The protosun had become a star. Nuclear reactions continue to the present day in the interior of the Sun and are the source of all the energy that the Sun radiates into space.

Snakes

Snakes differ from other living reptiles in terms of body shape and flexibility, and their evolutionary history has been a matter of ongoing debate. Whatever the precise relationships among snakes and other reptiles, it's clear that snakes arose from lizard-like ancestors. The many divergent features of snakes, including their elongate (lengthened) and limbless body form and unusual lidless eyes, have resulted in two main competing explanations for snake origins. The dominant hypothesis is that snakes underwent a burrowing (underground digging) phase in their immediate ancestry or early history. The supporting evidence lies in the observation that several lineages of lizards (for example, slow worms and dibamids) have become elongate and have lost or substantially reduced their limbs, and all these burrow in sand or soil. Some of the peculiarities of snake eyes are the result of evolutionary loss of some ancestral features, and this is also known to have occurred to varying degrees in other burrowing vertebrates.

The main competing hypothesis is that snakes passed through a profoundly water-dwelling phase in their immediate ancestry. Although some scientists claim that snake eyes are similar in several features to those of other aquatic vertebrates, the main supporting evidence for the marine hypothesis comes from several snake fossils that retain hind limbs and that were fossilized in marine sediments deposited in the Cretaceous period (145 to 65 million years ago). Proponents of this marine hypothesis argue that elongation of the body and reduction of the limbs arose as an adaptation to swimming in water rather than burrowing on land. The marine hypothesis has been weakened by evidence that the Cretaceous limbed snakes are not close to the immediate ancestry of living snakes, but are instead probably embedded more deeply within the evolutionary tree of living snakes, so that their limbs might have partly re-evolved from small stubs such as are still found in, for example, python snakes.

Although the marine Cretaceous snakes are known from some spectacular and complete fossil specimens, the fossil record of snakes generally is irregular and largely fragmentary. Many snake fossils consist of no more than isolated vertebrae. Interpreting these fossils can be difficult because even vertebral variation within and among individuals and species of living snakes is not well-known in detail. It is particularly tricky to identify the earliest member of any major group of organisms based on fragmentary fossils, and this is true also for snakes. Geologically the oldest snake reported thus far is from Cretaceous rocks in North America dating to about 98.5 million years ago. An older fossil find from Spain (between 132 and 124 million years old) was claimed previously to be a snake, but its vertebral anatomy is not distinct enough from that of lizards to be sure.

Over the course of evolution, elongation of the snake's body has necessitated the modification and rearrangement of its internal organs. The lungs in particular have undergone considerable modification. Some snakes have two lungs, but in these cases, the left is always smaller than the right. In some pythons, the left lung can be up to 85 percent of the length of the right lung, but in the majority of snakes the left lung has been either lost completely or become greatly reduced, as little as one percent of the right in some cases. In some aquatic snakes, such as the Asian/Australasian file snakes, the right lung is particularly large, extending backwards for nearly the entire length of the body. A possible additional respiratory (breathing) area in many snakes is provided by a special modification of the windpipe, known as the tracheal lung.

In most vertebrates, paired organs such as the kidneys and reproductive organs usually lie in the same position on either side of the body, but these are irregularly arranged in the elongate body of snakes. Among other organs affected by the radical change in body shape is the stomach, which has become greatly enlarged and in some snakes accounts for more than one-third of the total length of the body. There is also no urinary bladder; nitrogenous waste is eliminated not in a solution of liquid urea, as in humans and almost all other mammals, but in a semi-solid state as uric acid, as in lizards and birds.

The Cost of Venom

Venom, a poisonous substance that is produced by some animals and injected into an enemy or prey mainly by biting or stinging, is clearly an advantage for defense or hunting. But it is an expensive resource. Replenishing empty venom glands can greatly diminish an animal's energy supply. Even though the venom supply in snakes makes up only a tiny fraction of their weight, refilling venom glands after use can increase a snake's metabolic (energy use) rate by more than 20 percent. The cost of using venom is not restricted to the direct metabolic cost of replenishing it, however. Spending venom can result in temporary disarmament, so that, for instance, a predatory spider lacking its venom becomes defenseless prey. The speed of refueling venom glands varies considerably between groups. It can take theraphosid spiders - a group of mostly large and hairy spiders popularly known as tarantulas or bird-eating spiders - as long as 85 days to finish reloading their glands, while the tiny caterpillar-killing parasitoid wasp Bracon brevicornis can reload its glands in less than three hours. The North American centipede regenerates about 85 percent of its venom volume within two days.

The great cost of venom, direct or indirect, is also indicated by the phenomenon of "venom metering" (monitoring the amount of venom used). Snakes can control the amount of venom they release so they do not use more than is necessary. A larger, more dangerous, or more agile prey receives a larger dose of venom to ensure it is subdued effectively. Although venom metering has been poorly studied in spiders, researchers know that they are also able to make decisions about when and how much venom to deliver. The Central American hunting spider Cupiennius salei injects more venom into larger prey. When dealing with prey that have body armor, such as beetles, and the armor prevents the spider from injecting venom into an optimal location, the spider also increases the dose. Remarkably, this spider can also distinguish prey that are more or less sensitive to its venom (crickets and cockroaches, respectively) purely on the basis of smell, and choose the more vulnerable prey when its venom glands are depleted.

Venom is not just metered during prey capture. Snakes, scorpions, and spiders also show defensive venom metering. Females of the western black widow spider, for example, can adjust the amount of venom in their defensive bites depending on the threat level. Squeezing the spider's body provokes her to deliver almost twice the dose of venom than if her leg is pinched. Even so, more than half of her defensive bites contain no detectable venom at all. The defensive bites of some snakes and the defensive stings of scorpions can likewise be less costly. The South African thick-tail scorpion initially stings predators with a transparent prevenom that is poor in proteins but rich in pain-inducing potassium ions. The scorpion releases its milky venom, which is more protein-rich and toxic - but also more expensive to produce - only when its prevenom fails to do the job.

The cost of venom does not only affect how venoms evolve: it can also cause a loss of toxins. Some specialist venomous predators that have evolved to feed on only a limited number of species have streamlined their venoms by losing toxin types that are no longer useful. This removes the cost of producing obsolete toxins. Sea snakes, for example, have relatively simple venoms compared to their terrestrial relatives, with the venom consisting of only a few neurotoxins that effectively immobilize their fish prey. In a fascinating case of convergent evolution (in which species not closely related independently evolve similar traits), the venoms of sea kraits, which are related to sea snakes but independently adopted a marine fish-eating lifestyle, have undergone the same streamlining. In fact, the venom of sea kraits is so similar to that of sea snakes that it can be effectively neutralized by sea snake antivenom. Venom is such an expensive commodity that it is discarded if its ecological importance disappears. When the marbled sea snake switched to a diet consisting exclusively of fish eggs, it did not simply lose a few toxin types. It lost its venom, fangs, and even its venom gland.

The Prokaryotic Cell

Of all the types of cells revealed by the microscope, bacteria have the simplest structure and come closest to showing us life stripped down to its essentials. Indeed, bacteria contain essentially no organelles (recognizable substructures) - not even a nucleus to hold their DNA. This property - the presence or absence of a nucleus -- is used as the basis for a simple but fundamental classification of all living things. Organisms whose cells have a nucleus are called eukaryotes (from the Greek words eu, meaning "well" or "truly," and karyon, a kernel or "nucleus"). Organisms whose cells do not have a nucleus are called prokaryotes (from pro, meaning "before"). The terms "bacterium" and "prokaryote" are often used interchangeably, although we shall see that the category of prokaryotes also includes another class of cells so remotely related to ordinary bacteria that they are given a separate name.

Bacteria are typically spherical, rod-like, or corkscrew-shaped, and small - just a few micrometers long. They often have a tough protective coat, called a cell wall. In the electron microscope the cell interior typically appears as a matrix of varying texture without any obvious organized internal structure. The cells reproduce quickly by dividing in two. Under optimum conditions, when food is plentiful, a prokaryotic cell can duplicate itself in as little as 20 minutes. In less than 11 hours, by repeated divisions: a single prokaryote can give rise to 5 billion progeny. Thanks to their large numbers, rapid growth rates, and ability to exchange bits of genetic material by a process akin to sex, populations of prokaryotic cells can evolve fast, rapidly acquiring an ability to use a new food source or to resist being killed by a new antibiotic.

Most prokaryotes live as single-celled organisms, although some join together to form chains, clusters, or other organized multicellular structures. In shape and structure prokaryotes may seem simple and limited, but in terms of chemistry they are the most diverse and inventive class of cells. These creatures exploit an enormous range of habitats, from hot puddles of volcanic mud to the interiors of other living cells, and they vastly outnumber other living organisms on Earth. Some are aerobic, using oxygen to oxidize food molecules: some are strictly anaerobic and are killed by the slightest exposure to oxygen.

Virtually any organic material, from wood to petroleum, can be used as food by one sort of bacterium or another. Still more remarkable, some prokaryotes can live entirely on inorganic substances: they get their carbon from CO2 in the atmosphere, their nitrogen from atmospheric N2, and their oxygen, hydrogen, sulfur, and phosphorus from air, water, and inorganic minerals. Some of these prokaryotic cells, like plant cells, perform photosynthesis, getting the energy they need for biosynthesis from sunlight: others derive energy from the chemical reactivity of inorganic substances in the environment. In either case, such prokaryotes play a unique and fundamental part in the economy of life on Earth: other living things depend on the organic compounds that these cells generate from inorganic materials.

Traditionally, all prokaryotes have been classified together in one large group. But molecular studies reveal that there is a gulf within the class of prokaryotes, dividing it into two distinct domains, called the eubacteria (or simply bacteria) and the archaea. Remarkably, at a molecular level, the members of these two domains differ as much from one another as either does from the eukaryotes. Most of the prokaryotes familiar from everyday life - the species that live in the soil or make us ill - are eubacteria. Archaea are not only found in these habitats, but also in environments hostile to most other cells: there are species that live in concentrated brine, in hot acid volcanic springs, in the airless depths of marine sediments, in the sludge of sewage treatment plants, in pools beneath the frozen surface of Antarctica, and in the acidic, oxygen-free environment of a cow's stomach, where they break down cellulose and generate methane gas. Many of these environments resemble the harsh conditions that must have existed on the primitive Earth, where living things first evolved, before the atmosphere became rich in oxygen.

Towns in the High Middle Ages

Europe was fundamentally an agricultural society in the High Middle Ages (1000-1350). Powerful local lords controlled the countryside, and social and economic relationships were based on the realities of rural life; however, towns did exist during this period. One may call them towns because even though they varied enormously in numbers of inhabitants, and some were smaller than large villages in this respect, they shared certain characteristics that set them off from rural settlements.

Relatively speaking, there was a greater density of population in towns than villages, although the line separating the two is necessarily arbitrary. Relatively, too, there was a more highly developed specialization and greater diversity of labor, especially -though not exclusively- of craft labor, in urban settlements. To be called a town, a settlement with a relatively high density of population and specialization of labor also requires an economy based on money as opposed to barter (the simple exchange of goods) and one in which a sizeable proportion of income, again speaking relatively, comes from trade. The presence of a regulated periodic market (meeting every week, for example) did not define a town, for many a village had one, too. But a town without a periodic market is unthinkable for the Middle Ages, which is not the case for a village. Finally, towns were distinguished, if not fully distinct, from large villages by a concentration of diverse monumental buildings - large churches, bell towers, warehouses, permanent market halls, hospitals, town halls, and so on - although not every town had the full array of such buildings even by the end of the Middle Ages.

Some scholars would supplement the foregoing list of characteristics, for towns often had mints for making money, special legal status, and modes of landholding for the inhabitants, as well as autonomous criminal and civil court systems. Because what distinguished towns from villages was not so much size as the density of economic and social activities, one writer has argued that an important criterion for a town is the existence of traffic jams - the hustle and bustle of oxcarts; long lines of wagons bringing fruits and vegetables, raw materials, and finished goods to markets; and the parade of men and women coming to shop, visit, or attend meetings.

Towns in Europe around the year 1000 were few in number and very small in size, yet by the early fourteenth century, the number and size of urban settlements had increased enormously. Central Europe, which had only the thinnest sprinkling of settlements that might be deemed urban in the year 1000, witnessed the creation of 1,500 new towns from the eleventh century up until 1250, and 1,500 more in the 50 years or so after that. The Rhine River valley, which had no more than eight towns well into the twelfth century, boasted more than 50 in the thirteenth. In southern Europe there was more continuity from earlier settlement patterns: growth came less from the creation of new towns than from the flow of immigrants to old settlements.

Most of the early eleventh-century towns were weak in relation to the lords that ruled the countryside. In the majority of cases in the north, the lords controlled or owned the towns even though they usually did not live in them. Within the towns, the agents of the lords' authority contended with bishops and other church officials for domination. The urban commercial element was ordinarily the weakest of the three contenders for power, but the relative weight of each party would change dramatically in favor of the merchants over the course of the High Middle Ages. In southern Europe (Italy), where traditions of urban oligarchic domination (control by a small group) had not been completely erased, merchants and senior craftsmen had a stronger say in town government. But here, too, there was a three-way struggle for power among the commercial oligarchy, the church, and the lay lords, who more often than in northern Europe actually had their principal residences in the towns. The outcome of these struggles would vary from region to region over time. Throughout Italy, but particularly in the northern region, these struggles would dominate the political culture of the entire period from 1000 to 1350.

Ice Age Sculpture

For many years, scientists debated whether prehistoric humans in Central Europe lived alongside the mammoth and whether they hunted it with their stone tools. The mammoth, an extinct species of the elephant family, roamed across Europe during Earth's last ice age—a period during which large parts of Earth were covered by ice. Protected from the cold by its hairy coat, it survived even the coldest period some 26,500 years ago, when Earth's ice cover was most extensive. Archeologists did not initially believe that humans could live in this inhospitable climate. However, in the late nineteenth century, the remains of ancient human settlements that included numerous burned mammoth bones were found at the Moravian Gate, a mountain pass in what is now the Czech Republic. This suggested that humans survived even in frozen regions near the margins of the ice sheets by using mammoths for food and for raw materials used in clothing, blankets, and shelter walls. These ice-age people left behind numerous small sculptures of mammoths and other animals, formed of ceramic or carved out of stone, bone, or tusks. While these works usually show animals in still positions, there are some that depict animals in motion.

One such animated sculpture is the Predmosti mammoth. The fragments from which it is restored were found by geologist Martin Krz in 1895 when he took over the excavations started by mathematician Karel Maška in 1882 in the massive deposits of loess (accumulated windblown soil) on the Bečva River in Moravia. In the intervening period, Maška had exposed an area of about 200 square meters and found over 30,000 artifacts, human burials, and more than 1,000 kilograms of animal bones, most of which were those of mammoths, sometimes discovered in organized piles. It was also Maška who joined the broken pieces of ivory to reveal the sculpture that was a unique and important find at that time. Made on a large, flat segment from a tusk of considerable size, the realism of the animal, obviously observed in life by the sculptor, immediately disproved the opinion of the Danish archaeologist Japetus Steenstrup, who had visited in 1888 and suggested that the occupants of the site had arrived much later than the mammoths whose bodies they had found frozen in the permafrost and utilized. The indisputably ancient context of the sculpture found among mammoth bones and stone tools was used to further confirm the great age of cave art when it was compared to mammoths known from the walls of caves such as Les Combarelles and Font de Gaume in southwest France, the authenticity of which had only recently been accepted after many years of doubt. However, Steenstrup's view that the massive accumulation of mammoth remains at Predmosti and other Moravian sites was not due to hunting prevailed and still provokes questions about how we might interpret the sculpture.

Arguments against the hunting of mammoths in Moravia surfaced in the twentieth century when Wolfgang Soergel and later, Olga Soffer suggested that humans had exploited the remains of mammoths that had died natural deaths rather than hunting such potentially dangerous prey. Herds of animals migrating through the Moravian Gate Mountain pass stopped at mineral licks (exposed mineral deposits), which provided them with the salts essential to the diet, particularly of young animals and pregnant females, during late spring-early summer, but weak, tired animals would also die there. Humans just moved in to take advantage of them. This picture is contested by Martin Oliva, who reiterates the archaeological evidence for hunting, combined with some searching for available food, and in so doing proposes that culture as well as nature made the mammoth important to society in ways over and above bare survival. He points out that some humanly constructed accumulations of mammoth remains discovered within these sites have no apparent utilitarian function and notes that mammoth bones and teeth were placed in human burials. Such behavioral gestures, now incomprehensible, may have been influenced by engagement with and dependence on the animal, which consequently acquired special significance. Oliva also argues that people do not avoid hunting dangerous animals but take them on for the thrill of the competition in which they acquire social prestige and status.

Tyrannosaurus rex: Predator or Scavenger?

Tyrannosaurus rex was among the largest of the theropods, a group of mostly carnivorous (meat-eating) dinosaurs that roamed the Earth during the Cretaceous Period, 144 to 65 million years ago. For over a century, paleontologists have debated whether Tyrannosaurus was a predator that hunted living prey, as most scientists have assumed, or whether it was a scavenger that ate only the remains of animals that were already dead. Most recently, Jack Horner of the Museum of the Rockies has argued that the fossil record provides several clues that are inconsistent with a predatory lifestyle for Tyrannosaurus, suggesting instead a scavenging niche. These clues include (1) a relatively small eye that would have prevented long-distance spotting of potential meals; (2) an enhanced sense of smell, useful for detecting distant rotting carcasses (animal remains); (3) limb proportions suggestive of a slow, lumbering gait (manner of moving on foot), inconsistent with chasing down prey; (4) tiny forelimbs apparently useless for holding, let alone killing, prey; and (5) broad teeth unlike the narrow, bladelike teeth of most theropods.

Each of these points has been convincingly refuted by other paleontologists, including Jim Farlow (Indiana University-Purdue University) and Thomas Holtz (University of Maryland). These authors note that a predator does not need a large eye in order to see well. Indeed, although small for the enormous size of the animal, the eye of Tyrannosaurus was large in absolute terms, with increased light-gathering capacity and perhaps a high level of acuity. The fossil evidence is consistent with acute smelling ability, yet it seems reasonable that this sense could have served equally well to detect the living as well as the dead. With regard to foot speed, a number of studies support Horner's contention that Tyrannosaurus was closer to the lumbering end of the spectrum than the sprinting end. However, other studies demonstrate that the likely prey animals, including hadrosaurs and ceratopsids, were even slower. So, the giant predator appears to have been fully capable of catching its probable prey. It is hard to argue against the fact that Tyrannosaurus had ridiculously small arms for its size; the arms of a tall human are roughly the same length as those of the 12-meter-long dinosaur. Yet many carnivores—think of wolves—are highly effective predators without using their arms as weapons. Similarly, crocodiles and toothed whales have managed to remain at the top of their respective food chains for millions of years despite having broad teeth.

It is also important to consider the ecology of the predator-scavenger question and search for clues among the living. Today, carnivore species regularly compete for kills, often with the predator having its meal usurped by larger or more numerous challengers. Tyrannosaurus, many times larger than other predators in its ecosystem, would have been the undisputed bully in conflicts over carcasses. Yet there remains the question of quantity. Was there sufficient carrion (animal remains) available to sustain populations of 5,000-kilogram carnivores? It is difficult to answer this question in the face of important unknown variables such as the metabolic rate of Tyrannosaurus (which influenced daily caloric needs) and the sizes of the herbivore (plant-eater) populations. Nevertheless, a study by Graeme Ruxton and David Houston at the University of Glasgow attempted to estimate all the relevant parameters, modeling the Late Cretaceous ecosystem closely after the Serengeti. The authors' tentative conclusion, based on a number of untestable assumptions, is that there may have been sufficient carrion available to support populations of enormous carnivores like Tyrannosaurus.

Yet very few living carnivores are exclusive scavengers. Several species long regarded as scavengers, such as hyenas, obtain much of their diet from predation. The vast majority of carnivores turn out to be opportunistic, killing when necessary yet content to feed on a fresh carcass killed by some other animal. Among vertebrates, the only animals that subsist almost solely on carrion are twenty-three species of vultures, which are able to soar from carcass to carcass, expending a minimum of energy in the search for food. In contrast, Tyrannosaurus had to carry its substantial bulk overland, a nontrivial task even on level, open terrain.

Interestingly, in a follow-up paper, Ruxton and Houston concluded that animals able to use only scavenging to find food must be large, soaring fliers like vultures. Clearly, this strategy was not available to Tyrannosaurus.

Pastoralism as a Way of Life

Pastoralism is a form of food production that involves the keeping of domesticated herd animals and influences both the social and economic patterns of the societies in which it is practiced. Pastoralism is found in areas of the world that cannot support agriculture because of inadequate terrain, soils, or rainfall, however, these environments do provide sufficient vegetation to support livestock, provided the animals are able to graze over a large enough area. Thus, pastoralism is associated with geographic mobility, because herds must be moved periodically to exploit seasonal pastures.

Some anthropologists have differentiated between two types of movement patterns: transhumance and nomadism. With transhumance, some of the men in a pastoral society move their livestock seasonally to different pastures, while the women, children, and other men remain in permanent settlements. With nomadism, on the other hand, there are no permanent villages, and the whole social unit of men, women, and children moves the livestock to new pastures. But as the anthropologists Rada and Neville Dyson-Hudson have pointed out, the enormous variation even within societies renders such a distinction somewhat ineffective. For example, following seven Karamojong herds over a two-year period, the Dyson-Hudsons found that "each herd owner moved in a totally different orbit, with one remaining sedentary for a full year and one grazing his herd over 500 square miles."

Even though anthropologists tend to lump pastoralists into a single food-getting category, pastoralism is not a unified phenomenon. For example, there are wide variations in the ways animals are herded. The principal herd animals are cattle in eastern and southern Africa, camels in North Africa and the Arabian Peninsula, reindeer in the subarctic areas of eastern Europe and Siberia, yaks in the Himalayan region, and various forms of mixed herding (including goats, sheep, and cattle) in a number of places in Europe and Asia. In addition to variations in the type of animals, a number of other social and environmental factors can influence the cultural patterns of pastoral people, including the availability of water and pasturage, the presence of diseases, the location and timing of markets, government restrictions, and the demands of other food-getting strategies (such as cultivation) that the pastoralists may practice.

A general characteristic of nomadic pastoralists is that they take advantage of seasonal variations in pasturage so as to maximize the food supply of their herds. The Kazaks of Eurasia, for example, keep their livestock at lower elevations during the winter, move to the foothills in the spring, and migrate to the high mountain pastures during the summer. Such seasonal movement provides optimal pasturage and avoids climatic extremes that could negatively affect the livestock. This willingness to move their animals at different times of the year avoids overgrazing and enables them to raise considerably more livestock than they could if they chose not to migrate.

Anthropologists agree that pure pastoralists—that is, those who get all of their food from livestock—are either extremely rare or nonexistent. Because livestock alone cannot meet all of the nutritional needs of a population, most pastoralists need some grains to supplement their diets. Many pastoralists, therefore, either combine the keeping of livestock with some form of cultivation or maintain regular trade relations with neighboring agriculturalists. Moreover, the literature is filled with examples of nomadic pastoralists who produce crafts for sale or trade, occasionally work for the government, or drive trucks. Thus, many pastoralists have long engaged in nonpastoral activities, but they have always considered raising animals to be their normal, or certainly ideal, means of livelihood.

It is clear that in pastoral societies livestock play a vital economic role not only as a food source but in other ways as well. In addition to the obvious economic importance of meat and milk as food sources, cattle provide dung (used for fertilizer, house building, and fuel), bone (used for tools and artifacts), and skins (used for clothing). But in addition to these important economic uses, there are a number of important noneconomic or social functions of cattle. Livestock often influence the social relationships among people in pastoral societies. For example, an exchange of livestock between the families of the bride and the groom is required in many pastoral societies before a marriage can be legitimized.

Sensory Ecology

Some of the most important sensory information animals have to process comes from other animals. Interactions between predators and prey, parents and offspring, males and females, both shape and are shaped by the characteristics of sensory systems. Most animals are subject to two conflicting selection pressures: be inconspicuous to predators but be conspicuous to potential mates.

One of the best illustrations of how the trade-off between these pressures has influenced signals and behavior involves the color patterns and mating behavior of guppies. Guppies are small South American fish that live in clear tropical streams. Mature males sport-colored spots and patches that are used in courtship behavior. Male color pattern is heritable and varies in different populations. In experimental tests of the effectiveness of color patches, females are more likely to mate with males that have larger and brighter blue and orange, red, or yellow patches. Thus, female choice creates sexual selection pressure for conspicuous coloration. In contrast, predators create selection pressure for cryptic coloration (coloration that helps hide organisms): duller, smaller color patches and patterns that match the background.

The effects of predation have been established in several ways. In the field, guppies are found in streams that have different numbers and kinds of visually-hunting predators, mostly other fish. Males from populations with more predators are more cryptically colored. Prawns are thought to see poorly in the red end of the spectrum. As might therefore be expected, guppies in areas with heavy predation by prawns have more orange than guppies subject to predators with better red-orange vision. Predictions about the effects of predation have been tested directly by establishing guppies from a single genetic background and distribution of color patterns in laboratory "streams," and exposing them to different numbers and kinds of predators. In guppies' natural habitat of forest streams, the intensity and color of light varies with the time of day. Visually hunting predators are most active in the middle of the day, but in both the laboratory and the field, guppies engage in more sexual display early and late in the day, that is, in relatively dim light. Taken all together, the transmission characteristics of tropical streams and the visual capabilities of guppies and their most common predators indicate that at the times of day when they are most likely to be courting, guppies' colors are relatively more conspicuous to other guppies than to guppy predators.

The foregoing example illustrates how visual conspicuousness and crypticity vary by observer. Detailed sensory physiology may be needed to figure out whether color patterns that appear conspicuous or cryptic to us, appear that way to the animals that normally view them. A particularly good example involves camouflage of crab spiders. Crab spiders make their living sitting on flowers waiting to grab bees or other pollinators that happen by. By resting in such an exposed position, the spiders make themselves conspicuous to insect-eating birds. Clearly, they should be colored so as to be inconspicuous to both birds and bees, but this is not easy because bees and birds have different color sensitivities. Ecologist Marc Thery and colleagues collected crab spiders (Thomisus onustus) from the yellow centers of white marguerite daisies and measured the relative intensities of colors across the spectrum, including the ultraviolet, reflected by daisy petals and daisy centers and by crab spiders. The daylight reflectance spectra were then related to the color sensitivities of birds (blue tits, typical predators in the

French meadows where the spiders were collected) and honeybees. These computations showed that the spiders color did not contrast sufficiently with the flower centers for them to be detected by either predator or prey. Their contrast with the petals was well above both birds' and bees' thresholds, which presumably explains why spiders rest in the center. To make matters even more interesting individuals of this species of crab spider also match their color to pink flowers, and they are similarly of low contrast to both birds and bees on this background as well.

Lenape Horticulture

Older scholarship on Native Americans assumed that, at the time of their first contact with Europeans, Native Americans in the northeastern United States were farmers whose main food source was maize (corn). This view was based on evidence of extensive fields inland. It was also consistent with reports from early European explorers, including Henry Hudson, who sailed for the Dutch, about trading with Native Americans for "Turkish wheat," the explorers' term for maize. However, in the 1970s a persuasive archaeologist, Lynn Ceci, challenged this orthodoxy. She argued that for the people living in coastal New York, such as the Lenape, occupying Manhattan Island when Dutch explorers arrived in the early seventeenth century, the environment was so abundant with resources that horticulture was an adjunct to the diet, not the primary source of calories. She argued along several lines of evidence. First, she noted that "flotation tests"-in which prehistoric fire pits are flooded with water-caused relatively few seeds and corn kernels to float to the surface in coastal sites compared to inland sites. Second, she argued that the soils in coastal regions, generally sandy and rocky, are not particularly productive and would require a lot of work to cultivate (an observation also noted by the Dutch). Third, she argued from ecological evidence that it really was not necessary for coastal people to adopt an agricultural way of life, given the year-round abundance of shellfish, fish, mammals, nuts, and berries. Other archaeologists corroborated these views by testing Lenape bones from local archaeological sites and showing that the Lenape primarily ate local plants, not maize, and a lot of seafood, and were generally healthy. (Adopting agriculture often leads to a reduction in the health of a population.)

Research has added a couple of twists to Ceci's hypothesis. A Columbia University student used a modern crop model to simulate maize horticulture on Manhattan, using representative soil types, climate, and maize varieties. He found that when maize was grown by itself, productivity was uneven, and there was a modest but significant probability of total crop failure each year. Soil nitrogen levels appeared to be the limiting factor. But the Lenape did not plant maize in monoculture (as a single crop), rather, they practiced team planting (also known as multicropping); that is, they grew maize in combination with beans and squash-the traditional "three sisters" garden.

Beginning in late April, Lenape men cleared a plot of land by cutting down the large trees and burning the vegetation. Burning fertilized the soil with ash, which promoted plant growth by changing the soil chemistry from acidic to alkaline, releasing more nutrients. In early May the women created mounds and planted maize kernels saved from the year before. A few weeks later, they planted beans along the sides of the mounds and squash between them. Over the course of the summer, the beans climbed up the maize stalks, using the six-foot stems as a ladder to reach the sun; in the meantime, the beans, which have nitrogen-fixing nodules on their roots, transformed nitrogen from the atmosphere into a natural fertilizer, which fed the maize and the beans. The squash, growing its large green leaves between the mounds, kept the weeds down and held in soil moisture between rain showers, (the Lenape did not practice irrigation). Crops were harvested from late summer until first frost, usually in October.

Modern crop models are generally not written to include team planting, because they assume monocultures-but they will accept inputs of fertilizer. Researchers simulated nitrogen inputs from bean plants at three different levels and showed that even modest nitrogen support increased maize productivity, and, more importantly, enhanced the consistency of the crop. In model simulations by researchers, maize rarely failed when grown with beans. Productivity, though, was still generally low, at about 700–2,300 kilograms per hectare.

How many people that level of productivity would feed depends, of course, on how large the fields were. As mentioned, gardening was a woman's responsibility, but women had many other tasks that would limit their time in the garden. The Lenape were not immune to the human-wildlife conflicts that bedevil suburban gardeners today-deer, squirrels, raccoons, and assorted other wildlife were pests in the fields. Keeping the animals out required diligence throughout the growing season.

When Honeybees Become Foragers

Many species of insect, including the familiar honeybee, exhibit considerable flexibility in their behavioral development. Honeybee colonies usually contain tens of thousands of sterile female worker bees, laboring on behalf of a single queen, who lays most or all of the eggs in her hive. At first, the young workers feed the larvae that hatch from the queen's eggs (most of which will become new workers), but after about three weeks of nurse duty, they graduate to the job of collecting pollen and nectar outside the hive.

Research indicates that the age-related transition between nursing and foraging (collecting food) is regulated by hormonal changes. Young nurse workers have very low concentrations of juvenile hormone in their blood, whereas older foragers have much higher concentrations of this hormone. If young bees are treated with juvenile hormone, they become early foragers. In contrast, the onset of foraging behavior is delayed by removal of the corpora allata, the glands that produce juvenile hormone; furthermore, bees without corpora allata that receive hormone treatment regain normal timing of the switch to foraging. Bees that collect pollen and nectar also have bigger mushroom bodies, an anatomical feature of the honeybee brain. Neither juvenile hormone nor foraging experience appears to be necessary for this change in brain structure to occur, although both factors influence the rate at which the mushroom bodies grow. The expansion of the mushroom bodies takes place in anticipation of the needs of foragers, as they must be able to recognize spatial landmarks so they can travel back and forth between the hive and distant patches of flowers.

As it turns out, the changes in juvenile hormone concentrations that take place within the bodies of honeybee workers are not absolutely fixed with worker age. This conclusion is based on experiments with colonies that have been manipulated so that all the workers are the same, relatively young, age. Under these conditions, a division of labor still manifests itself, with some individuals remaining nurses much longer than usual, while others start foraging as much as two weeks sooner than average. As a result, the larvae are cared for continuously, while the colony also receives food supplies.

What enables bees to make these developmental adjustments? One hypothesis is that a deficit in social encounters with older foragers may stimulate the developmental transition from nurse to forager behavior. This possibility has been tested by studies in which groups of older foragers were added to experimental colonies made up of only young workers. The higher the proportion of older bees, the lower the proportion of young nurse bees that undergo an early transformation into foragers. The behavioral interactions between the young residents and the older transplants must inhibit the development of foraging behavior. Transplants of young workers have no such effect on young workers already at the site. Thus, the social environment of young honeybee workers influences their behavioral development by regulating the release of a key hormone. When released, the hormone becomes part of a bee's cellular environment and modifies the properties of the brain's nerve cells. After these hormonally induced changes in brain structure have occurred, workers change their behavior to match the special needs of their hive.

The interaction between genetic information and the environmental causes of task-switching in the honeybee has been highlighted by a study of how genetic variation affects the rate of behavioral development in this insect. If one keeps genetically different lineages of honeybees in identical conditions, some genotypes (genetic makeups) respond to exactly the same hive environment differently, by making the transition from nurse to forager more quickly than others. If the physiological cause of task-switching is the increase in juvenile-hormone concentrations caused by changes in the social environment, then the genes of fast-developing lineages can be predicted to influence things like the production schedule for juvenile hormone or the sensitivity of individuals to the age composition of their colony. As it turns out, some genotypes do tend to make more juvenile hormone sooner than usual, while other genotypes are especially sensitive to increases in the numbers of older workers in their hives. Thus, several different physiological routes can lead to fast (or slow) behavioral development in the honeybee, each dependent on a particular mix of genetic and environmental factors.

Dutch Art and the Middle Class

Throughout history, the wealthy have been the most avid art collectors. Indeed, the money necessary to commission leading artists to create major artworks can be considerable. During the seventeenth century in the Dutch Republic, however, the prosperity that a large proportion of the population enjoyed significantly expanded the group of art patrons, people whose wealth allowed them to support artists by purchasing art. As a result, one distinguishing feature of Dutch art production during this period was how it catered to the tastes of a middle-class audience, broadly defined. An aristocracy and an upper class of ship owners, rich businesspeople, high-ranking officers, and directors of large companies still existed, and these groups continued to be major patrons of the arts. But as the Dutch economy expanded, new patrons-traders, craftspeople, bureaucrats, and soldiers - also commissioned and collected art.

Young Woman with a Water Pitcher
by Johannes Vermeer

Although the financial success that the middle class increasingly enjoyed resulted in sharply higher investment in home furnishings and art, religious beliefs that disdained ostentation led these new Dutch collectors to favor small, low-key works-portraits of ordinary men and women, still lifes and scenes of everyday life, and landscapes. This focus contrasted with the seventeenth-century Italian Baroque fondness for gigantic ceiling frescoes and oil paintings with religious subjects. Stylistically, the art of northern Europe of this period, although also called "Baroque" by art historians, differs markedly from Italian Baroque art.

It is risky to generalize about the spending and collecting habits of the Dutch middle class, but some interesting facts are revealed by records of wills and estates, contracts, and archived inventories. These records suggest that an individual earning between 1,500 and 3,000 guilders a year could live quite comfortably. A house could be purchased for 1,000 guilders. Another 1,000 guilders could buy all the necessary furnishings for a middle-class home, including a significant amount of art. particularly paintings. Although there was, of course, considerable variation in prices, many artworks were very affordable. Prints, for example, were extremely cheap because of the high number of copies artists produced of each picture. Paintings of interior and everyday scenes were relatively inexpensive in the seventeenth-century Dutch Republic, perhaps costing one or two guilders each. Small landscapes could be purchased for between three and four guilders. Commissioned portraits were the most costly. The size of the work and quality of the frame, as well as the reputation of the artist, were other factors in determining the price of a painting, regardless of the subject.

With the exception of portraits. Dutch artists produced most of their paintings for an anonymous market, hoping to appeal to a wide audience. To ensure success, artists in the Dutch Republic adapted to the changed conditions of art production and sales. They marketed their paintings in many ways, selling their works directly to buyers who visited their studios. They also sold their works through art dealers, exhibitions, fairs, auctions, and even lotteries. Because of the uncertainty of these sales mechanisms (as opposed to the certainty of a binding contract for a commission from a church, king, or duke). artists became more responsive to market demands. Specialization became common among Dutch artists. For example, painters might limit their practice to portraits, still lifes, or landscapes-the most popular genres (types of art) among middle- class patrons.

Artists did not always sell their paintings. Frequently, they used their work to pay off loans or debts. Debts incurred in taverns (places where alcoholic drinks and food were served and rooms were rented to travelers), in particular, could be settled with paintings, which may explain why many art dealers, including Jan Vermeer (who himself became an acclaimed painter) and his father before him, were also innkeepers. This connection between art dealing and other businesses eventually solidified, and innkeepers, for example, often would have art exhibitions in their taverns hoping to make a sale. The institutions of today's open art market-dealers, galleries, auctions, and estate sales-owe their establishment to the emergence in the seventeenth century of a prosperous middle class in the Dutch Republic.

Reconstructing the Spread of the Grain Weevil

Perhaps the most common, most widespread, and arguably most destructive pest of stored grain in history is the grain weevil Sitophilus granarius. A member of the beetle family, it is small (3-5 mm), dark, heavily built, and cylindrical in shape. It has a long snout on the front of its head, at the very end of which are its jaws. It chews a deep drill-hole down into the center of a mature grain seed, then it turns around and lays a single egg at the bottom of the hole. Each grain of wheat or barley is large enough to feed one larva through to adulthood. The empty grain shell with a large, chewed hole through which the adult weevil emerges is a distinctive feature of this insect's damage. In archaeological digs the characteristic hollowed grain shells, as well as preserved weevil body parts, are clear evidence of the beetle's widespread occurrence throughout Asia, Africa, and Europe nearly 8,000 years ago. Because of the beetle's continuing importance as a major grain pest, there has been much interest and study in where it came from and how it came to be so closely associated with humans.

Various studies have tried to map the species' spread across the globe, shadowing the spread of grain agriculture. Records are sparse, and although intriguing, they remain inconclusive. The earliest known infestations were in what is now Turkey, in 5750–5500 B.C.E., and the eastern Mediterranean (5550 B.C.E.), and this fits neatly with the idea of a Middle Eastern origin for both grain and pest, but they were also in faraway Germany in 5140 B.C.E., long before they seemingly arrived in neighboring Egypt (2950 B.C.E.) or Greece (1650 B.C.E.). Our ideas of early agriculture are still hazy, but this disparate spread of grain weevil infestations actually gives us some interesting clues.

One suggestion is that the grain weevil was already a widespread wild species some 6,000–10,000 years ago and that it quickly moved in to take advantage of human stores of grain wherever agriculture was established. This is patently not the case, because the species cannot survive in temperate regions outside heated or at least sheltered buildings, and development from egg, through larva, to adult can only take place at the elevated temperatures found indoors.

An alternative scenario is that it became established as a pest somewhere in the warmer climate of the eastern Mediterranean or Southwest Asia, and had already, by the sixth century B.C.E., been spread far and wide by trade. Trying to track any prehistoric grain trade from archaeological evidence is now impossible, but European trade in other items like obsidian was well established at this time. This is beginning to look like the only possibility. Later evidence emphasizes the trade-route idea. The grain weevil seems to have arrived in Britain with the Roman invasion, around 47 C.E. in Londinium (present-day London), and shortly afterwards in various military outposts. The military machine offered good conditions for sustaining and transporting the beetle about the country. Notably, however, with the later withdrawal of the Romans, the beetle declined and was apparently absent from 400 C.E. until the arrival of another mainland European military force: the Norman invasion of 1066.

The likelihood is that it was climatic factors that kept the species under control, or at least subdued enough without the Roman trade network to constantly reinforce it (beetle fragments have been found in drowned grain cargoes from Imperial Roman shipwrecks dated to the second and third centuries C.E.). The grain weevil requires a temperature range of 15–35°C (optimum 26–30°C) to complete its life cycle, but Britain was just at the difficult edge of this natural limitation. Large grain stores, as existed under the Romans, might have helped insulate pockets to this comfortable temperature, but if the popular history books are to be believed, the presence of little to eat during the early Middle Ages (500–1000 C.E.) saw them disappear. The species was really a Mediterranean beetle, as was the wheat it invaded.

Even so, the original source of the grain weevil is still steeped in mystery. Unlike the biscuit beetles living in the natural habitat of spilled grass seed in birds' nests, grain weevils were never found outside storing areas, even where the original natural precursors of wheat and barley grow wild today.

The Deforestation of England

Today's typical English countryside of rolling green fields and few trees did not evolve naturally. England was originally wooded. Forests of oak and other hardwoods had filled the southern lands, while conifer stands populated the higher latitudes. Sheepherders over the centuries converted many of these to pastureland. Nevertheless, the domestic wood supply remained great enough to handle timber and firewood demands. Then, beginning in the 1540s, came new manufacturing industries that destroyed the forests for their fuel. This new wave of deforestation started with the iron industry—an early royal effort to boost manufacturing in accord with trade-based economic theory—since the production of iron required immense amounts of heat and, initially, used charcoal (which is derived from wood) as fuel. In 1543, Parliament first addressed the impending timber shortage with the Act for the Preservation of Woods, which restricted farmers from using woodlands more than two furlongs (about 400 meters) from their homes. British forests were rapidly disappearing.

The situation worsened during the long reign of Queen Elizabeth I (1558-1603). She promoted numerous other wood-fuel-driven manufacturing industries, including copper smelting (the process of obtaining metal from metallic mineral deposits), salt making, and glass production. (The coal industry, which was beginning, could not meet the rapidly increasing demand for fuel.) One writer from this period commented, "Never so much [oak] hath been spent in a hundred years before as in ten years of our time." The price of firewood doubled between 1540 and 1570. This pushed some citizens out of the firewood market, and it became commonplace for the poor to shiver through the winters. The timber shortage had turned a product once freely available for the cutting into a valuable commodity.

But fuel needs did not fully account for England's timber demand. Wood was also necessary in the construction of ships. And Queen Elizabeth, in addition to promoting domestic manufacturing, had championed shipbuilding, part of the Crown's long-term strategy to contest Spanish sea power and strengthen English commercial trade. Few industries in history have depended on wood quite like shipbuilding (at least before the conversion to iron and steel ship parts in the mid-nineteenth century). A large naval warship, known as a "ship of the line" and constructed almost entirely from wood, weighed over one hundred tons in the early seventeenth century. The bodies of such vessels required about two thousand mature oaks, which meant at least fifty acres of forest (200,000 square meters) had to be stripped. While oak supplied the timber for much of the ship, it was too inflexible and heavy for ship masts, the poles that supported the canvas sails. Instead, these required lighter and more shock-resistant softwoods, such as pines and firs. The largest masts were about a meter wide at their base and over 30 meters tall—roughly one meter in height per inch in width. To maintain these wooden cathedrals of the sea, shipwrights (skilled workers who build and repair ships) relied on a range of forest products, known as naval stores, extracted from pines as well as firs. Most notable were the tar, pitch, and turpentine used to seal the wood and protect it from seawater. These materials were essential for maintaining the integrity and longevity of the ships, ensuring that they were watertight and resistant to the rotting effects of the ocean. The procurement of these naval stores was itself a significant undertaking, requiring extensive knowledge of forestry and sustainable harvesting techniques to ensure a steady supply without depleting the forests.

The twin demands of shipbuilding and a manufacturing industry hungry for wood fuel had turned England into a wood importer. In particular, the country had to trade for masts and naval stores, since it had no suitably commercial conifer forests. The preferred mast trees, called Riga firs or Scotch pines, came from an Eastern European region around the city of Riga (in present-day Latvia), but several northern countries had giant spruce forests that were also exploited for naval stores. The trade centered on ports in Scandinavia and the Baltic Sea—the latter, which included Riga, accessible only through narrow straits (waterways) between

Denmark and Sweden. Rulers who controlled the various ports and access to the straits knew that England's sea power depended on forest products and, consequently, kept duties, taxes, and shipping fees high. The Danish, for example, collected tolls for each crossing. If England ever lost access to these ports, it would cripple the entire English shipping industry, and with it the Royal Navy.

The Azolla Event

During the early Eocene period, about 56 million years ago, Earth's atmosphere had high levels of carbon dioxide (CO2) which warmed the planet. But about 49 million years ago the climate began to cool, mostly as the result of activity in the Arctic. In the early Eocene the Arctic Ocean, unlike today, was almost completely surrounded by land, isolating it from the other oceans. One consequence of this isolation was that the Arctic Ocean was able to retain much of the freshwater that fell as rain and that entered along with the rivers that flowed into it from the nearby land. This freshwater accumulated on the ocean surface as a light, slightly salty layer floating on denser, saltier water a few meters below. Because rainfall was even more intense than it is today, and because there was less disruption by waves from other oceans, this layer of surface water occasionally became completely fresh. Those were the conditions that allowed rapid growth across the Arctic of Azolla, a freshwater fern.

Azolla

© Wanıda boon/Shutterstock

Azolla is a fascinating plant. Unlike the land-loving ferns most of us are familiar with, Azolla grows as free-floating mats on the surfaces of lakes and rivers. It can do this because of its mutually beneficial relationship with bacteria that are able to break apart the strong molecular bonds of atmospheric nitrogen to make the nitrate that plants need for growth. In other words, it obtains its food from the air. Azolla carries these useful companions around in specially formed growths, that provide it with the nitrate fertilizer that a free-floating plant cannot get by the usual approach of sticking roots into soil. In return for this invaluable service, the plant provides its bacterial guests with food in the form of glucose and this partnership allows Azolla to thrive in nutrient-poor waters where other aquatic plants struggle.

In the case of the Eocene Arctic Ocean with its very unusual freshwater surface layer, this ability occasionally allowed ocean-sized Azolla blooms to form in the polar summer. The entire Arctic Ocean became plant-green, rather than the ice-white one might have expected. But in the long, dark, polar winter, these weeds died or possibly just became dormant (inactive), and much of the vigorous summer growth sank into the deep ocean. This could well have been the cause of the mid-Eocene cooling, because large-scale burial of plant matter in ocean sediments removes CO2 from the atmosphere. Levels of the greenhouse gas CO2 would have fallen as Azolla remains accumulated at the seafloor, and the resulting temperature drop may have been large enough to begin the descent toward the icy world of the last 2.5 million years: a world of regular ice ages interspersed by relatively short but warmer interglacial phases such as the one we have been living through since about 9000 B.C.E. Many other factors played important roles in these changes to global climate, but Azolla blooms may have been the final trigger that set off a complex chain of events flipping the world from the warm climate of the early Eocene into the colder climate of more recent times.

This Azolla event was, geologically speaking, a brief episode lasting little more than a million years. Even then, Azolla blooms occurred only rarely and seem to have been associated with occasional drops in global sea level that further cut off the Arctic Ocean from the rest of the world. The world's oil industry is currently very interested in exploiting the oil that can be found in the Arctic Ocean, and an Azolla layer, which is one of the principal sources for this oil, has been found everywhere that drills have penetrated marine sediments of the right age. This discovery of the remains of a freshwater plant across an entire saltwater ocean was more than a little surprising, but the evidence for ocean-scale blooms of this particular aquatic weed is strong. The sediments show too much decomposed Azolla spread across too wide an area for it to simply be the result of river runoff from the land. The sediments also show no other terrestrial land remains, such as wood fragments, that would be expected in land-derived deposits.

The American Hospital

Few institutions have undergone as radical a transformation as the hospital has in its modern history. It is perhaps distinctive among social institutions in having first been built for and primarily used by poor people and only later used in significant numbers by wealthier people. In the United States the change began in the second half of the nineteenth century with the scientific redefinition of the hospital and its incorporation into medicine. We now think of hospitals as the most visible embodiment of medical care in its technically most sophisticated form, but in the past hospitals and medical practice had had relatively little to do with each other. From their earliest origins in pre-industrial societies, hospitals had been primarily religious and charitable institutions for tending to sick people, rather than medical institutions for curing them. In Europe, from the eighteenth century, hospitals played an important part in medical education and research, whereas systematic clinical instruction and investigation were neglected in the United States until the founding of Johns Hopkins University in 1876. Before the Civil War (1861-1865), an American doctor might have contentedly spent an entire career in practice without setting foot in a hospital ward. The hospital did not intrude on the worries of the typical practitioner, nor did the practitioner intrude on the routine of the hospital.

But in a matter of decades, roughly between 1870 and 1910, hospitals moved from the periphery to the center of medical education and medical practice. From refuges mainly for the homeless poor and insane, they evolved into doctors' workshops for all types and classes of patients. From charities, dependent on voluntary gifts, they developed into market institutions, financed increasingly out of payments from patients. What drove this transformation was not simply the advance of science, important though that was, but the demands and example of an industrializing capitalist society, which brought larger numbers of people into urban centers, detached them from traditions of self-sufficiency, and projected ideals of specialization and technical competence. The same forces that promoted the rise of hospitals also brought about changes in their internal organization. Authority over the conduct of the institution passed from the trustees (a group of people with high-level control) to the physicians and administrators. Nursing became a trained profession, and the division of medical labor was refined and intensified as conceptions of efficient and rational organization prevailing elsewhere in the economy were applied to care of the sick. The sick began to enter hospitals, not for the duration of an illness, but only during its acute phase, to have some work performed on them. The hospital took on a more activist posture: it was no longer a place of sorrow and charity but a workplace for the production of health.

The effects of this change rippled outward, altering the relationship of doctors to hospitals and to one another and shaping the development of the hospital system as a whole. Once the hospital became an integral part of medical practice, control over access to its facilities became a strategic basis of power within the medical community. The tight grip that a narrow elite long held over hospitals no longer seemed tolerable to other physicians, who responded by forming their own institutions or pressing for access to established ones. Under financial pressures and the threat of increased competition, the older hospitals gradually opened their doors to larger numbers of practitioners, creating a wider network of associations stratifying and linking together the profession in new and unexpected ways.

The access that private practitioners gained to hospitals without becoming their employees became one of the distinctive features of medical care in the United States, with consequences not fully appreciated even today. In Europe and most other areas of the world, when patients enter a hospital today, their doctors typically relinquish responsibility to the hospital staff, who form a separate and distinct group within the profession. But in the United States, private doctors follow their patients into the hospital, where they continue to attend them. This arrangement complicates hospital administration, since many of the people making vital decisions are not the institution's employees. Yet it also may encourage more private relationships between doctors and patients than exist where patients are attended solely by salaried hospital physicians.

Periglacial Mounds

Pingo

©Dreamstime/Timothy Epp

Periglacial regions are cold-climate areas that are located near the margins of glaciers—permanent ice sheets covering parts of the Arctic and other cold places. In some periglacial regions, huge mounds (hills) consisting of an outer layer of soil covering a core of solid ice may rise out of an otherwise flat landscape. These structures, called pingos, often develop over dried-out lake basins or along the sides of streams and rivers. Pingos become elevated by the growth of an ice core in the center of the mound, formed as a result of water pressure from below. Usually, they develop in sites where springs of water rise from the ground under pressure and feed the ice core, which freezes as it nears the surface of the ground. The upper part of the mound has a very different microclimate from the surrounding lowlands, and the southern slopes of the pingo mound may bear vegetation characteristic of a warmer climate, possibly even a tree cover. If the general climate becomes warmer, however, the ice core in the center of the pingo melts and the whole mass collapses to form a deep pool, surrounded by a circular rim or "rampart" formed from the soil and other organic material raised up by the mound. Circular ponds with ramparts can be found far to the south of the present-day Arctic regions, indicating the periglacial conditions that once prevailed there.

A similar type of structure is sometimes formed in those regions of the Arctic that are rich in peat (decomposed plant material), and this is termed a palsa. Palsas, like pingos, are large mounds that are formed by the development of ice beneath the surface of the ground. Generally smaller than pingos, palsas are only 2 to 3 meters in height and about 45 meters across. They often lie within wetland (water-saturated) areas, where they are interspersed with open pools and other palsas in various states of development. Palsas begin their formation as small irregularities on the surface of an area of low-lying land that is saturated in water. Even a very small elevation of a few centimeters can affect snow accumulation. The snow is blown from the slight mound and accumulates in the hollows between, so the more elevated spots have less snow cover in the winter. They therefore become cold, because snow acts as an insulating blanket, retaining some of the ground's heat that it gained in the summer sun. By contrast, the cold penetrates any small raised areas of ground, causing an ice core to develop within them. As the ice forms, it expands, pushing the mound even higher so that even less snow is retained in the high winds. Palsa growth is thus a self-propagating process: the higher the palsa grows, the colder it becomes, and the colder it becomes, the more the ice core expands and pushes the palsa upward. The ice core survives through the summer, so its growth continues for many years, even centuries.

As the palsa grows in height, drainage (the ability of water to move down) improves and the vegetation on its surface becomes drier. The surface of the palsa may then develop a vegetation cover of lichens (small plantlike organisms). Many lichens are white or pale gray, and these reflect the sunlight, keeping the mound cool in summer. In the course of time, dwarf shrubs replace these lichens, but these have a darker color and absorb more heat from the summer sun. This, coupled with the increasing height of the dome, eventually leads to gradual removal of the surface soil, revealing the peat beneath. The dark-colored peat absorbs even more heat, and its exposure leads to the meltdown of the ice core. The core collapses quite quickly, leading to the formation of an open pool of water that is available for recolonization by wetland plants as the cycle begins again. Finnish scientists, working in the 1980s, demonstrated experimentally how this process operates. They spent a whole winter visiting an area of the Arctic wetland where, using a broom, they swept an area clear of snow and kept it clear through the winter. The result was the development of a permanent ice core beneath the ground, and within a few years, the swept locality was developing into a palsa.

Opera in Seventeenth-Century Venice

Opera, a form of drama set to music and featuring vocal performance supported by an orchestra, was born in Italy at the very end of the sixteenth century. In the early seventeenth century, operas were performed in courts (residences of ruling members of the nobility), in front of small elite groups of spectators and created by composers and poets who were effectively court servants. These early operas were used to celebrate events—weddings, births, military victories—and generally to demonstrate the wealth of the court. Soon, however, opera was to reach a larger audience. This shift from private to public opera is clearly illustrated in the career of composer Claudio Monteverdi. While employed in the Mantua court, Monteverdi wrote his first opera, *Orfeo*, in 1607. Created with the rich instrumental resources available at the Mantua court, *Orfeo* was unprecedented in musical and dramatic expressiveness, guaranteed to impress political rivals of the court. In 1613, Monteverdi left the Mantua court and moved to Venice, where he took up a position at the Church of St. Mark's Basilica. Although much of his time was occupied with writing church music, as time went on, he acquired the freedom to branch out into writing secular music, including opera.

But opera in Venice was different. Venice was a republic, and without a single ruling family there was no place for court opera along the lines Monteverdi had experienced in Mantua. However, the city was an important commercial center that welcomed many foreign merchants, businessmen, dignitaries, and aristocrats, and the demand for entertainment was high, particularly during the famous Carnival of Venice, which took place each February. After a touring company brought an opera to Venice for the first time during the carnival season of 1637, some enterprising Venetian entertainment organizers came up with the idea of setting up opera houses and charging the public for seats on a subscription system (an agreement to purchase tickets in advance). However, it would be an exaggeration to depict seventeenth-century Venetian opera as opera "for the masses." The opera houses were owned by noble families, opera-goers were still from privileged backgrounds, and ticket prices were high.

Operas ceased to be one-off events and became regular occurrences, often playing for an entire season, and with opera now a commercial enterprise, public appeal became crucial. Although the earliest Venetian operas were based on Classical mythology, just as the court operas had been, composers increasingly turned towards historical subjects with more recognizably human characters. The priority for a composer was no longer to appeal to the vanity of a ruler; indeed, the new type of opera sometimes even depicted rulers as corrupt and greedy. They also had to be more economical than the early court operas—commercially run theaters could not afford the sort of elaborate sets that were used in the courts. Large-scale permanent orchestras were also too expensive for most theaters, meaning that composers had to scale back the orchestration that they called for. However, a benefit of using a smaller orchestra was that the voices were shown off to greater effect, and as the seventeenth century progressed, composers began to include more and more arias (self-contained songs typically sung by one voice) in their opera scores. Monteverdi embraced these new trends in his opera *The Coronation of Poppea*, which told the story of the love affair between the Roman emperor Nero and his mistress Poppea.

A "star system" for singers soon emerged in Venice that was similar to modern notions of celebrity—one of the most successful and celebrated singers of the era was the singer, Anna Renzi. The success of an opera depended increasingly upon the virtuoso qualities of the lead performers: parts were written to show off the strengths of an individual's voice, and the most successful singers acquired the power to dictate not only their role but how long their scenes should be and even which other singers should appear in the production. The most successful singers were typically paid more than the composers whose works they performed, often several times more, and their names appeared more prominently on publicity materials than those of composers or authors of opera texts.

Why Did Agriculture Arise

Sometime around 10,000 years ago, some hunter-gatherers took up agriculture. Though the advantages may seem obvious, there are also disadvantages: Farmers work harder and longer hours than hunter-gatherers, and domesticate plants and animals are more prone to catastrophic disease. So why did this change come about? By and large, some archaeologists suggest two moving principles: climate change and rising populations. In very general terms, it is argued that, as droughts intensified, food became harder to obtain and, at the same time, there were more people to feed. The first farmers were therefore acting rationally when they invented farming as a response to the challenge that the environment was presenting, or they began to mimic nature and encouraged seeds to grow near their settlements.

This changeover can be seen at, for instance, Abu Hureyra, an archaeological site overlooking the Euphrates River. Archaeologist Andrew Moore and botanist Gordon Hillman found that the layered mound left by many generations of humans living at Abu Hureyra had been occupied and abandoned, and then reoccupied. At about 9000 B.C.E., open forests were close to the settlement; people could gather nuts and fruits and stroll back to their mud houses. They also exploited wild einkorn wheat and rye. Sedentism (the lifestyle of living in settlements year-round) thus came about before agriculture.

After a few centuries, a drier climatic regimen caused the forests to retreat. The people of Abu Hureyra therefore concentrated on wild grasses that were still within walking distance of their homes. But even these more hardy plants disappeared as the drought intensified. By 8000 B.C.E., Abu Hureyra was no longer a viable settlement site, and the people abandoned it. Three centuries later, people returned and began to build new mud dwellings on the foundations of the old, decayed ones. Now they were depending not on wild plants but on domesticated emmer wheat, rye, and barley. They were agriculturalists.

An advantage of seeds is that they can be stored in large quantities. People can therefore remain in one place for longer; they do not exhaust the natural foods within walking distance and then have to move on to other locations that can be exploited, as hunter-gatherers do. Sedentary life in villages and towns becomes possible. And so, a chain of consequences, not all favorable, was triggered. Fields degraded natural soils, and erosion sometimes set in with disastrous results. If agriculture was indeed a form of human adaptation, it was sometimes questionably so. As fields grew in size, they demanded more water and, as a result, irrigation was developed in some parts of the world. There were also health hazards. Disease can rampage through closely packed towns with inadequate sanitation. Then, too, a starch diet can lead to tooth decay and other health problems. Mobile foragers have to keep their populations restricted because children are not easily transported from camp to camp, but sedentary populations do not suffer from this restriction. All in all, agriculture and its consequences were not always advantageous.

Some archaeologists offer a similar explanation for the domestication of animals. First of all, they look for environmental conditions that may have led to domestication. They cite such factors as growing aridity that could gradually change the relationship between people and animals that existed when hunting was the norm. Many species of animals live in herds. Therefore, all the people had to do was to control the herds to prevent them from wandering off to seek water and grazing; they thus interfered with the natural patterns of breeding and genetic change. Keep one herd separate from all others, and domesticated species are the result. But why would people wish to do this?

One answer that springs to the minds of many Westerners is that domestication of animals is labor-saving. But that is not necessarily true. Domesticated herds have to be tended, protected from predators, and taken to pasturage. Then, disease can sweep through herds more disastrously than it does through scattered populations of wild animals. A different kind of explanation is that hunters became too effective and, as a result, diminished the wild species they pursued. The logical answer was to corral them and to allow them to breed, culling them for food only when necessary.

The Origin of Mars's Moons

Mars has two tiny moons, Phobos and Deimos. The orbits of both these satellites are nearly circular and close to the planet: Phobos orbits at a distance of about 9,378 kilometers (5,627 miles) from the center of Mars, and Deimos at about 23,459 kilometers (14,075 miles). Their orbital periods are short: 7 hours, 37 minutes for Phobos, and 30 hours, 18 minutes for Deimos. Both satellites orbit in the same direction as the planet rotates, and they have low-inclination orbits (meaning their orbits are tilted at a small angle from Mars's equator). The origin of Phobos and Deimos is unknown, but there are two currently popular theories. One theory holds that these bodies are captured asteroids (rocky objects smaller than planets, that orbit the Sun). The other theory suggests that they are asteroid-sized pieces of debris left over from the initial formation of Mars from the cloud of rock and dust from which all the planets in the solar system formed.

Astronomers generally agree that small bodies (asteroids, comets, or other debris) that wander the solar system, if captured by a planet, would typically be trapped in a highly elliptical (oval-shaped), inclined orbit. Consequently, when astronomers see a small body that has a circular and low-inclination orbit around a planet, they generally regard this as evidence that the small satellite was not a captured wanderer but a piece of debris left over from the formation of the planet.

Some scientists argue that wandering solar-system bodies can be captured in an initially highly elliptical orbit and then later have their orbits circularized. This requires the delicate balance of many conditions to work, making many scientists uneasy about its validity. The circularization of an initially elliptical orbit can be done through a mechanism akin to aerobraking, used by modern-day space-flight controllers to slowly circularize highly elliptical orbits of some spacecraft. With this mechanism, under the right conditions, if an asteroid was initially captured into just the right highly elliptical orbit around Mars, the lowest part of its path could have carried it repeatedly through the upper part of the massive early Martian atmosphere, slowing it by atmospheric drag and slowly circularizing the orbit.

Circularization of an orbit can also be done through a mechanism involving tidal forces (gravitational forces whose strength periodically rises and falls) tugging on a satellite. Depending on the orbital period and spin period of Mars at the time of their capture, it has been calculated that tidal forces could act on these satellites in such a way that would eventually drag them from their initial highly elliptical orbits into circular ones.

Though these mechanisms require special conditions to work, some astronomers suggest that there is evidence that these moons are asteroids captured when they wandered from the outer solar system: their density and composition. The densities of Phobos and Deimos are surprisingly low, too low for them to be made of the materials that formed Mars. However, the density of some carbonaceous chondrite meteorites—meteorites rich in carbon and formed with particles present in the early solar system—is a much closer match, and with the addition of water ice, would be identical. Spectral measurements (taken from properties of reflected light) show that these moons are composed of nearly black rock, similar to carbonaceous chondrites and asteroids found in the outer part of the main asteroid belt, though very different from asteroids found closer to the Sun or from rocks on Mars.

The counter-argument to this holds that if Phobos and Deimos formed in place around Mars, then they should be composed of the same high-density material as the planet. To account for their low density, some scientists have speculated either that these moons are composed of high-density Mars rock and low-density ice trapped in their interiors, or that they contain enough pore spaces (air-filled sacs) to bring their density down to the observed values. In addition, to account for the spectral (reflected) properties of the bodies, these scientists argue that Phobos and Deimos are covered by a veneer (thin surface layer) of higher-density carbonaceous chondritic material delivered from the outer solar system, evidenced by the numerous impact craters seen on these moons.

The Moon's Origin

One of the earliest scientific theories about the Moon's origin postulated that Earth and its moon formed out of a dust cloud alongside one another, much as we see them today. However, when careful measurements of the lunar orbit were made, they showed that it was not steady but was changing with time. So, a new theory arose. This theory held that the Moon formed independently of Earth elsewhere in the solar system and was captured by Earth's gravity as it crossed Earth's orbit. If our Moon were formed elsewhere, such an alien moon would have a different geological history and structure than Earth, reflecting the region of the solar system in which it was formed. The Earth-capture theory fell out of favor when it was found that the Moon's geology and planetary characteristics closely resembled Earth's. However, in the 1980s, it was revived with the realization that some meteorites that had struck the Earth were of lunar origin and differed slightly from terrestrial rocks.

Towards the end of the twentieth century, a more advanced and informed theory developed, postulating that the Moon was born following a cataclysmic collision between early Earth and another planetary object with an irregular orbit that crossed Earth's orbit. This collision would have occurred around 4.5 billion years ago, when the smaller early (or "proto-") Earth was still largely molten and not quite yet solidified. The colliding object would have been roughly the size of Mars and has been given the name Theia. The collision was so cataclysmic that it knocked Earth off its axis, resulting in the current 23.5° tilt we see today. The tilt, now stabilized by the Moon, gives us our seasons during each orbit of the Sun. The impact was so catastrophic that the proto-Earth fractured and any early atmosphere was lost, with Theia being largely destroyed. The majority of the pulverized Theia mixed with Earth, and the remaining ejected debris formed a hot disc of vapor and material around the planet, much like the rings of Saturn. Then, over many centuries, some of the orbiting debris fell to Earth and mixed with the cooling surface, while the remainder coalesced to form the orbiting Moon. This created a Moon with a similar geological composition to that of Earth, with Theia's unique characteristics being completely obliterated in the process.

This theory has some limitations. While it requires the total destruction of Theia, it also requires that the orbiting ejected material would have to remain very hot for a long time to allow sufficient mixing to occur. It is very difficult to achieve this level of heating from a head-on collision. Thus, a more recent theory suggests that the collision was not head-on, but was more like a high-speed glancing blow. This sort of collision creates the heat required, and, if this theory is correct, then, somewhere in the solar system, a remnant of Theia may still survive.

The collision with Theia left Earth with a heavy molten-iron, radioactive core, giving it the ability to generate heat and a protective magnetic field. The Moon was left with only a small molten-iron core and a very weak magnetic field. Since then, the dynamic interaction between the liquid portion of Earth's core and its hard outer layer, the mantle, ensures that Earth's surface has been in continuous flux and the surface is always being subsumed or reformed through various geological processes somewhere in the world. While orbiting debris would have rained down onto the surface of the cooling Earth and Moon for millennia, there is scant evidence of this debris on Earth, as it has been lost in the continual geological reshaping and later weathering of the surface. Remarkably, on the Moon, a large proportion of the original surface remains just as it formed those many billions of years ago. While the surface is not flat and has undergone some volcanism of its own, there are plenty of valleys and mountains left over from this era. While moonquakes do occur, they do not alter the surface and are small—more like tremors. The Moon's surface therefore provides a record not only of the Moon of billions of years ago, but of all the impacts that have rained down on its surface throughout its lifetime.

Equestrian Portraits in the Eighteenth and Nineteenth Centuries

It was common for European rulers from the eighteenth and nineteenth centuries to be depicted in equestrian portraits, showing those rulers sitting on a horse. Two such artworks are Etienne Maurice Falconet's 1782 sculpture of the Russian ruler Peter the Great (died 1725) and Jacques-Louis David's painting of the French general (and later emperor) Napoleon (1800). These portraits promoted carefully crafted ideas about their subjects. Surmounting a 1,350-ton boulder, Falconet's bronze sculpture Peter the Great was a suitable monument to the ruler who oriented Russia toward European culture by building a new capital, St. Petersburg, on the western shore of Russia. Peter the Great serenely commands a rearing horse, beneath whose hooves the serpent of evil is about to be crushed. Falconet adopted the equestrian formula to equate Peter's greatness with that of ancient Roman emperors (although, as described below, Falconet attempted to show that Peter was a more enlightened ruler).

The status of equestrian portraiture can be traced to the famous second-century monument of Emperor Marcus Aurelius in Rome—the only surviving equestrian monument from ancient times, saved only because after the fall of the Roman Empire, people assumed that it represented the first Christian emperor, Constantine. Since then, there was a steady production of equestrian monuments, with kings, emperors, noblemen, and even important military leaders having their portraits astride a horse, either sculpted or painted. Falconet deviated from convention in several ways. He chose an innovative rising-up pose for the horse, and gave it a naturalistic character by placing it so that it appears to ascend a steep mountain. He also chose a purposely neutral cape (sleeveless garment) to avoid associations with Roman emperors, military leaders, or even Russia. Instead, Falconet wanted to emphasize Peter's identity as "a builder, a legislator, a benefactor of his country," a progressive ruler rather than a conqueror ruling with absolute control.

The painter Jacques-Louis David in *Napoleon at the St. Bernard Pass* (1800) glorified a historical moment during Napoleon's Italian war of 1800, when the general led troops through a dangerous mountain pass on the way to victory over the Austrians. The portrait was intended to provide the French public with a reassuring icon of control after the turmoil of the French Revolution. Aware of the urgent need to establish the legitimacy of Napoleon's regime—he came to power through a series of cleverly manipulated elections—David created a legacy that associated Napoleon with the only two previous historical figures capable enough to lead armies over the St. Bernard Pass in the Alps (mountains in Europe): Charlemagne, who united Western Europe under the Holy Roman Empire in 800, and the Carthaginian general Hannibal, who crossed the Alps with elephants during the Punic Wars with Rome in 218 B.C.E. and fought determinedly against Roman legions for 15 years. Their names are inscribed on the rock beneath the uplifting hooves of the horse, which Napoleon controls with the calm grandeur of a born leader. Here, David suggests that for Napoleon, no obstacle is insurmountable, no goal unattainable. Being true to historical fact did not really concern artists until the advent of the artistic movement of Realism in the 1840s, and it was then that the painter Paul Delaroche candidly represented Napoleon Crossing the Alps (1850) as it actually happened, with the general astride a sure-footed brown mule, led by a local peasant. The contrast in David's and Delaroche's images demonstrates how propagandistic intentions—distorting the truth to influence public opinion—steered the course of historical painting until well into the nineteenth century and also the extent to which truth can be manipulated in an image presumed to depict an event from history.

The French Academy of Fine Arts, which had largely controlled French art for centuries, forbade the representation of contemporary history in contemporary dress. David's *Napoleon*, in which Napoleon was dressed in a uniform from one of his own past battles, violates this boundary, inaugurating a practice of artistic rebellion that undermined the influence of the Academy over the next century. The unconventional character of David's *Napoleon* contrasts with the near-contemporary equestrian monument (1806) to Emperor Joseph II, who ruled Austria at the time. Reflecting Austrian art's conservative tendencies, sculptor Franz Zauner portrayed Joseph II in a Roman emperor's clothes.

The Early Iron Age

During the Bronze Age (roughly 3300–1200 B.C.E. in the Middle East), civilizations were limited in their use of iron. Most iron occurred not in an easily usable form, like copper and tin, but in ores (naturally occurring minerals or rocks) combined with other elements; the ore had to be smelted for the iron to be released. This was not easy. The relatively low melting point of bronze, around 1,922 degrees Fahrenheit (1,050 degrees Celsius), made it easy to work with in pottery ovens. But the smelting of iron ore required a temperature of 2,800 degrees Fahrenheit (1,538 degrees Celsius). Yet, if the technology for smelting iron were available, a huge, inexpensive metal supply would become available, since iron is very common, comprising 5 percent of the Earth's crust, whereas copper and tin are only trace elements.

Early iron production often is associated with the Hittites, who sometimes are said to have used it as a "secret weapon." But they, in fact, produced only small quantities of iron daggers and swords before their empire collapsed at the end of the Bronze Age. Not until then did the knowledge of iron smelting become widespread enough for iron to come into common use in western Asia. One can only speculate as to whether the movements of peoples at the end of the Bronze Age were a factor in the spread of this specialized knowledge. By 1100 B.C.E., knowledge of iron working had diffused into southwestern Europe, and it reached Britain by about 700 B.C.E. River valley civilizations, however, lagged behind; not until the seventh century B.C.E. did iron become widely used in Egypt.

The beginning of the Iron Age can be defined as the time when iron-working technology became sufficiently widespread and economical for commonplace, utilitarian items to be manufactured. Once this had happened, the transition from bronze to iron as the metal of choice for many items occurred quickly. Metal became available to a much larger segment of the population, rather than just to elites, and could be used for everyday objects such as household utensils and tools. Farming implements, in particular, such as plows, now could be made from iron rather than wood. As a result, large areas, such as the Rhine and Danube River valleys, whose tough soils hitherto had been difficult to work with earlier plows, were opened up to agriculture and thus to the spread of more complex cultures.

A misconception about iron is that it replaced bronze for weapons right from the beginning of the Iron Age. This was not the case. Early cast iron was soft, brittle, and did not retain a sharp edge. Bronze thus at first remained the most suitable metal for weapons, and it took a while for the development of a technology that could manufacture iron weapons of sufficiently good quality. It did not matter if a plow or axe broke because the iron shattered, but weapons had to be more reliable. Over time, methods for creating effective iron weapons were developed, such as hammering, carburizing (alloying, or mixing, iron with about 1 percent carbon), quenching (immersing heated iron suddenly in water), and tempering (reheating quenched iron and allowing it to cool slowly).

In addition, the characteristics that define the Iron Age go far beyond the use of iron. The civilizations that characterized the Bronze Age had been based on the extensive exploitation of agriculture in river valleys, but, lacking natural resources, they long since had exhausted their possibilities for further technological and economic expansion. By the beginning of the Iron Age, the centers of cultural and economic development had moved out of the river valleys, and, in general, they tended to move toward the west—first to the Levant, then to the Balkans, and finally to western Europe. Because cultures located outside the fertile river valleys could not seek economic expansion in agriculture, they found an alternative form of economic development in manufacturing and trade. Along with the luxury items that had been traded in the Bronze Age, Iron Age merchants also exploited trade in less expensive bulk materials, such as mass-produced pottery, textiles, and agricultural produce. For these societies, trade was not just an adjunct to agriculture but the primary means of economic expansion.

Mars's Disappearing Atmosphere

For about the first billion years of its life, Mars was probably a watery planet. There are signs of past water in surface features, such as dried riverbeds, and in minerals that could only have formed in the presence of water. For there to have been water, early Mars must have had a thick atmosphere secured by a magnetic field. While Mars and Earth seem superficially similar, Earth has a magnetic field that helps to protect and retain its atmosphere, whereas Mars lost its magnetic field long ago, and its atmosphere is about 100 times thinner than Earth's. Mars's atmosphere may have disappeared in two different ways: up into space, and down into the ground, to become locked up within rocks and soil. Observations by the spacecraft MAVEN have suggested that an "escape to space," in which the solar wind (electrically charged particles streaming out from the Sun) strips away material from the upper parts of a planetary atmosphere, is probably most to blame. This process is still taking place today with what is left of the Martian atmosphere.

The solar wind streams into Mars's vicinity, interacts with its atmosphere, and removes its uppermost ions (charged particles), dragging them out into space. The early Sun was far more volatile and active than the later Sun, flinging out bursts of radiation and producing solar storms far more frequently. This could have made any atmospheric stripping up to 20 times more devastating. Once Mars's magnetic field switched off, after half a billion years or so, the planet's upper atmosphere would have been naked, lacking its former shield. The solar wind could have streamed in and ionized molecules at a far higher rate, stripping them away faster.

There is also the fact that Mars is smaller, is less heavy, and thus has lower surface gravity than Earth. Accordingly, the planet simply struggles to keep hold of as much atmospheric material as Earth can. Mars's atmosphere also may have been blasted out into space by the disastrous collisions with large objects we know to have taken place during its early history, which in turn would have made subsequent impacts even more destructive due to the thinner Martian atmosphere, continuing and worsening the cycle. In a less exciting scenario, Mars's atmosphere might have just spun off into space of its own accord—the molecules may have swirled around and around, colliding, and transferring energy between themselves, growing faster and faster until they moved fast enough to escape.

At least some of Mars's atmosphere is thought to have headed down into the planet's surface layer, interacting with the soil and combining with the elements present there, a process known as sequestration. We see signs of this process in the few Martian meteorites we have found here on Earth and in the explorations of the United States space agency's Curiosity rover, which saw further signs when it arrived on Mars and began to study the soil and atmosphere in earnest. However, we have found that this is unlikely to have been a dominant process; we would expect the heavier parts of Mars's atmosphere to have disappeared if it were, as they would have sunk down and been the ones to interact with the Martian surface. We see the opposite: instead of an atmosphere filled with lighter forms of the constituent elements, we see heavier ones. This suggests that Mars's atmosphere was stripped from above and not below, thus removing more of the lighter elements and leaving the heavier ones behind. We would also expect to find carbonates, compounds that are a consequence of sequestration, in the Martian soil, but despite our various searches these do not seem to be present in large amounts.

We are still unsure of exactly what happened with Mars's atmosphere but are fairly confident that the planet was once wrapped in a far thicker atmosphere that allowed the planet to stay warm and pressurized enough to support bodies of running and standing water. Mars's ancient atmosphere was likely different in composition, with a far higher amount of oxygen, a necessary component of water. We know this because we have found minerals in old Martian rock that need either an oxygen-rich environment or one with microorganisms present in order to form, and we do not believe it to have been the latter.

Erratics and Drift

In the nineteenth century, scientists theorized that glaciers (ice sheets) covered much of Earth's surface during an ancient very cold period known as the Ice Age. Following efforts to map ancient glaciers, researchers concluded that there had been multiple ice advances and retreats. Evidence for multiple glaciations was tied to two types of glacial deposits: one called erratic boulders, often referred to simply as erratics, and a second, glacial drift—a general term for the loose, rocky debris distributed across the countryside by slowly moving glaciers. Erratic boulders are spread over large regions in Europe and North America; the most remarkable ones are very large, and they are often quite different in makeup from the local bedrock. How they got there was unknown before scientists understood how ancient glaciers transported rocks. In the Northeastern United States, where there are outcrops of very distinctive rock varieties, trails of erratics of a specific type can often be traced for hundreds of kilometers, fanning out over the countryside near their sources. Careful mapping of these erratic trails can provide an accurate picture of how the glaciers that carried the boulders moved across the land.

Before the Ice Age theory was generally accepted, most geologists and naturalists argued that the erratics had been transported by water to their current resting places. They realized that even fairly small boulders would sink instantly in normal streams, but they also understood enough about rare natural phenomena such as tidal waves and great storms to know that water could transport heavy objects under extreme conditions. But even extreme events could not easily explain the massive granite erratics in the European Jura Mountains, especially the ones that were perched high on valley walls, far above the streams below. Compounding the problem of interpreting these deposits, however, was the fact that at a few localities in Britain, where much of the most detailed research into the Ice Age controversy was being conducted, the fine-grained drift that accompanied the erratic boulders contained seashells. Critics of the hypothesis that glaciers had transported erratic boulders seized on this; they claimed it was conclusive evidence that the ocean was involved. They argued that the erratics must have been transported by great, violent floods coursing over the land, and they said that there were simply no modern-day counterparts. They knew that the sea had covered parts of the continents in the past because fossilized fish were found throughout Europe. The marine shells in "glacial" drift, they asserted, were proof that the sea had invaded the land yet again and left the drift behind when it receded. It was not until much later that the true origin of the seashells in drift was realized. James Croll, a Scottish scientist, deduced that they too had been transported by glaciers, scraped up by the ice along with sediments from the shallow seas around Britain and carried inland. However, before their origin was understood, the shells were a serious difficulty for those who argued that drift and erratics were Ice Age deposits.

Still, notwithstanding the seashell argument, even some of the opponents of the glacial theory had to admit that it would be impossible to transport large erratic boulders in water over long distances, no matter how violent the storm or flood. So, they came up with an ingenious solution: the erratics might indeed have been carried by ice, but ice that was floating on formerly more extensive seas, transporting boulders from a northerly source. If parts of the continents had been submerged in the past, they reasoned, the icebergs could have floated over the sunken land, dropping their rocky burden as they melted. That would explain the presence of ocean shells in the drift. It was the idea of drifting icebergs that first led to the use of the term "drift" for the characteristically chaotic sediments left behind by glaciers—sediments that have neither the well-defined layers nor the uniformity of grain sizes that characterize those deposited in water. The term is still in use today, although the theory that drifting icebergs deposited the seashells, erratics, and other material has been abandoned.

Jazz

Jazz music, an art form original to the United States, originated in the southern city of New Orleans at the turn of the twentieth century. It combined the elements of various music styles, including ragtime, folk songs, religious hymns, blues, and swing, into a commercially and artistically successful genre. New Orleans was a unique city for a variety of reasons. The location of the city at the end of the Mississippi River and as a major port city provided a great deal of commerce and trade nationally and internationally. The population was diverse, with a small percentage of white citizens in power, while most white citizens and all black citizens were at the lower end of the social and economic ladder. There was also a unique class of mixed-race people in New Orleans referred to as Creoles, who were very proud of their French heritage. The Creoles were responsible for the entertainment neighborhood of the community that is still known today as the French Quarter. Bars, nightclubs, and other entertainment outlets were the typical businesses found in the Creole section of town. Traders, merchants, and travelers were attracted to the nightlife of the French Quarter, much in the same way tourists are attracted there today.

The Creoles looked to talented black musicians to provide music for these establishments, and the opportunity for employment and music making was attractive to the performers. Most of these musicians were limited in their education and relied on their ear to perform. Musicians would get together and begin to play in a free, improvisatory fashion (spontaneously changing with each performer). None of this music was recorded and was probably a highly improvised variation of a single melody, known as monophony. When the musicians would attempt to perform the melody, mistakes, restrictions of their instruments or their abilities, the range, timbre, and flexibility differences between instruments, and creativity would make the monophony so varied that it would often be referred to as heterophony—music created by a number of musicians performing the same melody, with each player applying slight improvisation of the melody. This loose, highly rhythmic, improvisatory music became highly regarded in New Orleans, and soon jazz musicians and their music began to spread to other United States cities along or not far from the Mississippi River, such as Memphis, Kansas City, and Chicago.

The original jazz music of New Orleans is lost to us since it was never recorded or notated, but much of what is referred to commonly as Dixieland jazz (based on the popular name, Dixie, for the southern United States) was the style of music popular in the northern city of Chicago in the 1920s. Many of the musicians, such as King Oliver and his young protégé, Louis Armstrong, were recruited from New Orleans and were featured in nightclubs and speakeasies (unlawful liquor-selling establishments) that were part of the culture of Chicago in the 1920s, when the sale of alcoholic beverages was prohibited. This music began to attract a larger audience primarily through the rising popularity of the phonograph (an early sound-recording device). The first pieces of music to be recorded and sold for a considerable profit were "Dixie Jass Band One-Step" and "Livery Stable Blues" on each side of a 78-rpm record. These tunes were recorded in 1917 by a group of white musicians led by cornet player Nick La Rocca in New York City. The group was called "The Original Dixieland Jazz Band," but they were neither original, Dixieland in style, nor from New Orleans. This record sold over a million copies and set a standard for how New Orleans jazz was supposed to sound. This music was a fair copy of the music being performed in Chicago by black musicians at the time. The later recordings of King

Oliver and Louis Armstrong were stronger examples of the style, but this recording does mark an important era in jazz music and the beginnings of successful commercialization of popular music in the United States.

Parental Care by Frogs

Only an estimated 10 percent of anuran (frog and toad) species provide any kind of parental care. But within this 10 percent, there is very great diversity of caregiving behaviors. The behaviors are of five basic types. The most common type is attendance of eggs. In most cases, terrestrial eggs are attended. The other four types of parental care are transporting the eggs, attending young frogs or tadpoles (the immature stage of a frog's development), transporting the tadpoles or young frogs, and feeding the tadpoles.

The term parental care is generally used for any type of parental investment in offspring after the eggs have been deposited or the young have been born. Only a few studies have investigated the costs and benefits of parental care in amphibians, so, while parental care presumably increases the survivorship of the young and usually entails a cost to the caregiver, the costs and benefits for the vast majority of frog species that provide care to their offspring are unknown.

How does parental care increase the survivorship of offspring? The parents may protect the eggs from predators or disease, or they may aerate aquatic eggs. Because the terrestrial environment is not hospitable to the porous jelly-covered eggs of amphibians or to tadpoles, parental behaviors may be particularly beneficial in preventing drying out. In some species, the parent releases water from its bladder onto the eggs. Parents may turn the eggs, thus preventing developmental abnormalities. In New Guinea, male frogs of some species transport young on their backs. The tiny frogs hop off at different places in the habitat. Thus, the young may benefit from reduced competition for food, lower predator pressure, and reduced levels of inbreeding.

The benefits of egg attendance are best known in the Puerto Rican coqui frog. Female coquis lay eggs in a sheltered retreat within the male's territory, and the male attends the eggs. To determine the benefits of egg attendance, Daniel Townsend and his colleagues removed attending males from some clutches (groups of eggs produced at the same time) to determine the fates of attended and unattended eggs. Unattended clutches failed more frequently than attended clutches. The main causes of mortality of unattended clutches were being eaten by other male coquis and drying out. Male coquis are important predators of the eggs of other coquis, and attending males aggressively defend their retreats from intrusion by other males.

Attending males also protect their clutches from drying out. Most amphibians do not exhibit parental care. This observation suggests that parental care has costs to the parents that may outweigh the enhanced survival of offspring. Reduced reproductive output is one cost of parental care. Species that exhibit parental care usually produce fewer eggs per clutch than related species that do not have parental care. Furthermore, time and energy spent on parental care may limit opportunities for additional matings. Another cost may be decreased food intake for the caring parent. Parents typically do not eat when they are guarding nests or eggs, and females that remain with their clutch produce fewer clutches overall than do females that do not remain with their clutch. Reduced rates of parental survival are another potential cost of parental care. Remaining with the eggs could increase an individual's vulnerability to predation. Because most amphibians are small and have ineffective defenses against vertebrate predators, parental care could increase the risk of the parents' death and might not save their eggs or young.

What circumstances would favor the evolution of parental care? We would expect parental care to evolve if a parent is able to increase offspring survival enough to offset the costs involved. If environmental conditions are harsh, or if predation pressure on the eggs or larvae is high, the benefits might outweigh the costs. There are some circumstances where the costs of parental care might be minimal. For example, cost might be minimal where parents attend offspring in hidden nest sites not readily accessible by predators. For females, if reproductive output is already limited for other reasons, the cost of parental care might be minimal. If females generally only produce one clutch per season, caring for that clutch might not reduce reproductive output appreciably. Costs might be minimal for a male if he can provide care for his offspring and at the same time defend territory and attract additional females.

The Greek Revival

After the Mycenaean Greek civilization collapsed around 1100 B.C.E., Greece endured a three-century-long Dark Age about which relatively little is known. But the year 776 B.C.E., when the winners of the Olympic Games were first recorded, marks the beginning of a new period of Greek history, the Archaic Age (lasting to 500 B.C.E.), which brought a revival of culture, the economy, and political significance to Greece. During the relatively peaceful Dark Ages, populations had gradually increased to a point where Greece's rocky and hilly soil could not produce enough basic agricultural products. One solution was to import additional foodstuffs. As a consequence, Greek commerce and production of trade goods expanded. In exchange for grain imported from Egypt and lands around the Black Sea, the Greeks traded olive oil, fine pottery, and silver. This explosion of commerce brought the Greeks into direct conflict with the existing Mediterranean trading power, the Phoenicians. The warlike Greeks constructed fleets of maneuverable iron-beaked fifty-oared ships called galleys, for which the unwieldy Phoenician two-decked warships were no match. The Greeks soon wrested control of important Mediterranean trade routes from the Phoenicians.

A second solution to the overpopulation problem was to seek new farmland elsewhere. In a wave of colonization lasting roughly from 750 to 550 B.C.E., Greek cities established colonies on the shores of the Black Sea, Adriatic Sea, and Mediterranean Sea, as well as in North Africa, France, and Spain. Because colonists wanted access to the sea for trade, they occupied only coastal sites. The Greek city of Corinth, for example, founded the great city of Syracuse in Sicily. The most extensive immigration was into southern Italy and western Sicily, which became known as Greater Greece. Colonies had the same culture, social structure, and government as the cities that founded them. Although they maintained sentimental ties to their mother city, they were completely independent. Because the colonists were mostly male, they found wives and slaves among the local populations whose land they had also taken. A consequence of this intermingling was the spread of Greek culture, which became a common culture throughout the Mediterranean world and diffused into central Europe through the colony of Marseilles in southern France and into southern Russia via the Black Sea colonies.

The revival of Greek trade also brought Near East culture into Greece. From the Phoenicians, the Greeks borrowed the alphabet, for they now had a need to keep records. From the Lydians, a people from what is now Turkey, they picked up coinage, and the silver coins of Corinth and Athens became the standard currency throughout the Mediterranean world. The Greeks also assimilated artistic ideas. The simple geometric patterns of the Dark Ages gave way to Eastern influences. Greek sculpture assumed a very Egyptian look, and Greek pottery depicted many Eastern designs, such as sphinxes, lions, and bulls. Almost always, however, the Greeks modified what they borrowed to suit their own preferences. They were the first people to put designs on both sides of coins. They changed some letters of the Phoenician alphabet from consonants into vowels. And Greek potters and sculptors soon used designs from their own myths and legends.

Greek pottery was a particularly important trade good. Some pots, such as large urns used as grave monuments, were made for ceremonial purposes, but most were utilitarian. Four-foot-tall jars called amphorae stored wine, olives, and other edibles. Many kinds of bowls, plates, and cups served as tableware. Greek potters were soon the acknowledged masters of pottery making. All Greek cities produced pottery for local use, but the pottery of important trading cities was valued throughout the ancient world. During the sixth century B.C.E., Corinth was famous for pottery featuring black figures on a lighter background, but by 500 B.C.E., Athenian pottery with red figures on a black background had become the preferred style.

During the Archaic Age, the Greeks also began to build stone temples as centers of civic pride as well as to honor their primary gods. Temples usually were constructed in simple rectangular form, with a tile roof supported by external rows of stone columns, a style that also eventually spread throughout the Mediterranean.